

ECOLOGY

(Environmental Biology)





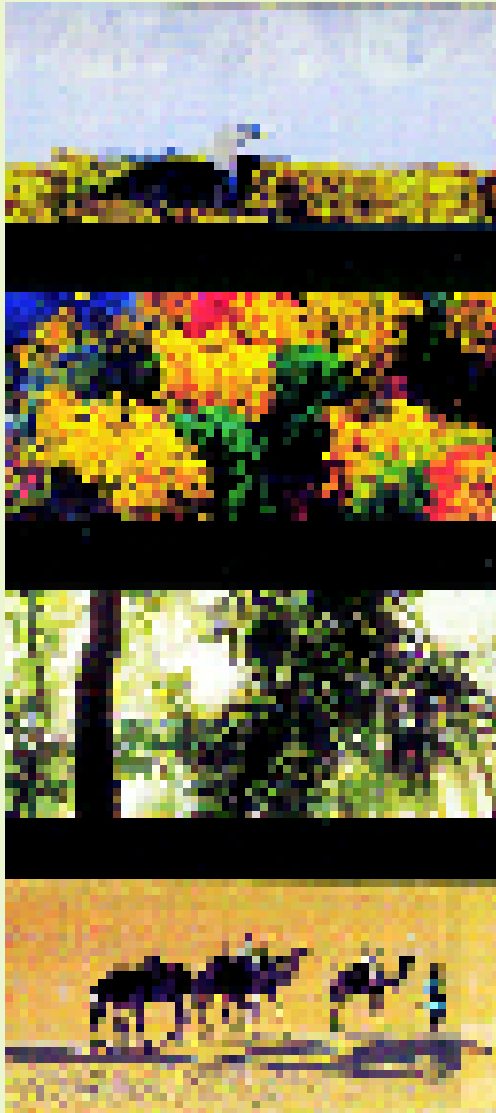
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CHAPTER

1

Introduction



Ecology is the study of inter-relationship of organisms with their physical and biotic environment.

No microorganism, plant or animal species including man is an isolated organism living in a void. Each of them is surrounded by a host of physical conditions that can be measured in terms of chemical composition, texture, pressure, temperature, and humidity, as well as being surrounded by a host of other living organisms which can be described in such terms as microorganism, plants, animals, food, parasites, and enemies. Studies of the inter-relationships of organisms with their physical and biotic environments are termed **environmental biology** or **ecology**. These words are very much in the public consciousness today as we become aware of some of the past and current ecological malpractices. It is important for everyone to know and appreciate the principles of this aspect of biology so that one can form an intelligent opinion regarding topics such as insecticides, detergents, mercury pollution, sewage disposal, power dams, urbanization and their effects on mankind, on human civilization, and on the world we live in.

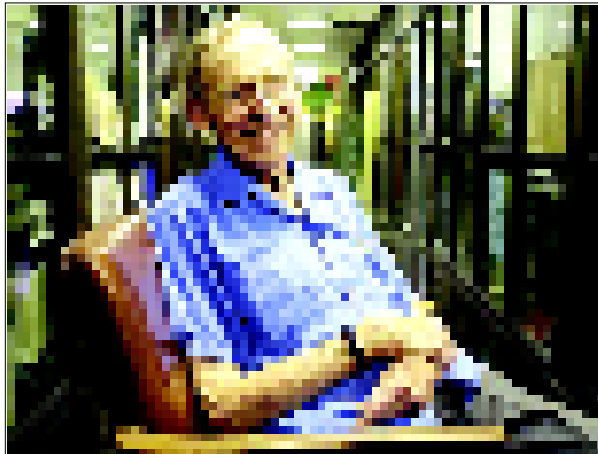
The term **ecology** (*oekologie*) is derived from two Greek words—*oikos*=means ‘house’ or ‘place to live’ and—*logos* means ‘a discussion or study’. Literally, ecology is the study of organisms ‘at home’, in their native environment. The term was first of all introduced by **Reiter** in 1868, but because the German biologist **Ernst Haeckel** (1869) first of all fully defined

this term and made an extensive use of this term in his writings, he is therefore, usually falsely credited for the coinage of the term ecology (see **C.B. Knight**, 1965).

DEFINITION OF ECOLOGY

The traditional definition of ecology is '*the study of an organism and its environment*', however, different ecologists have defined it variously. **Ernst Haeckel** (1869) defined ecology as, '*the total relation of the animal to both its organic and its inorganic environment.*' In 1936, **Taylor** defined ecology as '*the science of all the relations of all organisms to all their environments.*' **Charles Elton** (1947) in his pioneer book *Animal Ecology* defined ecology as '*scientific natural history.*' Although this definition does point out the origin of many of our ecological problems, yet it is much broad and vague like Haeckelian definition of ecology. **Allee et al.**, (1949), in their definition of ecology, clearly emphasize the all-encompassing character of this field of study. According to them ecology may be defined broadly as '*the science of the interrelation between living organisms and their environments, including both the physical and biotic environments, and emphasizing interspecies as well as intra-species relations.*' Though, **F.J. Vernberg** and **W.B. Vernberg** (1970) completely agree with **Allee et al.**'s definition, yet there are certain ecologists which are not satisfied with this definition and have provided their own definitions of ecology. For instance, **Andrewartha** (1961) defined ecology as '*the scientific study of the distribution and abundance of organisms.*' **G.A. Petrides** (1968) has defined ecology as '*the study of environmental interactions which control the welfare of living things, regulating their distribution, abundance, production and evolution.*' **Eugene Odum** (1963, 1969 and 1971) has defined ecology as '*the study of the structure and function of ecosystems*' or '*structure and function of nature.*' The Indian ecologist **R. Misra** (1967) defined ecology as the '*interactions of form, functions and factors.*' These three interacting aspects, i.e., form, functions and factors integrate together to construct the **triangle of nature**.

C.J. Krebs (1972) has shown his satisfaction over the definition of **Odum** for its emphasizing the form and function idea that permeates biology, but, he considered it vague. He, however, proposed the modified version of Andrewartha's definition of ecology, according to which '*ecology is the scientific study of the interactions that determine the distribution and abundance of organisms.*' According to **M.E. Clark** (1973) '*Ecology, the science of the environment, is a study of ecosystems, or the totality of the reciprocal interactions between living organisms and their physical surroundings.*' Two Indian authors **Ananthakrishnan** and **Viswanathan** (1976) have defined ecology as '*the study of the ways by which individual organisms, populations of species and communities of populations respond to diverse environmental pressures, i.e., physical and biotic pressures.*' This definition is merely the modified version of the definition of ecology which has been originally proposed by **T. Lewis** and **L.R. Taylor** (1967) in their book, *Introduction to Experimental Ecology*. According to **Lewis** and **Taylor** '*Ecology is the study of the way in which individual organisms, populations of same species and communities of populations respond to these changes.*' Lastly to avoid this ecological jargon we can rely upon the simple definition of ecology which has been provided by **Charles H. Southwick** (1976). According to him '*ecology is the scientific study of the relationships*



Eugene Odum.

of living organisms with each other and with their environments'. Thus, it is the science of biological interactions among individuals, populations, and communities. Ecology is also the science of ecosystems—the interrelations of biotic components with their non-living environments.

Ecology versus Environmental Studies

The continuing increase of human population and the associated destruction of natural environments with deforestation, pesticides and pollutants have awakened the awareness of the public to the word of ecology. However, much of this recent interest centres on the human environment and human ecology. Unfortunately the word ecology became identified in the public mind with the much broader problems of the human environment, and “ecology” came to mean everything and anything about the environment. The science of ecology is concerned with the environments of all plants and animals and is not solely concerned with humans. As such, ecology has much to contribute to some of the broad questions about humans and their environments. Apparently, ecology should be to environmental science as physics is to engineering. Just as humans are constrained by the law of physics when we build aeroplanes and bridges, so also we should be constrained by the principles of ecology when altering the environment (see **C. J. Krebs**, 1985).

HISTORICAL BACKGROUND OF ECOLOGY

The roots of ecology lie in natural history, which is as old as man himself. The vague beginning of ecology may be traced back to prehistoric man, who utilized environmental information to hunt food; fishing, trap animals, find edible vegetation, and locate shelter to survive the hardships imposed by nature. An increased knowledge of the importance of environmental conditions led quite naturally to religious rituals, worship of weather gods, and rain dances, all of which are quite prevalent in different social groups and tribes of Indian civilization. The establishment of agriculture increased the need to learn about the practical ecology of plants and domestic animals.

Further, the science of ecology has had a gradual but spasmodic development during the recorded history of human civilization. Consequently, the historical background of ecology can be traced as follows :

1. Ecology in Indian Classics

Early Indian philosophers and thinkers had some knowledge of ecology as has been revealed by the Indian classical writings, in the Vedic, Upanishadic, Pauranic, and Epic literature. **Charaka** considered the important factors of *Vayu* (air and gases), *Jala* (water), *Desha* (topography) and *time* in regulating the life of plants (see **R. Misra**, 1962). The great classical Indian Poet **Kalidasa** has displayed his ecological faculty in his classics—‘*Ritu Sanhar*’, ‘*Meghdoot*’ and ‘*Abhigyan Sakuntlam*.’

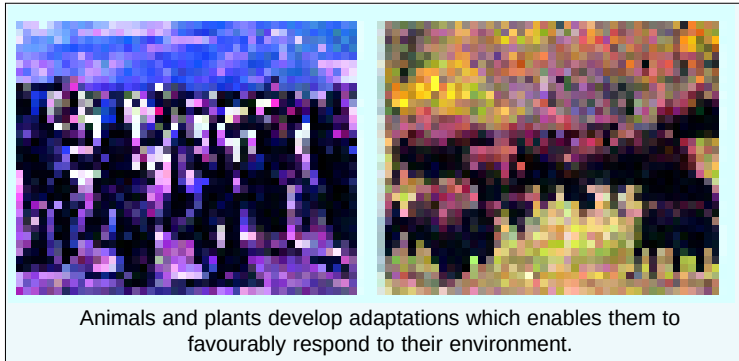
2. Early Greek's Ideas of Ecology

The early Greek philosophers too were well aware of the importance of environmental studies. One of the papers entitled *Airs, waters and places* of **Hippocrates** stressed on the need of ecological background for medical students. According to him—“whoever wished to investigate medicine properly, should proceed thus : in the first place to consider the seasons of the year, and what effects each of them produces.....” In the fourth century B.C. **Aristotle** made references about the habits of animals and environmental conditions prevailing in certain areas, in his writings on natural history. One of the Aristotle's friend and associate **Theophrastus** (370–250 B.C.) who is regarded as the first true ecologist, wrote about the association of plant communities and the relation of plants to each other and the non-living environment.

3. Ecology in 18th Century

After a considerable long gap of many centuries, the first naturalist who made a serious attempt

to systematise knowledge concerning relations of animals to environment was **Buffon** (1707–1788). Though such works as *Historia animalium* by **Konard Gesner** (1519–1565); 14 volumes of collected works of natural history by **Ulisse Aldrovandi** (1522–1605); the natural histories of fishes and birds by **Pierre Belon** (1517–1564); *The plants* by **Andrea Caesalpino** (1519–1603), the botanical work of **Gaspard Bauhin** (1550–1624); demographical studies about human ecology by **Graunt** (1664); studies of the sexuality of plants by **Rudolph Camerarius** (1665–1721); the works of English naturalist **John Ray** (1627–1705); studies on natural history of insects by French naturalist **Rene Reaumur** (1683–1757) and the works of many other naturalists doubtlessly laid the foundation upon which **Buffon** built his remarkable work. **Buffon** considered that animals and plants developed adaptations which enabled them to favourably respond to their environment and termed this **environmental induction**. He touched on many of our modern ecological problems and recognised that population of man, other animals and plants are subjected to the same processes. **Buffon** also explained how the great fertility of every species was counter-balanced by innumerable agents of destruction. He, thus, dealt with problems of population regulation.



Anton Van Leeuwenhoek, best known as a pioneer microscopist of the early 1700s, also pioneered the study of “food chains” and “population regulation”, which are two important areas of modern ecology (see **Odum**, 1971).

Malthus published one of the earliest controversial book on demography. In his *Essay on Population* (1798) he calculated that although the number of organisms can increase geometrically (1,2,4,8,16,32.....); their food supply may never increase faster than arithmetically (1, 2, 3, 4, 5....). Such a great disproportion between these two powers of increase led **Malthus** to infer that reproduction must eventually be checked by food production. Many workers questioned the ideas of **Malthus**. For example, in 1841 **Doubleday** brought out his true law of population. To contradict the **Malthus** idea, he believed that whenever a species was threatened, nature made a corresponding effort to preserve it by increasing the fertility of its members. Human populations that were undernourished had the highest fertility; those that were well fed had the lowest fertility. **Doubleday** explained these effects by the oversupply of mineral nutrients in well-fed populations.

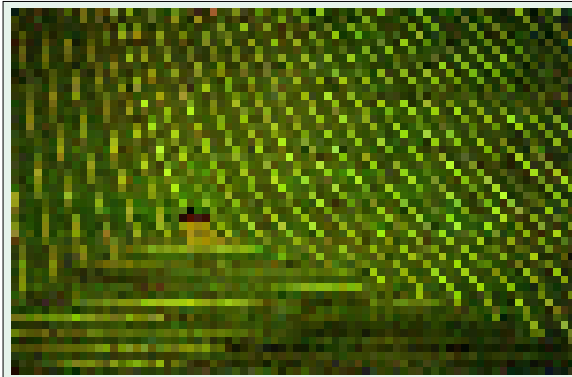
Further, many of the early developments in ecology came from the applied fields of agriculture, fisheries, and medicine. For example, work on the insect pests of crops has been one of the important source of ideas. The regulation of population size of obnoxious insects is a basic problem that has been under study. In 1762 the mynah bird was introduced from India to the island of Mauritius to control the red locust. Due to such **biological control**, by 1770 the locust threat was a negligible problem. **Forska** wrote in 1775 about the introduction of predatory ants from nearby mountains into date palm orchards to control other species of ants feeding on the palms in South Western Arabia. In subsequent years an increasing knowledge of insect parasitism and predation led to many such introductions all over the world in the hope of controlling introduced and native agricultural pests.

Similarly, medical work on infectious diseases such as malaria around the 1890's gave rise to the study of epidemiology and interest in the spread of disease through a population. Before malaria could

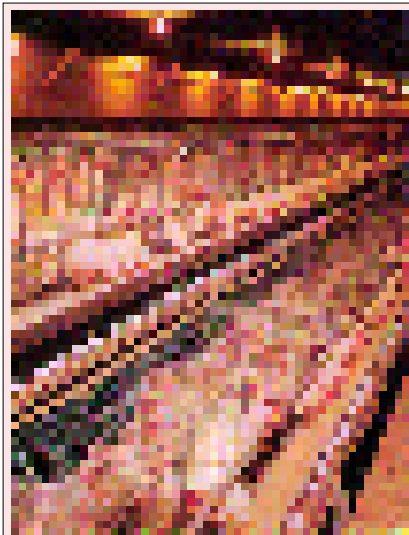
be controlled adequately it was necessary to know in detail the ecology of mosquitoes. Likewise, the production ecology had its beginnings in agriculture, and **Egerton** (1969) has traced this back to the eighteenth-century botanist **Richard Bradley**. Bradley recognized the fundamental similarities of animal husbandary and production, and he proposed methods of maximizing agricultural yields (and hence profit) for vineyards, trees, poultry, rabbits and fish.

4. Ecology in 19th Century

In 1807, **Humboldt** after extensive travelling of tropical and temperate South America, discussed the geographical distribution of plants and animals in relation to climate. **Edward Forbes** in 1844 described the distribution of animals in



Production ecology enables maximum agriculture yields.

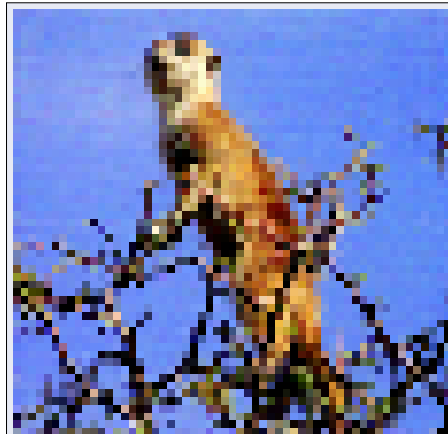


Poultry has a economic base.

British coastal waters and part of the Mediterranean sea, and he wrote of zones of differing depths which were distinguished by the associations of species they contained. He noted that some species are found only in one zone and that other species have a maximum of development in one zone but occur sparsely in other adjacent zones. Mingled in are stragglers that do not, fit in zonation pattern. **Forbes** recognized the dynamic aspect of the interrelations between these organisms and their environment. As the environment change, one species might die out, another might increase its abundance. English naturalist, **St. George Jackson Mivart** coined the term **hexicology** around 1859 and in 1894 he defined it as “devoted to the study of the relations which exist between the organisms and their environment as regards to the nature of the locality they frequent, the temperatures and the amounts of light which suit them, and their relations to other organisms as enemies, rivals or accidental and involuntary benefactors.” **Saint-Hilaire** (1859) used the term “**ethology**” for the study of relations of organisms to environment and outlined a volume, which

he never wrote, on instincts, habits, and other ecological matters. The term ethology, though was not accepted by early ecologists, but this term is now used as an important ecological discipline dealing with animal behaviour. In 1869, **Haeckel** proposed “**oecologie**” for the relations of organisms to organic and inorganic environments. The modern term *ecology* is derived from the word *oecologie*. The observations of **Priestley** and **Scheele** that plants produce oxygen led to an understanding of the inter-relationships of plants and animals. The discovery that green plants and animals use carbon dioxide and water to form organic matter for use by animals and result in the release of oxygen and water as wastes, provided new understanding of ecological food chains. **Justus von Liebig** (1803–1873) initiated the idea of carbon and nitrogen cycles in nature and his work (1840) led to **ecoclimatology** and **physiological ecology**. Further, out of such studies in aquatic environment developed the field of **ecological energetics**. **Louis Agassiz** (1807–1873) initiated the research in marine ecology by establishing many laboratories for marine ecological research. **Karl Mobius** (1877) introduced the word “**biocoenosis**” to designate a group of organisms as an ecological unit. **Charles Darwin** (1809–

1882) and **Alfred R. Wallace** (1823–1915) have increased our knowledge of island of insular life with their characteristically different types of environmental conditions. **Hensen** (1887) in a classical paper on “*The Lake as a Microcosm*” suggested that the species assemblage in a lake was an organic complex and that by affecting one species we exerted some influence on the whole assemblage. Thus each species maintains a ‘community of interest’ with the other species, and we cannot limit our studies to a single species. **Forbes** believed that there was a steady balance of nature, which held each species within limits year after year, even though each species was always trying to increase its number. The Russian pioneer ecologist **V.V. Dokuchaev** (1846–1903) and his chief disciple **G.F. Morozov** (who specialized in forest ecology) placed great emphasis on the concept of the “**biocoenosis**”, a term later expanded by Russian ecologist to “**geobiocoenosis**”, a term equivalent to ecosystem.



Ethology deals with particular behaviour patterns of animals.

Schroter (1896) introduced the terms “**autecology**” and “**synecology**”. **Herdman** (1896) indicated the economic possibilities of scientific agriculture. Danish botanist **Warming** (1895, 1909) raised questions about the structure of plant communities and the associations of species in these communities. In 1899, **H.C. Cowles** described *plant succession* on the sand dunes at the Southern end of Lake Michigan.

Thus, as a recognized distinct field of biology, the science of ecology dates from 19th century, and only in the mid 20th century has the word become the part of the general vocabulary.

5. Ecology in the 20th Century

At the beginning of the twentieth century ecology was a young, but an established science and such eminent ecologists as **Wasmann** (1901), **Dahl** (1901) and **Wheeler** (1902) were discussing whether **Saint Hilaire's** ethology or **Haeckel's** ecology should be used to designate the science of relations of organisms to environments. The latter term has gradually come into general usage. **Case** (1905) discussed environmental conditions of past geological periods and has continued to make contributions in that field. Two botanists, **Clements**, in his “*Research Methods in Ecology*” (1905) and subsequent writings, and **Warming** (1909, 1925) in his “*Oecology of Plants*” have done much to clarify ideas and contribute terms. **Adams** (1913) gave an excellent review of significant ecological literature, urged the usefulness of ecological surveys, and discussed methods to be used in making them, **Shelford** (1912) demonstrated excellent examples of ecological succession and has made notable contributions concerning animal communities (1913) and environmental factors (1929). In 1920 the German limnologist **Thienemann** introduced the concept of trophic levels in terms of producers and consumers. The theoretical approaches in population dynamics of **Lotka** (1925) and **Volterra** (1926) stimulated the experimental approaches by biologists. **Gleason** (1926) studied the development and dynamics of plant communities. **Shelford** (1913, 1937), **Adams** (1909) and **Dice** (1943) in America and **Elton** (1927) in England investigated the interrelations of plants and animals. In 1935 **A.G. Tansley** first proposed the term **ecosystem**.

Gause (1935) investigated the interactions of predators and prey and the competitive relationships between species. At the same time **Nicholson** studied interspecific competition. In 1954 **Andrewartha** and **Birch** and **Lack** provided a broader foundation for the study of regulation of populations. The discovery of the role of territory in bird life by **H.E. Howard** in 1920 led to further studies by **Nice** in the 1930s and 1940s. Out of such studies came the field of **behavioural ecology**.

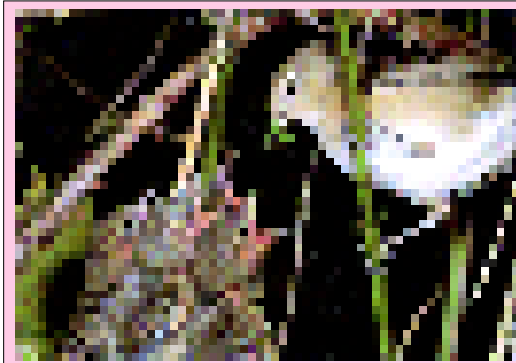
In the 1940s and 1950s **Lorenz** and **Tinbergen** developed concepts of instinctive and aggressive behaviour. In 1962, **Wynne-Edwards** of England explored the role of social behaviour in the regulation of populations.

In the field of ecological energetics, in 1920 the German limnologist **Thienemann** introduced the concept of trophic levels in terms of producers and consumers. Two American limnologists, **Birge** and **Juday** in the 1940s, through their measurements of the energy budgets of lakes, developed the idea of primary production. Out of their studies came the trophic-dynamic concept of ecology. Introduced by **Lindemann** in 1942, this concept marked the beginning of modern ecology. Out of **Lindemann's** study came further pioneering work on energy flow and energy budgets by **Hutchinson** and **H.T.** and **E.P. Odum** in the 1950s in America. Early work on the cycling of nutrients was done by **Ovington** (1957) in England and Australia and by **Rodin** and **Bazilevich** (1967) in Russia.

One of the interesting aspects of development of ecology during the first third of this century remains that a tendency towards increased specialization in ecology has occurred as is evidenced by the establishment of such disciplines as **palaeoecology**, a study of environmental conditions and life as it existed in past ages. Pollen analysis, radioactive dating, and paleontology have aided the palaeoecologists. Other specialized fields of ecology which took their birth during this time, are **zoogeography**, the scientific study of the geographic distribution of animals; **oceanography**, the study of the biotic and physical conditions existing on oceans, bays, and estuaries, and **limnology**, the study of the living and non-living components of inland waters. With the establishment of specialized disciplines of ecology, the value of application of ecology to such fields as agriculture, forestry, wildlife conservation, fisheries management, and pest control became apparent, and, thus, a new discipline called **applied ecology** came into its existence.

Further, in the second decade of 20th century great emphasis was laid on statistical studies on populations, sampling techniques, and community studies. During this period several important texts published—**Lotka's** *Elements of Physical Biology*; **Elton's** *Animal Ecology*; **Volterra's** *Animal Ecology*; **Gause's** *The Struggle for Existence*, and **Bodenheimer's** *Problems of Animal Ecology*. Certain distinguished ecologists of earlier half and present decade of this century are **Clements**, **Cowles**, **Shelford**, **Pearse**, **Oosting**, **A.M. Woodbury**, **Hutchinson**, **Deevey**, **Lindeman**, **F.E. Smith**, **Slobodkin**, **Hairston**, **T. Park**, **O.Park**, **Emerson**, **MacArthur**, **G.L. Clark**, **C.B. Knight**, **H.T.** and **E.P. Odum**, **R.L. Smith**, **F.J. Vernberg**; **C.J. Krebs**, **R. Misra**, **M.S. Mani**, etc.

Development of ecology during this century have been greatly stimulated by organization of ecologists. The British Ecological Society was founded in 1913 and this society sponsored two Journals namely *Journal of Ecology* (started in 1913)



Starling chicks forage for food without leaving the nest showing begging behaviour.



Limnology is the study of living and non-living components of inland waters e.g., pond, river.

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and *Journal of Animal Ecology* (started in 1932). Following the British, the Ecological Society of America was established in 1915 which also sponsored two publications, *Ecology* in 1920 and *Ecological Monographs* in 1931.

BRANCHES OF ECOLOGY

The science of ecology is often divided into autecology and synecology. **Autecology** deals with the study of the individual organism or an individual species. In this, life histories and behaviour as a means of adaptation to the environment are usually emphasized. **Synecology** deals with the study of groups of organisms which are associated together as a unit (*i.e.*, community). Thus, if a study is made of the relation of a white oak tree (or of white oak trees in general) to the environment, the work would be autecological in nature. If the study concerned the forest in which the white oak lives, the approach would be synecological (**Odum**, 1971). While autecology is experimental and inductive, the synecology is philosophical and deductive (**Smith**, 1974).

Synecology is often further subdivided into **aquatic** and **terrestrial ecology**. The aquatic ecology includes **freshwater ecology**, **eustuarine ecology** and **marine ecology**. Terrestrial ecology, subdivided further into such areas as **forest ecology**, **grassland ecology**, **cropland ecology** and **desert ecology**, is concerned with terrestrial ecosystems—their microclimate, soil chemistry, nutrient and hydrological cycle and productivity.

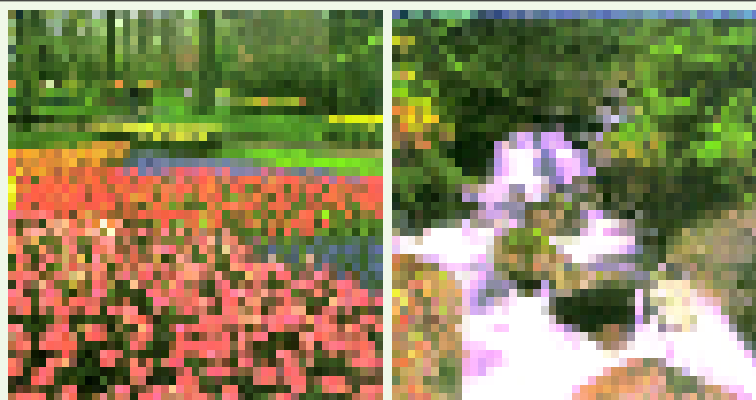
Further, **demecology** is that branch of ecology which deals with the ecology of populations. Early ecologists have recognized two major subdivisions of ecology in relation to plants and animals—**plant ecology** and **animal ecology**. But when it was found that in the ecosystems plants and animals are very closely associated and interrelated, then, both of these major subdivisions of ecology into plant ecology and animal ecology became vague. Besides these major subdivisions the ecology has been classified in the following branches according to the level of organization, kind of environments or habitats and taxonomic position :

1. Habitat ecology. It deals with the study of different habitats of the biosphere. According to the kind of habitat, ecology is subdivided into **marine ecology**, **freshwater ecology**, and **terrestrial ecology**. The terrestrial ecology too is further subdivided into **forest ecology**, **cropland ecology**, **grassland ecology**, etc., according to the kind of study of its different biomes.

2. Ecosystem ecology. It deals with the analysis of ecosystem from structural and functional point of view including the interrelationship of physical (abiotic) and biological (biotic) components of environment.

3. Conservation ecology. It deals with methods of proper management of natural resources such as land, water, forests, sea, mines, etc., for the benefit of human beings.

4. Production ecology. It is the modern subdivision of ecology which deals with the gross and net production of different ecosystems such as freshwater, sea water, cropfields, orchards, etc., and tries to do proper management of these ecosystems



Conservation of natural resources can be done through proper management.

so that maximum yield can be obtained from them.

5. Radiation ecology. It deals with the study of gross effects of radiations and radioactive substances over the environment and living organisms.

6. Taxonomic ecology. It is concerned with the ecology of different taxonomic groups and eventually includes following subdivisions of ecology—**plant ecology**, **insect ecology**, **invertebrate ecology**, **vertebrate ecology**, **microbial ecology** and so on.

7. Human ecology. It deals with the study of relationship of man with his environment.

8. Space ecology. It is a modern subdivision of ecology which remains concerned with the development of partially or completely regenerating ecosystems for supporting life of man during long space flights or during extended exploration of extra-terrestrial environments.

9. Systems ecology. The systems ecology is the most modern branch of ecology which is concerned with the analysis and understanding of the function and structure of ecosystem by the use of applied mathematics such as advanced statistical techniques, mathematical models and computer science.

RELATIONSHIP OF ECOLOGY WITH OTHER DISCIPLINES

Modern ecology is a multidisciplinary science which depends on a variety of disciplines such as **physics**, **chemistry**, **mathematics**, **statistics**, **meteorology**, **climatology**, **geology**, **geography**, **economics**, **sociology**, **agriculture science**, **forestry**, **horticulture**, **genetics**, **physiology**, etc. All these disciplines have helped in the better understanding of many ecological principles.

For instance, meteorological and climatological data for certain geographic localities allow for a more implied interpretation of results. A basic knowledge of forestry can be invaluable for a forest-ecologist to understand forest type distribution, floristic composition and prevalent environmental factors. Likewise, statistical data helps in interpreting the reasons for activity, population increases, migrations, probability of ecological events occurring in a particular area, sampling techniques and reliability of results. Palaeontology (geology) provides information about the ancestral organisms and environmental situations prevalent in past. Evolution and genetics are utilized to interpret the reason for organic changes when linked with environmental conditions, establishment of new populations and species, environmental effects on genetic populations and species and, so on. Such interdisciplinary approaches to ecology, consequently have given rise to following subdivisions of ecology :

1. Ecological genetics. An ecologist recognised a kind of genetic plasticity in the case of every organism. In any environment only those organisms that are favoured by the environment survive. The branch of ecology dealing with genetics in relation to ecology is called ecological genetics.

2. Palaeoecology. It deals with the movements of biotic elements based on palaeontological evidence, which provides information about ancestral organisms and environmental conditions existing in the past.

3. Ecophysiology. The factors of environment have a direct bearing on the functional aspects of organisms. The ecophysiology deals with the survival of populations as a result of functional adjustments of organisms with different ecological conditions of the ecosystem.

4. Chemical ecology. It deals with the adaptations of animals or preferences of particular organisms like insects to particular chemical substances.

5. Pedology. It is a branch of terrestrial ecology and it deals



Pedology is the study of soil and their influence on the organisms.

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with the study of soil, in particular their acidity, alkalinity, humus content, mineral contents, soil type, etc., and their influence on the organisms.

6. Ecogeography. It deals with the study of the role of the environment in animal distribution. It is related with **biogeography** which is concerned with the structural and functional relations of living organisms in space, which form the immediate environment of the individuals as well as populations. **Ecofloras** and **ecofaunas** are the lowest units of which a biogeographic flora or fauna is made up of.

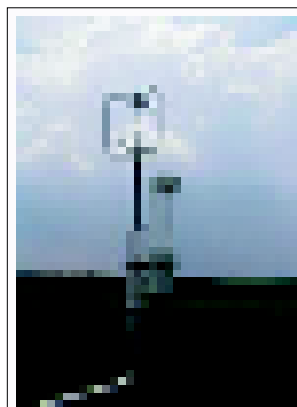
7. Ecological energetics. It deals with energy conservation and its flow in the organisms within the ecosystem. In it thermodynamics has its significant contribution.

Because of its far-flung involvements with so many fields, ecology is often regarded as a generality rather than a speciality. Visualizing this fact, an ecologist, **A. Macfadyen** (1957) wrote in his book *Animal Ecology : Aims and Methods*—“*The ecologist is something of a chartered libertine. He roams at will over the legitimate preserves of the plant and animal biologist, the taxonomist, the physiologist, the behaviourist, the meteorologist, the geologist, the physicist, the chemist, and even the sociologist; he poaches from all these and from other established and respected disciplines. It is indeed a major problem for the ecologist in his own interest, to set bounds to his divagations*”.

ECOLOGICAL TOOLS AND TECHNIQUES

Ecology has been studied on three broad lines—the **field**, **laboratory** and **mathematics**. Most of the ecologists have studied ecology from either a **descriptive** or **analytical** point of view. The descriptive point of view remains mainly natural history. A descriptive ecologist as a natural historian has asked the question “*What is present ?*” He has described the life histories and characteristics of organisms and also has done a correlated study of many of the responses of an organism, whether it was responding as an individual or as a member of a population, to the presence of physical forces in its habitat, such as weather, food, space and others. According to **Moen** (1973), a descriptive ecologist has to proceed a bit further than the natural historian in relating observed characteristics of the organism to observed characteristics of the habitat in a quantitative way. This has resulted in the formation of many ecological rules. **Bergman’s** rule is an example : animals living further north tend to be larger. These types of rules are generally applicable, although exceptions can be found in looking at detailed relationships. The analytical ecologist ask the question “*why*”? He is interested in the mechanism operating in the natural world. The recognition of simple relationship such as the condition of the range and conditions of the animal are persued further by analyzing the requirements of the animal through time and the ability of the range to satisfy these requirements.

An analytical ecologist has to face several practical problems because he has to work with living systems and more often he comes across the variables which are numerous and normally highly complex. Consequently the tool and techniques used by the physical scientists could not be easily applied in ecological investigations, nor were the results of ecological experiments so precise as those obtained in physics and chemistry. Despite these problems certain modern ecologists have made an active use of tools and techniques of chemistry, physics and mathematics in their ecological investigations. Physical and chemical measurements were taken to measure the various parameters of the environment. These may range from simple chemical determination of the various elements to the use of such sophisticated apparatuses such as **paper chromatogram**, **infrared gas analyzer**, **recording spectrophotometer**, and **microbomb calorimeter**. The use of statistical procedures such as **correlation**, **multiple regression**, and **matrix algebra** and the application of modern algebra, calculus and computer science to mathematical models simulating field conditions



Gas analyzer for H₂O and CO₂.

have provided new insights into population interactions and ecosystem functioning. With the use of **electronic equipment** and **biotelemetric techniques**, ecologists can sample and measure plant and animal populations without destroying them. **Radioisotopes** have enabled the investigators to follow the pathways of nutrients through ecosystem and to determine the time and extent of transfer. **Laboratory microcosms**—samples of both aquatic and soil microsystems—are useful in determining the rates of nutrient cycling and other parameters of ecosystems functioning.

SIGNIFICANCE OF ECOLOGY FOR MAN

Man is himself an organism within an environment. Like other animals man is influenced by the physical features of his environment, he is absolutely dependent upon other species for his food, clothing, medicine, and other similar aspects and he has to adjust to other individuals of his own species. Therefore, the basic laws of ecology apply well to him and its fundamental knowledge is must for man for his own existence on this planet (Earth).

Man almost always has a modifying influence, and without proper regulation he often has a destructive effect. For instance, by applying certain ecological principles to such fields as agriculture, biological surveys, game management, pest control, forestry, horticulture, and fishery biology, he has received tremendous economic gains. Its knowledge is found critically important for intelligent conservation whether in relation to soil, forest, wildlife, water supply or fishery resources. Further, though man has been known to control his environment successfully to meet his needs, but indiscriminate control of different pathogens or pests such as bacteria, fungi, weeds, insects, rodents, etc., through use of different chemical-poisons or pesticides (bacteriocides, fungicides, insecticides, etc.), the release of massive quantities of radioactive debris, **xenobiotics** (i.e., chemical compounds synthesized by humans which are not naturally found in living organisms and cannot normally be metabolised (broken down) by them; see **Green et al.**, 1990), discarded chemicals and industrial wastes into rivers result in atmospheric and aquatic pollution, which may have short and long term ecological effects. Further, rapid growth of urbanization and fast rate of multiplication of human population have resulted in fatal threat of scarcity of wild life, food, open space, and of survival.

There are certain other ecological problems. Agriculture and now forestry are concentrating upon monoculture—**single species ecosystems**—in spite of the difficulties and dangers associated with unnaturally simplified ecosystems that lack a diversity of species. Over much of the world, especially in the grasslands, we continue to disrupt the energy balance through overgrazing and end up with eroded mountain sides, silt-clogged streams and lakes, and a scarcity of water. Therefore, future of human life on earth demands more knowledge about the ecosystems and other ecological problems.

REVISION QUESTIONS

1. Define the term ecology. Why it has attracted the attention of modern man ? Explain.
2. Give the detailed account of historical background of ecology.
3. Enumerate different branches of ecology and write short account about each branch of ecology.
4. Ask your nonbiological friends how they would define *ecology*, and discuss the distinction between *ecology* and *environmental studies*.

CHAPTER

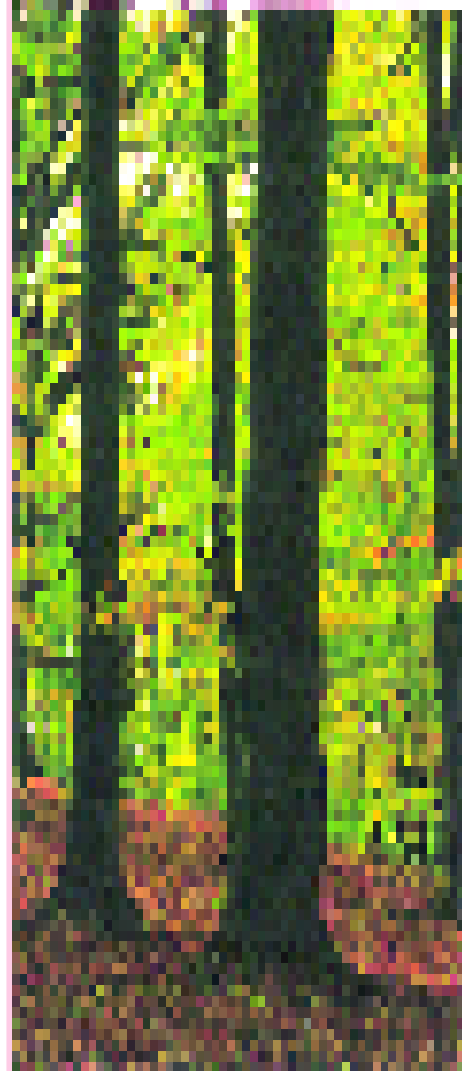
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Ecology in India

Growth of ecology in India is based on the earliest contributions (1875 to 1929) of some forest officers who provided purely descriptive accounts of forests. In fact, the first exhaustive ecological contribution was made by **Winfield Dudgeon** (1921) who published an ecological account of the Upper Gangetic Plains. He employed in his studies the concept of seasonal succession. **Saxton** (1922) elaborated this concept, however, **Misra** (1946, 1958, 1959) contradicted this view of succession. Instead of it, he concluded that the processes mentioned in Dudgeon's work might be better referred to as seasonality of communities rather than ecological succession.

GROWTH OF PLANT ECOLOGY

The plant communities were studied with phytosociological approach first time by **Agharkar** (1924) who studied the grasslands. **Bharucha** and co-workers (1930, 1937, 1957) at Bombay worked out phytosociology of grasslands. The study of forest vegetation was conducted by **Champion** (1929, 1935, 1937), **Troup** (1925), **Bor** (1947, 1948), etc. Ecological studies of autecological nature were performed for a variety of forest trees by **Champion** and **Pant** (1931), **Jagat Singh** (1925), **Phadnis** (1925) and **Champion** and **Griffith** (1947). Indeed, comprehensive autecological studies in India was initiated by **Prof. R. Misra** who worked out the autecology of herbaceous plants of different habitats. **Misra** and **Rao** (1948) studied the autecology of *Lindenbergia polyantha*, revealing



Physical and biotic conditions influences the responses of the organisms.

various significant aspects regarding the distribution of this species. Further growth of Indian Ecology can be contributed to extensive investigation into phytosociology of grasslands and mangroves by **Bharucha** (1941), forests by **G.S. Puri** (1950, 1951, 1960) and deserts by **Sarup** and co-workers.

Two pioneer centres of ecological research were established in 1930's in India, first by **R. Misra** at the Department of Botany, Benaras Hindu University, and second by **F.R. Bharucha** at the Institute of Science, Bombay. From 1937 onwards, **R. Misra** did extensive work at Benaras (Varanasi) on the ecology of grasslands, wastelands, playgrounds, lakes and ponds, low-lying lands, riparian eroded slopes and ecotypic differentiation. In 1950, Prof. Misra established another Ecological Research Institute at Sagar University (M.P.). In 1956, **Misra** moved to Varanasi and developed an internationally recognized centre of Ecological Investigation. At this centre, in between 1955–1962, investigations were carried on in the diverse areas such as ecophysiology (**V.Kaul**), root-soil relationships of grasslands (**S.S. Raman**) and riparian lands (**R.S. Ambasht**), ecotypic differentiation (**P.S. Ramakrishnan**) and reproductive capacity of herbage species (**H.R. Sant** and **V.G. Nelvigi**). During 1963–1971, Benaras centre of ecology carried mainly autecological work on the medicinal plants and the weeds (**R.S. Tripathi**), grassland productivity (**J.S. Singh**) and forest litter decomposition and productivity (**K.P. Singh**).

Autecological and synecological studies of forest communities were conducted by **Sharma** (1955), **Bhatia** (1954, 1955, 1956), **Misra** and **Joshi** (1952), **Arora** (1961–1964), **Puri** (1949, 1950), **Rao** (1967), etc. **Meher-Homji** (1989) did vegetation mapping in India. **Dhani** and **Behera** (1990) did quantitative analysis of forest community of Boudh-Khondmals (Phulbani, Orissa). **Joshi** and **Behera** (1990) did qualitative analysis of vegetation from a mixed tropical forest of Orissa. Productivity studies of forests have been performed by **Misra** (1969), **Raman** (1970), **Singh** (1971), **Bandhu** (1971), **Sharma** (1972), **Farugui** (1972), and **Pandeya et al.**, (1967, 1969, 1971). **Rajvanshi** and **Gupta** (1980) studied decomposition of litter in a tropical and deciduous forest. **Rajvanshi** and **Gupta**, in 1985, worked on mineral cycling in a tropical deciduous *Dalbergia sissoo* Roxb., forest. **Chaturvedi** and **Singh** (1978, 1987) described dry matter dynamics and nutrient dynamics of pine forests of Central Himalaya. **Ambasht** (1988) studied biomass productivity of some tropical grazing lands and plantation forests.

Autecological and phytosociological studies on several herbaceous plants of grasslands and wastelands have been made by **Pandeya** (1953, 1960, 1964, 1968), **Tiwari** (1955), **Ramakrishnan** (1960), **Ambasht** (1963, 1964), **Tripathi** (1965, 1977), **Kaul** (1959), **Sant** (1962, 1965), etc. Autecology of some medicinal plants have been made by **Pathak** (1967), **Kaul** (1965), **Shetty** (1967), etc. **Vimal** and **Talshikar** (1983) suggested various methods of recycling of agricultural wastes. Ecological investigations of specialized habitats were also done, such as of walls (**Varshney**, 1963),

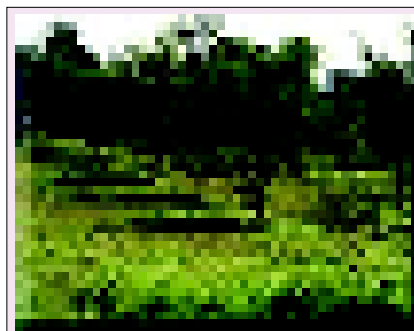


Ecologists have conducted studies on several herbaceous plants of grasslands and waste lands.

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of eroded river banks (**Ambasht**, 1968), of deserts (**Joshi**, 1956, **Mulay** and **Joshi**, 1964), freshwaters (**Misra**, 1946, **Das**, 1968, **Gupta**, 1968, **Jha**, 1968, **Gopal**, 1968 and **Sinha**, 1969).

Ecological studies of fungi were initiated at School of Ecology at Benaras by 1960's under the leadership of **R.Y. Roy**. His team made synecological studies on the rhizosphere mycoflora of different kinds of plants. His team mate, **R.S. Dwivedi** has also contributed to ecology of fungi active in the decay of herbaceous plants, forest litter, etc. **P.K. Khanna**, **P.D. Sharma** and **B. Rai** investigated the phenomenon of succession of air-borne fungi on decaying grasses. Some other active centres of fungal ecology were established at Madras (by **T.S. Sadasivan**), at Sagar (by **S.B. Saksena**) and at Lucknow (by **J.N. Rai**). In recent years (*i.e.*, 1972–1985), interaction of soil microfungi and earthworms has been investigated by **M.C. Dash** and his team of researchers comprising **J.B. Cragg**, **Senapati**, **Nanda**, **Behera**, **P.C. Mishra**, **H.K. Dash**, etc., at Sambalpur, Orissa.



Dalbergia sissoo.

GROWTH OF ANIMAL ECOLOGY

In comparison to plant ecology, relatively little work has so far been done for the enhancement of Indian animal ecology. Most animal ecologists of India have dealt with the ecology of regional fauna. Thus, **Baker** (1921, 1935) carried out some ecological studies on birds of India, Burma and Ceylon. **Salim Ali** had watched the Indian birds very closely and his work greatly contributed to the ethology and ecology of Indian birds. **Dharmkumar-Sinhji** (1937–1975) made extensive ecological studies on water fowls and other birds of Gujarat. **Mani** (1953–1974) worked on the ecology of some insects of Himalaya. In 1974, he also has given historical account of animal ecology of India. **Dr. Ishwar Prakash** at the Central Arid Zone Research Institute (CAZRI) has exhaustively dealt with ecology and conservation of wild life (vertebrate fauna such as reptiles, birds, and mammals including bats) of Indian hot deserts during span of more than 30 years (1958–1988). Ecology of mammals of Indian deserts were also made by **Purohit** (1968). **Choudhuri** (1989) worked on the conservation of Indian tigers.

Ramalingum (1961) made ecological observations on the larvae of trematode. **Southwick et al.**, (1961) carried a survey of populations of Rhesus monkeys in villages, towns and temples of North India. **Ananthasubramanian** and **Ananthkrishnan** (1963) studied ecology of microarthropods of pasture soil. **Nayar** and **Prabhu** (1965) studied the soil fauna of tea plantation of Kerala. Studies of ecological nature were conducted on village dwelling langurs (primates) of West Bengal by **Oppenheimer** (1973). **Singh et al.**, (1961) and **Bains** and **Shukla** (1975) made autecological studies on *Chilo partellus*. Ecological studies on the population of Thar in Nilgiri, Tamil Nadu were made by **Davidar** (1976).

In recent years, ecology of earthworms, soil nematodes and certain amphibians and reptiles has been extensively worked out. **M.C. Dash** (1978–1993) working at the School of Life Sciences, Sambalpur University, Orissa, has worked on diverse ecological aspects of earthworms and soil nematodes of the tropical pasture. He also worked on the energetics of anuran larvae and turtle conservation. **Dash** and **Senapati** (1986) published an extremely useful paper entitled “*Vermitechnology : an option for organic waste management in India*”. **Dash** and **Kar** (1987, 1990) suggested methods

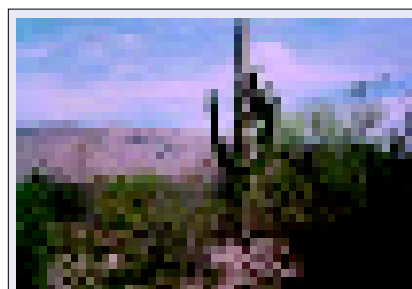


Salim Ali —a well known ornithologist of india.

of conservation and management of sea turtle resources at the Orissa coast. **Mohapatra** and **Dash** (1987) studied density effect on growth and metamorphosis of *Bufo stomaticus* larvae. **Rao** (1984) studied diapause induced physiological and biochemical changes in a tropical earthworm, *Octochaetoma surensis*. **B.K. Senapati et al.**, (1979–1985) studied various other ecological aspects of earthworms in the tropical pastures. **Sahu** and **Senapati** (1988) studied secondary production and energy utilization strategy of *r* and *k* selected earthworms from tropical pastures of Orissa.

GROWTH OF DESERT ECOLOGY

The hot arid zone occupies 2.86 lac km² or 8.7 per cent of the geographic area of India. It is characterized by distinct ecological features such as low and fluctuating rainfall, high solar radiation, low atmospheric humidity, strong wind regime and dominance of dunes and sandy plains amongst the landforms. Despite being the zones of scarcity and hardships, the arid zones of India exhibit a spectacular and vivid biotic diversity. Indian hot deserts have attracted the attention of several ecologists and significant contributions have been made in the field of desert ecology. **Krishnan** (1952) described geological history of Rajasthan to trace its present day conditions. **Wadia** (1960) studied the post glacial dessication of central Asia. **Lahiri** and coworkers (1961–1988) worked on the adaptations of arid zone plants to soil water deficit. **Abi Chandani** (1964) studied genesis, morphology and management of arid zone soils of India. **Sankhala** (1964) described wildlife sanctuaries of Rajasthan. **Roy** and **Sen** (1968) prepared a soil map of Rajasthan. **Mann** (1971) presented a report on operational research project on arid land management. **Seth et al.** (1971) studied micronutrient status of Rajasthan soil. **Gupta** and **Saxena** (1972) described potential grassland types and their geological succession in Rajasthan. **Mann et al.**, (1974) did a qualitative analysis of arid zones of India and analyzed the problem of spread of Indian desert. **Roonwal** (1975) described termites of Thar Desert; in 1982, he studied field biology and morphology of some central India grasshoppers. **Shankar et al.**, (1976) studied effects of Khejri tree (*Prosopis cineraria*) on the productivity of range grasses growing in its vicinity. **Shankar** (1977) suggested the means of measuring desertification through plant indicators. In 1983, he studied the depleted vegetation of the desert habitats and studied the process of natural regeneration. **Saxena** (1977) described vegetation and its succession in the Indian desert. **Shankarnaryan** (1977) analyzed impact of overgrazing on grasslands. In 1985, he and **Sen** suggested the means of combating desertification. **Dhir** (1977) analyzed the problem of soil degradation due to over exploitative human efforts. **Mann** and **Saxena** (1980) released monograph entitled “*Khejri (Prosopis cineraria) in the Indian Desert—its role in agro-forestry*”. **Dhabriya** (1982–84) performed drought monitoring/green biomass mapping in the arid and semi-arid parts of Rajasthan and Gujarat. He also suggested ways to check the desertification process. **Mathana et al.**, (1984) described root system of desert tree species. **Bhardwaj et al.**, (1984) reviewed ecological study of Rajasthan desert during years 1964–1984. **Bhandari** (1988) described floral wealth and plant adaptation of the Indian desert. **Mathur** and **Govil** (1988) suggested methods of greening of the Indian Desert. **Professor Ishwar Prakash** (1988) has published a research paper entitled *Wildlife, human-animal interactions and conservation in the Rajasthan desert*.



Indian deserts exhibit vivid biotic diversity.

GROWTH OF OCEANOGRAPHY AND LIMNOLOGY

The ecology of Indian seas and oceans (Indian oceanography) has been studied extensively, since it has an importance in defence. A variety of ships and vessels have been used from time to time to record data on physical, chemical and biological characteristics of Indian seas and oceans. Thus, various forms of life, meteorology, geology, and geophysics of the Indian seas and oceans have been studied. During 1960–1964, an International Indian Ocean Expedition (IIOE) was organized to study the Indian ocean

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in which India collaborated with other 20 countries. In 1966, the National Institute of Oceanography (NIO) was set up in 1966 with its Head Quarter at New Delhi as national laboratory under the Council of Scientific and Industrial Research (CSIR). In 1969, NIO's Head quarter was shifted to Dona Paula (Goa). NIO and its regional centres at Bombay and Cochin consist of seven divisions such as Indian Ocean Biological Centre (Cochin), Physical Oceanography, Chemical Oceanography, Geological Oceanography, Biological Oceanography, Data and Documentation and Oceanographic Instrumentation. Some noted oceanographers of India are **Jayaraman**, **Ramasastri**, **Mahadevan**, **Nair**, **Qasim**, **Gopinathan**, **Prasad**, etc.



Indian sea.

Pruthi (1933) reported seasonal changes in the physical and chemical conditions of the water in the tank in Indian Museum compound. **Philipose** (1940) studied ecology and seasonal succession in a permanent pool at Madras city. In the year 1959, **Philipose** made observations on the ecology of fresh water phytoplanktons of inland fisheries. **Ganapati** (1941, 1943, 1947) made divergent limnological investigations on the freshwater ponds of Madras City. **Banerjea** and coworkers (1954–1970) studied limnology of various fish ponds of India. **Chacko** and **Krishnamoorthy** (1954) studied ecology of planktons of three freshwater ponds of Madras City. **Biswas** and **Calder** (1955) made floral study on certain common water and marsh plants of India and Burma. **Mitra** (1955, 1966) studied autecology of certain aquatic plants such as *Limnanthemum cristatum*, *L. indicum* and *Hydrilla verticillata*. **Das** and **Shrivastava** (1955–1959) studied the limnology of various freshwater ponds of Lucknow. **Das** and **Moitra** (1955) worked on the feeding ecology of some common fishes of U.P. and classified them into surface feeders, mid feeders, and bottom feeders. **Mitra** and coworkers (1956, 1965) worked on feeding ecology of the Indian major carp fry. **Alikunhi** (1956) made certain ecological observation such as fecundity on *Labeo bata*. **Singh** (1956) studied limnological relations of Indian inland waters with special reference to algal blooms. **R.S. Ambasht**, Professor of Botany, Benaras Hindu University, Varanasi, (1958–1990) and his team of researchers have published more than 120 research papers dealing with primary production; forest ecology; aquatic ecology—ecology of pond ecosystem; Ganga river ecosystem (*e.g.*, riparian ecology), grassland ecology, aquatic pollution, weed's ecology and conservation of riparian wasteland. **Das** (1959–1967) studied ecological effects of micronutrients, vitamin B₁₂, density, etc., on the Indian freshwater carps.



Hydrilla verticillata.

George (1961–1966) made ecological observations on the physico-chemical nature of water and zooplanktons and rotifers of certain shallow ponds of Delhi. **Subhramanyam** (1962) published a monograph on certain common aquatic angiosperms of India. **Saigal et al.**, (1962) studied ecology of *Mystus seenghala* of Ganga river system. **Bhatt et al.**, (1963) reported mineral contents of freshwater and freshwater organisms of Kalyan, Kolhapur and Jaduguda areas. **Michael** (1964) studied limnology of planktons and macrofauna of a freshwater pond of West Bengal. **Sreenivasan** (1964) studied hydrology of tropical impoundment, Bhavanisagar Reservoir,

Madras state for the years 1956–1961. In 1968, **Sreenivasan** undertook the studies on the limnology of two upland impoundments of Nilgiris of Madras state. **Banerjee** and **Raychoudhury** (1966) worked out the physico-chemical nature of Chilka lake of Orissa. **Krishnamurthy** (1966) made preliminary studies on the macrofauna of the Tungabhadra Reservoir. **Verma** (1967) studied diurnal



Labeo bata.

variation in a fish pond in Seoni. **Vasisht** (1968) worked out ecology of a pond ecosystem of Benaras. **Moitra** and **Bhowmik** (1961) studied seasonal cycles of rotifers in a freshwater fish pond in Kalyani, West Bengal. **David et al.**, (1969) studied limnology and fishery of the Tungabhadra reservoir. **Nayar** (1970) studied ecology of rotifer populations of two ponds at Pilani, Rajasthan. **Khan** and

Siddiqui (1970–1974) investigated the diurnal and seasonal variations in the limnological features of a perennial fish moat of Aligarh, Uttar Pradesh. **Kaul** (1971) studied the production and ecology of some macrophytes of Kashmir lakes. **Patnaik** (1971) worked on the seasonal abundance and distribution of benthos of Chilka lake, Orissa. **Ganapati** (1972) compared the organic production of various types of aquatic ecosystems of India. **Kant** and **Kachroo** (1975) recorded the diurnal changes in the temperature and pH of water and diurnal movement of planktons in Dal lake of Shrinagar. **Pillai** and **Sreenivasan** (1975) estimated the carbon and nitrogen status of some lakes, reservoirs and ponds of Tamil Nadu. **Gupta** (1976) studied the limnology of macrobenthic fauna of Loni reservoir of Rewa of Madhya Pradesh. **Sreenivasan** (1977) studied the limnology and fisheries of Tirmoorthy reservoir, Tamil Nadu. **Rao** (1977) investigated the ecology of certain phytoplanktons such as diatoms, Euglenineae and Myxophyceae of three fresh water ponds of Hyderabad.

GROWTH OF POLLUTION BIOLOGY

C. K. Varshney at Delhi worked on pollution ecology with special reference to SO_2 pollution. **D.N. Rao** (1979) used plant leaf as a pollution monitoring device. **L. K. Dadhick** (1982) described certain indicator plants which may be useful in the recognition and monitoring of air pollutants. **Ambasht et al.**, (1979) made an assessment of effluents of a chemical and fertilizer factory for irrigation of agricultural lands. **Ambasht** (1982) wrote an essay entitled “Responses of aquatic plants to pollution.” **V. Venkateswarlu** (1982) studied the role of algae as indicators of river water quality and pollution. **Mahajan** (1982) described zooplankton as indicators for assessment of water pollution. **Kar et al.**, (1987) studied the problem of pollution in river Ib and described plankton population and primary productivity. **Palharya** and **Malvia** (1988) studied pollution of the Narmada river at Hoshangabad in Madhya Pradesh and suggested measures for its control. **Tiwari** and **Ali** (1988) prepared a water quality index for Indian rivers. **Shardendu** and **Ambasht** (1988) studied impact of urban waste waters on primary production and nutrient status of *Hydrilla verticillata*. **Ambasht et al.**, (1984) performed ecological investigation of conservation nature on the plant biomass and energy used for the burning of human dead bodies at Varanasi Ghat.

REVISION QUESTIONS

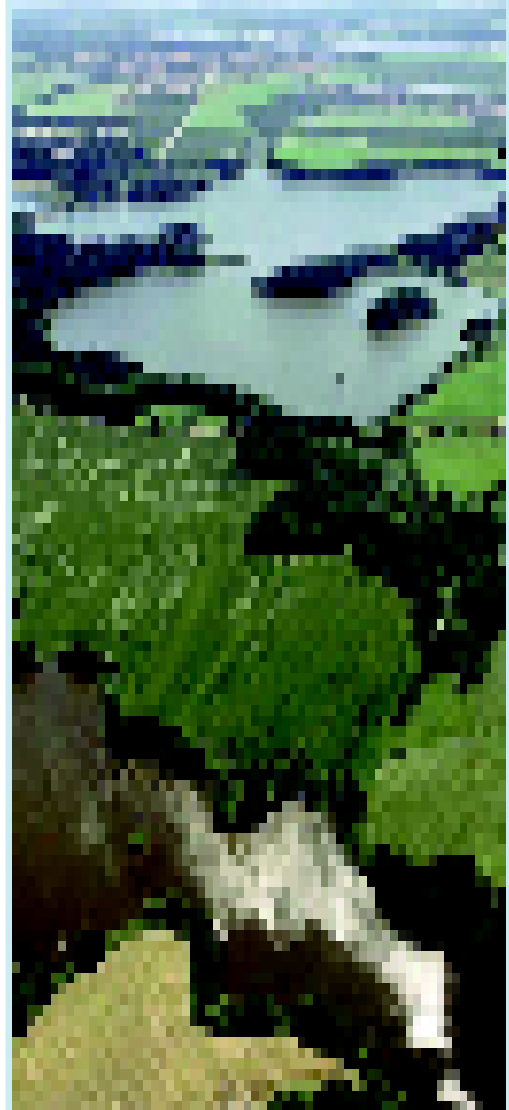
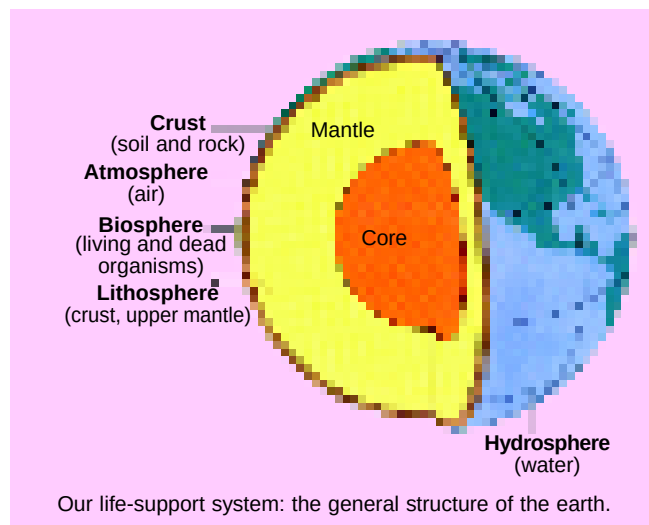
1. Describe the historical growth of plant ecology in India.
2. Write an essay on ‘growth of animal ecology in India’.
3. Trace the historical growth of oceanography in India.
4. Describe the history of limnology in India.
5. Describe the historical growth of desert ecology of India.

CHAPTER

3

Environment

Each and every living organism has its specific surrounding, medium of environment to which it continuously interacts and remains fully adapted. The **environment** is the sum total of physical and biotic conditions influencing the responses of the organisms (S.C. Kendeigh, 1974). The life supporting environment of planet earth—the **biosphere** is composed of following three chief media—**air**, **water** and **soil**, which are the components of three major sub-divisions of the biosphere—**atmosphere**,



Physical and biotic conditions influences the responses of the organisms.

hydrosphere and **lithosphere**, respectively. These media are not completely isolated from each other, however, some of the atmospheric gases are dissolved in all natural waters, and some moisture is present almost everywhere in the atmosphere. Each of these media can be discussed separately in the following manner :

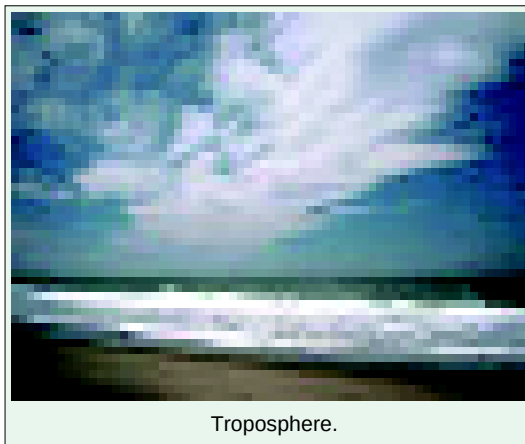
A. ATMOSPHERE (AIR)

The multilayered gaseous envelope surrounding the planet earth is called **atmosphere**. The atmosphere remains in contact with all the major types of environment of earth, interacting with them and greatly affecting their ability to support life. It filters sunlight reaching the earth, affect climate, and is a reservoir of several elements essential for life.

Various Zones of Atmosphere

The atmosphere is divided into five distinct layers or zones : troposphere, stratosphere, mesosphere, ionosphere and exosphere. The **tropopause** separates the stratosphere from the troposphere and **stratopause** separates mesosphere from stratosphere.

1. Troposphere. It is the lowest region of atmosphere which subjects to differential heating, temperature inversions and convection currents and which extends from the surface of the earth up to a height of 8 to 10 km at polar latitudes (poles), 10 to 12 km at moderate latitudes and 16 to 20 km at the equator. For the organisms, troposphere forms a most important zone of atmosphere. Many important climatic events such as cloud formation, lightening, thundering, thunder storm formation, etc., all take place in troposphere. In this zone the percentage concentration of different gases in air does not vary with an increase in height. But the water vapour content in air depends upon the weather (*e.g.*, part of the troposphere over an ocean carries more moisture than that over a land surface) and it decreases sharply with an increase in height as does the air temperature. Air temperature in this zone gradually decreases with height at the rate of about 6.5°C per km (more specifically 5°C per km (more specifically 5°C per km in the lower troposphere and 7°C per km in the upper troposphere). In fact, towards the upper layers of troposphere, the temperature may decrease up to – 60°C. Upper region of the troposphere has a narrow boundary called the **tropopause** which has a constant temperature.



Troposphere.

Further, the non-uniform heating of the ground surface by sun's radiations produces ascending and descending air currents, which cause turbulence and mixing of air masses vertically. Moreover, the average air pressure at the earth's surface is 1,014 millibars (*viz.*, 1 millibar= 1/1000 bar; 1 bar = 1.019 kg per cm² close to 1 atmosphere, 1.332 millibars=1mm of mercury (Hg). At an altitude of 5 km, the air pressure is half that at the surface; at 11 km it is 225 millibars and at 17 km it is only 90 millibars.

2. Stratosphere. Next to troposphere is the second zone, called **stratosphere**, which is about 30 km in height. This zone is free from clouds and aeroplanes usually fly in its lower zone. The temperature of stratosphere increases up to 90°C and such an increase in temperature is due to ozone formation under the influence of ultraviolet component of sunlight. Such a layer of ozone is called **ozonosphere**. In ozonosphere, the sunlight ionize oxygen to ozone by photochemical dissociation. The ozonosphere completely absorbs solar radiation, ultraviolet radiation from the sun and also a lot of the solar infra-red, thus, becoming warmer than adjacent layers above or below. There is a serious threat to this ozone layer now due to the harmful effects of gaseous pollutants. A big hole (*i.e.*, thinning of the O₃ layer) has occurred in it, above the antarctic region. Upper layers of stratosphere form **stratopause**.

3. Mesosphere.

Stratosphere is followed by next zone called **mesosphere** which is 40 km in height. In mesosphere, temperature shows again a decrease up to -80°C . Upper layers of this zone form the **mesopause**.

4. Ionosphere.

The remaining part of atmosphere above the mesosphere, up to the height of about 300 km above earth's surface, is called **ionosphere**. Ionosphere contains several layers of ionized air. Thus, most of the gaseous components which become ionized under the influence of radiant energy, remain as ions. Ionosphere reflects short radio waves, making telecommunication possible over long distances.

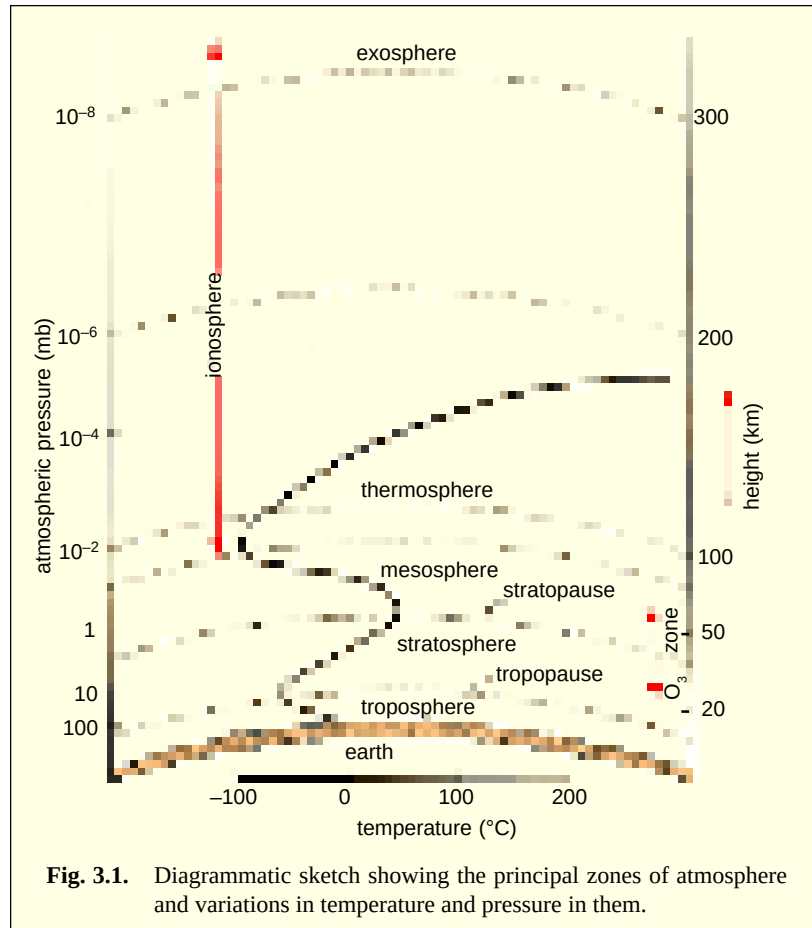


Fig. 3.1. Diagrammatic sketch showing the principal zones of atmosphere and variations in temperature and pressure in them.

5. Exosphere. The exosphere is the outer fringe of earth's atmosphere and outer space begins after it. The air density is very low in this zone; hydrogen being dominant element of it.

Air

The gaseous mixture of troposphere, is utilized by most organisms in respiration to liberate energy from food during oxidation and is called air. In atmosphere, about 95 per cent of the total air is present up to the height of about 20 km above earth's surface and the remaining 5 per cent in the rest, of about 280 km height. In the gaseous mantle, there is found a mixture of gases in different proportions (Table 3-1). Of these various gases nitrogen and oxygen are the major components of air.

Table 3-1. Composition of atmospheric air.

Component	% by volume	% by weight
Nitrogen (N_2)	78.09	75.54
Oxygen (O_2)	20.93	23.14
Argon (A)	0.93	1.27
Carbon dioxide (CO_2)	0.032	0.46
Miscellaneous	0.02	0.02

Miscellaneous component includes traces of hydrogen, ozone, radon, helium, neon, krypton, xenon, sulphur dioxide, hydrogen sulphide, ammonia, methane, etc.. Besides these gases, air may carry suspensions of liquids such as water in clouds and solids such as dust from the ground (soil), smoke from fires or salt from ocean spray. Air also contains microorganisms (viruses, bacteria, etc.), pollen grains and fungal spores, all forming biological constituents of the atmosphere.

In general, the composition of these gases in air and in other media on earth, such as water or soil, is in equilibrium with the atmosphere. In special ecological habitats such as the anaerobic regions of deep lakes or sand-mud flats, however, the composition of gases of air is altered dramatically. Slight differences occur in the atmosphere at different latitudes and, at places where gases are entering or leaving the atmosphere, such as volcanoes, fires, smelters, cities, metropolitan areas, and vegetation.

Physiologic-ecologic inter-relationships of gases and animals. The atmospheric composition is not always the most important factor to organism; rather, it is the partial pressure of a gas, especially oxygen and CO_2 , which influences the existence of life, as illustrated by altitudinal studies on organisms. The **partial pressure** of a gas is the product of the total barometric pressure times the concentration of gas in dry air.

The solubility characteristics of oxygen and carbon dioxide in water are different, for carbon dioxide is about 200 times more soluble than oxygen. Both temperature and salinity have evident effects on the solubility of these two gases; with either increased temperature or increased salinity, there is a decrease in their solubility in water as has been shown in table 3-2.

Table 3-2.

Coefficients of saturation of CO_2 in water at different temperatures and salinity (ml/litre in equilibrium with 760 mm Hg).

Temperature, °C		Salinity, %	
	0	28.91	36.11
0	1715	1489	1438
12	1118	980	947
24	782	695	677

In the process of photosynthesis, green plants use CO_2 and release O_2 during the day time; oxygen is used in respiration by all the organisms all the time. Photosynthesis, thus, regulates the oxygen and carbon dioxide balance in nature. Nitrogen which is abundantly present in the atmosphere is not used directly by plants or animals except by some bacteria and blue green algae. If gases are to be functionally important to an organism, there must be a mechanism whereby gases can enter and leave the body. Apparently the passage of oxygen across a membrane is accomplished by **diffusion**. The rate of diffusion can be calculated by **Fick's Law**, which is based on concentration coefficient, amount of surface thickness of the membrane, and time, and is expressed as the diffusion coefficient (**Davson**, 1959). The diffusion coefficient value is not the same for different animal tissues. For example, some representative values of diffusion coefficient of O_2 are : muscle, 0.000014; connective tissue, 0.000011; water, 0.000034; air, 11.0;



Green plants regulate the oxygen and carbon dioxide balance in the atmosphere by the process of photosynthesis.

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chitin, 0.000013. Not only is the diffusion rate different in various tissues, but the behaviour of each gas also differs. Carbon dioxide, for example, diffuses through water and animal tissues 25 times faster than oxygen.

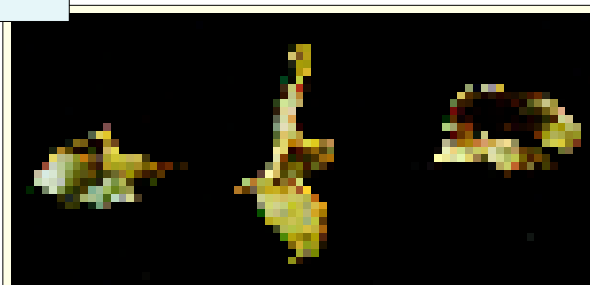
Air as a medium for living organism. Air is not an easy and suitable medium to support life (biota) and actually no organism ever originated in air, though, certain aquatic and terrestrial organisms



Draco volans.

have become secondarily adapted for aerial existence. Air has so much less **buoyancy** than water that organisms emerging from water or land are immediately subjected to a stronger pull of **gravity** which holds them to the earth. They are also exposed to **evaporation** of water from their bodies, which threatens loss of water from the protoplasm and death by desiccation. Changes in **temperature** are much more drastic in air and there is great danger of chilling or overheating. Further, periods of light exposure are much longer and much more intense in air than in water or upland. The problem of **mineral supply** becomes acute in the air.

Biota of the air. Only a few microorganisms, plants and animals have invaded the air. The most common examples of aerially adapted animals are insects, flying fish (*Exocoetus volitans*), flying frog, flying lizard (*Draco volans*), flying phalangiers (*Petaurus*), bats (*Eptesicus*), and birds (For volant adaptation see Chapter 18).



Bats.

B. HYDROSPHERE (WATER)

The oceans, rivers, streams, lakes, ponds, pools, polar ice caps, water vapour, etc., form the **hydrosphere**. About three-fourth of the earth's surface (75%) is covered with hydrosphere, the main component of which is **water**. Water is one of the most unusual natural compounds found on earth, and it is also one of the most important. The water remains in solid (snow), liquid (water) and gaseous (water vapour) forms. Life on earth began in the seas, and water in some form or the other is absolutely essential for the maintenance of all life. Water is one of the main agent in pedogenesis (soil formation) and is the medium for several different ecosystems. It permeates the atmosphere and the outer layers, of lithosphere and has uneven distribution on earth, so that, some of the great ocean depths are approximately six or seven miles (9750 meters). Further water in its two forms, salt water and freshwater, forms two chief aquatic environments—namely **marine environment** and **fresh water environment** of earth. The oceans holding marine environment are two and one half times more extensive than land and provide over 300 times the living space, since they are habitable throughout their entire depth by certain groups of organisms. Water is obviously heavier than air which imparts a greater buoyancy to the aquatic medium enabling organisms to float at variable levels. The most unique features of water concern its physical properties.

Physical Properties of Water

Henderson (1913, 1924) listed in his stimulating book, *The Fitness of the Environment*, the following characteristics of water which are favourable to biological systems :

1. A tremendous quantity of water exists on earth in three forms : gaseous, liquid, and solid. Not only is 75 per cent of the earth's surface covered by water but the atmosphere also contains an abundance of aqueous vapour, and the polar region is ice-covered. Moreover, at least 60 per cent of active protoplasm is water.

2. Water is an extremely inert body in relation to most other chemical substances.

3. It has unique thermal properties such as heat capacity, latent heat and higher freezing point.

(a) Heat capacity. Water has high heat capacity and it can withhold large amounts of heat. Because of the high heat capacity of water, oceans and lakes tend to maintain a relatively constant temperature, and therefore, the temperature of the biosphere is relatively stable. This property of water is functionally important to animal life. For example, a 165 lb (about 75 kg) man at rest produces sufficient heat to raise his body temperature more than 32°C, but if the heat capacity of his body were to correspond to that of many other substances, this amount of heat production would raise his temperature 100–150°C.

(b) Latent heat of melting and evaporation. The latent heat of melting is the number of calories required to convert 1 gram of solid at the freezing point into 1 gram of liquid at the same temperature. This value is about 90 for water, and means that the amount of heat necessary to melt ice is the same as that required to raise the temperature of the resulting ice water to 80°C. The latent heat of evaporation is defined as the number of calories required to change 1 gram of liquid into vapour. For water this value is 536; therefore, as much heat is required to boil away 1 gram of water as to raise the temperature of 536 grams through 1°C. These properties of water are important not only because they moderate the temperature of the biosphere, but also because they play a basic role in the evaporation of water and its precipitation as rain and as dew in the **hydrological (water) cycle**.

(c) Thermal conductivity. Although water is a poor thermal conductor compared to metals, among the common liquids it is excellent : for example, the conductivity value for silver is 1.10, for water, 0.0125, for alcohol, 0.00048, and for benzene, 0.00033.

(d) Expansion before freezing. The relationship between temperature and density, or mass per unit volume, is very unusual. When water is cooled from room temperature, it contracts, becoming denser until it reaches a maximum density at 3.94°C. If cooled further, it begins to expand again. At the freezing point (0°C) it expands markedly, unlike almost all other substances. Thus, ice will always float on the top of a lake or stream, and it is very unusual for an aquatic ecosystem ever to freeze solid, unless it is very small.

4. No other compound compares to water as a **solvent**. So many different substances can be dissolved in it that it is known as the **universal solvent**. More things, in fact, can be dissolved in water than in any other liquid. This is especially true for inorganic chemicals which split, or dissociate to form electrically charged entities termed **ions**. Ionization influences most electrical phenomena and many chemical phenomena of solutions. It is probable that all natural elements are soluble in water, at least in trace amounts, and that they are all found in natural water at some place or other on the earth's surface. In addition, many organic chemicals are water-soluble. Thus, water is the main medium by which chemical constituents are transported from one part of an ecosystem to the other. It is the only medium by which these constituents can pass from the abiotic portion of the ecosystem into the living portion. Even in the driest of terrestrial environments, nutrient materials pass into the roots of plants in aqueous solution ; when air is breathed by animals, oxygen is dissolved in water at the surface of the lung before it can cross the mucous membrane and be absorbed by the blood.

5. Water has the greatest **surface tension** of all common liquids, except mercury. The role of surface tension is most obvious in the way it allows certain things, such as pollen, dust, and water striders, to remain at the surface of a water body even though they are denser than the water. More important, however, the high surface tension of water allows soils to contain a significant amount of water through capillary attraction and to make it available to terrestrial plants.

Visualizing these unique physical properties of water, **Henderson** (1913) concluded that no other known substance could be substituted for water as a basic abiotic environmental factor. Now, ecologists have known certain other important physical properties of water which affect life in some way. Some of the additional important physico-chemical properties of water are the following :

(i) **Viscosity**. Water is a fairly viscous liquid. Animals that live and move in water need to be much more streamlined than those that move through air, because the resistance to motion in a viscous medium is high. But at the same time, the viscosity of water allows organisms to swim using relatively simple movements. Further, high viscosity of water protects the aquatic animals and plants from the mechanical disturbances.

(ii) **Buoyancy**. Water is buoyant medium. Organisms can exist in it without specialized supportive structures such as those that are needed by organisms that inhabit terrestrial environments.

(iii) **Transparency**. Water is transparent medium. Its transparency enables the penetration of light to the depths where it is ultimately absorbed. Different wavelengths are absorbed at different depths. The long heat waves are stopped near the surface. Shorter waves with more energy penetrate successively farther. The ultraviolet rays penetrate beyond 100 meters. The zone up to which light rays penetrate is called as **photic zone** and below this zone there is complete darkness and organisms that require light cannot live. The transparency of water is greatly affected by the presence of suspended particles, phytoplankton, etc., which absorb light and so penetration of light in turbid water is less.

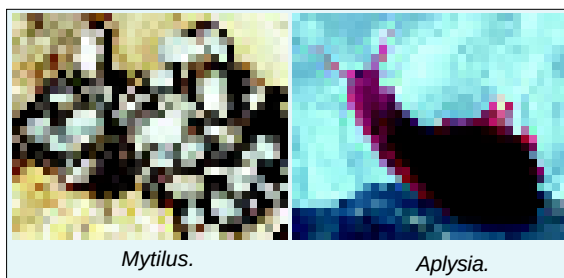
(iv) **Pressure**. Organisms living at sea level experience a pressure of about 15 psi, which is defined as 1 atm (760 mm of Hg). Pressure increases with increased depth of water at the rate of one atmosphere (1 atm) for every 10 meters of descent. Organisms inhabiting the floor of deep-sea areas at depth of 10,500 meters are exposed to hydrostatic pressure of about one ton per square centimeter. Pressure influences solubility, ionic dissociation, and surface tension, and water is slightly compressible with increased pressure.

(v) **Salinity**. Salinity has been defined as “the total amount of solid material in grams contained in one kilogram of the water, when all the carbonate has been converted into oxide, bromine and iodine replaced by chlorine and all organic matter completely oxidized.” All types of natural waters contain various amounts of different salts (ions) such as Na, K, Mg, Cl, SO₄, PO₄, CO₃, HCO₃, NO₃, etc., and all these salts are responsible for the **saltiness**, **salinity** or **salt content** of water (Table 3-3). The salinity of marine water is rather constant being about 3.5%. The salinity of fresh water varies greatly. Some salt lakes may have a salinity of 25% to 30% which greatly restricts life in them.

Table 3-3. Comparison of some of the principal ions found in different kinds of water.

Water	Na	K	Ca	Mg	Cl	SO ₄	CO ₃	Total per litre
1. Soft water	0.016	—	0.01	0.0005	0.019	0.007	0.012	0.065
2. Hard water	0.021	0.016	0.62	0.014	0.041	0.025	0.119	0.301
3. Sea	10.56	0.30	0.40	1.27	18.98	2.65	0.71	34.85
4. Great salt lake	65.54	3.76	0.065	4.47	110.08	13.04	—	197.51

Salinity of water acts as an important limiting factor for the distribution of a number of species of plants and animals. Certain animals such as, spider crab, *Maia*, etc., can tolerate only narrow fluctuations in salinity of water and are known as **stenohaline** animals. While some animals such as *Mytilus*, *Aplysia*, etc., can withstand wider ranges of salinity and are called **euryhaline** animals. However, there are certain animals, such as, *Anguilla*, *Salmon*, etc., which are both stenohaline and euryhaline.



Mytilus.

Aplysia.

Chemical Properties of Water

Water consists primarily of a single compound, H_2O . It is a universal solvent and most chemical compounds ionize readily in water and provide many radicals and considerable versatility in the rearrangement of chemical substances. It has following chemical properties :



Fishes use oxygen dissolved in water for respiration.

(a) Solubility of gases in water.

Most gases dissolve readily in water, most notably those that are essential for life. The concentration of any gas in water generally varies between zero and a theoretical maximum or **saturation** (Table 3-4). The latter is the amount of gas that can be dissolved in water when the atmosphere and the water are in equilibrium with one another. Except for water falls and very turbulent streams, the water in natural ecosystem is seldom in equilibrium with the atmosphere. A gas may show a **deficit**, if it is being utilized in the ecosystem faster

than it is going into solution across the air water interface, or it may be **supersaturated**, if it is being produced in the ecosystem faster than it is being released from solution across the interface. The concentration of important gases may vary widely in any ecosystem, both horizontally and vertically.

The saturation level of any gas in water depends on several variables, most notably temperature, salinity, (see Table 3-2) the concentration of the gas in the atmosphere, and its relative solubility in water. The greater the concentration of a gas in the atmosphere, the greater its concentration in the water will tend to be, depending on its relative solubility in water.

1. Oxygen. One of the most critical factor in an aquatic environment is the amount of oxygen in the water because most living organisms (excepting anaerobic forms) require this gas for respiration. In contrast to atmosphere, the oxygen becomes limiting factor for aquatic animals as the saturation concentration of oxygen in water is governed by temperature and salinity. As is evident in table 3-5, the lower the temperature, the greater the oxygen retaining capacity of water, whether it is freshwater or sea water.

Table 3-4.

Comparison between equilibrium concentrations of important gases in the atmosphere and in water (Clapham, Jr. 1973).

Gas	Atmospheric concentration	Saturation in water
Oxygen	210 cc./l. (21%)	7 cc./l. (32.9%)
Nitrogen	780 cc./l. (78%)	14 cc./l (65.7%)
Carbon dioxide	0.3 cc./l. (0.03%)	0.3 cc./l. (1.4%)

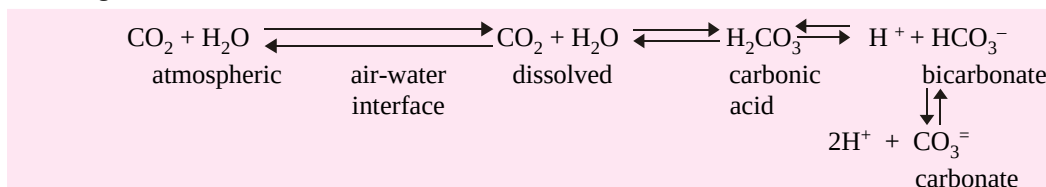
Table 3-5.

Comparison of the saturation concentration of oxygen in freshwater and salt water environment with varying temperature.

Water	Temperature, °C	Saturation point in millilitre per litre
Freshwater	0	10.27
Saltwater	0	8.08
Freshwater	30	5.57
Saltwater	30	4.52

2. Nitrogen. Nitrogen is significantly less soluble in water than oxygen. But because it constitutes 78 per cent of the atmosphere, it still accounts for about 65 per cent of the dissolved gases at equilibrium. It is fairly inert chemically and does not react with water, although some bacteria, fungi, blue-green algae, and so on, can use it to satisfy their nitrogen requirements; and other bacteria can produce it through reduction of nitrate under conditions of very low oxygen concentration.

3. Carbon dioxide. The decomposition of organic matter and the respiratory activity of aquatic plants and animals produce carbon dioxide. This gas is one of the essential raw materials necessary for photosynthetic activity by green plants. Carbon dioxide combines chemically with water to produce carbonic acid (H_2CO_3), which influences the hydrogen ion concentration (pH) of water. Carbonic acid dissociates to produce hydrogen (H^+) and bicarbonate (HCO_3^-) ions. The bicarbonate radical may undergo further dissociation forming more hydrogen (H^+) and carbonate ($\text{CO}_3^{=}$), as represented in the following reactions :



The amount of free or uncombined carbon dioxide in water is of ecological importance : it governs the precipitation of calcium in the form of calcium carbonate (CaCO_3). Calcium precipitates when temperature and salinity are high and the amount of uncombined carbon dioxide is low. This means more carbonate ($\text{CO}_3^{=}$) is present to combine with the calcium cation (Ca^{++}). These conditions exist in shallow tropical waters, where evaporation is high. This raises the salinity and photosynthetic activity of plants and reduces the quantity of free carbon dioxide in water. The precipitation of calcium carbonate in tropical areas as the Bahamas explains the preponderance of thick calcareous shells of shallow water tropical molluscs, plankton and algae. In deep oceanic water, temperature is low and there are no photosynthetic plants, consequently, the carbon dioxide content of the water is high. Deep water fauna (molluscs, crustaceans) possess very fragile skeleton because the precipitation of calcium carbonate is minimum.

4. Hydrogen sulphide. The deeper layers of many bodies of water, including ponds, lakes, and some estuaries, may contain significant amounts of the toxic gas, hydrogen sulphide, which is released by decaying organic matter. If concentration of the gas builds up, all life but anaerobic bacteria excluded from the area (e.g., deeper strata of Black sea).

Effect of Factor of Aquatic Environment on Aquatic Organisms

The aquatic environment is subject to water movements ranging from small vertical circulations to strong currents. The streams have a unidirectional movement and in seas the movement is reversible. Many aquatic animals have accordingly taken to sedentary or sessile lives depending on water movements. Radial symmetry is a characteristic of such animals. Transformation from a sessile to a

locomotive existence favours a bilateral symmetry. The water currents of water often abrade (= rub off) the inhabiting flora and fauna and varied modifications are encountered to withstand this abrasive action. Thick scales, strong shells and many attachment devices such as the holdfasts and suckers all are the results of this environmental stress. The ability to breathe air dissolved in water, at times even resorting to anaerobic existence, the modification of various senses to respond to stimuli characteristic of aquatic environments, the phenomenon of osmoregulation, and above all the phenomenon of external fertilization are other remarkable physiological adaptations to live in an exclusively aquatic medium.

Water and Ecological Adaptations

Water makes up a large proportion of the bodies of plants and animals, whether they live on land or in water. Active cytoplasm holds about 70–90 per cent of water. It has several important physiological properties. There exists a strong relationship between the water status of soil, plant and atmosphere. The rooting zone of the soil (zone of soil in which the water absorbing organs, roots, root hairs are present), the plant body and the lower layer of atmosphere behave as a continuum, called **spac** or **soil plant atmosphere continuum** in relation to water transfer (**Phillip**, 1966). Solar radiation is the primary energy source for the water transport process in the SPAC. On the other hand, animals obtain water (i) by drinking (ii) by absorbing it through their skin from contact with some damp ground, (iii) directly from their food or (iv) from water produced by metabolism. The method of obtaining water and the relation to the supply of liquid water as well as resistance to the drying effects of the surrounding atmosphere are important in determining the distribution of animals.

The scarcity or abundance of water brings about adaptations in living organisms. Plants which grow in areas where water is available in plenty, are classified as **mesophytes** and terrestrial animals living under such conditions are called **mesocoles**. Plants growing in water are called **hydrophytes**, while animals that live in the aquatic environment are called **aquatic animals** or **hydrocoles**. Some plants can grow in ecosystems where water is scarce and where the day temperature is very high. These plants are called **xerophytes** and the animals living in such xeric conditions are called **desert animals** or **xerocoles**. Xerophytes living in physiologically dry soils, i.e., saline soils with high concentrations of salts such as NaCl, Mg Cl₂ and Mg SO₄, are called **halophytes**. Based on their specific habitat, halophytes can be further classified into **lithophilous**, **psammophilous**, **pelophilous** and **helophilous** plants growing on rock and stones, sand, mud and swamp, respectively. Helophilous helophytes include mangroves of sea shores of Bombay such as *Rhizophora mucronata* and *Sonneratia*.

1. Hydrophytes and hydrocoles and their adaptations.

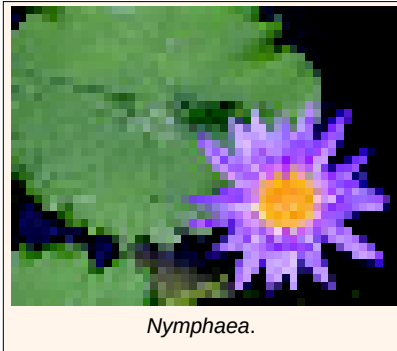


Hydrophytes include : (a) **free-floating hydrophytes** (e.g., *Wolffia*, *Lemna*, *Spirodella*, *Azolla*, *Eichhornia crassipes*, *Salvinia* and *Pistia*), (b) **rooted hydrophytes with floating leaves** (e.g., *Trapa*, *Nelumbo*, *Nymphaea*, *Marsilea*, etc.), (c) **Submerged floating hydrophytes** (e.g., *Ceratophyllum*, *Utricularia*, *Najas*, etc.), (d) **rooted submerged hydrophytes** (e.g., *Hydrilla*, *Chara*, *Vallisneria*, etc.), (e) **rooted emergent hydrophytes** or amphibious plants (e.g., *Sagittaria*, *Ranunculus*, etc.). **Tenagophytes** are amphibious plants—they grow in water bodies as well as in water logged soil.

These hydrophytes grow in hydric conditions and show the following general adaptive features: They possess poor mechanical, absorbing, conductive and protective tissues. They also contain an extensive development of air spaces (aerenchyma) in the tissues. Roots are either absent (e.g., *Wolffia*) or poorly developed (e.g.,

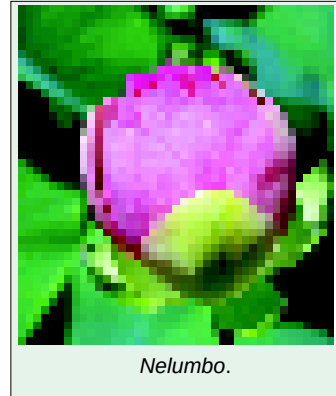
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Hydrilla). Roots may not have root hairs, root cap (instead of root cap *Eichhornia* has root pockets) and vascular tissue. Roots of hydrophytes are generally fibrous and adventitious, when present. The stem of hydrophytes is weak, slender and spongy. In some it is like a horizontal rhizome covered with mucilage, while it may be hard, as in *Nelumbo*. The aerial leaves may be broad but the



Nymphaea.

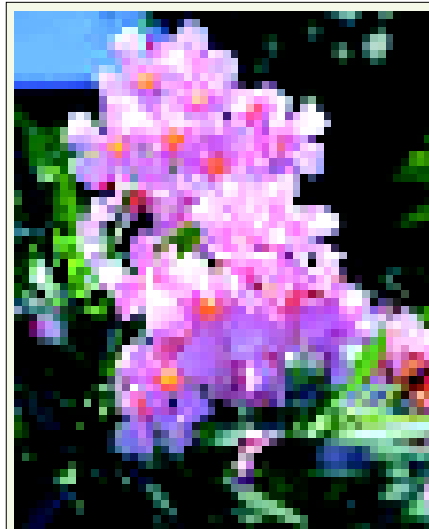
submerged leaves are thin, long or ribbon-shaped. Stomata are completely absent in submerged leaves (e.g., *Anacharis*), but in floating forms, stomata are confined only to the upper surfaces of leaves as in *Nymphaea*.



Nelumbo.

Aquatic animals or hydrocoles in general exhibit an elongated stream-lined body having a compressed head, body and tail. Hydrocoles include fishes, sea turtles, mammals such as whales, and many others. There are also amphibious forms such as frog, toad, crocodile, etc., and many birds which visit water bodies either for reproduction or for collection of food. (For further details of aquatic adaptations in animals see Chapter 18).

2. Xerophytes and xerocoles and their adaptations. Xerophytes grow in conditions of water scarcity, high temperature, strong winds, high transpiration rate and evaporation higher than precipitation. The soil is very dry and porous. The essential adaptations of xerophytes involve increased water absorption by roots, storing of water and retardation of transpiration. Thus, in search of water, xerophytic trees may go very deep in the soil and have extensive root hairs to absorb it. The roots of plants such as *Calotropis procera*, *Ficus*, and *Acacia nilotica* may go as deep as 10 to 16 metres and may reach the water table. As a consequence, these plants survive in deserts or arid conditions even if their rate of transpiration is higher. The storage of water is facilitated either by modifications of leaves, as in *Mesembryanthemum* and in the malacophyllous xerophytes (in which leaves contain turgescient parenchymatous cells) as *Aloe*, *Begonia*, *Bryophyllum*, *Agave*, *Yucca*, etc., or modification of stems, as in cacti such as *Opuntia* (phylloclade) and *Euphorbia*. In some xerophytes the water is stored in their roots as in *Asparagus* and *Ceiba parvifolia*. All these xerophytes are called **succulents** because they possess thick, fleshy, water storage organs such as stems, leaves and roots. Non-succulent xerophytes such as *Calotropis*, *Prosopis*, *Acacia*, *Zizyphus*, *Casuarina*, *Nerium*, *Saccharum* and *Pinus* possess other sort of xerophytic adaptations, viz., extensive root system, high osmotic pressure and other modifications in the leaves. Reduced transpiration is achieved by decreasing the leaf surface, as in *Casuarina*, *Acacia* and *Asparagus* or by modifying the leaves into spines and barbed bristles, as in cacti, or by having thick, leathery, thick cuticle or wax-coating bearing leaves with well-developed hypodermis and sunken stomata to reduce transpiration, as in *Calotropis* and *Nerium*. Halophytes (e.g.,



Nerium.

mangroves) resemble xerophytes and have high osmotic pressure; succulent organs; thin, evergreen, small leathery leaves with water storing tissues and thick cuticles and special air-breathing roots called **pneumatophores**.

Different animals have evolved the following adaptive features to live in arid environment :

1. Nocturnal life style. Most desert animals are nocturnal and seek shade or burrow deep in the soil in the day time to avoid excessive heat and dryness. Some xerophile rodents passively lose heat through conduction by pressing their bodies against the burrow walls.

2. Deceptive colouration. Desert animals are usually grey, brown or red matching with the colour of the sand or rock.

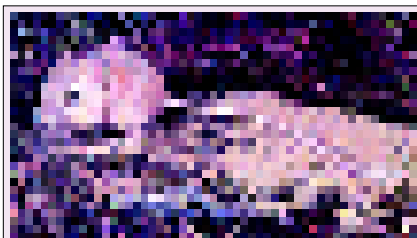
3. Suspended animation. Certain animals, usually with simpler organization, such as rotifers, nematodes, tardigrades, desert snails, etc., retain their vitality in long dry environment. Other forms (frogs, toads, etc.) **aestivate** during droughts and are active during moist season of the year.

4. Fast movement. Desert animals move much faster than other land animals, since they have to travel long distances in search of food and water.

5. Migration. Many birds and mammals of arid zones migrate when water becomes scarce or as a result of drought or for other reasons, the food supply is less.

6. Heat loss by radiation. Animals such as jack-rabbits (*Lepus*) and fox (*Vulpes velox*) have large ears that reduce the need of water evaporation to regulate the body temperature. Their ears function as efficient radiators to the cooler desert sky, which on clear days may have a radiation temperature 25°C below than that of the animal body. By seeking shade and sitting in depressions, *Lepus* could radiate 5 kcal/day through its two large ears (400 cm²).

7. Impervious skin. The drier habitats (deserts, etc., are invaded by only those animals which contain a thick impermeable body covering. Such integuments occur in many insects, birds and mammals. Some mammals such as men, apes and horses lose much water (and salt) through sweat glands in heat regulation. Most rodents and some ruminants such as antelope nearly or completely lack sweat glands. Moist skinned forms (most amphibians and earthworms), certain mites and soft-bodied insects are restricted to swamps, stream margins, moist soils and other similarly damp places.



This nocturnal kangaroo rat in a California desert is a master of water conservation. Instead of drinking water it gets the water it needs from its food and cellular respiration. It also conserves water by excreting dry faeces and thick nearly solid urine.

8. Upturned nostrils. Desert animals have nostrils directed upwards; this may provide a protection from clogging by wind blown sand.

9. Water from food and from metabolism. Most herbivores and carnivores live on the moisture obtained with food. Many insects utilize the high water contents of plants to meet their water requirements. In fact, most animals make use of water released during metabolism when fats and carbohydrates are broken.

Kangaroo rat, which seals its burrow by day to keep its chamber moist, can live throughout year without drinking water. It obtains its water from its own metabolic processes and from hygroscopic water in its food.

10. Internal lungs or tracheal system. Mode of respiration has some correlation with water. Crustaceans, with their gills covered by a water-retaining carapace, carry with them a liquid environment for their gills. The scaly body covering of a fish may be practically impermeable to water and exchange of gases may be limited to gills and gut. Internal lungs, whether in pulmonate snails, land isopods, spiders or higher vertebrates (amphibians, reptiles, birds and mammals) together with the

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internal tracheal system of insects are water saving. Much water is lost in breathing even in animals having internal lungs.

11. Dry excretion. A further water-saving device is the excretion of concentrated, relatively dry nitrogenous and faecal waste materials. Water-saving insects, reptiles and birds excrete nitrogenous waste as solid uric acid.

Camel and African antelope, *Oryx* provide good examples of xerocoles or drought-resistant animals. Camel can go without food and water as long as 10 days at a stretch. For water conservation, camel has following adaptations : (1) When camel drinks water, it can take in up to 50 litres in one gulp and this water is evenly distributed all over its body in tissues and not in specific pockets or organs (e.g., in stomach) as the common misconception holds. (2) Camel excretes highly concentrated urine. Its dung also contains very little water, compared to the dung of the donkey or the cow (both of which are xerocoles). (3) It perspires very little and its breathing rhythm is very slow. (4) Camel can withstand dehydration up to 25 per cent of body weight and it loses water from body tissues rather than from blood. (5) Body temperature of camel is labile, dropping to 33.8°C overnight and raising to 40.6°C by day, at which point it begins to sweat. Due to such thermoregulation, the amount of water loss by perspiration and in other ways is greatly reduced. (6) Camel accumulates its fat in the hump rather than all over body. This speeds heat flow away from the body and its thick coat prevents the flow of heat inward towards the body.



Oryx—a drought-resistant animal.

Snow as Habitat

Snow which is the solid state of liquid water, is an integral factor in the physical environment. It is important particularly in the higher latitudes and on the higher parts of mountains where snow cover may last throughout the year. Its significance is increased by the large amount of land surface in northern Eurasia and North America. Snow covers much of the available food for many animals and is a factor causing migration, both of birds and mammals, towards areas that are somewhat free from its direct influence. Certain mammals (rodents, etc.) tunnel through the snow near the underlying mosses and get access of food and often protection from winter cold. In the Tundra, the non-hibernating lemmings (*Dicrostonyx*) breed during winter beneath the snow. Snow prevents the locomotion of mammals and of running birds, and animals with long feet only are well adapted for moving on snow cover.



Lemming.

C. LITHOSPHERE (SOIL) (Edaphic Factors)

The earth is a cooled, spherical, solid planet of solar system, which rotates on its axis and revolves around the sun at a certain constant distance. The solid component of earth is called **lithosphere**. The lithosphere is multilayered including following main layers—**crust**, **mantle**, and **outer** and **inner core**. The **core** is the central fluid or vapourised sphere having diameter of about 2500 kms from the centre and is possibly composed of Nickel-Iron. The **mantle** extends about 2900 kms above the core. This is in molten state. The outer most solid zone of the earth is called **crust** which is about 8 to 40 kms above

the mantle. The crust is very complex and its surface is covered with **soil** supporting rich and varied biological communities, for living organisms find in the soil an environment providing food, shelter, and concealment from predators (**Wallwork**, 1970).

Soil

The word soil is derived from a Latin word '*solum*' meaning earthy material in which plants grow. The science which deals with the study of soil is called **Soil Science**, **Pedology** (*pedos*=earth) or **Edaphology** (*edaphos*=soil). The study of soil is significant for us due to various reasons. Soil is a natural habitat for microorganisms, plants and animals. Its knowledge is helpful in practices of agriculture, horticulture, forestry, etc., such as cultivation, irrigation, artificial drainage, and use of fertilizers. Pedology is important also for geology, petrology, mineralogy, palaeobotany and palaeozoology.

The soil can be defined as the weathered surface of the earth's crust which is mixed with organic material and in which microorganisms live and plants grow. Soil consists of the inorganic materials (the mineral matter) derived from parent rocks; the organic materials (the humus) derived from dead organisms; the air and water occupying the pores between the soil particles which are loosely-packed (the soil water and the soil atmosphere); small organisms (the biological systems) such as bacteria, fungi, algae, protozoa, rotifers, nematodes, oligochaetes, molluscs and arthropods, and higher plants which live in it. Soil is the ultimate source of all food production since plants form the base of all ecological pyramids and plants grow in soil and derive nutrition from them. It provides mechanical anchorage to plants, besides serving as a reservoir of food materials and water. It is the site where nutrient elements are brought into biological circulation by mineral weathering. The soil harbours the bacteria which incorporate atmospheric nitrogen into the soil. Plant's roots occupy a considerable portion of the soil; tie the vegetation to the soil; and pump water and its dissolved minerals to other parts of the plant for photosynthesis and other biochemical processes (see **Smith**, 1974).

Further, a **mature soil** is that state of soil that has assumed the profile features (*i.e.*, succession of natural layers) characteristic of predominant soils on the smooth uplands within the general climatic and botanic regions in which it is found (**Marbut**, 1926).

Soil formation or Pedogenesis

Soil is a stratified mixture of inorganic and organic materials, both of which are decomposition products. The mineral constituents of soil are derived from some parent material, the **soil forming rocks** by fragmentation or weathering, while, organic components of soil are formed either by decomposition (or transformation) of dead remains of plants or animals or through metabolic activities of living organisms present in the soil. Before discussing the soil-forming processes, let us describe in brief the nature of soil forming rocks as follows :

Soil forming rocks. Basically, there are following three kinds of soil forming rocks :

1. Igneous rocks which are formed due to cooling of molten magma or lava, *e.g.*, granite, diorite and basalt.

2. Sedimentary rocks which are formed by deposition of weathered minerals which are derived from igneous rocks, *e.g.*, shales, sandstone and limestone.

3. Metamorphic rocks which are formed by change of pre-existing rocks (*e.g.*, igneous or sedimentary rocks) through heat and pressure, *e.g.*, gneiss, schist, slate, quartzite and marble.

Chemistry of minerals of soil-forming rocks. Rocks are the chemical mixture of numerous kinds of minerals. The chemical nature of certain most common and abundant minerals of soil-forming rocks has been listed in table 3-6.

Table 3-6.

Chemical composition of some common soil minerals.

Minerals	Chemical constituents
A . Sand and silt minerals	
1. Quarts or silica	SiO_2
2. Feldspars	
a. Orthoclase	$\text{K}_2\text{Al}_2\text{Si}_6\text{O}_{16}$
b. Plagioclase	$\text{NaAlSi}_3\text{O}_8$
c. Calcium feldspar	$\text{CaAl}_2\text{Si}_2\text{O}_8$
3. Micas	
a. Muscovite	$\text{K}(\text{OH})_2\text{Al}_2(\text{AlSi}_3)\text{O}_{10}$
b. Biotite	K, Mg, Fe, Al silicate
4. Pyroxene	$(\text{Mg, Fe})\text{SiO}_3$
5. Amphibole	$(\text{Mg, Fe})_7(\text{Si}_4\text{O}_{11})_2(\text{OH})_2$
6. Olivine and serpentine	$(\text{Mg, Fe})_2\text{SiO}_4$
7. Calcite; magnesite; and dolomite	CaCO_3 , MgCO_3 ; and $(\text{CaCO}_3, \text{MgCO}_3)$
8. Iron oxides	
a. Haematite	Fe_2O_3
b. Magnetite	Fe_3O_4
c. Limonite	$\text{FeO}(\text{OH}), x\text{H}_2\text{O}$
B. Clay minerals	
1. Kaolin	$\text{Al}_2\text{O}_3, 2\text{SiO}_2, 2\text{H}_2\text{O}$
2. Montmorillonite	$(\text{Ca, MgO})\text{Al}_2\text{O}_3, 5\text{SiO}_3, 5\text{H}_2\text{O}$

Process of Soil Formation

The processes which are involved in the formation of mature soil can be studied under the following heads :

A. Weathering of Soil Forming Rocks

Soil formation is started by **disintegration** or **weathering** of parent rocks by some physical, chemical and/or biological agents, because of which the soil-forming rocks are broken down in small particles called **regoliths**. Regoliths are the basic materials which under the influence of various other pedogenic processes finally develop into mature soil.

(a) Physical weathering. The physical weathering agents are primarily climatic in character, exerting a mechanical effect on the substratum with the result that fragments are pulverised into progressively decreasing particle sizes (*i.e.*, regoliths). Such climatic weathering of rocks does not cause any chemical transformation of rock-minerals and commonly occurs in deserts, in high altitudes, in high latitudes, and in localities with marked topographic relief and sparse vegetation cover. The agents which are involved in climatic weathering of rocks, are **temperature, water, ice, gravity** and **wind** (see **Wallwork**, 1970).

The **temperature** causes break down of those rocks which have heterogeneous structure, due to the fact of differential expansion and contraction coefficient of materials composing the rocks. Minerals composing the rocks have got different degrees of expansion. These minerals expand in the high temperature of day and contract when the temperature falls. The differential expansion and contraction of different minerals of rocks set up internal tensions and produce cracks in the rocks and consequently, the rocks weather into finer particles.

In its liquid state **water** causes mechanical weathering of rocks by following methods : (i) **Rain water**. Natural water falling either in the form of rain drops or hail storm on the surface of rock with beating effect bring about abrasion of massive rocks into smaller particles. (ii) **Torrent water**. Rapidly flowing water rolls the heavy rock masses such as rock boulders along the bottom of stream and grinds them into finer particles. (iii) **Wave action**. The wave actions are most effective in sea shores. The rapidly striking water waves dislocate solid particles of varying diameters from sea shore rocks and the debris is then settled at the sea bottom to form **marine soil**. Water also acts as a mechanical carrier. Nearly 5 billion tons of mineral matter is annually carried away in solution from land to sea (see **Pearse**, 1939).

In its freezing and ice-melting states water causes rock-weathering by **frost action** and **glacier formation**. Water in the form of frost or ice, is an extremely effective physical weathering agent of rocks. It seeps into rock crevices, freezes due to sudden fall of temperature of rocks, expands about nine per cent of its original volume, exerts a pressure (expansion force) of approximately 150 ton/ft² and eventually cracks the rock into smaller pieces. Likewise, in summer when ice at mountain tops starts melting and glaciers (huge sliding masses of ice) move downwardly on the slope, then during the glacier movement, the rock over which they move is gradually worn down to produce fine particles which are deposited as **drift** or **till**, when glacier finally retreats.

Gravitation weathering action is most effectively demonstrated by land slides and rock slippages caused by earth quakes and faulting during which the rock is fragmented by abrasion and the forces of impact. Lastly, the rapid stormy **wind** carrying suspended sand particles causes the abrasion of exposed rock. It acts like a mechanical carrier in moving the particles over the surface of earth as **dunes** or **drifts** and in transporting large quantities of fine suspended particles long distances.



Biological weathering.



Torrent waterfall rolls the heavy rock masses and grinds them into finer particles.

(b) **Chemical weathering**. The physical weathering produces a greater surface area of rock exposed to the **chemical weathering**, which occurs simultaneously with physical weathering and continues much beyond that. During chemical weathering, a chemical transformation or decomposition of parental mineral materials into new mineral complexes or secondary products occurs. For example, primary minerals that contain aluminium and silicon, such as feldspar, are converted to secondary mineral such as clay. Because chemical weathering requires the presence of moisture and air as essential factors, therefore, chemical weathering is not effective in deserts. It includes reactions such as solution, hydrolysis, oxidation, reduction, carbonation and hydration.

(c) **Biological weathering**. Though the surface of bare rock is unsuitable for many forms of life, even then a number of microorganisms (bacteria, protozoans, fungi,

nematodes, etc.), lichens and mosses can gain a foothold. These early colonizers transform the rock into a dynamic system, storing energy and synthesizing organic material (Jacks, 1965). Their activities alter the mineral composition as well as the physical structure of the rock. For example, lichens are present in the initial stages of biological succession and their growth may cause cracking or flaking, exposing greater area of rock to further weathering. The lichens and mosses extract mineral nutrients such as P, S, Ca, Mg, K, Na, Fe, Si, and Al, from the rock. These elements are combined with organic complexes, and eventually return to the developing soil when the vegetation decomposes.

Products of Weathering and Soil Types

The soil which is formed by weathering of soil-forming rocks is called **embryonic** or **primary soil**. The **secondary** or **transported** soils are those which are carried to other places by carriers such as gravity, water, glacier and wind. Soil material transported from one area to another is known as **loess** or **eolian**; that transported by water as **alluvial**, **lacustrine** (lake soil) and **marine deposits**; that transported by gravity as **colluvial soil** and that transported by sand storms as **sand dunes**. In a few places soil material comes from accumulated organic matter as **peat**. The soil which develops *in situ* above parent bedrock is called **residual** or **sedentary soil**.

B. Mineralization and Humification

The weathering of inorganic soil matter is followed by decomposition of organic matter, which starts at an early stage of pedogenesis, and continues up to much later stages. During early stages of pedogenesis, the organic contents of the embryonic soil are not very high, as the vegetation and its associated fauna are not richly developed. Also, the products of organic composition frequently occur in combination with those of inorganic decomposition as complexes, not easily separable into component parts in these initial stages. However, as both the size of soil particles and the spaces between them become smaller with the result that the water holding capacity increases. This together with the increasing amount of plant nutrients and organic material, allows the soil to support higher plant life, such as grasses; the protective covering effect and binding action of root systems so provided, promotes a greater stability of soil structure. Thus, soil development begins under some influence of plants. They root, draw nutrients from mineral matter, reproduce and die. Their roots penetrate and further breakdown the regolith. The plants pump up nutrients from its depth and add them to the surface, and in doing so recapture minerals carried deep into the material by weathering processes. By photosynthesis plants capture the sun's energy and add a portion of it as organic carbon to the soil each year. This energy source, the plant debris, enables bacteria, fungi (such as *Rhizoctonia solani*, *Armiliaria mellea*, etc.), earthworms and other soil organisms to colonize the area.

The breakdown of organic debris into humus is accomplished by decomposition and finally mineralization. Higher organisms in the soil—millipedes, centipedes, earthworms, mites, springtails, grasshoppers, and others—consume fresh material and leave partially decomposed products in their excreta. This is further decomposed by microorganisms, the bacteria and fungi, into various compounds of carbohydrates, proteins, lignins, fats, waxes, resins and ash. These compounds are broken down into simpler products such as carbon dioxide, water, minerals and salts. This latter process is called **mineralization**. The residual amorphous, incompletely decomposed black coloured organic matter which undergoes mineralization is called **humus**. The process of humus formation is called **humification**. **Muller** (1879, 1884) has recognized two kinds of humus—mor and mull. **Mor humus** is acidic and supports an abundant fungal growth and low number of soil bacteria. Fungal mycelia may help to bind together particles of humus and decomposition litter into matted layers and in a well developed mor, three such layers can be distinguished, namely a surface **litter** or **L layer**, comprising undecomposed leaves and twigs, below which is the **fermentation** or **F layer**, in which decomposition has proceeded same way toward the development of humus, and beneath this, the **humus** or **H layer**, in which degraded humus fractions accumulate. Mor has low calcium content and developed on sandy soils under conifers. **Mull humus** is neutral or slightly alkaline and contain rich microflora of bacteria. It lacks distinct layering of the mor, largely because the presence of calcium compound favours the

development of a rich earthworm fauna which promotes a greater mixing of organic and mineral materials. The mull humus develops in brown forest soils under tree species having a relatively high calcium content, such as alder, elm, bass wood, aspen and few conifers (red cedar). Intermediate between the two types of humus is the '**moder**' which has richer and varied fauna.

In order to emphasize the part played by living plants and animals in soil formation, **Taylor** (1930) proposed the following formula :

$$S = M (C+V+VA+A) t+D$$

where, A signifies animals ; C = climate; D = erosion or deposition ; M = parent material, S = soil; t = time and V = vegetation.

Role of earthworms in soil formation and soil fertility. Earthworms play a significant role in soil formation. They may burrow two metres into the soil, make numerous transverse furrows and produce **worm casts** by eating the soil and soil organic matter (litter). These casts are very rich in nitrogen, water soluble aggregates and mineral substances and increase soil fertility. The burrows facilitate aeration and increase water holding capacity of the soil. **Dash** and **Patra** (1979) have calculated 77 tonnes of dry weight of earthworm cast production per hectare per year in an Indian grassland site. It is estimated that soil turn over due to earthworm action provides a stone-free layer about 15 cm deep on the surface.

C. Formation of Organo-Mineral Complexes

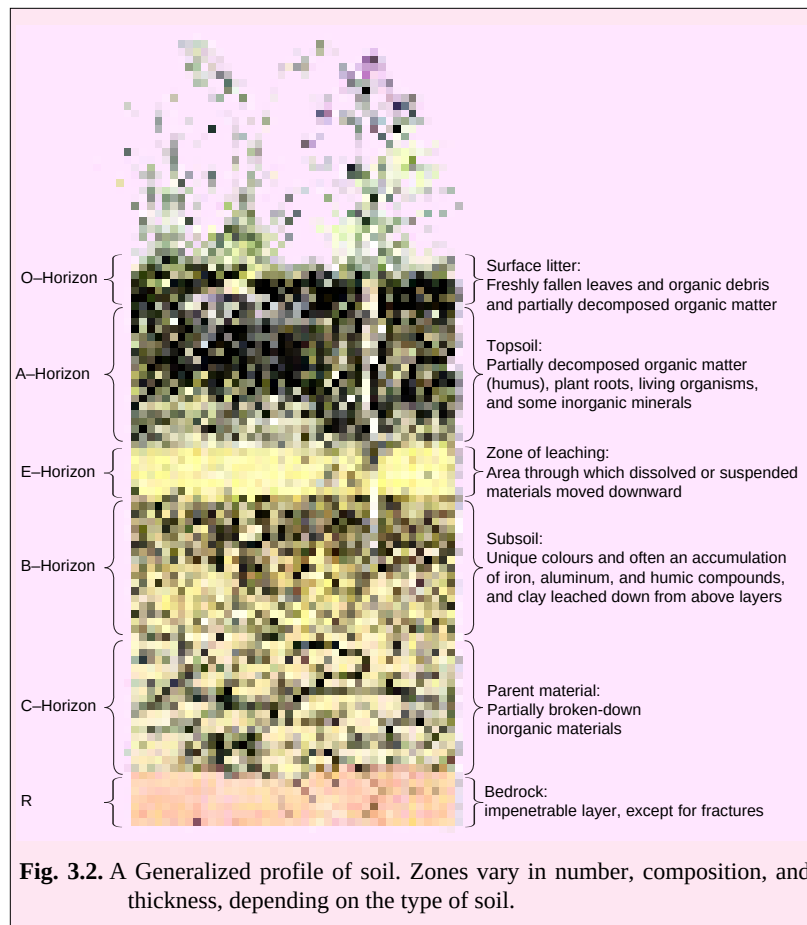


Fig. 3.2. A Generalized profile of soil. Zones vary in number, composition, and thickness, depending on the type of soil.

During final stage of pedogenesis, colloidal particles which are formed due to weathering, humification and mineralization accumulate and may aggregate into **crumbs** or **concretions**. Some colloidal humus particles may become associated with mineral particles to form organo-mineral complexes. According to **Wallwork** (1970) the crumbs or organo-mineral complexes are formed by following two mechanisms—**electro chemical bonding** and **cementing**. In electrochemical bonding method of crumb formation, aggregation of negatively charged colloidal clay and/or

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humus particles of water molecules and metallic ions, particularly calcium, takes place. The cementing mechanism of crumb formation involves the action of substances absorbed on the surface of soil particles which effectively glues them together.

The crumbs increase the total pore space in the soil, allowing good aeration and drainage. Eventually, a characteristic profile develops under the influence of climate, vegetation, parent material and the activities of the soil communities. The mature soil, thus, becomes a complex system of living and non-living materials not inert but active.

Soil Profile

Soil profile is the term used for the vertical section of earth crust generally up to the depth of 1.83 meter or up to the parent material to show different layers or horizons of soil for the study of soil in its undisturbed state. It is made up of a succession of horizontal layers or **horizons**, each of which varies in thickness, colour, texture, structure, consistency, porosity, acidity and composition. A **pedon** is the smallest three-dimensional volume of soil needed to give full representation of horizontal variability of soil.

In general, soils have following four horizons an **organic** or **O-horizon** and three **mineral (A,B,C) horizons**. Some workers recognized a **D-horizon**, in which rocks are in active weathering state, in between C and R-horizons. **R-horizon** is the consolidated bed rock on which a soil profile rests. A and B-horizons form the **true soil** or **solum**. Each horizon of soil profile is further subdivided. Horizon subdivisions are indicated by arabic numbers, e.g. O₁, O₂, A₁, A₂, etc. Different horizons of soil-profile have following characteristics :

1. O-horizon. The O-horizon, once designated as **L, F, H**, or Ao and Aoo, is the surface layer forming above the mineral layers and composed of fresh or partially decomposed organic material, as found in temperate forest soils. It is usually absent in cultivated soils and grasslands. O-horizon contains both kinds of humus (e.g., mull and mor) and is subjected to the greatest changes in soil temperatures and moisture conditions and contains most organic carbon. O-horizon and upper part of A horizon is the region where life is most abundant. The O horizon is divided into following two sub-layers:

(i) O₁ (Aoo) region. It is the uppermost layer which consists of freshly-fallen dead leaves, branches, flowers and fruits, dead remains of animals, etc. All these do not show evident breakdown.

(ii) O₂ (Ao) region. Below the O₁ region is the O₂ layer of partly decomposed organic matter. The process of decomposition of the litter is started in O₂ region. Thus, organic matter is found under different stages of decomposition and microorganisms such as bacteria, fungi, actinomycetes are frequently found in it. Upper layers contain detritus in initial stage of decomposition, in which material can be faintly recognized, whereas the lower layers contain fairly decomposed matter, the **duff**.

2. A-horizon. It is characterized by major organic matter accumulation, by the loss of clay, iron and aluminium and by the development of organo-mineral complexes, granular crumbs or platy structures. The A-horizon is divided into following two sub-layers :

(i) A₁ region. This region is dark and rich in organic matter and is called **humic** or **melanized region**. The amorphous, finely-divided organic matter here becomes mixed with the mineral matter, which is now known as **humus** and is dark brown or black-coloured.

(ii) A₂ region. It is light-coloured region where the mineral particles of large size as sand (silica) are more with little amount of organic matter. In areas of heavy rainfall, the mineral elements and organic chemicals are rapidly lost downwards in this region, making it light-coloured. A₂ region is, thus, also called **podsol** or **eluvial zone** or zone of leaching.

3. B-horizon. It lies below A-horizon and also called **sub-soil** or **illuviation** or **illuvial zone**, since, the nutrients received from A-horizon due to leaching are accumulated in this region. B-horizon is dark-coloured and coarse textured due to the presence of silica rich clay, organic compounds, hydrated oxides of aluminium, iron, etc. It is poorly developed in dry areas. B-horizon can be divided

into B₁ (A₃), B₂ and B₃ regions, depending upon the stages of soil development in the area.

4. C-horizon. Below B-horizon and above the surface of weathered parent rock, is the zone of regolith or C-horizon. It is a light coloured horizon containing weathered parent material.

5. R-horizon. Below all these horizons may lie the **R-horizon**, which is the parent, unweathered bedrock. The percolated soil water tends to collect at the surface of the bedrock.

Climate and Soil Types

Variation in climatic factors and rock properties, therefore, lead to different types of soil development in different climatic conditions, as follows :

1. Laterization. Laterite (L. *latus*=brick) soils are reddish brown in colour and are found in the warm and humid (=heavy rainfall) tropics. In the process of laterization, iron and aluminium oxides which are resistant to decomposition, do not leach down but remain in the surface soil. Silicic acid (*i.e.*, colloidal form of silica) on the other hand leaches to lower horizons. Laterites are comparatively less fertile soils and are found in Australia, India, some warm parts of Europe and North America.

Lotsol is a wider term than laterite and currently used to refer to soils of warm humid belt of the earth where silica usually leaches to lower horizons while oxides of iron and aluminium remain on the surface. Lotsol soil is also called **oxide** and **ferrasol** and is poor in bases and contains low organic content. It is rich in free quartz grains and kaolinite clays.

Vertisols are the other characteristic soils where there is a long dry season. These soils are formed from limestones, marl or ferromagnesium rocks and rich in montmorillonite clays. During dry summer, deep cracks are formed in vertisols and these cracks get filled with clay due to dust blow which in the rainy season on wetting swells so much as to form small bumps. Thus, vertisols are characterized by cracks in summer and bumps in rains.

2. Podsolization. It is the most actively studied soil forming process. It takes place in cold humid climate under forest vegetation (chiefly of genera of Ericaceae and conifers) where leaf litter and other organic matter decompose slowly, chiefly through fungal activity. Litter contains high lignin and low nutrient (particularly calcium) contents; it is also rich in phenolic compounds which inhibit microbial activities. In consequence, such a litter is acidic in nature. The water percolating through such litter, being acidic dissolves out with it minerals and humus contents (such as carbonates, sulphates and iron and aluminium compounds) from A-horizon of soil. These leached materials reach to the lower horizons, being collected in the form of a hard, distinct layer in the B-horizon. Leaching results into a grey ash-like surface called **podsol** (Russian *pod* = under, *zole* = ash). In the Indian conifer forests of the Himalayas true podosols are not met with, mainly due to alternate phases of wet and dry seasons.

3. Gleization. In humid climates, due to high ground level or retention of surface water in the soil, water-logging and consequent reducing conditions are fairly common. Under such reduced conditions firstly due to ferrous compounds, the soil colour becomes blue-grey or grey, and secondly the rates of decomposition of the organic matter are slow. These together result into the accumulation of a sticky compact layer of blue-grey colour at the bottom of B horizon. This process is called **gleization** and the soil as **gleys**. Gleization also takes place in ice covered regions of arctics (also tundra) where soil is not saline.

4. Melanization. The process is very common in regions of low humidity where humus is formed from the organic matter. As a result of melanization, the black humus along with water is mixed in the A-horizon of the soil which turn dark-coloured. Such a blackish brown top soil is called **chernozem** and it has a minimum leaching, so it is rich in organic detritus.

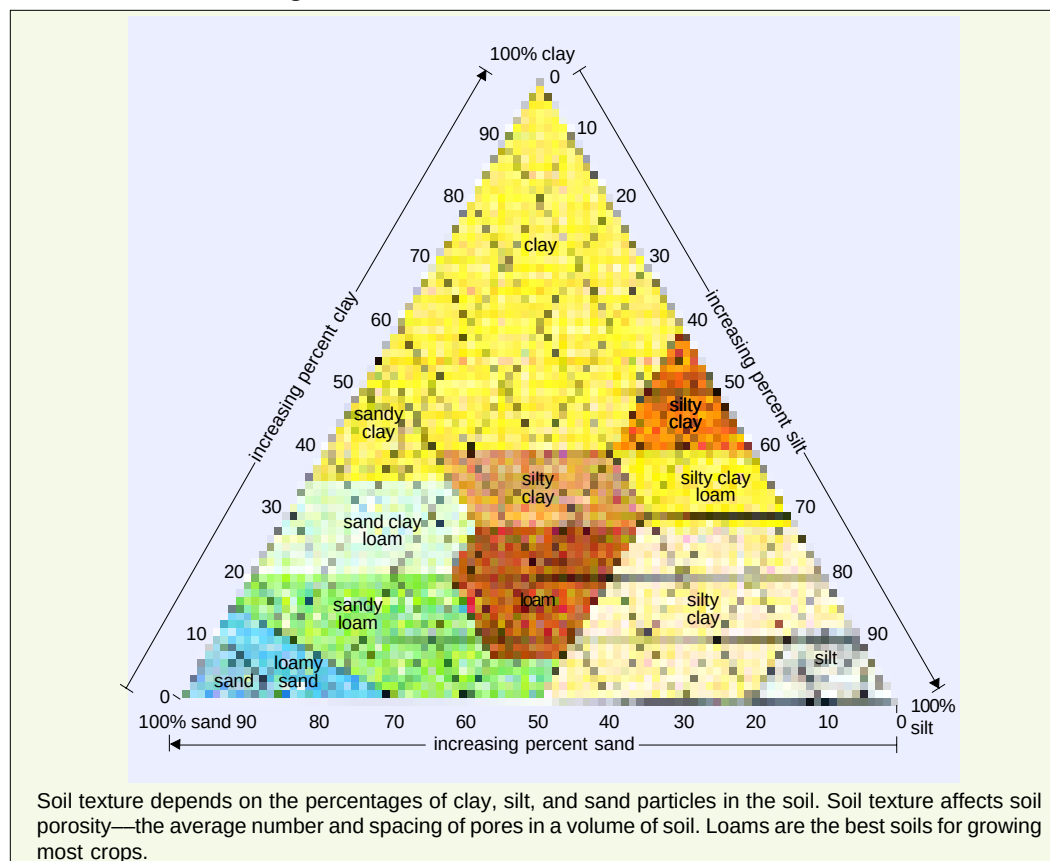
5. Calcification. It is a common process found in North India, where calcium in the form of calcium bicarbonate dissolves in water and leaches down escaping adsorption around dry particles. The leached CaCO₃ gets precipitated at the depths of about half to three metres, depending on the quantity

of rainfall and depth of infiltration. The precipitated calcium remains in the form of nodules or *kankar* and form a hard pan. This gradually reduces soil fertility. In the Northern India the kankar nodules are unearthed and are used in the building of roads and in the preparation of lime. Calcification is typical of grassland and a layer of calcium carbonate generally occurs at the base of B horizon. Sometimes, a layer of more soluble calcium sulphate (gypsum) is also present.

There are some other types of soils such as a hydromorphic, halomorphic and azonal. **Hydromorphic soils** are formed on poorly drained water filled regions with poor aeration, *i.e.*, they are highly reducing soils in which ferric ions are reduced to ferrous. But in dry seasons the soil becomes aerated and ferrous is again converted into ferric. The hydromorphic soils of tropical paddy fields are called **grey soil**. **Halomorphic soils** are characterized by the presence of high salt contents on the surface. Salinity and alkalinity are commonly found in big patches in the North Indian states resulting into the formation of *usar* or *reh* soils. In fact, such soils result due to upward capillary movement of dissolved salts from the rising water table in too frequently irrigated and fertilized regions, or from nutrient rich water received through stream flow from the surroundings during the periods of high rainfall. On evaporation, salts in different forms are left on the surface. Grazing animals often lick such salt deposits to fulfil their metabolic needs. **Azonal soils** lack horizon B and the horizon A is also thin. Among azonal soils, the common are **lithosols** found on mountain tops where parent rock is quite hard and resistant to weathering. **Regosols** are formed from volcanic deposits.

Morphology of Soil

In the field, differences among soils and among horizons within a soil are primarily reflected by variations in texture, arrangement, structure and colour.



1. Texture of soil. The texture of a soil is determined by the production of different sized soil particles. Mineral fractions or particles of soil are called “**soil fractions**” or “**soil separates**”. The soil particles have been classified into **gravel, sand, silt** and **clay** on the basis of their size differences. Gravel consists of coarse particles larger than 2.0 mm. Sand ranges from 0.02 to 2.0 mm in diameter and is easily seen and feels gritty. Silt consists of particles from 0.002 to 0.02 mm in diameter, which scarcely can be seen by the naked eye, feels and looks like flour. Clay particles range below 0.002 mm in diameter, are too fine to be seen even under the ordinary microscope, and are colloidal in nature. Clay controls the most important properties of soils, including plasticity and exchange of ions between soil particles and soil solution.

Most soils are mixture of these various particles. Based on the proportions of the various particles contained in them, soils can be grouped into 5 textural groups : **(1) Coarse textured soils** are loose, consist mainly of sand and gravel. They retain very little moisture and supply some plant nutrients. **(2) Moderately coarse soils** include sandy loam to very fine sandy loam. **(3) Medium textured soils** are mixture of sand, silt and clay, high enough to hold water and plant nutrients. **(4) Moderately fine textured soils** are high in clay. They are moderately sticky and plastic when wet, they may form a crust on the surface if organic matter is low. They have a high moisture-holding capacity. **(5) Fine textured soils** contain more than 40 per cent clay, may be sticky and plastic when wet and hold considerable water and plant nutrients, but may have restricted internal drainage. All these 5 textural soil-groups include 12 textural classes (see Table 3-7) which have been recognized on the basis of relative percentage of soil particles (*i.e.*, sand, silt and clay).

Table 3-7.

Textural classes of soils.

Soil classes or Textural names	Range in relative percentage of soil separates		
	Sand	Silt	Clay
1. Sandy soil	85-100	0-15	0-10
2. Loamy sand	70-90	0-30	0-15
3. Sandy loam	43-80	0-50	0-20
4. Loam	23-52	28-50	7-27
5. Silt loam	0-50	50-88	0-27
6. Silt	0-20	8-10	0-12
7. Sandy clay loam	45-80	0-28	20-35
8. Clay loam	20-45	15-53	27-40
9. Silty clay loam	0-20	40-73	27-40
10. Sandy clay	45-65	0-20	35-45
11. Silt clay	0-20	40-60	40-60
12. Clay	0-45	0-40	40-100

2. Structure of soil. Soil separates (sand, silt and clay) are held together in clusters or shapes of various sizes, called **aggregates** or **peds**. The arrangement of these peds in earth crust is called **soil structure**. Like texture, there are many types of soil structure. Soil peds may be classified as **granular, crumb-like, blocky plate-like** or **platy**, and **prismatic** or **columnar**. The properties of different soil peds have been tabulated in Table 3-8.

Table 3-8.

Characteristics of peds.

Types of ped	Properties
1. Granular	Small, spherical and non-porous.
2. Crumb-like	Small, porous and spheroidal.
3. Agranular blocky	Block-like with sharp ends, one end may be pointed.
4. Sub-granular blocky	Block-like but bounded by other aggregates.
5. Platy	Plate-like, sometimes plates are overlapped.
6. Prismatic or columnar	Prism-like but without rounded surface.

Structureless soil can be either single-grained or massive soil aggregates (peds) tend to become larger with increasing depth. Structure is influenced by texture, air, moisture, organic nutrients, micro-organisms, root-growth and soil chemical status.

3. Soil colour. Soils exhibit a variety of colour. Soil colour may be inherited from the parental material (*i.e.*, **lithochromic**) or sometimes it may be due to soil forming processes (**acquired** or **genetic colour**). Though colour has little direct influence on the functions of a soil, but is important in the identification of soil type. In temperate regions dark-coloured soils generally are higher in organic matter than light coloured ones. Well-drained soil may range anywhere from very pale brown to dark brown and black, depending upon the organic matter-content. Red and yellow soils are the result of iron oxides, the bright colours indicating good drainage and good aeration. Other red soils obtain their colour from parent material and not from soil-forming processes. Well-drained yellowish sands are white sands containing a small amount of organic matter and such colouring matter as iron oxide. Red and yellow colours increase from the cool regions to the equator. Quartz, kaolin, carbonates of lime and magnesium, gypsum and various compounds of ferrous iron give whitish and grayish colours to the soil. The grayest are permanently saturated soils in which the iron is in the ferrous form.

Soil colour and temperature. Soil colour influences the soil temperature. The dark coloured soils absorb heat more readily than light coloured soils. **Ramdas et al.**, (1936) showed that black cotton soils of Poona absorbed 86% of the total solar radiations falling on the soil surface as against 40% by the grey alluvial soil.

Physical Properties of Soil

Soil possesses many characteristic physical properties such as density, porosity, permeability, temperature, water and atmosphere, each of which can be studied under following separate headings:

1. Soil density. Average density of soil is 2.65 gms. per ml. Density of soil varies greatly depending upon the degree of weathering.

2. Porosity. The spaces present between soil particles in a given volume of soil are called **pore spaces**. The percentage of soil volume occupied by pore space or by the interstitial space is called **porosity** of the soil. Porosity of soil depends upon the texture and structure compactness and organic contents of soil. Porosity of the soil increases with the increase in the percentage of organic matter in the soil.

The pore spaces are of two types—**(1) Micro-pore spaces** (capillary pore spaces) and **(2) Macro-pore spaces** (non-capillary pore spaces). Capillary pore spaces can hold more water and restrict the free movement of water and air to a considerable extent, whereas macro-pore spaces have little water holding capacity and allow free movement of moisture and air in the soil under normal conditions.

3. Permeability of soil. The characteristic of soil that determines the movement of water through pore spaces is known as **soil permeability**. Soil permeability is directly dependent on the pore size, therefore, it is higher for the loose soil with large number of macro-pore spaces than it is for compact soil with numerous micro-pore spaces.

4. Soil temperature. Soil gets heat energy from different sources such as solar radiation, decomposing organic matter, and heat formed in the interior of earth. The temperature of soil is affected by its colour, texture, water content, slope, altitude of the land and also by climate and vegetational cover of the soil. Evaporation of water from soil makes it cooler. Black soils absorb more heat than white soils. Sandy soils absorb more heat and radiate it out quickly at night than clay or loam soils.

The soil temperature greatly affects the physico-chemical and biological processes in the soil. For example, the germination of seeds, normal growth of roots and biological activity of soil-inhabiting micro-and macro-organisms which require proper and specific temperature.

5. Soil water. In soil, water is not only important as a solvent and transporting agent but in various ways it maintains soil texture, arrangement and compactness of soil particles and makes soil habitat livable for plants and animals. It comes in soil mainly through infiltration of precipitated water (dew, rain, sleet, snow, and hail) and irrigation. According to improved classification method of **Bouyoucos** (1920), soil water is classified into following types— gravitational and ground water, capillary water, hygroscopic water, combined water and water vapour.

(i) Gravitational water. In a well-saturated soil, the accessory (extra) amount of water displaces air from the pore spaces between soil particles and percolates downwardly under gravitational influence and finally it is accumulated in the pore spaces. This accumulated excess water of large soil spaces is called **gravitational water**. When this gravitational water further percolates down and reaches to the level of parent rock, it is called **ground water**. Both kinds of these soil water are ecologically important in the leaching of nutrients.

(ii) Capillary water. The water which is held by capillary forces (*i.e.*, surface tension and attraction forces of water molecules) in smaller soil channels, when the gravitational water and ground water have been drained, is called **capillary water**. Capillary water occurs as a thin-film around soil particles in the capillary spaces and represents the normal available water to the plants. It remains in soil for long periods and carries with it nutrients in solution. Humus has more capillary water than soil minerals.

(iii) Hygroscopic water. Soil particles retain some water so firmly that the plants cannot absorb this. Such soil water is called hygroscopic water.

(iv) Combined water. Combined water is the water of chemical compounds held by chemical forces of molecules such as hydroxides of silicon, iron and aluminium. It is of no ecological significance.

(v) Water vapour. Some soil water occurs as moisture or water vapours in the soil atmosphere.

Further, the total amount of water present in the soil is called **holard**. The quantity of water that plant-roots can absorb out of holard is called **chresard** and that amount of soil water which cannot be absorbed by plant-roots is called **echard**. Moreover, there are some terms that reflect the water status of soil and are generally used for comparative studies of different soils.

(a) Soil water potential. This is an expression of the total reduction of water potential in the soil, due to mineral matrix, solubility, external pressure and gravitation effects.

(b) Field capacity. When a soil holds all the water it can, but no gravitational water, it is said to be at its field capacity. Field capacity is generally defined as the water content of an undisturbed soil (% oven-dry weight) after it is saturated by rainfall and drainage of gravitational water has completely stopped. Field capacity of soil may be taken as the total amount of capillary, hygroscopic and combined water plus water vapour.

(c) Moisture equivalent. It is defined as the water content (% oven dry weight) retained by the undisturbed soil.

(d) Water holding capacity (or storage capacity). This is the extent to which a soil can hold capillary water against gravity. It is equal to field capacity less hygroscopic water.

(e) Hygroscopic-coefficient. It is defined as the water vapour (%) absorbed by unit weight of dry soil, when placed in an atmosphere completely saturated with water vapours.

(f) Permanent wilting percentage (or wilting coefficient). It is the amount of water (% oven dry weight) that remains in a soil when permanent wilting is present in the plants growing in soil.

The status of soil water is also represented in terms of **capillary potential** (= metric potential). The relationship that exists between soil water content and water potential is usually determined by a pressure membrane apparatus, called **tensiometer**.

6. Soil atmosphere. Gases found in pore spaces of soil profiles form the **soil atmosphere**. The soil atmosphere contains three main gases namely O_2 , CO_2 , and N_2 . Soil air differs from atmospheric air in having more of moisture and CO_2 and less of O_2 . The soil atmosphere is affected by temperature, atmospheric pressure, wind, rainfall, etc. Loam soils with humus contain a normal proportion of air and water (about 34% air and 66% water) and therefore are good for majority of crops.

Course-textured or well-structured soils contain higher gaseous diffusive transfer rates than fine-textured or poorly-structured soils under **wet conditions**. This is because they contain many large pores which remain gas-filled. The water-filled pores of the fine-textured soils form potent barriers of gaseous diffusion; O_2 diffuses ten thousand times more slowly in water than in gas. In **dry soil**, the situation is reversed, as the fine textured soils have a greater total pore space and provide a larger gas-filled cross sectional area for diffusion. Soil aeration is very important in growth of roots, seed germination and microbial activity. Poor soil aeration suppresses root hair development, and may reduce rate of absorption of water and nutrients.

7. Soil solution. There exists a weak solution of various salts, alongwith other liquids and gases in the soil mass. This soil solution contains almost all the essential minerals. Complex mixture of minerals such as carbonates, sulphates, nitrates, chlorides, and organic salts of Ca, Mg, Na, K, etc., are found as dissolved in water. The chemical nature of soil solution depends on the nature of parent matter, chemical nature of organic matter and climatic factors and other factors involved in pedogenesis. The soil solution is the primary source of inorganic nutrients for plant roots. Soils with optimal concentration of various nutrient solutes are called **eutrophic**, whereas those with suboptimal concentration of these nutrient salts are called **oligotrophic**.

Chemical Properties of Soil

Soil is a mixture of various inorganic and organic chemical compounds and exhibits certain significant chemical properties, all of which can be discussed as follows :

(a) Inorganic elements and compounds of soil. The chief inorganic constituents of soils are the compounds of following elements—Al, Si, Ca, Mg, Fe, K and Na. Soil also contains smaller amounts of compounds of following inorganic elements—B, Mn, Cu, Zn, Mo, Co, I, F, etc.

(b) Organic matter of soil. The chief organic component of soil is **humus** which chemically contains amino acids, proteins, purines, pyrimidines, aromatic compounds, hexose sugars, sugar alcohols, methyl sugars, fats, oils, waxes, resins, tannins, lignin and some pigments. Further humus is black coloured, odourless, homogeneous complex substance.

(c) Colloidal properties. As soil is composed of crystalloids and colloids, therefore, it exhibits all the physico-chemical properties which are related with these two soil particles. Colloids for example exhibit absorption, electrical properties, coagulation, Tyndal phenomenon, Brownian movement, dialysis, etc.

(d) Soil pH. Many chemical properties of soils centre around soil reaction. As regards their nature, some soils are **neutral**, some are **acidic** and some **basic**. The acidity, alkalinity and neutrality of soils are described in terms of **hydrogen-ion concentrations** or **pH** values. A pH value of 7.0 indicates neutrality, a value above this figure (7.1–14.00) indicates alkaline condition and a value below (0–6.9) indicates acid conditions. Normally, the pH value of soils lies between 2.2 and 9.6. In

India, acidic soils (pH below 5.5 to 5.6) occur in the high rainfall areas of Western ghats, Kerala, Eastern Orissa, West Bengal, Tripura, Manipur and Assam. The saline, alkaline or basic soils (called 'Usar', contain pH upto 8.5) of India, occur in U.P., West Bengal, Punjab, Bihar, Orissa, Maharashtra, Madras, M.P., A.P., Gujarat, Delhi and Rajasthan.

Some pH values are ecologically significant for the plants (Pearsall, 1952); plants regarded as **calcicoles** usually occur in soil with pH 6.5, whereas **calcifuges** occur in soil with pH value below 3.8–4.0. Soils above pH 6.5 are generally cation-saturated (those containing free CaCO_3 called **calcareous soils**) while soil below pH 3.8–4.0 contain a considerable content of exchangeable hydrogen. These limits of pH value are also reflected in the nature of the soil's organic matter: raw humus or mor which is associated with soils below of pH 3.8, while mull is characteristic of the more cation saturated soils of pH 4.8–5.0 and above.

Highly acidic and highly saline or alkaline soils often remain injurious for plant growth, micro-organisms, etc. Soil pH strongly affects the microbial activities, as at below pH 5.0 bacterial as well as fungal activities are reduced. Neutral or slightly acidic soil, however, remains best for the growth of majority of plants.

Soil as Habitats for Animals

Soil is used by various animals as a hideout from enemies, to escape desiccation and extremes of temperature; as material for abodes, and as a highway. Clay is cold, dense, poorly aerated and though it holds much water, does not give up water readily to plants and animals. Sand is dry, loose (cave in readily) and has a variable temperature. Rock heats and cools quickly, has little available water and practically no air, and is usually too resistant to allow penetration by burrowing animals. Alkaline soils are "**physiologically dry**" because the salts present hold the water so that it is not readily available to organisms. Humus has a great water capacity and gives up water readily, is usually well-aerated and furnishes nitrogen as well as mineral salts and water. It remains rich in biota. Peaty soils are raw humus and are uninhabitable for most animals because they are poorly aerated and have an acid reaction. In general, the character of the vegetation cover serves as a good index of the habitability of soils for animals; conditions that make plants flourish are usually favourable for animals.

Soil Fauna and Soil Flora

Soil supports a wide array of organisms of different body-sizes and taxonomic groups. Generally, soil organisms are classified into three major groups namely **microfauna** and **microflora**, **mesofauna**, and **macrofauna**. Mesoflora and macroflora because occur above the surface of soil (land-surface), therefore, are excluded from this discussion :

1. Microfauna. It includes animals with body size within the range of $20\mu\text{m}$ to $200\mu\text{m}$. It includes all Protozoa and small-sized mites, nematodes, rotifers, tardigrades and copepode Crustacea. Soil inhabiting **protozoans** such as amoeba, ciliates, zoomastigine flagellates occur near the surface soils, while the testate forms such as *Thecamoeba*, *Euglypha* and *Diffugia*, have a wider vertical distribution. The common terrestrial **polyclad** is *Bipalium*. The **nematodes** such as *Rhabditis*, *Diplogaster*, *Tylenchus*, *Heterodera*, *Aphelenchoides*, *Mononchus*, *Pratylenchus*, *Xiphinema* and *Criconemoides*, abound by as much as 1–3 million in raw humus soils to 20 million/ m^2 in grassland soils.

2. Microflora. The microflora of soil includes bacteria, soil fungi, soil actinomycetes, blue green algae and algae. In soil, microflora bacteria form about 90 per cent of the total population. Fungi and algae together represent one per cent and actinomycetes cover only 9 per cent.

Soil bacteria grow fairly well in the neutral soils richly supplied with organic nutrients. Soil inhabitant bacteria fall into categories namely—**autotrophic bacteria** and **heterotrophic bacteria**. The autotrophic bacteria derive their energy from the oxidation of simple carbon compounds or from

inorganic substances and their carbon from the atmospheric CO_2 . The common autotrophic bacteria of soil are nitrifying bacteria, hydrogen bacteria, sulphur bacteria, iron bacteria, manganese bacteria, carbon monoxide bacteria and methane bacteria. Most of soil bacteria are heterotrophic bacteria depending upon the organic matter of soil for their energy source and are primarily concerned with the decomposition of cellulose, and other carbohydrates, proteins, fats and waxes. They bring about mineralization of organic matter of soil and release



Root nodules of leguminous plants have *Rhizobium* which are nitrogen - fixing bacteria.

considerable amount of nitrogen, phosphorus and other nutrients for plants. The common nitrogen-fixing bacteria of soil are *Rhizobium* (occurs in root nodules of leguminous plants); *Azotobacter* and *Clostridium pasteurianum* (the latter two are free occurring in soil).

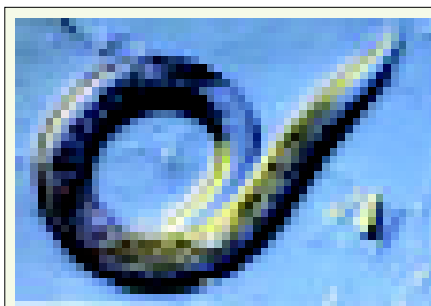
Majority of **soil fungi** are found in acidic soils. They may be parasitic, saprophytic and symbiotic. **Parasitic fungi** of soil infect roots of plants and cause plant diseases such as cotton root rot and many kinds of wilts, rusts, blights and smuts. Certain wilt-forming fungi produce toxins which are harmful, for example, *Fusarium lini*, which

causes wilt of flax (*Alsi*) and secretes HCN and *Fusarium udum*, a fungus causing wilt of pigeon pea (*Arhar*) secretes fusaric acid in the roots of host plants. However, certain parasitic fungi produce growth stimulating substances for host plant. *Fusarium* sp., for example, have been found to secrete gibberellin and gibberellic acid ($\text{C}_{19}\text{H}_{22}\text{O}_6$). **Symbiotic fungi** of soil live on the roots of certain plants and both fungus and plants are benefited. **Saprophytic fungi** depend on dead organic matter of soil and derive the energy from decomposition of the latter. They break down cellulose, lignin, and gum, sugars, starch, protein, etc., into simple gradients to be utilized by higher plants as nutrients.

Actinomycete fungi prefer saline soils and bring about the decomposition of organic matter such as cellulose. They produce a variety of antibiotics of great economic significance for man. The most important blue green algae of soil are those which fix nitrogen in soil. *Anabaena*, *Nostoc*, *Microcystis* are important nitrogen fixing blue green algae of soil. They also make soil aggregates because of having mucilage.

3. Mesofauna. Mesofauna include animals with body size within the range $200\mu\text{m}$ to 1 cm. The micro-arthropods Acari (mites) and Collembola (spring tails) are important members of this group which also include the larger nematodes, rotifers, and tardigrades, together with most of the isopods, Arachnida (spiders), Chelognathi (pseudoscorpions), Opiliones (harvestmen), Enchytraeidae (pot-worms), insect larvae and small millipedes (Diplopoda), isopods and molluscs.

Among **annelids** the microscopic enchytraeids are represented by *Enchytraeus fridericia* and *Achaeta lumbricellus*, which are more abundant in organic soils and forests than in grassland. *Oniscus*, *Porcellio*, and *Armadillidium* are the most common isopods (**crustacean**) of the tropics in the humid zone. Among the soil **arachnids**, mites are the most predominant. Mites flourish in moist organic soils and certain mites such as *Galumna*, *Cepheus*,



Nematode such as *Rhabditis* are found in raw humus soils.



Oniscus.

Hemorobates occur in lichens and mosses. Certain mites, such as *Schelorbates* and *Brachychthornus* live in humus. The mites are saprophagous, predatory and phytophagous and are involved in the process of organic decomposition and its resultant processes. Certain arachnids such as scorpions, *Thelyphonus*, *Galeodes*; and some spiders are crepuscular, hiding under rocks or in crevices in soil and in loose litter, and has no ecological significance in decomposition like other arachnids. Many **opiliones** or harvestmen occur in forest litter, frequently preying upon soil organisms.

Besides mites, only the pseudoscorpions or **chelognathi** occur in surface soils and most decaying vegetations. Of the xerophil litter inhabitants are *Stenatelmus indicus*, *Dhanus indicus*, *Fealla indica* and the hygrophil inhabitants living in the litter and under stones are—*Comsaditha indica*, *Tyrannochthonius madrasensis*, *Tyrannochthonius chelatus* and *Hygrochelifer indicus* (Murthy, 1964), feeding on Collembola, enchytraeids, etc.

The common millipedes or diplopodes of forest soil which are chief decomposers of soils are *Spirostreptus*, *Thyropygus*, *Glomeris*, *Arthrosphaera*, *Polydesmus*, *Iulis*, etc. **Tardigrades** or bear animalcules occur in surface layer of most soils in grassland being represented by the *Macrobiotus* and *Hypsibius*.

Among **insects**, apterygote Collembola form numerically the most important groups of soil insects. Other insects such as Dermaptera, Psocoptera, Dictyoptera, Isoptera, Coleoptera, a few Hymenoptera and some Diptera also occur in soil, sometimes as juveniles. The termites such as *Reticulotermes* and *Odontotermes* are important soil-dwellers of tropics and play an important role in the break up of organic materials and their mixing up with mineral soils. Among the Hymenoptera, ants are the most important soil dwelling forms. Among Collembola, Onchuridae, Isotomidae, Poduridae and Entomobryidae are richly represented in the soil both in number and species composition. Large-sized Collumbola such as *Tomo-cerus*, *Entomobrya* and *Orchesella* occur in surface layer, while the smaller *Onychurus*, *Tullbergia*, etc., occur in deeper layers of soils.



A dung beetle rolling away a ball of dung. This will provide the beetle and its young with food.

Among Diplura, *Anajapyx*, *Japyx* and *Campodea* are often found in small numbers in moist soils under stones and in humus. Proturans are more abundant than Diplura and very common in moist forest and grassland soils which abound in species of *Eosentomon*, *Acerentomon* and *Acerentulus*.

4. Macrofauna of soil. Macrofauna of soil includes those animals whose body size is greater than 1 cm. Here belong the majority of Lumbricidae, the Mollusca, the largest insects and arachnids and the soil-dwelling vertebrates.

Earthworms usually occur in abundance in alkaline and moist soils and sparse in acid soils. They have been proverbial for their influence on the process of decomposition of organic materials, breaking up litter fragments and mixing them thoroughly with mineral soils resulting in the formation of organic soils. Some of the



Earthworm (A) and wood ants (B) help make the soil fertile. They do this by constantly breaking up debris on the forest floor.

common Indian annelidan species of soil are *Megascolex*, *Pheretima*, *Octochaetus*, *Drawida* and *Moniligaster*. Among chilopods the carnivorous *Scolopendra* and *Lithobius* are common in moist soils feeding on leaf litter inhabitants.

Among soil vertebrates, following animals are well adapted for burrowing life in soils—*Ichthyophis*, *Cacopus systema*, *Breviceps* (Amphibia), *Sphenodon*, *Uromastix*, limbless lizards and snakes (Reptilia), *Talpa*, *Dasyurus*, *Notoryctes* and various insectivores and rodents (Mammalia).

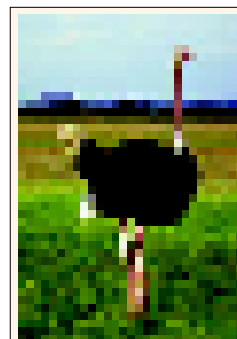
Adaptations of soil animals. Animals which are adapted for digging the burrows and for subterranean mode of life are called **fossorial animals**. These animals may dig either for their food or simply for retreat. Zoologically they are primitive, defenseless and unambitious animals. They have following adaptations : 1. The body contour is either cylindrical (e.g., *Ichthyophis*, limbless lizards, snakes, earthworms, *Scolopendra*, etc.), or spindle-shaped or fusiform (e.g., *Talpa*, *Echidna*, etc.), so



Mole is almost blind. Sensitive whiskers and a good sense of smell are more useful than eyesight underground. Powerful feet and claws are used by mole to dig through soil.

as to offer least resistance to subterranean passage. 2. The head tapers anteriorly to form a sort of snout for burrowing. 3. The tail is short or vestigial. 4. The eyes tend to become vestigial as they are of no use in dark habitat. 5. The external ears also tend to disappear since they would be obstructing in burrowing. 6. For digging, many structures may be found in different fossorial animals, e.g., hands are well adapted for digging. In the insect *Gryllotalpa*, the fore-legs are modified for digging purpose.

The ground-dwelling animals which may be **cursorial** (running), such as ostrich, rhea, ungulates, wolves, cats, bears, hyaenas, etc., **saltatory** (jumping) such as rodents, rabbits, wallabies, kangaroos, etc., or **graviportal** (heavy) such as turtles, armadillos, elephants, hippopotami, etc., exhibit different kinds of adaptations for different kinds of soils. For example, if the soil is firm and hard, the large animals inhabiting the ecosystem tend to have small hooves or paws; if the soil is wet and spongy, they tend to have broad hooves or paws.



Ostrich is a ground dwelling cursorial (running) bird.

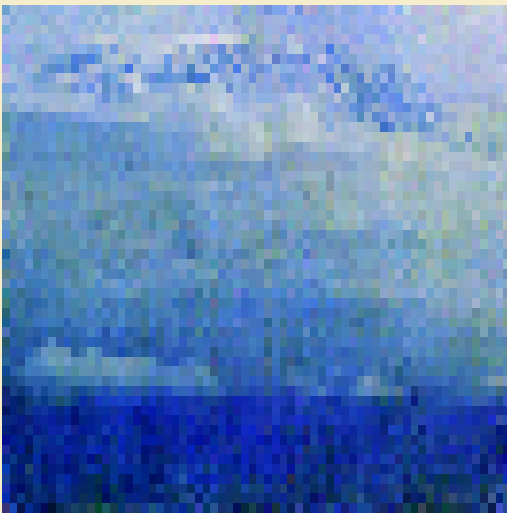
REVISION QUESTIONS

1. Describe physico-chemical nature of the atmosphere.
2. Define the following terms : environment ; atmosphere; ozonosphere; ionosphere; air; wind.
3. Describe physical and chemical properties of water and show, how they affect the organisms.
4. Describe different ecological adaptations of plants and animals in relation to water.
5. What is soil ? Describe its formation, morphology and profile (i.e., soil profile).
6. Discuss the role of edaphic factor in ecological studies.
7. Describe the soil-flora and soil-fauna of Indian soils.
8. Describe the influence of climate, vegetation and parent rock on soil development.
9. Describe the role of living organisms in soil formation and in increasing soil fertility.

CHAPTER

4

Abiotic Environmental Factors



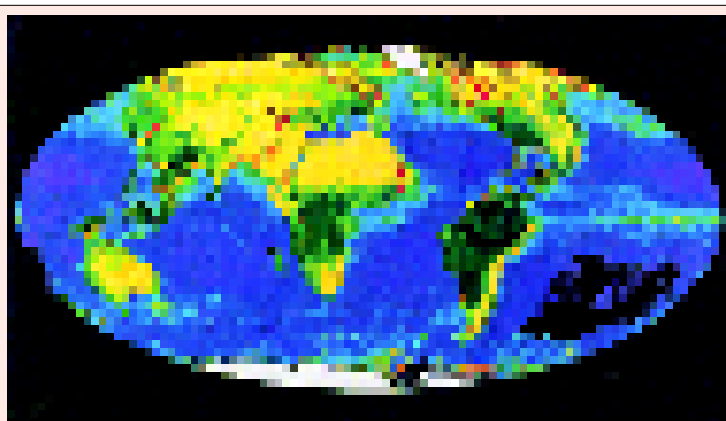
Non-living elements in an environment comprise the abiotic components of our ecosystem.

Environment of an organism has two components — **abiotic** and **biotic**. The first includes the **atmosphere** (air), **hydrosphere** (water) and **lithosphere** (land including soil). The abiotic components are characterized by physical and chemical factors such as light, temperature, rainfall, pressure, pH, the content of oxygen and other gases, and so on. These factors exhibit diurnal, nocturnal, seasonal, and annual changes. The biotic components include all living organisms which interact with each other and with the abiotic components. Earth's living organisms interacting with their physical or abiotic environment (including air, land and water) form a giant and vast ecosystem, called **ecosphere** or **biosphere** which is largest and most nearly self-sufficient biological system.

In physical terms, the biosphere is a relatively thin and incomplete envelope covering most of the world. It represents a mosaic of different biotic communities from simple to complex, aquatic to terrestrial, and tropical to polar. It does not exist in the extremities of the polar regions, the highest mountains, the deepest ocean troughs, the most extreme

deserts, or the most highly polluted areas of land and water. Its total thickness, including all portions of the earth where living organisms can exist, is less than 26 kilometres. Its zone of active biological production, in terms of photo-synthesis, is much narrower, and varies from a few centimetres to over 100 metres. This zone would, for instance, be only a few centimetres in muddy or turbid water, whereas in very clear ocean water, it could be more than 100 metres in thickness. On

land, the zone of biological production might be only a few millimeters in a desert or rock environment, whereas it might again be more than 100 metres in a sequoia or tropical rain forest.



The biosphere. Three years of satellite data were combined to produce this picture of the earth's biological productivity. Rain forests and other highly productive areas appear as dark green, deserts as yellow. The concentration of phytoplankton, a primary indicator of ocean productivity, is represented by a scale that runs from red (highest) to orange, yellow, green, and blue (lowest).

TYPES OF ABIOTIC ENVIRONMENTAL FACTORS

The distribution, abundance, growth and reproduction of the organisms comprising the individual members of populations are controlled by certain **environmental** or **ecological factors**. An environmental factor is any external force, substance or condition which surrounds and affects the life of an organism in any way. Abiotic environmental factors are customarily classified as follows :

1. Climatic factors

- (i) Light ;
- (ii) Temperature ;
- (iii) Water (including atmospheric water, rainfall or precipitation, soil moisture, etc.) ;
- (iv) Atmosphere (gases and wind) ;
- (v) Fire.

2. Topographic or physiographic factors

- (i) Altitude ;
- (ii) Direction of mountain chains and valleys ;
- (iii) Steepness and exposure of slopes.

3. Edaphic factors (soil formation, physical and chemical properties of soil, nutrients).

ESSENTIAL ELEMENTS AND LIMITING FACTORS

The individual organisms of species population, in order to grow and multiply, must be supplied with certain essential materials. Of the hundred four (*i.e.*, 104) naturally occurring chemical elements on the earth, all living organisms are believed to utilize only 16 different chemical elements for their survival. These are called **essential elements**. Several other elements are needed in small quantities by some species. They are summarized in Table 4-1.

Table 4-1.

Essential elements (Source : Clapham, 1973).

Macronutrients : Elements used in relatively large quantities	Micronutrients : Elements generally needed in relatively small quantities	Micronutrients : Elements needed by certain species in relatively small quantities
Carbon	Iron	Sodium
Hydrogen	Manganese	Vanadium
Oxygen	Boron	Cobalt
Nitrogen	Molybdenum	Iodine
Phosphorus	Copper	Selenium
Calcium	Zinc	Silicon
Magnesium	Chlorine	Fluorine
Sulphur		Barium

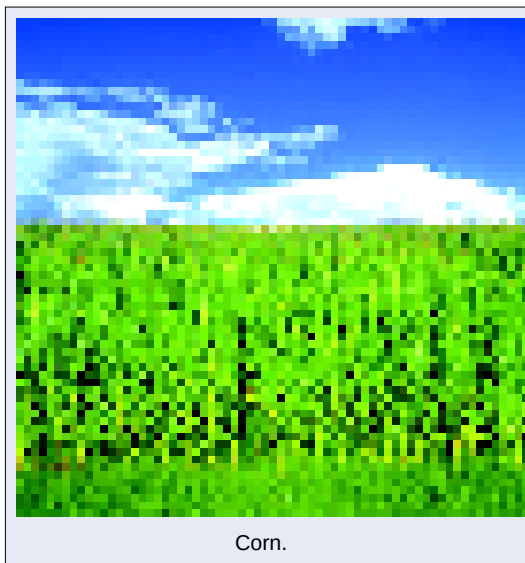
Some of these tabulated elements are used in relatively large quantities as the fundamental building blocks of organic tissues and are called **macronutrients**. Others, the **micronutrients**, are used in much smaller quantities. The micronutrients are also known as **trace elements**. For instance, a crop such as corn will remove over 45 kg. of nitrogen per acre from the soil but only 10–15 g. of boron. In addition to the nutrients required by all species, some organisms have special nutrient requirements.

Liebig-Blackman Law of Limiting Factors

An organism is seldom, if ever, exposed solely to the effect of a single factor in its environment. On the contrary, an organism is subjected to the simultaneous action of all factors in its immediate surroundings. However, some factors exert more influence than do others, and the attempt to evaluate their relative roles had led to the development of the law of the minimum by the German biochemist **Justus Liebig** in 1840.

Liebig while investigating the relationship between the available amounts of essential elements and plant growth, discovered that crop yield was frequently limited by elements other than those utilized in the largest quantity. Freely translated, a part of his statement on his experimental results is that “*growth is dependent on the amount of food stuff that is present in minimum quantity.*” This statement has come to

be known as **Liebig’s law of minimum**. It is now usually incorporated with a **law of limiting factors** developed by a British physiologist **F. F. Blackman** (1905), who at the beginning of this century investigated the factors affecting the rate of photosynthesis. He listed five factors involved in controlling the rate of photosynthesis : amount of CO₂ available, amount of H₂O available, intensity of solar radiation, amount of chlorophyll present, and temperature of the chloroplast. **Blackman** discovered that the rate of photosynthesis is governed by the level of the factor that is operating at a limiting intensity. Further work on limiting factors and substances has shown that a high level of one factor will modify the limiting effect of a second, a process described as **factor interaction** (**Shelford**, 1932). The same principle of limiting factors applies well to animal functions.



Corn.

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The law of minimum has been restated by **Taylor** (1934) in broad ecological terms, as follows : *“The functioning of an organism is controlled or limited by that essential environmental factor or combination of factors present in the least favourable amount. The factors may not be continuously effective but only at some critical period during the year or perhaps only during some critical year in a climatic cycle.”*

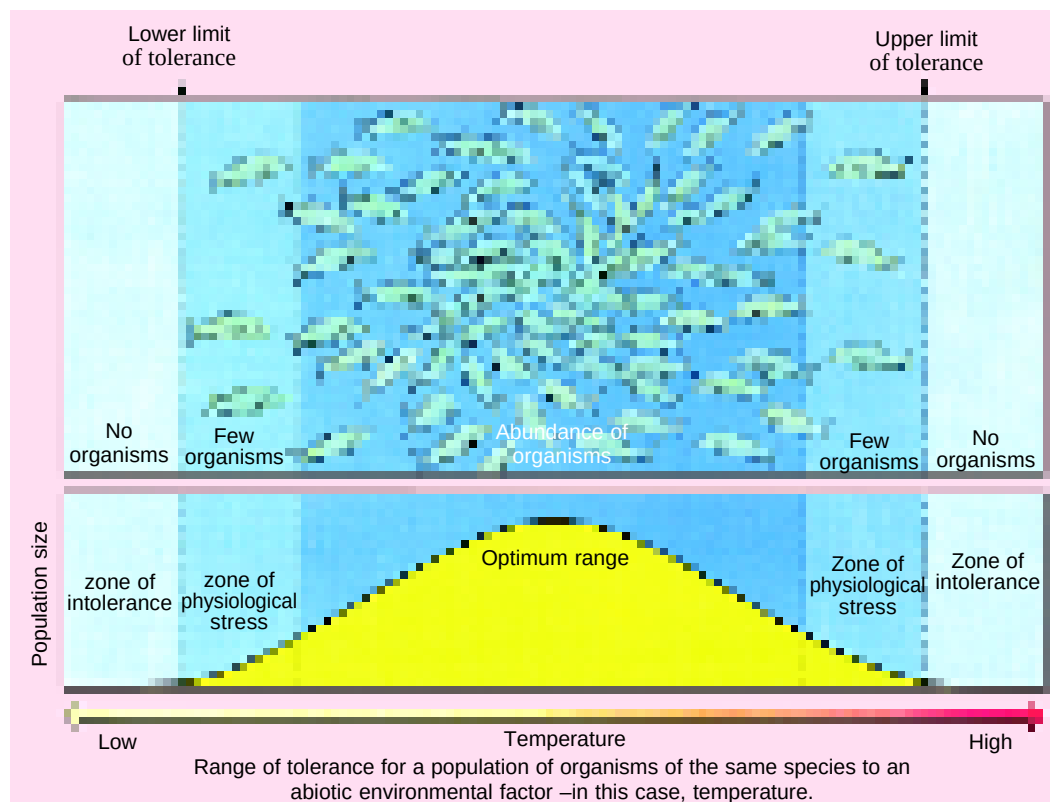
THRESHOLD AND RATE

Every environmental factor varies through a wider range of intensity than any single organism can tolerate characteristically. For each individual organism there is present a lower and an upper limit in the range of an environmental factor between which it functions efficiently. For any one factor, different organisms find optimal conditions for existence at different points along the range, hence their segregation into different habitats.

The **threshold** is the minimum quantity of any factor that produces a precipitable effect on the organism. It may be the lowest temperature at which an animal remains active, the least amount of moisture in the soil that permits growth of a plant, the minimum intensity of light at which photoreceptor is stimulated, and so forth. Above the threshold, the rate of a function increases more or less rapidly as the quantity of heat, moisture, light or other environmental factor is increased, until a maximum rate is attained. Above the maximum, there is usually a decline in the rate of a process either because of deleterious effect produced, the interference of some other factor or exhaustion.

SHELFORD'S LAW OF TOLERANCE

Organisms may be limited in their growth and their occurrence not only by too **little** of an element or too **low** an intensity of a factor but also by too **much** of the element or too **high** intensity of the factor. For example, carbon dioxide is necessary for the growth of all green plants, small increase in



concentration of carbon dioxide in the atmosphere will, under certain circumstances, increase the rate of plant growth, but very considerable increases become toxic. Likewise, small additions of arsenic to the human diet actually have a tonic effect, further increase in the dosage, however, soon proves fatal.

The idea that factors could be limiting at their maximum as well as minimum quantities was incorporated in **law of tolerance** formulated by **V.E. Shelford** in 1913. This law postulates that each ecological factor to which an organism responds has maximum and minimum limiting effect between which lies a range or gradient that is now known as the **limits of tolerance**.

Between the **lower** and **upper** limits of tolerance lies a broad middle sector of a gradient which is called the **zone of compatibility**, the **zone of tolerance**, the **biokinetic zone** or the **zone of capacity adaptation**. The region at either end of the zone of compatibility is called the **lethal zone** or the **zone of resistance** or **zone of intolerance**. The zone of compatibility too includes a broad **range of optimum** and narrow **zones of physiological stresses** in between the range of optimum and lethal zones.

Upper and **lower limits of tolerance** are intensity levels of a factor at which only half of the organisms can survive. These limits are sometimes difficult to determine, as for example with low temperature, organisms may pass into an inactive, dormant, or hibernating state from which they may again become functional when the temperature rises above a threshold. At high temperatures, there may be similar inactivation or aestivation before the lethal level is attained. Even without dormancy occurring, there are normally **zones of physiological stresses** before the limits of tolerance are reached.

The species as a whole is limited in its activities more by conditions that produce physiological discomforts or stresses than it is by the limits of tolerance themselves. Death comes close to the limits of tolerance, and the existence of the species would be seriously risked if it was frequently exposed to these extreme conditions. Therefore, in retreat before conditions of physiological stress there is a margin of safety, and the species adjusts its activities so that limits of tolerance are avoided. There is a variation in hardiness of individuals within a species, so that some hardy individuals find existence possible under conditions that disturb other individuals. The population level of a species becomes reduced before the limits of its range are actually reached.

Further, species vary in their limits of tolerance to the same factor. For example, the Atlantic salmon spends most of its adult life in the sea, but goes annually into freshwater streams to breed. Most other marine fishes are killed quickly when placed in fresh water, as are freshwater fishes when placed in salt water. The following terms (Table 4-2) are used to indicate the relative extent to which organisms can tolerate variation in environmental factors. The prefix *steno*—means that the species, population, or individual has a narrow range of tolerance and the prefix *eury*—indicates that it has a wide range.

Table 4-2. Terminology for ranges of certain abiotic environmental factors.	
Stenothermal–Eurythermal	Pertaining to temperature
Stenohaline–Euryhaline	Pertaining to salinity
Stenohydric–Euryhydric	Pertaining to water
Stenophagic–Euryphagic	Pertaining to food
Stenobathic–Eurybathic	Pertaining to depth
Stenoecious–Euryecious	Pertaining to niche or habitat selection

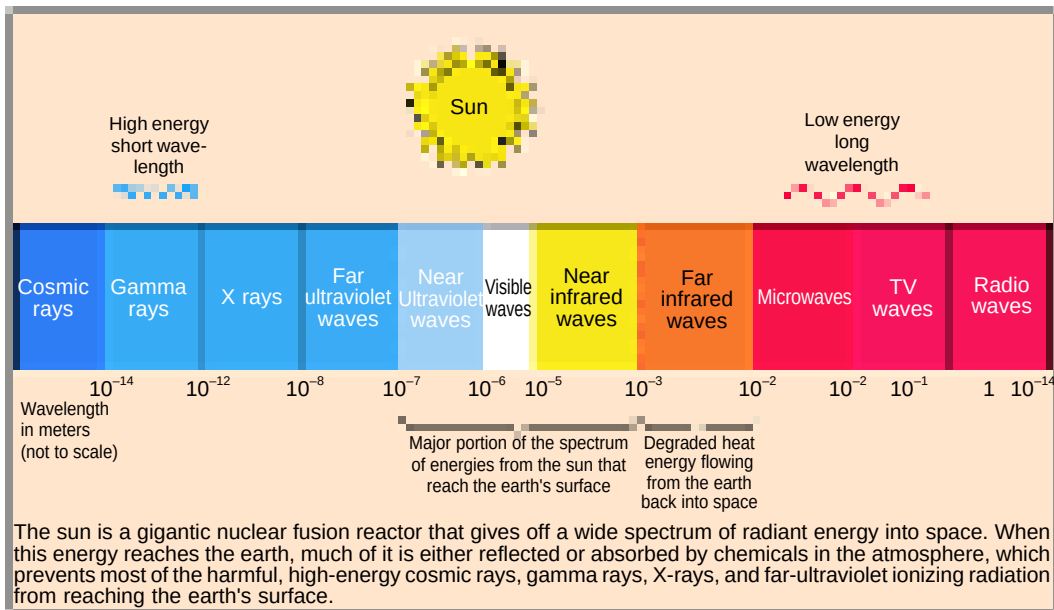
Moreover, a plant or animal may have a wide range of tolerance for one factor in the environment, but a relatively narrow range of tolerance for another condition. Thus, we find that some species of freshwater fishes are eurythermal but they are stenohaline.

Lastly, though climatic, edaphic and biotic factors affect plants/animals, their populations and community growth and dynamics take place in a holistic manner; but, it is difficult to understand the mechanism of environmental influences unless we study the different components of environment separately as follows.

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1. LIGHT AND RADIATIONS

The radiant energy from the sun is the basic requirement for the existence of life on the earth. This source of energy is of fundamental importance to the photosynthetic production of food by plants and as mentioned previously, the heat budget of the world is dependent on solar radiation. Although we generally, think only in terms of visible light, the sun emits other radiations of different wavelengths—cosmic rays, gamma rays, X-rays, ultraviolet rays, infra-red rays, heat waves, spark discharges, radar



waves, radio waves, slow electro-magnetic waves. Biologists have been primarily interested in only three regions near the centre of the electromagnetic spectrum—(1) the infra-red, (2) the visible light and (3) ultraviolet regions. The **infra-red wavelengths**, which are the longest of these three, are not visible to the human eye; they contribute to the warmth of the earth at the high altitudes in the terrestrial atmosphere. **Visible light**, is only a small fraction of the radiation spectrum and contains the frequency of wavelengths ranges from 390 to 700 millimicrons ($m\mu$). It is made up of a series of colours ranging from violet through indigo, blue, green, yellow, orange and red, all constituting the **visible spectrum**. Light energy, thus, reaches the earth as electro-magnetic waves of solar radiation with tremendous velocity and supplies most of the warmth the earth receives from the sun and also supplies the main source of energy which is utilized in photosynthesis of plants, oriented and rhythmic behaviour of ani-

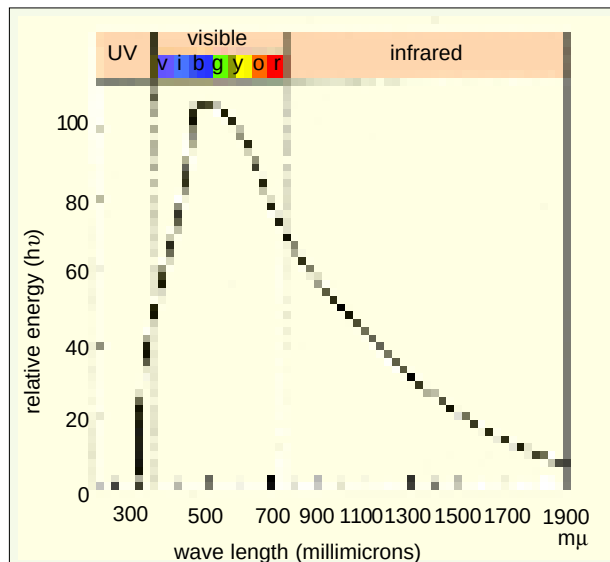


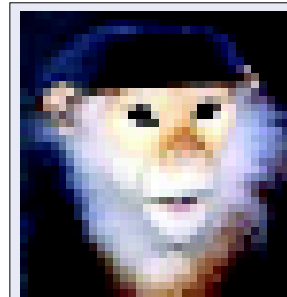
Fig. 4.1. Spectral distribution of ultraviolet light, visible light and infra-red radiation at the earth's surface.

mals, bioluminescence, periodicities of occurrence and periods of inactivity. Unlike temperature, light is a non-lethal ecological factor and it has a specific direction in its flow. The wavelength of **ultraviolet light** from the sun is shorter than that of visible light, and it produces the upper levels of the earth known as the ionosphere.

Light Receptors of Animals

Since animals depend on light for orientation, diurnal migrations and synchrony of rhythmic activities, light reception probably is the most important sensory modality in the exploration of the environment. It is not surprising, therefore, that receptors for light are common to almost all animals : in those with better developed photoreceptors, light greatly influences behaviour. Light receptors may be well-defined organs such as the vertebrate eyes, the compound eyes of Crustacea and insects, the simple eyes or ocelli of other arthropods and invertebrates, and the dermal light receptors.

Many homeotherm animals with normal vision do not commonly respond to ultraviolet or infra-red rays, since their retinas are not stimulated. They also do not properly distinguish between the different colours of the visible spectrum. Dogs, cats, hamsters, and opossums are colour blind, while horned cattle cannot identify red light. Horses, deer, sheeps, pigs and squirrels cannot distinguish close to red and green. Primates can distinguish colours. Some insects such as honey bee can see ultraviolet radiation.

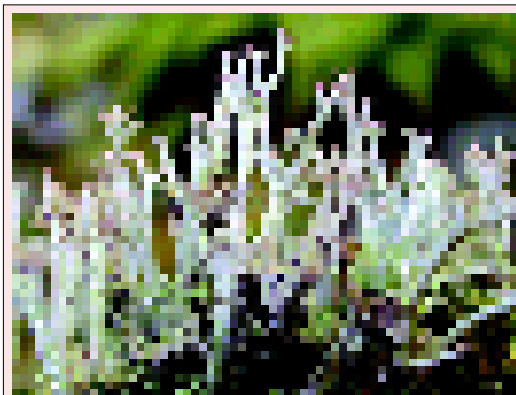


Primates can distinguish colours.

Light Variations in Different Environments

Light energy varies with different media. The transparency of air and water is important in regulating the amount and quantity of light that may be available in particular habitats. For example, the intensity of light reaching the earth's surface varies with the angle of incidence, degree of latitude and altitude, season, time of day, amount absorbed and dispersed by atmosphere and a number of climatic and topographical factors such as fog, clouds, suspended water drops, dust particles, etc., when the angle of incidence is smaller, light rays have to travel by a longer distance through the atmosphere, which results into relative reduction in intensity. Likewise, sun's altitude changes due to differences in latitude, changes in the season and in the time of day. When sun remains overhead, the intensity of sunlight over the earth's surface will be greatest. At higher latitudes the intensity of light becomes correspondingly reduced. The **illumination** or intensity of daylight is greatly diminished by moisture, clouds, and dust in the atmosphere and also by forest vegetation. The direction and slope of the mountain also affect light intensity. There will be no light on the one side of the slope. Illumination is measured in **lux** or **foot candle**. One foot candle is 10.7 lux. One lux is equal to one **lumen** per square meter. A lumen is equal to 1/620 watt. Most organisms respond to moonlight particularly on a full moon, which has an illumination of 0.25 lux in cloudless weather.

The light which enters in the aquatic media, comes from sun by passing through the atmosphere existing above the water surface and, hence, that is subjected to all kinds of atmospheric factors like that of terrestrial media. About 10% of the sunlight which falls over the water surface, is reflected back and rest 90% of that pass downward in the water and is modified in respect



Phyllophora.

to intensity, spectral composition, angular distribution (refraction) and time distribution. The phytoplankton, zooplankton, suspended organic and inorganic particles either reflect or absorb the light rays. Further, in water there is a selective absorption of light at various depths. The longer light rays are absorbed near the surface and in general the shortest light rays penetrate deepest. Thus, long infra-red rays are absorbed in the upper layers of water (about 4 metres); red and orange rays are completely absorbed up to the depth of 20 metres; yellow rays penetrate up to 50 metres and green and blue rays penetrate up to 80 to 100 metres deep. Violet and ultraviolet rays penetrate beyond 100 metres and no light ray penetrates beyond 200 metres depth. Depending upon the penetration of light, oceans are divided into **euphotic zone** (up to 50 metres depth), **disphotic zone** (up to 80 to 200 meters depth) and **aphotic zone** (below 200 meters of depth). In the ocean, algae are distributed according to length of light rays that their colours are best suited to absorb and to utilize. Photosynthesis in deeper waters occurs with blue and green rays, which are absorbed by the brown and red pigments of red algae (*Phyllophora*). The red and blue-green algae use pigments of photocyanin and photoerythrin for photosynthesis. Each plant performs optimum photosynthesis in complementary colours, *e.g.*, green pigment for red rays, brown pigment for green rays, and red pigment for blue rays. Further, since the photosynthesis is a function of the illumination, a variation in the amplitude of which depends on the earth's rotation, altitude, season, presence or absence of clouds and dust and so on.

Effect of Light on the Plants

Light energy influences almost all the aspects of plant life directly or indirectly. Thus, it controls plant's structure, form, shape, physiology, growth, reproduction, development, local distribution, etc. On the basis of light factor certain ecologists have classified plants into **sciophytes**, **shade loving**, or **photophobic plants** which have best growth under lower intensities of light, and **heliophytes** or **photophilous plants** which have best growth in full sunlight. The light factor affects following aspects of plant life :

A. Direct effects of light on plants. Light affects directly the following physiological processes of the plants— 1. It is an essential factor in the formation of **chlorophyll** pigment in chlorophyllous plants. 2. It has a very strong influence on the number and position of **chloroplasts**. The upper part of the leaf which receives full sunshine has larger number of chloroplast which are arranged in line with the direction of light. In leaves of plants which grow under shade, chloroplasts are very few in number and are arranged at right angle to the light rays, thus, increasing the surface of light absorption. 3. Light has its most significant role in **photosynthesis**. During photosynthesis, the green plants which are the “primary producers” of an ecosystem, synthesize their carbohydrate food from water and CO_2 in the presence of sunlight. Thus, during photosynthesis, the solar radiant energy is transformed into the chemical or molecular energy which remains stored in chemical bonds of carbohydrates and this chemical energy is utilized by other chlorophyllous and non-chlorophyllous parts of plants, all animals, bacteria and viruses in their different life activities. The rate of photosynthesis is greater in intermittent light than in the continuous light. At high intensity of light a photo-oxidation of chlorophylls and other enzymes takes place, which consequently reduces the rate of synthesis of carbohydrates and proteins. However, high intensity of light results in the formation of **anthocyanin** pigments. It is for this reason alpine plants have beautifully coloured flowers. 4. Light inhibits the production of **auxins** or **growth hormones** as a result of which it influences the shape and sizes of plants. Plants grown in insufficient light or in the total darkness, produce maximum amount of growth hormones, as a result of which they are elongated with weak pale yellow stems with very few branches. 5. Light also influences certain chemical compounds of plants which affect the differentiation of specialized tissues and organs. 6. Leaf structure too is influenced by the intensity of light. 7. The development of flowers, fruits and seeds is greatly affected by light intensity. Diffused light or reduced light promotes the development of vegetative structures and causes delicacy. For example, vegetative crops such as turnips, carrots, potato and beets give highest yield in regions with high percentage of cloudy days. Intense light favours the

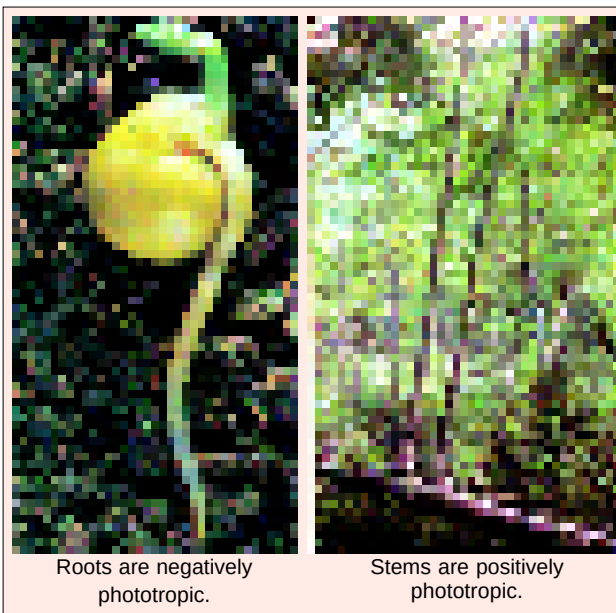
development of flowers, fruits and seeds. 8. Duration of light is also very important. Actual duration or length of the day (**photoperiod**) is a significant factor in the growth and flowering of a wide variety of plants. The controlling effect of **photoperiod** is called **photoperiodicity**. According to the response to length of photoperiods, the plants have been classified into following three groups :

(i) Short day plants. Which bloom when the light duration is less than 12 hours per day e.g., *Nicotiana tabacum* (tobacco), *Dahlia variabilis*, *Chrysanthemum indicum*, *Cosmos bipinnatus*, *Cannabis sativa* (hemp), etc.

(ii) Long day plants. Which bloom when the light duration is more than 12 hours per day, e.g., *Allium cepa* (onion), *Beta vulgaris* (beet root), *Daucus carota* (carrot), *Papaver somniferum* (opium poppy), *Vicia faba* (broad bean), *Brassica rapa* (turnip), *Avena sativa* (oat), *Secale cereale* (rye), *Sorghum vulgare* (sorghum), etc.

(iii) Day neutral plants. Which show little response to length of day light, e.g., *Cucumis sativus* (cucumber), *Gossypium hirsutum* (cotton), *Solanum tuberosum* (potato), etc.

9. Light also affects the movement in some plants. The effect of sunlight on the plant movement is called **heliotropism** or **phototropism**. The stems elongate towards light (positive phototropism) and the roots are negatively phototropic. The leaves grow transversely to the path of light. 10. The seeds when moist are very sensitive to light. In some cases the **germination of seeds** is delayed in light. 11. Light is an important factor in the distribution of plants. Some plants grow in full sunlight, while others prefer to grow in the shades.



B. Indirect effects of light on plants. Light affects opening and closing of stomata, influences the permeability of plasma membrane and has heating effect. All these in turn affect **transpiration** which in turn affects **absorption** of water. Light affects **respiration** of plants indirectly, as in the presence of light the respiratory substrates are synthesized. In many plants the respiratory rate increases with the increase in the light intensity (e.g., *Canna*, *Nerium*, *Bougainvillea*). However, in certain plants respiration rate is decreased slightly in intense light.

Effects of Light on Animals

Light affects divergent aspects of animal's life. It influences cellular metabolism, growth, pigmentation, locomotion, reproduction, ontogenetic development, and also controls the periodicity and biological clocks of animals. Some of its significant effects can be discussed as follows :

1. Effect of light on protoplasm. Though the bodies of most animals remain protected by some sort of body covering which save animal tissues from the lethal effects of solar radiations. But, sometimes sun rays penetrate such covers and cause excitation, activation, ionization and heating of protoplasm of different body cells. Ultraviolet rays are known to cause mutational changes in the DNA of various organisms.

2. Effect of light on metabolism. The metabolic rate of different animals is greatly influenced by light. The increased intensity of light results in an increase in enzyme activity, general metabolic rate and solubility of salts and minerals in the protoplasm. Solubility of gases, however, decreases at high

light intensity. Cave dwelling animals are found to be sluggish in their habits and to contain slow rate of metabolism.

3. Effect of light on pigmentation. Light influences pigmentation in animals. Cave animals lack skin pigments. If they are kept out of darkness for a long time, they regain skin pigmentation. The darkly pigmented skin of human inhabitants of the tropics also indicate the effect of sunlight on skin pigmentation. The skin pigment's synthesis is dependent on the sunlight. Light also determines the characteristic patterns of pigments of different animals which serve the animals in sexual dimorphism and protective colouration. Animals that dwell in the depths of the ocean where the environment is monotonous, though pigmented do not show patterns in their colouration.

4. Effect of light on animal movements. The influence of light on the movement of animals is evident in lower animals. Oriented locomotory movements towards and away from a source of light is called **phototaxis**. Positively phototactic animals such as *Euglena*, *Ranatra*, etc., move towards the source of light, while, negatively phototactic animals such as planarians, earthworms, slugs, copepodes, siphonophores, etc., move away from the source of light.

The light directed growth mechanisms are called **phototropisms** which occur in sessile animals. Phototropisms also include responsive movement of some body part of some active animals to the light stimulus, such as the movement of flagellum of *Euglena* towards light and movements of polyps of many coelenterates.

The velocity or speed of the movement of certain animals is also regulated by light. It has been observed that animals when responding to light reduce their velocity of movement and these movements which are non-directional are called **photokinesis**. Photokinesis may be a change in linear velocity (**rheokinesis**) or in the direction of turning (**klinokinesis**). During photokinesis when only a part of the body of an animal deviates away from the source of light, the reaction is termed **photoklinokinesis**. Larvae of *Musca domestica* show such movements. When animals are confronted with two lights of equal brightness they move towards or away to a position that is distanced between the two lights. This is termed **phototropotaxis**. Attraction of males towards the flash of the female is called **telotaxis**. Movement of animals at a constant angle towards the source of light is called **light compass reaction** or **celestial orientation**.

Celestial orientation. Some organisms, particularly arthropods, birds and fish, utilize their time sense as an aid to find their way from one area to another. To orient themselves, the animals use the sun, moon, or stars as a compass. To do this, they utilize both their biological clock and observations on the azimuthal position of the sun in relation to an established direction. The **azimuth** is the angle between a fixed line on the earth's surface and a projection of the sun's direction on the surface. Using the sun as a reference point involves some problems for animals because the sun moves. The target angle changes throughout the day. But animals which use the sun as a reference, correct their orientation somehow. Such celestial orientation has been observed in fishes, turtles, lizards, most birds, and such invertebrates as ants, bees, wolf spiders and sand hoppers.

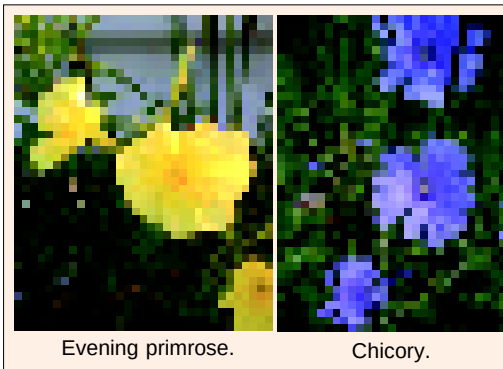
5. Photoperiodism and biological clocks (Biorhythms). During evolution, organisms have acquired a variety of endogenous rhythms, their periods are matched with the rhythmic events in the environment. A **rhythm** is a recurring process which is wave like in character, because maximum and minimum states appear at identical intervals of time. The time taken between two maxima (peaks) or two minima (troughs) is called a **period** or **cycle** and consists of two phases, a rise and fall in the biological process. The **amplitude** is the range of fluctuations from an average value. The response of different organisms to environmental rhythms of light and darkness is termed **photoperiodism**. Each daily cycle inclusive of a period of illumination followed by a period of darkness is called the **photoperiod**. The term **photophase** and **scatophase** are sometimes used to denote the period of light and the period of darkness respectively.

Some rhythms of organisms are matched to **24 hours** cycle of light and dark (**circadian**), other to the **12.4** or **24.8 hours** tide cycle (**circatidal**), the **29 days** lunar cycle (**circalunar** or **circasynodic**),

the **yearly seasons (circanular)** or the time between successive spring low waters (**14.7 days; semilunar** or **circasyzygic**). Prefixing word *circa* (which means about) with most types of biological rhythms is very necessary because all these internal clocks are only approximately matched, *e.g.*, 24 hours cycle or rhythm does not mean 24 hours by wrist watch, it means about 24 hour or circadian (L. *circa* = about + *di(em)* = day). This term was coined by an American ethologist **Franz Halberg** in 1959.

Now it is believed that all animals (and also plants) possess an internal (endogenous) and automatic **clock** that controls the rhythm of behaviour and keeps it going with the help of internal stimuli. This clock is entrained (*i.e.*, set and reset) by external environmental stimuli, called **zeitgebers** or **entrainers** such as day length, lunar phases, tides, temperature, humidity, etc. The biological clocks show following general characteristics: 1. They keep almost the same timings at high and low temperatures, *i.e.*, they are unaffected by metabolic inhibitors which are known to block biochemical reactions. 3. They are controlled by the physiological processes of body. 4. They keep normal cycle even in the absence of environmental cues and are self sustained in nature. 5. They function through nerves (brain) and hormones.

Types of biorhythms. (1) Circadian rhythms. Operating on an approximately **24-hour day-night cycle** of the earth's rotation, they are the most studied and well understood biological clocks or biorhythms. They are found in almost all the major taxonomic groups of animal kingdom. In their simplest form, circadian rhythms are reflected in the alternating periods of **activity** and **sleep** which correlate with the **light/dark cycle**. In this cycle some animals remain most active at sunrise



Evening primrose.

Chicory.



Poppy.

and sunset times, such animals are known as **crepuscular**, some animals are active during the night (**nocturnal**) but most animals are active during the day time, they are called **diurnal**. In the diurnal rhythm, the animal activity stage includes small cycles of flitting, flying, running and rest. Among plants showing diurnal rhythm, the flowers of some plants open up before the break of dawn (*e.g.*, at 4 to 5 a.m. in Dogrose, Chicory and Poppy) and in some others, with the onset of dusk (*e.g.*, evening primrose, *Matronalis*, etc.). In fact, when a flower closes up, it does so to protect its inner organs from the cold of the night and against extra moisture. In the day, flowers open up for pollination under most favourable conditions. Such periodic change in the position of the organs (*viz*, petals and leaves) is called **sleep**. Such sort of movements of plant organs are called **nyctinastic movements**. Circadian rhythms also have been observed in metabolic rates, cell division, growth, heart beat, rate of photosynthesis, cellular enzyme activity and a host of other activities.

2. Circatidal rhythms. The biological rhythms synchronized with the **low** and **high tides** (*i.e.*, the alternate rise and fall of the sea due to gravitational pull exerted by moon) in the sea are called **circatidal rhythms**. Thus, the organisms living in the intertidal zone of the sea shore are alternately submerged in water and exposed to air and in doing this, these animals become exposed to various ecofactors such as pressure, salinity, food supply, temperature and risk of predation. Animals inhabiting tidal areas show behavioural periodicity associated with the tides, for example, bivalve molluscs (*e.g.*, pearl oysters) increase filtration rates when high tide brings in more food and sea anemones increase the rate of body expansion and contraction during high tide, waving its tentacles actively to trap more food.

3. Circalunar rhythms. Biological rhythms which are synchronized with the phases of moon are called **circalunar rhythms**. For example, a marine polychaete *Platynereis dumerii* is a long worm-like non sexual creature which lives in a burrow. During its breeding season, this seaworm becomes brilliantly colourful, gets bigger in size, its parapodia, eyes and tentacles also get enlarged and it becomes a sexual form called **heteronereis**. Many *Heteronereis* swarm actively at sea shores during the **full moon**, perform nuptial dance and spawn.

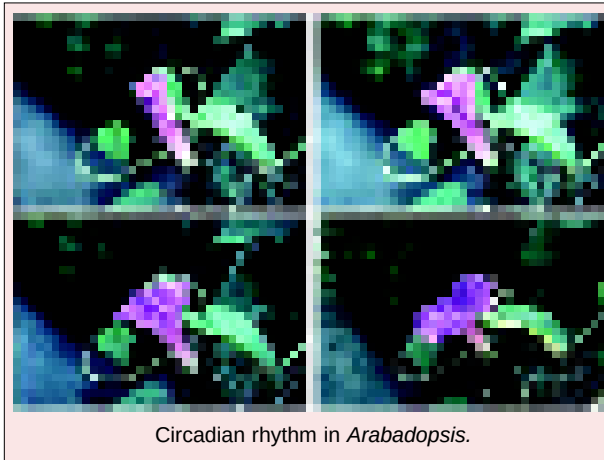
4. Semilunar rhythms. The biological rhythms which are synchronized with the fortnightly cycle of **spring tide** (i.e., high tide occurring a day or two after the new or full moon) and **neap tide** (i.e., low tide which occurs in the middle of the second and fourth quarters of the moon) are called **semilunar rhythms**. For example, the gastropod mollusc periwinkle, *Littorina rudis* shows a marked 14.5 day periodicity in its locomotory activity; the species lives high up on the shore and is only covered by the high water of spring tides, when it comes out of its burrow to move around and to feed.

5. Circannual rhythms. The activity of some animals and plants is influenced by the **seasons** occurring once in a year. They show circannual rhythms. For example, for most birds the height of the breeding season is the spring; for the deer the mating season is the fall. Likewise, brook trout spawns in the fall; bass and blue gills in late spring and summer. Many animals have a part of their life in the resting stage when they exhibit reduced metabolic activity. They either hibernate, aestivate or migrate from the place to avoid rigours of temperature. Migratory birds undertake long distance migrations every year and show cyclic changes in their body weight, gonad size, plumage and colouration. The larvae of many insects (e.g., parasitic wasp, flesh fly, etc.) show two phases every year (i) **active** (when the insect does work actively) and (ii) **inactive** or **diapause** (a period of arrested growth and development in insects which is under the control of endocrine system. This is an adaptation to avoid adverse conditions).

Mode of function of biological clocks. It is believed that in the course of evolution, DNA evolved the biological clock regulating growth, cell division, diurnal rhythmicity of metabolic processes and finally the activity pattern of organisms. Biological clocks show ecophysiological integrity of the DNA-cell/organism; the evolutionary mechanism has brought stability to these clocks through the heredity.

It is hypothesized that the intracellular (=endogenous) clocks are located in the nucleus of the cell and can be compared with a **spring** which is tightened in one phase and gradually released in another phase without ever reaching a zero level. Oscillatory motion occurs between the winding (tension) and unwinding (relaxed) phases. The winding of the spring requires energy, which is received in plants from reaction initiated by light and in mammals from oxidation and reduction reactions. Many biochemical studies suggest that in the **spring model**, the winding phase corresponds to the synthesis of RNA and enzymes and unwinding phase to the breakdown of proteins in the cytoplasm. Such a hypothesis may work well in case of unicellular organisms, but in the case of multicellular and complex organisms various processes occur simultaneously in different tissues and organ systems. It appears that in them a central mechanism may operate through the genes–hormones–organ systems.

Recent findings suggest that in invertebrates both **neural** and **neuroendocrine products** (hormones) are responsible for biological rhythms. In higher vertebrates, the central regulation of rhythms is largely performed by the **hypothalamus-pituitary** complex which is genetically controlled



and influenced by environmental factors such as photoperiod. For example, in birds and mammals, the **pineal gland** receives light stimuli and entrains to show biological rhythms. In amphibians, reptiles and birds the pineal gland is located under the skull but in some terrestrial forms of vertebrates, it is placed over the brain in the form of the third eye. The information of photoperiod is received first by the eye and then via neural pathways reaches the pineal body which secretes hormones such as **melatonin** (Reiter, 1980) which has antigonadotropic effects in various organisms such as rats, mice and many other mammals. Melatonin is believed to pass to the anterior **pituitary gland** and decrease secretion of gonadotropic hormones (see Guyton and Hall, 1996). Pineal body partly controls circannual rhythms of reproduction. In case of turtles, **serotonin** is synthesized during the day whereas melatonin during the night but this mechanism is completely lost during hibernation period.

6. Effect of light on reproduction. In many animals (e.g., birds) light is necessary for the activation of gonads and in initiating annual breeding activities. The gonads of birds are found to become active with increased illumination during summer and to regress during shorter periods of illumination in winter.

7. Effect of light on development. Light in some cases (e.g., *Salmon* larvae) accelerates development, whereas, in other (e.g., *Mytilus* larvae) it retards it.

Further, occasionally the output of sunlight is increased by the development of **sunspots**. As a result of this excess energy is radiated to space and this naturally increases the output of solar energy near the earth. A direct consequence of this is the greater evaporation of water which results in cloud formation to prevent more exposure to sunshine and thus to equalize temperature and modify climate.

2. TEMPERATURE

Temperature is one of the essential and changeable environmental factor. It penetrates into every region of the biosphere and profoundly influences all forms of life by exerting its action through increasing or decreasing some of the vital activities of organism, such as behaviour, metabolism, reproduction, ontogenetic development (viz., embryogenesis and blastogenesis) and death. Temperature is a universal influence and is frequently a limiting factor for the growth or distribution of animals and plants. Normal life activities go on smoothly at a specific temperature or at a specific range of temperatures. This is called the **optimum temperature** or the optimum range of temperature. Organisms react to any rise or fall of the optimum temperature range and biotic communities more often encounter alterations only due to extremes of temperature (viz., **minimum** or **maximum** temperatures). The interaction of temperature with certain other abiotic environmental factors such as humidity, etc., cause into many other climatic changes which influence the living organisms in one way or another.

Nature of Temperature

Temperature is a measure of the intensity of **heat** in terms of a standardized unit, and is commonly expressed as degrees on either the **Fahrenheit** or **Celsius scale**. Heat is a form of energy, and as such is necessary for the very existence of life. Heat may be received or lost by animals as molecular vibrations transmitted from one part of an environment to another by radiation, convection, or conduction. It may be received as radiant energy, ofcourse largely from the sun. Heat may be produced by mechanical work as in Hawk moth, the heat is generated by vibrating the wings before take-off ; or by chemical reactions.

Heat Budget

The temperature at the earth's surface is governed by the brightness of the sun; the constancy of brightness of sun has remained virtually unchanged for about 3 billion years. In fact, average terrestrial temperatures now probably do not differ radically from those at the earth's beginning (Schwarzschild, 1967). The total amount of heat entering the biosphere from the sun must balance the amount lost per unit time if temperatures are to remain unchanged, since the flow of geothermal heat from the interior of the earth is small by comparison and probably has been negligible for at least the past 500 million

years. The estimate of this energy flow is referred to as the **heat budget** (Vernberg and Vernberg, 1970). Heat budget has been projected for the total surface of the earth as well as for special environments.

Temperature Fluctuations in Different Environments

Environmental temperatures fluctuate both daily and seasonally. Different environments such as freshwater, marine and terrestrial environments, are subjected to varied responses to fluctuating temperature. Temperature fluctuations are comparatively less in the aquatic environment than in the terrestrial environment. The increase in depth of aquatic medium often increases the temperature fluctuations. There exists a distinct difference in the response of living organisms of the freshwater and sea to temperature fluctuations because of the presence of dissolved salts in sea water. The minimum temperature in the sea is -3°C , while in freshwater pond it never goes below 0°C . The maximum temperature of ocean generally goes up to 36°C , but in shallow pools of freshwater and tide of pools littoral zone, temperature may go higher. In deeper bodies of water, heating and cooling are restricted to the surface strata. But the deeper layers also get a lot of heat as a result of what is usually termed **vertical circulation**, where in, due to circulatory movement of water, surface waters are brought to the deeper regions, and vice versa. Studies on the vertical changes of temperature have led to the hypothetical classification of the freshwater media into three strata. The superficial layer of freshwater is constantly stirred by wind and is called **epilimnion**. It is the layer of warmer water and its temperature may rise up to 22°C during summers. The stagnant water of the bottom constitutes the **hypolimnion**. The hypolimnion has temperature of 5°C to 9°C . In between epilimnion and hypolimnion occurs an intermediate zone called **thermocline** or **metalimnion**, which has rapid vertical temperature changes (Fig. 4.2 B). During winter, the temperature of epilimnion of a freshwater lake becomes 0°C and the lake becomes ice covered. The process of differentiation of freshwater habitat into these three strata is called **thermal stratification**.

In the terrestrial environment the seasonal and daily fluctuations in temperature are varied and marked. The lowest temperature recorded for any land mass is -70°C (Siberia in 1947). Higher temperatures may likewise go often 85°C as in certain deserts at noon. However, the water in hot springs and geysers may approach 100°C , and even higher temperatures occur occasionally in the very special situation presented by volcanic areas. In Rajasthan the highest temperature exceeds 50°C . On land, diurnal variations of temperature are quite staggering. The fluctuation between day and night temperatures may be 17°C as in ordinary land masses or 40°C as in deserts. The Thar Desert of South Rajasthan (India) shows a diurnal range of $20\text{--}30^{\circ}\text{C}$ for all seasons. Difference in latitudes often cause variation in the annual temperature cycles with visible effects on the organisms. Further, with the increased altitude for every 150 meters, the decrease in temperature of 1°C takes place.

Range of Temperature Tolerance

Life in this universe exists within a range of -200 to 100°C . Though normal life persists within narrow temperature limits of about -10 to $+50^{\circ}\text{C}$. Individual species survive in a smaller range and

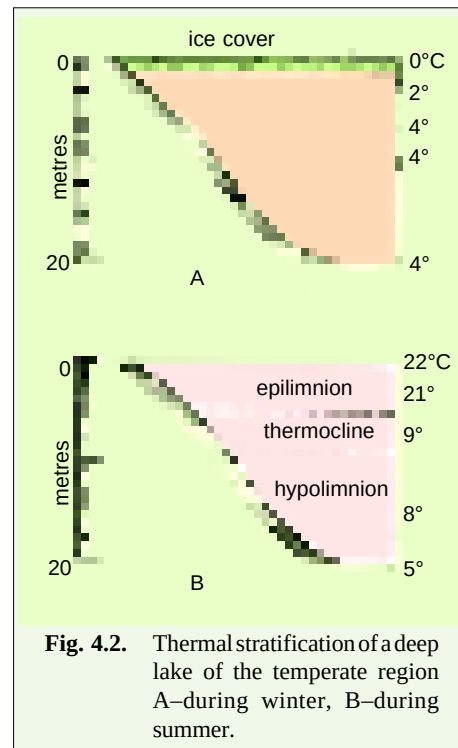


Fig. 4.2. Thermal stratification of a deep lake of the temperate region A—during winter, B—during summer.

are active within even narrower limits. Many nematodes, rotifers and tardigrades have shown to withstand cooling to -272°C without ill effects. Larvae of chironomids and certain other Diptera have been found to thrive at temperatures near 55°C . Praying mantis is reported to live in bare grounds at a temperature of 62°C in deserts. Some algae and bacteria are reported to live in hot springs at 88°C . Non-photosynthetic bacteria inhabiting hot springs can actively grow at temperatures greater than 90°C (**Bott** and **Brock**, 1969). On the other hand, some arctic algae can complete their life cycles in places where the temperatures barely rise above 0°C . Further, the eggs of the acanthocephalan *Macracanthorhynchus hirudinaceus* have been known to withstand temperatures from -10 to -45°C for about 140 days, and desiccation at temperatures up to 39°C for a period of 265 days.

The organisms (microbes, plants and animals) which can tolerate very large fluctuations in temperature are called **eurythermal organisms**. For example, cyclops, toad, wall lizard, grass snake, man, etc., are the eurythermal animals. The organisms which can tolerate only a small variation in temperature are termed **stenothermal organisms**. The common stenothermal animals are fishes, snails, coral reefs, etc. In organisms all metabolic processes necessary for life start at a certain **minimum temperature** and increase with rise in temperature until they reach the maximum level at a temperature called **optimum temperature**. Further rise in temperature beyond optimum brings about decrease in metabolic rate, until it ceases at a temperature called **maximum temperature**.

Thus, the favourable temperature range for any particular species is determined by the prevailing temperature at which normal physiological activities of the animals take place, as in the distribution of the rotifer *Keratella procura* (**Nayar**, 1970). This rotifer species is known to appear in the ponds of Pilani, Rajasthan (India), when the temperature is below 24°C and to disappear when it rises above 24°C , the frequency of distribution reaching a peak during months of October to March, with the fall in temperature.

Poikilotherms and Homeotherms

An animal's response to wide ranges in temperature is influenced by its physiology. All invertebrates, lower chordates (hemichordates, urochordates and cephalochordates), cyclostomes, fishes, amphibians and reptiles have no internal mechanism for temperature regulation, and their body temperatures vary with the surrounding environmental temperatures. Such animals are called **cold blooded, ectothermic** or **poikilothermic** organisms. The temperature range that a poikilothermic animal tolerates can be correlated closely with the environmental temperatures normally encountered in nature: tropical animals cannot withstand low temperatures; polar animals cannot withstand high temperature; temperate-zone animals survive a wide temperature range. Many poikilothermic animals show a rather precise discrimination of temperature. If the temperature decreases or increases appreciably, these animals may tolerate and adjust to change, may avoid the change by seeking a less exposed environment or may become inactive. Some animals respond by becoming dormant during periods of extreme temperatures. Such dormancy is called a **estivation** at high temperatures and **hibernation** at low temperatures. For example, most poikilotherms become inactive when the temperature of their surroundings goes below 8°C or rises to 42°C .

A few exceptional poikilotherms, especially insects, certain amphibians and reptiles, exercise a degree of thermoregulation by either physiological or behavioural mechanisms. For example, Hawkmoths can rise the temperature of their flight muscle to $32^{\circ}\text{--}36^{\circ}\text{C}$ by vibrating the wings before take-off and gregarious butterfly larvae may raise their temperature $1\frac{1}{2}\text{--}2^{\circ}\text{C}$ when clustered together. Locust, and



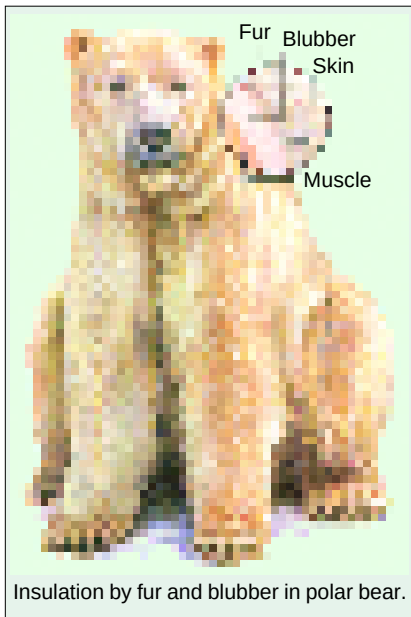
The dormouse hibernates in winter when food is scarce.

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grasshoppers may increase their temperature 10°C by basking, sideways in the sun. Ants move their larvae to warm or cool places within the nest and bees maintain temperatures within their hives between 13° and 25°C by fanning with their wings to evaporate water droplets when it is too hot, or releasing body heat through increased metabolic activity, when too cold.

When temperature drops, lizards bask in the sun to achieve the desired body temperature; once this thermal level is attained, they will divide their time between sun and shade to maintain it. Poikilotherms such as desert dwellers that live in environments where the temperature is apt to be very high are often nocturnal and, thus, avoid the highest temperatures of the day. Some poikilotherms, both vertebrates and invertebrates, lower their body temperatures slightly by evaporative cooling. In frogs and reptiles, evaporative cooling can occur through the skin or via the respiratory tract by panting (Warburg, 1967). Among the invertebrates, evaporative cooling has been reported in tropical intertidal-zone animals (Lewis, 1963).

In contrast to poikilotherms, birds and mammals can, within limits, maintain constant body temperatures, regardless of temperature variations of air and water. Such animals are termed **warm blooded, homeothermic** or **endothermic animals**. The life processes are adjusted to function at the animals's normal temperature, averging a little less than 38°C in mammals and 3 to 4° higher in birds. If its temperature control fails, the animal dies. The homeothermic animals are able to maintain the constancy of body temperature by a combination of several factors : (1) a thermoregulating centre in the brain (hypothalamus) (2) insulation (3) a peripheral vascular response to ambient temperature; and (4) metabolic compensation.



Insulation by fur and blubber in polar bear.

Among these factors, **insulation** is highly important particularly in enabling large arctic animals to withstand very low temperature. If the temperature is lowered, the oxygen consumption rate of tropical mammals increases, whereas that of arctic mammals remains basal. This response pattern is due to differences in insulation; some arctic animals are fur-insulated, whereas others are insulated with a layer of blubber. Many small animals, however, are not well insulated and must seek heat retention by other means, often by huddling together.

Homeothermic animals living in very hot environments cannot tolerate greatly elevated body temperatures and they utilize methods that facilitate heat transfer to the environment, including an increased peripheral blood flow and surface cooling by sweating or panting. Birds may accomplish this by rearranging their plumage so that more skin is exposed. There is however, no reduction in metabolic rate. Animals subjected to high temperature may exhibit diurnal patterns of behaviour—that is, they may reduce locomotor activity during the heat of the day or may move into the shade to avoid direct sunlight. Some animals such as monotremes and marsupials have a limited power of temperature regulation, they are called **heterothermic animals**; e.g., the pigmy mouse and the little pocket mouse, which respond to temperature extremes by a estivating or hibernating, and others such as the humming bird, which experience a nocturnal drop in temperature.

Plants too can be divided into the following three categories on the basis of their heat tolerating capacity: **megatherms**, **microtherms** and **mesotherms**. Megatherm plants occur in warm habitat (e.g. desert vegetation). Microtherm plants occur in cold habitat, (e.g., plants of high altitudes). Mesotherm plants are the plants of the habitat which is neither very hot nor very cold (e.g., aquatic plants). **Hekistotherms** include alpine vegetation which tolerate very low temperatures.

Factors or Variables Affecting Organismal Response to Temperature

The response of an organism to temperature is affected by a number of factors such as **thermal history, genetic differences, diet, size, stage in life cycle, sex, moulting, parasitism, hormones**, etc. 1. Generally, animals found in warmer environments can withstand higher temperatures than animals from colder situations. It is also generally true that animals from colder environments tolerate lower temperatures better than those from the warmer climates. These differences between warm-climate and cold-climate populations may reflect basic genetic differences or may be phenotypic expressions resulting from different thermal histories. 2. The thermal resistance of a species may be closely identified with the genetic composition of the parent stock, as illustrated by hybrid development in two anuran species, *Bufo valliceps* and *B. luetkeni* (**Ballinger** and **Mckinney**, 1966). The lower lethal temperature for development of *B. valliceps* is 18°C ; for *B. luetkeni*, 22°C. In their hybrid, the lower lethal limit is found to be in between that of the two parents at 19.5°C. 3. In some animals a relationship between the total intake of food and resistance to thermal stress has been demonstrated. Some species are more sensitive to elevated temperatures when they are starved for even short periods of time. Goldfish showed increased resistance to high temperature when placed on a high fat diet (**Hoar** and **Cottle**, 1952). 4. The size of animal body is found to have some correlation with thermal lethal limits of animals. In some species, the smaller animals are more resistant to higher temperatures than the larger ones; in some species the reverse is noted, and in other species size is not an apparent variable. Further, some smaller animals die faster at low temperatures than do larger animals of the same species, but at higher temperatures body size is not a factor. 5. Many species of animals have exceedingly complex life cycles during which the larval stages are not only morphologically dissimilar from the adult but also occupy different ecological niches. In the wharf crab, *Sesarma cinereum*, thermal requirements of the planktonic zoeal stages are different from those of adults because these larvae are limited to a smaller temperature range than are the adults. 6. The female sex is found to be more tolerant than males to the temperature fluctuations. 7. Among certain invertebrates, such as, in some crabs, moulting adversely affected heat resistance, but had no effect on other species. 8. Mud-flat snails, *Nassarius obsoleta*, which are heavily infected with trematode larvae (*i.e.*, parasites) cannot withstand high temperatures like the non-parasitized snails. 9. Hormones have been shown to influence cold and warm resistance as well as acclimatization (*viz.*, temperature adaptation) to temperature, especially in mammals and a few other invertebrates.

Effect of Temperature on Plants and Animals

Temperature has been found to affect the living organisms in various ways. Some of well-studied effects of temperature on living organisms are the following :

1. Temperature and cell. The minimum and maximum temperatures have lethal effects on the cells and their components. If too cold, cell proteins may be destroyed as ice forms, or as water is lost and electrolytes become concentrated in the cells; heat coagulates proteins (**Lewis** and **Taylor**, 1967).

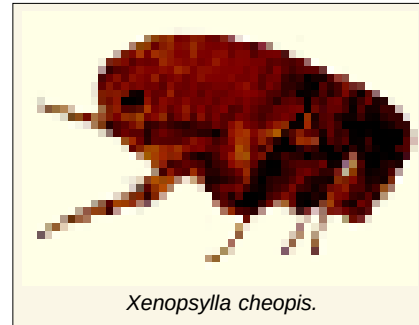
2. Temperature and metabolism. Most of metabolic activities of microbes, plants and animals are regulated by varied kinds of enzymes and enzymes in turn are influenced by temperature, consequently increase in temperature, up to a certain limit, brings about increased enzymatic activity, resulting in an increased rate of metabolism. For instance, the activity of liver arginase enzyme upon arginine amino acid, is found to increase gradually and gradually, with the simultaneous increase in the temperature from 17°C to 48°C. But an increase in temperature beyond 48°C is found to have an adverse effect on the metabolic rate of this enzymatic activity which retards rapidly.

In plants, the absorption rate is retarded at low temperature. Photosynthesis operates over a wide range of temperature. Most algae require lower temperature range for photosynthesis than the higher plants. The rate of respiration in plants, however, increases with the rise of temperature, but beyond the optimum limit high temperature decreases the respiration rate. The rate of respiration become doubled (like in animals) at the increase of 10°C above the optimum temperature, provided other factors are favourable (**Vant Hoff's law**). However, optimum temperature for photosynthesis is lower than that for respiration (**Smith**, 1974).

3. Temperature and reproduction. The maturation of gonads, gametogenesis and liberation of gametes takes place at a specific temperature which varies from species to species. For example, some species breed uniformly throughout the year, some only in summer or in winter, while some species have two breeding periods, one in spring and other in fall. Thus, temperature determines the breeding seasons of most organisms.

Temperature also affects fecundity of animals. **Fecundity** of an animal is defined as its reproductive capacity, *i.e.*, the total number of young ones given birth during the life time of the animal. For example, females of the insect, acridid *Chrotogonus trachypterus* became sexually mature at 30°C and 35°C than at 25°C, and the highest number of eggs per female was laid at temperature of 30°C. The number of eggs decreased from 243 to 190 when the temperature was raised to 30–35°C (**Grewal** and **Atwal**, 1968). Likewise, in grasshopper species—*Melanoplus sanguinipes* and *Camnula pellucida* when reared at 32°C produce 20–30 times as many eggs than those reared at 22°C (see **Ananthakrishnan** and **Viswanathan**, 1976). On the other hand, the fecundity of certain insects such as cotton stem weevil (*Pempherulus affinis*) was found to decline with an increase in temperature beyond 32.8°C (**Ayyar** and **Margabandhu**, 1941).

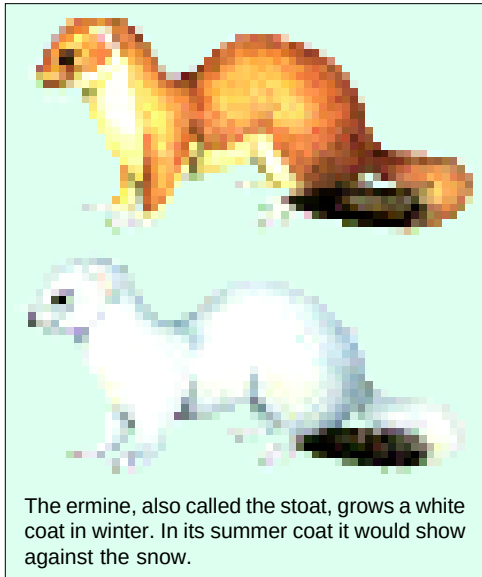
4. Temperature and sex ratio. In certain animals the environmental temperature determines the sex ratio of the species. For example, the sex ratio of the copepod *Macrocyclus albidu* is found to be temperature dependent. As the temperature rises there is a significant increase in number of males. Similarly in plague flea, *Xenopsylla cheopis*, males outnumbered females on rats, on days when the mean temperature remains in between 21–25°C. But the position becomes reverse on more cooler days.



Xenopsylla cheopis.

5. Temperature and ontogenetic development.

Temperature influences the speed and success of development of poikilothermic animals. In general, complete development of eggs and larvae is more rapid in warm temperatures. Trout eggs, for example, develop four times faster at 15°C than at 5°C. The insect, chironomid fly *Metriocnemus hirticollis*, requires 26 days at 20°C for the development of a full generation, 94 days at 10°C, 153 days at 6.5°C, and 243 days at 2°C (**Andrewartha** and **Birch**, 1954). However, the seeds of many plants will not germinate and the eggs and pupae of some insects will not hatch or develop normally until chilled. Brook trout grows best at 13°C to 16°C, but the eggs develop best at 8°C.



The ermine, also called the stoat, grows a white coat in winter. In its summer coat it would show against the snow.

6. Temperature and growth. The growth rate of different animals and plants is also influenced by temperature. For example, the adult trouts do not feed much and do not grow until the water is warmer than 10°C. Likewise, in the oyster *Ostrea virginica*, the length of the body increase from 1.4 mm to 10.3 mm when temperature is increased from 10°C to 20°C. Sea urchin *Echinus esculentus* shows maximum size in warmer waters. Corals flourish well in those waters which contain water below 21°C.

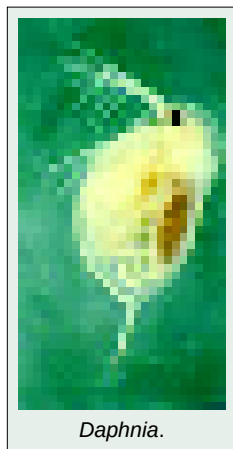
7. Temperature and colouration. The size and colouration of animals are subject to influence by temperature. In warm humid climates many animals such as insects, birds and mammals bear darker pigmentation than the races of some species found in cool and dry climates. This phenomenon is known as **Gloger rule**. In the frog *Hyla* and the horned toad

Phrynosoma, low temperatures have been known to induce darkening. Some prawn (crustacean invertebrates) turn light coloured with increasing temperature. The walking stick *Carausius* has been known to become black at 15°C and brown at 25°C.

8. Temperature and morphology. Temperature also affects the absolute size of an animal and the relative properties of various body parts (**Bergman's Rule**). Birds and mammals, for example attain a greater body size when they are in cold regions than in warm regions, and colder regions harbour larger species. But poikilotherms tend to be smaller in colder regions. Body size has played a significant role in adaptation to low temperature because it has influenced the rate of heat loss. According to **Brown** and **Lee** (1969), larger wood rats have a selective advantage in cold climates, apparently because their surface to air ratio and greater insulation permit them to conserve metabolic heat. For opposite reasons small-sized animals are favoured in deserts.

Further, the extremities of organism such as tail, ears and legs of mammals often appear to be shorter in colder climate (**Allen's Rule**). Mice reared at 31° to 33.5°C have longer tails than those of the same strain reared at 15.5° to 20°C. Moreover, the races of birds with relatively narrow and more acuminate (i.e., tapered to a slender point) wings tend to occur in colder regions, while those in warmer climates tend to be broader (**Rensch's Rule**). Temperature also influences the morphology of certain fishes and is found to have some relation with the number of vertebrae (**Jordon's Rule**). Cod which hatches off New Foundland at a temperature between 4° and 8°C has 58 vertebrae, while that hatches East of Nantucket at a temperature between 10° and 11°C has 54 vertebrae.

9. Temperature and cyclomorphosis. The relation between seasonal changes of temperature



Daphnia.

and body form is manifested in a remarkable phenomenon termed **cyclomorphosis** exhibited by certain cladocerans such as *Daphnia* during the warm months of summer. These crustaceans show a striking variation in the size of their helmet or head projection between the winter and summer month (**Coker**, 1931). The helmet develops on the *Daphnia* head in spring, it attains its maximum size in summer and disappears altogether in winter to provide usual round shape to the head. Such a kind of cyclomorphosis in the terms of size of the helmet is clearly showing a correlation to the degree of warmth of different seasons. These prolongations of the helmet have been interpreted as an adaptation aiding floatation since the buoyancy of water becomes reduced as the temperature increases (the **buoyancy hypothesis**). According to other interpretation (viz., **stability hypothesis**), the helmet acts like the rudder and gives greater stability to the animal. Besides temperature such structural polymorphism can be caused by other environmental factors including the food.

10. Temperature and animal behaviour. Temperature generally influences the behavioural pattern of animals. In temperate waters the influence of temperature on the behaviour of wood borers is profound. For example, in the winter months in general, both *Martesia* and *Teredo* occur in smaller numbers in comparison with *Bankia campanulata* whose intensity of attack is maximum during the winter months. Further, the advantage gained by certain cold blooded animals through thermotaxis or orientation towards a source of heat are quite interesting. Ticks locate their warm blooded hosts by a turning reaction to the heat of their bodies. Certain snakes such as rattle snake, copper heads, pit vipers are able to detect mammals and birds by their body heat which remains slightly warmer than the surroundings. Even in the dark these snakes strike on their prey with an unnerving accuracy, due to heat radiation coming from the prey. The arrival of cold weather in temperate zones causes the snakes to coil up and huddle together.

11. Temperature and animal distribution. Because the optimum temperature for the completion of the several stages of the life cycle of many organisms varies, temperature imposes a restriction on the distribution of species. Generally the range of many species is limited by the lowest critical

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temperature in the most vulnerable stage of its life cycle, usually, the reproductive stage. Although the Atlantic lobster will live in water with a temperature range of 0° to 17°C, it will breed only in water warmer than 11°C. The lobster may live and grow in colder water but a breeding population never becomes established there. Not only temperature affects on breeding in the geographical distribution but also temperature affects on survivality (*i.e.*, lethal effect of temperature), feeding, and other biological activities which are responsible in geographic distribution of animals. As noted earlier in this chapter, the animals from colder geographic regions are generally less heat tolerant and more cold tolerant than those animals from warmer regions ; for example, members of *Aurelia*, a jelly fish from Nova Scotia die at a water temperature of 29°–30°C, while *Aurelia* from Florida can tolerate temperatures up to 38.5°C. Thus, lethal limit of temperature may regulate the range of distribution of *Aurelia*.

Terrestrial invertebrates, particularly arthropods generally are distributed in all thermal environments where life is found. Many arthropods that have invaded the colder areas have one stage in their life cycle which is very resistant to cold, enabling them to over-winter until warmer weather returns (Salt, 1964). Birds and mammals are also adapted to live in nearly all thermal environments. The distribution of amphibians and reptiles, however, is limited to the relatively warmer thermal climates. Hock (1964) has listed three factors that limit the invasion of reptiles into cold environments : the daily ambient (=atmospheric) temperature must be high enough and long enough to allow breeding and to allow adults and young to acquire food for “overwintering” and there must be adequate sites for hibernation.

12. Temperature and moisture. The differential heating of the atmosphere resulting from temperature variation over the earth’s surface produces a number of ecological effects, including local and trade winds and hurricanes and other storms, but more importantly it determines the distribution of precipitation.

Thermal Adaptations of Plants and Animals

Most animals and plants of different ecological habitats have developed various sorts of thermal adaptations during the course of evolution to overcome the harmful effects of extremes of temperature. Some of the significant thermal adaptations of plants and animals are the following :

1. Formation of heat resistant spores, cysts, seeds, etc. Some of the animals and plants produce heat resistant cysts, eggs, pupae, spores and seeds which can tolerate extremes of temperatures. *Amoeba* in encysted conditions, can tolerate temperature below 0°C. Similarly, rye seeds remain active even at 0°C and can germinate at that temperature. As an adaptation against frost the starch of plants changes to fats or oils in the autumn. The fatty oils diminish the freezing points and, thus, increase the power of resistance in plants against frost. Many leaves, that grow in the coldest lands, store fats. Pentosans mucilage and pectic substances which have high moisture retaining power are abundant in many plants. They decrease the danger of plants from desiccation during extremes of heat and save them from death.

2. Removal of water from tissue. Dried seeds, spores and cysts avoid freezing because there remains no liquid in them that can freeze. Due to removal of water from seeds, the cold resistance of seeds of certain plants increases up to



North African jerboa, or desert rat, lives underground by day to protect themselves from the burning rays of sun. They come out to hunt in the cool of the night.

the extent that their exposure for 3 weeks to 190°C, does not diminish their germinating capacity.

3. Dormancy. Dormancy includes two already discussed phenomena namely **hibernation** and **aestivation**. During both kinds of dormancies metabolic rate becomes reduced, body temperature becomes low and heart beat rate is also reduced.

4. Thermal migration. Thermal migrations occur only in animals. The journeys taken by animals that enable them to escape from extremely hot or cold situations are referred to as **thermal migrations**. For example, desert animals move to shaded places to avoid burning heat of noon and some animals such as desert reptiles and snakes become nocturnal to avoid heat of the day. The frogs, toads, other amphibians, turtles, etc., make short trips into or out of water (or moist places) and this provides desired cooling and warming to the animal.

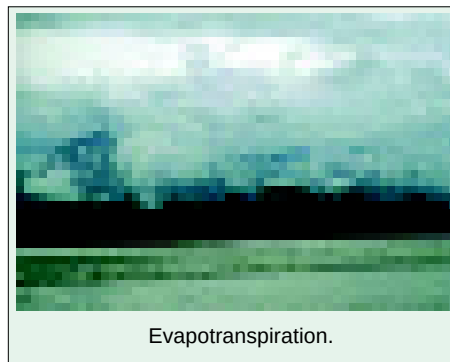
3. PRECIPITATION (RAIN FALL)

The moisture falling on an area in liquid, vapours or frozen form is termed as **precipitation**. Thus, precipitation includes all moisture that comes to earth in the form of rain, snow, hail and dew. Precipitation is the chief source of soil water. The water available to plants and animals from soil comes as a result of rainfall. Due to **water cycle** or **hydrological cycle**, there occurs an interchange of water between the earth's surface and the atmosphere. In this cycle following, two important events are involved : **precipitation** and **evapotranspiration**. The ecofactor of precipitation depends upon season, wind, air pressure and temperature.

Precipitation occurs as a result of the cooling and condensation of water vapour at high altitudes. The low temperature at high altitudes cools the air, which gets saturated and loses its water-holding capacity. As the temperature starts falling, the water vapour condenses and falls as rain due to gravity. Depending on the environmental conditions, precipitation falls as hail, snow or rain. In winter the ground temperature falls and as a result, atmospheric vapour gets condensed as **dew** or **frost** on the surfaces of objects, plants, animals, soils, etc. Dew becomes an important source of moisture to plants in the winter season. **Drizzle** involves minute drops appearing as to float in air. **Rain** is the drops of liquid water, which are larger than drizzle and also heavier. **Snow** is the moisture as solid state. **Sleet** is the form of small grains or pellets of ice, whereas **hail** consists of balls or lumps of ice.



Dew.



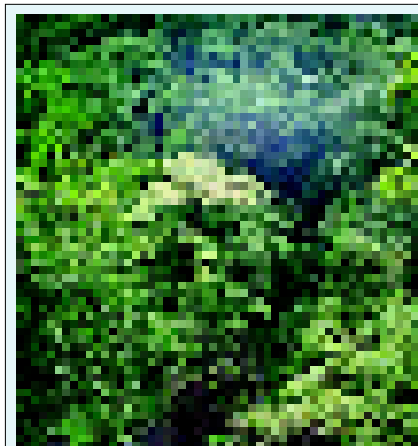
Evapotranspiration.

Light drizzle is of little importance as very little moisture penetrates the soil because much of it evaporates rapidly. Snow is injurious to plants, breaking tender branches, flowers and fruits. Hail and sleet also cause similar damage. Some sedges grow in snow patches. Of all the above forms of precipitation, the rain is most important. It is the source of soil water and also affects humidity of atmosphere. The total precipitation of the world is about 4.46×10^{20} g per year and of this amount 0.99×10^{20} g fell on land and the rest on oceans (**Hutchinson**, 1957). In India, rains are caused by monsoons; **monsoon** is a special pattern of moist air movement prevalent in India. During monsoon, the air masses moving from the Arabian sea to the West coasts of India and from the Bay of Bengal to the eastern part of country, become extremely moist. About 45 per cent of the water available during annual precipitation flows into river, 20 per cent percolates in the ground and the

remaining 35 per cent is lost by evaporation. Gentle steady rains are most effective because much of it penetrates the soil. Torrential rains on the land are most disastrous because they lead to flooding, soil erosion, destruction of vegetation and of animals.

The quantity, duration and intensity of rainfall profoundly regulate the vegetation of any place. For example, in tropical areas with heavy rainfall throughout the year, vegetation mainly include **evergreen forests**. In countries, with heavy rainfall during winter and low during summer, there are present **sclerophyllous forests**. The plants are shrubs, stunted in height, with leathery, thick, evergreen leaves. The areas with heavy rainfall during summer and low during winter are characterized by the presence of **grasslands**. The regions where rainfall is scanty, are seen with deserts and xerophytic vegetation.

The average annual rainfall of India is about 117 cm, the highest in the world. However, there exists great variations between the different regions in India with regard to rainfall. Thus, the average annual rainfall in Assam and the north-east is about 250 cm. In Cherrapunji in Meghalaya, the annual average of rainfall is about 1,100 cm, whereas in Jaisalmer in Rajasthan it is only 20 cm. The distribution of rainfall over India depends largely on the position of hills and mountains and the forest cover. Thus, monsoon winds strike the Western Ghats from a southwest direction and shed most of their water on the windward side. Bombay being on the windward side, receives 200 cm of rainfall annually during the monsoon, while, Pune being on the leeward side (*i.e.*, side of anything away from wind) receives only about 75 cm. If the rainfall to evaporation ratio is zero or less, deserts develop. Grasslands develop when this ratio is more than 0.2 and less than 1 and forests develop when the ratio is more than 1 (around 1.6 to 2). Often in forests, the floor receives rain from trees in the form of drips from tree leaves due to accumulation of condensed water on winter mornings.



An evergreen forest.

4. HUMIDITY OF AIR

Atmospheric moisture in the form of invisible vapour is known as **humidity**. The humidity of air is expressed in terms of values of **relative humidity** which is the amount of moisture in air as percentage of the amount which the air can hold at saturation at the existing temperature. Relative humidity is measured by the instrument called **psychrometer** or by paper strip **hygrometer** or a **thermo-hydrograph**. Humidity is greatly affected by intensity of solar radiation, temperature, altitude, wind exposure, cover and water status of soil. High temperature increases the capacity of the air to retain moisture and results in lower relative humidity. Low temperatures result in higher relative humidity by decreasing the capacity of air for moisture and reaches 100% or saturation point. Further cooling results in condensation of vapour into water and this temperature/moisture point is called the **dew point**. Daily variations in relative humidity values depend upon the type of habitat conditions. In plains and deserts, it may show variations during day, whereas in oceanic islands there is little variation, being same throughout the year.

Humidity plays an important role in the life of plants and animals. It affects the life processes such as transpiration, absorption of water, etc. The rate of evapo-transpiration is modified by the factors as saturation deficit (*i.e.*, degree of wetness or dryness), temperature and wind velocity. Parasitic fungi becomes abundant in moist weather and reduces primary production (see **Ambasht**, 1990). Some plants such as orchids, lichens, mosses, etc., make direct use of atmospheric moisture. In fungi, and other microbes, humidity plays an important role in germination of spores and subsequent stages in life cycle.

5. FIRE

Fire is an interesting ecofactor. Fire is of a common occurrence in natural vegetation all over the world; it is more common in drier habitats than the wet. Lightening is the commonest natural cause of fire initiation. Our earth's surface is hit by lightening every second in one or another part of the globe and many of these are of great magnitude. Other causes of fire are abrassive effects of falling rocks or of dried plant material such as bamboos, or spontaneous combination of very dry and hot material or by volcanic activities. Most forest fires are now man-caused, *i.e.*, by incendiaries (such as poachers), debris burners, smokers, campers, short-circuiting of high-tension electric lines, and nearby railway lines. In the Meghalay region, tribals and even modern farmers practice the *Jhum* cultivation by periodically slashing the forest, burning them and raising crops on the ash enriched soil for a few years and moving to another segment of forest as the earlier one becomes eroded and nutrient deficient.



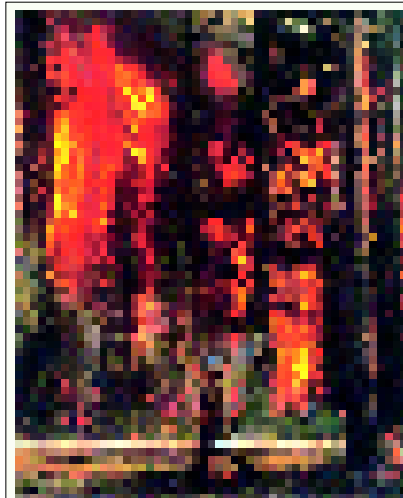
slash-and-burning subsistence farming in a small patch of cleared tropical rain forest in Costa Rica.



Surface fire in Ocala National Forest in Florida. Occasional fires like this burn deadwood, undergrowth, and litter and help prevent more destructive crown fires in some types of forests.

Types of Fire

Fires are generally classified as (i) **Ground fires** which develop in such conditions where organic matter (litter) accumulates richly as heaps and they catch fire which generally smoulder for longer periods. Thus, in dry litter, fire is rapid and extinguishes quickly, while in somewhat moist litter, the fire is slow and with its heat the inner parts of litter-heap also get dried and fire continues for a longer



Destructive crown fire in Yellowstone National Park during the summer of 1988. Wildlife that can't escape are killed and wildlife habitats are destroyed. Severe erosion can occur.

period. Ground fires are flameless and subterranean and kill almost all herbaceous plants rooted in the burning material except some woody species (shrubs and trees). (ii) **Surface fires** which sweep over the ground surface rapidly and their flames consume the litter, living herbaceous vegetation and shrubs and also scorching the tree bases if comes in contact. (iii) **Crown fires** which are most destructive, burning the forest canopy and are common in dense woody vegetations. They spread in the top layers from the canopy of one tree to the canopy of another and so on. Canopy fires produce a temperature up to 704.5°C, killing the trees, shrubs and herbs. However, in moist soil surface, the underground plant parts and buried seeds escape death.

Effects of Fire

Fire has direct (*e.g.*, lethal) as well as indirect effects on plants and wild-life. Some well confirmed indirect effects of fire on plants are as follows :

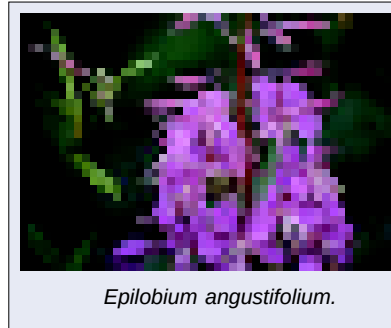
1. Fire causes injury to some plants, resulting large scars on their stems. Such scars may serve as suitable avenues of entry of parasitic fungi and insects.
2. Fire arrests the course of succession and modify the edaphic environment very much.
3. Fire brings about distinct changes of such ecofactors as light, rainfall, nutrient cycles, fertility of soil, litter and humus contents of soil, pH, water holding capacity and soil fauna (earthworms, nematodes, arthropods, etc.) Soil fungi are reduced while bacteria increase due to post-fire changes in the soil.

The microclimate too is greatly changed due to addition of ash, loss of shade, loss of raindrop interception, accelerated erosion, etc.

4. Fire plays an important role in the removal of competition for surviving species. **Fire tolerant plant** species generally increase in abundance at the expense of those killed by fire (**fire-sensitive plants**) due to considerable reduction in competition and possibly due to alteration in other conditions.

5. Some plants such as *Populus tremuloides* and *Epilobium angustifolium* are stimulated to growth by fires. A number of such grasses as *Aristida stricta*, *Cynodon dactylon*, *Paspalum notatum*, are stimulated by fire to produce large quantities of seeds. In some grasses and legumes, the seeds would germinate only after these get fire treatment (*e.g.*, seeds of *Themeda*, *Heteropogon*, *Andropogon*, *Rheus*, *Tephrosia* and *Stipa*).

6. Some fungi, mainly some ascomycetes, grow in soils of burnt areas. Such fungi are known as **pyrophilous** (*e.g.*, *Pyrophilous confluens*).



Epilobium angustifolium.

Adaptations to Fire

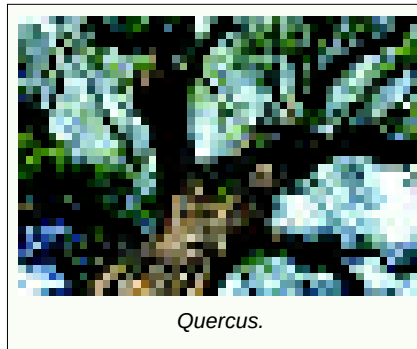
In frequent fire prone areas certain plants develop the following adaptations :

1. Certain trees, particularly conifers such as *Pinus* and *Larix* and dicots such as *Quercus* develop fire resistant bark with insulating effect against heat. These trees also have tall trunk with the crown restricted to upper zone only. This helps in escaping the destructions against surface and ground fires.

2. In some plants leaves are fire-resistant due to poor contents of such compounds such as resin or oil, and may check surface fires.

3. Some plants as *Pinus rigida* and species of *Eucalyptus* have adventitious or latent axillary buds which may develop into new branches. Similarly *Betula papyrifera* and *Vaccinium* sp., may develop new shoots after fire kills the older ones.

4. *Epilobium angustifolium* acts as a fire indicator species. It grows in patches and in dormant condition. In case of fire, these plants rapidly grow while other plants die due to fire.

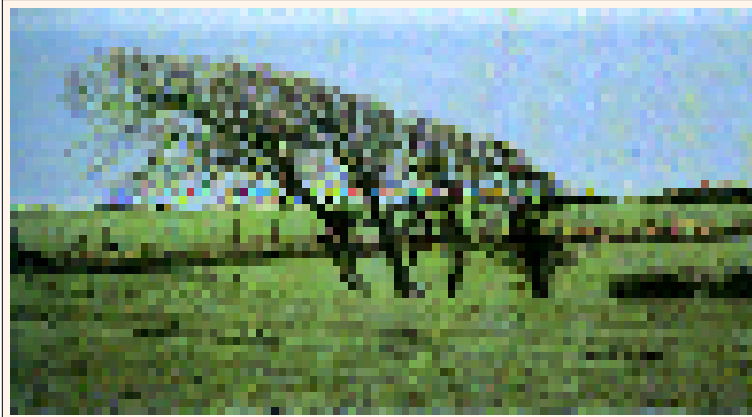


Quercus.

6. WIND FACTOR

The strong moving current of air is called **wind**. It is an important ecological factor of the atmosphere affecting variously the plant life on flat plains, along sea coasts and at high altitudes in mountains. Wind is directly involved in transpiration, in causing several types of mechanical damage

and in dissemination of pollen, seeds and fruits. Wind also modifies the water relations and light conditions of a particular area. The movement or velocity of wind is affected by such factors such as temperature, atmospheric pressure, geographical features (including topography), vegetation masses and position with respect to sea shores. Air moves from a region of high pressure to low pressure. The pressure differences are mainly due to differential heating of atmosphere. The equatorial regions



Strong prevailing winds can leave trees permanently bent.

receive more heat than north or south regions, thus, low pressure occurs at lower latitude. The air generally moves from poles towards equator. Winds result in various physical, anatomical and physiological effects on plants: 1. **Physical effects** such as breakage and uprooting; deformation; lodging or flattening of herbaceous plants such as wheat, maize, sugarcane, etc.; abrasion of buds

of plants (by wind-carried soil particles), erosion and deposition of soil around plant roots; plant injury due to salt spray along sea coasts 2. **Anatomical and physiological effects** such as formation of dense, reddish xylem called **compression wood** on the compressed side of wind deformation, desiccation due to increased rates of evaporation and transpiration which are caused by strong winds and **dwarfing** of trees on sea coasts, arctic or alpine timberline due to prolonged dehydration and consequent loss of turgidity under the influence of drying winds.

7. PHYSIOGRAPHIC FACTORS

Climate at any point on the globe depends upon the combined effects of three factors, the amount of solar heat (solar radiation), atmospheric circulation and physiographic factors. **Physiographic factors** are those which are introduced by the structure, conformity and behaviour of the earth's surface, by topographic or orographic features, such as elevation and slopes, by geodynamic processes such as silting and erosion and consequently by local geology. Mountains, hills, hillocks, valleys, etc., result from irregularities in earth's surface. These distinct topographic reliefs tend to produce marked local climate or microclimates. For example, in respect to the microclimate, the summits are different from the sides of mountains and narrow valleys are different from the open plains. Different physiographic factors can be grouped under the following headings :

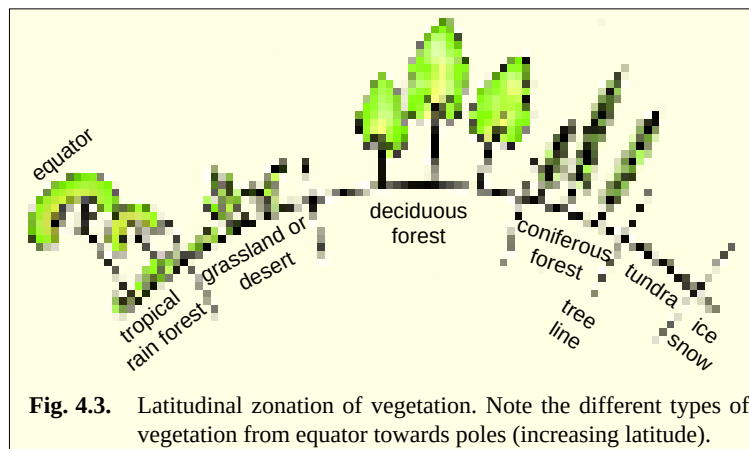


Fig. 4.3. Latitudinal zonation of vegetation. Note the different types of vegetation from equator towards poles (increasing latitude).

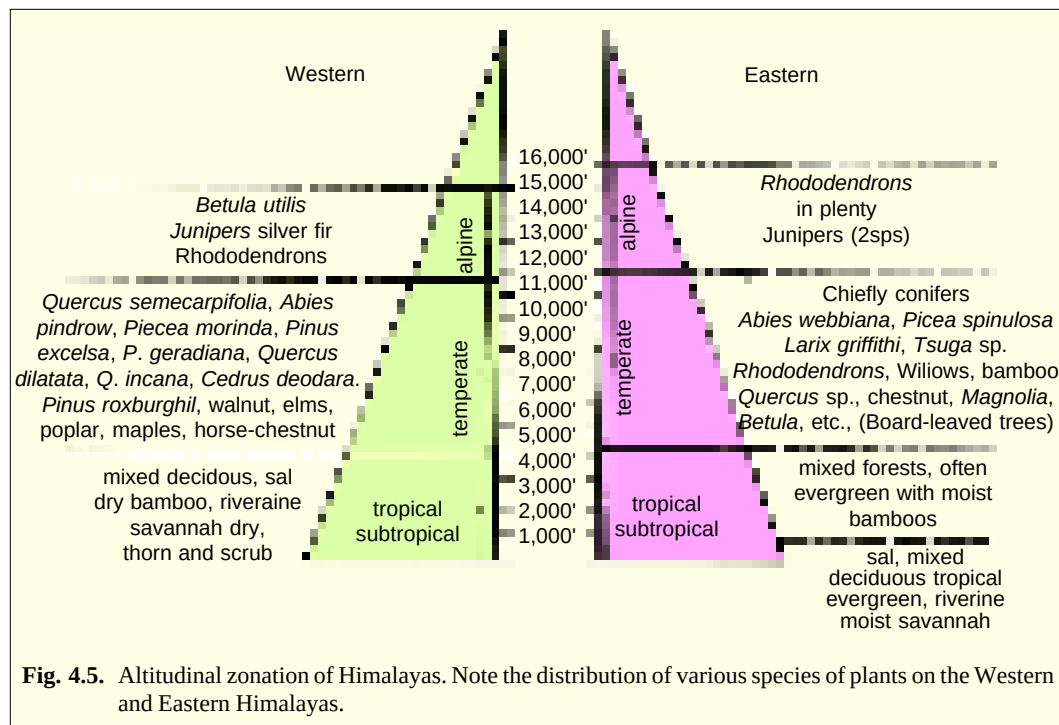
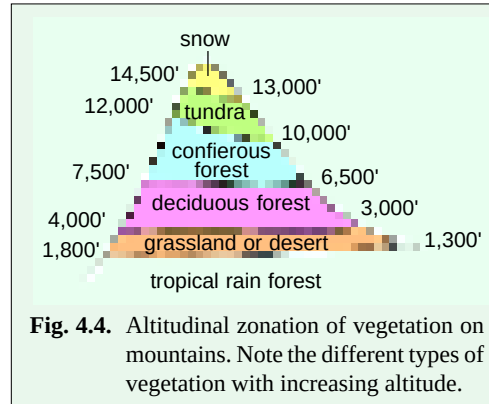
1. Latitudes and Altitudes

Latitude represents distance from the equator. Temperature values are maximum at the equator, decreasing gradually towards poles. Marked variations in temperature at different latitudes result in the division of earth's vegetation into different zones such as equatorial, tropical rain forests, desert or grasslands, deciduous forests, coniferous forests, alpine forests, tundra, ice and snow of poles (Fig. 4.3).

Height above the sea level forms the **altitude**. At high altitudes, the velocity of wind remains high, temperature and air pressure decrease, humidity as well as intensity of light increases. Due to these factors, vegetation at different altitudes is different, showing distinct zonation (Fig. 4.4).

Generally, mean temperature of air decreases about 10°F (-12.2°C) for every degree of latitude north of equator and for every 300' (or 91.44 metres) of altitude. Altitudinally, this temperature fall is more rapid on the leeward side of a mountain than on the windward side. For example, the fall in temperature in the Nilgiri hills is 1°C for every 100 metre rise on the leeward side and 122 metre rise on the windward side. Further, valleys and lowlands sometimes become much cooler than the mountain tops due to sinking of the heavier cold air.

Climatic changes in both, the latitude and altitude, exhibit identical effects upon the type of major vegetations of the world. Thus, communities at altitudes (alpine) roughly resemble those of polar regions (arctic tundra). Below the arctic line and the trees or timber line exists the coniferous forests or taiga which are followed by temperate zones which have quite cold conditions and support deciduous forests.



Lower to the temperate zone, the sunlight falls nearly at an angle of 90° , making the climate warm and dry. These places are known as **tropical regions** and lie on either side of the equator.

Temperature and rainfall are commonly said to determine the type of vegetation at different altitudes of mountain. In addition to these two factors edaphic conditions (soil characteristics) are also found important. At higher altitudes, generally temperature decreases and rainfall increases, and these certainly affect the development of soil and vegetation. Due to low temperature and high rainfall the organic matter content of the soil increases at high altitudes, resulting an increase in soil nitrogen and a decrease in soil's pH value. As a result, xerophytes are more common at lower altitudes and chamaephytes occur at higher altitudes.

Moreover, at sea level, the atmospheric pressure is 760mm Hg. The partial pressure of oxygen remains approximately one-fifth (*i.e.*, about 160 mm Hg) of the atmospheric pressure. With increasing altitudes, the atmospheric pressure and partial pressure of O_2 goes on decreasing. Since oxygen is thinner at higher altitudes, it becomes a stress factor for many animals, including mammals.

Lastly, in Himalayan mountains, temperature variations are quite evident, and there is a general zonation of vegetation from lower to higher altitudes. The successive zones of vegetation from base upwards are tropical and subtropical, temperate and alpine. But, there are practically no sharp boundaries between these vegetational zones due to differences in topography, soil and geology. Thus, after studying the vegetational zones at different altitudes on Himalayas, western as well as eastern, the effect of temperature together with altitude and other factors becomes quite apparent (Fig. 4.5).

2. Height of Mountain Chains

Effect of different heights (altitude) can be better seen on mountains. With an increase in altitude above sea level, there are changes in values of temperature, pressure, wind velocity, humidity, intensity of solar radiation, etc. Due to these changes, vegetation at different altitudes differ much.

3. Direction of Mountains and Valleys

The directions of mountain chains or ranges and high mountains act as wind barriers and affect the climate. Mountains also steer or deflect winds into different directions and capture moisture from wind on certain sides to cause precipitation. Thus, moisture-bearing winds may not be able to cross a mountain range and may discharge rain near it. The far side of the mountain may not get much rain and develop into an arid zone (having xeric conditions). For example, the Himalayas act as great barriers for moisture bearing sea winds, therefore, the ocean side of the land in the north eastern region and the Gangetic delta get good rains and have luxuriant vegetation, while, the middle and inner Himalayas are dry with poor vegetation. The absence of a high land masses may be one of the reasons for the development of desert conditions in Rajasthan. Similarly, southern side of Kulu Valley (in Himachal) is moist with rich vegetation, whereas at lower level into Lahul Valley, there is very poor vegetation.

High mountains not only act as climatic barriers to air masses between neighbouring areas or zones but also cause local air circulation, giving rise to mountain winds. The most common winds in such areas are **slope winds** which blow down-hill during night and up hill during the day. Such winds are frequent in Antarctica. Further, the relative humidity (*i.e.*, dryness, humidity or severity) of a climate depends on whether the air comes from the north or the south, from the sea, or deserts, mountains or forests and so on. The heating of earth's surface is directly proportional to the angle at which solar rays arrive on the ground. The closer their incidence to perpendicular, the more solar heat is delivered per square centimetre. Hence, mountain ranges affect the climate through rainfall and other factors which have a significant affect on the growth of vegetation and the distribution of animals.

4. Steepness of Slope

Slope is the characteristic feature of mountains. The steepness of a slope has a distinct effect on the climate of the area, *i.e.*, incidence of solar radiation, rainfall, wind velocity and the temperature of the region.

Generally, southern and western slopes of mountain display higher temperature than do the northern or eastern slopes. Most probably these differences arise due to different solar radiation received, differences in the amount of rainfall, snowfall, relative humidity and wind movement on the

two slopes. In India, all south-facing slopes are in general exposed to more solar radiation than north facing slopes. North facing slopes are generally cooler than south facing ones. This is due to the fact that the steep southern slopes receive the rays of the mid-day sun almost at right angles, whereas the northern slopes receive only oblique rays during morning and evening hours and sometimes none at all except for a short period during summer (*i.e.*, the duration of sunlight on the south facing slope is longest). East and west-facing slopes receive equal amounts of sunlight, but in the former, the angle of incidence of the sun's rays is greatest in the morning hours when the air is cool and the dew has not fully disappeared. In west-facing slopes, on the other hand, the sun's rays become scorching as the air is already heated up, causing a desiccation effect on the environment. The differences in solar radiation and consequent temperature values bring about changes in the vegetation on the slopes. For example, northern slope (which is protected from sun) support virgin forests with hygrophilous ground vegetation, whereas the southern slopes (which is heated by sun) have only a xerophytic vegetation. Various climatic factors of the slopes also affect the distribution of consumers (animals).

Steepness of slope decides the rapidity, with which water flows away from the surface and determines the characteristic of the soil. Thus, loss of water as run off is more with increase in steepness of the slope. This is why having same amount of rainfall bear different types of vegetations. On a steep slope, plants are unable to establish, even if there is sufficient rainfall. Any little vegetation occurring there will be xerophytic in nature. Soil erosion is more prevalent on steep slopes. The water moving over the slopes causes erosion of the top soil and consequent disappearance of vegetation from the areas. Eroded hills, steep slopes and rock may produce a special habitat where only particular species of plants can adapt themselves. Gravel, sand or clay eroded by water, wind, etc., from rock surfaces, are generally brought down by rivers and streams and these are ultimately deposited as silt at their mouths, providing new habitat (*e.g.*, salt marshes) which may support different kinds of plants. The areas subjected to quick and continuous erosion are completely devoid of vegetation.

Exposure of slope. A slope often remains exposed to sun and wind and this affects greatly the kind of plants growing there. Generally, the slope exposed to the sun and wind supports vegetation which may be entirely different from that which is less or not at all, exposed to sun and/or wind. Visualizing this fact, green houses and hot beds in temperate climates are always built in a way so as to face the sun or on southern slopes, which receive much heat from the sun.

REVISION QUESTIONS

- Write short notes on the following :
 - Essential elements,
 - Limiting factors,
 - Liebig's law of minimum,
 - Threshold,
 - Law of tolerance,
 - Cyclomorphosis,
 - Ciradian rhythms,
- Discuss the role of temperature as an ecological factor. Give an account of different morphological and physiological adaptations of plants and animals in relation to temperature.
- Describe the role of light as an ecological factor. Also add a note about photoperiodism.
- Describe the role of fire as an ecofactor.
- With suitable examples, discuss the effect of topographic factors on distribution of plants.
- Explain the following :
 - Outer Himalayas have luxuriant vegetation than inner Himalayas.
 - Vegetation differs at various altitudes of Himalayas.
- Describe the role of the following ecofactors :
 - Precipitation ;
 - Wind ;
 - Humidity ;
 - Steepness of slope.
- Write an essay on photoperiodism and biological clocks.
- Describe how the chemical factor such as O_2 , CO_2 and pH regulate distribution of the organisms.

CHAPTER

5

Biotic Environmental Factors



All organisms are interdependent on each other.

Organisms do not exist alone in nature but in a matrix of other organisms of many species. Many species in an area will be unaffected by the presence or absence of one another (This is often termed **neutralism**), but in some cases two or more species will interact. The evidence for such interaction is quite direct : populations of one species are different in the absence and in the presence of a second species.

INTERSPECIFIC INTERACTIONS

The interactions between species may have positive or negative results. Following six general types of interactions have been described :

- A. Positive interactions
 - 1. Mutualism ;
 - 2. Commensalism ;
 - 3. Protocooperation.
- B. Negative interactions
 - 4. Exploitation ;
 - (i) Social parasitism ;
 - (ii) Parasitism ;
 - (iii) Predation.
 - 5. Amensalism and antibiosis ;
 - 6. Competition.

A. Positive Interactions

In case of positive interactions, populations help one another and either one or both the species are benefited. This benefit may be regarding the food, shelter, substratum or transport. Further, such an association may be continuous or transitory, obligate or facultative and the two partners may be in close contact (*i.e.*, their tissues remain intermixed with each other) or one of them may live within a specific area of the other or attached to its surface. Moreover, for the positive interactions, how does one determine that what is the benefit to the species? Individuals that participate in the interaction should have a higher relative fitness than those that do not participate. At the population level, the growth rate of a population containing mutualistic individuals should be higher than that of a population lacking such individuals (**Boucher et al.**, 1982). Positive interactions may be of the following types :

I. Mutualism. Mutualism is an obligatory positive interspecific interaction that is strongly beneficial to both species. In past, it was termed **symbiosis** (see **Dash**, 1993). Such mutually beneficial interactions are more common in the tropics than elsewhere. In this case, both of the species derive benefit and there exists a close and often permanent and obligatory contact which is more or less essential for survival of each. In mutualism, two populations enter into some sort of physiological exchange and resulted in coevolution of both species (see **Krebs**, 1985). Some common examples of mutualistic association are the following :

1. Pollination by animals. Certain insects such as bees, moths, butterflies, etc., and birds derive food from the nectar, pollen or other plant products, and in return bring about cross pollination (Fig. 5.1A). To ensure the success of this function, various structural adaptations have occurred in both plants and animals, leading to coevolution.

2. Dispersal of fruits and seeds. Birds and mammals are of great importance as agents of plant distribution. Seeds, fruits, even entire plants become attached to feathers or fur or ingested seeds are eaten and eliminated unharmed with the faeces.

3. Mutual defence in ants and acacias. A mutualistic system of defence has been achieved by the swollen thorn acacias and their ant inhabitants in the New World Tropics. The ants (*Pseudomyrmex* sp.) depend on acacia (*Acacia* sp.) tree for food and a place to live and the acacia depends on the ants for protection from herbivores and neighbouring plants (**Janzen**, 1966). Swollen-thorn acacias (*e.g.*, *Acacia cornigera*) have large, hollow thorns in which ants live (Fig. 5.2). The ants feed on modified leaflet tips, called **Beltian bodies**, which are the primary source of protein and oil for the ants, and also on enlarged nectaries, which supply sugars. Swollen thorn acacias maintain year-around leaf production, even in the dry season, to provide food for the ants. The acacia ants continually patrol the leaves and branches of the tree and immediately attack any herbivore that attempts to eat acacia leaves or bark. The ants also bite and sting any foreign vegetation that touches an acacia, and they clear all the vegetation from the ground beneath the acacia tree. Some of the species of ants that inhibit acacia thorns are obligate acacia ants and live nowhere else.



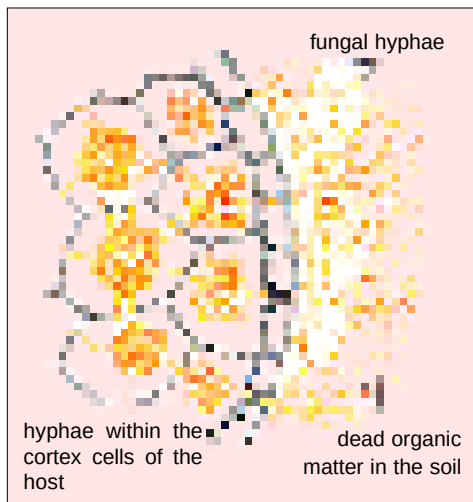
Beltian bodies.

4. Lichens. They form the examples of mutualism where contact is close and permanent as well as obligatory. The body of lichens is made up of a matrix formed by a fungus, within the cells of which an alga is embedded (Fig. 5.1B). The fungus makes available the moisture and minerals to the algae, which prepare food by photosynthesis. In nature, neither of the two can grow alone independently. Lichens tend to grow abundantly on bare rock surfaces.

5. Symbiotic nitrogen fixation. Here a bacterium *Rhizobium* forms nodules in the roots of leguminous plants and lives symbiotically with the host (Fig. 5.1 C). Bacteria get a protective space to live in and derive prepared food from the roots of higher plants and in return fix gaseous nitrogen,

making it available to the plants. The leguminous plants use this nitrogen in the protein synthesis and so legumes are rich in proteins. Nitrogen fixation-like association also occurs in root nodules of *Alnus*, *Alopecurus*, *Casuarina*, *Cycadaceae*, *Myrica*, *Podocarpus*, etc., and leaves of about 400 species of non-leguminous plants.

6. Mycorrhizae. In mycorrhizal associations, tree roots become infested with fungal hyphae (Fig. 5.1D). The fungi derive their food from the tree roots and in return their hyphae supply water and minerals that they absorb from the soil much like the root hairs of trees. It is believed that the fungus also regulates the pH and sugar level for a good growth of roots in acidic soils (e.g., conifers). Mycorrhizae may be on the surface of roots (**ectotrophic**) or inside between the cells of the roots (**endotrophic**). Ectotrophic mycorrhizae are common in nature on pines, oaks, hickories and beech and endotrophic ones occur in red maple and are common in roots and other tissues of many orchids and members of Ericaceae.



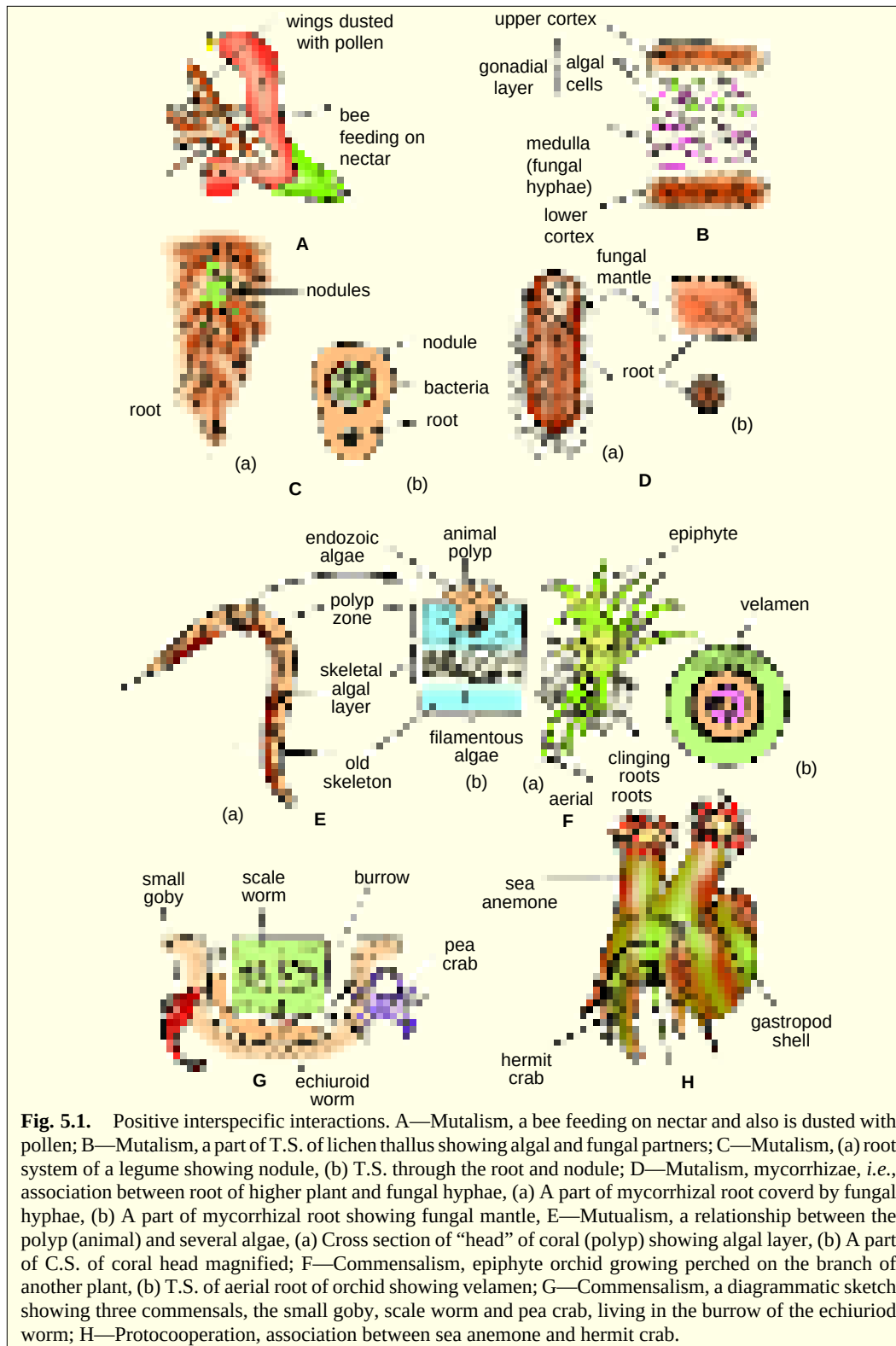
The other similar root associations are met with actinomycetes (actinorhizal) in form of nodules and with some blue-green algae (cyanobacteria) such as *Anabaena* and *Nostoc* forming coralloid roots of *Cycas*. Both of these kinds of associations are connected with nitrogen fixation by microscopic organisms present in the roots.

7. Zoochlorellae and Zooxanthellae. Some unicellular photosynthetic plants, especially algae, known as *zoochlorellae*, live symbiotically in the outer tissues of certain sponges, coelenterates, molluscs and worms (Fig. 5.1E). Some brown or yellow cells, probably flagellates (*zooxanthellae*) are also present. Algae are photosynthetic and produce oxygen and nitrogenous

compounds beneficial to hosts and in exchange, they obtain materials such as carbon dioxide and nitrogenous wastes released by metabolism of host animals. The unicellular green alga, *Chlorella vulgaris* lives within the gastrodermal cells of *Hydra*. Likewise, alga *Zoochlorella* lives in a planarian, *Convoluta roscoffensis*.

8. Microorganisms and cellulose digestion. Interspecific mutualism is nicely demonstrated by the flagellate protozoan, *Trichonympha* an obligate anaerobe in the gut of several species of wood-eating termites where it digests cellulose. *Trichonympha* also occurs in the alimentary canal of wood-eating roach *Cryptocerus*. The termite and roach reduce the wood to small fragments, passing them through the alimentary canal to hind gut, where the protozoans digest the cellulose, changing it into sugar. The host benefits the protozoa by removing harmful metabolic waste products and maintaining anaerobic conditions in the intestine.

9. Fungus gardens. There exist certain mutualistic associations between insects and fungi. For example, the insect-fungus-galls of gall insects (e.g., midge, *Lasioptera*) are lined by a fungus which is parasitic on the host plant. It is believed that the female midge deposits spores of fungus at the time she lays her eggs in the plant or when insect sucks plant sap. Gall fungus seems to assist the insect indirectly by partly breaking down the tissues so that the insect can digest it (Fig. 5.3). Besides these



gall insects, there are various cases of insect gardeners such as ambrosia beetles, tropical termites and American ants which do gardening of fungi in their nests to get a regular supply of their food directly or indirectly.

II. Commensalism. Commensalism defines the coaction in which two or more species are associated and one species at least, derives benefit from the association, while the other associates are neither benefited nor harmed. The concept of commensalism has been broadened in recent years, to apply to coactions other than those centering on food; cover, support, protection, and locomotion are now frequently included (Bear, 1951; Kendeigh, 1974). Some common examples of commensalism are the following :

1. Lianas. These plants are common in tropical rain forests where light at ground level is scarce because of the dense and multistoreyed growth of vegetation. Lianas are vascular (woody) plants rooted in the ground but which climb up with the support of other trees and reach almost to the top of the forest canopy. The woody stem of lianas is closely attached to the supporting tree but it is not involved with it in any nutritional relationship. Thus, with much economy of mechanical tissues, lianas are able to get better light. On the basis of the type of device used for climbing their supports, lianas are classified as leaners, thorn lianas, twiners or tendrils lianas. In North Indian tropical deciduous forests *Bauhinia vahili* is a common liana. Other common lianas are species of *Ficus* and *Tinospora*.



Lianas.

In North Indian tropical deciduous forests *Bauhinia vahili* is a common liana. Other common lianas are species of *Ficus* and *Tinospora*.

2. Epiphytes. Epiphytes are the plants growing on other plants (Fig. 5.1F). They use other plants only as support and not for water or food supply. Epiphytes differ from the lianas in that they are not rooted into the soil. They may grow on trees, shrubs or larger submerged plants. They grow either on the trunks or leaves. Certain common examples of epiphytes are orchids, bromeliads, hanging “mosses”, *Usnea* and *Alectoria*. Epiphytes are most common in tropical rain forests. They derive their supply of moisture and nutrient from the frequent rains and debris accumulated in bark crevices. Dust is also a source of the nutrients. Roots of epiphytes often store water in a special tissue, called **velamen**. For meeting water requirements, in the epiphyte *Dischidia*, the

leaves fold to form a jug-like structure having a narrow mouth where water accumulates; roots of the epiphyte enter and grow in these specialized cavities of leaf. Many epiphytes develop a thick network of roots upon which wind blown dust accumulates and provides the necessary edaphic environment.

3. Epizoans. Some plants grow on the surfaces of animals. For example, some green algae grow on the long, grooved hair of sloth. *Basycladia* (Cladophoraceae) grows on the backs of fresh water turtle.

4. Epizoite. Likewise, ectocommensals or **epizoite** animals are associated with another animal for the purpose of anchorage and protection. Many small animals, for example, become attached to the



Fig. 5.2. Swollen thorns of *Acacia cornigera* on a lateral branch. Each thorn is occupied by 20-40 immature ants and 10 to 15 workers ants. All the thorns on the tree are occupied by one ant colony (after Krebs, 1985).

outside of larger ones, such as the protozoans *Trichodina* and *Kerona* on *Hydra*, vorticellids (*Vorticella*, *Epistylis*) or various other aquatic organisms, branchiobdellid annelids on crayfish, the barnacles, which attach themselves to the backs of whales and shells of horse-shore crabs and so on. The remora fishes attach themselves to the bellies of sharks, swordfish, tunny, barracuda, or sea turtle by means of a dorsal fin highly modified into a suction disk on the top of the head. They are of small size and are not burdensome to the host. The host benefits the remora, however, for when the host feeds, the scraps of food floating back are swept up by the remora.

5. Commensals may also be internal. For example, many harmless protozoans occur in the intestinal tract of mammals, including man. Some microorganisms such as bacterium *Escherchia coli* is found in human colon. The pea crab *Pinnotheres ostreum*, lives as a commensal in the mantle cavity of certain sea mussels; the crab steals food collected by the host mollusc but does little if any other known injury. The hermit crab normally lives inside gastropod shell. The pitcher of the pitcher plant found in bogs furnishes a breeding site or home for certain species of midge flies, mosquitoes, and tree toads. Many kinds of microorganisms, plants and animals (such as small crabs, shrimps, polychaets, ophiuroids, ascidians, etc.) live in the canal system of sponges.

The nest of one species often furnishes shelter and protection for other species as well. Ant nests may contain guest species of various other insects; some birds place their nests close to wasps, bees or ants for the protection offered by these insects. Woodchuck burrows are used also by rabbits, skunks and raccoons, especially in winter.

6. Commensalism-like associations also exist between a variety of microorganisms and higher plants. For example, the zone of soil around the roots of higher plants characterized by intense microbial activity, called the **rhizosphere**; the surface proper of the roots growing in soil—the **rhizoplane**; the boundary layer of air over the green leaves with active microbes—the **phyllosphere**; and the leaf surface—**phylloplane**, constitute important ecological niches, where occur rich populations of microorganisms including fungi, actinomycetes and bacteria. The living roots and leaves of higher plants are known to continuously diffuse various metabolic products, mainly sugars and amino acids. These are sources of nutrition for microorganisms present therein.

III. Protocooperation. Protocooperation is less extreme sort of interaction than mutualism in which the interaction is clearly beneficial to both species, allowing the equilibrium population levels of both to be higher than they otherwise would be. However, it is not obligatory for either species. Examples of protocooperation are many, including the relationship between water moccasins and large birds such as herons and ibises on several islands off the west coast of Florida. The birds nest in the lower branches of relatively unprotected trees, while the snakes congregate around the bases. This protects the birds from tree climbing predators such as raccoons. In turn, the snake feed, in part, on fish dropped by the birds and the occasional baby birds that fall out of the nest (Ehrenfeld, 1970).

Another good example of protocooperation is of coelenterate, sea anemone (*Adamsia palliata*) attached to the empty molluscan shell harbouring hermit crab (*Eupagurus prideauxi*; Fig. 5.1H). The

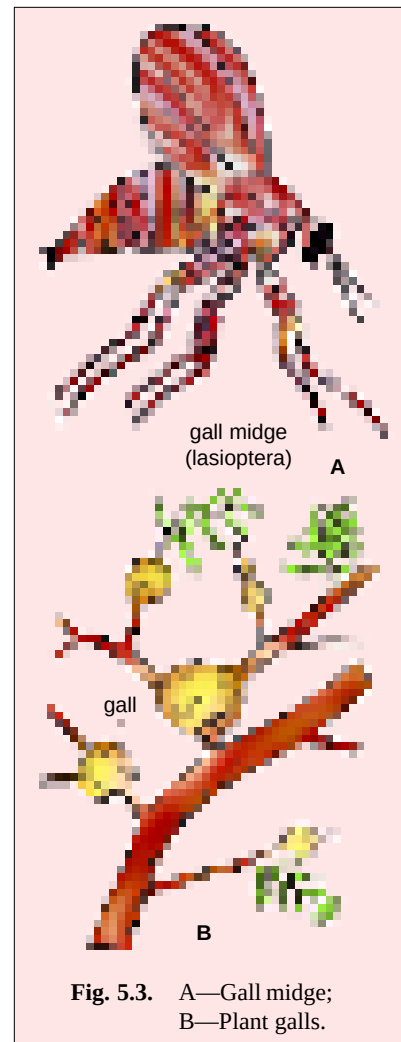


Fig. 5.3. A—Gall midge;
B—Plant galls.

sea anemone is carried by the crab to fresh feeding grounds and crab in turn is said to be protected from its enemies by sea anemone. Some ecologists prefer to include this association under commensalism.

B. Negative Interactions

In case of negative interactions, one or both species are harmed in any way during their life period. Some authors such as **Clarke**, 1954, prefer to call these types of interactions as **antagonism**. The negative interactions include the following three broad categories :

1. Exploitation. In exploitation, one species harm the other by making its direct or indirect use for support, shelter or food. It is of following types :

(i) Social parasitism. Social parasitism describes the exploitation of one species by another, for various advantages. It is a kind of parasitism in which the parasite foists the rearing of its young into the host. Social parasitism in various stages of development is found among some higher vertebrates and insects. For example, there occurs an **egg parasitism** in two species of birds— old world cuckoos and the brown headed cowbirds of North America, both of which do not build nests of their own, rather they deposit their eggs in nests of other species, abandoning eggs and young to the care of foster parents. Similarly, there occurs a **brood parasitism** in between two Indian birds—Indian koel and crow. The Indian koel (*Eudynamys scolopacea*), like many of its cuckoo cousins (Family Cuculidae), neither incubates its eggs nor feed its young. But these functions are secretly forced upon the house crow (*Corvus splendens*) and jungle crow (*Corvus macrorhynchus*) (**Davis, De, and Pal**, 1977). One species of ant attacks foraging workers of another species and snatches away the food they are transporting or the robber species may deliberately rob another nest for food. Some species of ants make slaves of the workers of other species. Such kind of dependency of one species on another also occurs in other social insects such as termites, wasps and bees. Social insects are apparently the only animals other than man to have succeeded in domesticating other species, and of cultivating plants particularly fungi for food.

(ii) Parasitism. Parasitism is a kind of harmful coaction (disoperation) between two species. It is the relation between two individuals wherein one individual called **parasite** receives benefit at the expense of other individual called **host**. Parasitism is mainly a food coaction, but the parasite derives shelter and protection from the host, as well. A parasite usually parasitizes a host which is larger in body size than it. Further, a parasite does not ordinarily kill its host, at least not until the parasite has completed its reproductive cycle. However, a host may die due to some secondary infection or suffer from stunted growth, emaciation, or sterility. The balance between parasite and host is upset if the host produces antibodies or other substances which hamper normal development of the parasite. In general the parasite derives benefit from the relation while the host suffers harm. But this does not mean that all parasites are harmful. The former category constitutes the **pathogenic parasites**.

Classification of parasites. Parasites exhibit a tremendous diversity in ways and adaptations to exploit their hosts. The parasites may be **viral parasites** (e.g., bacterial, plant and animal viruses), **microbial parasites** (e.g., bacteria, protozoa, fungi, etc.), **phytoparasites** (e.g., plant parasites) and **zooparasites** (e.g., animal parasites such as platyhelminthes, nematodes, arthropodes, etc.). They may parasitize microorganisms, plants and animals. They may occur on the outside of the host (**ectoparasites**) or live within the body of the host (**endoparasites**). The endoparasites usually live in the alimentary tract, body cavities, various organs or blood or other tissues of host. Ectoparasites may be parasitic only in the immature stages—the hairworm larvae, parasitic in aquatic insects; only the adult parasitic—fleas, on birds and mammals; or both larvae and adults may be parasitic—the blood sucking lice and flies, biting lice, mites, and ticks that occur on birds, mammals, and sometimes reptiles, and the monogenetic trematodes on fish. Similar relation occurs in endoparasites but in them usually all the stages of life are parasitic—entozoic amoebae, trichomonad flagellates, opalinid ciliates, sporozoans, pentastomids, nematodes, digenetic trematodes, acanthocephalons, cestodes, and some copepodes.

Further, endoparasites rely on various means of transport from one host to another so that their survival, range of distribution, and life cycle are unaffected. For the endoparasite, the interior of its host is its **microhabitat** and **microenvironment**, while the outside environment is the **macroenvironment**.

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The host acts as a buffer between the parasite and the outside environment. Changes in the microenvironment directly influence the parasite while alterations in the macroenvironment influence it directly. Thus, mutual adaptations and tolerance between the host and parasite are important factors facilitating successful transmission of endoparasites, and climatic fluctuations tend to alter the physiology of the parasite, host or vector. Mutual adaptations are the parasite's abilities to establish and maintain itself in a favourable location within the host and exist in a proper form for sufficient length of time enabling subsequent transport to another host and continuation of species.

Animals may also be parasitic to plants. Nematodes infest the roots of plants. Wasps or gnats form galls on plants such as oaks, hickories, willows, roses, goldenrods, and asters. Mites stimulate formation of witches brooms in hackberry. A variety of insects the larvae of which are leaf miners, wood borers, cambium feeders, and fruit eaters should be included here.

Lastly, parasites may be **full-time** (or **permanent**) **parasites** or **part-time** (or **temporary**) **parasites**. Mosquitoes and bugs that suck the blood of their hosts are temporary parasites. Some temporary parasites spend only a part of their life cycle as parasites. For example, glochidium larva of *Anodonta* (freshwater mussel) attaches itself to the body of the fish by means of its hooks and penetrates inside the fish integument to remain buried there for several weeks and finally emerges out as the young mussel to lead an independent existence. Permanent parasites, however, spend their life completely on other organisms. The common examples of permanent parasites are—*Plasmodium*, *Entamoeba histolytica*, and other protozoan pathogens, different platyhelminthes, nematodes, arthropods, etc.

Parasitic adaptations. While studying parasitism as a biotic factor influencing the life and activities of parasites and hosts we often encounter manifold adaptations, both offensive and defensive, being developed by the host, as well as by the parasite. An endoparasite entering a host often meets with the antibodies or phagocytic cells produced by the host. The ectoparasites, and endoparasites have following parasitic adaptations :

1. In parasitic animals, a reduction of organs of special sense of nervous system, and of locomotory organs occur.
2. Most ectoparasites develop some clinging organs such as hooks, suckers, etc., to get attached with the body of their hosts. They also develop special piercing and sucking organs to suck the blood of animals or sap of plants. Certain blood-sucking ectoparasites such as leeches, blood-sucking mosquitoes, etc., contain certain anticoagulant enzymes in their salivary secretions.
3. Most endoparasites exhibit anaerobic respiration, high rate of reproduction, parthenogenesis, hermaphroditism, polyembryony, intermediate host and a complicated life-cycle (**Lapage**, 1951). Some endoparasites, such as tapeworm, have become so adapted to the host that they no longer require a digestive system. They simply absorb their food directly through their body wall. Further, parasites that live within the bodies of plants and animals possess cuticles or develop cysts resistant to the digestive enzymatic action of the host.
4. Many parasites pass their entire existence in a single host; others require one, two, even three intermediate hosts. It is of ecological significance that both primary and intermediate hosts of a parasite occur in the same habitat or community. Even then the hazards to successful passage from one host to another are so great and mortality so high that enormous quantities of offspring are produced to ensure that at least a few individuals will complete the cycle.
5. Parasites are transferred from one host to another by active locomotion of the parasite itself; by ingestion, as one animal sucks the blood of or eats another; by ingestion as an animal takes in eggs, spores, or encysted stages of the parasite along with its food or drinking water; as a result of bodily contact between hosts; or by transportation from host by way of vectors. For example, the bacteria that cause tularemia in man are carried from rabbit to rabbit by ticks. Man contracts the disease when he handles infected rabbits but the incidence of infection is greatly reduced in the autumn when cold weather forces the ticks (*i.e.*, the vectors of bacteria) to leave the rabbits and go into hibernation.

Host specificity of parasites. Certain parasites, such as copepodes, infect a wide range of hosts

and they are ubiquitously present in various invertebrates and fishes. Most parasitic genera, however, are adapted to hosts of one phylum only. For example, each order of birds possesses its own particular species of tapeworms. This is true even when several order of birds live in the same habitats as do, for, instance, grebes, loons, herons, ducks, waders, flamingoes and cormorants (**Bear**, 1951). Species of flagellate protozoans that occur in termite alimentary canals are largely host-specific. Some species of gall wasps attack only one species of oak. Where a single species parasitizes two or more host species, the shape and structure of the gall formed around the egg and larva on both hosts is essentially similar. When several insects are found on the same oak, each kind of parasite produces its own characteristic gall form. The parasite specific gall formation is found to be related with parasite-specific enzyme (**Kinsey**, 1930).

Effects of parasite on the host : disease. Parasites may not cause immediate mortality but they cause damage to body structures, should it become excessive, may cause death. Because of these parasitically caused anatomical damages, the finely adjusted balance of different vital processes of host's body become disturbed and host is said to be diseased. There are a number of causes of diseases. Parasites are one; physiological stress, nutritional deficiency and poisoning are others. Some of the common parasitic agents of disease and consequent mortality of animals are following :

1. Viruses are the potent agents of several disastrous diseases of plants and animals including man. For instance, some viruses are the agents of hoof and mouth disease in deer, spotted fever in rabbit, encephalitis and distemper in foxes.

2. Bacteria may produce localized inflammatory changes in tissue, enter the blood stream, or produce powerful poisons known as **toxins**. They cause a variety of diseases, notably tularemia, paratyphoid, and tuberculosis among birds and mammals, as well as other diseases in lower types of organisms.

3. Fungus spores of *Aspergillus* may be drawn into the lungs of ground-feeding birds, where they germinate and grow, causing **aspergillosis disease**. Fungus may also develop on the external surface of animals.

4. Protozoan parasites are especially important in the alimentary tract and in the blood. A sporozoan species of *Eimeria* damages the walls of the intestine in upland game birds and causes **coccidiosis disease**; *Toxoplasma* becomes encysted in the brain of rodents; *Leucocytozoon* is a common blood parasite of waterfowl and game birds.

5. Worm parasites, such as tapeworms, nematodes, and acanthocephalans may wander through the host's body doing mechanical injury as well as destroying and consuming tissues. The host may respond by forming a fibrous capsule or cyst around an embedded parasite.

6. External parasites such as ticks, fleas, lice, mites, and flies do not commonly produce serious mortality by themselves, but they are often vectors, transmitting protozoa, bacteria and viruses from one animal to another. Heavy infestations of external parasites however lower the vitality of an animal and cause diseases of fur or feathers.

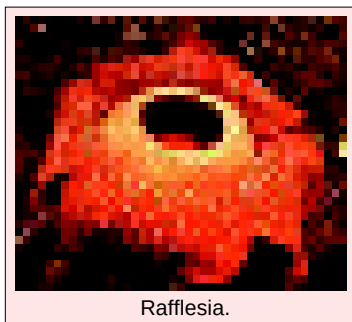
7. Nutritional deficiency in vitamins or minerals, or improper balance among carbohydrates, proteins, and fats may produce malformations, lack of vigour, or even death.

8. Food poisoning, **botulism**, occurs when certain food become contaminated with the toxins released by the bacterium *Clostridium botulinum*.

9. **Physiological stress** (**Selye**, 1955) is a term that has come to be applied to changes produced in the body non-specifically by many different agencies which may accompany any disease. Effects of stress include loss of appetite and vigour, ache and pains, and loss of weight. Internally, physiological stress causes following abnormalities, such as, acute degeneration of the lymphatic organs, decrease of the blood eosinophils, enlargement and increased secretory activity of the adrenal cortex, and a variety of changes in the chemical constitution of the blood and tissues.

Parasitism in the plants. There are some parasitic vascular plants. Species of *Cuscuta* (dodder) which are **total stem parasites**, grow on other plants on which they depend for nourishment. Young

stem twines around the host stem (Fig. 5.4) and it develops adventitious roots which finally penetrate the stem of host, establishing relationship with its conducting elements. The specialized roots are called **haustoria**. Other examples of such association are **total root parasites** as *Orbanche*, *Conopholis*, *Epifagus*, *Balanophora* and *Striga* which are found on roots of higher plants. *Orbanche* affects the yield of mustard and brinjal and is parasitic to the roots of Solanaceae, Cruciferae and Gramineae. *Rafflesia* is root parasite of *Vitis*. Partial root parasites belong to family Loranthaceae (*Viscum album*, *Loranthus* sp.) They grow rooted in branches of host tree. Examples of



Rafflesia.



Striga.

partial root parasites include *Santalum album* and *Thesium*. Fungi, bacteria, mycoplasmas, rickettsias and viruses are other common examples of plant parasites. Thus, fungi such as rusts and smuts are parasitic to many crop plants and reduce their production.

Parasitism may occur within the species. For example, in higher plants the growing pollen tube may be considered as a male plant parasitic on the tissues of the stigma and style of the flower (female plant).

Parasitoidism. Some Diptera and Hymenoptera deposit their eggs in the immature stages of other insects; the larvae on hatching feed on the host until they are fully grown. In such type of parasitism in which host generally dies of the larval destruction before the larva emerges and the parasitoid larva generally lives in spite of the host's death, is called **parasitoidism**. It stands in between parasitism and predation. The parasitoid may in turn be infested with **hyperparasitoids**. In the chichago, *Samia cecropia*, a saturniid moth, suffers the destruction of nearly 23% of its cocoon by an ichneumonid parasitoid, *Spilocryptus extrematis*, which deposits an average of 33 eggs on the inside of each cocoon or on the surface of larva. The host larva dies in a few hours after the parasitoid hatches and the ichneumonid larva moves about freely, feeding on the cuticle and burrowing into the tissues to drink the body fluids. *S. extrematis* is infested by another hyperparasitoid, *Aenoplex smithii* which causes about

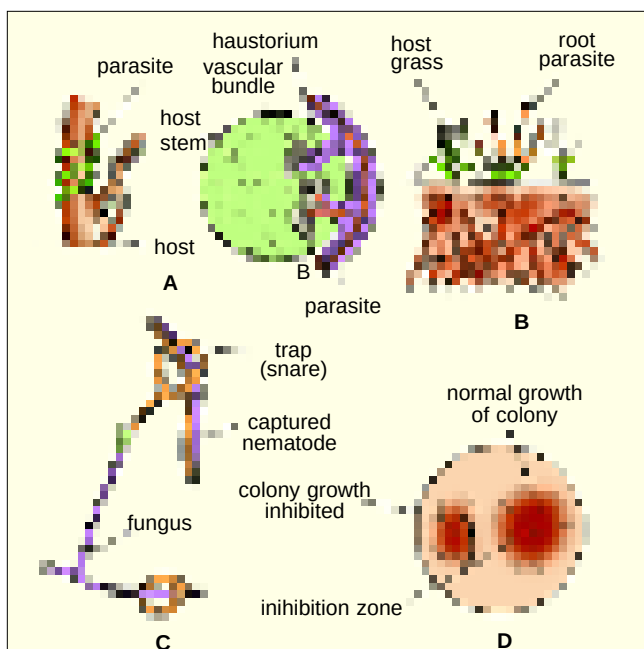


Fig. 5.4 Negative interactions. A—Parasitism, total stem parasites : a—*Cuscuta* twinning the host stem, b—T.S. of stems of host and parasite to show haustoria. B—A partial root parasite with its haustoria attached to the roots of host. C—Predation, a part of the mycelium of a nematophagous fungus (*Arthrobotrys dactyloides*) capturing the nematode as its prey. D—Antibiosis, inhibition of growth of one microbe colony by other colony.

13% destruction of *S.extrematis* cocoons. A chalcidid, *Dibrachys boucheanus*, feed both upon *S. extermatis* and as a tertiary parasitoid, upon *A. smithii*. A pycnidial fungus, *Cicinnobolus cesatii* is found as hyperparasite on a number of powdery mildew fungi.

(iii) Predation. Predation occurs when members of one species eat those of another species. Often, but not always, this involves the killing of the prey. Most of the predatory organisms are animals, but there are some plants (carnivores) also. Four types of predation have been recognized (see **Krebs**, 1985) : **Herbivores** are animals that prey on green plants or their seeds or fruits; often the plants eaten are not killed but may be damaged. Typical predation occur when **carnivores** prey on herbivores or on other carnivores. **Insect parasitism** is another form of predation in which the insect parasite lays eggs on or near the host insect, which is subsequently killed and eaten. Finally, **cannibalism** is a special form of predation in which the predator and prey are from the same species.

Predation is an important ecological process from three points of view. One effect of predation on a population may be to restrict distribution or reduce abundance. If the affected animal is a pest, predation appears useful. If it is a valuable resource such as caribou, the predation seems undesirable. Second, along with competition, predation is the second major process which affects community structure. Third, predation is a major selective force and many adaptations we see in organisms, such as warning colouration, have their explanation in predator-prey coevolution (Evolutionary changes in two or more interacting species are called **coevolution**.)

(A) Predation in animals. A predator animal tends to be larger than its prey. The main components of predation are predator and prey. A successful **predator** has following characteristics:

1. The hunting ability of a predator remains well developed.
2. By their hunting activities predators can be regarded as specialized or generalized. **Specialized predators** are those adapted to hunt only a few species. They are forced to move where the vulnerability of a staple prey item drops to a point where the predator population cannot support itself. For example, Peale's falcon shows a marked preference for ducks and pheasants. Deer exhibits a pronounced preference for certain species of browse (**Klein**, 1970). **Generalized predators**, not so restricted in diet, adjust to other food sources. The horned owl and buted hawk have a large range of collective prey available. Foxes can shift to vegetable and carrion diet, should conditions require it.
3. Hunting ability and success of predator involve the development of a searching image on the part of the predator. Once it has secured a palatable item of prey, the predator finds it progressively easier to find others of the same kind.
4. Though a predator may have strong preference for a particular prey, it can turn in the time of relative scarcity to an alternate, more abundant species that provides more profitable hunting. For example, if rodents are more abundant than rabbits and quail, foxes and hawks will concentrate on them instead of game animals.
5. Habitat preferences or overlapping territories can bring predator and prey into close contact, increasing prey risks. For example, predatory rainbow trout in Paul Lake, British Columbia, moves into the shoals when their prey, the red side shiner, are most heavily concentrated there.
6. Age, size, and strength of prey influence the direction that predation takes. Predators select food on the basis of size. Mountain lions, for example, avoid attacking large healthy elk, which they cannot successfully handle, and concentrate instead on deer and young or feeble elk (**Hornocker**, 1970).
7. Predators hunt only when it is necessary for them to procure food. The searching rate of predator is influenced by the speed of the predator relative to the speed and escape reactions of the prey, to the distance at which predators first notice and attack the prey, and to the proportion of attacks that result in successful capture.

Like the predators, the prey has its certain defensive specializations. The prey risk is determined

by density of prey population, availability of food and protective cover (concealment place), movement, activity, habits, size, age, strength and escape reactions of prey. Besides these striking morphological features, prey often develop the following two antipredator defence strategies : aposematic colouration and group living.

(i) Aposematic colouration. Many animals have evolved distastefulness as a means of predator defence. Toxins may be obtained from food or synthesized by the prey species. Many of these toxic prey species feature bright colours, termed **aposematic colouration** and these colours are usually regarded as a signal to predators. For example, Monarch butterflies advertise their noxiousness to blue jays by their large size and distinct colouration.

(ii) Group living. Group living is normally not advantageous for the individuals. Then why do birds form flocks and ungulates go about in herds ? One possible reason for living in groups is to reduce predation losses (**Bertram**, 1978). There are three main advantages a prey organism can obtain by living in a group. First, early detection of predators may be facilitated if many eyes are looking. Prey in groups may, thus, be able to spend more time feeding and less time looking for potential predators. Second, if the prey are not too much smaller than the predator, several of them acting together may be able to deter the predator from attacking. One possible example of this is mobbing in birds (e.g., crow). Primate troops will also mob predators in some circumstances and musk-oxen threatened by wolves gather into a circular defence formation. Third, if the predator is still able to attack a group, it must select one individual in the group to capture. By fleeing in confusion, the prey may confuse the predator, who may not be able to concentrate on any one individual. Alternatively, by being in the centre of a group, an individual may reduce its chances of being eaten.



Monarch butterflies.

Evolution of prudent predators. Coevolution of predator-prey system has produced **prudent predators** (a term coined by **Slobodkin**, 1961). If one predator is better than other at catching prey, the first individual will probably leave more descendants to subsequent predator generations. Thus, predators should be continually selected (= evolved) to become more efficient at catching prey. The problem, of course, is that by becoming too efficient, the predator will exterminate the prey and then suffer starvation. So a prudent predator should harvest prey with some constraint against overharvesting.

For example, a prudent predator would not eat prey individuals in their peak reproductive ages because that type of mortality would reduce the productivity of the prey population. Prudence dictates eating of those individuals that would die in the normal course and which contribute little to productivity. Often these are the oldest and the youngest individuals in a prey population which are hunted by the predator (**Slobodkin**, 1974). Old individuals may be post-reproductive, and young individuals often have a high death rate due to other cause.

Regulating effect of predation. It is commonly concluded that if the predator-prey interaction causes a reduction in the prey population, it is detrimental in same fashion to the prey. This belief has led to extensive "predator control" efforts conducted in the name of wildlife conservation. But the coevolution of species within natural ecosystems has in fact led to a dynamic balance between the

populations in a community, so that the population sizes of predator and prey species are inter-regulated by feed back mechanisms that effectively control the population of both.

(B) Predation in plants. Plants provide the following examples of their predation :

1. Carnivorous fungi. A number of fungi such as species of *Dactylella*, *Dactylaria*, *Arthrobotrys*, *Zoophagus*, etc., capture insects, nematodes and other worm-like animals. Such fungi use specialized structures such as **traps** or **snare**s (Fig. 5.5C) which are formed on their mycelia to capture the nematodes.

2. Carnivorous plants. A number of highly specialized plants such as *Nepenthes*, *Utricularia*, *Drosera*, *Dionaea*, *Darlingtonia*, *Sarracenia*, etc., are dependent for part of their nitrogen requirements upon small animals such as insects. These **insectivorous plants** have some specialized structures and mechanisms to attract, trap and ingest insects and then secrete some proteolytic enzymes to “digest” the protein contents of the insect body.

(C). Plants as prey (food) of animals: Browsing, grazing and scraping. Herbivores such as cattle and wild herbivores kill the plants and use unharvested herbs, shrubs or even trees as their food and sometimes pose great problem of management of natural and artificial vegetations (forests and grasslands). Different plants receive varying degree of damage as a result of browsing and grazing. Many insects and ruminants browse lightly over the vegetation. Cattle, camels, goats, etc., frequently browse the tender shoots of shrubs and trees and sheep graze the grasses. Generally, annuals suffer more due to grazing than the perennials, shrubs are damaged less than herbs.

Grazing has a profound effect on the composition, structure and physiognomy of vegetation.



Cynodon dactylon.

Heavy grazing reduces the photosynthetic parts so much that many plants succumb or their population decreases. This is true for palatable species. The unpalatable species avoided by grazing animals multiply and increase in number in absence of competition due to decrease in the population of palatable species. In a pasture with *Cynodon dactylon* and *Eragrostis amabilis* as dominant grass species, **Senapati** and **Dash** (1981) reported that a grazing pressure of 1.3 livestock/acre caused the accumulation of higher proportion of plant biomass (68%) underground compared to 46% in an ungrazed (i.e., protected) pasture. Plant diversity was more in the grazed plot and about 18% of NP (= nitrogen and phosphorus) was removed by grazing. Grazing also stimulated higher secondary production by soil invertebrate animals, particularly oligochaetes, but it also increased mortality due to hoof action and trampling.

In *Dichanthium annulatum*, there is a considerable loss of seeds and herbage due to grazing. The reduced reproductive capacity due to loss of seeds is compensated by a remarkable adaptation of increased reproductive buds on rhizomes (**Ambasht** and **Maurya**, 1970). The indirect effect of grazing is on the rate of soil erosion. This is certainly the greatest menace of grazing aspect of biotic factor. The decreased shoot cover ultimately affects root growth and exposes the soil surface to erosion due to rain and run off water and also to wind erosion.

Ambasht (1974) has indicated that grazing involves the large scale consumption of seeds, due to their high nutritive value, which results in poor successive crops, unless the plants adapt to vegetative



Nepenthes.

propagation. Rats selectively remove seeds from grasses and store them in their burrow in large quantities. Fruit-eating birds and animals have some advantageous effects also. Some seeds fail to germinate unless they have passed through the gut of birds or other animals. The seed coat in these seeds are hard and impermeable to water or oxygen. The seed coat is softened and often digested by the animals and the rest of the seeds come out and germinate quickly.

Further, in the outskirts of Indian forests, villagers tend to scrape grasses to feed their cattle in summer. **Scraping** is usually done right at the ground level. Completely exposed and loosened top soil is blown by strong summer wind. This is followed by rainy season, during which initial heavy showers result in great soil erosion.

2. Amensalism. Amensalism is a comprehensive term in biotic interactions to denote the adverse or depressing effect of one of the competing population or species while the other, remains stable. Thus, amensalism is a situation in which one population definitely inhibits the other while remaining unaffected itself. By so modifying the environment, the organism improves its own chance of survival. Ammensalism which commonly involves some type of chemical interaction, by which the organisms of one species affects the well-being and growth of individuals or the population biology of other, is called **allelochemic** (see **Whittaker**, 1970, **Smith**, 1974; **Kendeigh**, 1974). The inhibitor substance may be inorganic chemicals such as acids or bases which are produced by pioneering organisms (inhibitor species) and which reduce the competition for nutrients, light, and space, between the amensal species and inhibitor species. The production of relatively simple organic toxins is another source of chemical inhibitors. These toxins inhibit seedling growth in the vicinity. This may affect succession of plant species, especially important in the early stages (**Muller**, 1966). A third type of inhibitor chemical is the **antibiotic**—a potent antimicrobial agent. It is a substance produced by an organism, which, in low concentrations, can inhibit or kill the growth of another organism.

The majority of inhibiting chemicals are produced as secondary substances by plants and released into the soil through the roots or leaf wash. The suppression of growth through the release of chemicals by a higher plant is known as **allelopathy**. Thus, allelopathy (*allelon* = each other + *pathy* = suffering) means chemical control of distribution among plants. For example, walnut tree produces a non-toxic substance, juglone (**Bode**, 1958) which is found in its leaves, fruits and other tissues. When the leaves or fruits fall on the ground, juglone is released to soil, where it is oxidized to a substance that inhibits the growth of certain under story species and garden plants such as heaths and broad-leaf herbs, and favours others such as bluegrass and blackberries. In Southern California, *Salvia leucophylla* leaves emit some volatile oils which reach the soil surface and inhibit the germination of seeds of other species and the inhibitory effects persist for several months (**Muller**, 1966). Similar allelopathic effects are also reported to be exerted by *Adenostoma fasciculatum* (**Christensen** and **Muller**, 1975). **Datta** and **Chakrabarti** (1982) prepared weed plant extracts such as of *Croton bonplandianum* and *Clerodendrum viscosum* in water and tested their effects under different dilutions on seed germination and growth of crops as pea, mustard, rice and lettuce. *C. bonplandianum* inhibits germination of seeds of above crops except of pea at high concentration. They found that the active ingredient in *C. bonplandianum* leaves showing allelopathic potential are **abscisic** and **phaseic acids**, while in *C. viscosum* inhibiting mustard seed germination may be a **terpene**. *Clerodendrum* is also reported to exert allelopathic potential on several weeds such as *Abutilon indicum*, *Amaranthus spinosus*, *Cassia tora*, etc. (see **Ambasht**, 1990).

Agriculturists have recognized the action of **smother crops** as weed suppressors. The smother crops include barley, rye, sorghum, millet, sweet clover, alfalfa, soyabeans and sunflowers. Their inhibition of weed growth was assumed to be due to competition for water, light or nutrients. However,



Salvia leucophylla.

Overland showed that barley (*Hordeum vulgare*) inhibited the germination and growth of several weeds, even in the absence of competition for nutrients or water. He found that the adverse effect of barley is partly due to the secretion by its roots of chemicals (alkaloid) that reduce growth and germination of nearby weeds.

The term **antibiosis** generally refers to the complete or partial inhibition or death of one organism by another through the production of some substance or environmental conditions as a result of metabolic pathway. These substances and/or conditions are harmful (antagonistic) to other organisms. Antibiosis is more commonly referred to secretions by microorganisms that check the growth of others. Production of chemicals that are antagonistic to microbes—the **antibiotics** is well known. Antibiotics produced by bacteria, fungi, actinomycetes, and lichens are widespread in nature and may be one of the reasons why bacteria pathogenic to man cannot multiply well in soils. *Penicillium*, a fungus found in soil produces antibiotic substances that check the growth of a large variety of bacteria. A number of antibiotics such as penicillin, have been used extensively in human medicine.



Amaranthus spinosus.

Antagonistic substances are also reported in some algae, as for example in culture of *Chlorella vulgaris*, some substance accumulates which inhibits the growth of the diatom, *Nitzschia frustulum*. Substances produced by senescent cultures of *Chlorella* and of the diatoms, *Navicula* and *Scendesmus* inhibit the filter-feeding of *Daphnia* in laboratory. Pond “blooms” of blue-green algae especially *Microcystis* are known to produce toxins such as hydroxylamine which causes death of fish and cattle. In marine waters, populations of some microbes, popularly known as **red tide**, cause catastrophic destruction of fish and other aquatic animals.

Allelochemic effects are of great variety in both plants and animals : repellants, escape substances, suppressants, venoms, inductants, counteractants, attractants, signals, stimulants, autotoxins, autoinhibitors, and so on. **Pheromones** (ectohormones) are chemical messages between members of a species especially important in reproductive behaviour, social regulation and recognition, alarm and defense, territory or trail marking, food location, and so on. Many of these effects are beneficial to the individual, while other serve for competitive purposes (**Whittaker** and **Feeny**, 1971).

3. Competition. The presence of other organisms may limit the distribution of some species through **competition**. Such competition can occur between any two species that use the same sorts of places. Note that two species do not need to be closely related to be involved in competition. For example, birds, rodents and ants may compete for seeds in desert environments. Competition among animals is often over food, water and mates. Animals may also do competition for space, i.e., nesting sites, wintering sites, safer sites from predators. Plants can compete for light, water, nutrients, or even for pollinators and/or attachment sites. Competition is an important process affecting the distribution and abundance of plants and animals.

There are two different types of competition, defined as follows (**Birch**, 1957) :

(a) Resource competition (also called **scramble competition**) occurs when a number of organisms (of the same or of different species) utilise common resources that are in short supply.

(b) Interference competition (also called **contest competition**) occurs when the organisms seeking a resource harm one another in the process, even if the resource is not in short supply.

Note that the competition may be **interspecific** (between two or more different species) or **intraspecific** (between members of the same species).

(A) Intraspecific competition. It is an important density-dependent factor regulating populations.

The wildebeest (gnu) population is thought to be regulated by intraspecific scramble competition for a limited supply of grass of adequate quality in dry season.

Intraspecific competition is responsible for levelling off or fluctuations around a certain density of a population. Exact mechanism do vary in different species. Intraspecific competition is worked out in the beetle, the azuki bean weevil (*Callosobruchus chinensis*) which feed on stored legume seeds. Figure 5.5 shows results from a series of experiments begun with different numbers of adult weevils. Thus, at high density, female fecundity is reduced because overcrowding leads to fewer successful matings due to mechanical interference between individuals. These conflicts are the main factors which result in the decrease in population density. Minor effects include increased egg mortality and increased level competition for food at high density. The egg mortality is due to adult knocking eggs from the sites at which they were laid.

Dash and **Hota** (1980), **Misra** and **Dash** (1984) and **Mohapatro** and **Dash** (1984) studied the density effect on the growth and metamorphosis of three species of tropical anuran tadpoles *Rana tigrina* (= *tigerina*), *Polypedates* (= *Rhacophorus*) *maculatus* and *Bufo stomaticus*. For each of these tadpoles, they found the competition for food due to crowding which affects growth rate and the threshold size for metamorphosis. In very high density populations (with severe competition) some larvae were not able to attain the threshold size and could not metamorphose (see **Dash**, 1990).

(B) Interspecific competition. **A.J. Lotka** and **V. Volterra** in 1920 developed theoretical or mathematical models to explain prey-predator interactions which produce oscillatory patterns. They also discussed the problem of competition. If two different species populations require a common resource for living and if this resource is potentially or actually limiting, then the two species are said to be in competition for it. These models predict the following three types of outcomes :

1. One species only survives, it being the one with the greatest negative effect on its competitor. Growth of the surviving population to its carrying capacity is slower than if the second population had been absent.

Thus, only one species will survive and there will be an exclusion or displacement of one of the populations by competition. This is called **Gause's principle of competitive exclusion** which states that complete competitors cannot exist.

2. Both species coexist indefinitely. This occurs when interspecific competition is less intense than intraspecific one in both species. Neither population reaches the carrying capacity it would have in the absence of other species.

3. The species beginning at higher density persists, and the other is eliminated. This is a special case when the populations have equally negative effects on the growth of each other, but interspecific competition is stronger than intraspecific one.

Example. (i) **Gause** studied the growth patterns of two species of *Paramecium* (*P. aurelia* and *P. caudatum*) when cultured together and separately (Fig. 5.6). Each species population has a sigmoidal growth pattern when grown independently. However, when both species grown together, their growth patterns were sigmoid in the first week, but later there was a gradual increase of *P. aurelia* and gradual decrease of *P. caudatum*. *P. aurelia* too did not grow to the level it had done when grown separately. Thus, this experiment confirmed the competitive exclusion principle.

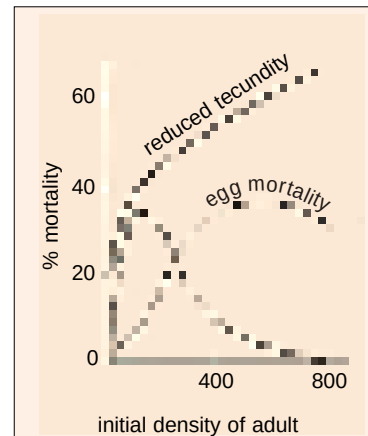


Fig. 5.5. Intraspecific competition between azuki bean weevils change as population density is increased in laboratory cultures. Fecundity is reduced and mortality among younger shows more complex and contrasting patterns.

(ii) The principle of competitive exclusion applied well to comparative studies of **Thomas Park** (1948-1962) using the grain beetles, *Tribolium confusum* and *Tribolium castaneum* under different environmental conditions.

(iii) Introduction of wasp parasite, *Opius oophilus* to control a fruit fly pest in Hawaii, apparently displaced the other two wasp species from the field.

(iv) The tramp ant, *Wasmannia auropunctata* has displaced all the native ants in some areas of Santa Cruz Island in the Galapagos group.

(v) **Misra** and **Dash** (1979) studied interspecific interactions in four species (*Lampito mauritii*, *Drawida calebi*, *Octochaetona surensis* and *Peryonix millardi*) of tropical earthworms. The rate of oxygen consumption of the different species of earthworms taken in mixed species pairs, showed significant reduction when compared to the sum of their individual rates of respiration.

(vi) **Ramakrishnan** and **Gupta** (1980) worked on the competition in related species of *Argemone* (*A. mexicana* and *A. ochroleuca*). They found that *A. mexicana* dominates over *A. ochroleuca* in mixed populations if seedling establishment occurs in the later part of the growing season and *A. ochroleuca* dominates if the seedling establishment occur in the early part. Thus, environmental conditions influence the competitive ability of the species.

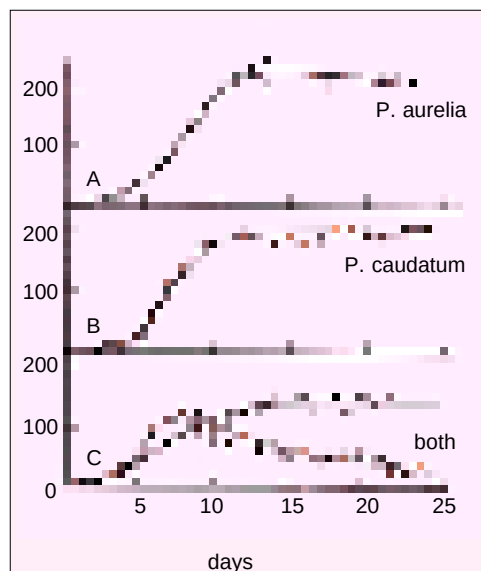


Fig. 5.6. Population growth and interspecific competition between two species of *Paramecium* in laboratory cultures. When cultured separately (A) and (B), the two species approximate to logistic growth. However, in mixed cultures (C) *P. aurelia* survives and *P. caudatum*

REVISION QUESTIONS

- Describe the phenomenon of mutualism by providing examples both from plants and animals.
- What is exploitation? Describe, with suitable examples, this negative ecological interaction.
- Discuss the role of parasitism as a biotic ecological factor and compare it with predation.
- Write an essay on 'prey-predation interaction.'
- Describe ecological effects of browsing, grazing and scrapping.
- Describe with appropriate examples the ecological aspects of competition.
- Discuss the phenomenon of ammensalism.
- Discuss with suitable examples the positive and negative interactions between two different populations.
- Write short notes on the following :
(i) Mutualism ; (ii) Commensalism ; (iii) Protocooperation ; (iv) Mycorrhizae ; (v) Epiphytes ; (vi) Social parasitism ; (vii) Fungus gardens ; (viii) Prudent predators ; (ix) Aposematic colouration ; (x) Antibiosis.

CHAPTER

6

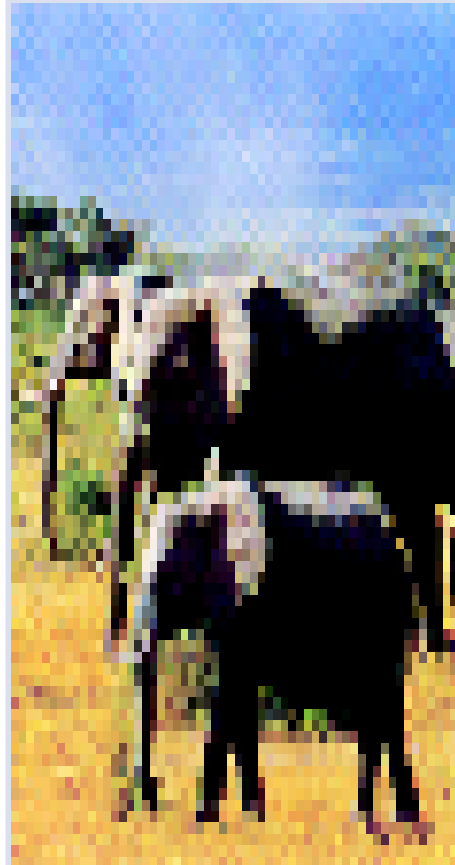
Population

(Population Ecology)

The term population has its origin in the Latin word *populus*, meaning people. In ecology, a **population** may be defined as a group of organisms of the same species occupying a particular space. Thus, we may speak of the deer population of Corbett National Park, the deer population of Kaziranga Wild Life Sanctuary, the human population of Delhi, or the human population of India. The ultimate constituents of the population are **individual organisms** that can potentially interbreed. The populations may be subdivided into **demes** or local populations, which are groups of interbreeding organisms, the smallest collective unit of a plant or animal population. Individuals in demes, thus, share a common gene pool. The boundaries of a population both in space and in time are vague and in practice are usually fixed arbitrarily by the investigator.

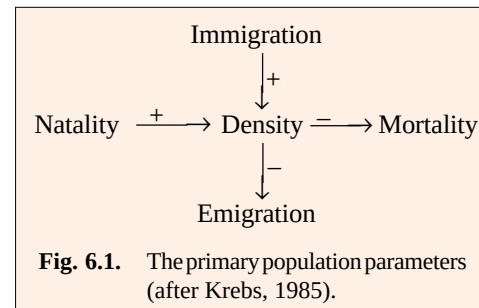
Some ecologists recognized following two types of populations : 1. **Monospecific population** is the population of individuals of only one species; and 2. **Mixed** or **polyspecific population** is the population of individuals of more than one species. Often ecologists use the term **community** for polyspecific population.

The population has various **group characteristics**, which are statistical measures that cannot be applied to individuals. These group characteristics are of three general types. The basic characteristic of a population is its **size** or **density** which is



Similar species of organisms constitute a population.

affected by four primary population parameters such as **natality** (births), **mortality** (deaths), **immigration** and **emigration** (Fig. 6.1). In addition to these attributes, one can derive secondary characteristics of a population, such as its **age distribution**, **genetic composition**, and **pattern of distribution** (distribution of individuals in space). These population parameters result from a summation of individual characteristics. Thus, group attributes of population are different from those of an individual, for instance an individual cannot have a density, it cannot have a rate of birth or death, although it occupies a place, is born and dies. The group attributes of a population with respect to rates of birth, death, reproduction, etc., form the science of **demography**. However, when they are studied from a ecological point of view, this forms a **population ecology** or **democology**. In other words, population ecology is the study of individuals of the same species where the processes such as aggregation, interdependence between individuals, etc., and the various factors governing such processes are emphasized.



The ecological study of populations include the following three main aspects :

1. Population characteristics;
2. Population dynamics;
3. Regulation of population.

POPULATION CHARACTERISTICS

The population has the following characteristics :

A. Population Size and Density

Total **size** of population is generally expressed as number of individuals in a population. The population size (N) at any given place is determined by the processes of **birth** (B), **death** (D), new arrivals from outside or **immigration** (I) and going out or **emigration** (E). Therefore, change in population size between an interval of time N_{t+1} is N_t (initial stage) + B – D + I – E.

Population density is defined as numbers of individuals per unit area or per unit volume of environment. Larger organisms as trees may be expressed as 600 trees per hectare, whereas smaller ones such as phytoplanktons (as algae) as 2 million cells per cubic metre of water. In terms of weight it may be 100 kilograms of fish per hectare of water surface. Density may be **numerical density** (number of individuals per unit area or volume) or **biomass density** (biomass per unit area or volume). When the size of individuals in the population is relatively uniform, as mammals, birds or insects, then density is expressed in terms of number of individuals (numerical density). But, when the size of



Cassia tora.

individuals is variable, such as true of fishes, trees or mixed populations (biomass density). To measure density of a population in terms of measurement of biomass will be satisfactory as a measure of density biomass one can take wet weight, dry weight, volume and carbon and nitrogen weight. Population density is also measured as **abundance** or absolute number of population. Generally, in an area smaller animals are more abundant than larger animals. But birds are less abundant than mammals of equivalent size (Krebs, 1985).

Further, since the pattern of distribution of organisms in nature is different it becomes important to distinguish between crude density and ecological density. **Crude density**

is the density (number or biomass) per unit total space. **Ecological density** (also called **specific** or **economic density**) is the density (number or biomass) per unit of habitat space, i.e., available area or volume that can actually be colonised by the population. In nature, organisms grow generally clumped into groups and rarely are uniformly distributed. For example, in plant species such as *Cassia tora* and *Oplismenus burmanni* individuals are found more crowded in shady patches, and few in other parts of same area. Thus, density calculated into total area (shady as well as exposed) would be crude density, whereas the density value for only shady area (where the plants actually grow) would be ecological density.

Fine distinction between the crude density and ecological density becomes more clear by the Figure 6.2, which is based upon **Kahl's** (1964) study on fish density in a water body. As shown in the figure the crude density of small fish in the area as a whole goes down as the water level drops during the winter dry season, but the ecological density in the contracting pool of water increases as fish are crowded into smaller and smaller water area. Thus, with changing time, water level and total number of fish, the two densities become different.

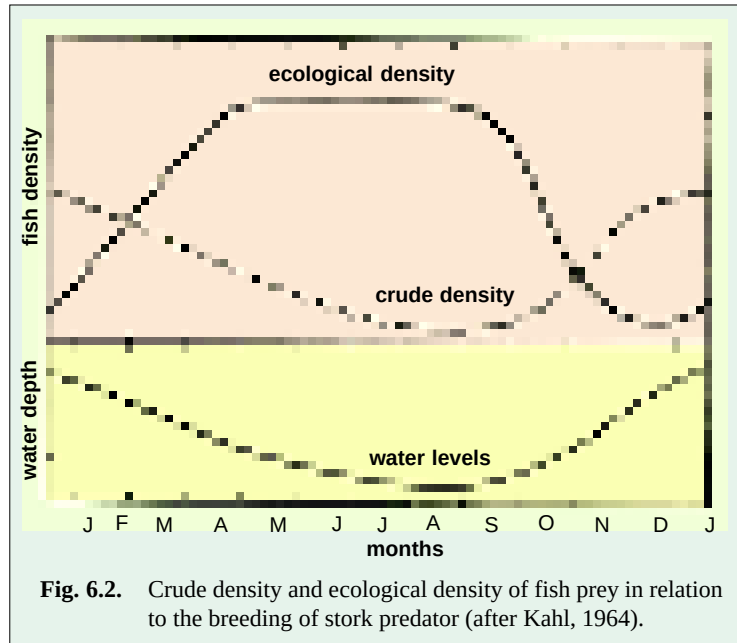


Fig. 6.2. Crude density and ecological density of fish prey in relation to the breeding of stork predator (after Kahl, 1964).

B. Patterns of Population Dispersion (or Spatial Distribution)

Individuals of a population arrange themselves in a manner that is specific for each population and these arrangements in space appear to be of great significance in the studies of the dynamics of the ecosystems. In nature, due to various biotic interactions and influence of abiotic factors, the following three basic population distributions can be observed : random distribution, uniform or regular dispersion, and clumped dispersion (Fig. 6.3).

C. Age Structure

In most types of populations, individuals are of different age. The proportion of individuals in each age group is called **age structure** or **age distribution** of the population. Age distribution of the population influences natality (birth rate) and mortality (death rate). Reproductive ability is determined by age of the females and, hence, natality denotes the numerical representation of young ones produced in unit time by the same age group. Similarly chances of death are more towards the earlier and later periods of life span and so mortality is under the control of age. Therefore,

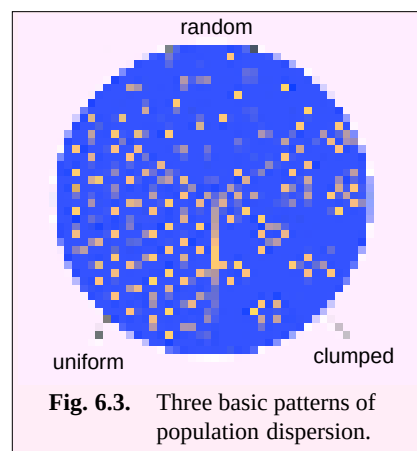


Fig. 6.3. Three basic patterns of population dispersion.

the ratio of the various age groups in a population determines the current reproductive status of the population and helps in anticipating its future. From an ecological point of view there are three major **functional** or **ecological ages** (age groups) in any population. These are (i) **pre-reproductive** (or juvenile or dependent phase), (ii) **reproductive** (or adult phase) and (iii) **post-reproductive** (or old age). The relative duration of these age groups in proportion to the life span varies greatly with different organisms. In human beings, the “three” ages are relatively equal in length, about a third of his life falling in each class. Many plants and animals have very long pre-reproductive period. For example, insects have extremely long pre-reproductive periods, a very short reproductive period and no post-reproductive period.

Age pyramids. A model representing geometrically the proportion of different age groups in the population of any organism is called **age pyramid** (or **age-sex pyramid**); (see **Ananthkrishnan** and **Viswanathan**, 1983).

An age pyramid is a vertical bar graph in which the number or proportion of individuals in various age ranges at any given time is shown from youngest at the bottom of the graph to oldest at the top. It is very useful in monitoring the commercial exploitation of food species by man among other things. According to **Bodenheimer** (1938), there are following three basic types of age-sex geometric figures (Fig. 6.4) :

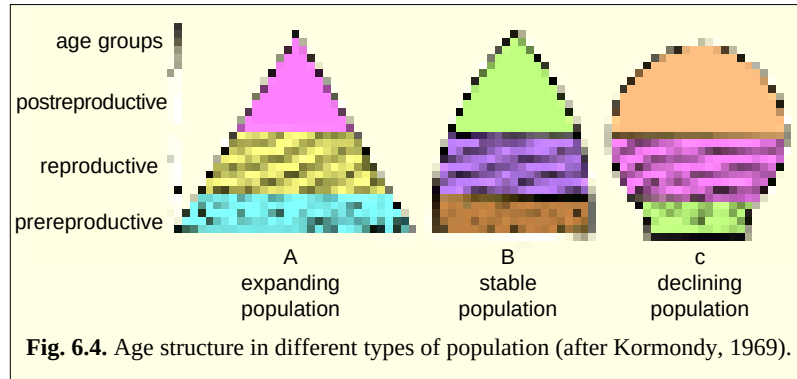


Fig. 6.4. Age structure in different types of population (after Kormondy, 1969).

1. Pyramid with broad base (or Triangular structure). It indicates a rapidly expanding population with a high percentage of young individuals and only few old individuals. Thus, in rapidly growing young population, birth rate is high and population growth may be exponential as in yeast, housefly, *Paramecium*, etc.

2. Bell-shaped polygon. It indicates a stationary population having an equal number of young and middle-aged individuals. As the rate of growth becomes slow and stable, *i.e.*, the pre-reproductive and reproductive age groups become more or less equal in size, post-reproductive group remaining as the smallest.

3. Urn-shaped structure. It indicates a low percentage of young individuals. It shows a declining population. Such an urn-shaped figure is obtained when the birth rate is drastically reduced, the pre-reproductive group dwindles in proportion to the other two age groups of the population.

D. Natality

Population increases because of natality. **Natality** is a broader term covering the production of new individuals by birth, hatching, germination, or fission. The **natality rate** may be expressed as the number of organisms born per female per unit time. In human population, the natality rate is equivalent to the ‘birth-rate’.

At this stage, one should be clear regarding two essential aspects of the reproduction which forms main component of natality : 1. **Fertility** is the physiological notion which indicates that *an organism is capable of breeding*. 2. **Fecundity** is an ecological concept that is based on *the numbers of offspring produced during a period of time*. Fecundity is of two types—**potential fecundity** and **realized fecundity**. For example, the realized fecundity rate for an actual human population may be only one birth per 8 years per female in the childbearing ages. Whereas the potential fecundity rate for human is one birth per 9 to 11 months per female in the childbearing ages (see **Krebs**, 1985).

Natality is of following two types :

1. Maximum natality. (Also called **absolute**, **potential** or **physiological natality**). It is theoretical maximum production of new individuals under ideal conditions which simply means that there are no ecological limiting factors and that reproduction is limited only by physiological factors. Hence, absolute natality is constant for a species population. This is also called **fecundity rate** (see **Sharma**, 1994).

2. Ecological natality. (Also called **realized natality** or simply **natality**). It refers to population increase under an actual, existing specific condition. Thus, ecological natality takes into account all possible existing environmental conditions. This is also called **fertility rate** (see **Sharma**, 1994). Ecological natality varies with the size and composition of the population, and with environmental conditions. It is expressed as—

ΔNn = production of new individuals

$\Delta Nn / Dt$ = the absolute natality rate (B)

$\Delta Nn / NDt$ = the specific natality rate (b) (i.e., natality rate per unit of population)

where, N = the reproductive part (mature female) of the population or initial number of organisms or total population

n = new individuals in the population

t = time

Δ = delta : a change in value.

Further, the rate at which females produce offsprings is determined by the following three population characteristics :

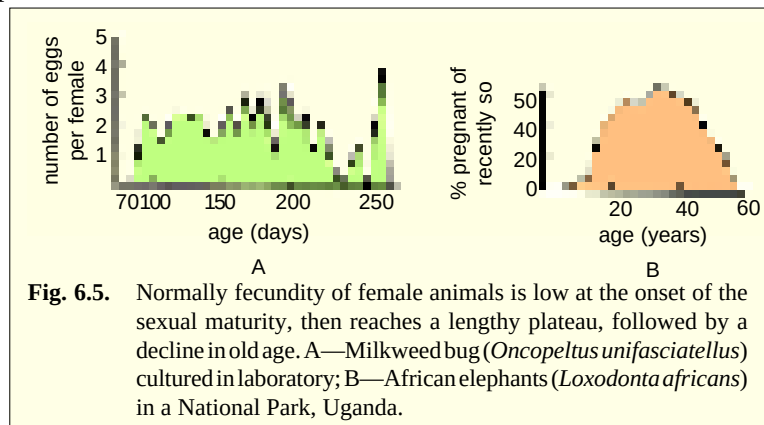
1. **Clutch size** or the number of young produced on each occasion;
2. the time between one reproductive event and the next, and
3. the age of first reproduction.

Thus, natality usually increases with the period of maturity and then falls again as the organism gets older (Fig. 6.5). But there are some trees which continue to increase fruit production as they get older. Natality patterns differ in tropical and temperate populations.

Fecundity. Species vary greatly in the

characteristic number of generations, broods or litters produced per year, and in the size of them. Protozoans often divide so rapidly that they produce a new generation every few hours. Plankton organisms, less fecund, may produce a new generation every few days. Many vertebrates breed but once a year, some large animals only once every two or three years. Several species of small birds and mammals have two or more broods per year. Rodents may continue to breed throughout the winter under favourable environmental conditions, so that their reproductive potential is enormous.

The maximum size of a litter is determined by the physiological and morphological characteristics of the species. In mammals, which produce viviparous young, the size of uterus and body cavity as well as the number of mammary glands for suckling the young after birth acts as limiting factors for the size of the litter. In birds there is a limit on the number of eggs that one individual can cover and successfully



incubate. However, in species that do not take care of their eggs after laying, the number produced may be limited only by the energy resources of the parent.

The number of eggs or young produced per litter is correlated inversely with the amount of attention (*viz.*, parental care) that they require. When parental care is altogether lacking, invertebrates may lay 1,000 to 500,000,000 eggs at one maturation; where there is some protection afforded by brood pouches, 100 to 1,000 eggs may be laid; with a high degree of brood protection, 1 to 10 or more eggs may be laid. Mammals seldom have more than a dozen young in a single litter and in larger species usually only one. Characteristic clutch size among birds varies from 1 to 15; rarely, 20.

Further, there is a limit on the size of the brood or litter that adult warm blooded animals can successfully feed and raise to maturity. There is no advantage, for example, for starlings to have broods larger than five. In larger broods, each individual receives less food, and, hence, has less vigour and weight on leaving the nest. In such weak individuals, mortality increases either before fledging or in subsequent months (Perrins, 1965). In those species that feed their young in the nest, the clutch size has evolved through natural selection to the greatest number that can be hatched and raised successfully through efforts of the adults (Lack, 1967). In those species whose young leave the nest and feed themselves at hatching, the clutch size depends in large part on the capability of the female to mobilize energy in her body to produce eggs of a particular size (Ryder, 1970). The variability in clutch and litter size for most species allows them to take advantage of temporarily improved conditions.



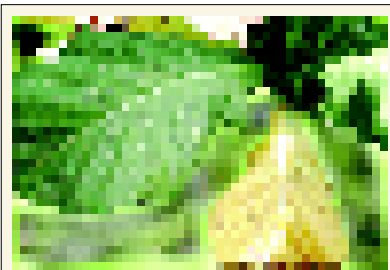
White tail deer.

The size of brood, litter or clutch is also regulated by weather. For example, the clutches laid by birds during periods of hot weather are usually smaller than those laid when temperature is moderate (Kendeigh, 1941). Clutches laid by related species in temperate latitudes tend to be larger than those laid in the tropics. Likewise, the fecundity of white-tailed deer is higher with good forage than with poor forage. The reproduction is generally more successful after periods of high mortality than during years of abundance.

E. Mortality

Mortality means the rate of death of individuals in the population. It is a negative factor for population growth. Like natality, mortality may be of following types :

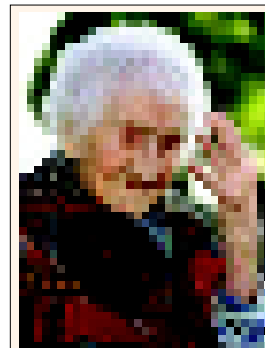
1. Minimum mortality. It is also called **specific** or **potential** mortality. Minimum mortality represents the theoretical minimum loss under ideal or non-limiting conditions. It may be constant for a population.



Senescence in plant leaves.

2. Ecological mortality. It is also called **realised mortality**. It is the actual loss of individuals under a given environmental condition. Ecological mortality is not constant for a population and varies with population and environmental conditions, such as predation, disease and other ecological hazards.

All organisms have a **physiological** or **potential longevity** representing the age (called **old age**) up to which the organisms can live under ideal conditions without stresses or diseases and death occurs purely due to aging or **senescence**.



Aging.

Like natality, mortality may be expressed as the number of individuals dying in a given period (death per time), or as specific rate in terms of units of the total population or any part thereof. The death rates vary among species and are correlated with rates of reproduction (natality). The death rate of a species is influenced by a number of factors such as destruction of nests, eggs or young by storms, wind, floods, predators, accidents, and desertation of parents, but of fundamental importance is the number of young that are born in relation to the carrying capacity of the habitat. When more young are born than the habitat can support, the surplus must either die or leave the area. Mortality rate is usually expressed as a percentage of individuals dying within a given time. The mortality rate in many species varies from one age level to another; thus, a mean death rate has only general significance. In birds, however, the death rate is nearly

constant once they become adult, and it is then apparent that it varies inversely with adult longevity. In adult penguins, pelicans, shore birds, gulls, and swifts, the annual mortality rate is commonly between 12 and 30 per cent; in herons, hawks, and owls it is about 30 per cent; in ducks, doves, and song birds it is between 40 and 68 per cent, while in gallinaceous birds it is the highest, 60 to 80 per cent. As mortality varies positively with age in the majority of organisms, the specific mortalities at particular ages can be illustrated in the form of life table.

Vital index and survivorship curves. A birth-rate ratio ($100 \times \text{births/death}$) is called **vital index**. For a population the surviving individuals are more significant for a population than the dead ones, the death rate can be represented by **survivorship curves** or **life tables**, both of which provide an estimate of organisms surviving at various ages.

(i) Survivorship curves. The pattern of mortality with age is best illustrated by survivorship curves which plot the numbers surviving to a particular age. There are following three types of survivorship curves (Fig. 6.6) which represent the different nature of survivors in different types of populations :

(a) Highly convex curve. This type of curve (pattern I) in the figure is the characteristic of the species in which the mortality rate of the population is low until near the end of the life span. Thus, such species tend to live throughout their life span, with low mortality rate in young and adult. Many species of large animals as deer, mountain sheep, man and small rotifers show such curves. Among various large-sized perennial plants, generally the plant die after reaching reproductive phase within a period of old age.

(b) Highly concave curve. This type of curve (pattern III) is the characteristic of such species where mortality rate is high during the young stages. Some birds, blacktail deer, oysters, shell fish, oak trees, short-lived weedy annuals, etc., exhibit pattern III type curve.

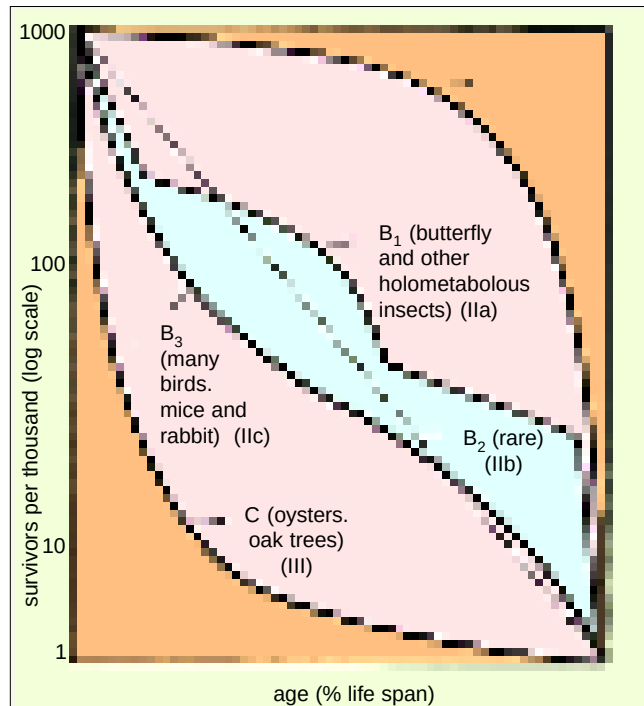


Fig. 6.6. Different types of survivorship curves plotted on the basis of survivors and age. Examples of each has been given within the brackets.

(c) Diagonal straight-line curve. This type of curve (pattern II b) indicates an age-specific constant survivorship, *i.e.*, a constant rate of mortality occurs at every age. Some animals such as hydra, gull, American robin, etc., exhibit this type of curve.

In fact, no population in the real world has a constant age-specific survival rate throughout the whole life span. Thus, a slightly concave or sigmoid curve (pattern IIc) is characteristic of many birds, mice and rabbits. In them, the mortality rate is high in the young but lower and almost constant in the adult (1 year or older). In some holometabolous insects (*i.e.*, insects with complete metamorphosis), such as butterflies, the survival rate differs in successive life-history stages and the curve becomes the **stair-step** type survivorship curve (curve IIa), the initial, middle and final steep segments represent the egg population and short lived adult stages and the two middle flatter segments represent the larval and pupal stages which exhibit less mortality. Some earthworms also exhibit stair-step type survivorship curve (Dash, 1987). Lastly, crowding (high density) in certain populations such as black tail deer population may influence the shape of survivorship curve (Taber and Dasmann, 1957).

The survivorship curve of human population is highly convex. This has become possible because of increased medical care, better hygiene, improved nutrition, and so on. The average life span of an Indian male has gone up from 30 years (before 1947) to 58 years now. But maximum longevity has not gone up very much in any society.

(ii) Life tables. Species differ widely in the number of young produced each year, in the average age to which they live and in their average rate of mortality. When sufficient facts about a species are known, a **life table** that tabulates the vital statistics of mortality and life expectancy for each age

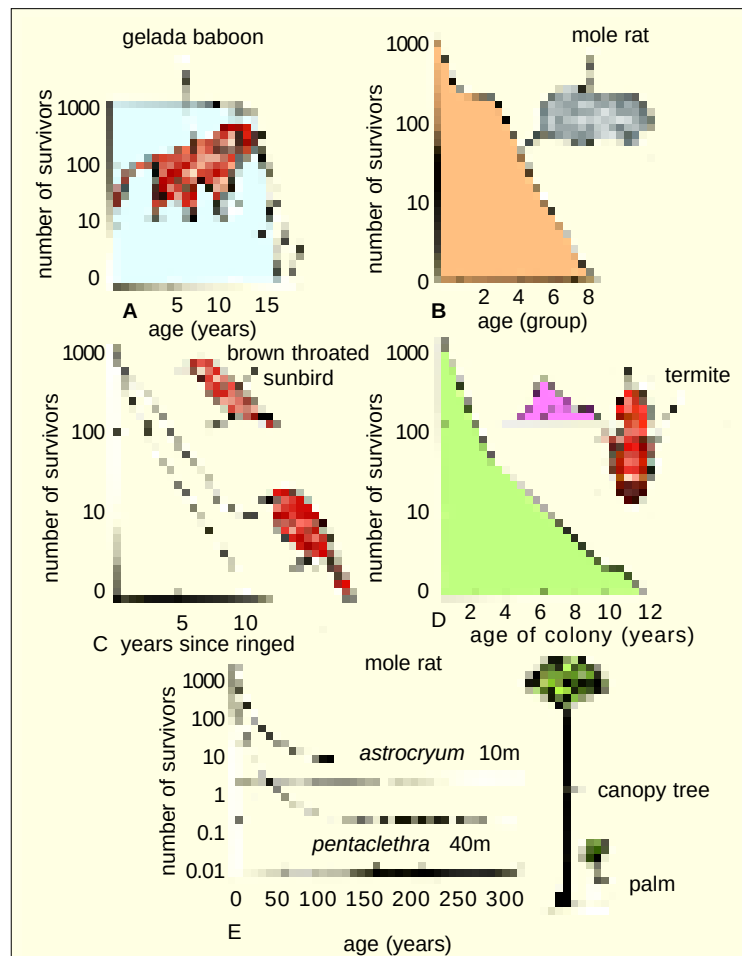


Fig. 6.7. Some examples of survivorship curves of real population. A—Convex curve in gelada baboons, (montane Ethiopia); B—Diagonal curve of mole rat (Nairobi, Kenya); C—Diagonal curve of brown-throated sunbird (forests of south-east Asia); D—Concave curve of whole termite colonies (Nigerian savanna); E—Concave curves of two rainforest trees (solid line is for understory palm in Mexico; broken line for a canopy tree in Costa Rica).

group in the population may be formulated. In such tables, age is usually represented by the subscript index x and is some convenient fraction of a species mean life span, such as year or stage of development. The life table is set up on the basis of an initial cohort of 100, 1000, or 100,000 individuals; and the number living to the beginning of each successive age interval is symbolized as l_x . Plotting these data gives a **survivorship curve** for the species and a number of survivorship curves have been worked out for different species. The number dying with each age interval is designated as dx and gives a mortality rate. The rate of mortality during each age interval is commonly expressed as the percentage of the number at the beginning of the interval 100 (dx/l_x) and is indicated as qx . Survival rate is the difference between the mortality rate and 100 per cent ($100-qx$) and is expressed as sx . Life expectancy (ex) is the mean time that elapses between any specified age and the time of death of all animals in the age group. For example we can consider in Table 6-1 the life table for budworms of Canada.

Uses of life tables. Life tables are useful for computing the average longevity of a population, for showing the age composition of a population, for indicating critical stages in the life cycle at which mortality is high, for showing differences between species, for showing the success of the same species in different biotopes, for furnishing information of value in game and fish exploitation (yield) and in control of pests.

F. Biotic Potential

Charles Darwin (1859) had shown in his natural selection theory that all organisms reproduce many more offspring than survive. Thus, he estimated that a single pair of elephants would have over 19 million offspring alive after 750 years. **Kormondy** (1978) mentioned that a single pair of English sparrows after being introduced into USA could give rise to 275, 716, 983, 698 descendants in ten years. **Howard** estimated that the female *Musca domestica* on the average produces 120 eggs at a time and there are seven generation per year. Assuming that all survive one year and all the females (50 : 50 sex ratio) produce each generation, than at the end of the seventh generation, there would be more than six trillion flies. **Dash** and **Kar** (1990) found that *Lepidochelys olivacea* (the olive ridley sea turtle) female lays, on the average 110 eggs in the Orissa Coast. Assuming a 50 : 50 sex ratio, a female after one egg-laying session is replaced by 55 newborn females. A female lays eggs many times in her life time. If all the offspring survive, one female can be replaced by a few hundred females. All these examples indicate the potential of organisms to reproduce. This reproductive capacity of the organism is called **reproductive** or **biotic potential**.

The term biotic potential was coined by **Chapman** (1928) to designate maximum reproductive power. He defined it as “*the inherent property of an organism to reproduce to survive, i.e., to increase in numbers. It is a sort of algebraic sum of the number of young produced at each reproduction, the number of reproduction in a given period of time, the sex ratio and their general ability to survive under given physical conditions.*” Thus, with the term of biotic potential, one is able to put together natality, mortality and age distribution.

Each population has the inherent power to grow when the environmental conditions are not limiting with regard to space and food; and interspecific competition is absent. The growth rate per individual in the population may reach a maximum and remain constant under specific environmental conditions. In a population with a stable age distribution, the reproductive capacity or growth rate is called the **intrinsic rate of natural increase**, represented by the symbol r . The r is the exponent in the differential equation for population growth in an unlimited environment under specific physical conditions. In fact, the index r is the difference between the instantaneous specific natality rate (i.e., rate per time per individual) and the instantaneous specific death rate and may be expressed as :

$$r = b - d$$

Table 6-1. Life table for the 1952-53 generations of the spruce budworm in New Brunswick, Canada (after Morris and Miller, 1954, source Kendeigh, 1974).

x	lx	Factors responsible for dx	dx	qx
Eggs	1000	Parasites	17	2(—)
		Predators	86	9(—)
		Others	6	1(—)
		Total = 109		12
Instar I	891	Dispersal, etc.	428	48
Hibernacula	463	Winter	79	17
Instar II	384	Dispersal	242	63
Instars III-VI	142	Parasites	51	36
		Disease	3	2
		Birds	20	14
		Others	61	43
		Total = 135		95
Pupae	7	Parasites	0.6	8
		Predators	0.7	10
		Others	1.3	18
		Total = 2.6		36
Moths	4.3	Total for generation	= 995.7	99.5

The r depends on the age composition and the specific growth rates due to reproduction of component age groups. Thus, there may be several values of r for a species depending upon population structure (Table 6-2) :

Table 6-2. Intrinsic rate of natural increase.

Organism	Approximate biotic potential (r) per year
1. Large mammals	0.02 – 0.5
2. Birds	0.05 – 1.5
3. Small mammals	0.3 – 8
4. Larger invertebrates	10 – 30
5. Insects	4 – 50
6. Small invertebrates (including large protozoans)	30 – 800
7. Protozoa and unicellular algae	600 – 2,000
8. Bacteria	3,000 – 20,000

Under natural conditions biotic potential is a rare phenomenon, since environmental conditions do not permit unlimited growth of any population. The difference between the maximum r (biotic potential) and the rate of increase which occurs in an actual laboratory or field conditions is often taken as a measure of the **environmental resistance**, which is the sum total of environmental limiting factors which prevent the biotic potential from being realized. Environmental resistance is often low when a species is first introduced into a new territory so that the species increases in number at a fantastic rate as when the rabbit was introduced into Australia and English sparrow and Japanese beetle were brought into the United States. But as a species increases in number the environmental

resistance to it also increases, in the form of organisms that prey upon it parasitize it, and the competition between the members of the species for food and living space, and consequently, population decreases. An equilibrium will be reached either by decreasing the birth rate (natality) or by increasing the mortality rate.

POPULATION DYNAMICS

Populations have characteristic pattern of increase which are called **population growth forms**. Such growth forms represent the interaction of biotic potential and environmental resistance. The study of population dynamics is done by three approaches (1) mathematical models, (2) laboratory studies and (3) field studies.

The growth is the most fundamental dynamic feature that a species population displays. Populations characteristically increase in size in a sigmoid, S-shaped or logistic fashion. When a few organisms are introduced into an unoccupied area, the growth of the population is at first slow (**positive acceleration phase**), then becomes very rapid (**logarithmic phase**) and finally slows down as the environmental resistance increases (**the negative acceleration phase**) until an equilibrium level is reached around which the population size fluctuates more or less irregularly according to the constancy or variability of given environment. The level beyond which no major increase can occur represents the saturation level or **carrying capacity**. The carrying capacity or equilibrium density is represented by the letter **K**. It is often useful to define the maximum rate of growth of the population. This parameter, generally termed the **intrinsic rate of natural increase**, is symbolized **r₀** and represents the growth rate of a population that is infinitely small. Accordingly such type of population growth can be described by following **logistic equation** :

$$dN/dt = r_0 N (K - N) / K$$

where r_0 = innate capacity of population to increase (birth rate without resource limitation), N = population size and K = highest population density that can be maintained in real environment, i.e., at carrying capacity.

There are two main types of population growth forms. (1) J-shaped and (2) S-shaped or sigmoid forms (Fig. 6.8). The growth forms are due to the nature of species and prevailing environmental conditions. In **J-shaped curve** there is a rapid increase in density with the passage of time (called **exponential growth**). The density values when plotted against time give a J-shaped growth curve and at the peak the population growth ceases abruptly due to environmental resistance. For example, the population growth curve in human populations and growth of yeast,

Drosophila and rabbit under laboratory conditions show an initial slow rate and then it accelerates and finally slows giving the growth curve which is **sigmoid** or **S-shape**. The peak constant level represented by K or upper level (called **asymptote**) of the sigmoid curve is called the **maximum carrying capacity**. It marks the limit to which the environment can support the population.

Plant Population Dynamics

In many respects, plant populations behave like the animal population, but, they have some unique features such as follows : Most higher plants are **modular** organisms, developing from a single zygote but producing an indeterminate number of repetitive structures, called **modules** vegetatively. A clump of herbs, grasses or trees may be product of one zygote. Plants cannot move to mate or disperse.

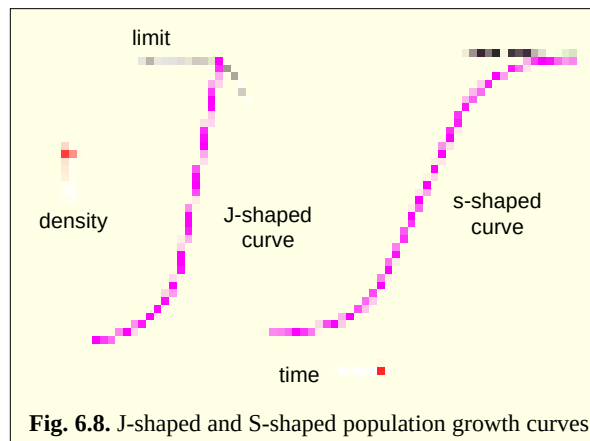


Fig. 6.8. J-shaped and S-shaped population growth curves.

Thus, they have evolved means as gravity, wind, water flow or animals for dispersal of pollen, seed, vegetative parts, etc. The **seed population** present in the soil for different species are referred to as **seed bank** or **seed pool** (Silvertown, 1987). All these seeds do not germinate or all the seedlings do not establish. Some die due to environmental stresses and this is called **environmental sieve** which allows only the stronger individuals to survive. In most cases, the seeds germinate in batches and seedlings of one lot is known as **cohort**. In this way, from a huge seed bank through ecological selection, cohorts are formed and these in turn result into adult population. This process is referred to as **recruitment**. Further, a plant may originate from a vegetative part, called **ramete** (or **tiller**) or from seed called **genet**. Thus, ramete and genet form two levels of population structure. The term **clone** is normally used to designate the population derived from ramete of the same parent plant.

The 3/2 thinning law. Most aspects of growth of population are density related. One important generalisation applied is **3/2 thinning law**. If we plot the relationship between the dry weight and density of shoots (known number of individuals of) in plant population, the line relating weight of each individual to density has a slope of -1.5 (or $-3/2$). The slope would be -1 , if increasing density has been exactly compensated by reduction in weight of individuals. Thinning is normally inversely density dependent, but does not always occur if the growth of the plants is extremely plastic. The $3/2$ is universal and applied well in a wide variety of plants from mosses to trees. Exact reason of its occurrence is still not clear.

GROWTH RATE OF POPULATION

The rate of growth of a population is expressed as the number of individuals by which the population increases divided by the amount of time that elapses while this population increase is taking place :

$$\text{Growth rate } (r) = \frac{\text{number of births } (b) - \text{number of deaths } (d)}{\text{average population in time interval}}$$

The actual change in population number (ΔN) over any span of time (Δt) is equal to rN (Δ is the entity that is changing). This can be written $\Delta N / \Delta t = rN$ or, using the symbology of the calculus, the rate of change of the population at any instant time (dN/dt) can be expressed $dN/dt = rN$. This is equivalent to saying that the number of individuals at any arbitrary time t , or N_t , is related to the number of individuals at the beginning, N_0 , by the equation $N_t = N_0 e^{rt}$, where $e = 2.71828$, the base of the natural logarithms.

If r is constant, the growth of population will be exponential. If r is positive ($b > d$), the population shows an exponential increase to indefinite density and if r is negative ($b < d$), it shows an exponential decay to extinction. It is impossible for a population to change at an exponential rate indefinitely. However, there are many cases in which conditions are such that b is substantially large than d for a period of time, following which conditions change so that d becomes much larger than b . The responses of populations to variations of this sort is an exponential "**population explosion**" during favourable conditions, followed by a "**crash**" when conditions change. Diatom populations in Lake Michigan, USA, for example, undergo such exponential increases at different times of years, triggered by variations in abiotic factors within the lake, followed by equally rapid declines.

POPULATION DISPERSION

Populations have a tendency to disperse, or spread out in all directions, until some barrier is reached. Accordingly, population dispersion is the movement of individuals into or out of the population area. It takes three forms : **emigration**—one-way outward movement; **immigration**—one-way inward movement; and **migration**—periodic departure and return. Dispersion supplements natality and mortality in shaping population, growth form and density, and also it plays a significant role in the distribution of plants and animals even to the areas previously unoccupied by the members of the population. Most types of population dispersion occur due to a number of reasons such as for

obtaining food, avoiding predators, preventing overcrowding, result of action of wind and water, environmental factors as light and temperature, breeding behaviour, physiological reasons as secretion of some hormone or for interchange of genetic material between populations.

Emigration

Emigration under natural conditions occurs when there is overcrowding in the migratory locust, lemming, grouse, snowy owl, snowshoe rabbit, Arctic fox, gray squirrel and occasionally in other species. This is generally regarded as an adaptive behaviour that regulates the population on a particular site and prevents over-exploitation of the habitat. Further, it leads to occupation of new areas elsewhere. By dispersing into new localities, there is opportunity gained for interbreeding with other populations leading to more genetic heterozygosity and adaptability.

It is, of course, population pressure that is responsible in large part for the dispersal of the young and extension of ranges into new areas. Under normal conditions adult animals, especially among the higher vertebrates, are well established on their territories and the youngs are forced to seek homes elsewhere. Among insects, there is a relation between emigration and inherited behaviour tendencies. Individual tent caterpillars, both larvae and adults, differ innately in the extent to which they show activity even within same colony. In the development of populations of excessive size, spread of infestations of the insect into new regions is largely by the more active individuals. The outbreak finally terminates when the proportion of sluggish individuals comes to predominate in the population (Wellington, 1966). Continuous emigrations are rare and when they occur, result in **depopulation**. Equilibrium of populations is maintained in such circumstances by enhancing the reproductive ability as well as by decreased mortality among the populations.

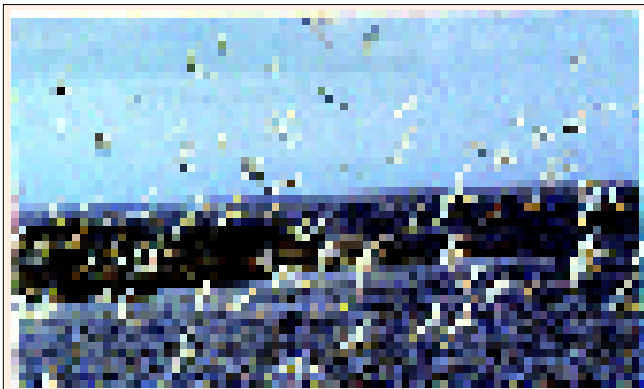
Immigration

Immigration leads to a rise in population level, causing an **overpopulation** which may lead to an increase beyond the carrying capacity. These immigrations result in increased mortality among the immigrants or decreased reproductive capacity of the individuals. Both emigration and immigration are initiated by weather and other abiotic and biotic environmental factors.

Migration

Migration is a peculiar kind of population dispersion which involves the mass movement of entire population. This can occur only in mobile organisms and best developed in insects such as desert locust, *Schistocerca gregaria*, migratory locust, *Locusta migratoria*, butterfly, *Danaus plexippus* and in the migratory dragonflies, *Libellula quadrimaculata* and *Pantala flavescens*; in fishes like eels, in birds and in certain mammals. Most two-way migratory movements are rhythmic processes of population and regular periodicity is a common feature. Very often, environmental periodicities control these migratory movements, as for example day and night rhythm, lunar periods, tides and changing seasons. The monarch butterflies, *Danaus plexippus*, travel very long distances and their migrations are found to be pathed every year through their same conventional routes, and their migratory movements are initiated by the oncoming winter and the return trip being influenced by spring.

In most cases, migration of population may occur for food, shelter, or reproduction. Better utilization of uninhabited or hitherto untouched habitats and their resources are the greatest benefits



Migration of population may occur for food, shelter, or reproduction.

derived from the migratory movements. However, during migration of population, mortality of numerous individuals may occur due to different ecological hazards such as temperature fluctuations, scarcity of food, predation, etc. Anyhow, migration has certain benefits for populations—as it enables wider dispersion of populations; it avoids intraspecific competition for food, shelter or any other means.

REGULATION OF POPULATION SIZE

The inherent tendency of all animal populations is to increase in number. But this increase in number is, however, not infinite since the carrying capacity of the environment always imposes a restriction upon it. Thus, after reaching the carrying capacity level, the population density tends to fluctuate above and below this level and such fluctuation in population between upper and lower limits tend to give some stability to the population. The nature of the processes that regulate the numbers of plants and animals is a major problem of population ecology.

Many groups of animals and plants are provided with unique and intrinsic self-regulatory mechanism such as failure of reproduction and self-inflicted mortality for controlling the size of the population. Populations grow when natality exceeds mortality, and they decline when mortality exceeds natality. However, limitation of animal number in a population is brought about by the action and interaction of two basic regulatory processes namely **density independent** and **density dependent factors**. Density independent factors are the extrinsic factors which tend to regulate the density of a population under different conditions, appearing to act on the population and inflict loss of individuals irrespective of the population density. Variations in space or cover, favourable weather and food occur independently of population densities and may cause drastic changes in the abundance of animals. Such ecological or environmental factors influence negatively or positively all the individuals of a population irrespective of density.

The density dependent factors are intrinsic or biotic factors and they depend on coaction between individuals within the same population or between populations of different species. Density-dependent factors may stabilize populations at an asymptote, the level of which is determined by the carrying capacity of the environment. Some of the important density dependent factors are competition, reproductivity, predation, emigration and disease. The combination of factors or any specific factor involved in the density dependent action may vary from species to species. Further these species have been known to display the property of intercompensation. According to intercompensation, if there is a change in the environment to relieve the population from the pressure of an existing effect, then the population increases till it reaches a level when the second effect takes over. Thus, if the predators that normally keep herbivorous animals down are removed, the population of herbivores may increase to become overcrowded and result is starvation. A supply of abundant food at this junction would make the individuals susceptible to diseases due to the intensity of crowding (**Wilson and Bossert**, 1971). Generally, the population fluctuations controlled by extrinsic factors tend to be irregular and correlated with the variation in one or more major physical limiting factors such as temperature, food, water, etc., while fluctuations of populations controlled by intrinsic factors exhibit regularity and population cycles.

Population Cycles

Populations are said to be cyclic when they alternatively erupt and subside in a more or less uniform manner between high and low levels of density. Different animals exhibit population cycles different times. The best established cycles of population density of fluctuations are those of periodicities of 3–4 years and 9–10 years. The 3–4 year cycles are most commonly observed in many birds such as snowy owl, willow ptarmigan, capercaillie, Blackgame, Hazel grouse etc.; mammals such as lemmings, voles, arctic foxes, etc., and fishes. 9–10 years cycles of population density fluctuations have been observed in birds such as ruffed-grouse, sharp-tailed grouse, willow ptarmigan, etc., and mammals such as snowshoe rabbit, muskrats, Canada lynx, etc. Among invertebrates, insect pests of

coniferous forests in Germany fluctuate in periods variously from 6 through 18 years. *Asterias forbesi* is found to contain periodicity of 14 years.

Attempts to explain these vast fluctuations in numbers on the basis of climatic changes have been unsuccessful. At one time it was believed that these were caused by sunspots, and the sunspots and lynx cycles do appear to correspond during the early part of the nineteenth century. However, the cycles are of slightly different lengths and by 1920 were completely out of phase, sunspot maxima corresponding to lynx minima. Attempts to correlate these cycles with other periodic weather changes and with cycles of disease organisms have been unsuccessful.

The snowshoe hares die off cyclically even in the absence of predators and in the absence of known disease organisms. The animals apparently die of “shock” characterized by low blood sugar (hypoglycemia), exhaustion, convulsions and death symptoms which resemble the “alarm responses” induced in laboratory animals subjected to physiological stress. Visualizing this similarity, **J.J. Christian** (1950) proposed that their death, like the alarm response is the result of some upset in the adrenal-pituitary system. As population density increases; there is increasing physiological stress on individual hares owing to crowding and competition for food. Some individuals are forced into poorer habitats where food is less abundant and predators more abundant. The physiological stresses stimulate the adrenal medulla to secrete **epinephrine** or **adrenaline** which stimulates pituitary via the hypothalamus to secrete more **ACTH** (adrenocorticotrophic hormone). This in turn, stimulates the adrenal cortex to produce corticosteroids, an excess or imbalance of which produces the alarm response or physiological shock.

In the latter part of the winter of a year of peak abundance, with the stress of cold weather, lack of food and the onset of the new reproductive season putting additional demands on the pituitary to secrete gonadotropins, the adrenal-pituitary system fails, becomes unable to maintain its normal control on carbohydrate metabolism, and low blood sugar (hypoglycemia), convulsion and death follows.

According to **Kendeigh** (1974) several intrinsic factors such as disease, predation, food factor and natural selection affect animal cycles in addition to several extrinsic factors such as weather, solar radiation, and so on.

POPULATION ECOLOGY AND EVOLUTION

Population parameters have been used by population ecologists to draw conclusions about evolution of species :

r-selection and k-selection. In this regard an important synthesis was done by **MacArthur** and **Wilson** (1967) who declared that populations are outcome of *r*- or *k*- selection. The **r-selected** populations have a high intrinsic rate of growth (*r*) and tend to “boom” when environmental conditions are favourable and “best” when these conditions deteriorate. In consequence, they show great fluctuations in density, and incidentally have the potential for large genetic change through the founder effect. The **k-selected populations** have relatively constant density at or near the carrying capacity (*K*) of the environment.

REVISION QUESTIONS

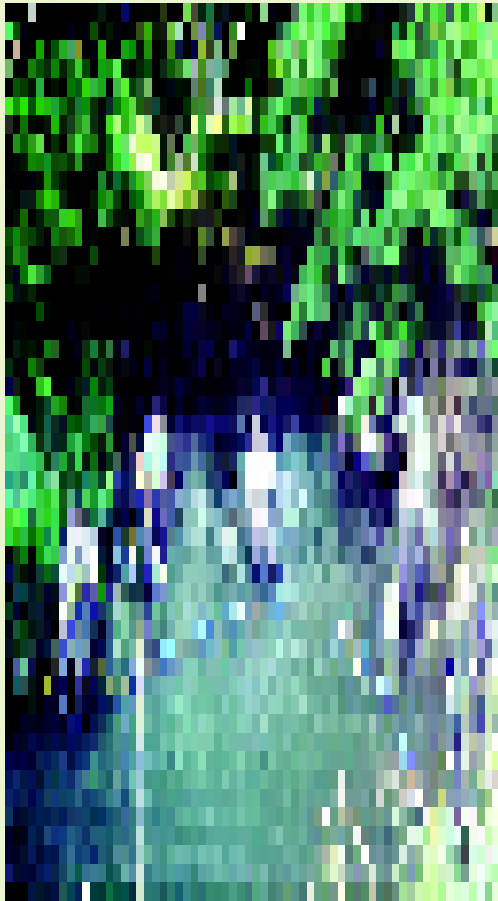
1. What is population ? Describe its different characteristics with suitable examples.
2. Describe the *r* and *k*- selection.
3. Describe in brief various techniques of measurements of population density of animals.
4. Discuss that in what way a population is a self-regulating system.
5. Write short notes on the following :
(i) Fecundity, (ii) Natality, (iii) Mortality of population, (iv) Survivorship curve, (v) Age pyramids, (vi) Population dispersion.

CHAPTER

7

Biotic Communities

(Community Ecology : Communities, Niche, and Bioindicators)



Living creatures of an ecosystem form a biotic community.

All the living entities of an ecosystem form a single biotic component, the **community** or **biotic community**. All the organisms of a community live together, share same habitat, influence each other's life directly or indirectly and have reached a survival level within a given radiant energy. Thus, a community is any assemblage of populations of living organisms in a prescribed area of habitat. A community is claimed to have one or more of the following attributes : (1) co-occurrence of species, (2) recurrence of groups of the same species, and (3) homeostasis or self-regulation (**Krebs**, 1985).

Community is a larger unit than the population and it achieves many characteristics that are not found in its constituents, *i.e.*, the organisms and the populations. Communities may have a wide range of sizes, ranging from a small patch of land or water-body to extensive forests. **Minor communities** are greatly influenced by inputs from adjacent communities, while **major communities** are relatively independent and self-sufficient of their habitat. Communities differ from place to place and at the same place at different times.

The approach of botanists and zoologists to community studies is quite different. While the zoologist is mainly concerned with the functional relationships within a community,

involving both the plants and animals, the botanists are generally concerned with community structure and the changes that undergo in time and space.

Characteristics of a Community

Like a population, a community has a series of characteristics such as follows :

1. Species diversity. Various species of plants and animals live in a community and exhibit species richness or species diversity. The study of species diversity is an essential component of community study. For animal communities, a study of age structure and growth pattern is important, while for plant communities, floristics, study of taxonomy, life forms such as herbs, shrubs, climbers, trees are important. Since seasonal changes occur in the appearance of plant structure and growth, periodicities and phenology (= seasonal succession in natural communities) are significant parameters.

2. Growth form and structure. The type of the community is described by major categories of growth forms (e.g., trees, shrubs, herbs, mosses, etc.) These different growth forms determine the stratification, or vertical layering of the community.

3. Dominance. Among several species present in a community, a few exert a major controlling influence by virtue of their size, numbers, or activities. These are called as **ecological dominants** or **dominant species**.

4. Relative abundance. Different populations in a community exist in relative proportions and this idea is called as **relative abundance**.

5. Trophic structure. Who eats whom ? The feeding relations of the species in the community will determine the flow of energy and materials from plants to herbivorous animals to carnivorous animals.

CLASSIFICATION OF THE COMMUNITIES

Communities have been classified by different ecologists from different view points. In terms of the general growth, composition, shape, etc., of vegetation, and organisms associated with them, communities may be classified as **forests, deserts, grasslands, tundra** and so on. Likewise, according to the amount of water in the habitat, communities may be divided as **hydrophytic** in predominantly aquatic habitats, **mesophytic** in moderately moist soils and **xerophytic** in arid or dry conditions. Communities growing in condition of abundant light are called **heliophytic** and those growing in shade are called **sciophytic**. **Clements** (1916) recognized the fact that plant communities are not always the same at any place and he classified the communities on two parallel lines : One in the process of change which are called **seral communities** and the others are called **stable** or **climax communities**.

Further, the **global community** is an enormous mass of life, comprising all the plants and animals in the world. The global community is further divided into : **continental communities** and **oceanic communities**. Since due to great variability in climatic factors, an exhaustive study in such vast areas is practically impossible, therefore, communities are often studied as **biotic province**. A biotic province can be defined as a considerable geographic area, over which the climate is relatively uniform though often modified by physiographic features (**Dice**, 1952). Since biotic province is an abstract community, so based on ecological criteria, the associations are studied as **concrete communities**. A concrete community can be defined as a specific area which can be observed directly and which is an assemblage of plants and animals that actually exists and from which some ecological data can be collected.

Lastly, a **stand** is the largest concrete community, for example, a particular forest, river, swamp, meadow or lake that can be seen, observed, measured and worked over by the ecologist. A **microstand** is a small localized area within stand. Thus, each individual plant or animal with its associated parasites, epiphytes and commensals are good examples of the microstands.

COMPOSITION OF COMMUNITY

Composition of community comprises the following parameters :

1. Size. Communities may be large or small. Larger one extends over areas of several thousands of square kilometers, as forests, others such as deserts, etc., are comparatively smaller with dimensions in hundreds of kilometers, and still others such as meadows, rivers, ponds, rocky plateaus, etc., covering a more restricted area. Very small-sized communities are the groups of microorganisms in such microhabitats as leaf surface, fallen log, litter, soil, etc.

2. Number of species. The number of species and population abundance in communities vary greatly. **Charles Elton** (1927) calculated that the communities of certain British rivers included 131 species of invertebrates, in addition to fish, amphibians, populations of algae, protozoa, bacteria, and rooted aquatic plants. In contrast to this, he found a community on a sandy beach consisting of only five species of invertebrates. A meadow (=a rich pasture ground especially beside a stream) on clay near Oxford, England included 93 species of invertebrates in the soil and on above ground vegetation. In India too, various ecologists recently have estimated the number of species in the composition of various Indian communities. **M.C. Dash** and his coworkers (1980, 1981 and 1984) studied the composition of a biotic community in some tropical pastures of Sambalpur, India and found that it consisted of 22 grass and other herbaceous species (the primary producers), 5 species of earthworms, 25 of test-bearing protozoa, 15 of nematodes and 7 of microarthropods. The team of **Dash** (1980, 1984) reported 9 species of grasses and other herbaceous species, 1 of earthworms, 11 of testate protozoa and 16 of nematodes in a tropical hill ecosystem of Sambalpur. At one site of river Ib, Orissa, the community consisted of 1 species of rooted plant, 32 of phytoplanktons, 20 of zooplanktons and 7 of fish (**Kar et al.**, 1987; **Dash et al.**, 1988).

3. Dominants. In each community there occurs diverse species. All these species are not equally important but there are only a few over topping species which by their bulk and growth modify the habitat and control the growth of other species of the community, thus, forming a kind of characteristic nucleus in the community. These species are called the **dominants**. Generally in most of the communities, only a single species is particularly conspicuous and dominant and in such case the community is called by the name of dominant species, *e.g.*, spruce forest community. In other cases, there may be more than one dominants in a community, *e.g.*, oak-hickory forest community.

According to **Clements** and **Shelford** (1939), dominance is most commonly expressed in the reactions of an organism on its habitat. Community dominants sustain the full impact of the climate or the environment but modify this effect for other organisms within the community by tempering light, moisture, space, and other conditions. Only those other organisms that find these modified physical conditions tolerable can exist within the community. Furthermore, dominants are ordinarily the most prominent species in the community, make up its greatest mass of living material (**biomass**), and serve as the major source of food, substrate, and shelter for the animals that are present. In a forest community, trees are dominant. They decrease light intensity, increase the relative humidity, intercept precipitation, monopolize most of the moisture and nutrients in the soil, decrease wind velocity and furnish shelter and food for animals. Grasses play a similar, though less conspicuous, role in prairie communities; sedges, rushes and cattails in marsh communities : sagebrush in the arid habitat of the Great Basin; mussels and barnacles on a rocky seashore; and so forth.

5. Ecological amplitude. The range of environmental conditions which a taxon can tolerate is called **ecological amplitude**. The composition of a biotic community in any habitat is dependent upon the frequency of environmental conditions in that habitat and the ecological amplitude of species populations. Thus, the climate and other abiotic and biotic conditions of a habitat determine the type of community which survives and develops.

STRUCTURE OR STRATIFICATION OF COMMUNITY

The communities exhibit a **structure** or recognizable pattern in the spatial arrangement of members of the communities. **Stratification** forms one of the outstanding general principles of community structure. Organisms are distributed unevenly throughout the biotic community, resulting in stratification. The stratification in a community may be resulted by the following factors : 1 Specific tolerations and adaptations. 2. Chemical reactions taking place between the by-products and physical stratification. 3. Other organisms take up temporary or permanent residence as a direct response to the presence of initial residents. The stratification in community may be horizontal or vertical and is different in aquatic and terrestrial habitats.

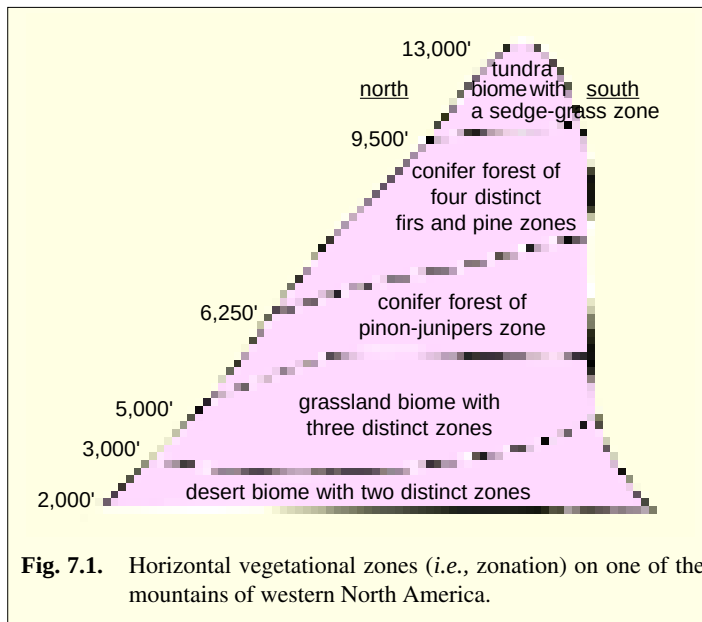


Fig. 7.1. Horizontal vegetational zones (i.e., zonation) on one of the mountains of western North America.

1. Horizontal Stratification

A community may be divided horizontally into **subcommunities** which are units of homogeneous life forms and ecological relations. This horizontal division constitutes the **zonation** in the community. Figure 7.1 shows the zonation of different distinct vegetational types on a mountain. Latitudinal as well as altitudinal zonations of vegetation have also been shown in relation with climatic factors described in Chapter 4.

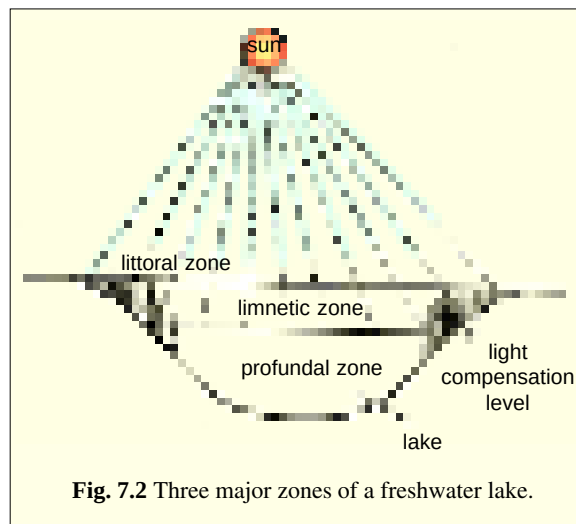


Fig. 7.2 Three major zones of a freshwater lake.

In shallow ponds, the zonation is very little. However, in deep ponds and lakes, there may be recognized three zones, viz., **littoral zone**, **limnetic zone** and **profundal zone** (Fig. 7.2). In each zone, organisms differ from each other. Further, in case of animal communities of rocky sea shore, there are following four distinct zones : 1. **Balanus zone**. This zone includes the animals such as *Balanus* and *Patella* which live on exposed rocks. 2. **Littoral zone** or **intertidal zone**

(i.e., zone between high and low tides). This habitat develops in the sheltered niche beneath loose stones and contain a variety of crustaceans and collembola (springtail). Lower on the shore, where the tides always flood are living beneath the stone some annelids and nemertine worms (e.g., *Lineus*). Adjacent to low water mark are certain species of *Neries*, shore crabs, etc. 3. Holes and cracks in the rocks are

occupied by various annelid worms, crustaceans, small sea-cucumbers and rock-boring bivalves. 4. There are rock-pools that enclose those animals which escape direct current of water.

Sandy-shore dwelling marine animals are in great part burrower such as spinoid (*Prionospio*, *Aricia*), annelids (*Glycera*, *Nerina*), archiannelids (*Protodrilus*) and crustacean (*Emerita*, *Albunea*). Populations which inhabit this substrata adjust to both sand and mud particles. This becomes evident in their respiratory adaptations and locomotory and feeding behaviour.

2. Vertical Stratification

Vertical gradients in environmental factors such as the availability of sunlight, the temperature and so on bring about a recognizable stratification in water bodies, particularly in marine ecosystem (Fig. 7.3) such as deep water lakes. The **oceanic region** (i.e., the region of open sea beyond continental shelf) comprises two zones : 1. **bathyal zone** (i.e., region of continental slope and rise) and 2. **abyssal region** (i.e., area of ocean “deeps”). In terms of light penetration, the ocean region is vertically also divided into an upper thin **euphotic zone** (= the light compensation zone) and a vastly thicker permanently darker zone, the **aphotic zone** (including bathyal and abyssal zones). The oceanic region also includes two more vertical zones : the **benthic** (bottom) and the **pelagic** (whole body of water). The community of pelagic zone includes (1) **phytoplanktons** (the producers such as diatoms, dinoflagellates, flagellates, green algae, brown algae such as *Sargassum*); (2) **zooplanktons** (the consumers such as Foraminifera, Radiolaria, annelid worms, swimming snails, jelly fishes, ctenophores, etc., (3) **nektons** (these are free swimming marine animals which are essentially independent of water movements). Nektons are consumers such as cephalopods (*Sepia*, *Loligo*, *Nautilus*) bony fishes, sharks, sea turtles, whales, seals and sea birds. Minute organisms that float or swim on surface water are called **neustons**. **Benthos** are bottom dwellers of benthic environment. They are sessile forms such as limpets, chitons, mussels, oysters, sponges, corals, hydroids, anemones, bryozoans, crabs, lobsters, some echinoderms and flat fishes.

Vertical stratification in communities is also observed in terrestrial ecosystems. For example, in a **grass-land community**, the euphotic zone consists of the **herbaceous stratum** consisting of grasses and other herbs as producers and rodents, cattle and insects as consumers. The **subterranean stratum** consists of different layers of soil containing litter, humus, mineral fraction of soil, plant

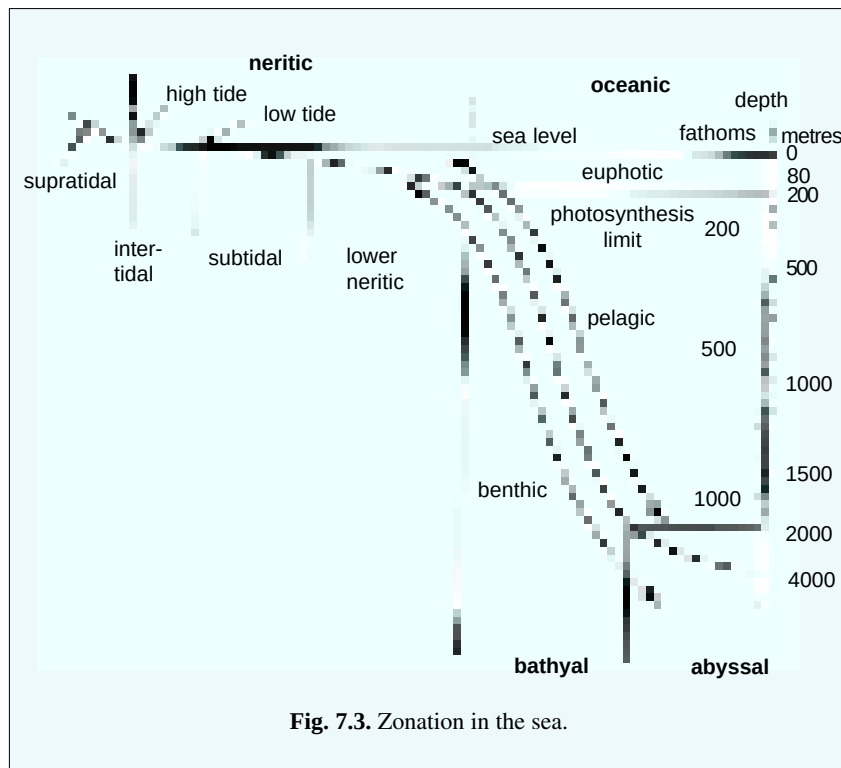


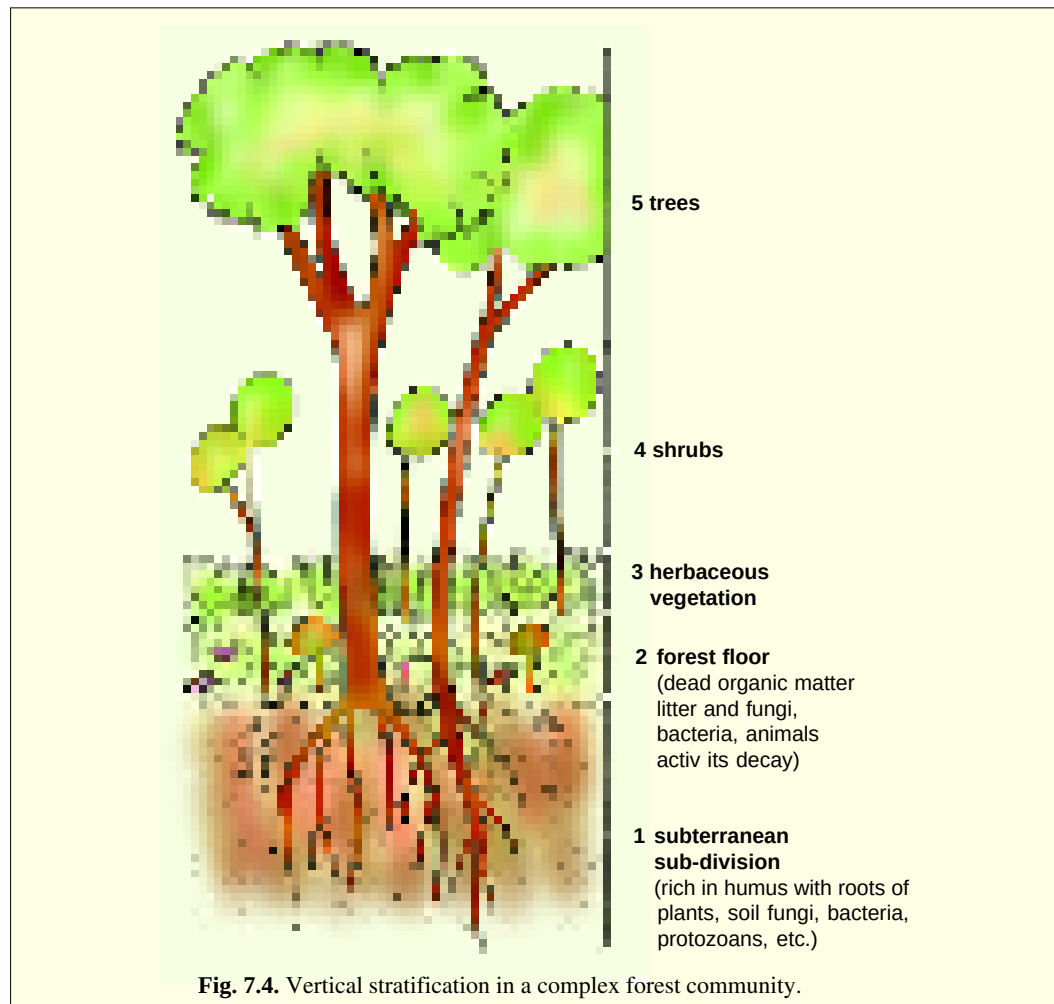
Fig. 7.3. Zonation in the sea.

roots, rhizomes and organisms such as bacteria, fungi, protozoa, microarthropods, insect larvae, pupae, oligochaetes and molluscs. In **forest community**, however, the vertical stratification is most complicated and includes following five subdivisions: 1. **subterranean subdivision**; 2. **forest floor**; 3. **herbaceous vegetation**, 4. **shrubs** and 5. **trees** (Fig. 7.4). In some **tropical rain-forests** three more subdivisions exist in the forest floor such as leaf litter layer (**L layer**), partially decomposed leaf litter layer (**F layer**) and a humus layer. Thus, based upon the light and relative humidity requirements, we find stratification in above ground parts.

Ecological potential of a zone or strata determines the distribution of community. **Fenton** (1947) has classified these communities as (1) **Micro-biota** which include bacteria, algae, fungi, protozoans, rotifers, nematodes and mites. (2) **Meso-biota** which include many nematodes, centipedes, millipedes, spiders, insects, etc. (3) **Macro-biota** which include plant roots, earthworms, toads, lizards, snakes, rodents and large mammals including man. He also classified these communities according to their feeding habits such as **chemophagous**, **ectophagous**, **entophagous**, **predators** and **shelterers**.

CHARACTERS USED IN COMMUNITY STRUCTURE

Structure of plant communities can be determined by quantitative and qualitative features as follows :



A. Quantitative Structure of Plant Communities

Quantitative characters of the community include such characters as frequency, density, abundance, cover and basal area, etc.

1. Frequency. Various species of the community are recorded by different phytosociological methods, by taking any sampling unit such as **transect**, **bisect**, **quadrat**, **point centre** or **point quarters**, etc. Frequency is the number of sampling units (as %) in which a particular species occurs. Thus, the frequency of each species is calculated in the following way :

$$\text{Frequency (\%)} = \frac{\text{Number of sampling units in which the species occurred}}{\text{Total number of sampling units studied}} \times 100$$

After determining the percentage frequency of each species, various species are distributed among **Raunkaier's** (1934) five frequency classes depending upon their frequency values (Table 7-1)

Table 7-1. Raunkaier's frequency classification of plant communities.	
Frequency class	Species with frequency range (%)
A	0–20
B	21–40
C	41–60
D	61–80
E	81–100

The **relative frequency (RFR)** is determined by the use of the following formula (data collected using the quadrant method).

$$\text{Relative frequency (RFR) of the species} = \frac{\text{Number of occurrence of a species}}{\text{Number of occurrence of all species}} \times 100$$

Since environmental conditions vary, the frequency values differ in different communities. **Raunkaier** observed the following relationships between different frequency classes : A = 53, B = 14, C = 9, D = 8 and E = 16. He generalized his findings in the **law of frequency**, which showed that

$$A > B > C \begin{matrix} \geq \\ < \end{matrix} D < E$$

Such a relationship occurs only in undisturbed communities or communities where disturbance is minimum.

2. Density. Density represents the numerical strength of a species in the community. The number of individuals of the species in any unit area is its density. It gives an idea of degree of competition. Samples are taken by the help of different types of samples and sampling techniques (**Southwood**, 1966) and estimation are made of the numerical strength of a species population and of the total community. The data can be multiplied with an appropriate factor to express the total number of organisms per unit of area. The density of a species is calculated as follows :

$$\text{Density} = \frac{\text{Total number of individuals of the species in all the sampling units}}{\text{Total number of sampling units studied}}$$

The **relative density (RDE)** of a species is calculated in the following way :

$$\text{RDE} = \frac{\text{Number of individuals of the species in all quadrates}}{\text{Number of individuals of all species in all quadrates}} \times 100$$

3. Abundance. This is the number of individuals of any species per sampling unit of occurrence. Abundance is calculated as follows :

$$\text{Abundance} = \frac{\text{Total number of individuals of the species in all the sampling units}}{\text{Number of sampling units in which the species occurred.}}$$

4. Cover and basal area. The above ground parts of plants (such as leaves, stems and inflorescence) cover a certain area—if this area is demarcated by vertical projections, the area of the ground covered by the plant canopy is called **canopy cover**, **foliage cover** or **herbage cover**. Canopy cover is a good measure of the herbage availability and is estimated by chart quadrat, line intercept or point frame.

Basal area refers to the ground actually penetrated by all stems and is readily seen when the leaves and stems are clipped at the ground surface. It is either measured 2.5 cm above ground or actually on the ground level. Basal area of a plant is measured by callipers, line-interception or point-centered quadrat method. It is one of the chief characteristics to determine dominance.

Since plants differ in their growth form, so the relationship between herbage cover and basal area differs in different types of plants as shown in Figure 7.5.

For the measurement of crown cover, the diameter is taken at ground level, across the canopy perimeter passing by the main stem. The cover is expressed as a percentage. Based on their cover values, species are grouped into the following six groups : Group 1 (less than 5%), Group 2 (5–25%), Group 3 (25–50%), Group 4 (50–75%), Group 5 (75–95%) and Group 6 (95–100%).

B. Qualitative Characteristics of Plant Communities

These involves parameters such as physiognomy, phenology, stratification, abundance, sociability or gregariousness, vitality and vigour, life form (growth form), etc. All these parameters can be described and may be grouped in point scales.

1. Physiognomy. Physiognomy is the general appearance of vegetation as determined by the growth form of dominant species. Such a characteristic appearance is often expressed by single term. For example, observation of a plant community having trees and some shrubs as the dominants, clearly forced us to conclude that it is a forest. Similarly on the basis of appearance of a community, it may be a grassland, desert, etc.

2. Phenology. The life history of a plant species involves seed germination, vegetative growth, flowering, fruit formation, seed maturation, leaf fall, seed dispersal and death. These events are different for different species and are recorded for the individual species. A study of the date and time of occurrence of these events is called **phenology**. In other words, phenology is the calendar of events in the life history of the plant. Environmental factors tend to influence the phenological behaviour of a species population. These phenological events are customarily recorded diagrammatically (Fig. 7.6), monthwise and seasonwise and provide valuable information. These diagrams are called **phenograms**. It will be

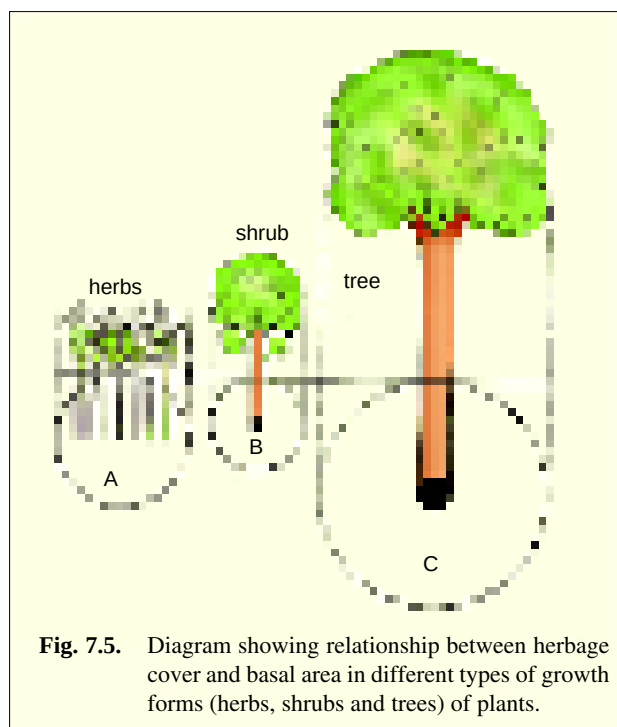


Fig. 7.5. Diagram showing relationship between herbage cover and basal area in different types of growth forms (herbs, shrubs and trees) of plants.

significant to apply phenology to those ecosystems where seasonal changes are not sharply defined, e.g., deserts. Phenology also includes the periodicity and aspection of the community.

Periodicity of communities. As already described, all the major phenomena on earth are cyclic. The communities being dynamic entities respond constantly to seasonal changes and exhibit a variety of periodicities such as seasonal rhythms, lunar and tidal rhythms and many periodicities associated with the cycle of day and night (e.g., circadian rhythms). Ecologists tend to recognize six phases (seasons) in the annual cycle—prevernal, vernal, extival, aestivation, autumnal and hiemal periods. Such cyclic fluctuations can be observed in both abiotic and biotic factors of a particular environment such as follows : 1. Abiotic variability , i.e., cycle changes in temperature, moisture, wind speed and a number of other factors. 2. Biotic variability, i.e., flora and fauna influenced by the immediate abiotic environment.

3. Stratification. As described elsewhere, stratification of communities is the way in which plants of different species are arranged in different vertical layers in order to make full use of the available physical and physiological requirements.

4. Abundance. Plants are not found uniformly distributed in an area. They are found in smaller patches or groups, differing in number at each place. Abundance is divided into following five arbitrary groups, depending upon the number of plants : 1. **very rare**, 2. **rare**, 3. **common**, 4. **frequent**, and 5. **very much frequent**.

5. Sociability. Sociability or gregariousness expresses the degree of association between species. It denotes the proximity of plants to one another. Plants generally grow as isolated individual, in patches, colonies or groups.

Braun -Blanquet (1951) classified plants into the following five sociability groups :

- (i) **S₁ plants.** Plants (stems/shoots) found quite separately from each other, growing singly.
- (ii) **S₂ plants.** Plants growing in small groups (4 to 5 plants).
- (iii) **S₃ plants.** Plants growing in small scattered patches.
- (iv) **S₄ plants.** Several bigger groups of many plants at one place.
- (v) **S₅ plants.** Pure populations of plants covering large areas.

Interdependence is another community feature which includes inter-and intraspecific competition, parasitism, symbiosis and commensalism, all of which have been discussed else-where in this book.

6. Vitality. It is the capacity of normal growth and reproduction, which are important for successful survival of a species. In plants, the vitality is determined by the following parameters : stem height, root length, leaf area, leaf number, number and weight of flowers, fruits, seeds, etc.

7. Life-form (Growth form). Ecologists generally use **Christen Raunkiaer's** classification (1934) of plant life forms. A **life form** is the sum of the adaptation of the plant to the community (it is

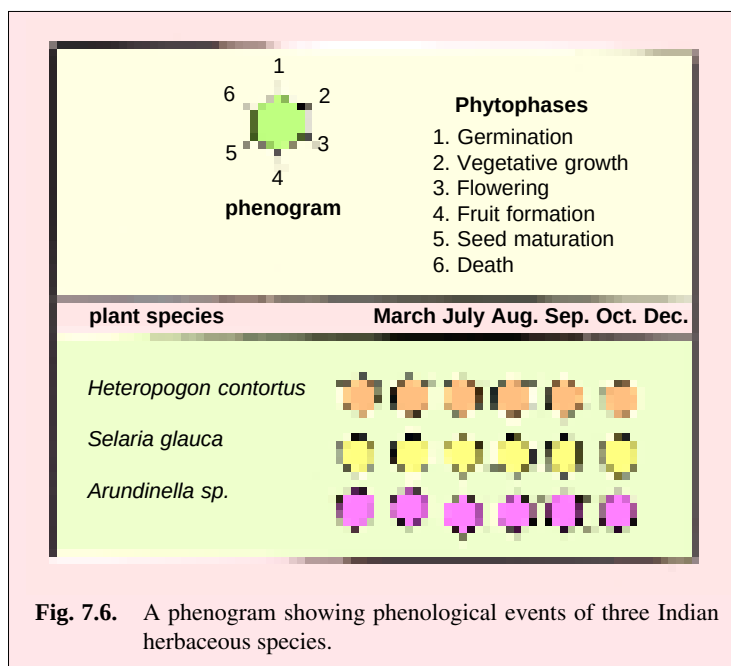


Fig. 7.6. A phenogram showing phenological events of three Indian herbaceous species.

determined by the stature of the plant species, their spread and characters of life-form). **Raunkiaer** classified plants into following five broad life-form categories (Fig. 7.7); his classification is based on the position of perennating buds on plants and the degree of their protection during adverse conditions.

(i) Phanerophytes.

The growing buds of these plants are naked or covered with scale (*i.e.*, they are not well protected), and situated in upright shoots much above the ground surface. Phanerophytes include trees, shrubs and climbers (also woody lianas). Phanero-



Woody lianas is a phanerophyte.

phytes are found mostly in tropical regions and decrease progressively from the tropics to the temperate to the polar regions.

(ii) Chamaephytes.

In these plants the buds are situated close to the ground surface and these buds get protection from fallen leaves and snow cover. Chamaephytes commonly occur in high altitudes and latitudes, *e.g.*, *Trifolium repens* which is found in North America.

(iii) Hemicryptophytes. These are mostly found in cold temperate zone. Their buds are hidden under soil surface, protected in soil itself. In the warm season the growth of aerial parts is marked. Their shoots generally die each year, *e.g.*, most of the biennial and perennial herbs including grasses.

(iv) Cryptophytes or geophytes. In these plants, the buds are usually buried in the soil or in bulbs and rhizomes where food is stored to withstand long periods of adverse climatic conditions (*i.e.*, freezing and drying). Cryptophytes include the hydrophytes (buds remaining under water), helophytes (marsh plants with rhizomes under the soil) and geophytes (terrestrial plants with underground rhizomes or tubers).

(v) Therophytes. These are seasonal plants, completing their life cycle in a single favourable

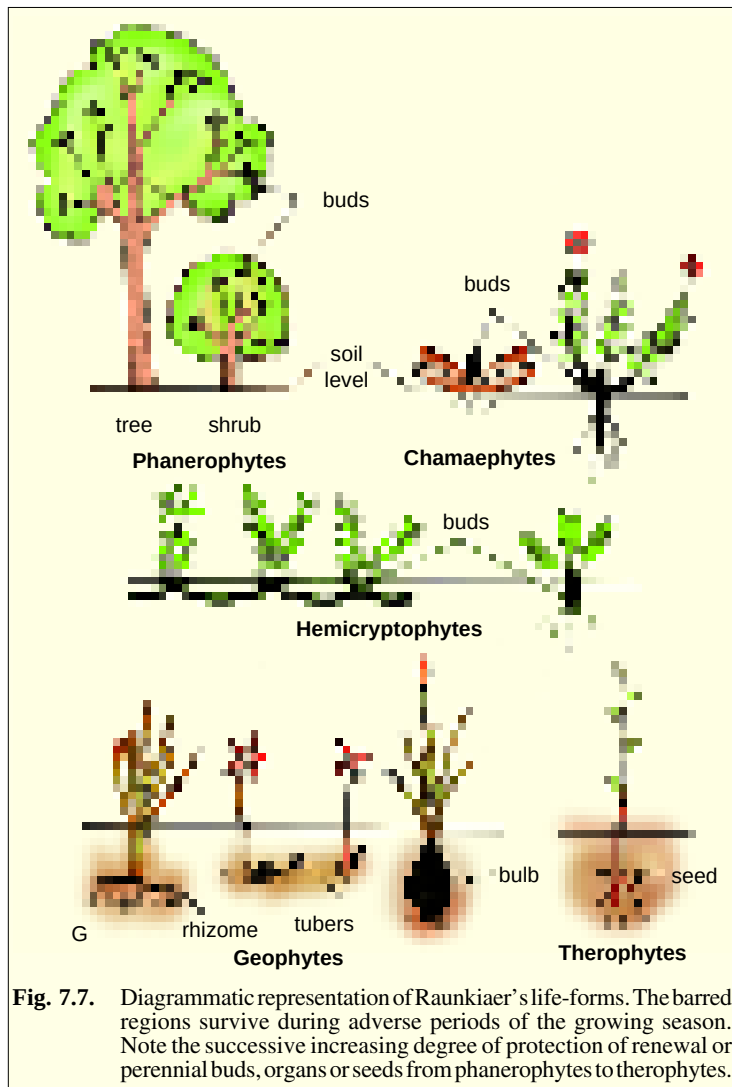


Fig. 7.7. Diagrammatic representation of Raunkiaer's life-forms. The barred regions survive during adverse periods of the growing season. Note the successive increasing degree of protection of renewal or perennial buds, organs or seeds from phanerophytes to therophytes.

season, and remain dormant throughout the rest unfavourable period of year in the form of seeds. Therophytes are commonly found in dry, hot or cold environments (*i.e.*, deserts).

Life-forms of animals. There have been several attempts to classify the life-forms of animals, but no definite system has resulted (Remane, 1952 ; Krivolutskii, 1972). The major life-forms more often agree with their taxonomy than do plants, but some life-forms include representatives from several different taxonomic groups. These can be recognized **encrusting forms**, such as the freshwater bryozoan *Plumatella* and some sponges ; **coral forms**, including grass, leaf, or shrub forms; **radiate forms**, such as coelenterates and echinoderms; **bivalve forms**; **snail forms**; **slug forms**; **worm forms**; **crustacean forms**; **insect forms** ; **fish** **snake**, **bird** and **four-footed forms**. Each of these major types may be divided into narrower structural or behavioural types, for example, Osburn *et al.*, (1903) subdivided four-footed mammalian forms into following types :

1. **Aquatic (swimming).** Seal, whale, walrus.
2. **Fossorial (burrowing).** Mole, shrew, pocket gopher.
3. **Cursorial (running).** Deer, antelope, zebra.
4. **Saltatorial (leaping).** Rabbit, kangaroo, jumping mouse.
5. **Scansorial (climbing).** Squirrel, opossum, monkey.
6. **Aerial (flying).** Bat.

C. Synthetic Characters

Synthetic characters are determined after computing the data on analytical characters (*i.e.*, quantitative and qualitative characters) of the community. For comparing the vegetation of different areas, community comparison needs the calculation of their synthetic characters. Synthetic characters are determined in terms of the following parameters :

1. Presence and constance. The presence expresses the extent of occurrence of the individuals of a particular species in the community. The species on the basis of its percentage frequency may belong to any of the following five presence classes that were first proposed by Braun Blanquet :

- I. **Rare**—present in 1 to 20% of the sampling units.
- II. **Seldom present**—present in 21–40% of the sampling units.
- III. **Often present**—present in 41–60% of the sampling units.
- IV. **Mostly present**—present in 61–80% of the sampling units.
- V. **Constantly present**—present in 81–100% of the sampling units.

A species with high frequency is considered constant. The term constancy is used when equal sample areas are taken in each stand, and presence is used when the area of the sample varies from stand to stand.

2. Fidelity. Fidelity or “faithfulness” is the degree with which a species is restricted in distribution to one kind of community. Such species are sometimes known as **indicators**. The species have been grouped into the following fidelity classes which were first formulated by Braun Blanquet:

(i) **Fidelity 1 or Strangers.** Rarely occurring species which either have arrived from other communities or are relicts from earlier stages of succession.

(ii) **Fidelity 2 or Indifferents (or ubiquitous).** Species which occur in any community without exhibiting any preference to any particular kind of community.

(iii) **Fidelity 3 or Preferentials.** Species which occur in several kinds of communities but are predominant in one.

(iv) **Fidelity 4 or Selectives.** Species occurring frequently in one kind of community and sometimes also in others. Sometimes preferential and selective species are considered together as **characteristic**.

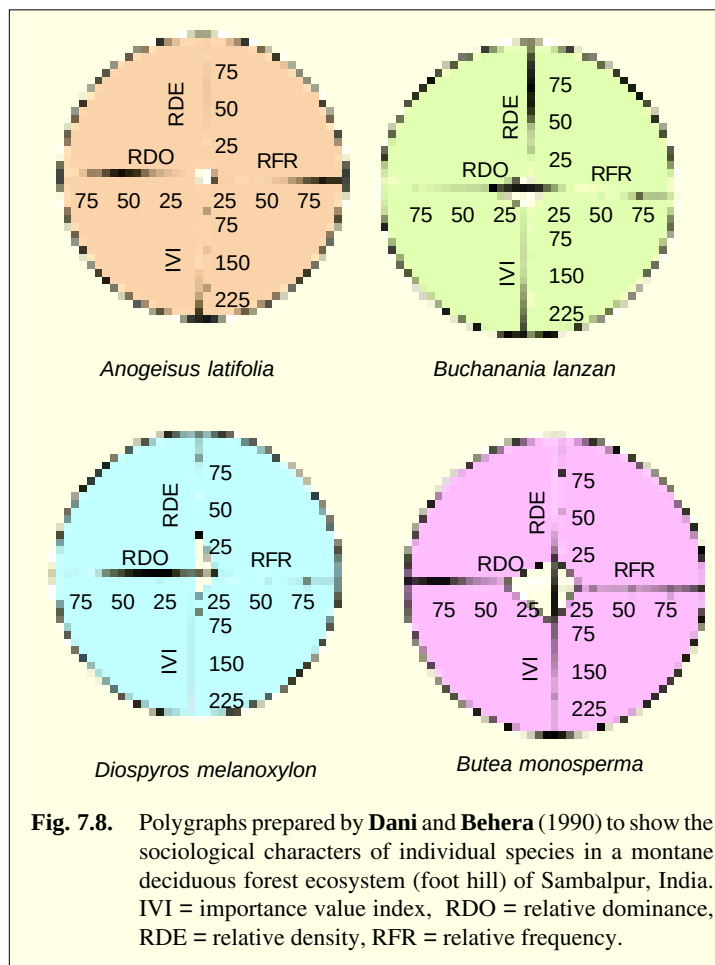
(v) **Fidelity 5 or Exclusive or True.** Species which are completely restricted to one kind of community.

3. Dominance. It is used as a synthetic as well as analytical character (**Daubenmire**, 1959). The number of organisms sometimes may not give correct idea of the species. If one bases his conclusion on number, a single or few trees in a grassland, or few grasses in a forest should be of little value. But if he considers the species on the basis of area occupied or weight (biomass), the situation may be different. Thus, in dominance, cover is included as an important character. The single tree in grassland may occupy fairly a large area and may have much mass. Relative dominance (cover; RDO) is calculated in the following way :

$$\text{Relative dominance (cover, RDO)} = \frac{\text{Dominance (cover) of the species}}{\text{Total dominance (cover) of all the species}} \times 100$$

4. Importance value index (IVI) and polygraph construction. This index is used to determine the overall importance of each species in the community structure. In calculating this index, the percentage values of the relative frequency, relative density and relative dominance are summed up together and this value is designated as IVI or importance value index of the species. The IVI provides an idea of the sociological structure of a species in its totality in the community but does not indicate its position separately with regard to other aspects, such as frequency, density, and so on. In consequence, plant ecologists often use **polygraphs** to show the individual and combined aspects of the position of each species in the structure of the community.

The method of polygraph preparation involves drawing a circle and dividing it into four equal segments, by drawing two lines at right angles to each other passing through its centre. Three of the radii are divided into 100 equal parts from the centre to the circumference and the fourth is divided into 300 equal parts to mark the IVI value. On the 0–100 scale in the three radii, the values of relative frequency, relative density and relative dominance are marked. Figure 7.8 shows some of the polygraphs (or phytographs) which are generally used by plant ecologists (e.g., **Dani** and **Behera**, 1990).



Animal ecologists use an **importance value** (IV) as follows (Sajise *et al.*, 1976) :

$$IV (\%) = \frac{W_a}{W_t} \times 100$$

where W_a is the average dry weight (dry weight in grams per square meter per month) of the species and W_t is the total dry weight of all species of the community (of the area).

HABITAT AND NICHE

The term **niche** was for the first time used by **Joseph Grinnel** (1971) to explain microhabitats of California thrashers. According to him, “niche is the ultimate distributional unit, within which each species is held by its structural and instinctive limitations.....no two species in the same general territory can occupy for long identically the same ecological niche.” The **area** of a species or any other taxon refers to the total extent of its geographical range of dispersal; this can be plotted on a map. The **habitat** of a species described in a single word or phrase is the totality of abiotic factors to which the species is exposed in this area. Thus, one can talk of marine habitats, coastal habitats, marsh habitats, forest habitats, disturbed habitats or even dry habitats.

In many cases, however, the habitat of the species is highly specialized. Certain species of leaf miners, for example, live only in the upper photosynthetic layer of the leaves of certain species of plants, while other species live in the lower cell layer. These patterns of location are so consistent that it is clear that the habitats of the species are different. Subdivisions of the environment on this scale are commonly called **microhabitats**. Thus, the leaf constitutes a microhabitat for leaf miners within the total forest, and the different cell layers of leaf constitute different microhabitats within the leaf for different species of leaf miner. Conditions within the leaves are quite different from the general conditions in the forest. The specific environmental variables in the microhabitat of a population is called **microenvironment** or the **microclimate** (Clapham, Jr., 1973).

Though habitats and microhabitats indicate the places where organisms of a community live, but, if one wishes to understand how a species fit into the fabric of the ecosystem, it is not enough to merely describe the creature’s habitat or microhabitat, he must also understand its functional role within the community—“address” of the organism but its “profession” as well. This includes what it eats, what it does, where it lives—everything about it that influences the community. Thus, the ecological niche is the property of the community and it represents the place of the species in the formal community structure.

In fact, the ecological niche is an inclusive term that involves not only the physical space occupied by an organism, but also its functional role in the community (*i.e.*, trophic position occupied) and its position in environmental gradients including other conditions of existence. These three aspects of ecological niche are generally designated as (i) the **spatial** or **habitat niche** (the physical space occupied); (ii) the **trophic niche** (functional role, *i.e.*, trophic position), and (iii) the multidimensional or **hypervolume niche** (position in the environmental gradients).

1. Spatial or Habitat Niche

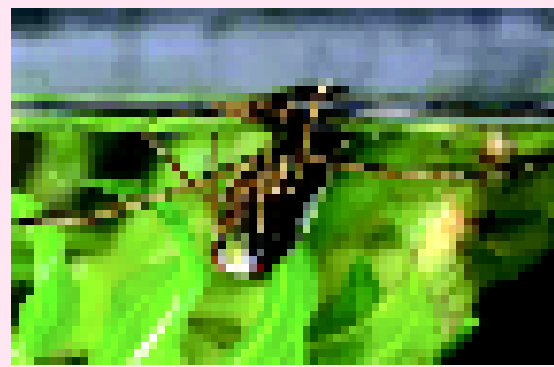
This type of niche pertains to the physical space occupied by an organism. **O’Neill** (1967) has cited a good example of spatial niche in the distribution of seven species of millipedes in forest floor of a maple-oak forest. All the seven species occur in the same general habitat, *i.e.*, forest floor and all belong to the same basic trophic level (*i.e.*, all are decomposers in their role as detritivores). Each species of millipedes predominates in its own specific different microhabitat. In fact, there exist several gradients in decomposition stage from the centre of log to the position underneath the leaf litter, and these gradients constitute distinct microhabitats in the same general habitat, the forest floor. Thus, *Euryurus erythropygus* predominates in the heartwood at centre of logs, *Pseudopolydesmus serratus* in superficial wood of logs, *Narceus americanus* in the outer surface of the logs beneath the bark, *Scytonotus granulatus* under log but on the log surface, *Fontaria virginiensis* under the log but on

ground surface, *Cleidogonia caesioannularis* within leaves of litter, and *Abacion lacterium* beneath litter on ground surface.

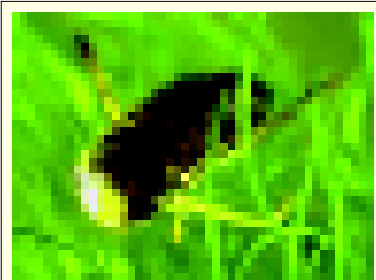
2. Trophic Niche

In this case two species live in the same habitat but they occupy different trophic niches, because of differences in food habits. For example, two aquatic bugs, *Notonecta* and *Corixa* live in the same pond but occupy different trophic niches. The former is an active predator that swims about grasping and eating other animals, but the latter feeds mainly on decaying vegetation.

Finches of Galapagos Islands in America provide another good example of trophic niche. These birds belong to three genera, viz., *Geospiza* (ground finche), *Camarhynchus* (tree finche) and *Certhidia* (warbler finch). These all birds live in the same general habitat, but differ in terms of their trophic position. For example, in tree finches, one of the species, *Camarhynchus crassirostris* has a parrot-like beak and is basically vegetarian, feeding on buds and fruits. Rest of the tree finches are insect eaters, three of which feed on insects of different sizes. Another species, *C. heliobates* feeds on insects in mangrove swamps. The most interesting is wood-pecker finch, *C. pallidus* which climbs tree in search of insects in the cracks of bark. Although, it has wood-pecking beak, it lacks long extensive tongue of wood-pecker.



Notonecta.



Corixa.

3. Multifactor or Hypervolume Niche

This concept, where niche is considered as an abstract n-dimensional inhabited hypervolume, was developed by **G.E.Hutchinson** (1965) of Yale University. Suppose that we measure the range of some environmental factor (e.g., temperature) over which a particular species can live and reproduce (in effect, its range of tolerance) and we put this on a graph. Suppose that we then do the same for another environmental factor (e.g., humidity) and put this on the second axis of the graph. The space that is enclosed will represent the niche of the species. If the effects of two variables (or factors) (i.e., temperature and humidity) are independent, the space would be two-dimensional box (Fig. 7.9 A). However, in fact, temperature and relative humidity are not independent in their biological effects. Thus, tolerance to higher temperature may be related with increase in relative humidity. Under such condition of interacting variables or factors the niche instead of being the box-like would be an oval in shape (Fig. 7.9 B). Now suppose that the tolerance to levels of a third factor (i.e., available phosphorus) was affected by interactions with both temperature and relative humidity. We would now have a niche representing three variables or three-dimensional, volumetric figure (Fig. 7.9 C). If there is present a fourth variable, the space enclosed would be a hyper-volume with four dimensions. Since there are a large number (n) of other environmental factors, both abiotic and biotic, that affect the population, the niche is n-dimensional hypervolume, an abstraction since one is able to draw only with respect to three dimensions. This is the **fundamental niche** of the species. When the fundamental niches of two species overlap them, the two species are competing. Competitive exclusion would be called into play only in

the complete overlapping of n-dimensional niches (Fig. 7.9 D). Thus, this concept refers to the totality of abiotic and biotic factors to which a given species is uniquely adapted.

Advantages of niche segregation. The major advantage which organisms gain by occupying different niches is escape from continuous intense competition. In fact, the niche occupied is favourable to the species physically in furnishing suitable substratum and microclimate, although many species have the ability to live elsewhere where competition is not involved. Automatic segregation of a species into its niche through inherited behaviour patterns or imprinting avoids the great expenditure of energy and loss of time that would be required if this segregation had to be worked out anew each year or

each generation. Segregation into niches also avoids confusion of activities between organisms in the community and permits a more orderly and efficient life cycle on the part of each species. Furthermore, the segregation of each species into different niches permits the occupancy of the area by a large number of species, since they will better divide the available resources between them. Similarly, the more distinct the niche of a species is, the more it can avoid conflict with its neighbours and lead a life that is orderly, productive and quite efficient (Kendeigh, 1974).

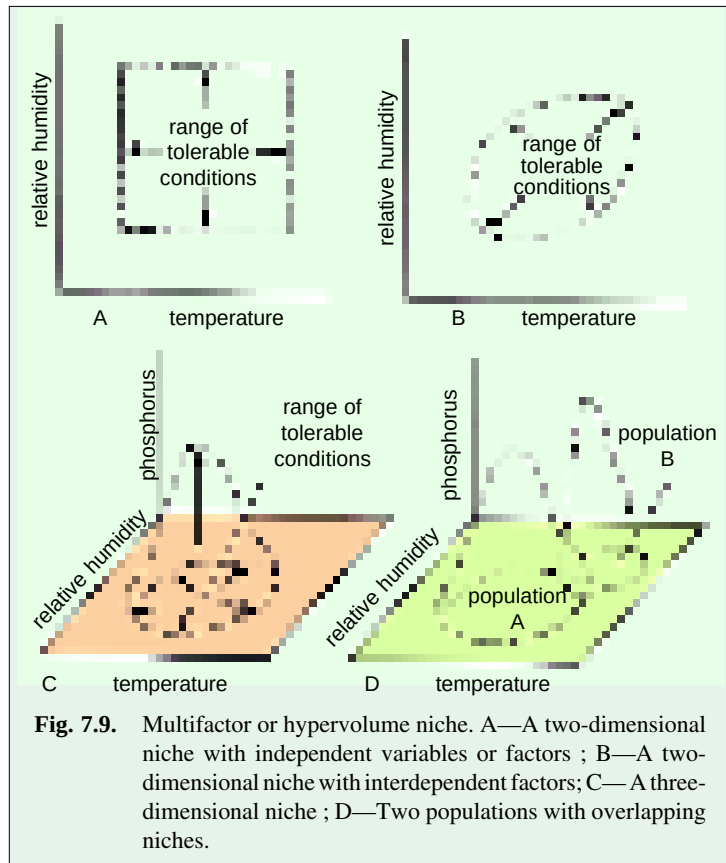


Fig. 7.9. Multifactor or hypervolume niche. A—A two-dimensional niche with independent variables or factors; B—A two-dimensional niche with interdependent factors; C—A three-dimensional niche; D—Two populations with overlapping niches.

COMMUNITY METABOLISM

Once a basic idea of community structure is gained it becomes necessary to ask the question how does a community function? The study of functional organisation or community metabolism is fundamentally a study of food relationship within the community. Such studies involve investigations on the flow of energy and cycling of nutrients within a community. Through the **food web** one can make the graphic depiction of the actual flow of food materials from population to population through the community.

COMMUNITY STABILITY

One of the most important considerations about communities is their stability, or lack of variation in time. There are two discrete senses in which this term is used. The first, **community stability**, refers to the degree of fluctuation in population size of the populations comprising the community. The second, **environmental stability**, refer to the fluctuations in the abiotic factors of the ecosystem. The

two types of stability are closely related, and stable communities are generally to be found in stable environments, while unstable environments are more often characterized by unstable communities.

The global range of variation in community stability is extraordinarily broad. Some communities such as the tropical rainforest or the coral reef, appear exceedingly stable, and the densities of their populations are constant over long periods of time. Others, such as the tundra, are extremely variable, and they are made up of populations that fluctuate widely in abundance. Much of this fluctuation is due to variations in environmental factors which the community cannot avoid. However, genetically determined adaptations that have arisen through natural selection also play a part in determining the responses of a population to environmental fluctuations.

Further, the structure of community in which a population is found, has a major role in determining the degree of environmental variability to which the species is subjected. There are at least two mechanisms by which community structure affects the stability or variability of a community. The first mechanism involves the relationship between community structure and microenvironment. Many organisms are protected from the full impact of environmental fluctuations because of the presence of other members of the community. For example, the floor of a conifer forest is protected from winds, evaporation, and temperature shifts, making it a much less harsh and variable environment for organisms than the forest canopy or an open field adjacent to the forest. The second mechanism by which community structure affects community stability is community diversity. But about the correlation of community stability and community diversity, many ecologists have expressed their doubts (Hairston *et al.*, 1968).

ECOTONE AND EDGE EFFECT

In the ecocline (*i.e.*, gradient of ecosystems), the line of demarcation (boundary-line) between two communities is often very difficult in view of the chances of overlapping of one community over another. Such demarcation will be conspicuous only when the dominants of the adjacent communities show clear and characteristic differences. The transition zone between two or more diverse communities is called the **ecotone**. The common examples of ecotone are following—the border between forest and grassland, the bank of a stream running through a meadow or between a soft bottom and hard bottom marine community. An ecotone, thus, presents conditions that are intermediate to the communities which are on either side of it. The ecotones may be narrow or very wide, extending to large areas. The community on either side of the ecotone may have a typical structure, but the ecotone is strikingly different. The ecotone has a higher diversity than either of the main communities, a diversity that is not directly controlled by the climate or further fundamental environmental factors, but because of the migrations of individuals of different species-populations from both communities. Further, a number of special populations can become adapted specifically to the ecotone, even when both of the major communities are too simple or are otherwise unsuited for successful colonization by the species. This potential for the ecotone to act as habitat for species found in neither major community is termed as the **edge effect**. A common example of the edge effect in action can be seen in those species of owl that live in or near ecotones between forests and grassland. They depend on forest trees for nesting, and they do their hunting in the grassland, where they depend on field rodents for food. In man-made communities such as agricultural fields, the



Ecotone.

transition zone (ecotone) between the field and the forest may act as a refuge for animal species formerly found in the plowed area, as well as for other plants such as weeds. Ecotones of this type are also the prime habitat of many species of insect, game bird, and mammal.

FACTOR COMPENSATION AND ECOTYPES

Organisms are not just 'slaves' to the physical environments, they adapt themselves and modify the physical environment so as to reduce the limiting effects of temperature, light, water, and other physical conditions of existence. Such **factor compensation** is particularly effective at the community level of organization, but also occurs within the species. Species with wide geographical ranges almost always develop locally adapted populations called **ecotypes** that have different genetically based tolerance ranges and optima to correspond to the environments within which they normally live (Odum, 1971; Clapham Jr., 1973). The term ecotype was coined by a Swedish botanist, **Göte Turesson** in 1912 and in 1922 he defined it as the product arising as a result of genotypical response of an ecospecies to a particular habitat. Ecotypes are especially common in plants, but are also shown by animals and microorganisms.

The ecotypes of the plant species though remain genetically distinct, but are interfertile and are put into one taxonomic species. The different ecotypes of a particular species may differ in their edaphic, biotic, or microclimatic requirements. In India **Misra** and **Rao** (1948) first time reported ecotypic differentiation in *Lindenbergia polyantha*. Ecotypes have also been reported in following Indian plant species as *Euphorbia hirta*, *Euphorbia thymifolia*, *Cassia tora*, *Xanthium strumarium*, etc.

Among animals, ecotypes have been reported in jelly fish, *Aurelia aurita* (Bullock, 1955), European oyster, *Ostrea edulis* (Korringa, 1957), and frog. Some frogs, for example, undergo rapid development at lower temperatures at one extreme of their range of distribution, while others show a more retarded development rate at higher temperatures at another extreme of their range.

ECOLOGICAL INDICATORS

Since, specific factors often determine rather precisely what kinds of organisms will be present in a particular habitat, we can turn the situation around and judge the kind of physical environment from the organism present. It is found that certain species of microorganisms, plants and animals have one or more specific requirements and they become very much limited in their distribution. Thus, the occurrence of such species in a particular area indicates special habitat conditions, and such species

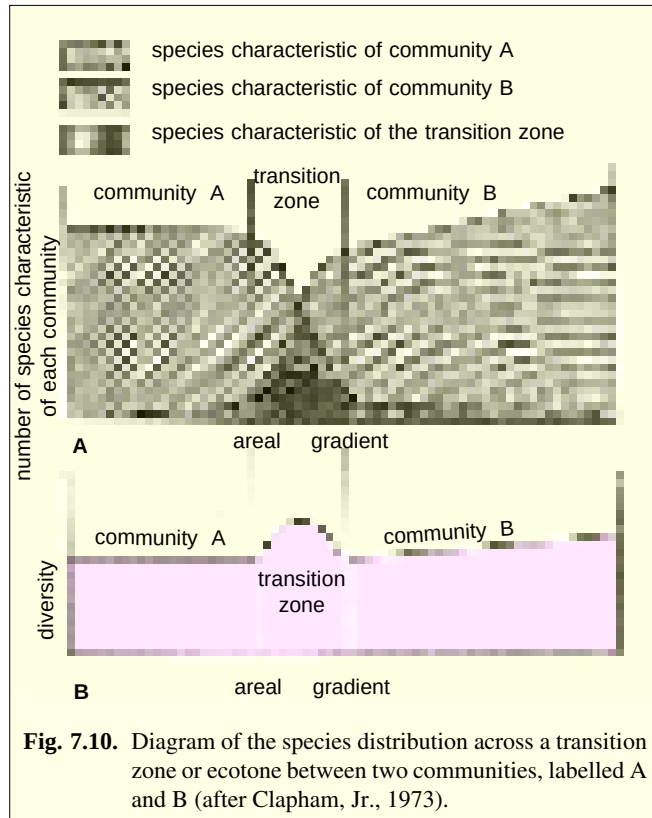


Fig. 7.10. Diagram of the species distribution across a transition zone or ecotone between two communities, labelled A and B (after Clapham, Jr., 1973).

are called **biological** or **ecological indicators**, since they indicate some very specific condition of the environment. Some of the Indian plant species which serve as ecological indicators have been tabulated in Table 7-2.

Table 7-2. Certain ecological indicators of plant species.

Name of species	Indicator (s) of —
1. <i>Utricularia</i> , <i>Chara</i> , <i>Wolffia</i> , <i>Ottelia alismoides</i>	Water pollution
2. <i>Petridium</i> sp.	Burnt and highly disturbed coniferous forest
3. <i>Argemone maxicana</i>	Recently disturbed or flooded soil
4. <i>Rumex acetosella</i>	Acid grassland soil
5. <i>Carissa spinarum</i> and <i>Capparis spinosa</i>	Intense soil erosion
6. <i>Enicostema littorale</i>	Well drained soil
7. <i>Saccharum spontaneum</i>	Poorly drained soil
8. <i>Zizyphus rotundifolia</i>	Soil deposition
9. <i>Andropogon scoparium</i>	Sandy loam type soil
10. <i>Lippia nodiflora</i> and <i>Rumex</i> sp.	Nitrate rich soil
11. <i>Woodfordia foribunda</i> and <i>Chloris virgata</i>	High lime content of soil
12. <i>Atriplex</i> , <i>Salsola</i> , and <i>Suaeda</i>	Saline water condition
13. <i>Hydrilla verticillata</i> and <i>Ceratophyllum demersum</i>	Hard water
14. <i>Waltheria indica sterculiaceae</i>	Copper, lead and zinc rich soil
15. <i>Equisetum plebejum</i> and <i>E. arvense</i>	Gold rich soil
16. <i>Silene cobalticola</i> and <i>Crotalaria cobalticola</i>	Cobalt rich soil
17. <i>Spermacoce stricta</i>	Iron rich soil
18. Pines and junipers	Uranium rich soil

Besides plants, certain microorganisms serve as ecological indicators. For example, *Escherichia coli*, a bacterium, is an index organism for aquatic pollution. Some species of diatoms (Araphidinae) indicate the degree of sewage pollution of lakes. The presence of fusilinids (protozoans) in the soil indicate the petroleum deposits.

Similarly, certain animal species serve as ecological indicators. Chironomid larvae and maggots increase in number as the pollution increases. Burrowing may fly *Hexagenia* serves as indicator for well oxygenated aquatic environments. Emigration and consequent disappearance of certain Indian fish species such as *Catla catla*, *Labeo gonius*, *Labeo bata*, *Labeo rohita*, *Notopterus*, etc., from Kalinadi river near Mansurpur Sugar Factory, U.P. indicate industrial pollution of water, as these fishes are found to be very sensitive to aquatic pollution (Verma, 1971).

REVISION QUESTIONS

- Define biotic community and describe in detail its different characteristics.
- What is community stratification ? Explain the phenomenon with suitable examples of plants and animals.
- Write short notes on the following :
 - Ecotone,
 - Edge effect,
 - Ecotype,
 - Ecological indicators,
 - Ecological niche,
 - Life or growth forms.

CHAPTER

8

Ecological Succession



The process of directional change in vegetation is termed as succession.

When stripped of its original vegetation by fire, flood, or glaciation, an area of bare ground does not remain devoid of plants and animals. Beginning with plants, area is rapidly colonized by a variety of both plant and animal species that subsequently modify one or more environmental factors. This modification of the environment may in turn allow additional species to become established. The transitional series of communities which develop in a given area are called **sere** or **seral stages**, while the final stable and mature community is called the **climax**. The development of the community by the action of vegetation on the environment leading to the establishment of new species is termed **succession**. Succession is the universal process of directional change in vegetation during ecological time. It can be recognized by the progressive change in the species composition of the community. Retrogression in community development does not occur unless succession is disturbed or halted by fire, grazing, scraping or erosion.

CAUSES OF SUCCESSION

Since succession involves a series of complex processes, so there exist many causes of its occurrence. Ecologists have recognized the following three primary causes of succession :

1. Initial or Initiating causes. These are climatic as well as biotic in nature. The climatic causes include factors such as

erosion and deposits, wind, fire, etc., which are caused by lightening or volcanic activity. The biotic causes include various activities of organisms. All these causes produce the bare areas or destroy the existing populations in an area.

2. Ecesis or Continuing causes. These are processes as migration, ecesis, aggregation, competition, reaction, etc., which cause successive waves of populations as a result of changes, chiefly in the edaphic (soil) features of the area.

3. Stabilising causes. These include factors such as climate of the area which result in the stabilisation of the community.

TRENDS OF SUCCESSION (Functional Changes)

Trends of changes during ecosystem development from a young to mature stages include the following features :

1. A continuous change occurs in the kinds of plants and animals.
2. An increase in the diversity of species takes place. The general appearance of the community or the physiognomy keeps on becoming more and more complex as the succession proceeds.
3. There is a progressive increase in the amount of living biomass and dead organic matter. Such an increase occurs in gross as well as net primary production in the initial and seral stages. Thus, there is more biomass accumulation, gradually reaching a huge biomass structure in the climax.
4. Green pigment (Chlorophyll) goes on increasing during the early phase of primary succession. The ratio of yellow/green pigments remains around 2 in the early stages and increases to 3 to 5 in the climax stage. Pigment diversity also increases.
5. The community respiration increases but the P/R (*i.e.*, Production/Respiration) ratio remains more than 1 in the seral stages. The huge living biomass respire a lot in the climax stage and the P/R ratio equals 1 (*i.e.*, $P/R = 1$). Thus, in the early stages $P > R$ and in the climax stage $P = R$.
6. The food chain relationships become more complex as succession proceeds.
7. Nutrients in the young stage are allocated mostly in the soil, but as the seral stages advance, nutrients get allocated more in the vegetation and less in soil. Further the nutrient cycling becomes more closed or intrabiotic with an efficient cycling mechanism whereas in the young stage the nutrients easily leak out from the system, *i.e.*, the cycling is more of an open type.
8. The role of detritus becomes progressively more and more important.
9. The quality of the habitat gets progressively modified to a more mesic condition from either too dry or too wet condition, in the early seral stage.
10. The niche specialization increases, *i.e.*, different functions are more effectively performed by specialist species in mature seral stage, whereas in early stage many functions are performed but less efficiently by a few species.
11. The life cycle of mature community species are longer and more complex.
12. The importance of macroenvironment becomes less in later stages.
13. Relationship among component species becomes more and more mutualistic or helpful even though there exist enough competition and allelochemic activities to prevent invasion of outside elements.
14. Dispersal of seeds and propagules is by wind in young stage, while by animals in mature stage.

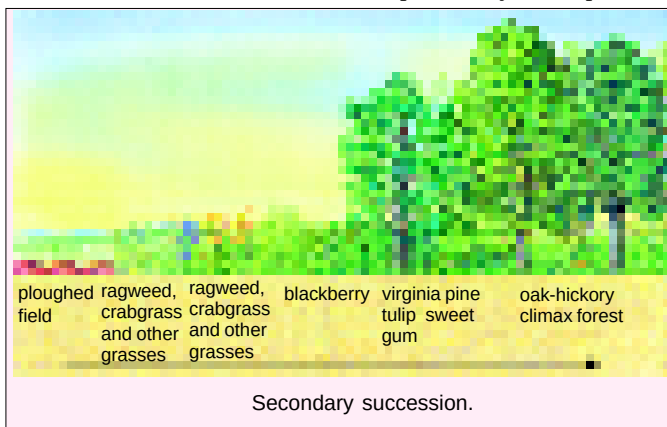
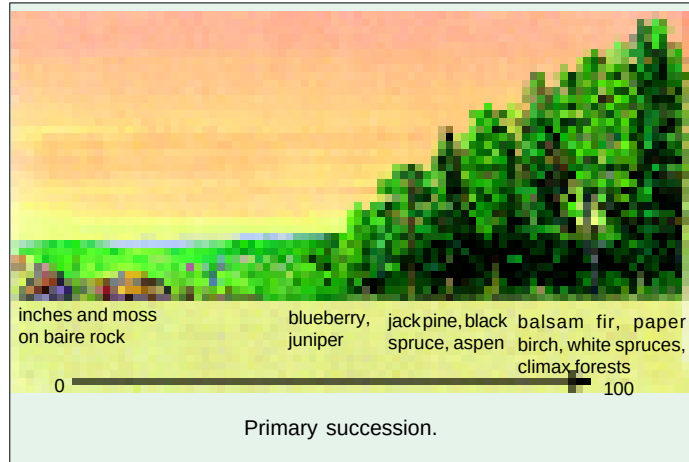
BASIC TYPES OF SUCCESSION

Based on different criteria, there are following kinds of succession :

1. Primary succession. If an area in any of the basic environments (such as terrestrial, fresh-water or marine) is colonized by organisms for the first time, the succession is called **primary succession**. Thus, primary succession begins on a sterile area (an area not occupied previously by a

community), such as newly exposed rock or sand dune where the conditions of existence may not be favourable initially.

2. Secondary succession. If the area under colonization has been cleared by whatsoever agency (such as burning, grazing, clearing, felling of trees, sudden change in climatic factors, etc.) of the previous plants, it is called **secondary succession**. Usually the rate of secondary succession is faster than that of primary succession because of better nutrient and other conditions in area previously under plant cover.



3. Autogenic succession.

After the succession has begun, in most of the cases, it is the community itself which, as a result of its reactions with the environment, modifies its own environment and, thus, causing its own replacement by new communities. This course of succession is known as **autogenic succession**.

4. Allogenic succession.

In some cases replacement of one community by another is largely due to forces other than the effects of

communities on the environment. This is called **allogenic succession** and it may occur in a highly disturbed or eroded area or in ponds where nutrients and pollutants enter from outside and modify the environment and in turn the communities.

5. Autotrophic succession. It is characterized by early and continued dominance of autotrophic organisms such as green plants. It begins in a predominantly inorganic environments and the energy flow is maintained indefinitely. There is gradual increase in the organic matter content supported by energy flow.

6. Heterotrophic succession. It is characterized by early dominance of heterotrophic organisms such as bacteria, actinomycetes, fungi and animals. It begins in a medium which is rich in organic matter such as small areas of rivers, streams; these are polluted heavily with sewage or in small pools receiving leaf litter in large quantities.

7. Induced succession. Activities such as overgrazing, frequent scraping, shifting cultivation or industrial pollution may cause deterioration of an ecosystem. Agricultural practices are retrogression of a stable state to a young state by man's deliberate action.

8. Retrogressive succession. It means a return to simpler and less dense or even impoverished form of community from an advanced or climax community. In most cases, the causes are allogenic, i.e., forces from outside the ecosystem become severe and demanding. For example, most of our natural forest stands are degrading into shrubs, savanna or impoverished desert-like stands by the severity of grazing animals brought from surrounding villages. Excessive removal of wood, leaf and twig litter also leads to retrogressive succession.

9. Cyclic succession. It is of local occurrence within a large community. Here *cyclic* refers to repeated occurrence of certain stages of succession whenever there is an open condition created within a large community.

Further, in ecological literature, there are mentioned still so many other kinds of succession, depending mainly upon the nature of the environment where the process of succession has begun. Thus, it may be a **hydrosere** or **hydrarch** when succession occurs in regions where water is in plenty, as ponds, lakes, streams, swamps, bogs, etc., a **mesarch** when there are present adequate moisture conditions; and a **xerosere** or **xerarch** when there is present minimum amount of moisture, *e.g.*, deserts, rocks, etc. Sometimes, few more categories are recognized such as **lithosere** – succession initiating on rocks, **psammosere** – succession initiating on sand and **halosere** – succession occurring in saline water or saline soil.

GENERAL PROCESS OF SUCCESSION

The entire process of primary autotrophic succession is completed through the following sequential steps :

1. Nudation

This is the development of a bare area without any form of life. Exposure of new surface may occur due to several causes such as landslide, erosion, deposition, or other catastrophic agency. These causes of nudation are of three main types :

(i) Topographic. Due to soil erosion by gravity, water or wind, the existing community may disappear. Other topographic causes include deposition of sand, landslide, volcanic activity and other factors.

(ii) Climatic. Glaciers, dry period, hails and storm, frost, fire, etc., may also destroy the community.

(iii) Biotic. Man forms a most important biotic factor; he is responsible for destruction of forests, grasslands for industry, agriculture, housing, etc. Other factors are disease epidemics due to fungi, viruses, etc., which destroy the whole population.

2. Invasion

Invasion is the successful establishment of a species in a bare area. The species actually reaches this new site from any other area. Invasion includes the following three steps :

(i) Migration (dispersal). The seeds, spores, or other propagules of the species reach the bare area. This process is called **migration** and is generally brought about by air, water, etc.,

(ii) Ecesis (establishment). After reaching to new area, the process of successful establishment of the species, as a result of adjustment with the conditions prevailing there, is known as **ecesis**. Ecesis is followed by full-scale **colonization**. In plants, after migration, seeds or propagules germinate, seedlings grow, and adults start to reproduce.

(iii) Aggregation. Colonization by successive offspring and new migrants help increase the population, a process called **aggregation**. Thus, after ecesis, as a result of reproduction, the individuals of the species increase in number and they come close to each other. Plants or autotrophic organisms which are the first to colonize and aggregate are called **pioneers**. The pioneer communities are likely to be more dynamic and have low-nutrient requirements and to take minerals in comparatively more complex forms. They are small-sized and make less demand from environment.

3. Competition and Coaction

Due to aggregation of a large number of individuals of the species at the limited place, there develops **competition** (*i.e.*, interspecific and intraspecific competition) for space and nutrition. Individuals of a species affect each other's life in various ways and this is called **coaction**. The species which fail to compete with other species are ultimately discarded. The reproductive capacity, wide

ecological amplitude, etc., help the species to withstand the competition.

4. Reaction

Reaction includes mechanism of the modification of the environment through the influence of living organisms on it. Due to this very significant stage, changes take place in soil, water, light conditions, temperature, etc., of the environment. As a result of reaction, the environment is modified and become unsuitable for the existing community which sooner or later is replaced by another community (seral community). The whole sequence of communities that replaces one another in the given area is called a **sere**, and different communities constituting the sere are called **seral communities**, **seral stages** or **developmental stages**.

5. Stabilization (Climax)

Finally, there occurs a stage in the process, when the final terminal community becomes more or less established for a longer period of time and it can maintain itself in equilibrium with the climate of the area. The final community is not replaced and is known as **climax community** and the stage as **climax stage**.

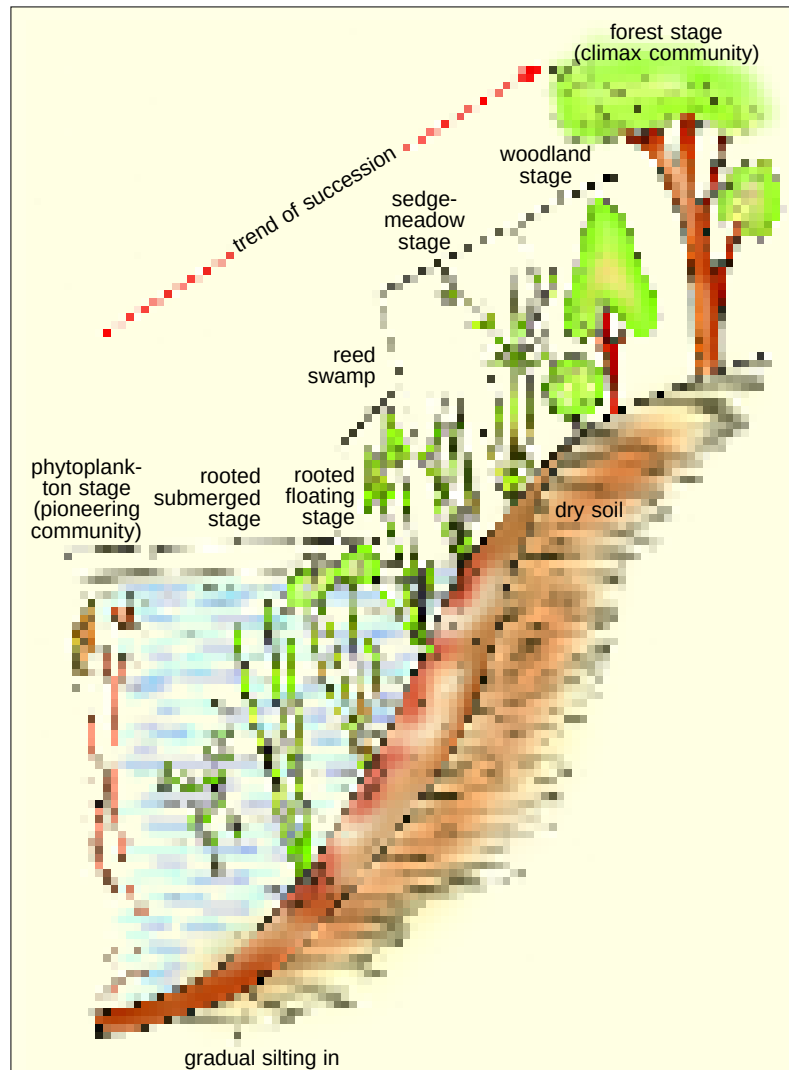


Fig. 8.1. Diagrammatic representation of succession through autogenic processes. Seral communities modify the environmental complex which in turn changes the vegetation and the process continues until the dynamic equilibrium is reached. Note the pioneer, seral and climax communities.

SOME EXAMPLES OF SUCCESSION

1. Hydrosere. A good example of succession is the **hydrarch succession** or **hydrosere** (Fig. 8.2), in which a pond and its community are converted into a land community. In the initial stage, phytoplankton (e.g., some blue green algae (cyanobacteria), green algae (e.g., *Spirogyra*, *Oedogonium*), diatoms and bacteria) are the pioneer colonizers. They are consumed by zooplankton (e.g., protozoans such as *Amoeba*, *Paramecium*, *Euglena*, etc.), fish such as blue gill fish, sun fish, large mouth, etc. Gradually these organisms die and increase the content of

dead organic matter in the pond. This is utilized by bacteria and fungi, and minerals are released after the decomposition. The nutrient-rich mud then supports the growth of rooted hydrophytes such as *Hydrilla*, *Elodea*, *Vallisneria*, *Ceratophyllum*, etc., in the shallow water zone. This submerged stage is also inhabited by the animals such as dragon flies, may flies and crustaceans such as *Asellus*, *Gammarus*, *Daphnia*, *Cypris*, *Cyclops*, etc. The hydrophytes die and are decomposed by microorganisms, thus, releasing nutrients. In addition to this, due to silting, the water depth of the pond is reduced and at the margin of the pond grow rooted floating vegetation (i.e., the plant species whose leaves reach the water surface and roots remain in the mud). Plants such as *Nelumbo nucifera*, *Trapa*, *Monochoria*, etc., grow in these conditions. In floating stage faunal living space is increased and diversified. Hydras, frogs, salamanders, gill-breathing snails, diving beetles (*Dysticus*) and host of new insects capable of utilizing the under surfaces of floating leaves appear. Some turtles and snakes also invade the pond.

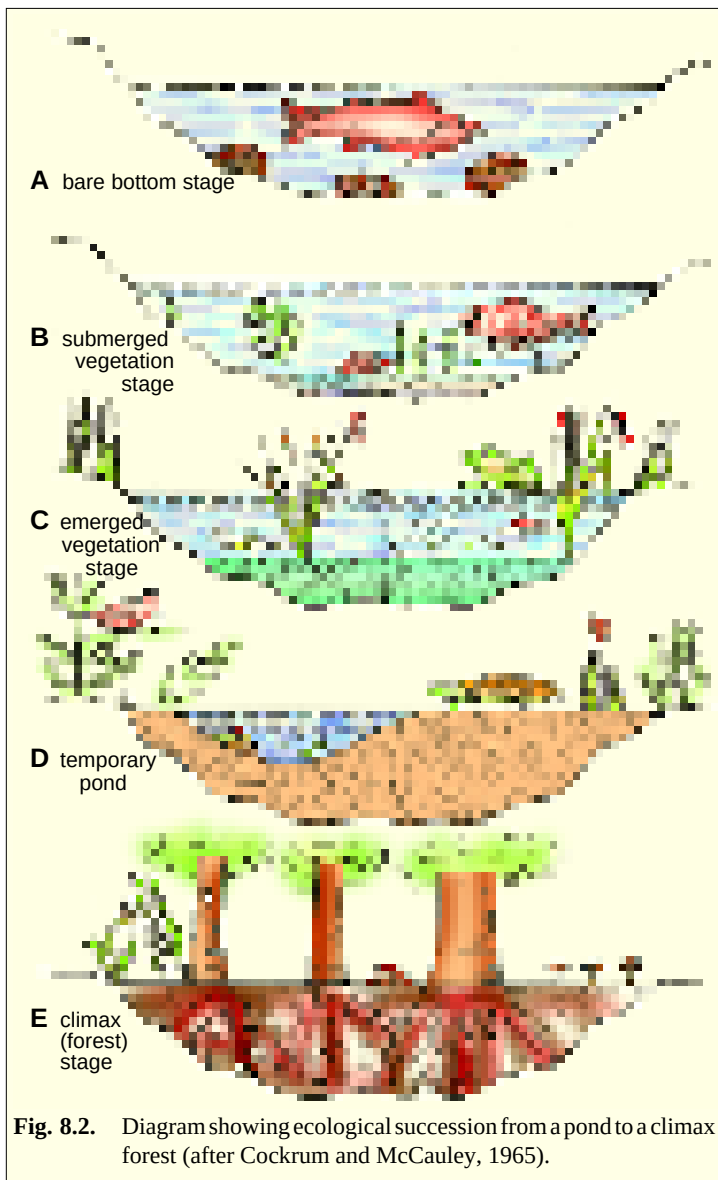
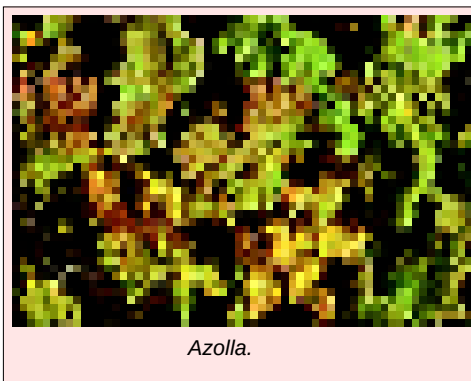


Fig. 8.2. Diagram showing ecological succession from a pond to a climax forest (after Cockrum and McCauley, 1965).



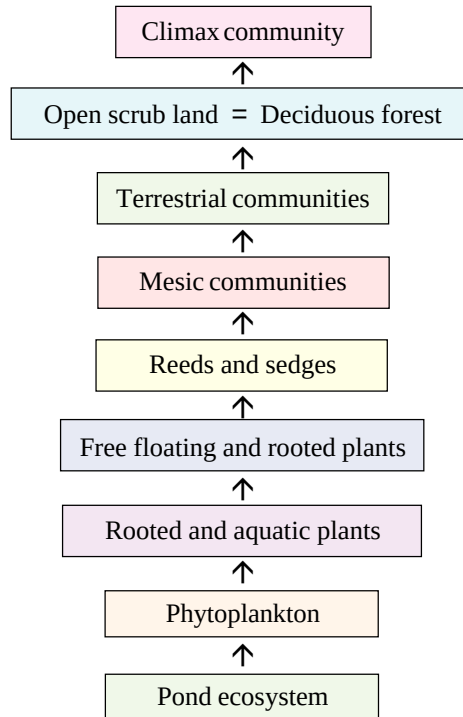
Azolla.

Gradually the water depth in the pond decreases due to evaporation and the deposition of organic matter and the concentration of the nutrients increases. Free-floating plants such as *Azolla*, *Lemna*, *Pistia*, *Wolffia*, *Spirodella*, etc., increase in number because of the high nutrient availability. Gradually their dead parts fill up the pond ecosystem, resulting in the further build up of the substratum.

At this stage, the pond becomes a swampy ecosystem. The reed swamp species (such as *Scirpus* or bulrushes, *Typha* or cattail, *Phragmites* (reed grass), *Rumex*, etc.) and sedges (e.g., *Carex*, *Juncus*, *Cyperus*,

etc.) invade the pond and the latter are gradually replaced by mesic communities as the water depth is reduced greatly. Gradually land plants, such as, shrubs (*Salix*, *Cornus*) and trees (*Populus*, *Alnus*) invade ending in the climax community such as deciduous forest (Fig. 8.2). In association with the changes in water depth and vegetation, the aquatic fauna also change and ultimately gets replaced by land animals.

Thus, possible trend of succession in the aquatic environment is as follows :



Climax communities vary from place to place. For example, in low lying lands in some parts of Kashmir, the climax community has trees such as *Salix* (Ambash, 1988). In Indian upland plateaus, the climax woody species consist of *Diospyros*, *Butea* and *Zizyphus* and ground vegetation of *Eragrostis*, *Sporobolus*, *Bothriochloa*, etc. Climax vegetation of lowlands and valleys which provide a mesic environment includes *Terminalia*, *Ficus*, *Sterculia*, *Salix*, etc.

2. Succession in xeric habitat. Xerosere or xerarch succession begins on exposed parent rocks (**lithosere**) or dry sand (**psammosere**). A lithosere (Fig. 8.3) involves the following stages : crustose lichens stage (pioneers) → foliose lichens stage → moss stage → herbs stage → shrub stage → forest stage (climax stage). Thus, pioneer plants are lichens, mosses and *Selaginella*, which help in soil formation by accelerating erosion. In course of time grasses, annuals and herbaceous vegetation grow on the soils deposited on rocks.

Like the hydrosere, the lithosere involves successive changes in animal life. The pioneer animals of the lichen stages are few species of mites, ants and spiders. These animals are exposed to harsh environment such as extreme fluctuations in temperature. During the moss stage, many new species of mites, small spiders, tardigrades and springtails invade the community. The herb stage is characterized by nematodes, mites, collembola, ants and various insect larvae. During the shrub and forest stages great qualitative and quantitative modifications occur in the fauna. Thus, there occur numerous kinds of animals such as slugs, snails, wire worms, millipedes, centipedes, mites, ants, sow bugs, springtails, amphibians such as salamanders, frogs, etc., reptiles such as turtles, skinks and other lizards, snakes, birds such as flycatcher and grouse, and mammals such as mole, mouse, shrews, squirrels, chipmunk and fox.

CONCEPT OF CLIMAX

The concept of climax has since long been a subject of much controversy and discussion. There are following three theoretical approaches to the climax :

1. Monoclimax theory. This theory is developed largely by **Frederick Clements**. This theory recognizes only one climax, determined solely by climate, no matter how great the variety of environmental conditions is at the start. All seral communities in a given region, if allowed sufficient time, would ultimately converge to a single climax. The whole landscape would be clothed with a uniform plant and animal community. All other communities than the climax are related to the climax by successional development and are recognized as subclimax, disclimax, preclimax, postclimax and so on. A **subclimax** is a stage in succession of forests just preceding the climate climax community. The **disclimax** is the particular type of vegetation maintained in an area as a result of recurrent disturbance, chiefly the biotic, thus preventing a successful establishment of climate climax community. The **preclimax** is a vegetation of lower-life forms than the one adjacent to it and results from different edaphic conditions. The **postclimax** is a strip of vegetation of higher life-forms occurring within a climate climax, for example, a forest along a stream in a grassland community constitutes postclimax community. Later on numerous other terms such as coclimax, superclimax, quasiclimax, anticlimax, pseudoclimax, etc., were coined by some post-Clements ecologists to describe specific situations.

The monoclimax theory is supported by **Cowles**, **Ranganathan**, and **Puri**, but strongly objected by **Daubenmire** (1968).

2. Polyclimax theory. This theory was developed by **Tansley**. This theory considers that the climax vegetation of region consists of not just one type but a mosaic of vegetational climaxes controlled by soil moisture, soil nutrients, topography, slope exposure, fire and animal activity. The advocates of polyclimax theory preferred to call each stable community as a climax and described them with a prefix as edaphic climax, topographic climax, biotic (zootic, grazing or anthropogenic) climax and fire climax. For example, grassland communities of central India, Sri Lanka and parts of California have developed under the influence of fire, grazing and other biotic factors and so have been considered as biotic climaxes by **Misra**, **Pandeya** and **Holmes**.

3. Climax pattern hypothesis. This theory was developed by **Whittaker**, **MacIntosh** and **Sellack**. According to this theory, the composition, species structure and balance of a climax community are determined by the total environment of the ecosystem and not by one aspect, such as climate alone. Involved are the characteristics of each species population, their biotic interrelationships, availability of flora and fauna to colonize the area, the chance dispersal of seeds and animals, and the soils and climate. The pattern of climax vegetation will change as the environment changes. Thus, the climax community presents a pattern of populations that corresponds with and changes with the pattern of environmental gradients, intergrading to form ecoclines. The central and most

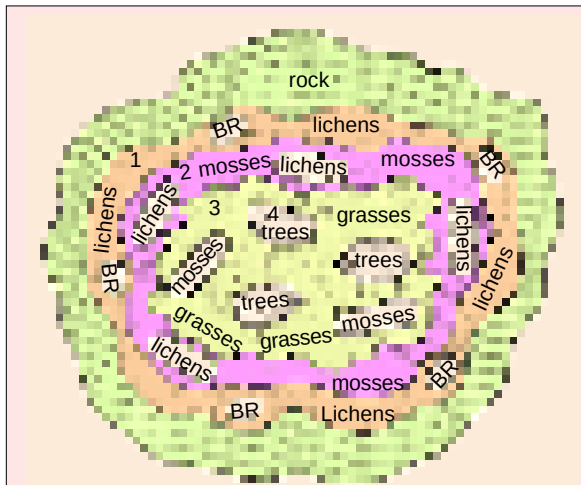


Fig. 8.3. Diagrammatic representation of different plant communities of a lithosere appearing on a rock. The various zones from outside towards the centre of rock are (1) lichens (pioneers), (2) ring of mosses, (3) grasses (broad zone) and (4) trees (seedlings). Note the pioneer community of lichens around the outer edge and more advanced stage of trees located in the centre. BR=bare rock.

widespread community in the pattern is the prevailing or climatic climax. It is the community that most clearly expresses the climate of the area.

4. Information theory. This theory was proposed by **Leith, Odum** and **Golley**. It considered succession and climax in terms of ecosystem development. In autotrophic succession (ecosystem development), diversity of species tends to increase with an increase in organic matter content and biomass supported by the available energy. Thus, in a climax community, the available energy and biomass, which is called **information content**, increase. In contrast to it, in a heterotrophic succession occurs a gradual depletion of energy, because the rates of respiration always exceed production rates. However, in an ecosystem both the autotrophic and heterotrophic successions operates in a co-ordinate manner. The autotrophic individuals derive mineral elements from the soil and atmosphere, while the heterotrophic individuals carry on the return of the nutrients to soil and atmosphere, through decomposition of complex dead organic matter. Thus, succession reaches a stage, the climax stage, when the amount of energy and nutrients received from the environment by the plants is again returned in more or less similar amount to the environment by decomposition through heterotrophs.

Certain Recent Models of Succession

Connell and **Slatyer** (1977) have proposed the following three models to accommodate different possible pathways of succession :

1. Facilitation model. It is based on the Clements ideas of relay communities in which the seral community is believed to modify. According to this model each new community in course of time prepares suitable ground to facilitate its own replacement by another better suited community. So each community like a 'relay process' delivers the habitat to next or higher status community. This model has not been supported by any proper evidence.

2. Tolerance model. This model is based on the concept of **IFC** or **initial floristic composition** which suggests that arrival of new invaders of higher life form types necessarily does not eliminate pioneers. According to this model, only such higher succession or climax species are able to join which can be tolerated by the early settlers. So, the tolerance model differs from Clement's relay succession model in the sense that as the succession proceeds, more and higher life form plants are tolerated to join and co-exist, than necessarily replacing the earlier component species. With the passage of time, species which mutually tolerate each other gain control over the habitat to form the climax vegetation.

3. Inhibition model. According to this model, the early arrived species (populations) on a new habitat may develop counter-mechanism to normal replacement process. For example, the allelopathy may be the common counter-acting adaptation to thwart or inhibit the entry of late arriving species. This kind of highly adapted early stage communities may not be common on a wide range of habitats. In such a case, relay succession gains control only after the death of the allelopathic plants. Since inhibition model lacks universality, so it was also criticised.



Clements gave the idea of relay community.

Resource-Ratio Hypothesis of Succession

This model was proposed by **Tilman** (1985) who was agreed to Clement's idea of replacement of communities. According to him, the resources of the habitat are regarded as the key factor. Limited resource leads to competition. Due to competition a resource ratio is created for each kind of resource. Depending on the newly created ratio level, new species adapted to it succeeds.

In conclusion, succession is directional and progresses towards the climax. In some cases, it may be non-directional, *i.e.*, it returns to a particular stage and it is cyclic. It takes hundreds and thousands of years to complete primary succession, while the secondary succession is quicker (takes 10 to 200 years for its completion). In some cases, the initial stage community may inhibit altogether the process of succession by allelochemic inhibition of new arrivals.

COMMUNITY EVOLUTION

Like the responses of communities to changing abiotic conditions, community evolution involves progressive changes in climax communities. Because the evolution is exceedingly slow, it cannot be observed in operation, and few instances from the fossil record are sufficiently complete to show the process in action.

The example that best demonstrates evolution of the basic structure of the community is that of the development of a terrestrial community of a modern type by early reptiles some 250 million years ago. Between the time when vertebrates (amphibians) first became able to lead a predominant terrestrial existence some 350 million years ago and the establishment of an essentially modern type food web some 100 million years later, the structure of terrestrial community was decidedly different from what it is now. Development of the modern type of community structure required not only a complete rearrangement of the niche structure of the community but also the evolution of new species that could fill the new niches (**Olson**, 1961, 1966).

Attainment of the adaptations needed for terrestrial life by the first amphibians did not in itself establish a land-based vertebrate community. These early amphibians were carnivores, and the only animals inhabiting the land environment were insects. It is unbelievable that the clumsy locomotor system of early amphibians would have allowed them to prey effectively on animals such as insects. Thus, the first communities inhabited by terrestrial vertebrates are best regarded as extensions of aquatic communities, with the land habit as an adaptation to improve the capabilities of organisms whose prime food supply was aquatic invertebrates and fish.

By some 300 million years ago, reptiles had evolved that could feed effectively on terrestrial invertebrates. An entirely land-based community was theoretically possible in which all herbivore niches were assumed by invertebrates and some of the carnivore niches by vertebrates. However, the palaeoecological evidences suggest that most contemporary carnivorous vertebrates were unable as yet to realize an entirely terrestrial carnivore niche, so that the great majority of the energy flow through the community continued to pass through the aquatic route. The typical food chain to the highest terrestrial vertebrate carnivore was plant → aquatic invertebrate → aquatic-invertebrate-feeding vertebrate → semi-aquatic predator → terrestrial predator.

By 250 million years ago terrestrial herbivorous vertebrates had evolved and a fully terrestrial vertebrates community could come into being. From this time onward the basic structure of the terrestrial community was of an essentially modern sort, with all consumer trophic levels occupied by a wide range of animals, both vertebrates and invertebrates.

Such evolutionary changes in the structure of communities are caused by a large number of factors. One factor is changes in the regional climate. It became progressively drier during the period under consideration, and the development of a land-based community reasonably responded to this sort of change. Indeed many evolutionary changes in community structure can be explained on the basis of responses of major changes in the regional abiotic factors of the environment (**Axelrod**, 1950, 1958). But other chief factors of evolutionary change in community include reorganization of the community's structure in response to the realization of niches that had not previously existed in the community.

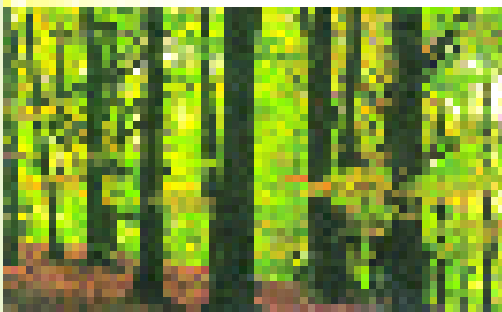
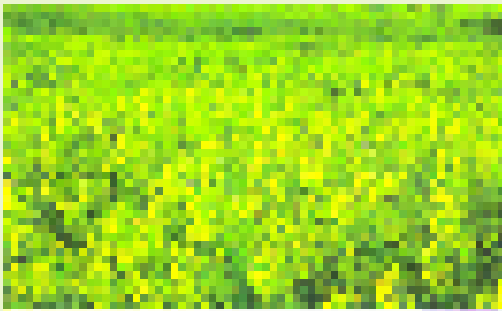
REVISION QUESTIONS

1. What is ecological succession? Describe the process of ecological succession in pond or on a bare rock. Add a note about climax.
2. Describe the causes, trends and basic types of succession.
3. Write short notes on :
(i) Ecesis ; (ii) Climax ; (iii) Allogenic succession ; (iv) Retrogressive succession.
4. Describe various models explaining the nature of climax communities.
5. Give an illustrated account of the sequential stages of a typical hydrosere.

CHAPTER

9

Ecosystem : Structure and Function



Our earth is a giant ecosystem where abiotic and biotic components react with each other bringing functional changes.

Any ecological unit that includes all the organisms (*i.e.*, the communities in a given area) which interact among themselves and with the physical environment, so that a flow of energy leads to clearly defined trophic structure, biotic diversity and material cycle (*i.e.*, exchange of materials) within the system, is known as **ecological system** or **ecosystem**. There exist nutritional relationships (or food links) amongst the living organisms of such a system. Keeping this in view, the earth can be considered as a giant ecosystem where abiotic and biotic components are constantly acting and reacting upon each other bringing forth structural and functional changes in it. This vast ecosystem — the **biosphere** is however, difficult to handle and, thus, for the sake of convenience, we generally study nature by making its artificial subdivisions into units of smaller ecosystems (such as **terrestrial** — forest, desert, grassland; **aquatic** — fresh water, marine; and **man-made** — cropland, etc.). An ecosystem may, thus, be as small as a single log, a pond, a cropland, or as large as an ocean, desert or forest. Though these unit ecosystems are separated from each other with time and space, but functionally they all are linked with each other, forming an integrated whole.

The term **ecosystem** was proposed by **A.G. Tansley** in 1935. There are many other parallel terms or synonyms for the ecosystem which have been proposed by various ecologists, *e.g.*, **biocoenosis** (**Karl Mobius**, 1877), **microcosm** (**S.A. Forbes**, 1887), **holocoen** (**Friederichs**, 1930), **biosystem** (**Thienemann**, 1939), **geobio-coenosis** (**Sukhachev**, 1944), **bioenert body** (**Vernadsky**, 1944) and **ecosom**, etc.

In recent years, ecological studies of ecosystems undertake besides structure, the similarities and differences in food and energy relationships among living components of ecosystem. This is called **bioenergetic approach** of modern ecology.

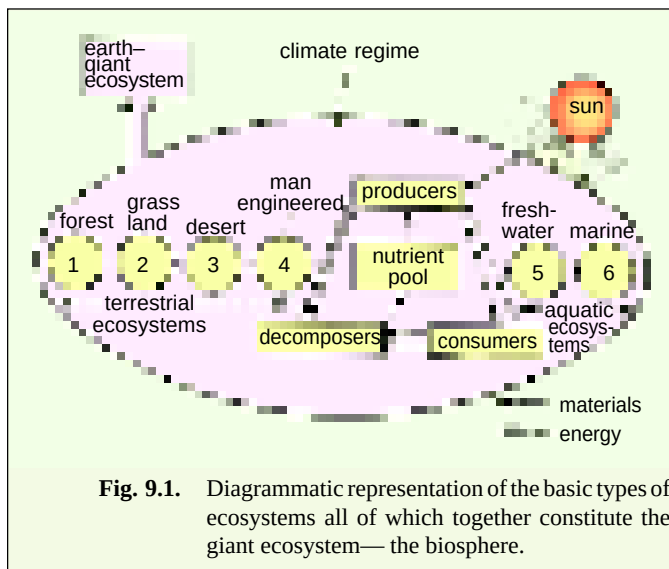


Fig. 9.1. Diagrammatic representation of the basic types of ecosystems all of which together constitute the giant ecosystem—the biosphere.

KINDS OF ECOSYSTEM

An ecosystem can be natural or artificial, temporary or permanent and large or tiny. Thus, various constituent ecosystems of the biosphere fall into the following categories :

1. Natural ecosystems. These types of ecosystems operate by themselves without any major interference by man. Based upon the particular kind of habitat, these are further classified as :

(i) **Terrestrial ecosystems** such as forests, grasslands, deserts, a single log, etc.

(ii) **Aquatic ecosystems** which may be further distinguished as follows :

(a) **Fresh water ecosystems.** These may be **lotic** (running water as spring, brook, stream or river) or **lentic** (standing water as lake, pond, pool, puddle, ditch, swamp, etc.).

(b) **Marine ecosystems.** These include salt water bodies which may be deep bodies as an ocean or shallow ones as a sea or estuary.

2. Artificial ecosystems. These are also called **man-made** or **man-engineered ecosystems**. These are maintained artificially by man where, by addition of energy and planned manipulations, natural balance is disturbed regularly, *e.g.*, croplands such as sugarcane, maize, wheat, rice-fields ; orchards, gardens, villages, cities, dams, aquarium and manned spaceship.

Microecosystems

Natural ecosystems are usually large in size. In them numerous variables (factors) operate at one time leading to great complexity. As a result, it becomes usually difficult to study them with the normal scientific methods. Thus, to reduce the number of variables and to work in a system with a discrete boundary, ecologists are trying to simulate microecosystems in the laboratory, which can be replicated and manipulated at will. The microsystems can be laboratory systems build by taking one or very few species, at a time, from axenic cultures (An **axenic culture** is a bacterial culture that consists of only one species) and studying them in desired combinations. Recently, the concept of polyaxenic cultures has been developed. **Odum** (1971) and others have elucidated the microecosystem concept.

STRUCTURE OF ECOSYSTEM

The structure of an ecosystem is basically a description of the species of organisms that are

present, including information on their life histories, populations and distribution in space. It is a guide to who's who in the ecosystem. It also includes descriptive information on the non-living (physical) features of environment, including the amount and distribution of nutrients. An ecosystem typically has two major components :

A. Abiotic or Non-living Components

Abiotic component of the ecosystem comprises three sort of components : (1) **Climatic condition** and **physical factors** of the given region such as air, water, soil, temperature, light (*i.e.*, its duration and intensity), moisture (relative humidity), pH, etc. (2). **Inorganic substances** such as water, carbon (C), nitrogen (N), sulphur (S), phosphorus (P) and so on, all of which are involved in cycling of materials in the ecosystem (*i.e.*, biogeochemical cycles). The amount of these inorganic substances, present at any given time in an ecosystem, is designated as the **standing state** or **standing quality**. (3). **Organic substances** such as proteins, carbohydrates, lipids, humic substances, etc., present either in the biomass or in the environment, *i.e.*, **biochemical structure** that link the biotic and abiotic components of the ecosystem.

B. Biotic or Living Components

In the trophic structure of any ecosystem, living organisms are distinguished on the basis of their nutritional relationships, which are discussed as follows :

1. Autotrophic component. Autotrophic (*auto* = self ; *troph* = nourishing) component of ecosystem includes the **producers** or **energy transducers** which convert solar energy into chemical energy (that becomes locked in complex organic substances such as carbohydrate, lipid, protein, etc.) with the help of simple inorganic substances such as water and carbon dioxide and organic substances such as enzymes. Autotrophs fall into following two groups : (i) **photoautotrophs** which contain green photosynthetic pigment **chlorophyll** to transduce the solar or light energy of sun, *e.g.*, trees, grasses, algae, other tiny phytoplanktons and photosynthetic bacteria and cyanobacteria (=blue green algae). (ii) **Chemoautotrophs** which use energy generated in oxidation - reduction process, but their significance in the ecosystem as producers is minimal, *e.g.*, microorganisms such as *Beggiatoa*, sulphur bacteria, etc.

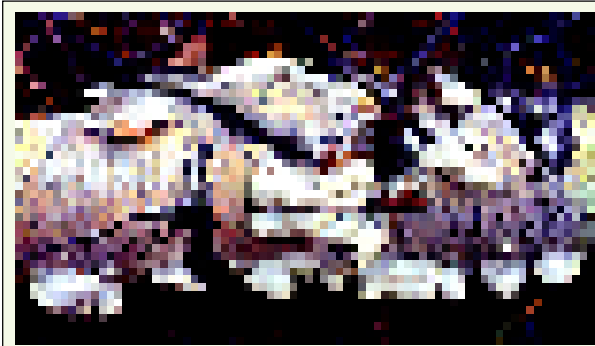
2. Heterotrophic component. In the heterotrophic (*hetero* = other; *trophic* = nourishing) organisms predominate the activities of utilization, rearrangement and decomposition of complex organic materials. Heterotrophic organisms are also called **consumers**, as they consume the matter built up by the producers (autotrophs). The consumers are of following two main types :

(a) Macroconsumers. These are also called **phagotrophs** (*phago* = to eat) and include mainly animals which ingest other organisms or chunks of organic matter. Depending on their food habits, consumers may either be **herbivores** (plant eaters) or **carnivores** (flesh eaters). Herbivores live on living plants and are also known as **primary consumers**, *e.g.*, insects, zooplanktons and animals such as deer, cattle, elephant, etc. **Secondary** and **tertiary consumers**, if present in the food chain of the ecosystem, are carnivores or omnivores, *e.g.*, insects such as preying mantis, dragon flies; spiders and large animals such as tiger, lion, leopard, wolf, etc. Secondary consumers are the carnivores which feed on primary consumers or herbivores. Carnivores are, often, recognized as carnivore order - 1 (C_1), carnivore order - 2 (C_2) and so on, depending on their food habits.

Ticks and mites, leeches and blood-sucking insects (mosquito, bed-bug) are dependent on herbivores, carnivores and omnivores.

(b) Microconsumers. These are also called **decomposers**, **reducers**, **saprotrophs** (*sapro* = decompose), **osmotrophs** (*osmo* = to pass through a membrane) and **scavengers**. **Wiegert** and **Owen** (1971) have coined the term, **biophages** for heterotrophic decomposers which feed on the dead organic matter. Microconsumers include microorganisms such as bacteria, actinomycetes and fungi. Microconsumers breakdown complex organic compounds of dead or living protoplasm, absorb some

of the decomposition or breakdown products and release inorganic nutrients in the environment, making them available again to autotrophs or producers. Some invertebrate animals such as protozoa, oligochaeta such as earthworms, etc., use the dead organic matter for their food, as they have the essential enzymes and, hence, can be classified as decomposer organisms. Some ecologists believe that micro-organisms are **primary decomposers**, while invertebrates are **secondary decomposers**.



Detritivores act on organic detritus and break it down to release inorganic nutrient.

The disintegrating dead organic matter is also known as **organic detritus** (Latin word *deterere* means to wear away). By the action of **detritivores** (=decomposers), the disintegrating detritus results into particulate organic matter (POM) and dissolved organic matter (DOM) which play an important role in the maintenance of the edaphic environment.

EXAMPLE OF ECOSYSTEM

A pond as a whole serves as a good example of an aquatic and freshwater ecosystem (Fig. 9.2). In fact, it represents a self-sufficient and self-regulating system. It has the following components:

1. Abiotic Component

The chief non-living or abiotic substances are heat, light, pH value of water, and the basic inorganic and organic compounds, such as water itself, carbon dioxide gas, oxygen gas, calcium, nitrogen, phosphates, amino acids, humic acid, etc. Inorganic salts occur in the form of phosphates, nitrates and chlorides of sodium, potassium and calcium. Some proportion of nutrients exist in solution state but most of them are present as stored in particulate matter as well as in living organisms.

2. Biotic Component

It includes various organisms which are classified into the following types :

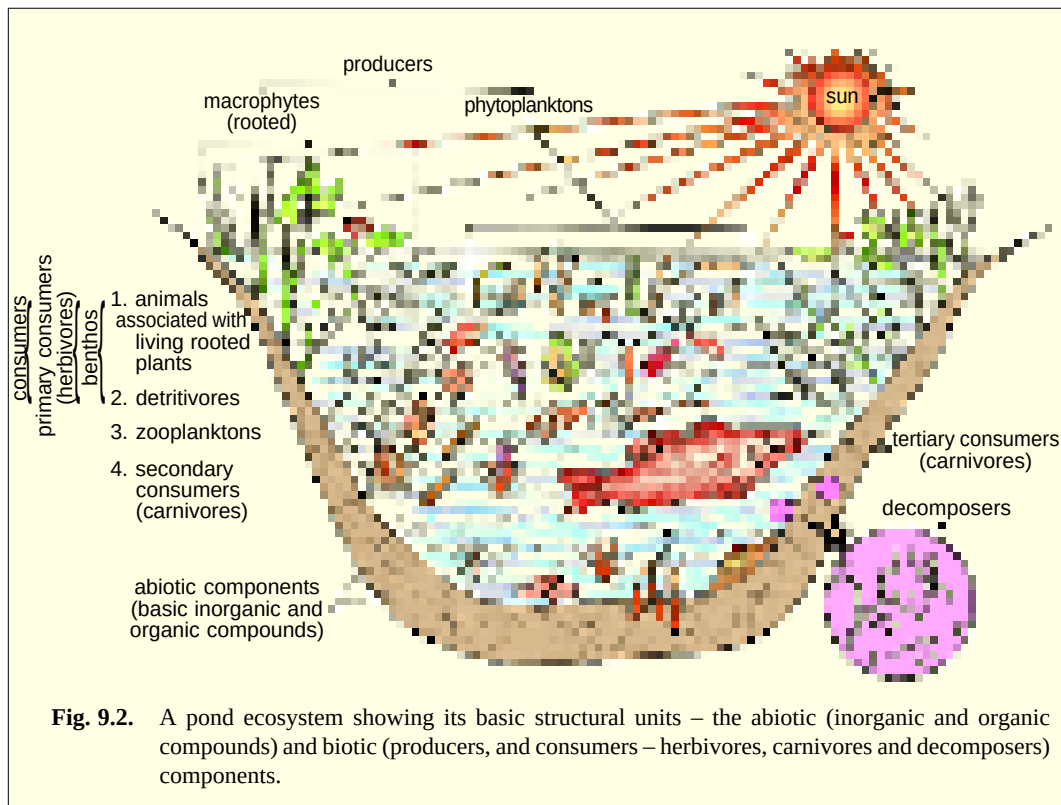
(a) Producers. These are photoautotrophic green plants and photosynthetic bacteria. The producers fix radiant energy of sun and with the help of minerals derived from water and mud, they manufacture complex organic substances as carbohydrates, proteins and lipids. Producers of pond are of the following types :

(i) Macrophytes. These include mainly the rooted large-sized plants which comprise three types of hydrophytes : partly or completely submerged, floating and emergent aquatic plants. The common plants are species of *Trapa*, *Typha*, *Eleocharis*, *Sagittaria*, *Nymphaea*, *Potamogeton*, *Chara*, *Hydrilla*, *Vallisneria*, *Utricularia*, *Marsilea*, *Nelumbo*, etc. Besides these plants, some free floating forms also occur in the pond ecosystem, e.g., *Azolla*, *Salvinia*, *Wolffia*, *Eichhornia*, *Spirodella*, *Lemna*, etc.

(ii) Phytoplanktons. These are microscopic (minute), floating or suspended lower plants (algae) that are distributed throughout the water, but mainly in the photic zone. Most of them are filamentous algae such as *Spirogyra*, *Ulothrix*, *Zygnema*, *Cladophora* and *Oedogonium*. There also occur some chlorococcales (e.g., *Chlorella*), *Closterium*, *Cosmarium*, *Eudorina*, *Pandorina*, *Pediastrum*, *Scenedesmus*, *Volvox*, Diatoms, *Anabaena*, *Gloeotrichia*, *Microcystis*, *Oscillatoria*, *Chlamydomonas*, *Spirulina*, etc., and some flagellates.

(b) Macroconsumers. They are phagotrophic heterotrophs which depend for their nutrition on the organic food manufactured by producers, the green plants. Macroconsumers are of the following three types :

(i) Herbivores (Primary consumers). These animals feed directly on living plants (producers) or plant remains. They may be large or minute in size and are of following two types : 1. **Benthos** which are the bottom dwelling forms such as fish, insect larvae, beetles, mites, molluscs (e.g., *Pila*,



Planorbis, *Unio*, *Lamellidens*, etc.), crustaceans, etc. 2. **Zooplanktons** which feed chiefly on phytoplanktons and are chiefly the rotifers as *Brachionus*, *Asplanchna*, *Lecane*, etc., although some protozoans as *Euglena*, *Coleps*, *Dileptus*, etc., and crustaceans such as *Cyclops*, *Stenocypris*, etc., are also present in the pond.

Besides these small-sized herbivores, some mammals such as cow, buffaloes, etc., also visit the pond casually and feed on marginal rooted macrophytes. Some birds also regularly visit the pond to feed on some hydrophytes.

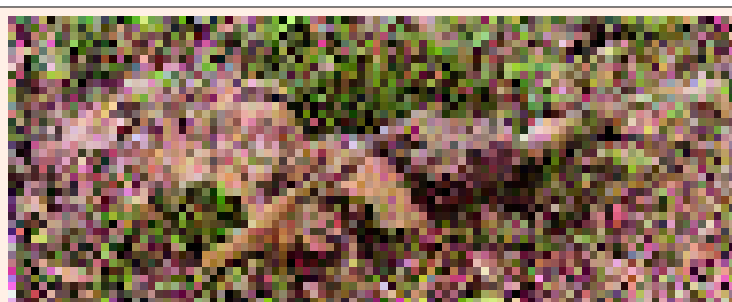
(ii) Carnivore order-1 (Secondary consumers). These carnivores feed on the herbivores and include chiefly insects, fish and amphibians (frog). Most insects are water beetles which feed on zooplanktons; some insects are the nymphs of dragonflies which feed upon aquatic insects.

(iii) Carnivore order-2 (Tertiary consumers). These are some large fish as game fish that feed on the smaller fish and, thus, become the tertiary (top) consumers.

(c) Decomposers. They are also called microconsumers, since they absorb only a fraction of the decomposed organic matter. They bring about the decomposition of dead organic matter of both producers (plants) as well as macroconsumers (animals) to simple forms. Decomposers help in returning of mineral elements again to the medium of the pond and in running biogeochemical cycles. Decomposers of pond ecosystem include chiefly bacteria, actinomycetes and fungi. Among fungi, species of *Aspergillus*, *Cephalosporium*, *Cladosporium*, *Pythium*, *Rhizopus*, *Penicillium*, *Thielavia*, *Alternaria*, *Trichoderms*, *Circinella*, *Fusarium*, *Curvularis*, *Paecilomyces*, *Saprolegnia*, etc., are most common decomposers in water and mud of the pond.

FUNCTION OF AN ECOSYSTEM

When we consider the function of an ecosystem, we must describe the flow of energy and the cycling of nutrients. That is, we are interested in things like how much sunlight is trapped by plants in a year, how much plant material is eaten by herbivores, and how many herbivores are eaten by carnivores. Thus, the producers, the green plants, fix radiant energy and with the help of minerals (such as C, H, O, N, P, Ca, Mg, Zn, Fe, etc.) taken from their edaphic (soil) or aerial environment (the nutrient pool) they build up complex organic matter (carbohydrates, fats, proteins, nucleic acids, etc.). Some ecologists prefer to refer to the green plants as **converters** or **transducers**, since in their view, the most popular and prevalent term ‘producer’ from energy view point is somewhat misleading. Their view point is that green plants produce carbohydrates and not energy and since they convert or transduce radiant energy into chemical form, they must be better called **converters** or **transducers**. The two ecological processes of energy flow and mineral cycling involving interaction between the physico-chemical environment and the biotic communities, may be considered the ‘heart’ of ecosystem dynamics. In an ecosystem, energy flows in non-cyclic manner (**unidirectional**) from sun to the decomposers via producers and macroconsumers (herbivores and carnivores), whereas the minerals keep on moving in a cyclic manner.



The dead organic matter is decomposed to simple forms by micro-organisms.

Productivity of Ecosystem

The productivity of an ecosystem refers to the rate of production, *i.e.*, the amount of organic matter accumulated in any unit time. It is of following types :

1. Primary productivity. It is defined as the rate at which radiant energy is stored by photosynthetic and chemosynthetic activity of producers. Primary productivity is of following types:

(i) Gross primary productivity. It refers to the total rate of photosynthesis including the organic matter used up in respiration during the measurement period. GPP depends on the chlorophyll content. The rate of primary productivity are estimated in terms of either chlorophyll content as chl/g dry weight/unit area or photosynthetic number, *i.e.*, amount of CO₂ fixed/g chl/hour.

(ii) Net primary productivity. It is the rate of storage of organic matter in plant tissues in excess of the respiratory utilization by plants during the measurement period.

Primary production is measured by following methods—harvest method, oxygen measurement method (or light or dark method), oxygen diurnal curve method, carbon dioxide measurement method (enclosure method), the aerodynamic method, the pH method, radioisotope method, chlorophyll estimation method (see **Dash**, 1993).

2. Secondary productivity. It is the rate of energy storage at consumer’s levels—herbivores, carnivores and decomposers. Consumers tend to utilise already produced food materials in their respiration and also convert the food matter to different tissues by an overall process. So, secondary productivity is not divided into ‘gross’ and ‘net’ amounts. Due to this fact some ecologists such as **Odum** (1971), prefer to use the term **assimilation** rather than production at this level – the consumers level. Secondary productivity, in fact, remains mobile (*i.e.*, keeps on moving from one organism to another) and does not live *in situ* like the primary productivity.

3. Net productivity. It is the rate of storage of organic matter not used by the heterotrophs or consumers, *i.e.*, equivalent to net primary production minus consumption by the heterotrophs during the unit period as a season or year, etc.

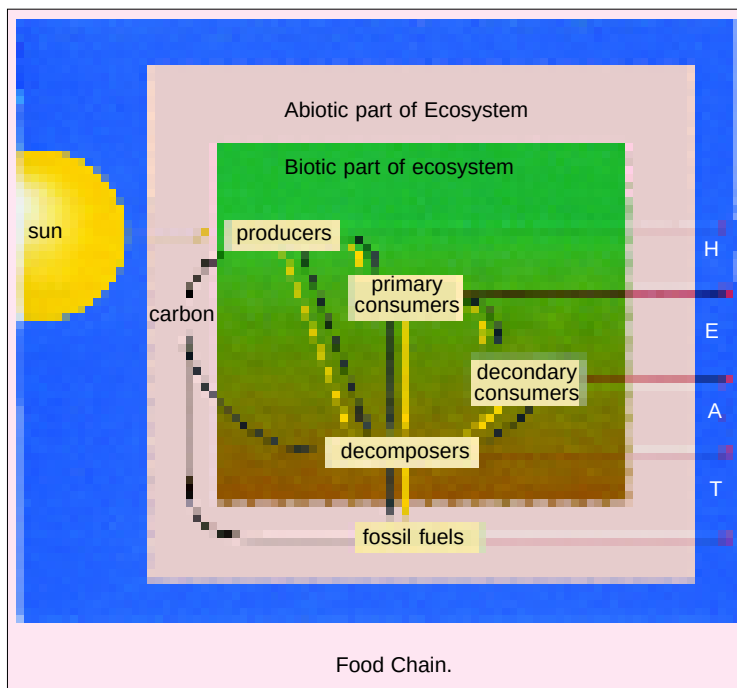
Food Chains in Ecosystems

In an ecosystem one can observe the transfer or flow of energy from one trophic level to other in succession. A **trophic level** can be defined as the number of links by which it is separated from the producer, or as the n th position of the organism in the food chain. The patterns of eating and being eaten forms a linear chain called **food chain** which can always be traced back to the producers. Thus, primary producers trap radiant energy of sun and transfer that to chemical or potential energy of organic compounds such as carbohydrates, proteins and fats. When a herbivore animal eats a plant (or when bacteria decompose it) and these organic compounds are oxidized, the energy liberated is just equal to the amount of energy used in synthesizing the substances (first law of thermodynamics), but some of the energy is heat and not useful energy (second law of thermodynamics). If this animal, in turn, is eaten by another one, along with transfer of energy from a herbivore to carnivore a further decrease in useful energy occurs as the second animal (carnivore) oxidizes the organic substances of the first (herbivore or omnivore) to liberate energy to synthesize its own cellular constituents. Such transfer of energy from organism to organism sustains the ecosystem and when energy is transferred from individual to individual in a particular community, as in a pond or a lake or a river, we come across the food chains. The number of steps in a food chain are always restricted to four or five, since the energy available decreases with each step. For example, in a typical food chain of an Indian river, a diatom may be eaten by a copepod which is eaten by a small fish, which forms the food source of large fish and so on (Dash, 1993) :

Scenedesmus boligues → *Brachionus falcatus*
 (phytoplankton) (zooplankton)
 ↓
Homo sapiens ← *Wallago attu* ← *Amblypharyngodon* sp.
 (man) (a large fish) (a small fish)

In an Indian pasture, the following food chain operates :

Cynodon dactylon → *Melanoplus differentialis*
 (a grass species) (a grasshopper)
 ↓
 Hawk ← *Zamensis mucosus* ← *Bufo melanostictus*
 (a rat snake) (a toad)



One may ask—why is the number of trophic levels in a food chain limited? In a simple food chain (Fig. 9.3), out of 1000 calories of energy reaching a plant only 10 calories (1%) are stored by the plant. The remaining calories of energy (99%) are lost to the environment or for plant's own maintenance. Of the 10 calories which are available to the herbivore, 9 calories (99%) are lost at its level and only 1 calorie is passed down to the carnivore. Thus, at each trophic level in a food chain, a large portion of energy is used for its own maintenance and ultimately lost as heat. Consequently, organisms in each trophic level pass on less and less energy than they receive. This tends to limit the number of steps or trophic levels to four or five. The longer the food chain, the less is the energy available to the final member.

Many direct or indirect methods are employed to study food chain relationships in nature. They include gut content analysis, use of radioactive isotopes, precipitin test, etc.

In nature, basically two types of food chains are recognized—grazing food chain and detritus food chain.

1. Grazing food chain. This type of food chain (Fig. 9.4) starts from the living green plants, goes to grazing herbivores and on to the carnivores. Ecosystems with such type of food chain are directly dependent on an influx of solar radiation. Thus, this type of food chain depends on autotrophic energy capture and the movement of this energy to herbivores. Most of the ecosystems in nature follow this type of food chain. These chains are very significant from energy standpoint. The phytoplanktons → zooplanktons → fish sequence or the grasses → rabbit → fox sequence are the examples of grazing food chain. Further the producer → herbivore → carnivore chain is a **predator chain**. **Parasitic chains** also exist wherein smaller organisms consume larger ones without outright killing as the case of the predators.

2. Detritus food chain. The organic wastes, exudates and dead matter derived from the grazing food chain are generally termed **detritus**. The energy contained in this detritus is not lost to the ecosystem as a whole; rather it serves as the source of energy for a group of organisms (**detritivores**) that are separate from the grazing food chain, and generally termed as the **detritus food chain** (Fig. 9.5). The detritus food chain represents an exceedingly important component in the energy flow of an ecosystem. Indeed in some ecosystems, considerably more energy flows through the detritus food chain than through the grazing food chain. In the detritus food chain the energy flow remains as a continuous passage rather than as a stepwise flow between discrete entities. The organisms of the detritus food chain are many and include algae, bacteria, slime molds, actinomycetes, fungi, Protozoa, insects, mites, Crustacea, centipedes, molluscs, rotifers, annelid worms, nematodes and some vertebrates. Some species are highly specific in their food requirements and some can eat almost anything. Many Protozoa, for instance, need certain specific organic acids, vitamins, and other nutrients before they can thrive; on the other hand, the guts of small Collembola (a group of tiny soil insects) have been reported to contain decaying plant material,

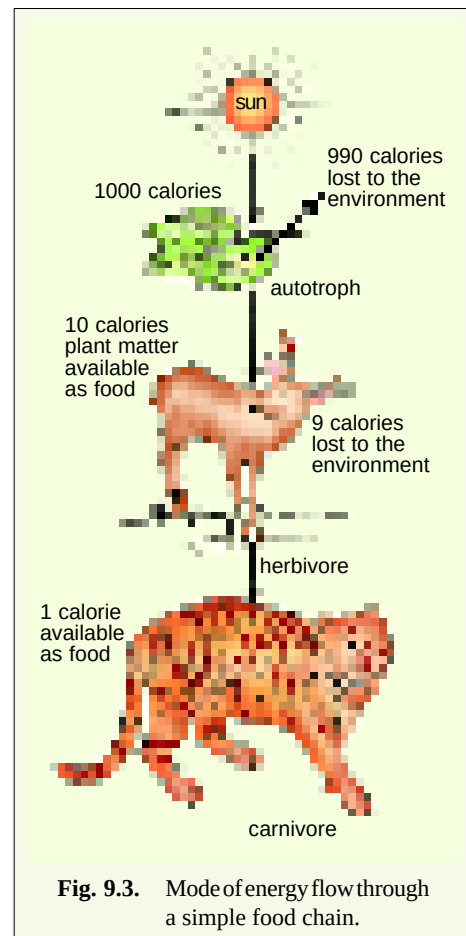


Fig. 9.3. Mode of energy flow through a simple food chain.

fungus fragments, spores, fly pupae, other Collembola, parts of decaying earthworms, and cuticle from their own faecal casting (Hale, 1967). In contrast to the grazing food chain, in which energy storage is entirely within the tissues of living organisms, energy storage for the detritus food chain may be largely external to the organisms, and in the detritus itself.

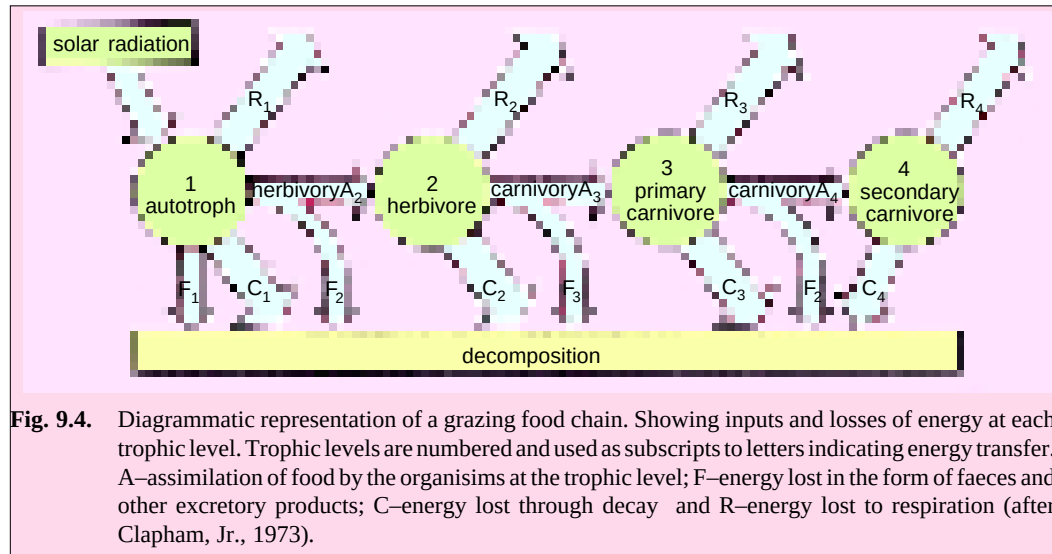


Fig. 9.4. Diagrammatic representation of a grazing food chain. Showing inputs and losses of energy at each trophic level. Trophic levels are numbered and used as subscripts to letters indicating energy transfer. A—assimilation of food by the organisms at the trophic level; F—energy lost in the form of faeces and other excretory products; C—energy lost through decay and R—energy lost to respiration (after Clapham, Jr., 1973).

Significance of food chain. The food chain studies help understand the feeding relationships and the interaction between organisms in any ecosystem. They also help us to appreciate the energy flow mechanism and matter circulation in ecosystem, and understand the movement of toxic substances in the ecosystem and the problem of biological magnification (*e.g.*, DDT; for details see Chapter 14).

Food web. In nature simple food chains occur rarely. The same organism may operate in the ecosystem at more than one trophic level, *i.e.*, it may derive its food from more than one source. Even the same organism may be eaten by several organisms of a higher trophic level or an organism may feed upon several different organisms of a lower trophic level. Usually the kind of food changes with the age of the organism and the food availability. Thus, in a given ecosystem various food chains are linked together and intersect each other to form a complex network called **food web**.

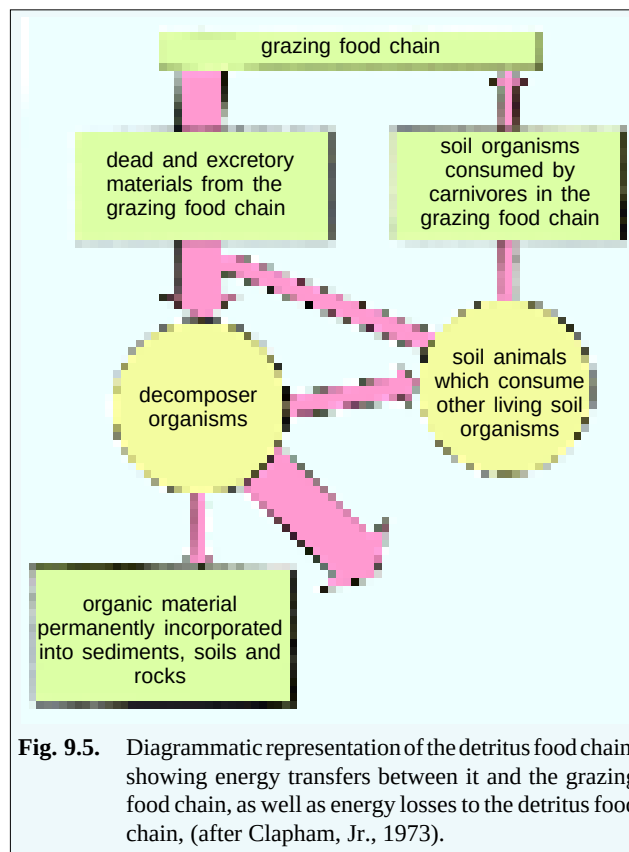


Fig. 9.5. Diagrammatic representation of the detritus food chain, showing energy transfers between it and the grazing food chain, as well as energy losses to the detritus food chain, (after Clapham, Jr., 1973).

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In any complex food web, one can recognize several different trophic levels :

1. Producers	Green plants	First trophic level
2. Primary consumers	Herbivores	Second trophic level
3. Secondary consumers	Carnivores, insectivores	Third trophic level
4. Tertiary consumers	Higher carnivores, insect hyperparasites	Fourth trophic level

A classification of organisms by trophic levels is one of function and not of species as such. A given species may occupy more than one trophic level. The complexity of food web can vary greatly, and we express this complexity by a measure called the **connectance** of the food web :

$$\text{Connectance} = \frac{\text{Actual number of interspecific interactions}}{\text{Potential number of interspecific interactions}}$$

Generalizations about food web. Generally, food webs are not too complex. As more and more species are involved in a food web, the connectance falls. Except in insect communities, omnivores are scarce, and when they occur, they usually feed on species in adjacent trophic levels. Within habitats, food webs are rarely broken up into discrete compartments. The number of species of predators in a food web typically exceeds the number of species of prey by an average of 1.3 predator species per prey species.

ECOLOGICAL PYRAMIDS

In the successive steps of grazing food chain—photosynthetic autotroph, herbivorous heterotroph, carnivores heterotroph, decay bacteria—the number and mass of the organisms in each step is limited by the amount of energy available. Since some energy is lost as heat, in each transformation the steps become progressively smaller near the top. This relationship is sometimes called “**ecological pyramid**”. The ecological pyramids represent the trophic structure and also trophic function of the ecosystem. In many ecological pyramids, the producer form the base and the successive trophic levels make up the apex.

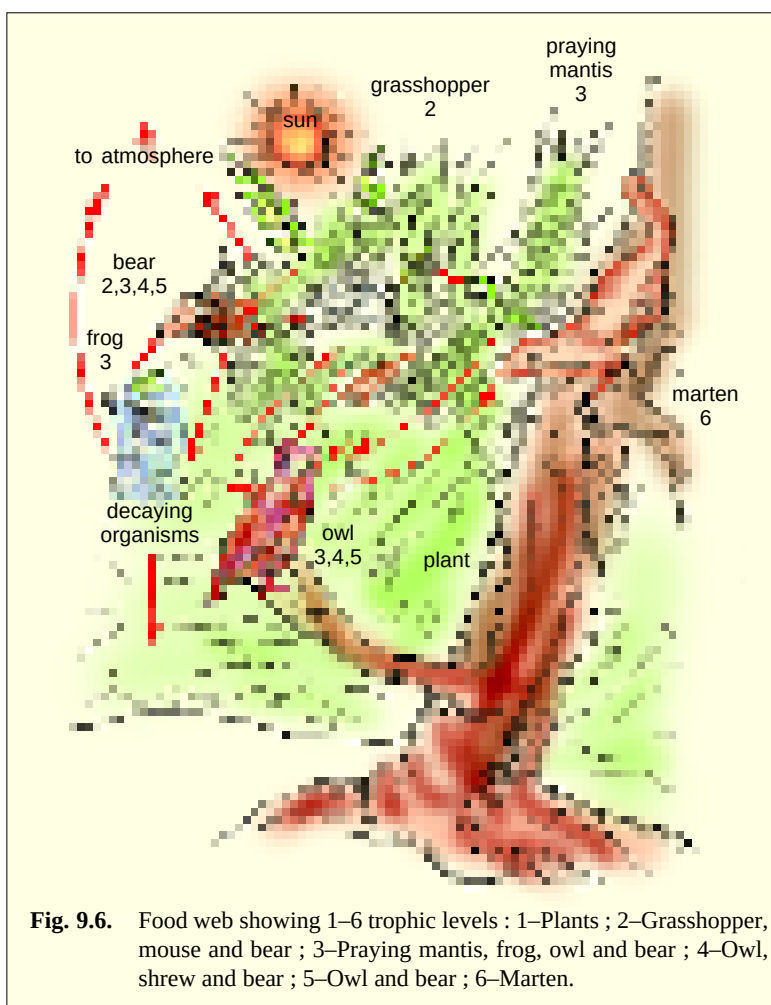


Fig. 9.6. Food web showing 1–6 trophic levels : 1–Plants ; 2–Grasshopper, mouse and bear ; 3–Praying mantis, frog, owl and bear ; 4–Owl, shrew and bear ; 5–Owl and bear ; 6–Marten.

Thus, communities of terrestrial ecosystems and shallow water ecosystems contain gradually sloping ecological pyramids because these producers remain large and characterized by an accumulation of organic matter. This trend, however, does not hold for all ecosystems. In such aquatic ecosystems as lakes and open sea, primary production is concentrated in the microscopic algae. These algae have a short-cycle, multiply rapidly, accumulate little organic matter and are heavily exploited by herbivorous zooplankton. At any one point in time the standing crop is low. As a result, the pyramid of biomass for these aquatic ecosystems is inverted: the base is much smaller than the structure it supports.

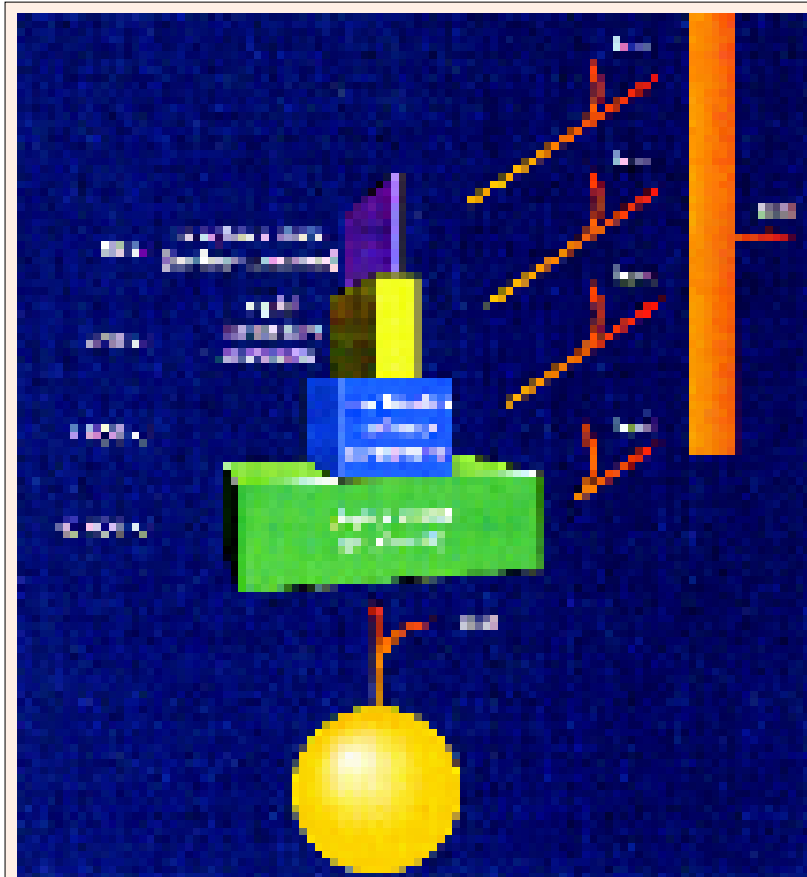
Types of Ecological Pyramids

The ecological pyramids may be of following three kinds :

1. Pyramid of number. It depicts the number of individual organisms at different trophic levels of food chain. This pyramid was advanced by **Charles Elton** (1927), who pointed out the great difference in the number of the organisms involved in each step of the food chain. The animals at the lower end (base of pyramid) of the chain are the most abundant. Successive links of carnivores decrease rapidly in number until there are very few carnivores at the top. The pyramid of number ignores the biomass of organisms and it also does not indicate the energy transferred or the use of energy by the groups involved. The lake ecosystem provides a typical example for pyramid of number.

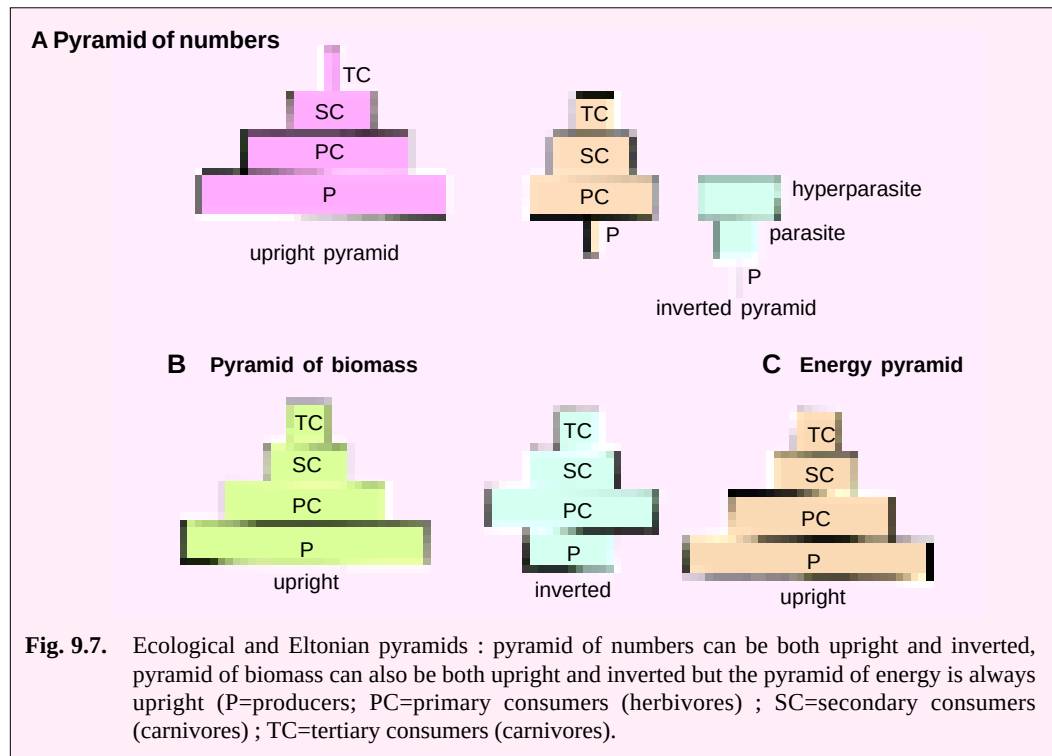
2. Pyramid of biomass. The biomass of the members of the food chain present at any one time forms the pyramid of the biomass. Pyramid of biomass indicates decrease of biomass in each trophical level from base to apex. For example, the total biomass of the producers ingested by herbivores is more than the total biomass of the herbivores in an ecosystem. Likewise, the total biomass of the primary carnivores (or secondary consumer) will be less than the herbivores and so on.

3. Pyramid of energy. When production is considered in terms of energy, the pyramid indicates not only the amount of energy flow at each level, but more important, the actual role the various organisms play in the transfer of energy. The base upon which the pyramid of energy is constructed is the quantity of organisms produced per unit



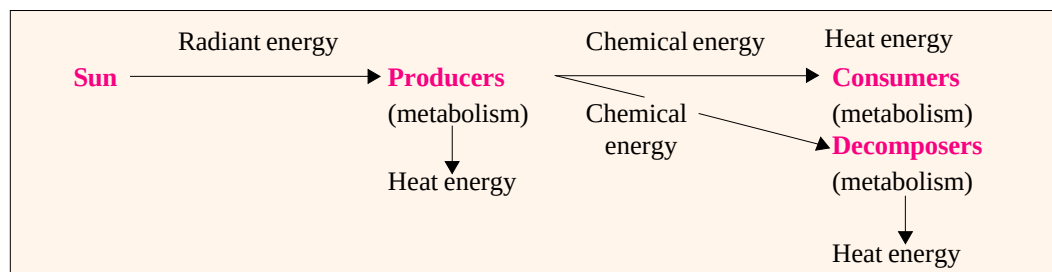
Pyramid of energy.

time, or in other words, the rate at which food material passes through the food chain. Some organisms may have a small biomass, but the total energy they assimilate and pass on, may be considerably greater than that of organisms with a much larger biomass. Energy pyramids are always sloping because less energy is transferred from each level than was paid into it. In cases such as in open water communities the producers have less bulk than consumers but the energy they store and pass on must be greater than that of the next level. Otherwise the biomass that producers support could not be greater than that of the producers themselves. This high energy flow is maintained by a rapid turn over of individual plankton, rather than an increase of total mass.

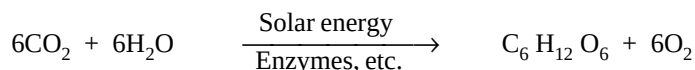


ENERGY FLOW IN ECOSYSTEMS

1. Concept of energy. Energy is the capacity to do work. Biological activities require consumption of energy which ultimately comes from the sun. Radiant energy of sun (or **solar energy**) is transformed into **chemical energy** by the process of photosynthesis—this is stored in plant tissues and then transformed into **mechanical** and **heat** form of energy during metabolic activities. In the biological world, the energy flows from the sun to plants and then to all heterotrophic organisms, such as microorganisms, animals and man in the following manner :



Mechanical energy has two forms, namely kinetic or free energy and potential energy. The energy a body possess by virtue of its motion is called **kinetic energy**, and is measured by the amount of work done in bringing the body at rest. **Potential energy** is stored energy (the energy at rest) and becomes useful after conversion into kinetic energy. All organisms require a source of potential energy, which is found in the chemical energy of food. The oxidation of food releases energy which is used to do work. Thus, chemical energy is converted into mechanical energy. Food means material containing energy that organism can use. Food is the means to transfer of both matter and energy in the living world. Plants synthesize food with the help of solar energy and inorganic substances such as nutrients, CO_2 and H_2O in a biochemical process called **photosynthesis** :



(Chlorophyll) Glucose; food

2. Unit of energy. The unit of measurement of energy is **erg**; the work done in lifting 1 gram of weight to a height of 1 cm against the force of gravity is equal to 981 ergs. One crore ergs (10^7 ergs) is equal to one **Joule**. All forms of energy can be completely converted into heat energy. For a better and uniform expression in ecology, therefore, energy is measured not in terms of ergs but joules or units of heat measurement. Heat is measured in calories. One calorie is equal to the heat energy required to raise the temperature of 1 gram of water from 14.5°C to 15.5°C , and one calorie is equal to 4.2 joules or 4.2×10^7 ergs. One thousand calories (10^3) makes one kilo calories or a kilogram calories (Kcal or Cal). Now, there is a trend of expressing energy in ecological literature in terms of kilojoules.

3. Ecological energetics. Ecological energetics includes energy transformation which occur within ecosystems. In ecological energetics, we consider (i) quantity of energy reaching an ecosystem per unit of area (say a square metre) per unit of time (say one hour, day or year); (ii) quantity of energy trapped by green plants and converted to a chemical form (photosynthesis) and (iii) the quantity and path of energy flow from green plants to organisms of different trophic levels over a period of time in a known area (*i.e.*, energy flow from producers to consumers).

The energy used for all plant life processes is derived from solar radiations. A fraction, *i.e.*, about 1/50 millionth of the total solar radiation reaches the earth's atmosphere. Solar radiation travels through the space in the form of waves, their wavelength ranging from $0.03 \text{ \AA}$ to several kilometres. While most radiations are lost in space, those ranging from $300 \text{ m}\mu$ to $10 \text{ m}\mu$ and above 1 cm (radiowaves) enter the earth's outer atmosphere (which is about 28 km altitude). The energy reaching the earth's surface consists mainly of visible light ($390\text{--}760 \text{ m}\mu$) and infrared component. On a clear day radiant energy reaching the earth's surface is



Only about 0.02% of the sunlight reaching the atmosphere is used in the process of photosynthesis.

about 10% UV, 45% visible and 45% infra-red. Green plants absorb strongly the blue and red light (400 to 500 mμ and 600 to 700 mμ respectively).

About 34% of the sunlight reaching the earth's atmosphere is reflected back (by clouds and dust), 10% is held by ozone layer, water vapour and other atmospheric gases. The rest, 56% reaches the earth's surface. Only a fraction of the energy reaching earth's surface (1 to 5%) is used by green plants for photosynthesis and rest is absorbed as heat by ground vegetation or water. In fact, only about 0.02% of the sunlight reaching the atmosphere is used in the process of photosynthesis.

The amount of radiant energy of all wavelengths that cross unit area per unit time is called **solar flux**. The solar flux is about 8.368 J (2 cal)/cm² min. At a given place, it varies diurnally because of the earth's rotation on its axis.

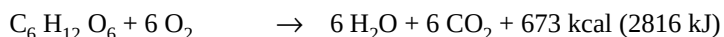
4. Laws governing energy transformation. Energy transformation in ecosystems can also be explained in relation to the laws of thermodynamics, which are usefully applied to closed systems. The **first law of thermodynamics** is the law of conservation of energy, which says that *energy may be transformed from one form into another but is neither created nor destroyed*. If an increase or decrease occurs in the internal energy (*E*) of the system itself, work (*W*) is done and heat (*Q*) is either evolved or absorbed. Thus,

$$\begin{array}{rcccl} \Delta E & = & W & + & Q \\ \text{Decrease in the internal energy} & & \text{Work done by the} & & \text{Heat given off} \\ \text{of the system} & & \text{system} & & \text{by the system} \end{array}$$

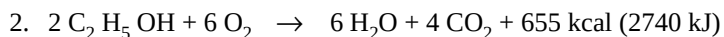
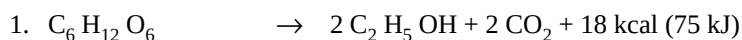
The sign Δ refers to a change in quantity.

The total amount of heat produced or absorbed in a chemical reaction, either occurring directly or in stages, always remains the same. This is called the **specific law of constant heat sums** and included in the first law.

For example, combustion (a direct chemical reaction) of food release the following amount of energy :



The fermentation is a two-step reaction, but releases the same amount of energy :



This law explains the interconvertibility of all forms of energy but does not refer to the efficiency of transformation or conversion. In ecological systems solar energy is converted into chemical energy stored in food materials which is ultimately converted into mechanical and heat energy. Thus, in ecological systems, the energy is neither created nor destroyed but is converted from one form into another. Thus, when wood is burned the potential energy present in the molecules of wood equals the kinetic energy released, and heat is evolved to the surroundings. This is an **exothermic reaction**. In an **endothermic reaction**, energy from the surrounding may be paid into a reaction. For example, in photosynthesis, the molecules of the products store more energy than the reactants. The extra energy is acquired from the sunlight, but even then there is no gain or loss in total energy.

The second law of thermodynamics states that *processes involving energy transformation will not occur spontaneously unless there is degradation of energy from a non-random to a random form*. In man-made machines (closed systems), heat is the simplest and most familiar medium of energy transfer. But in biological systems, it is not a useful medium of energy transfer, as living systems are essentially isothermal and there are no significant differences in temperature between different parts of a cell or between different cells in a tissue.

5. Concept of free energy, enthalpy and entropy. Free energy is that component of the total energy of a system which can do work under isothermal conditions. All physical and chemical processes proceed with a decline in free energy until they reach an equilibrium where the free energy of the system is at a minimum.

$$\Delta G = \Delta H - T \Delta S$$

Where ΔG = Change in the free energy of the system

ΔH = Change in **enthalpy** (i.e., a change in the amount of energy in the form of heat liberated or absorbed by the system during physical or chemical changes)

ΔS = Entropy change of the system (**Entropy** is the name of a quantity in thermodynamics representing the degree of disorder in a physical system or the extent to which the energy in a system is available for doing work)

T = Absolute temperature

Thus, a decline in G is accompanied by an increase in $T \Delta S$. These are equal if there is no heat transfer between the system and the surrounding. If a reaction proceeds with a decline in free energy, we call it spontaneous.

6. Lindeman's trophic-dynamic concept. As **Lindeman** (1942) pointed out, the amount of energy reaching each trophic level is determined by the net primary production (NPP) and the efficiency with which food energy is converted to biomass energy within each trophic step. Of the light energy assimilated by plants, 15 to 70 per cent is used for maintenance and, therefore, is unavailable to consumers. Most herbivores and carnivores are more active than plants and spend correspondingly more of their assimilated energy on maintenance. As a result productivity of each trophic level is usually no more than 5 to 20 per cent that of the level below it. The percentage transfer of energy from one trophic level to the next is called both the ecological efficiency and food chain efficiency.

7. Maintenance cost of secondary producers. In general, 55 to 75% of the assimilated energy is spent on maintenance of secondary producers. Temperature, moisture conditions of the habitat and the type of species determine the maintenance cost. The drier and hotter the habitat, the higher is the maintenance cost, irrespective of species. For example, the average maintenance cost of some Indian earthworm species was found to be 6.48, 9.96 and 20.54 kJ/g dry tissue/month in the winter, rainy and summer seasons respectively in tropical pastures (**Dash**, 1987). The maintenance cost varies seasonally and is three times more in summer than in winter.

8. Assimilated energy and respiration energy. Once food is eaten, its energy follow a variety of paths through the organisms (Fig. 9.8). Regardless of an organism's source of food, what it digests and absorbs is referred to as **assimilated energy**, which supports maintenance, builds tissues, or is excreted in unusable metabolic byproducts. The energy used to fulfill metabolic needs, most of which is lost as heat, is known as **respired energy**. A smaller fraction of assimilated energy is excreted in the form of

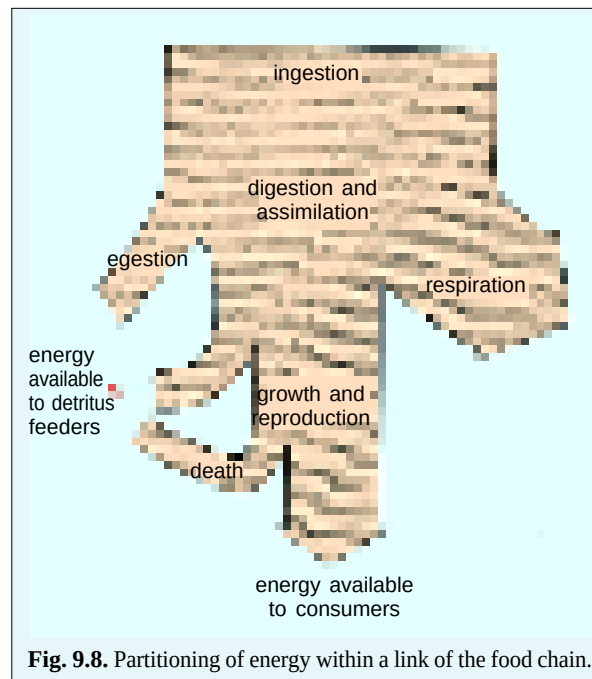


Fig. 9.8. Partitioning of energy within a link of the food chain.

organic, nitrogen –containing wastes (primarily ammonia, urea, or uric acid) produced when the diet contains an excess of nitrogen. Assimilated energy retained by the individual organism is available for the synthesis of new biomass (**production**) through growth and reproduction, which may then be consumed by herbivores, carnivores, and detritivores.

Further, not all food can be fully assimilated; hair, feathers, insect exoskeletons, cartilage and bone in animal foods, and cellulose and lignin in plant foods resist digestion by most animals. These materials are egested either by defecation or by regurgitation of pellets of undigested remains. Some egested wastes are substances that have been relatively unaltered chemically during their passage through an organism, but nearly all have been mechanically broken up into fragments by chewing and by contractions of the stomach and intestines, which makes them more readily usable by detritus feeders (**Edwards** and **Heath**, 1963).

9. Ecological efficiency. Ecological efficiency is the product of efficiencies with which organisms exploit their food resources and transform them into biomass which becomes available to the next higher trophic level. Because most biological production is consumed, **exploitation efficiency** is 100 per cent overall, and ecological efficiency depends on two factors : the proportion of assimilated energy incorporated in growth, storage and reproduction. The first proportion is called the **assimilation efficiency** and the second, the **net production efficiency** (Table 9-1). The product of the assimilation and net production efficiencies is the **gross production efficiency** : the proportion of food energy that is transformed into consumer biomass energy.

Net production efficiency for the plants is defined as the ratio of net to gross production. This index has been found to vary between 30 and 85 per cent, depending on habitat and growth form. Rapidly growing plants in temperate zones, whether trees, old-field herbs, crop species, or aquatic plants, have constantly high net production efficiencies (75 to 85 per cent). Similar vegetation types in the tropics exhibit lower net production efficiencies, perhaps 40 to 60 per cent respiration increases relative to photosynthesis at low latitudes.

Table 9-1. Definition of several energetic efficiencies.

1. Exploitation efficiency =	$\frac{\text{Ingestion of food}}{\text{Prey production}}$
2. Assimilation efficiency =	$\frac{\text{Assimilation}}{\text{Ingestion}}$
3. Net production efficiency =	$\frac{\text{Production (growth and reproduction)}}{\text{Assimilation}}$
4. Gross production efficiency = (2) X (3) =	$\frac{\text{Production}}{\text{Ingestion}}$
5. Ecological efficiency = (1) X (2) X (3) =	$\frac{\text{Consumer production}}{\text{Prey production}}$

In comparison to the plant food, animal food is more easily digested. Assimilation efficiency of predatory species vary from 60 to 90 per cent. Vertebrates prey are digested more efficiently than insect prey because the indigestible exoskeletons of insects constitute the larger proportion of body than the hair, feathers and scales of vertebrates. Assimilation efficiencies of insectivorous vary between 70 and 80 per cent, whereas those of most carnivores are about 90 per cent.

The nutritional value of plant foods depends upon the amount of cellulose, lignin, and other indigestible materials present. Herbivores assimilate as much as 80 per cent of the energy in seeds, and 60 to 70 per cent of that in young vegetation (**Chew** and **Chew**, 1970). Most grazers and browsers (*e.g.*,

cattle, elephants and grasshoppers) assimilate 30 to 40 per cent of the energy in their food. Millipedes, which eat decaying wood composed mostly of cellulose and lignin (and the microorganisms that occur in decaying wood), assimilate only 15 per cent (O'Neil, 1968).

Maintenance, movement and in warm-blooded animals, heat production require energy that otherwise could be utilized for growth and reproduction. Active warm-blooded animals (homeotherms) exhibit low net production efficiency—birds less than 1 per cent, small mammals with high reproductive rates up to 6 per cent. More sedentary, cold-blooded animals (poikilotherms), particularly aquatic species, channel as much as 75 per cent of their assimilated energy into growth and reproduction (Welch, 1968). The extreme high value approaches the biochemical efficiency of egg production and tissue growth, between 70 to 80 per cent in domesticated animals (Ricklefs, 1974).

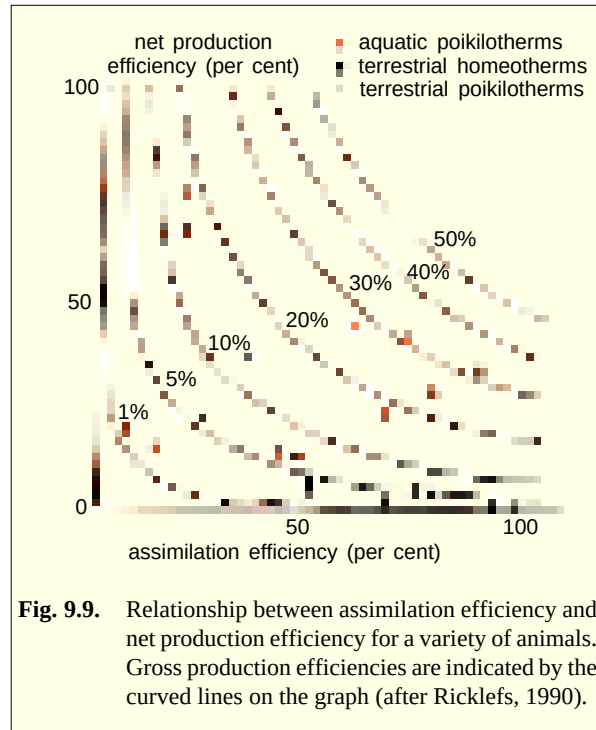


Fig. 9.9. Relationship between assimilation efficiency and net production efficiency for a variety of animals. Gross production efficiencies are indicated by the curved lines on the graph (after Ricklefs, 1990).

The **gross production efficiency** (i.e., biomass production efficiency within a trophic level) is the product of assimilation efficiency and net production efficiency. Gross production efficiencies of warm-blooded terrestrial animals rarely exceed 5 per cent, and those of some birds and large mammals fall below 1 per cent (Turner, 1970). Gross production efficiencies of insects lie within the range of 5 to 15 per cent, and those of some aquatic animals exceed 30 per cent (Fig. 9.9).

REVISION QUESTIONS

1. Describe the structure of an ecosystem by taking an example of pond ecosystem.
2. Describe the function of ecosystem.
3. Into how many types the ecosystems can be classified ?
4. How does energy flow in an ecosystem ?
5. Describe different kinds of ecological pyramids.
6. What is food chain ? Describe different food chains with examples.
7. What do you mean by ecological efficiency ? Discuss it briefly.

CHAPTER

10

Biogeochemical Cycles

The total mass of all the organisms that have lived on the earth in the past 1.5 billion years is much greater than the mass of carbon and nitrogen atoms present. According to the **law of conservation of matter**, matter is neither created nor destroyed; obviously the carbon and nitrogen must have been used over and over again in the course of time. The earth neither receives any great amount of matter from other parts of the universe nor



There is a constant movement of chemical elements of the biosphere between the organism and the environment.

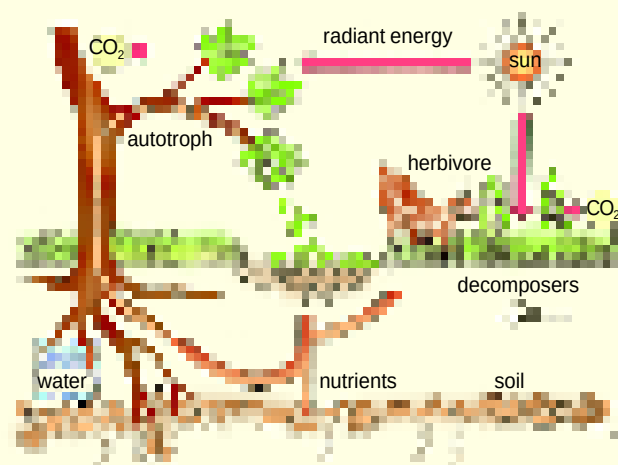


Fig. 10.1. Biogeochemical or nutrient cycle in a forest ecosystem.

does it lose significant amount of matter to outer space. The atoms of each element as carbon, hydrogen, oxygen, nitrogen, phosphorus, calcium, and the rest are taken from the environment, made a part of some cellular component of an organism and finally, perhaps by a quite circuitous route involving several other organisms, are returned to the environment to be used over again. The cyclic movements of chemical elements of the biosphere between the organism and the environment are referred to as **biogeochemical cycles**, after **Vernadsky** (1934). “*Bio*” refers to living organisms and “*geo*” to the rocks, soil, air and water of the earth.

Organic and Abiotic Phases of Geochemical Cycles

Since an element is necessary for the maintenance of life, its movement through biotic communities (organisms) can be viewed in the terms of food chain. The flow of a chemical element through the food chain can be viewed as the **organic phase** of the biogeochemical cycle. Further, the biogeochemical cycles also include **abiotic phases** which are the functions of the chemistry of the elements in question. These abiotic phases are of critical importance to the ecosystem, as the major reservoirs for all nutrient elements are external to the food chains, and flow in the abiotic phases tends to be much slower than in the organic phase. The rapidity and direction of nutrient cycling through the abiotic phases determine not only the distribution of the element in the total environment, but also its availability of living systems. There are two classes of abiotic phases in biogeochemical cycles, a **sedimentary phase**, which is part of all cycles, and an **atmospheric phase**, which is possessed by some. In some cycles, such as nitrogen, the atmospheric phase is more important than the sedimentary. In others, such as phosphorus, the atmospheric phase is essentially non-existent. In still others, such as sulphur, both phases are present and their relative importance depends on other environmental factors. Biogeochemical cycles that have dominant atmospheric phases are often called **atmosphere-reservoir cycles**; those whose sedimentary phase is dominant are termed **sediment-reservoir cycles**.

TYPES OF BIOGEOCHEMICAL CYCLES

There are two types of biogeochemical cycles, the **gaseous** and the **sedimentary**. In gaseous cycles the main reservoir of nutrients is the atmosphere and the ocean. In sedimentary cycles the main reservoir is the soil and the sedimentary and other rocks of the earth’s crust. Both involve biotic and abiotic agents, both are driven by the flow of energy and both are tied to the **water cycle**.

A. Water Cycle

Living organisms, atmosphere and earth maintain between them a circulation of water and moisture, which is referred to as **water cycle** or **hydrologic cycle**. As we have already discussed in Chapter 3, water forms a very significant factor of environment and without the cycling of water, biogeochemical cycles could not exist, ecosystems could not function, and life could not be maintained. Water is important for an ecosystem for several reasons— it is the medium by which nutrients are introduced into autotrophic plants; it is an important part of living tissue, either as liquid water or as part of essential organic molecules; it serves as a means of thermal regulation for both plants and animals; it is the medium by which sediments — a prime source of mineral nutrients— are removed from or added to local ecosystem; it covers the great majority of the earth’s surface, and is the dominant feature of all aquatic ecosystems.

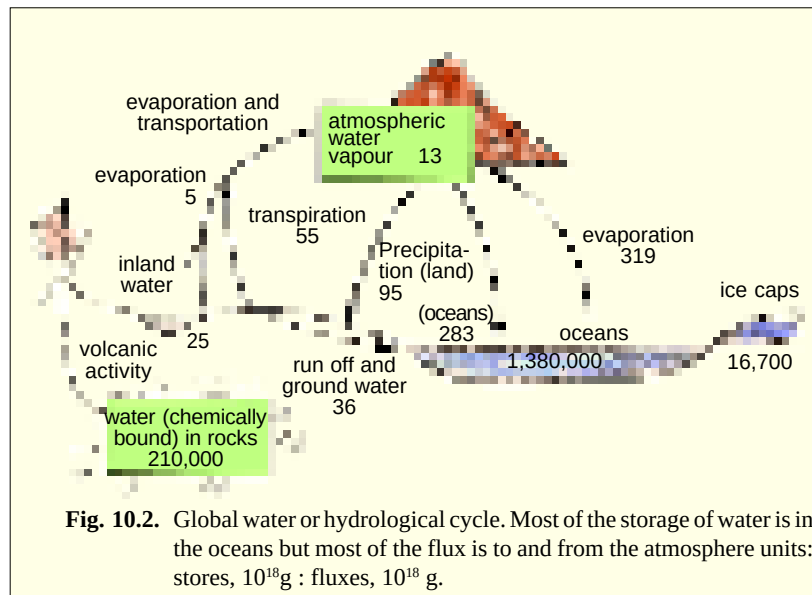
The hydrologic cycle is driven by solar energy and gravity. More than 80 per cent of the total insolation that is not lost immediately as electromagnetic radiation goes to **evaporate** water. The atmospheric water vapour produced by this means can then condense around particles of dust in the atmosphere, often called **nucleation particles**. The atmosphere possesses a limited capacity for holding water vapour, thus, the droplets formed by this means are heavy enough to fall as precipitation, under the influence of gravity. Eventually, the hydrologic cycle can be defined as an alteration of evaporation and precipitation, with the energy used to evaporate the water being dissipated as heat in the atmosphere as the water condenses.

Distribution of water in earth's surface. Water is not evenly distributed throughout the earth. Almost 95 per cent of the total water on earth is chemically bound into rocks and does not cycle. Of the remainder, about 97.3 percent is in the ocean, about 2.1 per cent exists as ice in the polar caps and permanent glaciers, and the rest is fresh water, present in the form of atmospheric water vapour, ground water, soil water, or inland surface water (Nace, 1967).

Table 10-1. Distribution of water in the earth's crust and surface (Clapham, Jr., 1973).

A. Chemically bound water of rocks : Does not cycle	
1. Crystalline rocks	$250,000 \times 10^{17}$ Kg.
2. Sedimentary rocks	$21,00 \times 10^{17}$ Kg.
B. Free water : Moves via hydrologic cycle	
1. Oceans	$13,200 \times 10^{17}$ Kg.
2. Ice caps and glaciers	292×10^{17} Kg.
3. Ground water to a depth of 4000 m	83.5×10^{17} Kg.
4. Freshwater lakes	1.25×10^{17} Kg.
5. Saline lakes and inland seas	1.04×10^{17} Kg.
6. Soil moisture	0.67×10^{17} Kg.
7. Atmospheric water vapour	0.13×10^{17} Kg.
8. Rivers	0.013×10^{17} Kg.

The rate of cycling of water. The rate of cycling between surface and atmosphere is very rapid. The amount of water vapour in the atmosphere is sufficient, on the average, to cover the entire earth to a depth of 2.25 cm. But the average annual rainfall for the earth is about 81.1 cm (Furon, 1967) and in some places it ranges up to 1200 cm. This means that the average turnover time for atmospheric water is about 11.4 days, or that the equivalent of all the water vapour in the entire atmosphere falls as precipitation and is re-evaporated more than 32 times per year.



Further, the distribution of evaporation and rainfall is quite uneven. If we compare the annual evaporation and precipitation over land and sea, we find that relatively more water precipitates on land than evaporates from land. This is fortunate from the point of view of terrestrial organisms. Even so, the amount of rainfall on the open ocean is proportionately greater than that on land, taking into account the relative percentages of the earth's surface covered by land and sea.

Nature of hydrologic cycles. The hydrologic cycle over the oceans is extremely simple—the water is evaporated from the surface of the ocean and water vapours form the clouds which when cool

down precipitate the water as rain fall. But several routes are open to precipitation that falls on land—direct evaporation, transpiration, entry of water into ground water system and runoff. Consequently, the routes of hydrologic cycles on land can be divided into following three main categories—the rapidly cycling portion, or **evapotranspiration**, which includes the evaporation and transpiration, the less rapidly cycling water, or surface runoff, and very slowly cycling ground water that seeps into the soil can end up in any one of these three categories.

1. Evapotranspiration. Evapotranspiration includes evaporation and transpiration. **Evaporation** refers to water that is evaporated directly from any surface other than a plant, such as a lake, soil surface, or animal skin. In most cases, the main effects of direct evaporation are to moderate the temperature of local area and to allow the hydrologic cycle to continue. In some ecosystems, evaporation also leads to a concentration of salts in the water of soil which may be a critical environmental factor. **Transpiration** is water that evaporates from the surface of leaves of plants. Transpiration acts to move the biogeochemical cycles for all mineral nutrients that enter the food chain via the roots of plants.

2. Surface runoff. If transpiration is related to the mechanism of nutrient uptake, the gross movement of soluble and solid particles in the ecosystem is accomplished largely by runoff. Nutrients that have accumulated in sediments or soils can be eroded by streams and removed altogether from a local ecosystem, or soluble nutrients may be carried by soil seepage into surface waters, where they are removed from the area. Streams may carry sediment particles which can be chemically altered through additional weathering so that the nutrient elements they contain may be utilized by organisms. Finally moving water acts as an agent of **erosion** which removes soil and allows weathering of the underlying rock to make their nutrients available to plants.

3. Ground water. Ground water is water that saturates either sediment or rock below the water table. In general, it is not trapped by plants for transpiration and it is too deep to be directly evaporated from the soil surface. It is an exceedingly important reservoir for water which moves from one place to another under the influence of gravity. The area where the net water movement is from the surface into the ground water systems is termed a **catchment area**; areas where ground water reaches the surface and runs off are termed **springs**. A rock body through which ground water flows is called an **aquifer**. A well drilled into an aquifer that has sufficient hydrostatic pressure to force water up into it is called an **artesian well**.

The hydrologic cycle on land, thus, includes evapotranspiration of water from earth's surface and leaf surface → formation of clouds → precipitation → surface runoff + accumulation of water as ground water → return of water to sea via streams or direct evaporation and cloud formation, and so on.

B. Gaseous Cycles

The gaseous geochemical cycles are of following types :

1. The oxygen cycle. Oxygen (O_2), the by-product of photosynthesis, is involved in the oxidation of carbohydrates with release of energy, carbon dioxide and water. Its primary role in biological oxidation is that of a hydrogen acceptor. The break-down and decomposition of organic molecules proceeds primarily by dehydrogenation. Hydrogen is removed by enzymatic activity from organic molecules in a series of reactions and is finally accepted by the oxygen, forming water. Though oxygen is necessary for life, but being very active chemically, molecular O_2 may be toxic to living

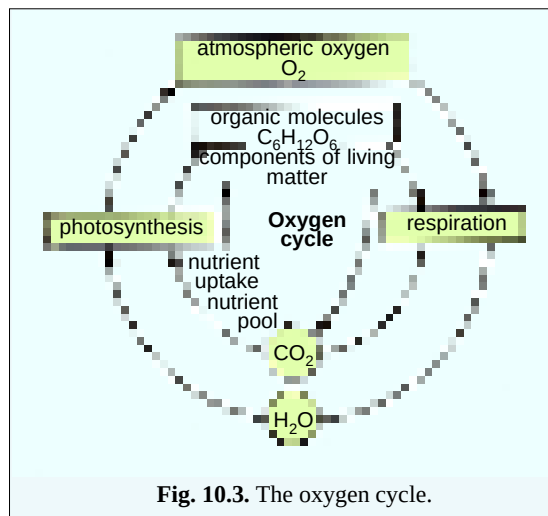
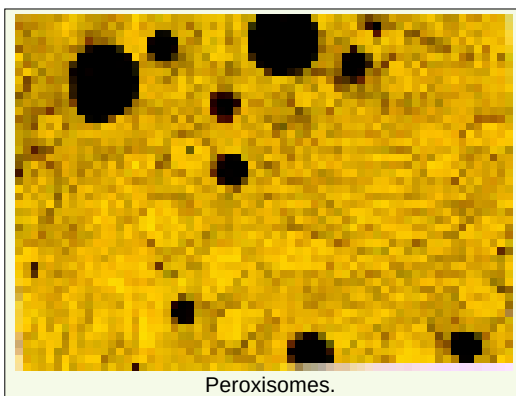


Fig. 10.3. The oxygen cycle.

body cells. Therefore, for the protection from toxic effects of molecular O_2 , cells possess the cellular organelles called **peroxisomes** which mediate oxidative reactions resulting in the production of hydrogen peroxide which in turn is used through the mediation of other enzymes as an acceptor in oxidizing other compounds.



The major supply of free oxygen which supports life occurs in the atmosphere. There are two significant sources of atmospheric oxygen. One is the **photodissociation** of water vapour in which most of the hydrogen released escapes into outer space. The other source is **photosynthesis**, active only since life began on earth. Because photosynthesis and respiration are cyclic, involving both the release and utilization of oxygen, one would seem to balance the other, and no significant quantity of oxygen would accumulate in the atmosphere. However, at some time in the earth's history the amount of oxygen introduced into the

atmosphere had to exceed the amount used in the decay of organic matter and that tied up in the oxidation of sedimentary rocks. Part of the atmospheric oxygen represents that portion remaining from the unoxidized reserves of photosynthesis—coal, oil, gas, and organic carbon in sedimentary rocks. The amount of stored carbon in the earth suggests that $150 \times 10^{20}g$ of oxygen has been available to the atmosphere, over 10 times as much as now present, $10 \times 10^{20}g$ (Johnson, 1970). The main non-living (abiotic) oxygen pool consists of molecular oxygen, water, and carbon dioxide, all intimately linked to each other in photosynthesis and other oxidation-reduction reactions, and all exchangeable in such compounds as nitrates and sulphates utilized by organisms that reduce them to ammonia and hydrogen sulphide.

The cycling of oxygen is very complex. As a constituent of CO_2 , it circulates freely throughout the biosphere. Some carbon dioxide combines with calcium to form carbonates. Oxygen combines with nitrogen compounds to form nitrates, with iron to ferric oxides, and with many other minerals to form various other oxides. In these states oxygen is temporarily withdrawn from circulation. In photosynthesis the oxygen freed is split from the water molecule. This oxygen is then reconstituted into water during plant and animal respiration. Some part of the atmospheric oxygen that reaches the higher levels of the troposphere is reduced to ozone (O_3) by high energy ultraviolet radiation.

2. The carbon cycle. The carbon being a basic constituent of all organic compounds and a major element involved in the fixation of energy by photosynthesis, is so closely tied to energy flow that the two are inseparable. The source of all the fixed carbon both in living organisms and fossil deposits is

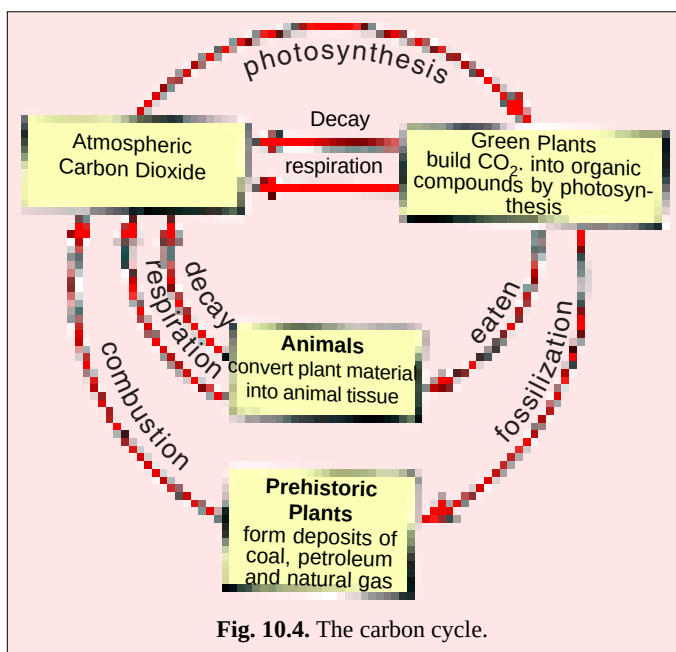
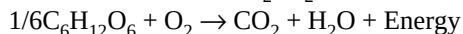


Fig. 10.4. The carbon cycle.

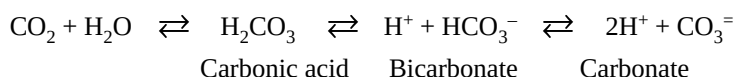
carbon dioxide CO_2 , found in the atmosphere and dissolved in the waters of the earth. During photosynthesis, carbon from atmospheric CO_2 is incorporated into the production of the carbohydrate, glucose, $\text{C}_6\text{H}_{12}\text{O}_6$, that subsequently may be converted to other organic compounds such as polysaccharides (sucrose, starch, cellulose, etc.), proteins and lipids. All the polymeric organic compounds containing carbon are stored in different plant-tissues as food and from them the carbon is passed on to the trophic levels of herbivores or phytoparasites, or retained by the plant until it serves as food for decay organisms (viz., decomposers). Some of the carbon is returned to the atmosphere (or the enveloping aqueous medium) in the form of CO_2 a by-product of plant respiration, in which, a considerable portion of glucose is oxidized to yield CO_2 , H_2O and energy as follows :



The CO_2 which is released as the by-product of plant respiration is again used by plants in photosynthesis. This is the basic carbon cycle which is simple and complete. Decomposing micro-organisms are important in breaking down dead material with the release of carbon back into the carbon cycle.

Similarly, carbon taken up by herbivores or phytoparasites may travel a number of routes. It may be incorporated into protoplasm (assimilation) and stored until the organism dies, where upon it is utilized by decomposers; it may be released through animal respiration; it may serve as live food for other organisms; or finally it may be stored in the environment as CO_2 . Similar fates await carbon at the carnivore trophic levels. In fact, all the carbon of plants, herbivores, carnivores and decomposers is not respired but some is fermented and some is stored. The carbon compounds that are lost to the food chain after fermentation, such as methane, are readily oxidized to carbon dioxide by inorganic reactions in the atmosphere. As for the storage of carbon in sediments, just as deposition works to store materials, erosion may uncover them, and inorganic chemical weathering of rock can oxidize the carbon contained there. Some carbon is permanently stored in sediments and not uncovered by weathering; it may be replaced by carbon dioxide released from volcanoes and other similar examples of intense geological activity. In modern age, man has greatly increased the rate at which carbon is passing from sedimentary form to carbon dioxide. The combustion of fossil fuels is a significant means of recycling sedimentary carbon much faster than natural weathering.

Small portion of carbon, especially in the sea, is found not as organically fixed carbon, but as carbonate (CO_3^{2-}), especially calcium carbonate (CaCO_3). CaCO_3 is very commonly used for shell construction by such animals as clams, oysters, some protozoa, and some algae. Carbon dioxide reacts with water to form carbonate in the following three step reaction :



The precise amount of each of these constituents in the water depends on the pH of the water. Organisms such as clams can combine bicarbonate or carbonate with calcium dissolved in the water to produce calcium carbonate. After the death of the animal, this calcium carbonate may either dissolve or remain in sedimentary form.

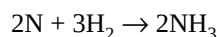
Certain control mechanisms are inherent in the carbon cycle. The rate of carbon utilization is dependent on its availability. If excessive amounts of carbon are taken up in any one phase of the cycle, other phases of activity may be inhibited or slowed down. For example, if the pH of water is alkaline, more carbon is tied up in a carbonate and less is in solution. This removal of carbon in solution would upset the equilibrium established between the atmospheric and the dissolved CO_2 and the net effect would be a movement of CO_2 into solution until equilibrium was reached.

Peculiarities of carbon cycle. Though carbon-cycle exhibits basic similarity with other biogeochemical cycles, yet it is unusual in that the organic phase is not essentially a complete cycle within itself. The organic (biotic) and atmospheric (abiotic) phases, however, are so closely intertwined that the rapid cycling typical of the organic phase is present. The multiplicity of paths along which

carbon can flow is typical of biogeochemical cycles in general, and provides a well-buffered system with adequate feedback mechanisms to insure an adequate supply of the carbon. It is significant that all phases of the cycle yield carbon dioxide at some time, and carbon dioxide is the raw material for them. Thus, despite its relative low concentration in the atmosphere (0.03 per cent), carbon in a form in which it can be used by living organisms is virtually always present.

3. The nitrogen cycle. Nitrogen is an essential constituent of different biologically significant organic molecules such as amino acids and proteins, pigments, nucleic acids and vitamins. It is also the major constituent of the atmosphere, comprising about 79 per cent of it. The paradox is that in its gaseous state, N_2 is abundant but is unavailable to most life. Before it can be utilized it must be converted to some chemically usable form.

To be used biologically, the free molecular nitrogen has to be fixed and fixation requires an input of energy. In the first step molecular nitrogen, N_2 has to be split into two atoms : $N_2 \rightarrow 2N$. The free nitrogen atoms then must be combined with hydrogen to form ammonia, with the release of some energy:



This fixation comes about in two ways. One is by high-energy fixation such as cosmic radiation, meteorite trails, and lightning that provide the high energy needed to combine nitrogen with oxygen and hydrogen of water. The resulting ammonia and nitrates are carried to the earth in rain water. The second method of nitrogen-fixation which contributes about 90 per cent of fixed nitrogen of earth, is biological. Some bacteria, fungi, and blue-green algae can extract molecular nitrogen from the atmosphere and combine it with hydrogen to form ammonia. Some of this ammonia is excreted by the nitrogen-fixing organism, and, thus, becomes directly available to other autotrophs. Some of these nitrogen-fixing organisms may be free-living, either in the soil (e.g., bacteria — *Azotobacter* and *Clostridium*) or in water (e.g., blue-green algae—*Nostoc*, *Calothrix* and *Anabaena*) and produce vast quantities of fixed nitrogen. In other cases, certain symbiotic bacteria of genus *Rhizobium*, although unable to fix atmospheric nitrogen themselves, can do this when in combination with cells either from the roots of legumes (e.g., peas, beans, clover and alfalfa) and of other angiosperms such as *Alnus*, *Ceanothus*, *Shepherdia*, *Elaeagnus* and *Myrica*, or from the leaves of African genera of *Rubiaceae* and *Pavetta*. The bacteria invade the roots or leaves and stimulate the formation of root-nodules or leaf-nodules, a sort of harmless tumor. The combination of symbiotic bacteria and host cells remains able to fix atmospheric nitrogen and for this reason legumes are often planted to restore soil fertility by increasing the content of fixed nitrogen. Nodule bacteria may fix as much as 50 to 100 kilograms of nitrogen per acre per year, and free soil bacteria as much as 12 Kilograms per acre per year. Further both free soil bacteria (*Azotobacter* and *Clostridium*) produce ammonia as the first stable product and like

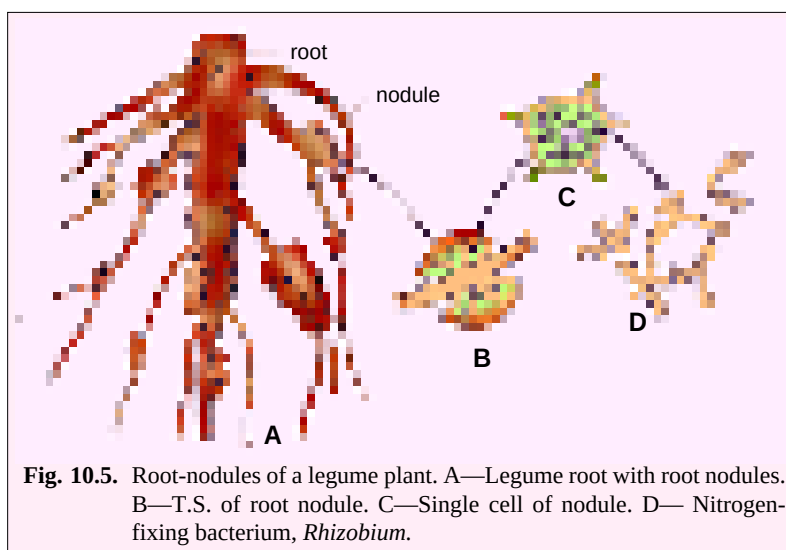


Fig. 10.5. Root-nodules of a legume plant. A—Legume root with root nodules. B—T.S. of root nodule. C—Single cell of nodule. D— Nitrogen-fixing bacterium, *Rhizobium*.

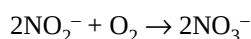
the symbiotic bacteria, they require molybdenum as an activator and are inhibited by an accumulation of nitrates and ammonia in soil.

Recently, certain lichens (*Collema tunaeforme* and *Peltigera rufescens*) were also implicated in nitrogen fixation (Henriksson, 1971). Lichens with nitrogen-fixing ability possess nitrogen-fixing blue green species as their algal component.

Nitrogen fixed by symbiotic and non-symbiotic microorganisms in soil and water is one source of nitrogen. Another source is organic matter. The nitrogenous wastes and carrion of animals are degraded by the detritus organisms, nitrogen is converted to the amino form (e.g., L-Alanine). The amino group ($-\text{NH}_2$) is liberated from organic molecules to form ammonia; this process is called **deamination**. Certain specific bacteria, most notably of the genus *Nitrosomonas*, can oxidize ammonia to nitrite (NO_2^-) by the reaction.

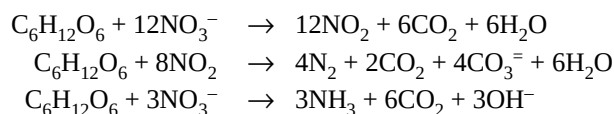


This reaction takes place in the soil, in lake or sea water or sediments, and whenever ammonia is being released and oxygen is present. As fast as nitrite is produced, other bacteria, such as *Nitrobacter*, can combine nitrite with oxygen to form nitrate (NO_3^-) by the reaction :



Both of these reactions which are performed by two nitrifying bacteria — *Nitrosomonas* and *Nitrobacter* are the parts of a single biological process called **nitrification**. In nitrification process, thus, ammonia is oxidized to nitrate and nitrite yielding energy. This energy is used by the bacteria to make their organic materials directly from carbon dioxide and water. Nitrate can be taken up by autotrophs at the beginning of food chain.

Under certain circumstances, nitrate is either not produced in the nitrogen cycle or it is degraded before it can be utilized by autotrophs. Degradation of nitrate is called **denitrification**, and may be important when oxygen concentration is low. Denitrifying bacteria such as *Pseudomonas* can use the energy of the nitrate ion to drive their metabolism, and in so doing, they break the nitrate down to nitrite, ammonia, or molecular nitrogen :



If denitrification is significant in an ecosystem, nitrite is transitory and is also degraded into either ammonia or molecular nitrogen.

Cycling of nitrogen in the ecosystem. The sources of inputs of nitrogen under natural conditions are the bacterial fixation of atmospheric nitrogen, addition of inorganic nitrogen in rain from such sources as lightning fixation and fixed “juvenile” nitrogen from volcanic activities, ammonia absorption from the atmosphere by plants and soil, and nitrogen accretion from windblown aerosols, which contain both organic and inorganic forms of nitrogen. In terrestrial ecosystems, nitrogen, largely in the form of ammonia and nitrates is taken up by plants, which convert it into amino acids and proteins. Animals (primary macro-consumers) may eat the plants and utilize the amino acids from the plant proteins in the synthesis of their own proteins and other cellular constituents. When animals and plants die, the decay bacteria convert the nitrogen of their proteins and other compounds into ammonia. Animals excrete several kinds of nitrogen-containing wastes—urea, uric acid, creatinine, and ammonia and the decay bacteria converts these wastes to ammonia. Ammonia may be lost as gas to the atmosphere, may be acted upon by nitrifying bacteria, or may be taken up directly by plants. The nitrates may be utilized by plants, immobilized by microbes, stored in decomposing humus, or leached away. This material is carried to streams, lakes, and eventually the sea, where it is available for use in aquatic ecosystems. There nitrogen is cycled in a similar manner, except that the large reserves contained in the soil humus are largely lacking. Life in the water contributes organic matter and dead organisms that

undergo decomposition and subsequent release of ammonia and ultimately nitrates. In aquatic ecosystem atmospheric nitrogen is fixed by numerous blue-green algae.

Under natural conditions nitrogen lost from ecosystems by denitrification, volatilization, leaching, erosion, wind blown aerosols, and transportation out of the system is balanced by biological fixation and other sources. Both chemically and biologically, terrestrial and aquatic ecosystems constitute a dynamic equilibrium system in which a change in one phase affects the other.

C. Sedimentary Cycles

Mineral elements required by living organisms are obtained initially from inorganic sources. Available forms occur as salts dissolved in soil water or lakes, streams,

and seas. The **mineral cycle** varies from one element to another, but essentially it consists of two phase; the **salt-solution phase** and the **rock phase**. Mineral salts come directly from the earth's crust by weathering. The soluble salts then enter the water-cycle. With water they move through the soil to streams and lakes and eventually reach the sea, where they remain indefinitely. Other salts are returned to the earth's crust through sedimentation. They become incorporated into salt beds, silts and limestones; after weathering they again enter the cycle.

Plants and animals, the living components of the ecosystems, fulfil their mineral requirements from mineral solutions in their environments. Other animals acquire the bulk of their minerals from plants and animals they consume. After the death of living organisms the minerals are returned to the soil and water through the action of the organisms and process of decay.

There are different kinds of sedimentary or mineral cycles, depending on the kinds of elements, but following two cycles are very significant for a ecosystem :

(i) Sulphur cycle. Sulphur, like nitrogen, is an essential part of protein and amino acids and is characteristic of organic compounds. It exists in a number of states—elemental sulphur, S, sulphides, sulphur monoxide, sulphite and sulphates. Of these three are important in nature : elemental sulphur, sulphides and sulphates.

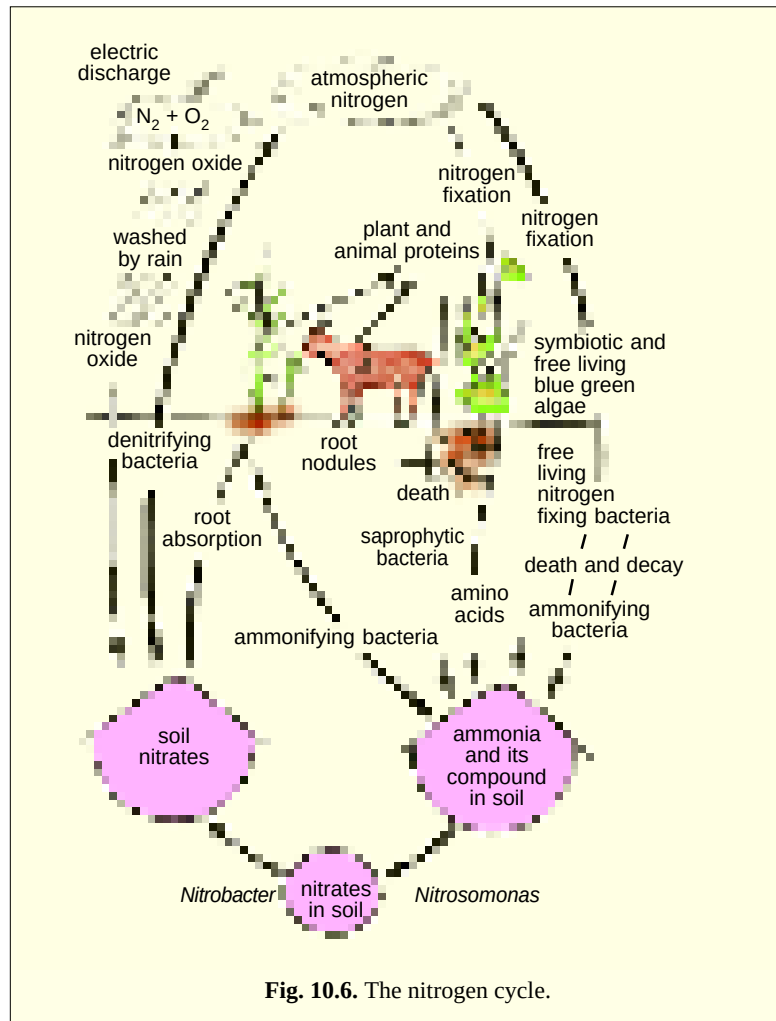


Fig. 10.6. The nitrogen cycle.

The sulphur cycle is both sedimentary and gaseous (*i.e.*, it includes gaseous phase and sedimentary phase). The sedimentary phase of sulphur cycle is long-termed and in it sulphur is tied up in organic and inorganic deposits. From these deposits, it is released by weathering and decomposition, and is carried to terrestrial and aquatic ecosystems in a salt solution. Atmospheric (gaseous) phase of sulphur-cycle is less pronounced and it permits circulation on a global scale.

Sulphur enters the atmosphere from several sources—the combustion of fossil fuels, volcanic eruption, the surface of the oceans and gases released by decomposition. Initially sulphur enters the atmosphere as hydrogen sulphide, H_2S , which quickly oxidizes into another volatile form, sulphur dioxide, SO_2 . Atmospheric sulphur dioxide, soluble in water, is carried back to earth in rainwater as weak sulphuric acid, H_2SO_4 . Whatever its source, sulphur in a soluble form, mostly as sulphate (SO_4^{2-}) is absorbed through plant roots, where it is incorporated into certain organic molecules, such as some amino acids (*e.g.*, cystine) and proteins. From the producers the sulphur in amino acids is transferred to the consumer animals, with excess being excreted in the faeces.

Excretion and death carry sulphur in living material back to the soil and to the bottoms of ponds, lakes, and seas where the organic material is acted upon by bacteria of detritus food chain. Within the detritus food chain, the sulphhydryl group ($-\text{SH}$) of amino acids (*e.g.*, L-cysteine) is separated from the rest of the molecule as hydrogen sulphide (H_2S) by most decomposing bacteria as a normal part of the degradation of proteins. In an aerobic environment, the hydrogen sulphide is oxidized to sulphate by bacteria specially adapted to perform this conversion :

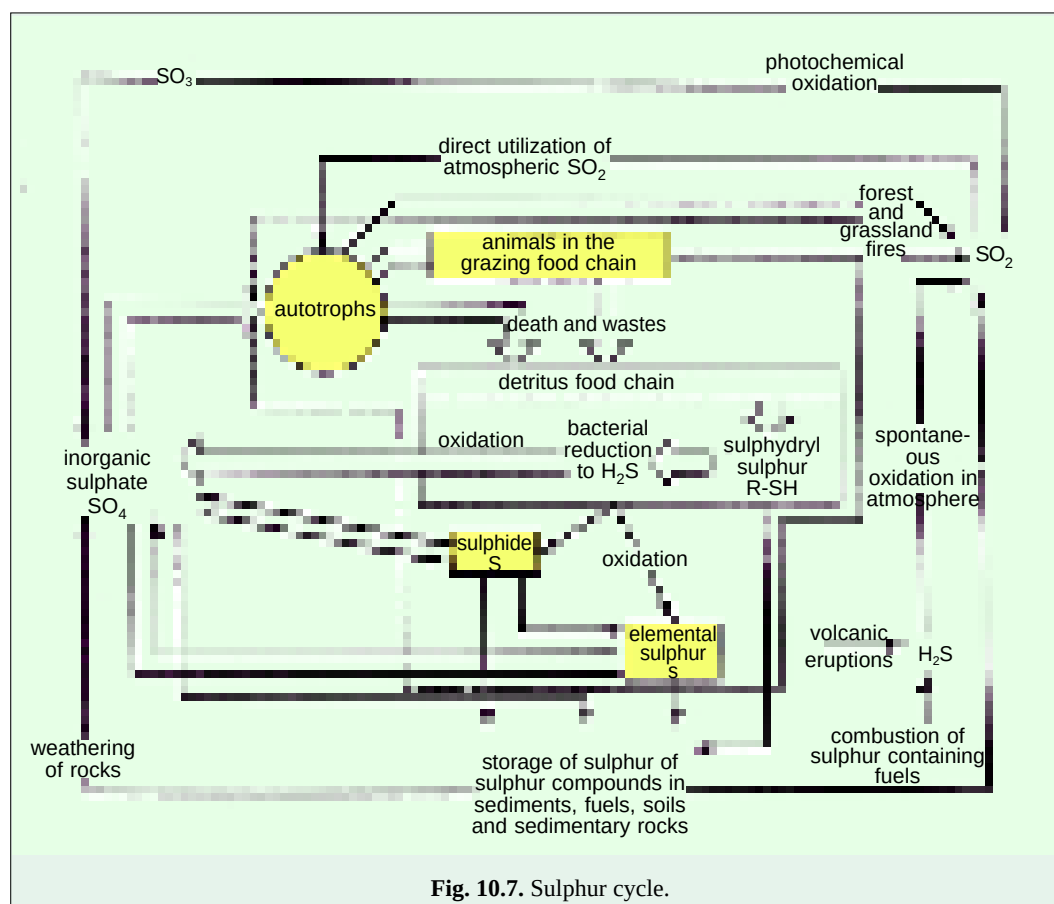
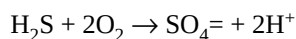
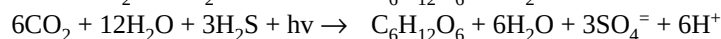
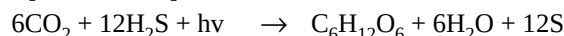
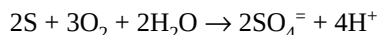


Fig. 10.7. Sulphur cycle.

The sulphate produced then can be reused by the autotrophs. In anaerobic environments, such as bottom of certain lakes, it is impossible to oxidize sulphide by this means, because the process of oxidation requires oxygen. But if infrared radiation is present in these environments, there are photosynthetic bacteria that can use it to manufacture carbohydrates and oxidize sulphide either to elemental sulphur or to sulphate :



Elemental sulphur can also be utilized by other bacteria to form sulphate. If oxygen is present, the reaction is quite rapid.

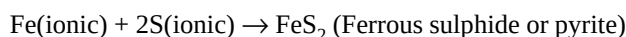
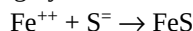


Under anaerobic conditions, elemental sulphur can still be oxidized to sulphate by certain bacteria if nitrate is present :

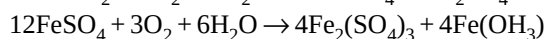
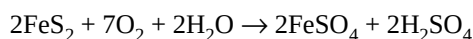


None of these bacterial reactions is unidirectional; under certain conditions, sulphate can also be reduced either to sulphide or to elemental sulphur by bacteria. This series of reactions operating within the organic phase of the sulphur cycle provides a rather finely tuned mechanism for regulating the availability of sulphur to autotrophs.

The sulphur is removed from the organic phase in the form of elemental sulphur which is insoluble and accumulates in sediments. If iron is present in the sediment, it can combine with sulphide to form iron sulphides, all of which are highly insoluble :



FeS_2 is highly insoluble under neutral and alkaline conditions and is firmly held in mud and wet soil. Some ferrous sulphide is contained in sedimentary rocks overlying coal deposits. Exposed to the air in deep and surface mining, the ferrous sulphide oxidizes and in the presence of water produces ferrous sulphate and sulphuric acid :



In this manner sulphur in pyrite rocks, suddenly exposed to weathering by man, discharges heavy slugs of sulphur, sulphuric acid, ferric sulphate and ferrous hydroxide into aquatic ecosystems. These compounds destroy aquatic life and cause acidic water.

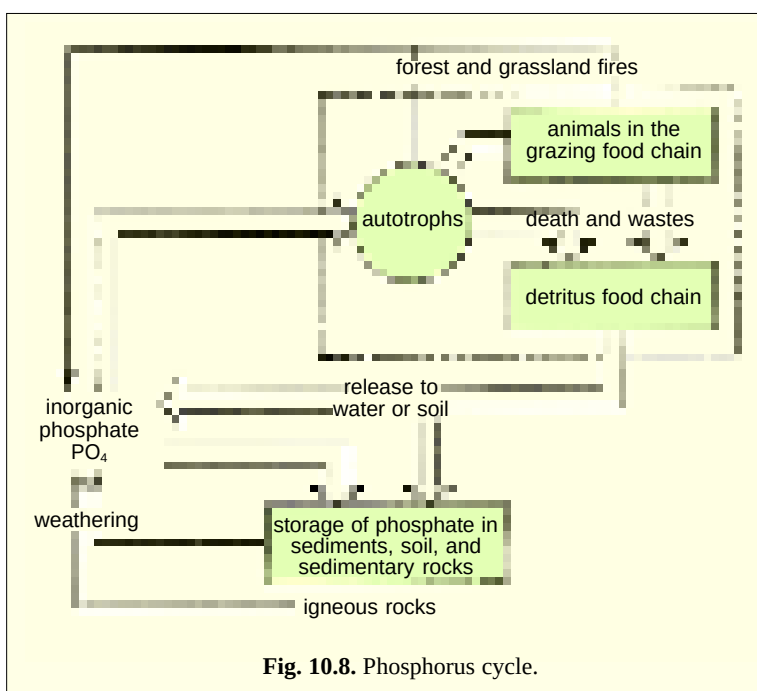


Fig. 10.8. Phosphorus cycle.

(ii) Phosphorus cycle. Phosphorus cycle has no atmospheric phase. It occurs naturally in environment as phosphate (PO_4^- , or one of its analogues, HPO_4^- or H_2PO_4^-), either as soluble inorganic phosphate ions, as soluble organic phosphate (*i.e.*, as a part of a soluble organic molecule), as particulate phosphate (*i.e.*, as part of an insoluble organic or inorganic molecules) or as mineral phosphate (*i.e.*, as part of a mineral grain as found in a rock or sediment). The ultimate source of phosphate in the ecosystem is crystalline rocks. As these are eroded and weathered, phosphate is made available to living organisms, generally as ionic phosphate. This is introduced into autotrophic plants through their roots, where it is incorporated into living tissues. From autotrophs, it is passed along the grazing food chain in the same fashion as nitrogen and sulphur, with excess phosphate being excreted in the faeces. An extreme example of faecal phosphate is the tremendous guano deposits built up by birds on the desert west coast of South America. Phosphates can also be released as particulate matter from forest and grassland fires.

In the detritus food chain, as large organic molecules containing phosphate are degraded, the phosphate is liberated as inorganic ionic phosphate. In this from it can be immediately taken up by autotrophs, or it can be incorporated into a sediment particle, either in the soil of a terrestrial ecosystem or in a sediment of an aquatic ecosystem. The sedimentary phase of phosphorous cycle remains comparatively slow than the organic phase.

Besides phosphorus, there are biogeochemical cycles for all the other nutrients (minerals) used by living organisms, as well as some that are not. Most of them has complete cycles in sedimentary phase. The availability depends on their solubility in water and availability of water as solvent.

Thus, the geochemical cycles of different chemical substances are closed : the atoms are used over and over again. To keep the cycles going does not require new matter but it does require **energy**, for the energy cycle is not a closed one. Further, the patterns of flow, both of energy and of chemical substances, are of great significance. The simpler pat-

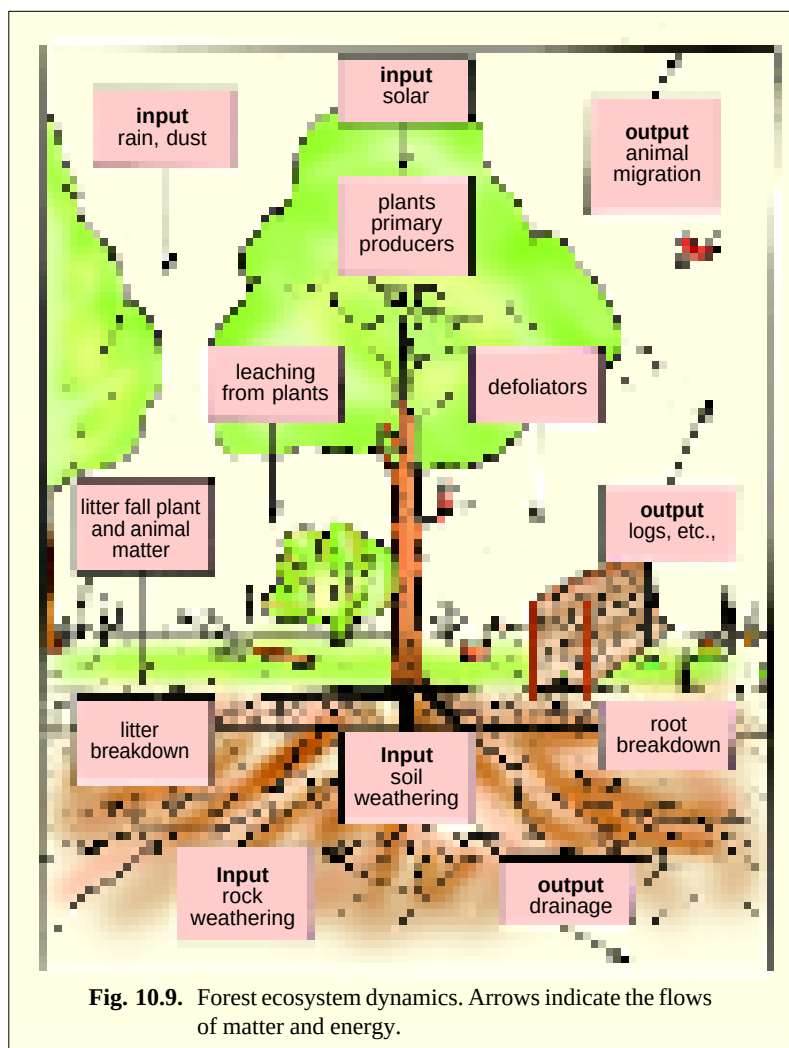


Fig. 10.9. Forest ecosystem dynamics. Arrows indicate the flows of matter and energy.

terns involve energy, as the sources of energy are external to the ecosystem, and flow is unidirectional through it. Chemical substances, on the other hand, are finite and have their origin inside the ecosystem, thus, they must continuously cycle within the system.

Biogeochemical Cycle of Micronutrients

Many other micronutrients (elements) such as calcium, potassium, sodium, etc., have a similar pathway of cycling like the other nutrients. They also involve the abiotic (soil, water, etc.) and biotic phases (living organisms and their decomposition). These elements usually cycle within the same ecosystem and in small quantities globally. Usually, the rocks are their reserve pools and the weathering process enriches the soil and water with these elements. Though, the dust of these nutrients may be blown away by the wind and carried across ecosystems and some of their amount is lost in the runoff to lakes and oceans and then to deep sediments. For example, in a tropical rain forest on average, more than 80% of total store of these micronutrients is found in the soil, with progressively less in vegetation, litter and the fauna (Fig. 10.9). Though, the concentration of trace elements tends to increase along the food chains in the sequence such as soil – plants – herbivores – carnivores and litter – detritivores. Thus, concentration of iron in a rain forest were 6 ppm in soil, 218 ppm in vegetation, 3054 ppm in herbivores and 3387 ppm in the carnivores. Many of these micronutrients become toxic to plants in high concentration (see **Sharma**, 1994).

The nutrient concentration of the leaf litter of some plants have been tabulated in Table 10-2.

Table 10-2. Concentration of nutrients (in %) in the leaf litter of some plant species (Source : Vyas and Galley, 1975 ; Rajvanshi and Gupta, 1985).

Species	Calcium (Ca)	Magnesium (Mg)	Potassium (K)	Sodium (Na)	Nitrogen (N)	Phosphorus (P)
1. <i>Terminalia arjuna</i>	4.42	0.37	0.54	0.06	0.85	0.24
2. <i>Terminalia tomentosa</i>	3.73	0.54	0.40	0.06	0.93	0.52
3. <i>Tectona grandis</i>	2.54	0.25	0.24	0.20	0.78	0.18
4. <i>Diospyros melanoxylon</i>	1.90– 2.10	0.51– 0.69	0.57– 0.90	0.05– 0.10	0.56– 0.61	0.008– 0.02
5. <i>Butea monosperma</i>	1.85	0.56	0.75	0.06	2.26	0.25
6. <i>Shorea robusta</i>	1.03– 1.58	0.36– 0.48	0.26– 0.59	0.05– 0.08	0.71– 0.81	0.41– 0.50
7. <i>Dalbergia sissoo</i>	1.02	0.96	0.25	0.07	2.23	0.025

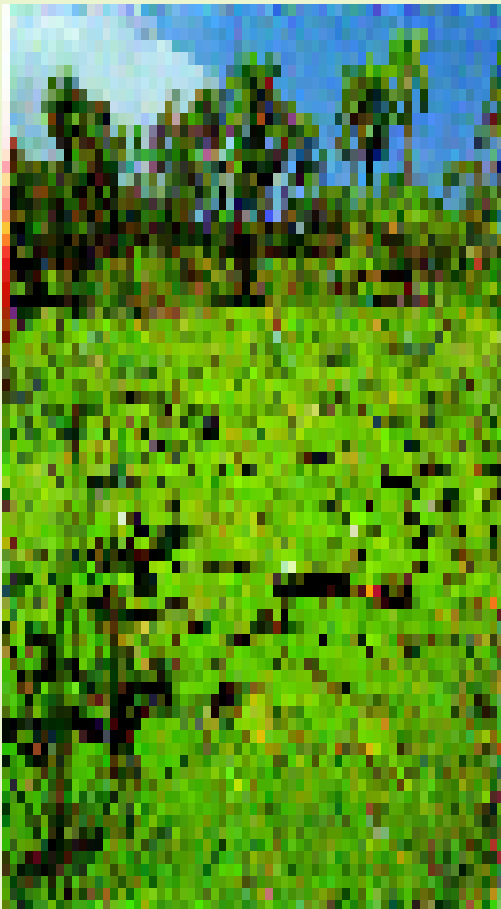
REVISION QUESTIONS

1. What are biogeochemical cycles ? What role they play in the ecosystem ? Discuss.
2. Describe nitrogen and phosphorus biogeochemical cycles with suitable examples.
3. Write short notes on the following :
 - (a) Carbon cycle ;
 - (b) Water cycle ;
 - (c) Sedimentary biogeochemical cycles ;
 - (d) Gaseous biochemical cycles.

CHAPTER

11

Aquatic Ecosystems : Freshwater Communities



Aquatic ecosystems are the source of life on earth.

The **habitat approach**, describes the distinctive features of the major habitats and their subdivisions, how they are organized, the organisms present in each and the ecologic role of these organisms in that region (*i.e.*, the identity of the major producers, consumers and decomposers). According to **Elton** (1949) habitat is definable as an area possessing uniformity of physiography, vegetation, climate or any other quality the investigator assumes important. Four major habitats can be distinguished : **marine, estuarine, freshwater** and **terrestrial**. No plant or animal is found in all four major habitats and, indeed, no animal or plant is found everywhere within any one of these. According to recent ecological trend only two major habitats or more specifically ecosystems namely **aquatic ecosystems** and **terrestrial ecosystems**, are recognized.

AQUATIC ECOSYSTEMS

Liquid water covers about three quarters of the earth's surface either as oceans or as freshwater. Virtually all these

waters contain life in one form or other; hence, aquatic ecosystems would be important for their sheer volume, if for nothing else. But aquatic ecosystems are historically the source of life on earth. Even now, tiny single-celled marine plants are the main source of the earth's oxygen supply, because most photosynthetic oxygen is derived from these plants (Clapham, Jr., 1973). In addition, aquatic ecosystems are simpler in many ways than terrestrial ecosystems. The reason for this is that the omnipresent environmental factor which sets the tone for all aquatic systems, regardless of their biotic complexity, is water. It is the medium within which all aspects of the ecosystem coexist, both living and non-living; it is the source of all nutrients for aquatic life, including the gaseous nutrients such as oxygen and carbon dioxide; it is the medium by which organic and inorganic wastes and sediments are distributed throughout the ecosystem. The amount of light energy reaching the community is determined by the way in which light is absorbed by water; the heat properties of water determine much of the circulation patterns within the ecosystem and have a major control over the structure of the aquatic community that exists in any one place.

SUBDIVISIONS OF AQUATIC ECOSYSTEMS

Global aquatic system fall into two broad classes definable by salinity, or amount of material dissolved in water—the **freshwater ecosystems** and **salt-water ecosystems**. The latter may include inland brackish water, as well as marine and estuarine habitats. In fact, estuary represents a transitional zone between a river and the sea and it contains dissolved solid content intermediate between those of fresh and marine waters. Freshwater ecosystems, the study of which is known as **limnology**, are conveniently divided into two groups—**lentic, standing or still water habitats** and **lotic or running water habitats**. Both can be considered on an environmental gradient. The lotic follows a gradient from springs to mountain brooks to streams to rivers. The lentic involves a gradient from lakes to ponds to bogs, swamps and marshes.

FRESHWATER ECOSYSTEMS : PHYSICO-CHEMICAL NATURE OF FRESHWATER

The freshwater of both kinds—lentic and lotic, has low percentage of dissolved salts and is subjected to the influence of a wide array of physical and chemical factors. The rise and fall of these factors very frequently affect the fauna, altering their number and diversity. Some of the important factors of freshwater environment are following :

1. Pressure, Density and Buoyancy

The pressure imposed on a lake-dwelling organism is the weight of the column of water above it plus the weight of the atmosphere. In all freshwater environments maximum pressure is much less than in the ocean, and organisms appear to adjust to them readily. The absence of animal life from deep water is ordinarily a consequence of low oxygen supply, or low temperature, rather than pressure.

The density of water varies inversely with temperature and directly with the concentration of dissolved substances. Water is most dense at approximately 4°C and becomes progressively less dense as it cooled below +4°C. Ice also expands markedly the colder it gets. It is because the coldest water is at the surface in winter that ice forms there, rather than at the bottom of a lake. In summer, the coldest waters of deep lakes are at the bottom. Dissolved salts increase the density of water; the density of most inland water-bodies is much less than that of the ocean. However, when great evaporation occurs in a lake having no outlet, the lake may come to contain a higher percentage of salts (*i.e.*, hypersalinity) than the ocean. The few species capable of living in these very salty lakes (*e.g.*, Great Basin of USA) include some algae and Protozoa, the brine shrimp *Artemia gracilis* and the immature stages of two brine flies, *Ephydra gracilis* and *E. hians*.

According to the **law of Archimedes** the buoyancy of an object is equal to the weight of the water it displaces. Buoyancy varies with the density of water, and is influenced by the factors that affect density. Viscosity, the measure of the internal friction of water, varies inversely with temperature and also influence buoyancy. Most aquatic organisms keep stations by swimming movements or have special adaptations to decrease the specific gravity of the body and take advantage of any turbulence in the water. For this purpose freshwater aquatic organisms have some swimming adaptations, clinging organs (in case of animal of lotic habitat) or following adaptations—absorption of large amounts of water to form jelly-like tissues; storage of gas or air bubbles within the body; formation of light-weight fat deposits within the body or oil droplets within the cell; increase of surface area in proportion to body mass, which increase frictional resistance (**David**, 1955). When an organism so equipped dies, the special mechanism quickly cease to function, and it sinks to the bottom.

2. Temperature

The unique thermal properties of water are best demonstrated by freshwater environment. Diurnal and seasonal variations of temperatures are very much evident in these environments than in marine environments. A diurnal variation range of 4.8–5.0°C has been recorded by **Sreenivasan** (1964) in a tropical pond, with an average depth of 3.0 meters. In shallow water habitats, difference between day and night temperatures remain more conspicuous. For example, in a polluted moat with an average depth of 1.5 meters, the lowest night time temperature was 26.6°C, the highest day time temperature was 32°C with a variation of 5.4°C (**Sreenivasan**, 1964). However, the Kodaikanal lake in South India showed a diurnal variation of only 2.8°C (**Sreenivasan**, 1964). Flowing lotic waters of streams and rivers lack such wide fluctuations in temperature.

Further, the lentic water of lakes and ponds undergo thermal stratification phenomenon according to the seasons. Thermal stratification has been reported most frequently in the lakes of tropical countries such as Java, Sumatra, etc. In fact, according to their temperature relations, lakes have been classified into three types : (1) **Tropical lakes** in which surface temperatures are always maintained above 4°C ; (2) **Temperate lakes** in which surface temperature vary above and below 4°C and (3) **Polar lakes** in which surface temperatures never go above 4°C. The seasonally regulated thermal stratification of lentic habitats has a significant influence on their inhabitant biotic communities. Decreasing temperatures often cause a fall in metabolism, resulting in a lower rate of food consumption. The extremes of lower and higher temperature have lethal effects on the aquatic organisms. So fluctuations in temperature of aquatic media regulate the breeding periods, initiate hibernation, gonadal activation and a number of other biological phenomena such as thermally oriented migration, etc., of freshwater biota. On the basis of their ability to tolerate thermal variations, most freshwater organisms are **stenothermic** with a narrow range temperature tolerance, but some are **euthermic** with a wide range of temperature tolerance. For example, the stenothermic oligochaets includes **steno-minimothermal forms** (narrow range of temperature, e.g., *Aeolosoma*, *Megasclex mauritii*), **steno-maximothermal** (e.g., *Dero limosa*) or **steno-optimothermal forms** (e.g., *Branchiodrilus semperi* and *B. menoni*) (**Sitaramiah**, 1966).

3. Light

Light influences freshwater ecosystems greatly. The freshwaters often have a lot of suspended material. While affording protection to the light sensitive species, these substances more often obstruct the light that normally reaches the water. The degree of such obstruction of light influence the productivity of the freshwater ecosystems. A shallow lake receive light to its very bottom resulting in an abundant growth of vegetation both phytoplankton and rooted vascular plants. These plants in living or dead states form nice food for consumers of grazing food chain or organisms of detritus food chain, respectively. The running water contains little plant or animal plankton not due to the lack of sunlight

but because of the action of the currents in washing it away. Further, light controls the orientation and changes in position of attached species and their nature of growth and it also causes the diurnal migration of planktonic species of freshwater.

4. Oxygen

Chemically pure water is biologically uninhabitable and all freshwaters contains an array of chemical substances. The oxygen, which is a most essential chemical component of life processes, remain dissolved in freshwaters. The aquatic environments which remain in close proximity with atmosphere contain an abundance of oxygen, that reaches the water either by direct diffusion or by movements of water such as wave action or water circulation. Lotic (moving) water of streams and rivers often have a high percentage of oxygen. Aquatic plants supply water with oxygen that is formed as a product of photosynthesis. Rooted vegetation of shallow water zones and floating phytoplankton of open waters also produce oxygen. The amount of photosynthetically produced oxygen remain high at warm temperatures and at greater light intensities.

The oxygen level in a tropical pond exhibits diurnal variation—it remains at peak between 14.00 and 17.00 hours of day (Sreenivasan, 1964). Oxygen contents of a freshwater body are depleted in numerous ways. Primarily oxygen is utilised in the respiration of organisms and decomposition of dead organisms in the aquatic environment. While photosynthesis remains restricted to the surface layer of water containing phytoplankton and exposed regions of rooted vascular plants, respiration and decomposition occur at all levels. In stagnant pools with a lot of decaying vegetation oxygen content often reaches a stage of complete depletion. The reduction in dissolved oxygen is magnified by the release of many gases as end products of decomposition or by the mixing up of waters of low oxygen content reaching the habitat as an inflow. Aquatic animals with very few exception, *i.e.*, those that breath air, utilize the oxygen dissolved in water. Certain freshwater inhabitants such as many anaerobic bacteria and insect larvae of chironomids perform anaerobiosis and requires no oxygen.



Oxygen the most essential chemical component is dissolved in fresh waters.

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5. Carbon dioxide

Aquatic vegetation and phytoplankton require carbon dioxide for photosynthetic activity. The carbon dioxide of freshwater environments is produced as the end product of respiration and of decomposition. Carbon dioxide also diffuses directly from the atmosphere and is readily dissolved in water to result in carbonic acid (H_2CO_3) which affects the pH of water. It is also present in the freshwater as carbonates and bicarbonate of calcium, magnesium and other minerals. The growing plants and lime deposition bacteria and other animals may cause a depletion in carbon dioxide resources. Photosynthesis is the major cause for its drain. The high saturation levels of O_2 and CO_2 have been found to have toxic effects on aquatic biota.

6. Other Gases

Streams and lakes contaminated by sewage and stagnant pools with decaying vegetation show an abundance of the gas, hydrogen sulphide which is a decomposition product. This gas is highly toxic to living organisms and results into complete denudation of bottom fauna. Methane and carbon monoxide are other toxic gases which are the products of decomposition. Nitrogen, hydrogen, sulphur dioxide and ammonia are some of the other gases which are found dissolved in freshwaters.

7. Dissolved Salts and Salinity

Freshwater being efficient solvent contains many solutes in solution, but even then its salt contents remain under 1/5% than marine water which contains about thirty-five parts per thousand (%) dissolved salts (see **Clapham, Jr.**, 1973). Different dissolved salts reach the water by erosion, inflow and decay of aquatic forms. Dissolved substances have peculiar significance for floating aquatic vegetation and phytoplankton, since these organisms do not depend on the substratum for the supply of nutrients. Compounds of nitrogen, phosphorus and silicon are most important substances found dissolved in freshwater. Nitrate, nitrites and ammonium salts are essential for the food of aquatic vegetation such as algae and water weeds. Nitrate always remain available due to nitrogen cycle occurring between nitrogen fixing bacteria and nitrogen consuming plants. Ammonium salts in excess have a lethal effect on the fauna. Dissolved silicates of freshwater are readily utilized by diatoms and sponges in constructing their body structures such as shell in case of diatoms and spicules in case of sponges.

All freshwater environments also contain small amounts of phosphorus which more often acts as a limiting factor. Utilization of phosphorus by plankton during the periods of abundance may result in a total elimination of other plants that require the element (phosphorus). Many other elements such as calcium, magnesium, manganese, iron, sodium, potassium, sulphur, and zinc are found dissolved in water and influence the fauna variously. Iron being a growth promoting element for plants exists as the compound of oxygen (ferrous oxide) or sulphur (ferrous sulphide) in different freshwater bodies. Its influence is often modified by the pH of water. Calcium is an essential element for plants. The abundance and scarcity of carbonate of calcium determine the faunal composition. Deposition of calcium carbonate in water called **marl** is produced by the activity of plants. External coverings of arthropods and the shell of molluscs and tubes of some worms need calcium carbonate. Snails are found to develop a heavy shell if the waters in which they lived contained excess of calcium. Bryozoans, sponges, and cladocerans prefer an increased calcium content.

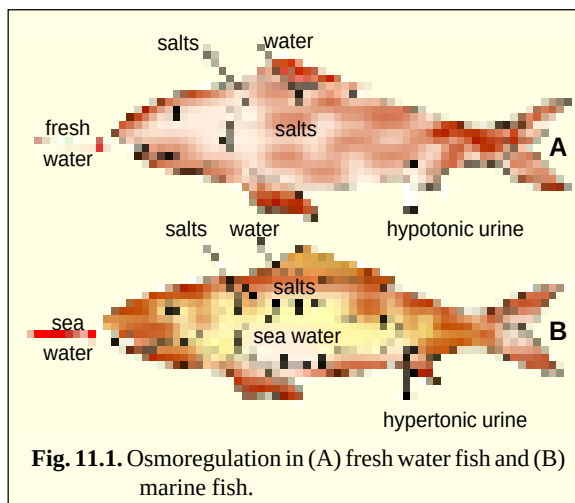


Fig. 11.1. Osmoregulation in (A) fresh water fish and (B) marine fish.

Due to low salinity of freshwater, animals face the problem of **osmoregulation**. Because the salt concentration of body fluids of animals remain higher than the freshwater, hence, the water continuously tend to enter the body which should be readily removed. Most aquatic animals (*e.g.*, Protozoa and fishes) have the means to excrete extra amount of water of body by osmoregulation. For this purpose Protozoa employ contractile vacuoles and other multicellular invertebrates and chordates use excretory organs, such as nephridia, kidney, etc.

8. pH or Hydrogen Ion Concentration

In freshwater environments pH is a determining factor for the biota by becoming a limiting factor. The pH value of different freshwater bodies may fluctuate seasonally and annually. The pH of surface waters and deeper waters exhibit marked differences. **Sreenivasan** (1968) has reported a marked pH variation of 2.2 units between surface water and deeper water in the Sandynulla reservoir in the Nilgiris (India). Though pH range is species specific, yet lower aquatic forms in general showed little reaction to alterations in pH, while higher aquatic organisms (*e.g.*, fishes) responded quickly to little pH variations.

LENTIC ECOSYSTEMS

Lentic ecosystems include all standing water (freshwater) habitats such as lakes, ponds, marshes, swamps, bogs, meadows, etc. Some of the important lentic ecosystems or wetlands have been tabulated in Table 11-1.

Table 11-1. Classification of wetlands or lentic ecosystems (excluding lakes and ponds) (Smith 1974).

A. Inland freshwater areas		
1.	Seasonally flooded	Soil covered with water or waterlogged during variable periods but well-drained during much of the growing season. In upland depressions and bottomlands. Bottomland hardwoods to herbaceous growth.
2.	Freshwater meadows	Without standing water during growing season ; water logged to within few inches of surface. Grasses, rushes, sedges, broadleaf plants.
3.	Shallow freshwater marshes	Soil waterlogged during growing season ; often covered with about 15 cm or more of water. Grasses, bulrushes, spike rushes, cattails, arrowhead, smartweed, pickerelweed. A major water-fowl-production area.
4.	Deep freshwater marshes	Soil covered with 15 cm to 91 cm of water. Cattails, reeds, bulrushes, spike rushes, wild rice. Principal duck-breeding area.
5.	Open freshwater marshes	Water less than 3 m deep. Bordered by emergent vegetation : pondweed, naiads, wild celery, water lily. Brooding, feeding, nesting areas of ducks.
6.	Shrub swamps	Soil waterlogged; often covered with 15 cm or more water. Alder, willow, buttonbush, dogwoods. Ducks nesting and feeding to limited extent.
7.	Wooded swamps	Soil waterlogged; often covered with 30 cm of water. Along sluggish streams, flat uplands shallowlakes, basins. North of USA : tamarack, arborvitae, spruce, red maple, silver maple. South of USA : water oak, overcup oak, tupelo, swamp black gum cypress.
8.	Bogs	Soil waterlogged; spongy covering of mosses. Heath shrubs, sphagnum, sedges.
B. Coastal freshwater areas		
9.	Shallow freshwater marsh	Soil waterlogged during growing season ; at high tides as much as 15 cm of water on landward side, deep marshes along tidal rivers, sounds, deltas. Grasses and sedges. Important water-fowl areas.
10.	Deep freshwater	At high tide covered with 15 cm to 1.8 m of water. Along tidal rivers and bays. Cattails, wild rice, giant cutgrass.
11.	Open freshwater	Shallow portions of open water along fresh tidal rivers and sounds. Vegetation scarce or absent. Important water-fowl areas.

1. Lakes and Ponds

Lakes are inland depressions containing standing water. They may vary in size from small ponds of less than a hectare to large seas covering thousands of square kilometers. They may range in depth from a few centimeters to over 1666 meters. **Ponds**, however, are considered as small bodies of standing water so shallow that rooted plants can grow over most of the bottom. Most ponds and lakes have outlet streams and both are more or less temporary features on the landscape because their filling is inevitable.

The aquatic habitats of lake and pond remain vertically stratified in relation to light intensity, wave length absorption, hydrostatic pressure, temperature, etc. In a lake, for example there are three well recognized horizontal strata namely ;

(i) Shallow water near the shore forms the **littoral zone**. It contains upper warm and oxygen rich circulating water layer which is called **epilimnion**. The littoral zone includes rooted vegetation.

(ii) **Sublittoral zone** extends from rooted vegetation to the non-circulating cold water with poor oxygen zone, i.e., **hypolimnion**.

(iii) **Limnetic zone** is the open water zone away from the shore. It is the zone upto the depth of effective light penetration where rate of photosynthesis is equal to the rate of respiration. (iv) **Profundal zone** is the deep-water area beneath limnetic zone and beyond the depth of effective light penetration.

(v) **Abyssal zone** is found only in deep lakes, since it begins at about 2,000 meters from the surface. Ponds have little vertical stratification. In them littoral zone is larger than the limnetic zone and profundal zone. In a small pond the limnetic and profundal zone are not found. Lakes, thus, differ from ponds in having relatively larger limnetic zone and profundal zone than littoral zone.

Further, lentic water of a lake or pond is also classified on the basis of the depth of light penetration enabling photosynthesis into **trophogenic zone** (includes littoral plus sublittoral zones), and a **tropholytic zone** (upper part of profundal zone). The former is often distinguished by abundant plant growth and dependent fauna, while the latter denotes a general absence of vegetation and harbours mostly saprobes. In between the two zones is the **compensation level** which forms a boundary between two zones. It exhibits perfect equilibrium between respiration and photosynthesis.

Physico-chemical properties of lakes and ponds. Lakes have the tendency to become thermally stratified during summer and winter to undergo definite seasonal

periodicity in depth distribution of heat and oxygen. Light too penetrates only to a certain depth, depending upon turbidity. These gradations of oxygen, light and temperature profoundly influence the life in the lake, its distribution and adaptation.

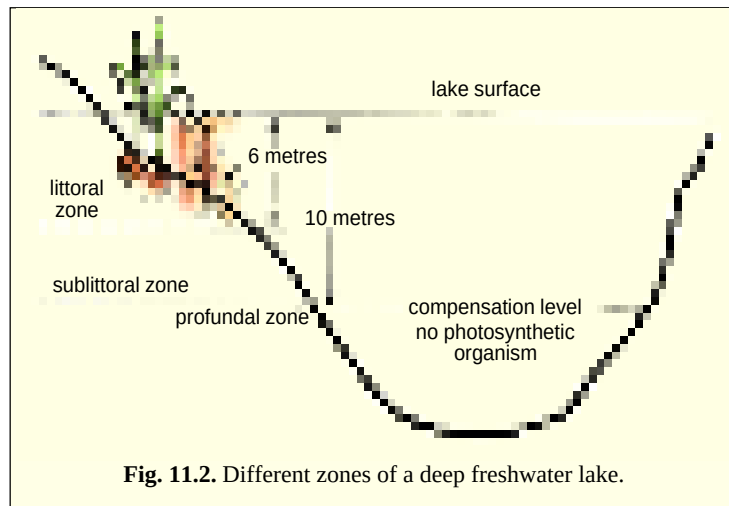


Fig. 11.2. Different zones of a deep freshwater lake.

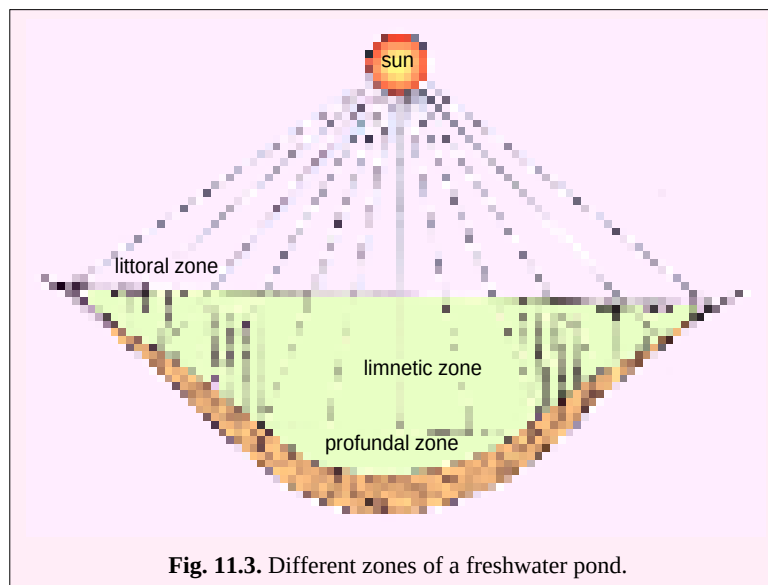


Fig. 11.3. Different zones of a freshwater pond.

Kinds of lakes. Different classifications of lakes on the basis of physical factors, productivity, etc., exist. Based on temperature **Hutchinson** (1957) classified lakes into **dimictic**, **monomictic** and **polymictic** lakes. The dimictic lakes exhibit two overturns every year, while monomictic lakes may be **cold monomictic** and **warm monomictic**, the former being characterized by a circulation only during summer, while the latter has a complete circulation in winter as well. Polymictic lakes present circulations throughout the year. Based on the humic acid contents the lakes of world have been classified into **clear water lakes** and **brown water lakes**, brown water contains high humus content. Clear water lakes may be divided into two types—the **oligotrophic type**, whose water is poor in nutritive plant material and shows nearly equal distribution of oxygen during summer and winter months and whose mud bottom contains little organic material; and the **eutrophic type**, which is rich in nutrients. At greater depth below the thermocline in summer, eutrophic lakes show a considerable reduction in oxygen content and their mud bottom is composed of typical muck. Eutrophic lakes are polluted lakes and support the growth of the microorganisms. With regards to above features, an intermediate type of lake is called **mesotrophic lake** (see **Dash**, 1993).

Special types of lakes. There also occur the following five types of lakes :

1. Dystrophic lakes. These lakes contain high concentration of humic acid in water. The margins of these lakes have a low pH and develop into peat bogs.

2. Deep ancient lakes. Lake Baikal in Russia is the most famous of ancient lakes. It is the deepest lake in the world. This lake is often nicknamed as “**Australia of Freshwater**”, since it contains various unique animals. It contains about 384 species of arthropods, 98 per cent of which are **endemic** (*i.e.*, found no where else). Likewise, 81 per cent of its 36 species of fish are also endemic.

3. Desert salt lakes. These lakes exist in sedimentary drainages in arid climates where evaporation exceeds precipitation, *e.g.*, Great Salt Lake of Utah, USA. Biotic community of these lakes comprises only those species which can tolerate high salinity, *e.g.*, *Artemia* (Brine shrimp).

4. Desert alkali lakes. These lakes occur in igneous drainages in arid climates, *e.g.*, Pyramid lake, Nevada, USA. High pH and concentration of carbonates are the characteristic features of these lakes.

5. Volcanic lakes. Acid or alkaline lakes associated with volcanic regions are called **volcanic lakes**, *e.g.*, Japanese and Philippine lakes. In these lakes exist the extreme chemical conditions.

Types of ponds. Ponds too are of the following four types :

(i) Flood plain ponds. These are ponds which are found in stream flood plains. These are quite productive habitats due to accumulation of organic matter in them.

(ii) Temporary ponds. These ponds remain dry for most part of the year but support a unique community. Organisms in such ponds must be able to survive in a dormant stage during dry periods or be able to move in and out of ponds.

(iii) Artificial ponds. These ponds are the result of damming of a stream or basin by man. They are artificially managed habitats.

(iv) Beaver ponds. Ponds constructed by beavers (=amphibious rodents) are called beaver ponds. The beaver often does not construct pond but live in holes in the bank of streams and, thus, becomes essentially the stream animals.

Origin of lakes and ponds. Lakes and ponds arise in many ways. For example, North American lakes were formed by glacial erosions and deposition and a combination of the two. Glacial abrasions of slopes in high mountain valleys engraved basins, which filled with water from rain and melting snow to produce tarns *i.e.*, small mountain lakes. Retreating valley glaciers left behind crescent-shaped ridges of rock debris, which dammed up water behind them. Lakes are also formed by the deposition of silt, driftwood, and other debris in the beds of slow-flowing streams. Craters of extinct volcanos may fill with water and landslides can block off streams and valleys to form new lakes and ponds. Further man intentionally creates artificial lakes by damming rivers and streams for power, irrigation, and water

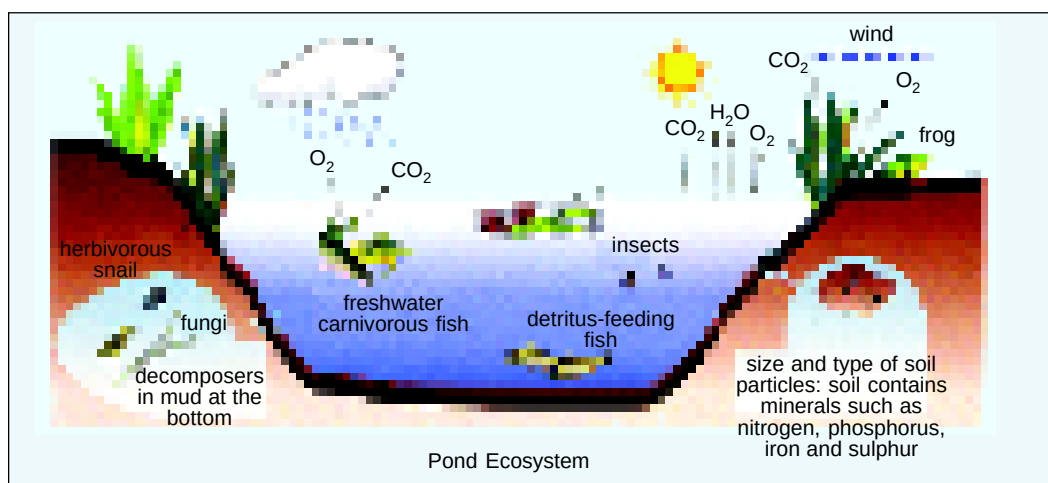
storage, or by constructing small ponds and marshes for water, fishing and wild life. Man made lakes are often called **impoundments**. They may be a 'close' type or 'open' type depending on the discharge of water into it. Sandynulla is a closed impoundment, while Pykara is an open impoundment (**Sreenivasan**, 1957). Impoundments differ from natural lakes in thermal and oxygen variations and are often characterized by a low percentage of bottom dwellers or benthos.

Ponds occur in most regions of adequate rainfall. Temporary ponds, which are often shallow and quick drying contain a characteristic lentic environment. Permanent ponds contain water throughout the year, while flood-plain ponds are formed when a stream shifts its position, leaving the formed bed isolated as a body of standing water.

Biotic Communities of Lakes and Ponds

Different organisms of the lentic environment can be ecologically classified based on whether they are dependent on the substratum or free from it. Organisms depending on the substratum are called **pedonic forms** and those that are free from it are the **limnetic forms**. Further, the aquatic organisms may also be classified into following groups depending upon their sizes and habits :

1. Neuston. These are unattached organisms that live at the air-water interface. They may include floating plants such as duckweed, as well as many types of animals. Animals that spend their lives on top of the air-water interface, such as water striders, are termed **epineuston**, while others, including



insects such as diving beetles and back swimmers, which spend most of their time on the underside of the air-water interface and obtain much of their food from within water, are termed **hyponeuston**.

2. Plankton. These are forms which are found in all aquatic ecosystems except for fast-moving rivers. They are small plants and animals whose powers of self-locomotion are so limited that they cannot overcome currents. Thus, their distribution is controlled largely by the currents in their ecosystems. Most planktons (phytoplanktons and zooplanktons) can move a bit, however, either, to control their vertical distribution or to seize prey. Certain animal planktons or zooplanktons are exceedingly active and move relatively great distances considering their small size, but they are so small that their range is still controlled largely by currents and such planktons are also known as **nekto plankton**.

3. Nekton. Nektonic animals are swimmers and are found in all aquatic systems except for fastest-moving rivers. In order to overcome currents, these animals are relatively large and powerful; they range in size from the swimming insects of quiet water, which may be only about 2 mm long, to the largest animal that has ever lived on earth, the blue whale.

4. Benthos. The benthos include the organisms living at the bottom of the water mass. They occur virtually in all aquatic ecosystems. The benthos organisms living above the sediment-water interface are termed **benthic epifauna** and those living in the sediment itself are termed **infauna**.

Neustons of lentic aquatic environment. The water surface of lake or pond contains certain free-floating hydrophytes such as *Wolffia*, *Lemna*, *Spirodella*, *Azolla*, *Salvinia*, *Pistia* and *Eichhornia*. These plants remain in contact with water and air, but not soil.

Biota of littoral zone. Lentic aquatic life is most prolific in the littoral zone. The littoral zone of a lake remains rich in pedonic flora especially up to the depth to which effective light penetration is possible facilitating the growth of rooted vegetation. At the shore proper are the **emergent vegetation** which remain firmly rooted in the shore substratum but their tops with thin chlorophyll bearing regions are exposed. Certain rooted emergent plant species of littoral zone are *Ranunculus*, *Monochoria*, *Cyperus* and *Rumex*. Interspersed with these plants are the cattails (*Typha*), bulrushes (*Scirpus*), arrowheads (*Sagittaria*) and pickerel-weeds. Slightly deeper are the rooted plants with floating leaves such as the water lilies—*Nymphaea*, *Nelumbo*, *Aponogeton*, *Trapa*, and *Marsilea*. Still deeper in the littoral zone of lake are the agile thin-stemmed water weeds, rooted but totally submerged. Such a submerged vegetation includes plants such as *Elodea*, *Vallisneria*, *Myriophyllum*, *Isoetes*, *Hydrilla*, *Chara*, *Potamogeton*, etc.

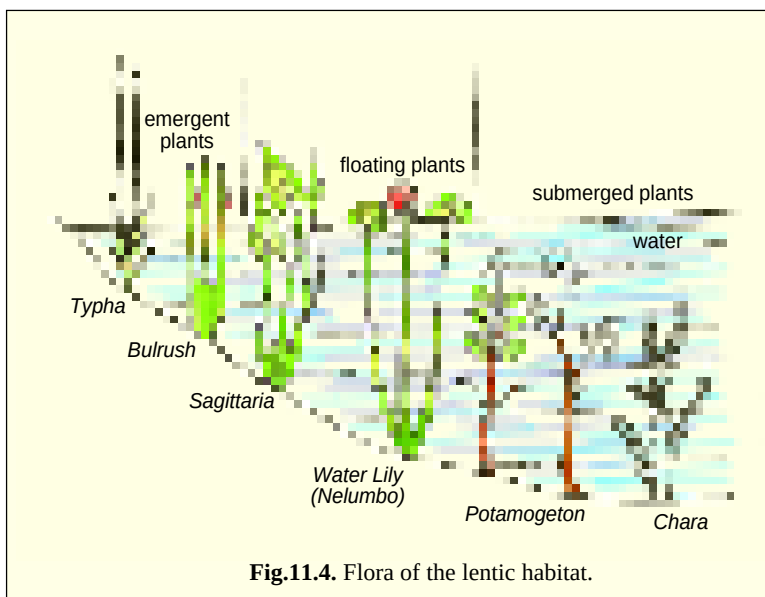


Fig.11.4. Flora of the lentic habitat.

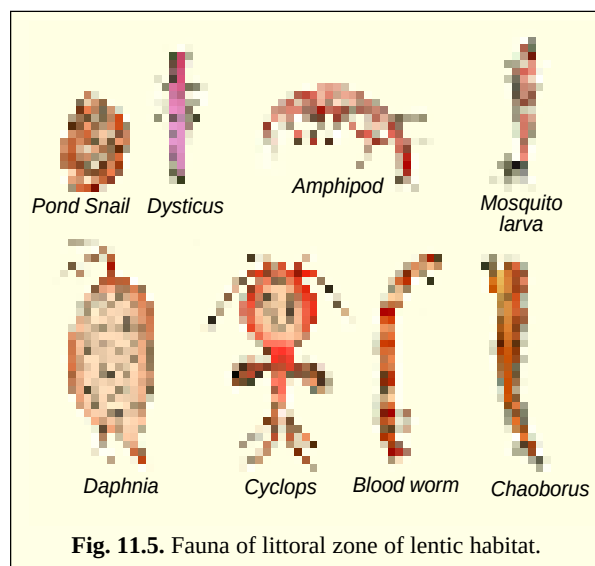


Fig. 11.5. Fauna of littoral zone of lentic habitat.

The phytoplankton of littoral zone of lake is chiefly composed of diatoms (e.g., *Navicula*, *Cyclotella*), blue green algae (e.g., *Microcystis*, *Oscillatoria*), green algae (e.g., *Cosmarium*, *Staurastrum*) and holophytic flagellates. **Sreenivasan** (1970) has observed the abundance of marine dinophycean flagellate *Ceratium hirudinella* in some impoundments of India.

The littoral zone also contains great concentration of animals, which remain distributed in recognizable communities. In or on the bottom are various dragonfly nymphs, crayfish, isopods, worms, snails and clams. Other animals live in or on plants and other objects projecting up from the bottom. These include protozoans such as *Vorticella*, *Stentor*; larvae of *Dytiscus*, *Laccotrophes*, leach-like *Glossophonia*,

climbing dragonfly, damsel fly nymphs, rotifers (*Brachionus*), flatworms, Bryozoa, Hydra, snails (*Lymnaea*) and other. With the exception of a few rotifers such as *Keratella*, *Brachionus*, *Diurella*, and *Trichocerca* and crustaceans such as *Cyclops*, *Daphnia* and *Moina*, most freshwater animals of lentic habitat depend in one way or other on the aquatic vegetation. Larvae of *Chironomus* are found underneath the leaves of floating plants. The zooplankton of littoral zone consists of water fleas such as *Daphnia*, rotifers and ostracods. The free swimming fauna (Nekton) includes *Paramecium*, *Euglena*, *Ranatra*, *Corixa*, *Dytiscus*, larvae of *Culex* and of *Chaoborus*, *Gerris*, *Gyrinus*, etc. Among the vertebrates frogs, salamanders, snakes and turtles are the nektons of littoral zone. Floating members of the community (neuston) are whirligig beetles, water spiders and numerous Protozoa. Many pond fishes such as sun-fish, top minnows, bass, pike and gar spend much of their time in the littoral zone.

Biota of limnetic zone. Limnetic zone is the region of rapid variation, with the water level, temperature and oxygen content varying from time to time. In certain Himalayan lakes which have a glacial origin, change in water level is a conspicuous phenomenon, the onset of summer bringing forth a rapid and sudden rise in water level. Various protozoans which are capable of encystment during adverse ecological conditions, tardigrades such as *Macrobiotus*, rotifers such as *Rotaria*, *Philodina*, copepods, snails and frogs occupy the limnetic zone. Sedentary and slow moving forms are excluded from this zone because of predators and lack of permanent substratum for attachment. Many microscopic plants (*Volvox*, *Euglena*) and fishes also occupy this zone. The limnetic zone has autotrophs in abundance.

Biota of profundal zone. The deep (profundal) life consists of bacteria, fungi, clams, blood worms (larvae of midges), annelids and other small animals capable of surviving in a region of little light and low oxygen. In the profundal zone, autotrophs cannot produce food, and the main source of energy is detritus that rains out of the limnetic zone. All the organisms of this zone are heterotrophs, either as detritus feeders or as carnivores. Most of our largest lake fishes inhabit the dark waters of hypolimnion for most of their lives. In addition to detritus from the limnetic zone, food washed into lakes by rivers can settle out and serve as food for profundal animals.

Benthos of lake bottom. The lake bottom in young lakes may be of the original rock; in older lakes, it will have been covered with sediment to form a uniform substrate of mud or sand. The benthos community includes several species of insect larvae, including those of small mosquito such as midges, burrowing mayflies, clams, snails, and tubeworms.

Distribution of Oxygen and Dissolved Nutrients in Lakes

The productivity of a lake depends on the amounts of oxygen and available nutrients dissolved in the water. As a general rule, the littoral zone behaves as a normal ecosystem, with oxygen being produced by photosynthesis within the community, as well as being constantly mixed into the water at the surface. Productivity within the littoral zone is directly controlled by the nutrients cycling within the littoral zone ecosystem and biogeochemical cycles apparently operate as do in terrestrial ecosystems, with sediments serving as a storage place for inorganic nutrients such as phosphate, and some rooted plants being able to draw nutrients out of the sediments through their roots (**McRoy and Barsdate**, 1970). But in the pelagic zone (limnetic and profundal zones), the distribution of oxygen and nutrients is exceedingly uneven, and this has a profound effect on the behaviour of the lake as a whole, especially in those whose pelagic zone is much larger than the littoral.

The epilimnion of a lake is typically well oxygenated, both because of the production of photosynthetic oxygen in the epilimnion and because of the oxygen uptake at the surface. The amount of nutrients dissolved in the water is highly variable, mainly because most of the nutrients in the epilimnion have been incorporated into living tissue. Thus, production of epilimnion is controlled by amount of available nutrients, which, in turn is controlled primarily by the rate of cycling within the epilimnion, the rate of loss to the hypolimnion of nutrient-rich detritus, and the rate of return of soluble nutrients from the hypolimnion during overturn. The fact that only heterotrophic metabolism proceeds in the hypolimnion means that the level of production is determined by the amount of detritus raining into it. As the decomposition of detritus releases soluble nutrients, therefore, there is an inverse relationship between oxygen concentration and concentration of vital nutrients. It is paradoxical that

the greater the productivity of a lake the lower the oxygen concentration in its hypolimnion. For example, oligotrophic lake which is virtually sterile biologically has low production but its water remain well saturated with oxygen from top to bottom (**orthograde oxygen curve**). Likewise, a eutrophic lake has high productivity, but there is substantial oxygen depletion from top of the hypolimnion to the bottom, with depletion most extreme in the bottom layers (**clinograde oxygen curve**). However, in most lakes, oxygen is thoroughly mixed and is essentially at saturation during overturn.

LOTIC ECOSYSTEMS

Running-or moving-water or lotic ecosystems include rivers, streams, and related environments. They are remarkably variable, ranging in size from Ganga, Yamuna, Hindon, Kali Nadi, Sutlaj, Gomti, Brahmaputra, Narmada, Mahanadi, Kaveri, Krishna, Godavari, etc., to the trickle of a small spring. They vary from raging torrents and waterfalls to rivers whose flow is so smooth as to be almost unnoticeable. A given river varies considerably over its length, as it changes from a mountain brook to a large river. This is most noticeable in the abiotic factors of the environment, but all features of the ecosystem vary in response.

Characteristics of Lotic Environment

Moving waters differ in the three major aspects from lakes and ponds : current is a controlling and limiting factor; land-water interchange is great because of the small size and depth of moving water systems as compared with lakes and oxygen is almost always in abundant supply except when there is pollution. Temperature extremes tend to be greater than in standing water. Besides these, the most distinctive features of moving water ecosystems are those related to their motion, *i.e.*, the **rate of flow** and the **stream velocity**. The rate of flow or discharge refers to the volume water passing a given observation point during a specific unit of time; it is measured in units such as m³/sec., ft³/sec., or acre-feet/sec., one acre-foot is equivalent to a volume of water 1 acre in area by 1 foot deep). It tends to increase steadily growing down stream, as tributaries join with the main river. The **velocity of flow** is the speed at which the water moves, and is measured in m./sec., ft./sec., or mi./hr. Velocity is variable, but it also tends to increase downstream with increasing discharge. A factor that may be even more important to the biotic community is the **turbulence**, or the irregularity of the motion of the water particles. Perfectly even flow, in which water particles move parallel to one another, is called **laminar flow**. It is contrasted with **turbulent flow**, in which the movement of water particles is highly irregular. In highly turbulent water the erosive power is great, the sheer forces at the water-sediment interface at the bottom of the river are powerful, and the amount of oxygen incorporated into the water is very high. In streams with laminar flow the erosive power is lower, the sheer forces at the bottom of the stream are lower, and relatively less oxygen is incorporated into the water than turbulent waters.

The moving-water ecosystems can be divided into following two ecosystems depending upon the velocity and rate of flow of water body :

1. Rapidly flowing water and 2. Slowly flowing water.

A. Rapidly Flowing Water

Rapidly flowing water (of fast moving torrential streams, etc.) can be defined as the portion of the stream in which the flow is both rapid and turbulent. Everything that is not attached or weighted is swept away by the current ; this includes organisms and sediment particles alike. The substrate tends to be rock or gravel, and the fragments are rounded and smoothed by the water. The habitat itself is an extremely diverse one. Physical parameters such as sheer force and rate of water movement tend to be quite different on top of a rock fragment, between rock fragments, or beneath rock fragments, and different species can exploit these differences in microhabitats. Aquatic plants, in addition, provide microhabitats for some torrential animals. The rarely occurring phytoplankton of **rapid** or **riffle zone** of stream includes diatoms, blue-green and green algae (*e.g.*, *Cladophora*, *Ulothrix*) and water moss (*e.g.*, *Fontinalis*).

1. Plant inhabiting torrential forms. Some animals of rapidly flowing water streams live among the mosses and flowering plants such as *Eriocaulon miserum*, *Hydrolyum lichenoides*, and *Duroea wallichii*. Animals inhabiting these plants have a torpedo-like body which enables them to offer minimum resistance to the current. Some animals such as funnel mouthed tadpoles of *Megalophrys* and tipulid larvae live entangled in the roots of these plants. They possess devices for fixation and have hydrostatic organs.

Some animals such as *Cephalopteryx*, *Helodes*, *Phalacroceras*, *Gammarus*, etc., live among leaves and stems of aquatic plants. All these animals possess hook-like structures on their body helping them in anchoring on leaves and stems. The larvae of simuliids and chironomids, however, live on the exposed surfaces of plants.

2. Rock inhabitant forms. The animals that live on the tops of exposed rocks have an efficient mechanism for staying in one place, otherwise, they will be swept away. The organisms in this microhabitat are universally flattened. Some, like the fresh-water limpet, *Ferrissia*, are virtually flat, offering little resistance to the current they hold themselves in place with a very large powerful foot that extends over almost the entire area of the shell. Others, such as water penny, the larvae of the riffle beetle, and *Baetis* larvae, are not only almost flat, but each of their legs possesses hooked claws that enable them to hold onto the substrate more firmly. Larvae of *Simulium* and chironomids cling to rock top by grappling hooks at the posterior end of the body. Caddisfly worms contain both of these structures (claws on legs and hooks at posterior end of body). The mayfly nymphs *Iron* and *Psephenus* attach themselves to the rocks by means of functional pads. Some animals, including freshwater sponges, actually cement themselves to the surface of the rocks. Others, such as caddis flies, build “houses” out of sediment or wood fragments, which are then cemented firmly to the rocks. Exposed surfaces of rocks also contain few sessile algae and few plants. A significant amount of organic detritus is washed into rapid-water ecosystems from up stream and adjacent terrestrial areas, however. This is the source of the most of energy in fast water ecosystems, and is much more important as an energy source than primary production within the stream itself. In consequence, the majority of primary consumers of such a microhabitat are detritus-eaters.

3. Inhabitant forms of spaces between rocks. Numerous different kinds of animals live in the spaces between the rocks. Many of them such as mayfly and stonefly naiads, are flattened and have certain behavioural adaptations to hold them in place. These include **thigmotaxis**, by which they cling instinctively to any hard surface such as rock or another insect larva. **Rheotaxis**, by which the animals orient themselves to face into current and move upstream is another adaptation of this sort. In many of these naiads, the combination of the organism’s shape and behaviour is such that the current presses the insect tightly against the rock, increasing the friction between the animal and its substrate. Other insect larvae, such as the hellgramite, are large and covered with spines. Their size make it somewhat harder for the current to sweep them away, while the spines help in holding the larvae in places between stones.

4. Inhabitants occurring beneath rocks. Many species such as flatworms, annelid worms, other insect larvae, clams and some species of snails, live beneath the rocks. The current is weaker here, and animals are less likely to be carried away. They lack any special adaptation.

Finally, if the current is sufficiently slow, certain swimming organisms such as fish, will be present. The fish of fast-water ecosystems tend to be stenothermal cold-water fish, such as trout. The fishes of fast-water ecosystems have small body size and streamlined body. Trout and masheer are capable of moving against the current or waterfall by muscular effort. Majority of torrential fishes, however, live on or among rocks and boulders, over which water flows very swiftly. Small loaches (*Noemacheilus*) or loach-like fishes (*Ambyceps olya*) are met with at the bottom and limpet-like fishes (e.g., *Glyptosternum*, *Balitora*) or fishes with special modifications (*Garra*, *Glyptosternum*, *Balitora*) occur on the rocks. The loaches and carps adhere to rocks like the limpets, with the ventral surface, inclusive of paired fins forming one large sucker, e.g., *Gastromyzon* of Borneo, the Indian loach *Baletora* of the eastern Himalayas and Assam hills (Hora, 1947)

Physico-chemically, the fast-water ecosystems resemble with cold, deep lakes. Water temperatures tend to be quite low, productivity is also quite low, and diversity is high. In the fast-water ecosystem, the main control of productivity is the current, which seriously limits the amount and type of autotrophs production that can take place.

B. Slowly Flowing Water

A slowly flowing water ecosystem is a very different type of system from the fast streams. Because the flow is both slower and more likely to be laminar, the erosive power of the stream is greatly reduced, and smaller sediment particles (silt) and decaying organic debris, instead of being carried away by the stream, are deposited on the bottom. In addition to these the slow streams have a higher temperature, consequently, planktonic organisms, especially protozoans, occur in large numbers in this ecosystem. The detritus-feeding benthos of slow water ecosystems include those which either live on the bottom, such as isopods (sowbugs), molluscs (*Sphaenium*, *Pisidium*, and *Anodonta dominata*), and mayfly and damselfly naiads, or which burrow into the sediment, such as tubeworms, naiads of the burrowing mayflies, *Sialis* (alderfly), midge *Chironomus*, and several other insect larvae, as well as, clams, nematodes, snails and rotifers. Swimming organisms are also abundant, including not only fishes, such as carps, cat fishes, suckers, stingers, spoonbills, etc., which tend to be different species from those of fast-water areas, but also larger Crustacea such as amphipods (fast-water shrimps). Finally, several insects spend most of their time at the surface of the stream. These include forms such as water striders, water boatmen, backswimmers, and predaceous diving beetles. Zooplankton is abundant, including a rich assemblage of Protozoa and smaller Crustacea such as Cladocera (water-fleas) and copepods. The richness of food afforded by slow streams attract a large number of reptiles (water-snakes, crocodiles, turtles) and amphibians.

Plant life is also abundant in a slow-water ecosystem. It includes rooted vascular plants such as pond weeds and grasses, firmly attached aquatic mosses and large multicellular filamentous algae. Motile algae such as diatoms and flagellates may abound in the open water. In all, the primary productivity of slow-water ecosystem is higher than that of the rapid-water ecosystem, and the community is relatively less dependent on food materials from outside. Further, the detritus food chain consisting of organisms such as bacteria and fungi is much better developed in this community as the partially decomposed organic debris that comprises the main food source for these organisms collects in the mud bottom. In some slow-moving streams, in fact, the bottom muds contain more organic material than mineral fragments.

In slow-water streams oxygen concentration remains main limiting factor. The high level of animal activity coupled with an active detritus food chain, can withdraw a large amount of oxygen from water of a slowly moving stream. In addition, the low level of turbulence means that less oxygen is incorporated into the water at the surface. Thus, the dissolved oxygen content of a slowly moving stream is likely to be much lower than that of a fast-moving stream. This is commonly reflected in the fauna of the stream. Fishes tolerant of low oxygen levels such as carps and catfishes are the most common fishes in slow water, while species with high oxygen demands, such as trout are found in fast water.

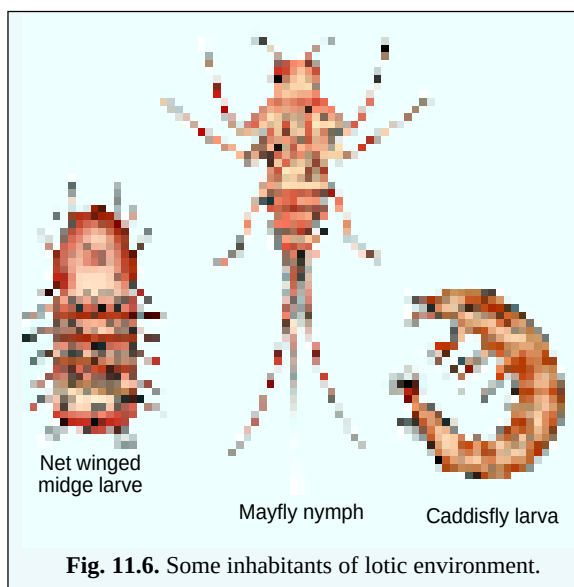


Fig. 11.6. Some inhabitants of lotic environment.

REVISION QUESTIONS

1. Classify the freshwater environment. Describe lentic ecosystem with suitable example.
2. What is lotic habitat ? Describe its physico-chemical characteristics and flora and fauna.
3. Write short notes on the following :
(a) Limnology ; (b) Lakes ; (c) Biotic zonation of a deep lake ; (d) Thermal stratification of lake.

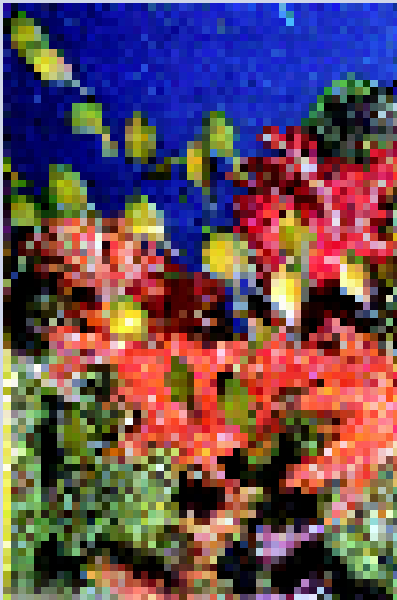
CHAPTER

12

Aquatic Ecosystems : Estuaries and Marine Communities

ESTUARINE ECOLOGY

Waters of all streams and rivers ultimately drain into the sea ; the place where this freshwater joins the salt water is called an **estuary**. An estuary (L. *aestus*—tide) is the wide lower tidal part of a river which is strongly affected by tidal action. Estuaries are the transitional zones (ecotones) between the sea and rivers and are the sites of unique ecological properties possessing a characteristic biological make up. They are semienclosed coastal bodies of water that have a free connection with the open sea and within which sea water is measurably diluted with freshwater from rivers. Examples of estuaries are river mouths, coastal bays, tidal marshes and water bodies behind barrier beaches. Not all rivers open into estuaries : some simply discharge their runoff into the ocean. Estuaries differ in size, shape and volume of water flow, all influenced by the geology of the region in which



Habitat of marine fish.

they occur. As the river reaches the encroaching sea, the stream carried sediments are dropped in the quite water. These accumulate to form **deltas** in the upper reaches of the mouth and shorten the estuary. When silt and mud accumulations become high enough to be exposed at low tide, **tidal flats** develop, which divide and braid the original channel of the estuary. At the same time, ocean currents and tides erode the coast line and deposit material on the seaward side of the estuary, also shortening the mouth. If more material is deposited than is carried away, then **barrier beaches**, **island** and **brackish lagoons** appear.

Type of Estuaries

Estuaries are classified in various ways depending on diverse criteria such as geomorphology, water circulation and stratification and systems energetics. Thus, on the basis of geomorphology, estuaries include (1) drowned river valleys, (2) ford-type estuaries, (3) bar-built estuaries, (4) estuaries produced by tectonic processes, (5) river delta estuaries, etc. Based on water circulation and stratification, estuaries are classified as (i) highly stratified or salt-wedge estuaries, (ii) partially mixed stratified estuaries, (iii) completely mixed or vertically homogeneous estuaries and (iv) hypersaline estuaries. On the basis of the ecosystem energetics, the estuaries have been classified by **Odum et al.**, (1969) as (1) physically stressed systems of wide latitudinal range, (2) natural arctic ecosystems with ice stress, (3) natural temperate ecosystem with seasonal programming, (4) natural tropical coastal ecosystems of high diversity, and (5) emerging new systems associated with man.

Physico-Chemical Aspects of Estuaries

Current and **salinity**, both very complex and variable, shape life in the estuary, where the environment is neither freshwater nor salt. Estuarine currents result from the interaction of a one-direction stream flow, which varies with the season and rain fall, with oscillating ocean tides, and the wind. The salinity of estuaries varies vertically and horizontally and fluctuates amazingly between 0.5–0.35 per cent. Due to low salinity freshwater has a lower density than sea water (± 1.00 , as opposed to 1.03). Were there no tides in an estuary to mix fresh and salt-water the lighter freshwater would simply flow over the heavier sea water and dissipate in the ocean. However, the tidal action acts as a plunger to thoroughly flush the estuary and mix the fresh and salt water. This has several results. The water level in the estuary fluctuates regularly, unlike that of river. For this reason the habitat that is covered at high tide and uncovered at low tide is a prominent one in the estuary, and has no analogue in any purely freshwater ecosystem. Further, the salinity is exceedingly variable and may change by a factor of ten over the course of a day at any location. At low tide, most of water passing through the estuary is fresh-river water, and the salinity is correspondingly low. At high tide, most of the water may be of marine origin, and the salinity correspondingly high. Flow may be somewhat stratified, or mixing may be incomplete, and the salinity at any time may vary considerably from one location to another. The intensity of current, and, hence, of the degree of mixing, is a function of the intensity of the tides and of the rivers flow rate. In other words, an estuary is an exceedingly variable environment. Further, salinity of estuaries remains highest during the summer and during periods of drought, when less freshwater flows into the estuary. It is lowest during the winter and spring, when rivers and streams are discharging their peak loads. No stenohaline organism (such as Echinodermata, Cephalopoda, and other molluscs) could hope to survive in the estuaries. Likewise, no organism that could not tolerate strong currents and the turbid water that results from strong currents could live there.

The temperatures in estuaries fluctuate considerably diurnally and seasonally. Waters are heated by solar radiation and inflowing and tidal currents. High tide on the mud flats may heat or cool the water, depending on the season. The upper layer of estuarine water may be cooler in winter and warmer in summer than the bottom, a condition that, as in a lake, will result in a spring and autumn overturn.

Anyhow, all estuaries have high productivity. As estuary's high productivity is exceeded among aquatic communities only by coral reefs. The reason is that although an estuary may be a harsh

environment in some ways it is not in others. Because the typical estuary is shallow and turbulent, the amount of dissolved oxygen tends to be fairly high although it may be low in bottom layers of those estuaries where water does not mix from top to bottom. More important, the tidal action acts to accumulate (concentrate) the nutrient and energy materials that wash in from upstream or, in some cases, that enter from the nutrient rich bottom waters of the sea.

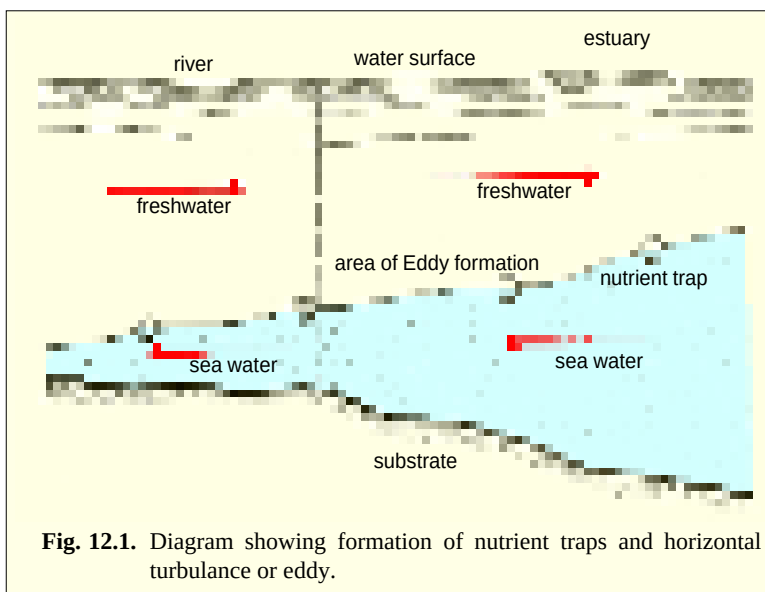


Fig. 12.1. Diagram showing formation of nutrient traps and horizontal turbulence or eddy.

The mechanism of nutrient concentration is very simple—particulate nutrient material enters the estuary at its upper end is carried seaward by the falling tide and is brought back through the estuary by the rising tide, and so on for several cycles. The length of time it takes from a nutrient particle to traverse the estuary is substantially greater than it would take for it to traverse a similar length of even the most slowly flowing river. Thus, the estuary acts as a nutrient trap, with an average nutrient level significantly higher than either the river or the sea that it connects. Likewise, the concentration of energy-rich organic materials remain high in estuaries.

The primary result of the concentration of nutrients and fixed carbon is a very high level of production within the detritus food chain. This has two aspects. First the nutrient material is broken down bacterially at a very high rate, and recycled back into soluble form. This allows a very high rate of gross primary production for estuarine plants. More important, perhaps, the amount of nutrient-rich organic detritus allows a level of productivity for detritus-eating animals much higher than could be maintained on the basis of the primary production by estuarine plants.

Biotic Communities of Estuaries

Carikker (1967) has classified the regions of estuaries into the **head**, where freshwater enters the estuary, **upper**, **middle** and **lower reaches** with increasing range of salinities and the **mouth** with salinity nearly equal to the sea. He has also classified the animals inhabiting the estuarine region into **oligohaline** (0.5 to 5 per cent), **mesohaline** (5 to 18 per cent) in the upper reaches, **polyhaline** (18 to 25 per cent) in the lower reaches, 25 to 30 per cent in the middle and **euhaline** (30 to 50 per cent) in the mouth of the estuary. Communities of estuaries are a mixture of endemic species and those which come in from sea. **Krishnamoorthy** (1963) in fact reported that the extent of penetration of the polychaets, *Marphysa gravelyi*, *Diopatra uariabilis*, *Clymene insecta*, *Loimia medusa*, *Glycera embranchiata*, *Onuphis eremita* into the Adyar estuary in Madras from the Bay of Bengal vary with varying salinities at constant temperatures.

Furthermore a estuary being a transitional zone between the freshwaters and seas is an ecotone and, therefore, typical estuarine forms are unique in their habitat. Thus, the estuarine community is a mixture of three components: the marine, the freshwater and the brackish water. However, the diversity of both freshwater and marine components reaches a maximum. Total diversity of the estuarine community is lower than that of either of the more normal environments. The number of interactions between species is not high and some estuarine populations may fluctuate greatly in size. The plant of

the estuary are of four basic sorts : phytoplankton, marginal and marsh vegetation, mud-flat algae, and epiphytic plants-growing on the marginal marsh vegetation. Because of the turbid water found in estuaries, phytoplanktons are normally uncommon. However, in Hooghly-Matla estuarine system in West Bengal, **Gopalakrishnan** (1971) has reported an abundance of phytoplanktonic forms—several species of diatoms, *Synedra*, *Navicula*, *Rhizostoma*, *Fragilaria*, *Asteriobella*, *Biddulphia*, *Planktoniella*, *Hemidiscus*, *Chaetoceros*, *Cyclotella*, *Stephanodiscus*, *Triceratium*; several species of green algae such as *Pediastrum*, *Spirogyra*, *Eudornia*, *Tribonema*, *Closterium*, *Zygonema*, *Pandorina*, *Volvox*, *Chlorella*; and blue-green algae such as *Microcystis*, *Oscillatoria*, *Anabaena*, and *Trichodesmus*. Most estuarine algae are of marine origin. Further, the most significant estuarine plants are marsh grasses such as *Spartina*, *Salicornia*, and *Scirpus*, as well as some submerged filamentous algae such as *Cladophora*, *Chara* and *Enteromorpha*. Very few animals feed on these plants directly, but a very large amount is consumed as detritus.

The estuarine animal communities include zooplanktons and other animals. For example, the planktonic animal forms of Hoogly-Matla estuary of India include flagellate protozoans such as *Euglena*, *Ceratium*, *Peridinium*, *Noctiluca*; other protozoans such as *Diffugia*, *Arcella*, *Vorticella*; rotifer species such as *Branchionus*, *Keratella*, *Asplanchna*, etc.; copepods such as *Diaptomus*, *Pseudodiaptomus*, *Cyclops* and *Paracalanus*; cladocerans of the genus *Bosmina*, *Bosminopsis*, *Ceriodaphnia*, *Moina*, besides a number of isopods (**Gopalakrishnan**, 1971). All these forms remain confined to water and lead a pelagic existence. The best known estuarine animals are detritus feeders such as oysters (*Ostrea* sp., of Lake Chilka), clams, lobsters, and crabs. Several insect larvae, annelid worms, molluscs, enter the estuary from the freshwater, while most marine phytoplankton, Crustacea, annelid worms, anemones and Bryozoa enter an estuary from marine ecosystems. For example, out of 130 species of fishes and 30 species of prawns of Hoogly-Matla estuary, several fishes such as *Hilsa*, *Harporodon*, *Mugil*, *Trachyurus*, etc., migrate from sea in to the mouth of the estuary, to form important fish catches, while some such as palaemonid prawns are freshwater inhabitants and come to live in estuaries.

Adyar estuary of Madras coast is found rich in invertebrate and vertebrate fauna (**Aiyar and Panikkar**, 1937). The common invertebrates of this estuary are sea anemones (e.g., *Phytocoetes gangeticus*, *Phytocoeteopsis ramunnii*, *Stephensonaetis ornata*, *Pelocoetes exul*, *Bolocractis gopalai*, etc.); hermaphrodite nereids (e.g., *Lycastis indica*); the tubicolous polychaet *Diopatra variabilis*; the burrowing polychaet *Marphysa gravelyi*, the oligochaet *Pontodrilus bermudensis*; the mysids *Rhopalophthalmus egregius* and *Mesopodopsis orientalis*; the crabs *Uca annulipes*, *Uca triangularis*, *Neptunus pelagius*, *Scylla serrata*, *Veruna litterata*, *Sesarma tetragonum*; the hermit crabs — *Clibanarius olivaceus* and *Clibanarius padavensis*, and several molluscs (e.g., *Ostraea arkanensis*, *Modiolus undulatus*, *Meretrix casta*, *Cuspidaria cochinchensis*, *Stenothyria blanfordiana*, *Amnicola stenothyroides*, *Potamides cingulatus*, *Styliger gopalai* and *Bursatella leachii*). The vertebrate fauna of Adyar estuary includes the fishes such as *Etroplus maculatus*, *Etroplus suratensis*, *Acentrogobius viridipunctatus*, *Acentrogobius neilli*, *Panchax parvus*, *Aplocheilus melastigma*, and mud skipper *Periophthalmus koelreuteri* and snakes such as *Natrix piscator* and *Cerberus rhyncops*.

Subsystems of Estuaries

An estuary consists of several basic subsystems linked together by the ebb and flow of water that is driven by the hydrological cycle and the tidal cycle. Some of its main subsystems are the following:

1. Shallow water production zones. In these zones, the rate of primary production exceeds the rate of community respirations. The producers are reefs, seaweed or sea grass beds, algal mats and salt marshes. This system exports energy and nutrients to deeper waters of the estuary and adjacent coastal shelf. Being rich in diverse types of producers these remain “programmed” for virtually year-round photosynthesis and derive benefit from tidal action in creating a subsidised fluctuating water level ecosystem. They have all the three types of producers such as **macrophytes** (e.g., see weeds, sea grasses and marsh grasses), **benthic microphytes** and **phytoplanktons**. Certain significant macrophytes are *Spartina*, *Zostera* and *Thalassia*. Benthic algae grow on macrophytes, sessile animals,

rocks, sand and mud. Some estuaries have distinct blooms such as **red tides** of large blooms of red pigmented dinoflagellates (*i.e.*, *Gonyaulax* and *Gymnodinium*).

2. Sedimentary subsystems. These subsystems occur in the deeper channels, sounds (=straits) and lagoons in which respiration exceeds production and in which particulate and dissolved organic matter from the production zone is used. In these habitats, nutrients are regenerated, recycled, stored and vitamins and growth regulators are manufactured. The consumers of these zones are versatile in their feeding habits.

3. Plankton and nekton. These organisms move freely between the two abovesaid fixed subsystems. They continue producing, converting and transporting nutrients and energy while responding to diurnal, tidal and seasonal periodicities. Holoplanktons tend to comprise relatively few species, while the meroplanktons being more diverse reflecting a variety of benthic habitats.

MARINE ECOSYSTEMS

Marine ecosystem includes the salt water mass of seas and oceans that covers 70.8% of the earth's surface and contain about 97% of the planet's water; a volume of $1370 \times 10^6 \text{ km}^3$. The average depth of oceans is 3730 m. Oceans strongly influence climate by acting as a reservoir of solar heat, thus, improving temperature extremes, and as the source of water which falls as precipitation (*i.e.*, hydrologic cycle). The **Pacific ocean** is the largest ($179.7 \times 10^6 \text{ km}^2$), coldest (average 3.36°C), least saline (34.6 parts per thousand) and deepest (average depth 4028m). The **Atlantic ocean** is warmer (average 3.73°C), more saline (average 34.9 parts per thousand), and relatively shallow (average depth 3310 m); it also has the longest coastline providing many natural harbours. The smallest and shallowest ocean is the **Arctic**. The **antarctic (Southern) ocean** shows wide variations in temperature (-1.8 — 10°C). The **Indian ocean** covers $77 \times 10^6 \text{ km}^2$ to an average depth of 3872 m.

The volume of surface area of marine environment lighted by the sun is small in comparison to the total volume of water involved. This and the dilute solutions of nutrients limits production. All the seas are interconnected by currents, dominated by waves, influenced by tides and characterized by saline waters. Not only the seashore and banks which are the homes of many organisms but the open ocean, many hundreds of miles away from land, supports plant and animal communities of great diversity and complexity.

Physico-Chemical Aspects of Marine Environment

In the marine environment, the most important physical factors which influence marine life are light, temperature, pressure, salinity, tides and currents.

1. Light. Light is a very significant factor in regulating the pattern of distribution of marine animals and it contributes significantly to organic production. The autotrophic primary producers exploit the light energy in the photosynthetic production of food for primary macroconsumers of marine ecosystem. The amount of light exploited in photosynthesis depends on intensity of light and turbidity of water. For example, a ten meters deep turbid coastal zone receives an equal amount of light energy as a hundred meters deep clear oceanic zone, but the maximum intensity at a ten meters coastal water is greater. Light determines diurnal migrations of marine organisms, and it also regulates colour pattern of marine animals. The deep sea fauna which lives in total darkness exhibit either colourlessness or uniform colouration. It is also somehow related with the development of visual sense organs as shown by absence of functional eyes in deep-sea animals.

2. Temperature. The range of temperature in sea is far less than that on land. Arctic waters at 27°F (-2.8°C) are much colder than tropical waters at 81°F (27.2°C), and currents are warmer or colder than the waters through which they flow. Seasonal and daily temperature changes are larger in coastal waters than on the open sea. The surface of coastal waters is the coolest at dawn and the warmest at dusk. In general, sea water is never more than 2° to 3° below the freezing point of freshwater or higher than 81°F (27.2°C). At any given place the temperature of deep water is almost constant cold, below the freezing point of freshwater. Unlike freshwater, sea water does not have a density maximum at 4°C ;

rather it becomes continuously denser as it gets colder.

3. Pressure. Pressure in the ocean varies from 1 atm at the surface to 1,000 atm at the greatest depth. Pressure changes are many times greater in the sea than in terrestrial environments and have a pronounced effect on the distribution of life. Certain organisms are restricted to surface waters when the pressure is not so great, whereas other organisms are adapted to life at great depths. Some marine organisms, such as the sperm-whale and certain seals, can dive to great depths and return to the surface without difficulty.

4. Zonation of marine environment. Just as lakes exhibit stratification and zonation, so do the seas. The ocean itself is divided into two main divisions, the **pelagic**, or whole body of water, and the **benthic**, or bottom region (Fig. 7.3). The pelagic region is further divided into two provinces: the **neritic** (near shore), water that overlies the continental shelf, and the **oceanic provinces**. The **continental shelf** is the underwater extension of the continent and it generally extends to a depth of roughly 125—200m. The edge of the continental shelf may be within a few kilometers of the shore, or it may be several hundred kilometers from shore. From the edge of the continental shelf there is a more rapid descent, the **continental slope**; to the broad flat **abyssal plain** that underlies most of the ocean at a depth of 4,000—5,000m. On the abyssal plain, there are extensive mountain ranges, or **midoceanic ridges**, some of which have tips projecting above sea level. In addition, there are very deep troughs, which drop down below 11,000 m., below sea level. Life exists from the very top of sea to the bottom in virtually all areas, although its abundance is exceedingly variable.

The benthic life zones are defined in terms of following physical subdivisions: The **littoral**, or **intertidal** zone, is the zone between high tide and low tide levels. The **sublittoral** extends from the low tide mark to the edge of the continental shelf, with the **bathyal zone** comprising the continental slope. The **abyssal zone** includes the abyssal plains, and the **hadal zone** includes any life in the deep trenches below 5,000 m.

The pelagic zone is divided into three vertical layers or zones: from the surface to about 200 m is the **photic zone**, in which there is sharp gradients of illumination, temperature, and salinity. The region below this zone is called **aphotic zone**. From 200 to 1,000m where very little light penetrates and where the temperature gradient is more even and gradual and without much seasonal variation, is the **mesopelagic zone** of aphotic zone. It contains an oxygen-minimum layer and often maximum concentrations of nitrate and phosphate. Below mesopelagic is the **bathypelagic zone**, where darkness is virtually complete except for bioluminescence, and where temperature is low and pressure is great.

5. Stratification of marine environment. The upper layers of ocean water exhibit a stratification of temperature and salinity. Depths below 300m are usually thermally stable. In high and low latitudes, temperatures remain fairly constant throughout the year. In middle latitudes temperatures vary with the season, associated with climatic changes. In summer the surface waters become warmer and lighter, forming a temporary seasonal thermocline. In subtropical regions the surface waters are constantly heated, developing a marked permanent thermocline. Between 500 and 1,500m a permanent but relatively slight thermocline exists.

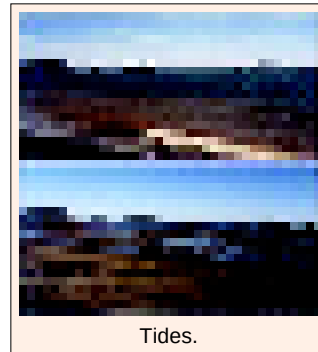
Associated with a temperature gradient is a salinity gradient or **halocline**, especially at the higher latitudes. There the abundant precipitation reduces surface salinity and causes a marked change in salinity with depth. Thus, in the middle latitudes in particular, the two produce a marked gradient in density. Water masses form density layers with increasing depth. Because density of seawater does increase in depth and does not reach its greatest density at 4°C as with freshwater, there is no seasonal overturn. This results in a normally stable stratification of density known as the **pycnocline**. Because there are marked changes in temperatures with depth in the open ocean and, thus, with density, the pycnocline is the deep zone which is oxygen rich, cold and comprises 80 per cent of the ocean.

6. Salinity. The sea is salty, with an average salinity of 35 parts of salt (weight basis) per 1000 parts of water or 3.5 percent, that is usually written as 35‰, i.e., parts per 1000 (in contrast the salinity of freshwater is 0.5‰). The main salts are chloride, sulphates, bicarbonates, carbonates and bromides of sodium, magnesium, calcium and potassium. Of these salts, sodium chloride is present in maximum

amount. The salinity of marine water fluctuates from place to place. The marine animal life has specific osmoregulatory adaptation for high saline sea water. The absence of many animal species in marine environment has been related to their inability to tolerate the high salt contents of sea water. Except few insects such as *Halobates* and shore collembolans such as *Isotoma* and *Sminthurus*, most insects do not occur in marine ecosystems.

7. Currents and tides. Sea water is never static and waves, currents, and tides are the regular features of sea water. All these phenomena are controlled by winds, cosmic forces and varying water densities. Most world-wide water currents occur at the surface and at great depths of sea water. These currents determine the interchange between the surficial and deep water masses, as well as horizontal movements. Both the horizontal and vertical movements of ocean water are significant, but for different reason. Within a given current, such as warm Gulf stream, for instance, the water mass retains its identity for great distances, and the warm-sea community ranges far north of what have been anticipated as its northernmost limits. In addition, the climate of terrestrial ecosystems is strongly affected by the nature of adjacent water masses. The prevailing winds blow across the ocean and are heated or cooled by the waters they traverse. In marine ecosystems, however, the oceanic current system controls the distribution of productivity of marine communities through vertical mixing of water masses. Because most of the ocean is permanently stratified, any essential nutrient that settles out of the upper layers, either as an inorganic mineral or as organic detritus, is not returned to the upper layers at the same location. This had led to such an improvisation of the surficial waters of the oceans that productivity in most parts of the open oceans is about the same as that of desert, because of the very low concentrations of nutrients, especially phosphorus. At the same time, the lowest waters of the ocean are nutrient-rich. There are certain regions in the world, notably at the west coasts of continents and in subpolar latitudes in both hemispheres, where deep currents rise to the surface. These are called **zones of upwelling**, and even though their waters are very cold, they are sites of intense production because of their relatively high concentration of nutrients. Nutrients are then cycled back to the rest of the ocean via surface currents.

Tides are water movements which are caused by some astronomical factors (*i.e.*, due to a pull by moon and sun). They represent a rhythmic rise and fall of water and often waves of long wave lengths characterize the process. The chief astronomic force behind the tidal rhythm is the attraction of the moon and sun which depends on the movement of earth in relation to the moon and the sun. Tides influence the sea-shore fauna variously, as low tides expose the shore and high tides flood the substratum.



Tides.

MARINE COMMUNITIES

The sea biota is not abundant but contains well marked diversity. Every major group of algae and almost every major group of animals can be found some where in the oceans. The only striking omissions are the vascular plants and insects, which are abundant in estuaries but which have few or no marine representatives. Communities of marine environment includes : (1) **producers** such as phytoplanktonic diatoms and dinoflagellates, green algae, brown algae, red algae and kelps; (2) **consumers** such as zooplanktons, benthos, nekton and neuston and bacteria. The marine biotic communities can be studied separately for different life zones of the ocean as follows :

A. Biotic Communities of Oceanic Region

The oceanic region or pelagic zone is less rich in species and numbers than the coastal areas, but it has its characteristic species. Many of these are transparent or bluish and since the sediment-free water of the open sea is marvelously transparent, these animals are nearly invisible. Animals that are too thick to be transparent frequently have smooth shiny and silvery bodies which make them invisible

by mirroring the water in which they swim. Further, the animals of the pelagic zone of ocean encounter stable and uniform environmental conditions due to the continuity of the waters of the sea. This continuity of sea waters causes the individuals of the open ocean to stay away temporarily or permanently from any solid objects, and they are provided with different locomotor organs. However, the planktonic forms lack a locomotor organ and float freely in the vast expanse of the sea.

(i) Biotic communities of pelagic zone. The pelagic plankton also called **epiplankton** are exceedingly diverse. **Pytoplankton** includes diatoms and dinoflagellates, which together produce most of the organic carbon in the sea (and most of the oxygen in the atmosphere), as well as other forms of golden-brown algae and flagellated green algae. Some seaweeds, such as the large brown algae *Sargassum*, may have a floating stage. The **zooplankton** of pelagic zone

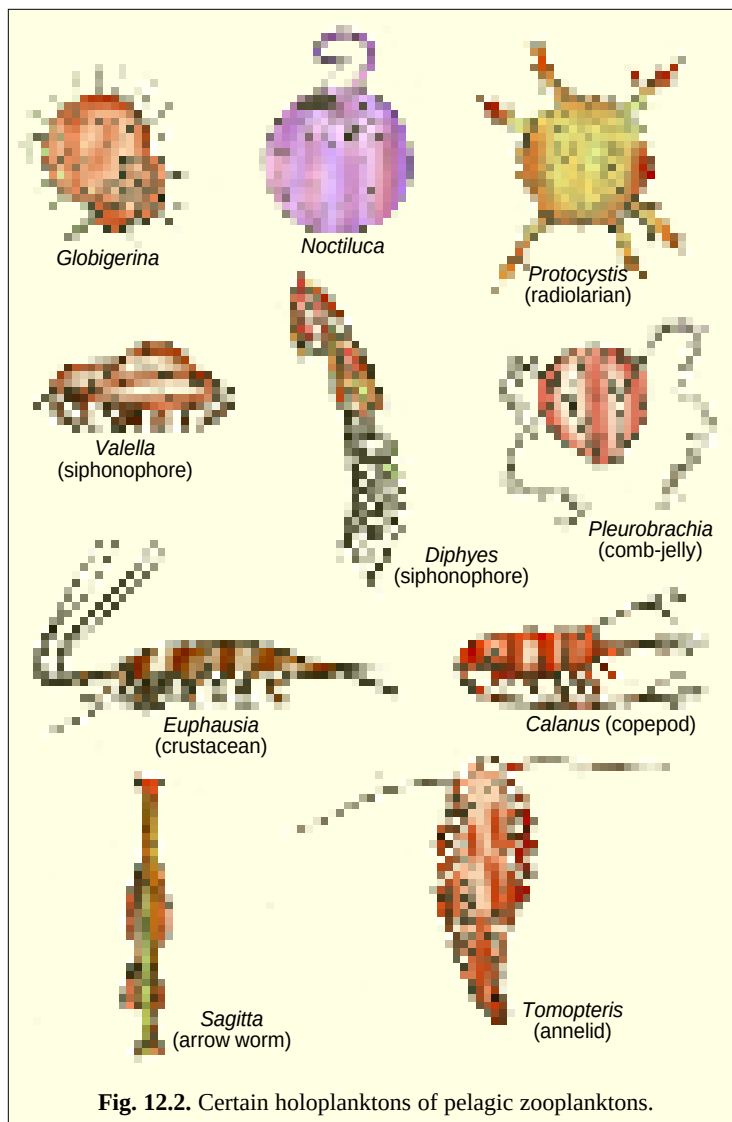


Fig. 12.2. Certain holoplanktons of pelagic zooplanktons.

includes representatives of every major phylum and most minor phyla, either as permanent members of the plankton community (**holoplanktonic forms**) or as transients during their larval stages (**meroplanktonic forms**). Common among permanent planktonic forms are the Foraminifera and Radiolaria, arrow worm (*Sagitta*), certain annelid worms, swimming snails, jelly fishes and most abundant of all are the crustaceans such as shrimps, copepods and cladocerans. Among the temporary zooplanktons are the larva of large animals from all marine environments and even some freshwater environments. For example, in the zooplanktons of inshore water of Mandapam (South India) **Prasad** (1956) reported following animal species—flagellate protozoan *Noctiluca miliaris*; larval forms of coelenterates such as *Planula*, *Ephyra*, *Semper's* larva and *Cerianthus* larva; medusae of certain coelenterates such as *Bougainvillia*, *Obelia*, *Charybdea*, *Rhizostoma* and *Rhopalina*; ctenophores such as *Pleurobrachia* and *Beroe*; larvae of many polychaet annelids and annelids such as *Tomopteris*, *Alciopa* and *Autolytus*, several species of chaetognaths such as *Sagitta* and *Spadella*; heteropod and pteropod molluscs such as *Carinaria*, *Pterotrachea*, *Cliona*, *Creseis*; larvae of

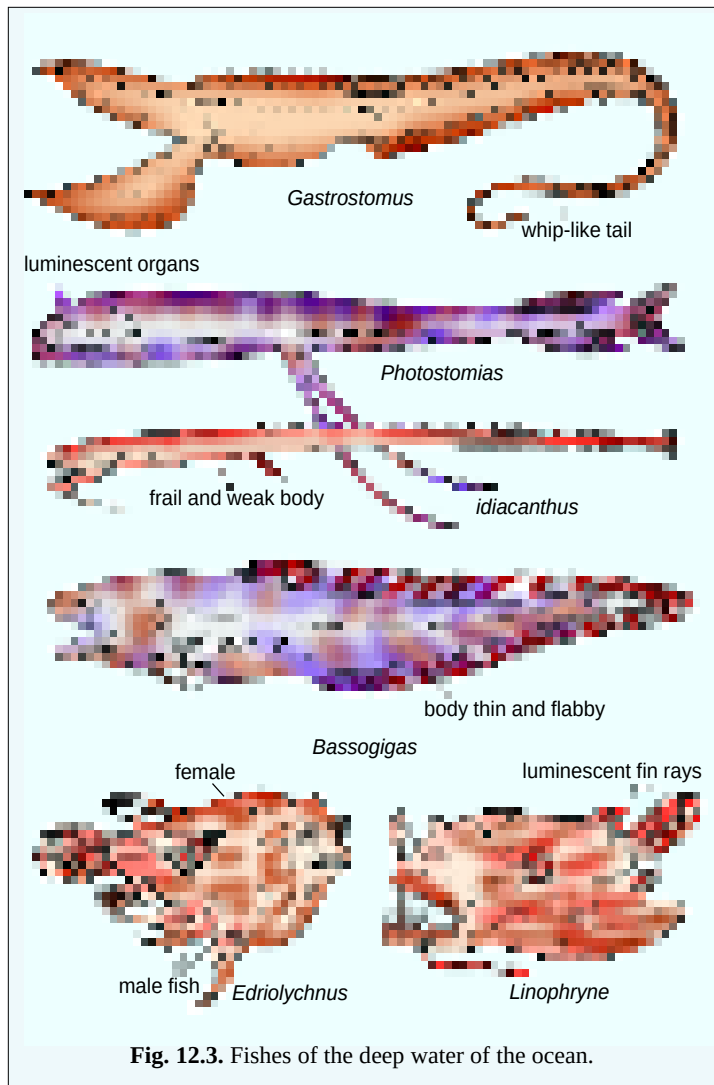


Fig. 12.3. Fishes of the deep water of the ocean.

echinoderms such as bipinnaria, ophio-pluteus, echinopluteus, auricularia; tornaria of *Balanoglossus*; species of copepods such as *Acartia*, *Calanopia*, *Labidocera*, *Centropages*, *Paracalanus*, *Canthocalanus*, *Acrocalanus*, *Eucalanus* and *Microsetella*; larvae of decapod *Lucifer*; protochordate species of Larvacea, Thaliaceae and Ascidiaceae such as *Thalia democratica*, *Jasis zonaria*, *Salpa cylindrica*; appendicularian *Oilopleura diocia*, *O. parava* and *O. cephalocera* and also cephalochordate *Amphioxus*.

The largest animals in the pelagic region are the nekton. These include cephalopods such as squid and nautili among the invertebrates, as well as many marine vertebrates, such as bony fishes, sharks, sea turtles and whales. In addition, sea birds also feed on many of the same food types as these large carnivores. Air-breathing nekton, such as the turtles, and whales, are found

mainly in the photic zone, but fishes extend from the sea surface to the bottom. Certain fishes such as tuna, shark, sardine, mackerel, herring, bonito, and anchovy live near the surface. Those fishes which occur at greater depths are often bizarre and unlike any fish found at the surface. They tend to be small—15 to 20 m is large for the ocean deeps—and they are exceedingly dispersed. Many have luminescent appendages with which to lure prey, or mouth that looks too large for the rest of the body. Food is not plentiful in the deep waters of the sea, so these fishes must go for long periods of time without food, and then consume as much as they can when they have a chance.

(ii) Biotic communities of abyssal benthic zone. The abyssal benthic zone or deep sea of oceanic region is pitch dark and universal absence of light in this environment excludes the possibility of any growth of vegetation or other photosynthetic organisms. Consequently, most of the deep sea dwellers depend upon the detritus (dead body and excreta of surface forms) which sink to the sea bottom while a large number of species are active predators (carnivores). However the deep ocean benthos are surprisingly diverse. Food is a little more plentiful on the bottom than it is in the deep waters of the

ocean, because the bottom sediments are the ultimate resting place for all detritus raining out of the upper layers (oceanic surface and pelagic zones). The bottom of the sea is a soft ooze, made of the organic remains and shells of Foraminifera, Radiolaria, and other animals and plants. Sea cucumbers, brittle stars, crinoids (sea lilies), sea urchins, certain benthic fishes, and several types of crustaceans, as well as sea anemones, clams, and similar animals are all found on the bottoms of the deepest ocean trenches. The biomass of carnivorous brittle stars is often higher than that of the detritus feeders that serve as their food source; however, their metabolic rate is much lower. The great diversity of sea bottom fauna has been related with great constancy in the physical environment of the deep sea.

B. Biotic Communities of Continental Shelf

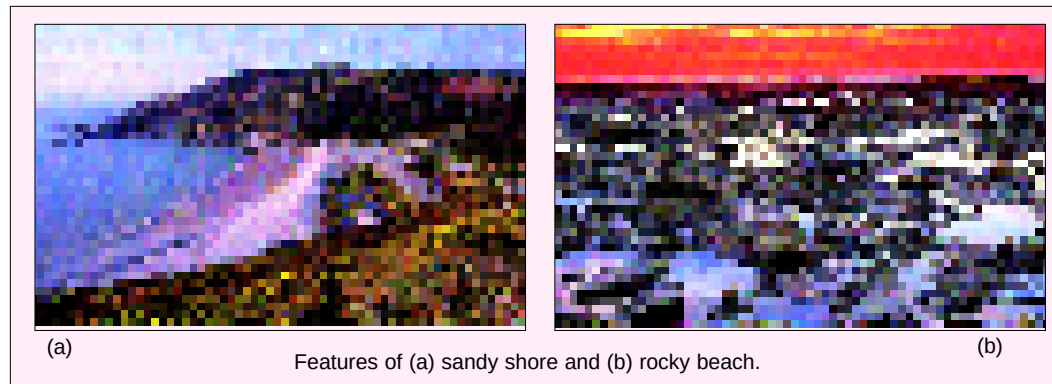
The communities of continental shelf are both richer and more diverse than those of open ocean. Diatoms and dinoflagellates are still the most productive phytoplanktons, but in the shallower regions green, brown, and red algae anchored to the bottom may be of great importance. Several kinds of these seaweeds are harvested from rocky shores as human food or for some commercial purpose in eastern countries, USA and India. The zooplankton of continental shelf are generally the same as in the pelagic region, but some purely pelagic species have been replaced by neritic species, and the overall diversity is somewhat higher. The temporary zooplanktons are much more abundant over the shelf region than in the open ocean.

The nektons of the **neritic sea** are both diverse and well known. The significant nekton species of the region are large squid, whales, seals, sea otters, and sea snakes. The most numerous nektonic forms, of course, are the fishes, which include many species of shark as well as herring-like species such as menhaden, herring, sardine and anchovy; cods and their relatives, such as haddock and pollack; salmon and sea trout; flounder and other flatfish such as sole, plaice and halibut; and mackerels, including tuna and bonito.

Communities of sea shore or intertidal zone. The littoral, intertidal or eulittoral zone, is the region of sea shore which exists between the high and low tide lines. The region of the high tide mark is called **supratidal** or **supralittoral zone**. The region of low tide is called **subtidal zone**. **Davenport** (1903) has divided sea shore or intertidal zone into **submerged zone** which is the portion of the shore below the low tide exposed by very low tides, a **lower beach** occupying the area between the mean low tide and mean high tide and an **upper beach** which is reached by the highest of high tides. The intertidal zone is the most variable zone in the entire sea. It is completely covered at high tide, and is completely uncovered at low tide except for tide pools. Any organism that is to survive in the intertidal must be either resistant to periodic desiccation or able to burrow to water level. For example intertidal polychaets escape desiccation by resorting to a tubicolous mode of life whereby they can resist desiccation by retaining sufficient moisture till the occurrence of the next wetting. Barnacles can survive desiccation by remaining in a state of suspended activity during dry periods. Many plants of sea shore contain jellylike substances such as agar, which absorb large quantities of water and retain it while the tide is out. Further, the majority of shore animals utilize oxygen dissolved in water and when the low tide exposes them during the periods of drought, they utilize the oxygen stored in the blood plasma, *e.g.*, nereid *Arenicola*. Barnacles achieve this by enclosing an air bubble between the shells.

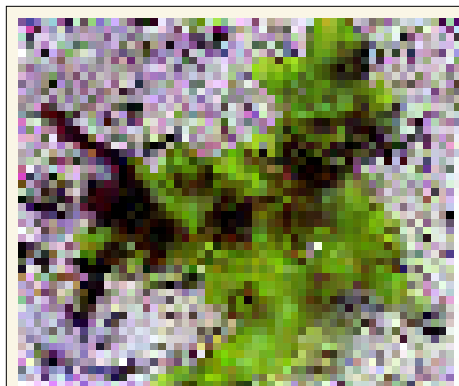
One of the notable characteristics of this region is the ever present action of the waves, therefore, the organisms living on a sandy or rocky beach have had to evolve ways of resisting wave action. Many seaweeds have tough pliable bodies, able to blend with the waves without breaking, while the animals either are encased in hard calcareous shells, such as those of mollusks, Bryozoa, starfish, barnacles and crabs, or are covered by a strong leathery skin that can bend without breaking, like that of the sea anemone and octopus. Many sedentary shore animals (*e.g.*, sponges, tunicates, etc.) have special modes of adhesion which keep them firmly attached to the substratum. However, animals of sandy beaches escape wave action by taking burrowing mode of life, because, these habitats lack a hard substratum for the attachment of the animals.

The biota of intertidal zone also have to overcome the wide fluctuations in salinity and temperature. Thus, like the estuary, this variable environment is a zone in which an organism must be adapted to a broad range of environmental conditions. Like the estuary also, it is an area of very high productivity with a simple community, many of whose members may be exceedingly abundant. The diversity of biota of intertidal zone is determined by nature of substratum whether loose sand or mud or rocky coast. Ecologists have recognized three major types of intertidal zones or seashores, namely, **rocky**, **sandy** and **muddy**. **Eltringham** (1971) has included a fourth type of sea shore, the **pabble shore** in this classification. Each type of sea shore possesses specific biota each of which remains specifically adapted for its peculiar habitats. Here we will discuss only rocky shore and sandy beach habitats.



1. Biotic communities of rocky shore. The rocky shore presents solid substratum for the attachment of many sessile animals which often remain abundant here. The animals which are sedentary and inhabit rocky shore are limpets such as *Patella*, *Haliotis*, *Fissurella*, and oysters, barnacles, tunicates and bryozoans. Sessile organisms such as the sponges, the colonial hydrozoans and anthozoans such as *Gammaria* and *Zoanthus* attached to rocks. Certain animals such as some sponges, annelids and molluscs either bore the soft rocks like limestone or seek protection in the crevices of hard rocks. All these sedentary sessile animals are adopted for filter feeding.

Further, all these animals of rocky shore occurs in successive zones. At the uppermost end is a **zone of bare rock** marking the transition between land and sea. Next is a **spray zone** with dark patches of algae on which the periwinkles (*Littoria*) graze. Below this is the zone called **barnacle zone** regularly covered by the high tide; rocks in this zone are encrusted with acorn barnacles, limpets and mussels (*e.g.*, *Mytilus*). This zone is a vulnerable zone being exposed to the stress of wave action and active predation by starfishes and gastropods. The mussels act like hard substratum for tunicates, sponges and small mussels. The next one is the **zone of oysters** which is less vulnerable and includes green algae (*Enteromorpha* and *Ulva*) and barnacles. The next zone includes mussels, *i.e.*, boring bivalves, which remain within crevices with exposed siphons only to be protected from waves and predation. The chief species of borer bivalves are *Pholas*, *Martesia*, *Hiatella*, *Tridacna*, etc. Besides bivalves a variety of filter feeder bryozoans, and brachiopods, and tubicolous polychaets such as *serpulids*, *terebellids*, *sabellids* and *cirratulids* which live in tubes



Green alga: *Ulva*.

are the chief inhabitants of this zone. The subtidal zone includes red algae, brown algae (Laminarians and fucoids), sea anemone, sea urchin, corals, etc.

For example, the Okha coast of Gujarat in India is a typical rocky sea coast which has well defined fauna and flora (**Gopalakrishnan**, 1970). Dominant weeds of subtidal zone of Okha coast are *Sargassum*, *Dictyopteris*, *Gracilaria*, *Padina*, *Ulva*, *Polysiphonia*, etc. This subtidal zone also contains sea urchins, serpulid tubes of species *Pomatoleius crosslandi*, and mollusc species such as *Trochus radiatus*, *Cypraea arabica*, *Thais alveolata*. The upper limit of the dense weed growth of subtidal zone marks the lower end of eulittoral or intertidal zone, where occurs a dense cluster of *Crassostrea cucullata* and *Littorina* sp. The intertidal zone includes three subzones, i.e., 1. the subzone of patchy weed growth; 2. the subzone of limpets such as *Sellana radiata* and barnacles and 3. the subzones of oysters and barnacles. The supratidal zone is inhabited by green algae and molluscs. Chlorophyceae such as *Enteromorpha*, *Ulva*, *Bryopsis*, *Codium*, *Halimeda*, *Caulerpa*, *Dictyosphaeria* and *Cladophora*; Phaeophyceae such as *Padina*, *Sargassum*, *Colpomenia*, *Iyengaria*, *Spathoglossum*, *Dictyopteris* and *Turbinaria*; Rhodophyceae such as *Gracilaria*, *Hypnea*, *Acanthophora*, *Polysiphonia*, *Cryptonemia* are the chief algae of the area. The common animals of this area are coelenterate species *Stoichactis*, *Ixalactis*, and *Gyrostoma*; Nereid (e.g., *Eunice*, *Tubifex*); molluscs such as *Chiton*, *Nerita rumphii*, *Cypraea integerrimus*, *Pilumnus vespertilio*, seaweed grazer (nudibranch) molluscs, *Aplysia benedicti*, *Onchidium*, *Elysia grandifolia*, *Eolis*, *Doris* and floating mollusc *Janthina*, etc. The midlittoral reef community of Okha coast consists of fishes such as *Epinephelus*, *Petrachus*; the crab *Pilumnus vespertilio*, the annelid *Eurythoe complanata* and molluscs such as *Murex*, *Bursa*, etc. Rock crevices of intertidal zone harbour within them crabs such as *Atergatis*, *Grapsus*, *Pilumnus*, *Matuta*, *Charybdea*, *Gelasimus*, squillids (e.g., *Sesarma* and *Gonodactylus*); coelenterates (e.g., *Stoichactis*, *Ixalactis*, *Zoanthus*, *Lobophytum*; the hydroid species such as *Lytocarpus*, *Sertularia* and *Plumularia*; and annelids such as *Sepula*, *Polynoe*, *Eurythoe*, *Sabella*, etc.

(ii) Biotic communities of sandy shore. The sandy shore may be even more harsh than the rocky shore. It is subjected to all the extremes of the latter (temperature, salinity, turbidity, wave action, etc.) plus inconvenience of a constantly shifting substratum. The last factor makes life on the surface almost impossible; life has retreated below the surface. Generally sandy beaches are characterized by gentler wave action. Because of the prolonged time taken for drying up, these beaches are suitable for animal life. Decaying seaweeds and dead remains of animals result in the addition of organic matter to the sand. The coastal subsoil water is the environment of a special interstitial brackish water fauna. Most of the interstitial organisms are small, elongate, vermiform with transparent bodies. They have no eye but possess well developed adhesive organs and sense organs. They are negatively phototactic and gregarious in habit. They may be herbivore, carnivore, omnivore or detritus feeder. Large animal forms which are few in number prefer coarse sand grains and smaller one which are very abundant, prefer finer sand. The principal animal groups of interstitial water are the Nematoda, Turbellaria, Annelida, Gastrotricha and Acarina. Certain animal groups such as Protozoa, occur sporadically. The top 20 cm of sand between low and midtide levels of the beach is occupied by the inhabitants such as Hydrozoa, Turbellaria, Nemertina, Rotifera, Archiannelida, Polychaeta, Ostracoda, Halacaridae and Nudibranchiata. Species of Gastrotricha, Kinorhyncha, and Isopoda occur at mid-tide levels preferably in deep layers of sand. Oligochaets are sporadic and are found at different levels. Harpacticoid copepods remain restricted to mid-tide level. Tardigrades occupy the region between mid-tide and high-tide. Thus, sandy beaches also possess characteristic successive zones or zonation of animals.

For example, according to studies of **Ganapathy** and **Rao** (1962) sandy beach fauna of Visakhapatnam coast though lacks or contains small number of tubicolous polychaets and Crustacea due to very unstable substratum, but include following animal species—spinoid *Prionospio krusadiensis*, *Aricia* and *Lumbriconeries*, all of which possess the capacity to secrete mucus and formation of cover of sand grain over the body for protection, *Glycera lancadivae*, *G. alba*, *Nerina bonnierii*, *Pisionidens*

indica, *Pisone complexa* and archiannelids *Saccocirrus minor* and *Protodrilus* also occur. The crustacean fauna of this zone includes *Emerita asiatica*, *Albunea symnista* (burrowing), amphipod species *Harpinia* sp., isopod *Sphaeroma walkeri*, mysid shrimp *Gastrosaccus spinifer*, hermit crabs *Clibanarius aretheustus* living within the shells of *Thais* and *Cerithium*, the eight-oared swimming

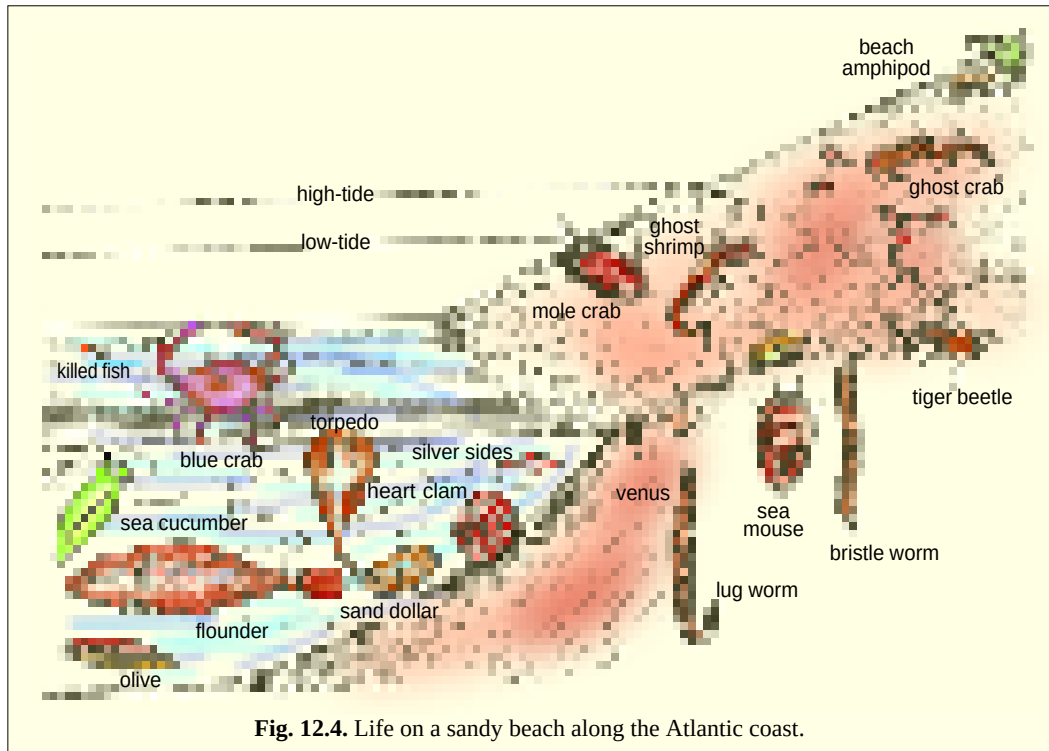


Fig. 12.4. Life on a sandy beach along the Atlantic coast.

crab *Matuta victor*, the spider crab, *Philyra scabruescula*, *Ocypoda platytarsis* and *O. macrocera*. Lamellibranchs such as *Donax cuneatus* and *Pamphila textile*, burrowing snails (e.g., *Sinun neritoideum*), common gastropods (e.g., *Oliva gibbosa* and *Terebra* sp.), *Bullia vittata* are also found along the coast of Visakhapatnam. This beach also contains eel *Ophichthys orientalis* and siphonophore coelenterates such as *Physalia*, *Velella* and *Porpita*; and medusae such as *Aequoria* and *Chiropsalmus*.

Coral Reef as a Specialized Oceanic Ecosystem

Coral reefs are the specialized ecosystems of ocean which are among the most productive of all ecosystems anywhere, with a diversity equalled only by tropical rainforests. The reef environment is very rugged one because it is constantly subjected to the pounding of the waves, and parts of the reef are often above water at low tide. But reef is also a very rich environment for those species that can withstand the physical pounding of water. The amount of oxygen is very high. During the day, it may reach 250 per cent of saturation because of the production of oxygen by algae in the reef structure. Likewise, the tremendous amount of CO₂ produced by the metabolism can be carried off before it becomes too high.

Reefs can be found wherever a suitable substrate exists in the lighted waters of the tropics, away from areas of continental sediments which would tend to bury the reef in mud or cold upwellings which would tend to lower the temperature too much. They may form **barrier reefs** (e.g., barrier reef in Australia which contains canal or lagoon between them and coast) adjacent to continents; **atolls** or ring shaped or horse-shoe-shaped islands on the top of submarine heights, or **fringing reefs**, as in

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Mandapam, Appa Islands and Hare Islands. In all modern reefs, the main reef structure is formed by madreporian corals of following families—Poritidae (*Porites minicoiensis*, *Porites mannarensis*), Acroporitidae (*Acropora intermedia*), Astreaeidae and Milliporidae. Secretions of calcareous algae go into the building up of reefs. Equally important are calcareous Bryozoa, lamellibranchs and gastropod molluscs and Foraminifera, which become significant elements in the total reef structure. Although the name ‘coral reef’ implies that corals should be the most abundant organisms, algae of many sorts comprise over three times the biomass of the corals. In addition, the calcareous algae are not only producers; their skeletons act to cement the entire reef to make it more resistant to the pounding surf.

Coral reefs normally exist in water with mean temperature 20.5°C, but they can withstand very high temperatures. The growth of coral reef is influenced by salinity and reduced tides. Moderate water movement around the coral reefs help in the circulation of food.

Biotic Communities of Coral Reef

Coral reef ecosystems harbour an abundant and well diverse biota. The well known Krusadi Island is the coral island of gulf of Manaar. It's dominant corals are *Porites*, *Montipora*, *Poecilopora*, *Acropora*, *Madrepora*, *Favia*, *Symphylia*, *Psammocora*, *Platygyra*, *Merulina* and *Leptastrea*. In the Krusadi area there is a luxuriant growth of sea weeds such as *Sargassum*, *Caulerpa*, *Turbinaria*, *Spatoglossum*, *Gracilaria*, *Ulva*, *Enteromorpha*, *Chaetomorpha* and *Polysiphonia*. The abundant sponges of this reef are *Spirastrella inconstans* and *Callyspongia fibrosa*. The other inhabitants of this reef are following—hydroid coelenterates such as *Syncoryne*, *Sertularia*, *Eudendrium*, *Pennaria*, *Tubularia*, *Thyrosocypris*, *Plumularia*; alcyonarians such as *Clavularia*, *Alcyonium*, *Lobophytum*, *Scelerophytum*, *Telesto*, zooantharians such as *Zoanthus*, *Gemmamaria*; polychaet annelids such as *Nereis*, *Polynoe*, *Eunice*, *Lepidonotus*, *Iphion*, *Eurythoe*, *Hesione*; sipunculid species such as *Dendrostomum*, *Aspidosiphon* and *Phascolosoma*; molluscs such as *Doris*, *Dendradoris*, *Acanthochiton*, *Ischnochiton*, *Haliotis*, *Emarginula*, *Trochus*, *Turbo*, *Nerita*, *Natica*, *Cypraea*, *Thais*; and echinoderms such as *Holothuria atra*, *H. scabra*, *H. edulis* and starfish *Proteaster lincki*.

Beautifully pigmented fishes such as *Holocentrus*, *Lutianus* sp., *Pomacanthoides*, *Chaetodontopsis*, *Chaetodon*, *Linophora*, *Thallasoma* and *Zanclus* and the eels *Gymnothorax undulatus* and *G. punctatus* are commonly frequent in the coral reefs. These inhabitants of coral reef community, when exposed to low tides are subjected to predation by large number of birds such as the crested tern *Sterna bergii*, golden plover *Pluvialis dominica fulva*, etc.

According to **Clapham, Jr.**, (1973) the great productivity of reefs is obviously caused by something more than the abundance of light and oxygen. He considers nutrient cycles which runs very quickly and efficiently between the components of reef ecosystem to be main cause of great productivity of reef ecosystems. In fact, no organic matter is incorporated permanently into the sedimentary system portion of reef. The conditions of rapid nutrient cycling allow a mechanism by which the natural paucity of nutrients in the ocean can be effectively circumvented. It may also allow a gradual increase in nutrient availability by incorporating the nutrients of the non-reef organisms that wash into the reef, while still continuing to prevent the reef's own nutrients from being lost to the deeper seas.

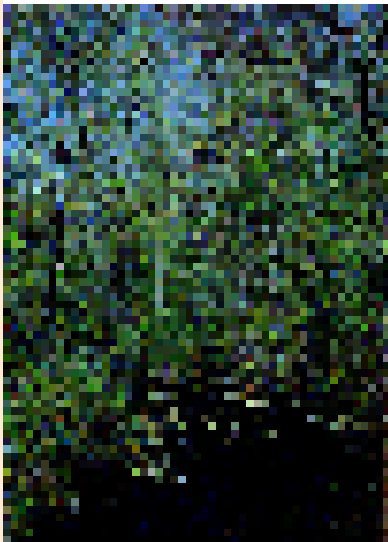
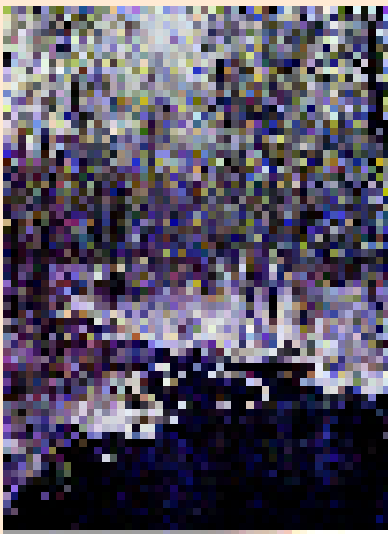
REVISION QUESTIONS

1. What are the physico-chemical characteristics of marine environment ? Classify the marine environment and describe in detail the flora and fauna of the pelagic region.
2. Describe physico-chemical characteristics and biota of estuaries.
3. Describe the “coral reef as a specialized oceanic ecosystem.”
4. Write short notes on the following ;
 - (a) Sublittoral zone ;
 - (b) Deep sea fauna ;
 - (c) Intertidal sandy shore.

CHAPTER

13

Terrestrial Ecosystems



Terrestrial A Ecosystem.

Only about a quarter of the earth's surface is dry land, yet the complexity and variegation of terrestrial ecosystems are much greater than those of aquatic ecosystems. There is no unifying theme in terrestrial ecosystems analogous to the physical properties of water. Rather, the variety of climates, the diversity of all lithosphere, and heterogeneity of terrestrial biotic communities all conspire to give rise to such a variety of themes that they are woven together in nature to give an enormous diversity of biotic communities and terrestrial ecosystems.

Physico-Chemical Nature of Terrestrial Ecosystems and their Comparison with Aquatic Ecosystems

An aquatic system is essentially a **single-phase system**, where water sets the tone for entire habitat. A terrestrial ecosystem, on the contrary, is a **three-phase system**, where the characteristics of the habitat are a function of the atmosphere and climate, the soil, and the biotic community itself. The atmosphere is the source of oxygen for animals and carbon dioxide for plants. Both are sufficiently common and well-mixed in the atmosphere that they are essentially never limiting. Air as a medium of support is much less buoyant than water. Thus, an organism need relatively little support to

survive in water, but it needs a fairly rigid skeleton to live and move on land. Climate in a terrestrial ecosystem is much more variable than it is in water. The most important aspects of climate are the temperature and water relations of ecosystems. These relations include mean annual temperature, amount of temperature fluctuation, annual rainfall, degree and time of fluctuation in rainfall and potential evapotranspiration. Rainfall tends to be somewhat irregular even in the most predictable terrestrial ecosystems, and in arid regions or during periods of drought, lack of water may be a source of extreme stress. The temperature is much more variable on land than in water. This is caused by the fact that aquatic ecosystems are heated by absorption of radiation by water, and water has a high specific heat; terrestrial ecosystems, on the other hand, are heated by the absorption of heat by soil, rock, and vegetation, whose specific heat are much lower (Table 13-1). A given quantity of heat energy absorbed by solid material can change its temperature two to five times as much as the same energy absorbed by water. Thus, heat is gained and lost rapidly by terrestrial ecosystems, with a resulting wide fluctuations of temperature, both diurnally and seasonally.

Table 13-1. Specific heat of several common substances (Clapham, Jr. 1973).

1. Freshwater	1.00
2. Sea water	0.93
3. Wet mud	0.60
4. Moist sandy clay	0.33
5. Solid rock	0.20
6. Copper	0.09

In terrestrial ecosystems, the soil serves two major functions—it provides support for living organisms, and it is the source for all essential nutrients except for carbon, oxygen and hydrogen (for detail see Chapter 3). Even its supportive role is different from the way in which a lake sediment supports rooted vegetation. Because the entire weight of all living organisms is borne by the soil, there is no partition of support between the soil and the air. The role of soil as a source of nutrients is unique to terrestrial environments. It is the site of the entire detritus food chain (see Chapter 3) and, thus, is central to the biogeochemical cycling of nutrient materials. Different types of soils have different properties that affect the availability of nutrients to plants, so the productivity of the terrestrial ecosystems is very closely tied to the chemistry of the soil. Further, in terrestrial biotic communities interactions between species are real but they are simple, involving relationships such as predation, mutualism, parasitism and the like. In contrast to aquatic ecosystems, living organisms of terrestrial ecosystems leave a permanent mark on the system. For example, the role of plants in breaking down and weathering of rock and building soil is much more evident than it is in almost all aquatic ecosystems. In addition, the succession that takes place in a terrestrial community is almost entirely a function of the organisms within the ecosystem, unlike aquatic successions such as lake eutrophication, which are dependent in large part on materials being washed in from outside.

Lastly, in each of these two ecosystems, namely aquatic and terrestrial ecosystems, several types of major subdivisions can be recognized: thus, one can distinguish freshwater, estuarine and marine aquatic ecosystems, and several major types of terrestrial ecosystems such as prairies, forests, tundra, etc. The former are distinguished on the basis of a major chemical difference (*i.e.*, salt content), the latter generally on the basis of the predominant type of vegetation (grass, tree, etc.; see Kormondy, 1976).

CLASSIFICATION OF TERRESTRIAL ECOSYSTEMS

The earth surface—the continental land masses have been classified by biogeographers and ecologists into following regions:

- (i) Biogeographic realms ; (ii) Biomes.

(I) Biogeographic Realms or Regions

Careful studies of the distribution of plants and animals over the earth have revealed the existence of six major biogeographic realms, namely **Palaearctic realm**, **Nearctic realm**, **Neotropical realm**, **Ethiopian realm**, **Oriental realm** and **Australian realm**, each characterized by the presence of certain unique organisms. Each of these realms embraces a major continental land mass and each remains separated by oceans, mountain ranges, or desert.

(II) Biomes

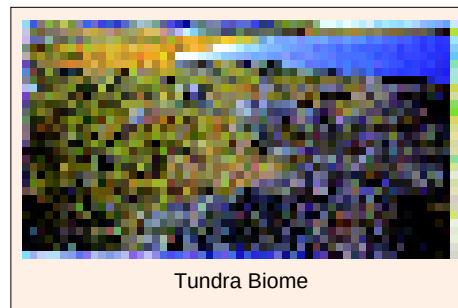
Within these biogeographic realms and established by a complex interaction of climate, other physical factors and biotic factors, are large community units called **biomes**. A biome is a large community unit characterized by the kinds of plants and animals present. In each biome the **kind** of climax vegetation is uniform—grasses, conifers, deciduous trees—but the particular **species** of plant may vary in different parts of the biome. The kind of climax vegetation depends upon the physical environment and the two together determine the kind of animals present. The definition of biome includes not only the actual climax community of a region, but also the several intermediate communities that precede the climax community. Further, there is usually no sharp line of demarcation between adjacent biomes; instead each blends with the next through a fairly broad transition region termed as **ecotone**. There is, for example, an extensive region in Northern Canada where the tundra and coniferous forests blend in the tundra-coniferous ecotone. Ecologists have identified the following biomes :

1. Tundra Biome

Tundra presents the most common example of “fragile ecosystem” (**Clapham, Jr.**, 1973). Tundra which means “marshy plain”, lies largely north of latitude 60°N (*i.e.*, between the Arctic ocean and polar ice-caps and the forests to the south) and is characterized by the absence of trees, the presence of dwarfed plants, and an upper ground surface that is wet, spongy and uneven, or hummocky, as a result of freezing and thawing of this poorly drained land (**Kormondy**, 1976). Some five million kilometers areas of Tundra stretch across Northern America, Northern Europe and Siberia. Although there is variation from place to place within the biome, temperature, precipitation, and evaporation are characteristically low, the warmest months averaging below 10°C and the wettest with about 25 mm of precipitation. Despite the small amount of precipitation, water is usually not a limiting factor because the rate of evaporation is also very low.

The ground usually remains frozen except for the uppermost 10 to 20 cm, which thaw during the brief summer seasons. The permanently frozen deeper soil layer is called **permafrost**. The permafrost line which may exist at a depth of a few centimeters to several meters, is the ultimate limit of plant root growth, but the immediate control is the depth to which soil is thawed in summer. The rather thin carpet of vegetation of tundra biome includes few species : grasses and sedges are characteristic of numerous marshes and poorly drained areas, but large areas consist of grasses, sedges, mosses and lichens, with occasional occurrence of dwarf birches (*Betula*) and willows (*Salix*). Perhaps the most characteristic arctic tundra plant is the lichen known as “reindeer moss” (*Cladonia*). Tundra is also characterized by heath-plants such as *Erica* and other members of Ericaceae family. The net primary productivity is extremely low.

The animals that have adapted to survive in the tundra are caribou or reindeer, musk ox, the arctic hare, arctic fox, polar bear, wolves, lemmings, snowy owls, ptarmigans and during the summer, swarms of flies (Dipteran black flies), mosquitoes and a host of migratory birds.



Tundra Biome

Tundra biomes cover large areas of the arctic zone. There are two tundra biomes one in the palaeartic, and another in the nearctic region. Tundra-like areas, called the **alpine tundra** are also quite similar to some of Arctic tundra, but in the absence of permafrost and in the growing season, mosses and lichens are less prominent, flowering plants more so.

2. High Altitude or the Alpine Biome

The region of mountain above the timber line contains a distinct flora and fauna and is referred to as the **alpine zone**. Alpine zone remains conspicuous in those mountains whose peaks reach up to the nival or snow zone, as in the Himalayas. The alpine zone (zone which lies between timber line and snow zone) includes in the descending order, a **sub-snow zone** immediately below the snow zone, a **meadow zone** in the centre and a **shrub zone** which gradually merges into the timber zone. According to **Mani** (1957) snow zone of Himalayas lies over 5100 m above mean sea level and alpine zone exists at a height of 3600m. From an ecological view point, the zone above the limits of tree growth (timber line) exhibits extreme environmental conditions which greatly influence the biota of this region.

The characteristic features of high altitude environment are following—a low air density, low oxygen and carbon dioxide contents and water vapour, high ozone content, high atmospheric transparency affording greater penetration of light, cold, snow cover, increased rate of desiccation, high wind velocity, insolation of high intensity during the day and rapid radiation during the night, the high glare from the sky and snow, high intensity of ionizing radiation and the exclusion of trees.

In the western Himalayas of Jammu and Kashmir, Himachal Pradesh and U.P. and the eastern Himalayas of Assam and Arunachal Pradesh, **subalpine forests** dominated by trees of such species as *Abies*, *Pinus*, and so on, occur at altitudes of 2900 to 3500 meters, while **alpine scrub** forests are found at altitudes of 36,000 meters and more. These scrub forests contain the broad-leaved *Betula utilis*. Alpine zone of Himalayas is also characterized by a sparseness of animal groups, the scarcity being relatively important as far as tropical Indian, South Chinese, Indo-Chinese and Malayan derivatives are concerned. Important constituents of the fauna of this zone are cold-adapted palaeartic forms (**Mani**, 1968).

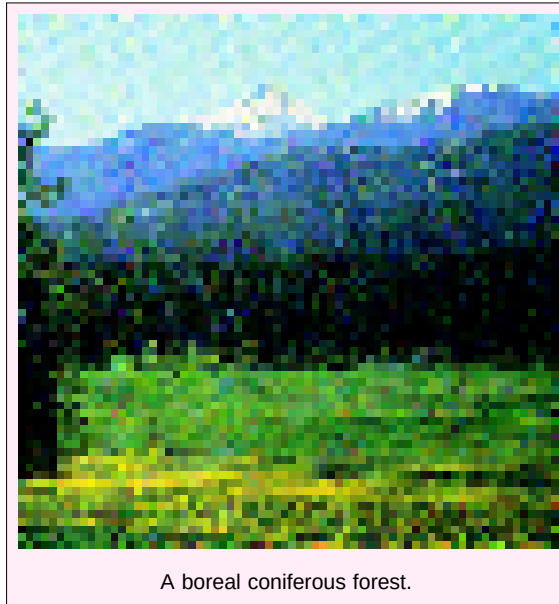
Many invertebrates of alpine zone are predatory and occur in lakes, streams and ponds. Among vertebrates fishes and amphibians are totally lacking and reptilian fauna is greatly impoverished. However, pelobatid frog, *Aleurophryne mammata* is found to occur at 4500 meters of height in Himalayas and viviparous *Mabuya varia* has been reported to live at 4000 meters of height in Mount Kilimanjaro. Some of the representative vertebrates of Himalayan alpine zone are—crow, *Corvus corax*; snow partridge, *Lerwa nivicola*; snow leopard, *Felis unica* and *Felis lynx*; European leech marten, *Mustela foina*; the vole, *Lagomys*; Tibetan yak, *Bos grunnius*; Tibetan sheep, *Ovis hodgsoni*; pamiir sheep, *Ovis poli*; ibex, *Capra sibirica*; markhor, *Capra falconeri*; and Persian wild goat, *Capra aegagrus*.

A large number of insects and arachnids remain best adapted to Himalayan alpine zone and at 6900 meters height. The torrential streams which are the products of melting snow contain many mayflies, stoneflies and caddisflies. Some of common insects of this zone are—stonefly *Rhabdiopteryx lunata*; grasshoppers such as *Bryodema* and *Gomphomastax*; apterous species *Conophyma*; *Dicranophyma*; *Spingonotus*; tettigoniid *Hypsinomus fasciata*; earwigs such as *Anechura*; bugs such as *Dolmacoris*, *Tibetocoris*, and *Nysius ericae*; aquatic beetle *Amphizoa*; Carabids such as *Bembidion*, *Carabus*, *Calosoma*, *Harpalus*, *Trechus*, *Calathus*, *Bradytus*, *Broscus*, and *Dyschirius*; staphylinids such as *Atheta*, *Aleochara*, *Geodromicus*, *Oxypoda*, *Philonthus* and *Tachinus*; tenebrionids such as *Bioramix* and *Chianalus*; ants such as *Formica picea*; bumble bees such as *Lapidariobombus separandus*; butterflies such as *Calias*, *Argynnis* and *Parnassius*; the fly *Deuterophlebia*; collembolans such as *Entomobrya*, *Isotoma*, *Proistoma*, *Hypogastrura*, *Isotomurus*, *Tomocerus* and *Onychiurus*; and numerous simuliids blepharocerids, syrphids, anthomyids tachinids and sarcophagids (**Ananthankrishnan** and **Viswanathan**, 1976).

3. Forest Biomes

The word **forest** is derived from Latin '*Foris*' meaning outside, the reference being to village boundary fence, and must have included all uncultivated and uninhabited land. Today a forest is any land managed for the diverse purposes of forestry whether covered with trees, shrubs, climbers, etc., or not (see **S.P. Sagreiya**, 1967). The forest biomes include a complex assemblage of different kinds of biotic communities. Optimum conditions of temperature and ground moisture responsible for the growth of trees contribute greatly to the establishment of forest communities. In addition, 50 mm rainfall is a pre-requisite for the trees. The nature of soil, wind and air currents determines the distribution (abundance or sparseness) of forest vegetation. Normally ecologists recognise among forest communities such features as their evergreen nature, whether deciduous or indeciduous, as well as their shape, whether broad-leaved as in temperate forests or more needle-like as in the conifers. On the basis of these features the forest biomes of the world have been classified into following biomes—**coniferous forest**, **tropical forest**, and **temperate forest**. All these forest biomes are generally arranged on a gradient from north to south or from high altitude to lower altitude.

(i) Boreal coniferous forest. Cold regions with high rainfall and strongly seasonal climates with long winters and fairly short summers are characterized by boreal coniferous forest which is transcontinental. For example, adjacent to the tundra region either at Latitude or high altitude is the **northern coniferous forest**, which stretches across both North America and Eurasia just south of the tundra (*i.e.*, Canada, Sweden, Finland, Siberia and Mussoorie). The term **taiga** is applied to the northern range of coniferous forests. This is characterized by evergreen plant species such as spruce (*Picea glauca*), fir (*Abies balsamea*) and pine trees (*Pinus resinosa*, *Pinus strobus*) and by animals such as snow shoe hare, the lynx, the wolf, bears, red fox, porcupines, squirrels, amphibians like *Hyla* and *Rana*, etc. Large size is common, both in the trees, which range upto some 40 m in height, and in vertebrates, which include the giants of several groups of animals, such as moose, caribou, elk, grizzly bear, wolverine, beaver, and several large species of birds. Species diversity is low, and pure stands of trees and shrubs are common, as are outbreaks of defoliating insects of several sorts (*e.g.*, bark beetles, saw flies, geometrid moths, etc.). Understory trees are uncommon as a result of the continual low light penetration. Among common understory associates are orchids and ericaceous shrubs like the blue berry. The thalloid mosses and lichens being very rich understory vegetation.



A boreal coniferous forest.

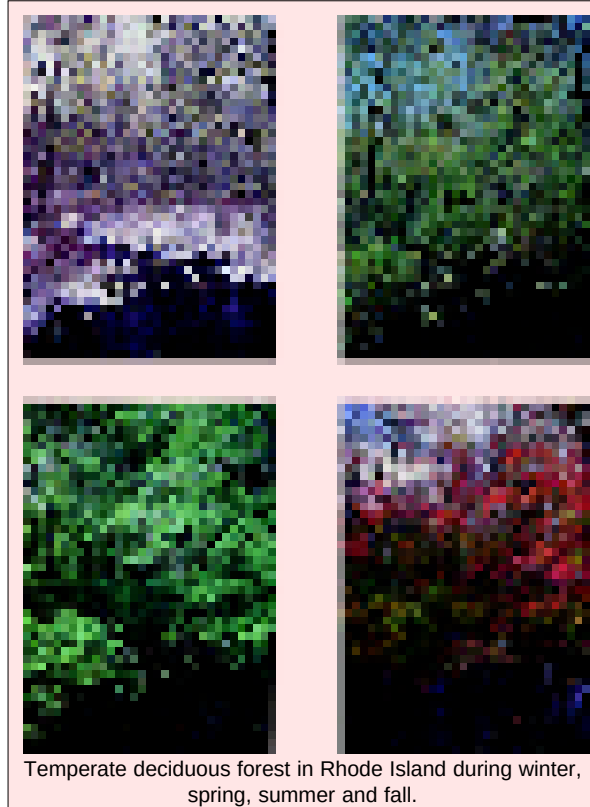
Boreal forest soils are thin podozols, and are rather poor both because the weathering of rock proceeds slowly in the cold environments, and because the litter derived from conifer needle is broken down very slowly and is not particularly rich in nutrients. These soils are acidic and mineral deficient, the result of the movement of a large amount of water through the soil; in the absence of a significant counter upward movement of evaporation, soluble essential nutrients such as calcium, nitrogen, and potassium are leached sometimes beyond the roots thereby leaving no alkaline-oriented cations to encounter the organic acids of the accumulating litter (**Kormondy**, 1976). The productivity and community stability of a boreal coniferous forest are lower than those of any other biome.

(ii) Temperate deciduous forest. The temperate forest biomes are characterized by a moderate climate and broad-leaved deciduous trees, which shed their leaves in fall, are bare over winter and grow new foliage in the spring. These forests are the characteristics of North America, Europe, Eastern Asia, Chile, part of Australia and Japan, with a cold winter and an annual rainfall of 75—150 cm and a temperature of 10—20°C. In these biomes the precipitation may be fairly uniform throughout the year. In India, at elevations of 9000' (=2743.2m)—12,000' (=3655.6m) in Himalayas occur temperate vegetation including pines, fir, yew and juniper trees with an undergrowth of scrubby rhododendrons.

Soils of temperate forests are podzolic and fairly deep. Trees are quite tall—about 40—50m in height and their leaves are thin and broad. The predominant genera of this biome are maple (*Acer*), beech (*Fagus*), oak (*Quercus*), hickory (*Carya*), basswood (*Tilia*), chestnut (*Castanea*), cottonwood (*Populus*), sycamore (*Platanus*), elm (*Ulmus*) and willow (*Salix*). In some locations, coniferous vegetation may be quite predominant and that includes white pine (*Pinus strobus*), hemlock (*Tsuga canadensis*) and red cedar (*Juniperus virginianus*). There are not many epiphytes or lianas save for some species of mosses, algae and lichens growing on tree trunks, and a few vines, notably *Vitis*, the grape. The understory of shrubs and herbs in the deciduous forest is typically well-developed and richly diversified, with a considerable portion of the photosynthesis and flowering attuned to the short day of the spring season, prior to the leafing out of and consequent shading by the tree canopy. Accordingly, there are often two separate herb assemblages. One consists of spring flowers, which bloom before the trees have expanded their leaves and are gone by summer, and the other is adapted to the low light levels of the forest floor and lasts into the fall.

The animals originally present in temperate forests are deer, bears, squirrels, gray foxes, bobcats, wild turkey and wood peckers. Other common animals of this region are invertebrates such as earthworms, snails, millipedes, Coleoptera and Orthoptera and vertebrates like amphibians such as newts, salamanders, toads, and cricket frogs; reptiles such as turtles, lizards and snakes; mammals such as racoon opossum, pigs, mountain lion, etc., and birds like horned owl, hawks, etc. All these animals and plants show a profound seasonality; some may even hibernate throughout the winter. The range of animal size and adaptations is wide; the largest animals include such forms as the deer and black bear. The dominant carnivores are large, including the wolf and mountain lion, although smaller carnivores such as fox and skunk are also common. Diversity of fauna is lower than in any of the rain forests and a few species seems clearly to be dominant.

(iii) Temperate evergreen woodland (Chaparral). Many parts of the world have a Mediterranean-type climate with warm, dry summers and cool, moist winters. These are commonly inhabited by low evergreen trees with small hard needles or slightly broader leaves. The most important area of



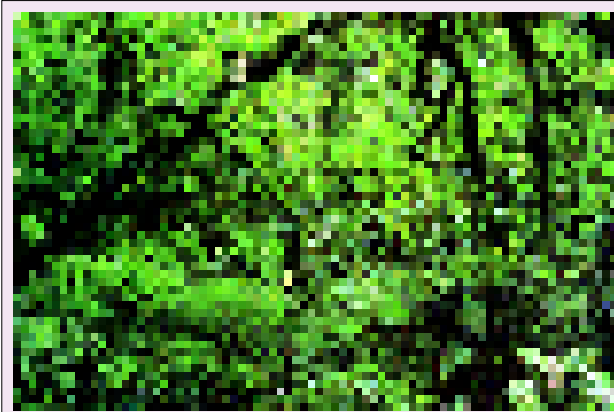
Temperate deciduous forest in Rhode Island during winter, spring, summer and fall.

tropical evergreen woodland in North America is the '**chaparral**' of the Pacific coast, the Mediterranean '**maquis**', Spanish '**encinar**' and '**melle scrub**' on Australia's South coast are the same type of community. In such a woodland, trees are essentially lacking, although shrubs may range upto 3—4 m in height. Species diversity is roughly intermediate between that of a temperate deciduous forest and a drier grassland. Fire is an important factor in this ecosystem, and the adaptations of the plants enable them to regenerate quickly after being burned. The characteristic animals of temperate evergreen woodland or chaparral are mule deer, brush rabbits, wood rats, chipmunks, lizards, wem-tits and brown towhees. Small-hooved cursorial ungulates are the dominant herbivores. Saltatorial (jumping) animals and many fast moving ungulates are also common in this fauna.

(iv) Temperate rainforests. The temperate rainforest is a colder ecosystem than any other rainforest. Such a forest has a definite seasonality, with both temperatures and rainfall varying throughout the year. Rainfall is high, but fog maybe very heavy and actually more important as a source of water than rainfall. The diversity is much lower, both in plants and animals, in comparison to warmer rainforests, yet it remains still higher than other temperate forest types. The diversity is much lower, both in plants and animals, in comparison to warmer rainforests, yet it remains still higher than other temperate forest type. The dominant trees (canopies) are coast redwood (*Sequoia sempervirens*) of the Pacific coast of North America and the alpine ash (*Eucalyptus regnans*) of Australia and Tasmania, both of which reach more than 100m in height. Epiphytes and lianas are common but are not abundant like those of other rainforest. The animals of temperate rainforests are similar to those of deciduous forests, but show a somewhat higher diversity.

(v) Tropical rainforests. Tropical rainforests occur near the equator in Central and South America, Central and Western Africa (Congo, Zambesi river), Southeast Asia (parts of India and Malaysia), Malaya, Borneo, New Guinea and Northwest Australia. Tropical rainforests are among the most diverse communities on earth. Both temperature and humidity are high and constant. The annual rainfall which exceeds 200 to 225 cm is generally evenly distributed throughout the year.

The flora is highly diversified : a square mile may contain 300 different species of trees, a diversity unparalleled in any other biome. The extremely



A Tropical rainforest.

dense vegetation of the tropical rainforests remains vertically stratified with tall trees often covered with vines, creepers, lianas, epiphytic orchids and bromeliads. Under the tall trees is a continuous evergreen carpet, the canopy layer, some 25 to 35 meters tall. The lowest layer is an understory of trees, shrubs, herbs like ferns and palms, all of which become dense where there is a break in the canopy. Nearly all plants are evergreen, and those that do lose their leaves entirely do so at irregular intervals with no apparent regard to the climatic regime. The leaves of most plant are of moderate size, leathery and dark green in colour. Their roots are often shallow and have swollen bases or flying buttresses.

Soils of tropical rainforests are red latosols, and they may be exceedingly thick. The high rate of leaching makes these soils virtually useless for agriculture purposes, but if they are left undisturbed the extremely rapid cycling of nutrients within the litter layer (due to decomposition) can compensate for the natural poverty of the soil. It is the nature of the soil, both its potential for high leaching as well as its chemical composition that promotes a rock-like quality when exposed to air, that largely has prevented western style agriculture from being applied to the tropical forests (**Kormondy**, 1976).

Invertebrate density and abundance are very high in tropical rainforests, but while vertebrates are diverse, they are not as abundant as in many other communities. The common invertebrates of these forests are worms, snails, millipedes, centipedes, scorpions, isopods, spiders, insects, planarians and leeches. Amount insects, heteropterans, orthopterans, blattids, mantids, phasmids, bees, termites and ants are most common. The common vertebrates of tropical rainforests are the arboreal amphibian *Rhacophorus malabaricus*; aquatic reptiles, chameleons, agamids, geckoes, and many species of snakes; many species of birds, social birds being predominant; and a variety of mammals. Nocturnal and arboreal habits are most common in many mammals such as insectivores, leopards, jungle cats, ant-eaters, giant flying squirrels, monkeys and sloths. But in New Guinea and Northern Queensland, where monkeys are absent, there are arboreal kangaroos (*Dendrolagus* sp.), despite the fact that the basic anatomy of the kangaroo is not particularly well-suited to arboreal life. Further, in the foot hills of the forest zone of peninsular India covered with dense tropical vegetation, we have the tiger (*Panthera tigris*), the elephant (*Elephas maximus*), sambar deer (*Rusa unicolor*), muntjac (*Muntiacus muntjak*), the gaur (*Bibos gaurus*), the chital or spotted deer (*Axis axis*), and the swamp deer (*Rucervus duraucelli*) as the major ground dwelling mammals (see **Ananthakrishnan** and **Viswanathan**, 1976).

(vi) Tropical seasonal forests. Tropical seasonal forests occur in region whose total annual rainfall is very high, but segregated into pronounced wet and dry periods. Tropical seasonal forests are found in Southeast Asia, Central and South America, Northern Australia, Western Africa and the tropical islands of the specific as well as India and Southeast Asia. In exceedingly wet tropical seasonal forests, commonly known as **monsoon forests**, the annual precipitation may be several times higher than that of the tropical rainforests. Trees may reach heights over 40 m, but are more commonly 20-30m high. Stratification is of a relatively simple type with a single understory tree layer, canopy is deciduous and understory is evergreen. Teak is often a major large tree in the best-known tropical seasonal forests, those of India (Central India) and Southeast Asia. Bamboo is also an important climax shrub in these areas although in other areas it is important only in earlier stages of the succession.

(vii) Subtropical rainforests. In regions of fairly high rainfall but where temperature differences between winter and summer are less marked, as in Florida (USA), the broad-leaved evergreen subtropical biome is found. The vegetation includes mahogany, gumbo limbo, bays, palms, oaks, magnolias, tamarinds, all laden with epiphytes (of pineapple and orchid families), ferns, vines and strangler fig (*Ficus aureus*). This stratification is simpler, with only one understory tree horizon. All these plants tend to be evergreen, but may lose their leaves during the dry season. Animal life of subtropical forest is very similar to that of tropical rainforests.

Rainforests of India. In India, patches of rainforests are found in Kerala, Assam and the Gandhamardan hills of Orissa. They are of following three types :

(a) Moist tropical forests. They include the southern tropical wet evergreen forests found in Assam and West Bengal, the northern semievergreen forests of Assam and Orissa, and the southern tropical semievergreen forests of Andamans.

(b) Montane subtropical forests. They include the northern subtropical broad-leaved wet hill forests of Assam and West Bengal, the southern subtropical broad-leaved hill forests of Orissa and Kerala and the subtropical pine forests of UP, Himachal Pradesh, Assam, Manipur, etc. The *Chir* forests of UP and HP are also example of this type of forests.

(c) Montane wet temperate forests. They include the forests of Kodaikanal and Udagamandalam in Tamil Nadu and Kerala, the northern wet temperate forest of the north-eastern region and West Bengal and the Himalayan moist temperate forests.

4. Tropical Savanna Biomes

Savannas are tropical grassland with scattered, drought resistant trees which generally do not exceed above ten meters in height and do not form a canopy. Thus, a savanna is an intermediate between a forest and a grassland. Savannas constitute extensive areas in Eastern Africa, which support the

richest diversity of grazing mammals in the world, and also occur in Australia, South America and Asia. The climate is generally characterized by a rainy (May through October) and dry (November through April) season; in the llanos of Venezuela, for example, nearly 90 per cent of the annual rainfall of 130 cm falls during the rainy season. The latosol soils



A tropical savanna biome.

of savannas are nutrient poor owing to heavy leaching. Quite widely distributed in savannas are soils called laterite which when dried harden to rock-like consistency thereby precluding their use in western style agriculture (Kormandy, 1976). Although climate and soil are significant regulating factors in this biome, the controlling factor appears to be fire, which gives grass and certain species of trees a powerful advantage over other tree species. As a result of this latter factor, species diversity is quite low in comparison to adjacent tropical forests; in some situations a single species of both grass and tree may be dominant over large areas.

The trees of savannas are resistant to desiccation, and may be either deciduous or evergreen. Their leaves are often hard and drought-resistant. Grasses are the most conspicuous plants, and may reach heights of 1½–2m. Gigantism of certain animal groups in these tropical savannas is as pronounced as it is in the boreal forest including such giants as many antelopes, giraffes, elephants, buffalo and lions (Clapham, Jr., 1973). In addition, a rich insectan fauna especially those with strong mandibles capable of mastication are conspicuous in this biome. Grasshoppers and termites are encountered in large numbers.

Savannas of India. The savannas of India are dominated by grasses and sal trees and the consumers are cattle, rodents, insects, jackals, hyaenas, etc. They are classified into the following types :

(1) High savanna. It occurs in the Brahmaputra valley and constitutes open stands of low branching trees which are usually 2 to 3 metres tall, such as *Syzigium cerasoideum* and *Emblica officinalis*. Its common grasses are *Imperata*, *Saccharum spontaneum*, *Ophiurus* and *Vetiveria*.

(2) Moist sal savanna. It occurs in the Gangetic plain and consists of open forests of sal (*Shorea robusta*) having tall grasses. Its common grasses are *Imperata cylindrica*, *Themeda arundinacea*, *Cymbopogon nardus*, *Erianthus* and *Apluda*.

(3) Low alluvial savanna woodlands. It occurs in the Gangetic plain and on riverine flats. The soil of this savanna is sandy and alluvial and contains patches of clay in depressions. The common trees of this region are *Dalbergia sissoo*, *Butea monosperma*, *Albizia*, *Adina cordifolia* and *Zizyphus procerum*, *Arundinella*, *Themeda gigantea* and *Erianthus* sp.

(4) Dry savanna. This region is found in Punjab, Haryana, Bihar, Orissa and eastern Tamil Nadu. It is characterized by trees which stand far apart, singly or in small groups such as *Acacia lenticularis*, *Emblica officinalis*, *Gardenia turgida*, *Crotalaria hirta* and *Pterocarpus marsupium*. The common and abundant grasses of this region are *Themeda triandra*, *T. quadrivalves*, *Apluda mutica* and *Arundinella setosa*.

(5) Saline alkalina scrub savanna.

It occurs throughout the Indo-Gangetic Plain. The common tree of this region are the *Phoenix sylvestris*, *Acacia* sp., *Tamarix* sp., *Calotropis procera*, *C. gigantea* and *Kochia indica*.

5. Grassland Biomes

The grassland biome is found where rainfall is about 25 to 75 cm per year, not enough to support a forest, yet more than of a true desert. The seasonality of grasslands is pronounced, both with respect to rainfall, which is concentrated in the summer and to temperature. Grasslands typically occur in the interiors of continents and include the **tall grass prairies**, **short grass prairie** or great plains and **arid grassland** of North America as well as the

steppes of Eurasia (Southern Russia, Siberia and Asia), **puszta** in Hungary, the **veldt** of Africa and the **pampas** of South America (Argentina). The grassland communities are open land communities with limited moisture conditions, irregular rainfall, sharp seasonal and diurnal variations and very high radiation. Vegetation of grassland is dominated by grasses, legumes and composites. Since tall trees or other thick vegetations are excluded from these communities there is a free movement of air. These winds carry the particles of sand or dust. These open habitats provide natural pasture for grazing animals (herbivore) which are excluded from predation by predators which hide in thick vegetation and prey upon them.

In grasslands, the stratification is reduced essentially to a single story, but within that level, species diversity may be as high as in a deciduous forest—especially for the tall grass prairies. Only along the streams are trees to be found, but the “gallery forests” within a few meters of stream bank are characteristics of grasslands. The grasses comprising of the grassland can be divided into two basic groups, the tall grasses more than 1m high, which are found in moist portions of the grassland, and the short grasses less than 1 m high which are found in the drier regions. For example, prior to its conversion to agriculture and urban development, the tall grass prairie toward east of North America was dominated by species of bluestem (*Andropogon*) forming dense covers 122 cm to 183 cm tall. Westward, Buffalo grass (*Buchloe dactyloides*) and other grasses but a few centimeters high dominated the landscape. Flowering herbs including many kinds of composites are common, but much less important than grasses.

Since the precipitation—evaporation ratio is below one in the grassland, leaching is considerably less. The soils of the grasslands are rich, fertile prairie soils (black-earths) and chernozems. Organic matter (humus) accumulates in the upper portion of the soil, rendering it dark; this upper portion remains neutral to slightly alkaline because of the continued replenishment of cations like calcium and potassium through the upward movement associated with evaporation.

Typical animals of grasslands tend to be quite small, with the exception of a few very large cursorial herbivore mammals such as the bison and pronghorn in North America, the wild horse, ass, and saiga antelope of Eurasia and zebra and antelops of Southern Africa. The large herbivores are no where near as diverse as they are in savanna areas. Likewise, carnivores are relatively small such as are coyotes, weasels, badgers, foxes, ferrets, owls and rattle snakes. Rodents such as prairie dogs, rabbits, and ground squirrels are common. Most herbivores characteristically aggregate into herds or colonies; this aggregation provides some protection against predators. The characteristic birds of grasslands are prairie chickens, meadow larks, and rodent hawks. Further, saltatorial motion is widespread, not only



Sheep grazing on a temperate grassland
(short grass prairie).

in mammals such as rabbits and kangaroo rats, but also in insects, such as grasshoppers and crickets. This probably has to do with increasing rate at which animal can move through or over the tall grasses, as well as making it harder for a predator to catch the animal. Visibility in the grasslands is exceedingly good and the premium placed on efficiency including predators is very high.

The common insects of steppes are termites, locusts (*Locusta migratoria*, *Schistocerca gregaria* and *Melanoplus* sp.), bees, burrowing wasps, mutilid wasps, bumble bees and blister bees. The steppes also harbour an undisturbed reptilian fauna. Lizards and snakes are met with in large numbers and possess, remarkable diversity. Most of them are fossorial, insectivorous and carnivorous.

Types of grasslands. Grasslands tend to occupy about 20% of the earth's land surface and are of three types :

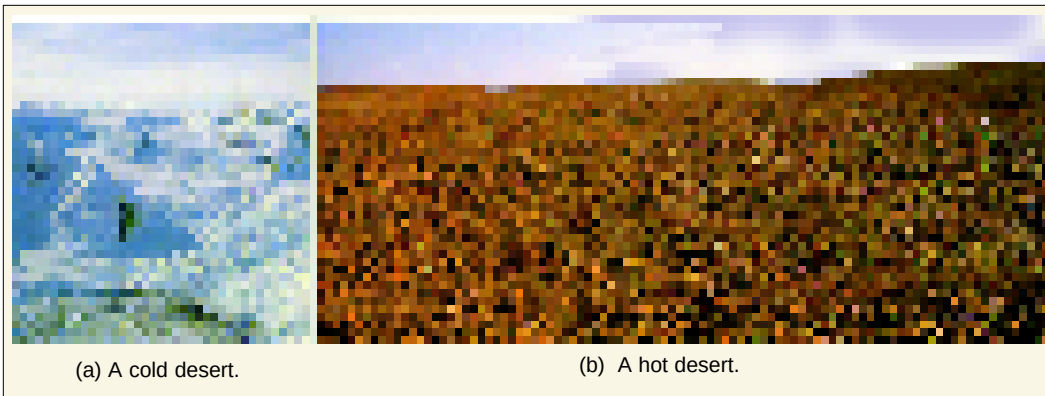
(i) Tropical grasslands. These are situated 20 degrees away from the equator and the rainfall varies from 40 to 100 cm. Tall grasses rise to a height of about 1.5 to 3.5 metres. The tropical grasslands of Africa include ungulates, deer, antelopes, giraffes, lions, etc.

(ii) Temperate grasslands. They are found in Europe, Asia and North America. Their rainfall varies from 25 to 75 cm per year.

(iii) Alpine grasslands. They occur at higher latitudes. They are of the meadow type and many flowering herbs also grow here.

6. Desert Biomes

Deserts are the biomes formed in the driest of environments. Temperature may range from very hot as in **hot deserts** to very cold as in **cold deserts**. Major hot deserts of the world are situated near the tropics of cancer and capricorn, with a rainfall of less than 10 mm. The most important hot deserts



of world is the Sahara-Arabia-Gobi desert complex extending from Africa to Central Asia and contains highly irregular and very insignificant rainfall, and low humidity due to excessive evaporation. Fairly extensive hot deserts also occur in India (Sind-Rajasthan deserts), South America (Chile), North America and Australia. The cold deserts occur at high elevations where the temperatures are low and rainfall scanty as the air loses all its moisture content as it ascends higher and higher. Cold deserts occur in Ladakh regions of Himalayas, Tibet, and Bolivia Arctic. The hot and cold deserts may also be distinguished by differences in plant population which are mostly succulent type (e.g., cactus, palo verde trees, creosote bush, etc.). Most cold deserts have sage brush.

Low abnormal precipitation coupled with soil and air temperatures that are extremely high by day and drop abruptly by night, low humidity, and high insolation are the major desiccating environmental factors to which desert vegetation and animals have adapted. Desert plants which tend to be shrubs are adapted to drought conditions through reduced leaf size and the dropping of leaves in dry conditions, both reducing water loss via evapotranspiration. The roots of most desert plants remain well developed and occur in the top meter of the soil to take maximum possible advantage of any rainfall. Further, the

root hairs on many desert plants are ephemeral, drying back under drought conditions and thereby reducing potential water loss by osmosis. Yet other species are short lived annuals that complete their life cycles during the short-moist period. In most hot deserts, there occur plants such as cacti, water-storing succulents such as acacias, euphorbias, cacti, prickly pears, etc., which are adapted by their protoplasmic colloids, which enable the accumulation of substantial water reserves, as well as by a reduced leaf surface, which obviates water loss via evapotranspiration.

The animals present in the desert are reptiles, insects and burrowing rodents. All these animals possess special morphological, physiological and ethological adaptations for deserts. In general, large animals are very uncommon except mule deer and some species of gazelle and all animals have cursorial, fossorial and/or saltatorial adaptations. Some desert animals are nicely adapted for high extremes of temperature. For example, the lethal temperature of different species of insects found to be following—for the canal spider *Galeodes granti* it is 50°C, for the *Gryllus domesticus* it is 40°C and for the forficulid *Labidura riparia* it is 38°C (Cloudsely-Thompson, 1962). Diurnal rhythms are perhaps the best method of avoiding the heat. While some desert plants close their petals at night, many blossom only at night. Some insects such as tenebrionid beetle *Akis spinosa* remain active during the day, and the centipede *Scolopendra clavipes* restricts its activities to night time.

Further, certain reptiles and certain insects are well adapted for survival in deserts because of their thick, impervious integuments and the fact they excrete dry waste matter. A few species of mammals have become secondarily adapted to the desert by excreting very concentrated urine. They avoid the sun by remaining in their burrows during the day. Kangaroo rat and pocket mouse, both are able to live without drinking water by extracting the moisture from the seeds and succulent cactus they eat. The camel and the desert birds (ostrich, etc.) must have an occasional drink of water but can go for long periods of time using the water stored in the body. Most insects of deserts are herbivores and as a correlation the number of small insectivorous lizards found in the desert is usually high.

Fauna of Indian deserts. Prakash (1974) have recognized four habitat types in the Indian desert based on land forms namely **aquatic, sandy, rocky** and, **ruderal habitats**. The aquatic habitats of Indian deserts which remain confined to the perennial lakes, are inhabited by various species of fishes such as *Labeo nigripinnis*, *Oxygaster clupeoides*, *Tor khurdee*, *Puntius amphibia*, *Noemacheilus denisonii*. Among amphibians of the Indian desert are a species of toad (*Bufo andersonii*) and five species of frogs (Indian bull frog *Rana tigrina*, cricket frog *Rana limnocharis* and burrowing frog *Rana breviceps* being dominant species). Among reptiles, there occur two species of testudines (Loricata), 18 species of lizards and 18 species of snakes. Of the lizards, some species such as *Acanthodactylus cantoris*, *Calotes versicolor*, *Uromastix hardwicki*, *Ophiomorus tridactylus* are predatory on the desert locust inhabiting localized areas in the Thar desert. The avian fauna of the Indian desert includes species such as painted partridge *Francolinus pictus pallidus*, grey partridge *Francolinus pondicerianus*, common quail *Coturnix coturnix*, black breasted quail *Coturnix coromandelicus*, rock bush quail, the little bustard quail *Lurnix sylvatica*, button quail *Turnix tanki*, great Indian bustard *Choriotes nigriceps*, florican *Sypheotides indica*, sandgrouses such as *Petrocles exustus erlangeri*, *Petrocles alchata*, *P. senegallus*, and *P. indicus indicus*. During winter numerous aquatic birds become abundant in Indian deserts. Such birds are white fronted goose *Anser albifrons*, wigeon *Anas penelope*, garganey *anas querquedula*, red crested pochard *Netta fufina* and tufted duck *Aythya futigola*. Among the predominant predatory birds are two species of the vultures namely *Gyps bengalensis* and the white vulture *Neophron percnopterus*.

The mammalian fauna of Indian deserts includes many species, some of which have been tabulated in Table 13-2.

All these mammals have well specialized adaptations for survival in thermal extremes and low humidity. Many of them possess diurnal rhythmic activities.

7. Wetland Biomes

Wetlands may be defined as submerged or water saturated lands, natural and artificial, permanent or temporary with water that is static or flowing and fresh, brackish or saltish. They are

Table 13-1. Certain common mammals of Indian deserts (**Prakash, 1974**).

	Order	Species	Popular Name
1.	Chiroptera	<i>Rasetius arabicus</i>	Fruit bat
		<i>Pteropus giganteus giganteus</i>	Fruit bat
		<i>Rhinopoma kinneari</i>	Rat-tailed bat
		<i>Rhinopoma hardwicki</i>	The lesser rat-tailed bat
		<i>Taphozour perforatus</i>	Tomb bat
		<i>Megaderma lyra</i>	Indian false vampire
		<i>Rhinolophus lepidus lepidus</i>	Horse-shoe-bat
2.	Insectivora	<i>Hemiechinus auritus lepidus</i>	Long ear hedgehog
		<i>Paraechinus micropus</i>	Long ear hedgehog
		<i>Suncus muinus sindensis</i>	House Shrew
3.	Pholidota	<i>Manus crassicaudata</i>	Scaly ant eater
4.	Lagomorpha	<i>Lepus nigricollis caudatus</i>	Hare
		<i>Lepus nigricollis dayanus</i>	Hare
5.	Rodentia	<i>Funambulus pennanti</i>	Northern palm squirrel
		<i>Hystrix indica indica</i>	Indian crested porcupine
		<i>Rattus rattus</i>	House rat
		<i>Mus musculus rufescens</i>	House mouse
		<i>M. musculus bactrianis</i>	House mouse
		<i>Gerbillus gleadovi</i>	Indian hairy footed gerbil
		<i>Tatera indica</i>	Gerbil
		<i>Merriones hurrianæ</i>	Gerbil
6.	Artiodactyla	<i>Sus scrofa cristata</i>	Wild boar
		<i>Boselaphus tragocamelus</i>	Blue bull
		<i>Antilope cervicapra rajputani</i>	Black buck
		<i>Gazella gazella bennetti</i>	Gazelle
7.	Carnivora	<i>Canis lupus</i>	Wolf
		<i>Canis aureus</i>	Asiatic jackal
		<i>Vulpes vulpes</i>	Red fox
		<i>Vulpes bengalensis</i>	Bengal fox
		<i>Lutra perspicillata</i>	Indian otter
		<i>Viverricula indica</i>	Small Indian civet
		<i>Herpestes aeropunctatus</i>	Small Indian mongoose
		<i>Felis chaus prateri</i>	Jungle cats
		<i>Felis libyes ornata</i>	Jungle cats
		<i>Panthera pardus sindica</i>	Panther

characteristic transitional conditions between terrestrial land and deep water bodies and are represented by **swamps**, **marshes** (or palustrine lands), **fens** (i.e., tract of marshy land), **Peatlands**, **lagoons** (i.e., a shallow lake, especially one near and communicating with the sea or river), **lakes**, etc. Wetlands are also represented by paddy fields, fishery ponds, *Trapa* cultivation ponds, riverine flood plains, lacustrine (*L.lacus*= a lake) marshy lands, marine backwaters and coastal wetlands. Wetlands are often seasonal, i.e., with free overlying water in the rainy season, marshy and often almost dry in

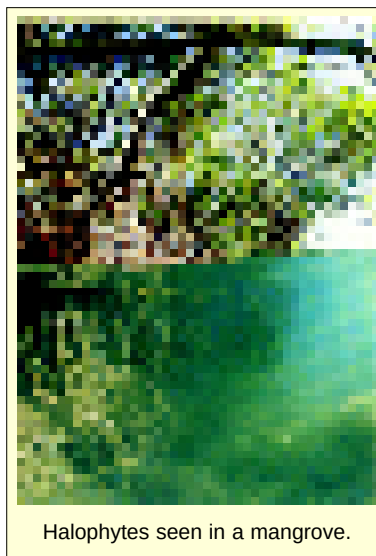
summer. Wetlands support specialized vegetation and fauna; they are suitable niche for fish and other aquatic fauna, as breeding and nursery ground for water fowls and as filters for sediments and pollutants. Flood plains and marshes are sometimes not easily accessible and, hence, serve as a secure place for many kinds of wild life. Sunderbans is one such habitat in the coastal belt of the Bay of Bengal. Since wetlands contain supraoptimal water, they are quite productive. In tropics (e.g., India), because of associated bright sunshine and warm condition their primary productivity becomes very high. Rice which is the major crop of India, is best grown in wetlands. Biomass of wetlands is well distributed between the above ground foliage and underground rhizomatous and cormous parts. *Typha* and *Phragmites* have very high production efficiency. *Euryale ferox*, a prized dry fruit crop is most extensively grown in wetlands of north Bihar. Wetlands are fast disappearing in many parts of world because of dumping of wastes and reclamation for agriculture, housing, forestation, etc. The flowing or lotic wetlands are most easily polluted by city sewage, industrial effluents and runoff pesticides and other agrochemicals applied in the drained upland crop fields.

8. Mangroves in India

Halophytes growing in the muddy swamps of the estuaries of tropical and subtropical regions and sea coasts flooded by rivulets and tides form the **mangroves**. Mangroves are, thus, the salt-tolerant forest ecosystems found mainly in tropical and subtropical inter-tidal regions. They comprise the swamps, forest land within and its water spread areas. In India, the total area of mangroves is about 6740 sq. km., which is nearly 7% of the world's mangroves. The mangrove areas of Sunderbans (W. Bengal) and Andaman and Nicobar Islands constitute about 80% of the Indian mangroves, the rest are scattered in the states of Andhra Pradesh, Tamil Nadu, Orissa, Maharashtra, Gujarat, Goa and Karnataka.

Mangroves are dense forests of low trees, especially adapted to (a) fixation, (b) respiration, (c) vivipary, (d) migration, and (e) xeric characters. In mangroves, the plant and animal diversity is very rich. Typical plant associations of Indian mangroves include *Avicennia officinalis*, *Avicennia alba*, *Aegiceras majus*, *Aegialitis rotundifolia*, *Acanthus ilicifolius*, *Kandelia rheedii*, *Bruguiera caryophyllodes*, *Sonneratia griffithii* and *Carapa obovata*.

Mangroves are highly productive ecosystems. The net primary production is usually high. Secondary productivity and decomposition activity of the mangroves are also very high. The terrestrial consumers are snakes, lizards, turtles, jackals, wild boars, hyaena, tigers and so on. In the Bhitarkanika sea coast of Orissa adjacent to the mangroves exists the largest sea turtle (*Lepidochelys olivacea*) rookery. Every year 300,000 to 500,000 female turtles come to the sea beach (arribada) in late December-January and again in March-April to lay about 50 million eggs (**Dash and Kar**, 1990). Birds are also the most colourful and conspicuous inhabitants of the mangroves. Mangroves of the Florida key support a fauna of mostly insects and spiders, along with scorpions, isopods and other arthropods (see **Krebs**, 1985). In recent years, the Indian mangroves have been subjected to reckless exploitation and other biotic interference.



Halophytes seen in a mangrove.

REVISION QUESTIONS

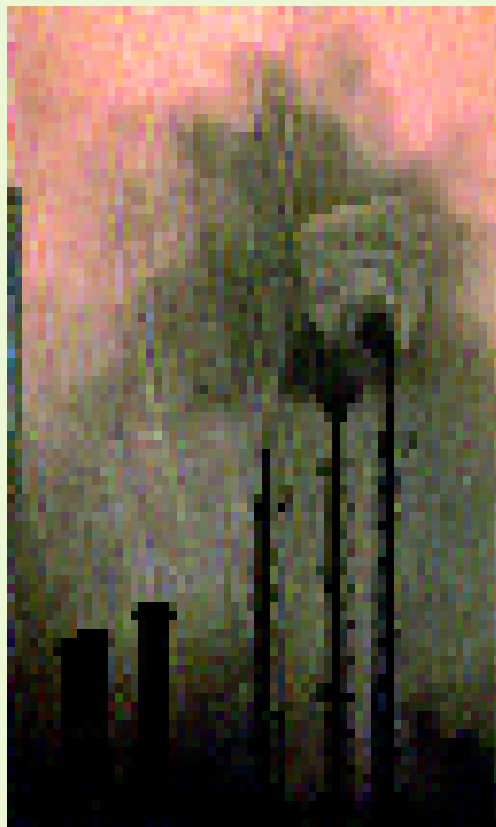
1. Define biome and describe the physio-chemical characteristics and biota of tundra biomes.
2. Describe the biota of tundra biomes.
3. Write short notes on the following :
 - (a) Alpine biome ;
 - (b) Forest biome ;
 - (c) Grassland biome ;
 - (d) Wetlands ;
 - (e) Mangroves.

CHAPTER

14

Pollution

(Environmental Pollutants
and Toxicology)



Harmful materials introduced by man into the environment is the cause of pollution.

Recently, first time in his entire cultural history, man has faced one of the most horrible ecological crisis—the problem of pollution of his environment which sometime in past, was pure, virgin, undisturbed, uncontaminated and basically quite hospitable for him. **Pollution** is an undesirable change in the physical, chemical or biological characteristics of our air, land, and water that may or will harmfully affect human life or that of desirable species, our industrial processes, living conditions, and cultural assets (**Odum**, 1971). In other words, pollution is the unfavourable alteration of our environment, largely as a result of human activities (**Southwick**, 1976).

By and large, the term pollution is used to refer to harmful materials introduced by man into the environment. Thus, in a way, pollution is the release in excess of permissible limits of foreign substances within the environment (see **Michael**, 1984).

ORIGIN OF POLLUTION

Some authors (such as **Southwick**, 1976), associate the human population explosion with the pollution problem. They point out that with more people there is more sewage, more solid wastes, more fuel being burned, more fertilizers and insecticides being used to produce more food for hungry mouths. But, there are certain writers who have pointed out that in underdeveloped countries, pollution is not the severe problem as it is in technologically developed countries and yet the populations may be very dense. They feel that it is the wasteful aspects of our technology which strive always to produce more convenient products (“disposable” items) which pollute our environment.

There are certain modern ecologists such as **Odum** (1971), **Southwick** (1976), **Smith** (1977), etc., who sought many factors such as human population explosion, unplanned urbanization and deforestation, profit oriented capitalism and technological advancement, which may have originated pollution crisis on earth. In fact, in countries where there have been the greatest technological advances, the worst pollution occurs. In these countries, whether capitalist, socialist, or communist, 'there has been an emphasis on growth. Thus, in well-developed nations on a per capita basis, citizens consume more food, use more pesticides, fertilizers, fuel, minerals, cars, and other manufactured products of all kinds. Most of these products are manufactured in one or other kinds of industries, all of which in their turn add some pollutants in our environment and cause pollution.

POLLUTANTS: THE CREATORS OF POLLUTION

Every human society, be it rural, urban, industrial and most technologically advanced society, dispose of certain kinds of byproducts and waste products which when are injected into the biosphere in quantities so great that they affect the normal functioning of ecosystems and have an adverse effect on plants, animals, and man are collectively called **pollutants** (**Smith**, 1977). A pollutant is a constituent in the wrong amount, at the wrong place or at the wrong time. For example, nitrogen and phosphorus are essential nutrients for living organisms and are extensively used in agriculture to increase crop yields but they can also cause pollution in lakes and rivers when found in excess by promoting undue algal growth (eutrophication).

Types of Pollutants

Pollutants primarily are grouped into the following two types:

1. Natural pollutants. Certain pollutants such as carbon dioxide, carbon monoxide, sulphur dioxide, lead, mercury and other trace elements are the consequence of life processes being produced through respiration, faeces, urine and body decomposition. With an increase in human population, the pollutants are increasing with alarming rate.

2. Synthetic, man-made, anthropogenic or xenobiotic pollutants. A vast array of synthetic pollutants are increasing continuously with urbanization and industrial growth. They include pesticides, detergents, pharmaceuticals, cosmetic products, organic acids, aerosols, and metals, etc. Several of these compounds are extremely stable and persist in the environment for a considerable period posing serious environmental hazards.

From the ecosystem viewpoint, these pollutants can be classified into two basic types: **nondegradable pollutants and biodegradable pollutants** (**Odum**, 1971). The materials and poisons, such as aluminium cans, mercurial salts, long-chain phenolic chemicals and DDT that either do not degrade or degrade only very slowly in the natural environment, are called **nondegradable pollutants**. Such nondegradable pollutants not only accumulate but are often "biologically magnified" as they move in biogeochemical cycles and along food chains. Also they frequently combine with other compounds in the environment to produce additional toxins. **The biodegradable pollutants** include domestic sewage, heat, etc. The domestic sewage can be rapidly decomposed by natural processes or in engineered systems (such as a municipal sewage treatment plant) that enhance nature's great capacity to decompose and recycle. Problems arise with the biodegradable pollutants when their input into the environment exceeds the decomposition or dispersal capacity.

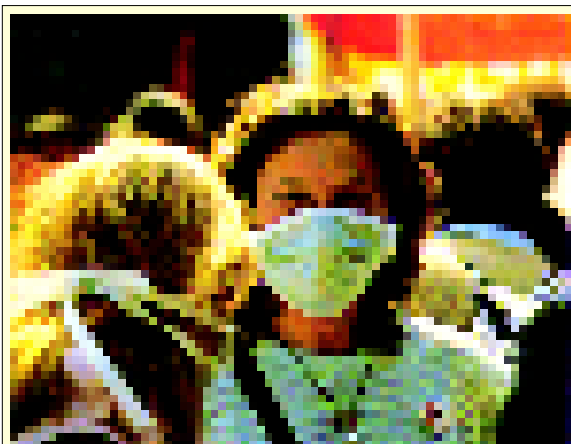
1. AIR POLLUTION

Air pollution is the presence in the atmosphere, or injection into it, of substances that are not present naturally, or present naturally but in much smaller concentrations, and that may harm living organisms directly or indirectly (**Allaby** 1995). Thus, when due to some natural processes or human activities the amount of solid waste or concentration of gases other than O₂ increase in the air which

normally has constant percentage of different gases in it, the air is said to be polluted and this phenomenon is referred to as **air** or **atmospheric pollution**. Air pollution is one of the most dangerous and common kind of environmental pollution that is reported in most industrial towns and metropolitans of India and abroad such as Delhi, Bombay, Calcutta, Kanpur, Madras, Hyderabad, Jaipur, Ahmedabad, Nagpur, Firozabad and also in London, New York, Tokyo, Pittsburg, etc.

Air Quality

In our country, data on air quality have been collected by NEERI (1978-79, 1980-81). Although there exist several parameters to determine air quality, but only three, i.e., SO_2 , NO_x and SPM (= suspended particulate matter) are used; these give fair idea of load of pollution carried by the air.



Vehicle exhausts release a collection of chemicals including carbon monoxide, sulfur dioxide, nitrogen oxides and hydrocarbons. Some gasoline also contains lead, which appears in the exhaust fumes and can have adverse effects on brain development in children. Throughout the world's cities, many people – such as cyclists, who have to breathe deeply while in close proximity to vehicle exhausts – have begun to wear masks to filter the air they breathe. Here, a Green Party protestor against the poor quality of city air in Rome emphasizes the point.

NEERI has collected data from ten Indian cities/Metropolitans such as Delhi, Kanpur, Calcutta, Nagpur, Hyderabad, Madras, Cochin, Bombay, Ahmedabad and Jaipur to assess the extent and nature of degradation of air quality due to industrialization and urbanization. Each of these ten Indian cities was surveyed for residential, commercial and industrial situations.

Thus, for sulphur dioxide (SO_2), Calcutta is found to be most polluted city, followed in a descending order by Bombay, Delhi, Ahmedabad, Kanpur, Hyderabad, Madras, Nagpur and Jaipur. However, yearly averages did not exceed $80 \mu\text{g}/\text{m}^3$. Levels of oxides of nitrogen (NO_x) ranged from $4 \mu\text{g}/\text{m}^3$ in residential and industrial areas of Jaipur to the highest $40 \mu\text{g}/\text{m}^3$ in industrial areas of Ahmedabad (1980) and commercial areas in Kanpur (1980). Level of SPM was highest in Delhi and Calcutta and the lowest in Madras and Bombay (coastal cities). In general, SPM levels in all cities were much above the international levels. Resi-

dents of Delhi are exposed to some of the highest levels of air pollution in the country and perhaps the world.

Methods of Detection and Measurement of Air Pollution

Air pollution is usually measured by sampling of air by *thermal* and by *electrostatic precipitation*, *Sonkin impactor* and *electrostatic dust collectors*. The particulate pollution is measured by the instrument called *deposit gauge* or by *Owen's dust counter*. The thickness of the smoke is measured by *Liegean sphere* and by *Ringelmann chart*. The rough estimation of SO_2 in air can be made by chemical analysis of the dust collected in a deposit gauge or by a *bubbler method*. Fluorides are estimated by colour reactions.

Sources of Air Pollution

1. Air Pollution by Natural Means

Nature adds few natural pollutants such as pollen, hydrocarbons released by vegetation, dusts from deserts, storms, and volcanic activity. Thus, **volcanic eruptions** may eject large amounts of gases

and particulate matter. Settling volcanic ash can kill vegetation by coating leaves and preventing photosynthesis and transpiration (*e.g.*, following the 1980 eruption of Mt. St. Helens, USA). Fine particles, mainly of sulphates, may penetrate the stratosphere, spread widely and reflect significant amounts of solar radiation, leading to climatic cooling (*e.g.*, the 1991 eruption of Mt. Pinatubo, Philippines). Likewise, **dust storms** sometimes carry fine sand for thousands of kilometers and favourable weather conditions stimulate the release of pollen, affecting people sensitive to it.

2. Air Pollution by Human Activities

(a) Industrial chimney wastes. There are a number of industries which are potent sources of air pollution. **Petroleum refineries** are the major sources of gaseous pollutants (*e.g.*, SO_2 , NO_x , etc.) Mathura-based petroleum refinery has been accused to aggravate the pollution-related decay of Taj Mahal in Agra and other historical monuments of Fatehpur Sikri Complex. **Industrial processors** such as metallurgical plants and smelters, chemical plants, petroleum refineries, pulp and paper mills, sugar mills, cotton mills, and synthetic rubber manufacturing plants are responsible for about one fifth of the air pollution. **Cement factories** emit plenty of dust, which is potential health hazard. Stone crushers and hot mix plants also create a menace. The SPM levels in such stone crushing areas are found to be five times the industrial safety limits. **Chemical manufacturing industries** emit acid vapours in air.

(b) Thermal power stations. The coal consumption of thermal power stations of India (*e.g.*, Delhi has three thermal power stations, one at Indraprastha Estate, others at Rajghat and Badarpur) is several million tonnes. The chief pollutants of coal burning are fly ash, SO_2 and other gases (CO , NO_2), aldehydes and hydrocarbons (**Chaudhuri**, 1982).

(c) Automobiles. The transportation industry exclusive of automobiles and including railroads, ships, aircrafts, trucks, buses, tractors, etc., contribute the same type of pollutants as cars (see **Smith**, 1977). The vehicular exhausts are toxic being a source of considerable air pollution, next only to thermal power plants. The ever increasing vehicular traffic density poses continued threat to the surrounding air quality. At the global level, there are over 300 million cars, trucks and buses and their number is increasing rapidly. India too has millions of vehicles, of which more than 65% are two wheelers operating on petrol.

The sources of emission in the automobiles are (i) exhaust system; (ii) fuel tank and carburettor and (iii) crankcase (Fig. 14.1). The exhaust produces many air pollutants including unburnt hydrocarbons, CO , NO_x and lead oxides. There are also traces of aldehydes, esters, ethers, peroxides and ketones; these are chemically active and combine to form smog in presence of light. Due to volatile nature of petrol, evaporation from fuel tank goes on constantly and results in emission of hydrocarbons. The evaporation through carburettor occurs when engine is stopped and heat builds up and as much as 12 to 40 ml of fuel (petrol/diesel) is lost during each long stop causing emission of hydrocarbons. Some gas vapour escapes between walls and the piston, which enters the crankcase and then

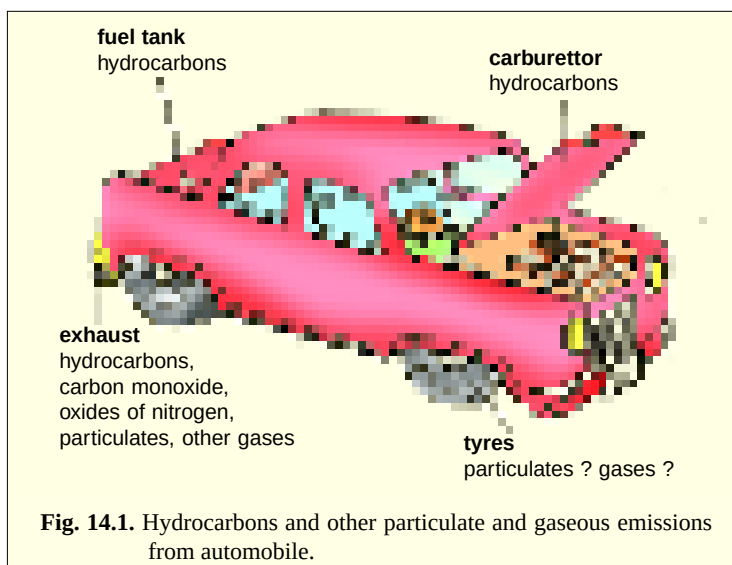


Fig. 14.1. Hydrocarbons and other particulate and gaseous emissions from automobile.

discharges into the atmosphere. Thus, the total hydrocarbon emission of an engine reaches upto 25%.

Other sources of air pollution are minor in quantities but bear significance due to the harmful substances they release, these are **agriculture**, which is responsible for pesticides, dust from agriculture practices and field burning, and the construction industry.

Types of air pollutants. All the just described sources of air pollution release the following air pollutants: 1. **Carbon compounds** (*e.g.*, CO₂, CO); 2. **Sulphur compounds** (*e.g.*, SO₂, H₂S and H₂SO₄); 3. **Nitrogen oxides** (*e.g.*, NO, NO₂ and HNO₃); 4. **Ozone** (O₃); 5. **Fluorocarbons**; 6. **Hydrocarbons** (*e.g.*, benzene, benzopyrene, etc.); 7. **Metals** (*e.g.*, lead, nickel, arsenic, beryllium, tin, vanadium, titanium, cadmium, etc.); 8. **Photochemical products** (*e.g.*, olefins, aldehydes, photochemical smog, PAN, PB₂N, etc.); 9. **Particulate matter** (*e.g.*, fly ash, dust, grit and SPM); and 10. **Toxicants**.

Some of the most common air pollutants, their sources and their effects on human health have been tabulated in Table 14-1.

Table 14-1.

Common air pollutants, their sources and pathological effects on man
(Source: Southwick, 1976)

Pollutants	Where they come from (source)	Pathological effect on man
1. Aldehydes	Thermal decomposition of fats, oil, or glycerol.	Irritate nasal and respiratory tracts.
2. Ammonias	Chemical processes—dye-making; explosives, lacquer; fertilizer.	Inflame upper respiratory passages.
3. Arsines	Processes involving metals or acids containing arsenic soldering.	Break down red cells in blood, damage kidneys; cause jaundice.
4. Carbon monoxides	Gasoline motor exhausts; burning of coal.	Reduce oxygen-carrying capacity of blood.
5. Chlorines	Bleaching cotton and flour; many other chemical processes.	Attack entire respiratory tract and mucous membranes of eyes; cause pulmonary edema.
6. Hydrogen cyanides	Fumigation; blast furnaces, chemical manufacturing; metal plating.	Interfere with nerve cells; produce dry throat, indistinct vision, headache.
7. Hydrogen fluorides	Petroleum refining; glass etching; aluminium and fertilizer production.	Irritate and corrode all body passages.
8. Hydrogen sulphides	Refineries and chemical industries; bituminous fuels.	Smell like rotten eggs; cause nausea; irritate eyes and throat.
9. Nitrogen oxides	Motor vehicle exhausts; soft coal.	Inhibit cilia action so that soot and dust penetrate far into the lungs.
10. Phosgenes (carbonyl chloride, COCl ₂)	Chemical and dye manufacturing.	Induce coughing, irritation, and sometimes fatal pulmonary edema.
11. Sulphur	Coal and oil combustion.	Cause chest constriction, headache, vomiting, and death from respiratory ailments.
12. Suspended particles (ash, soot, smoke)	Incinerators; almost any type of manufacturing.	Cause emphysema, eye irritations and possibly cancer.

Ecology of Air Pollution

Once injected into the atmosphere, pollutants enter the biogeochemical cycles by different routes. The air above many cities can assimilate and disperse great quantities of fine particulate and

gaseous pollutants as long as air can move and disperse (Fig. 14.2). But if air masses over cities become stagnant, pollutants accumulate quickly and deteriorate air quality which cause many respiratory diseases in man and other animals. Air pollutants also accumulate during **tem-**

perature inversions, when cooler surface layers of air become trapped under warmer upper layers (Fig. 14.3). In these situations, the upper layers of warm air prevent the vertical rise and dispersal of pollutants which are held near the ground. Temperature inversions commonly occur in cities surrounded by mountains or bordered by mountains on the leeward side.

Further a portion of air pollutants reaches land as dry fallouts; it may then enter various nutrient cycles and food chains through water and soil. Other contaminants of air react chemically or

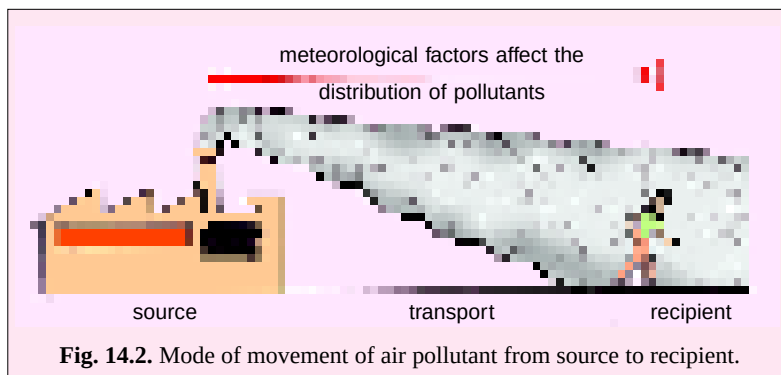


Fig. 14.2. Mode of movement of air pollutant from source to recipient.

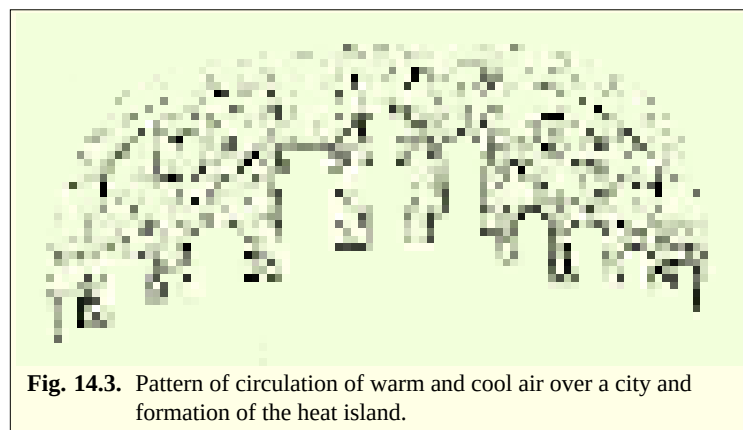


Fig. 14.3. Pattern of circulation of warm and cool air over a city and formation of the heat island.

photochemically with each other and produce such secondary pollutants as sulphuric acid, ozone, and, peroxyacetyl nitrate or PAN. Aerosols and other forms of fine particulate matter act as **condensation nuclei**, to which water vapours present in the air quickly surround to form droplets of fog or rain.

Moreover, different air pollutants adversely affect flora, fauna and climate

of a given area variously and some of the common air pollutants and their specific effects on man, vegetation, climate, etc., have been discussed as follows:

A. Gaseous Pollutants

1. Sulphur oxides and hydrogen sulphide. These gaseous pollutants are naturally released by the biological decomposition and from volcanic eruptions. They are also released artificially due to human activities such as smelting of sulphide-containing ores, combustion of sulphur-containing fuels such as coal and oil, petroleum refining and obtaining of geothermal energy.

According to a report the sulphur dioxide (SO_2) content of the atmosphere in Delhi city has reached the level of 0.233 ppm, whereas in USA and West Germany the permissible limit is only 0.1 and 0.05 ppm. Concentrations as low as 0.3 ppm may damage plants. Lichens are particularly sensitive to SO_2 and in polluted regions one does not find lichens growing on the tree trunks. Thus, lower quantities of sulphur dioxide suppresses the overall vegetative as well as reproductive growth and yield. Its high atmospheric concentrations produce various injuries to leaves such as interveinal and blade damage, necrosis of leaves and cellular collapse (Fig. 14.4). However, moderate SO_2 pollution results in chlorosis of leaves without cellular collapse. Pine trees are more susceptible than broad leaved trees and react by partial defoliation and reduced growth. Plant's exposure to hydrogen sulphide results in

leaf lesions, mottling, defoliation and reduced growth.

Sulphur dioxide pollution causes in human beings various types of injuries such as eye irritation, chest constriction, headache, vomiting and death from respiratory ailment (Fig. 14.5). It paralyzes or destroys bronchial cilia in air passages of man, constricts bronchii, damages lungs, lowers resistance to pneumonia and influenza and causes bronchitis, emphysema and irritation of the mucous membranes (*i.e.*, an increase in cough and sputum). In fact, SO_2 and other pollutants bring about coalescence of alveoli (Fig. 14.6) and reduce the amount of surface area available for the transport of oxygen and also reduce the rate at which air is exchanged. When there occurs severe pollution of SO_2 , the death rate and bronchial asthma are found to increase and in past it caused such disasters as Meuse Valley in Belgium in 1930; Donora, in 1938; London, in 1952; and New York and Tokyo in 1960s (see Southwick, 1976).

Further, in the atmosphere, SO_2 does not remain in the gaseous state for long time, but very soon it reacts with moisture to form sulphuric acid or H_2SO_4 . Sulphuric acid causes many respiratory diseases in man and also produces **acid rainfalls** over parts of the earth. In Scandinavia, downwind from the industrial centers of Britain and the Ruhr Valley, the acidity of the rainfall has increased 200 fold since 1966, with pH values as low as 2.8 being recorded (Oden and Ahl, 1970). This acid rainwater has increased the acidity of Scandinavian streams, interfering with salmon reproduction and destroying salmon runs. It has reduced forest growth and increased the amount of calcium and other nutrients leached from agricultural soil. The acid rainfall is also involved in the erosion of building materials as limestone marble, the slate used in roofing, mortar and deterioration of statues.

2. Carbon monoxide. It is released chiefly from gasoline engines and burning of coal in the defective furnaces. In man, CO produces headache, dizziness, inability to distinguish time intervals, nausea, ringing in the ears, heart palpitation, pressure in chest and difficulty in breathing. This gas combines with haemoglobin of blood to form carboxy-haemoglobin in RBC which reduces its oxygen carrying capacity to all parts of body and, thus, causing asphyxia. The higher concentrations of carbon monoxide (CO) may be even fatal. CO and nicotine from the cigarette smoke increase the tendency for the blood to clot and so block the coronary arteries. Carbon monoxide also increases the rate at which the fatty materials are deposited in the arteries.

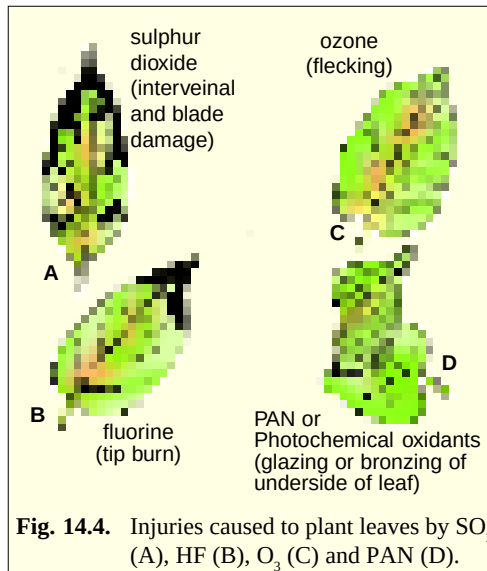


Fig. 14.4. Injuries caused to plant leaves by SO_2 (A), HF (B), O_3 (C) and PAN (D).

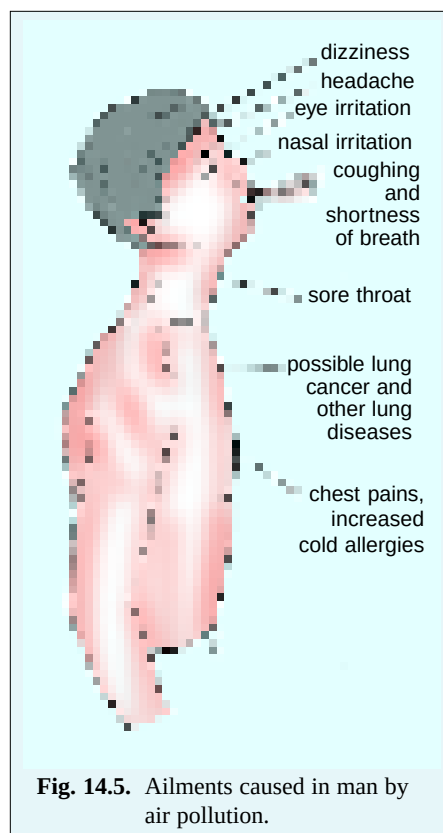


Fig. 14.5. Ailments caused in man by air pollution.

3. Nitrogen oxides. The natural sources of this pollutant are anaerobic bacterial breakdown of nitrogenous compounds, forest fires and lightning. Its chief anthropogenic sources are power generators and automobile exhausts (motor vehicles). Burning of organic wastes, and manufacture of explosives and nitrogenous fertilizers further add to nitrogen oxide pollution.

Nitrogen dioxide (NO_2), a pungent gas that produces a brownish haze, causes nose and eye irritations and pulmonary discomfort in man. It also produces general and pulmonary oedema and hemorrhage. In plants, NO_2 brings about bifacial necrosis leading to collapse of leaves, enhancement of green colour followed by chlorosis and extensive leaf drop. Ultimately there occur an increase in fruit drop and decrease in fruit crop.

Photochemical smog. In the atmosphere nitrogen dioxide is reduced by ultraviolet light to nitrogen monoxide and atomic oxygen :



Atomic oxygen reacts with oxygen to form ozone :

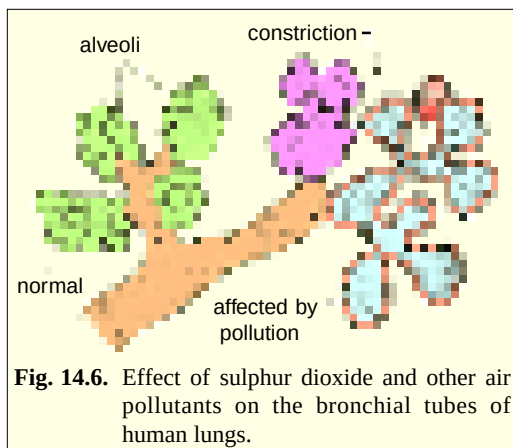


Fig. 14.6. Effect of sulphur dioxide and other air pollutants on the bronchial tubes of human lungs.

and NO_2 . As a result nitrogen dioxide is regenerated, nitric oxide disappears, ozone accumulates and a number of secondary pollutants are formed such as formaldehyde, aldehydes and peroxyacetyl nitrate or PAN ($\text{C}_2\text{H}_3\text{O}_5\text{N}$). All of these collectively form **photochemical smog**.

Ozone, PAN and nitrogen dioxide severely injure many forms of plant life, destroying the cells of leaves, damaging the chloroplasts, and interfering with the plant's metabolic processes.

PAN is known to block "Hill reaction" of photosynthesis. It results in bronzing and glazing of abaxial leaf surface (Fig. 14.4D) which is due to plasmolysis and collapse of mesophyll cells around substomatal chambers (Fig. 14.8 B). Epidermal and guard cells are not injured by PAN.

4. Ozone. Levels of ozone (O_3) may rise in atmosphere due to human activities. It is also

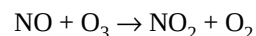


Acid deposition of corrosive

This limestone statue at Rheims Cathedral in France is being dissolved away by acid deposition.



Ozone reacts with nitrogen monoxide to form nitrogen dioxide and oxygen, thus, closing the cycle :



Sometimes, in the presence of sun light, atomic oxygen from the photochemical reduction of NO_2 also reacts with a number of reactive hydrocarbons (such as methane, ethane, toluene, etc., all of which originate from burning of fossil fuels or directly from plants) to form reactive intermediates called **radicals**. These radicals then take part in a series of reactions to form still more radicals that combine with oxygen, hydrocarbons,

formed by NO_2 under UV-radiations effect. Minor amounts of ozone are also added to the atmosphere by electric discharges such as lightning flashes, by vertical flux of stratospheric ozone and by tropospheric storms.

Ozone near the earth's surface in the troposphere creates pollution problem. Increase in O_3 concentration near earth's surface is toxic to plants reducing crop yields significantly. It also has adverse effects on human health (see Table 14-2). Thus, while higher levels of O_3 in the atmosphere protects us, it is harmful when it comes in direct contact with us and plants at earth's surface.

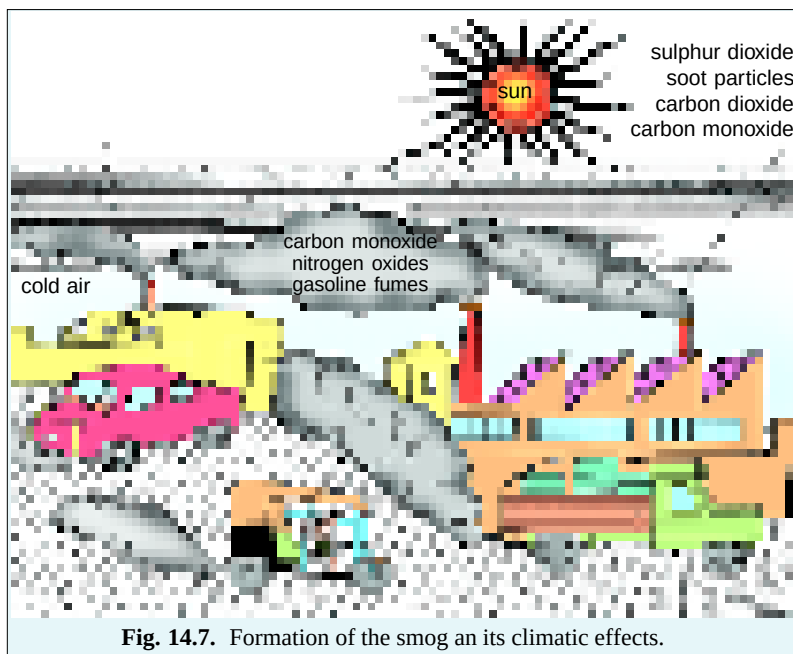


Fig. 14.7. Formation of the smog and its climatic effects.

Table 14-2. Effects of ozone (O_3) on human health.

Concentration (ppm)	Effects observed
0.2	No ill effects
0.3	Nose and throat irritation
1.0–3.0	Extreme fatigue after 2 hrs.
9.0	Severe pulmonary oedema

In plants, O_3 enters through stomata and produces visible damage to leaves and results in decrease in yield and quality of plant products. Thus, O_3 results in necrotic flecking of upper surface of leaf (Fig. 14.4C), general chlorosis and bronzing, premature senescence of plants, precocious dropping of older leaves, reduced growth of shoots and roots, suppression of nodulation, reduction in seed set and yield. Ozone causes shrinkage of nuclei and cytoplasm or mesophyll cells which become granular and results in increased intercellular space (Fig. 14.8A). At 0.02

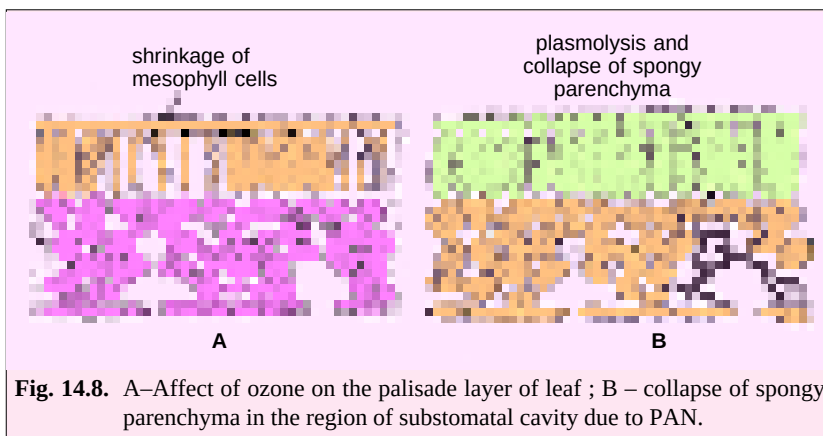


Fig. 14.8. A—Affect of ozone on the palisade layer of leaf ; B – collapse of spongy parenchyma in the region of substomatal cavity due to PAN.

ppm, O_3 damages tobacco, tomato, bean, pine and other plants. It is now known to be the cause of several widespread diseases of plants such as weather fleck of tobacco, leaf tip burn of carnations, tip burn of onions, bronzing of bean, speckle-leaf of potato and brown leaf of grapes. In human beings, O_3 at its 9.0 ppm concentration result in severe damage of pulmonary organs. Its lower concentrations result in nose and throat irritation and severe tired feeling (see Table 14-2).

5. Fluorocarbons (Hydrogen fluoride). Natural sources of fluorides in the atmosphere are active volcanoes. Their man-made or artificial sources are petroleum refining, aluminium, steel and electrochemical reduction plants, blast furnaces, brick-kilns, and tile, glass etching and superphosphate fertilizer industries and combustion of coal.

Fluoride burns the tip of plant leaves (Fig. 14.4B). Its low amounts impair plant growth, result in excessive dropping of bloom and young fruits, development of small, partially or completely seedless fruits and premature formation of soft red flesh and splitting of peach. In human being, it irritates and corrode all body passages.

6. Hydrocarbons. Biological decomposition of organic matter, spill and seepage from natural gas and oil fields and volatile emissions from plants are some major natural causes for the release of hydrocarbons such as methane, terpenes, ethylene and aniline. Incomplete combustion of fuels, automobile exhaust (Fig. 14.1), petroleum-refineries, agricultural burning, motor fuel marketing, manufacture of explosives and cracking of natural gas in petrochemical plants (as a blow-off emissions) constitute the man-made sources that emit hydrocarbons.

The hydrocarbon **ethylene** causes yellowing and occasional necrosis of leaves, chlorosis of floral buds, inhibition of terminal growth, epinasty of leaves, shortening of internodes, thickening of stems, lack of apical dominance, stunted growth, dry sepal disease of orchids and decrease in the amount of chlorophyll and carotenoids. Another hydrocarbon **aniline** results in the appearance of bands on leaves as if they are water-soaked, necrotic spots and abscission of leaves. In man, hydrocarbons bring about irritation of mucous membrane, bronchial constriction and eye irritation. One of the hydrocarbons released due to incomplete combustion, is **3-4 benzpyrene** which is said to cause lung cancer. **Methyl isocyanate** gas when accidentally leaked out from the storage tanks of the pesticide factory in Bhopal on December 2, 1984, had killed over 3,000 persons and seriously affecting lakhs of residents.

7. Hydrogen chloride. This pollutant is released from combustion of coal, paper, plastics, chlorinated hydrocarbons, accidental spills from the chemical manufacturing plants and ignition of solid-fuel rocket engines in plants, hydrogen chloride causes plasmolysis and collapse of epidermal cells of leaves and thereby results in abaxial glazing of leaves.

8. Ammonia. The main anthropogenic sources of this gaseous pollutant are refrigerator, pre-cooler systems of cold storages, manufacture of dyes, explosives, lacquers (varnishes) and anhydrous ammonium fertilizers and nitric acids and domestic incineration. Ammonia causes in plants, bleaching of leaves, rusty spots on leaves and flowers, reduction of root and shoot growth, browning and softening of fruits, development of dark, corky lenticels in apples, and reduction in the rate of seed germination. In man, it inflames upper respiratory passages.

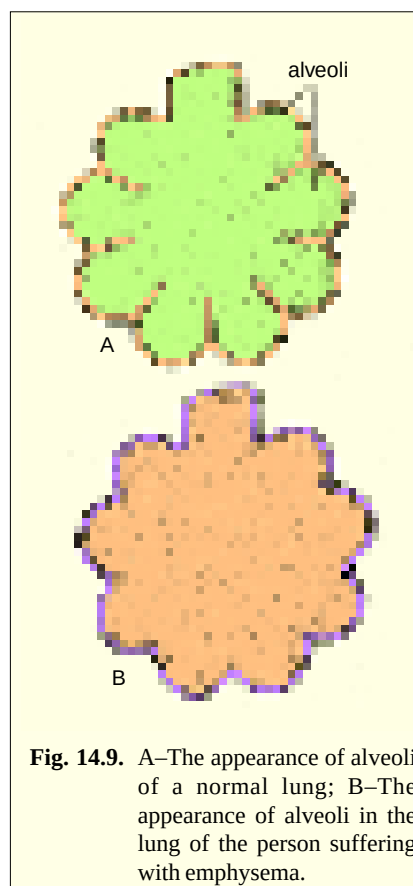
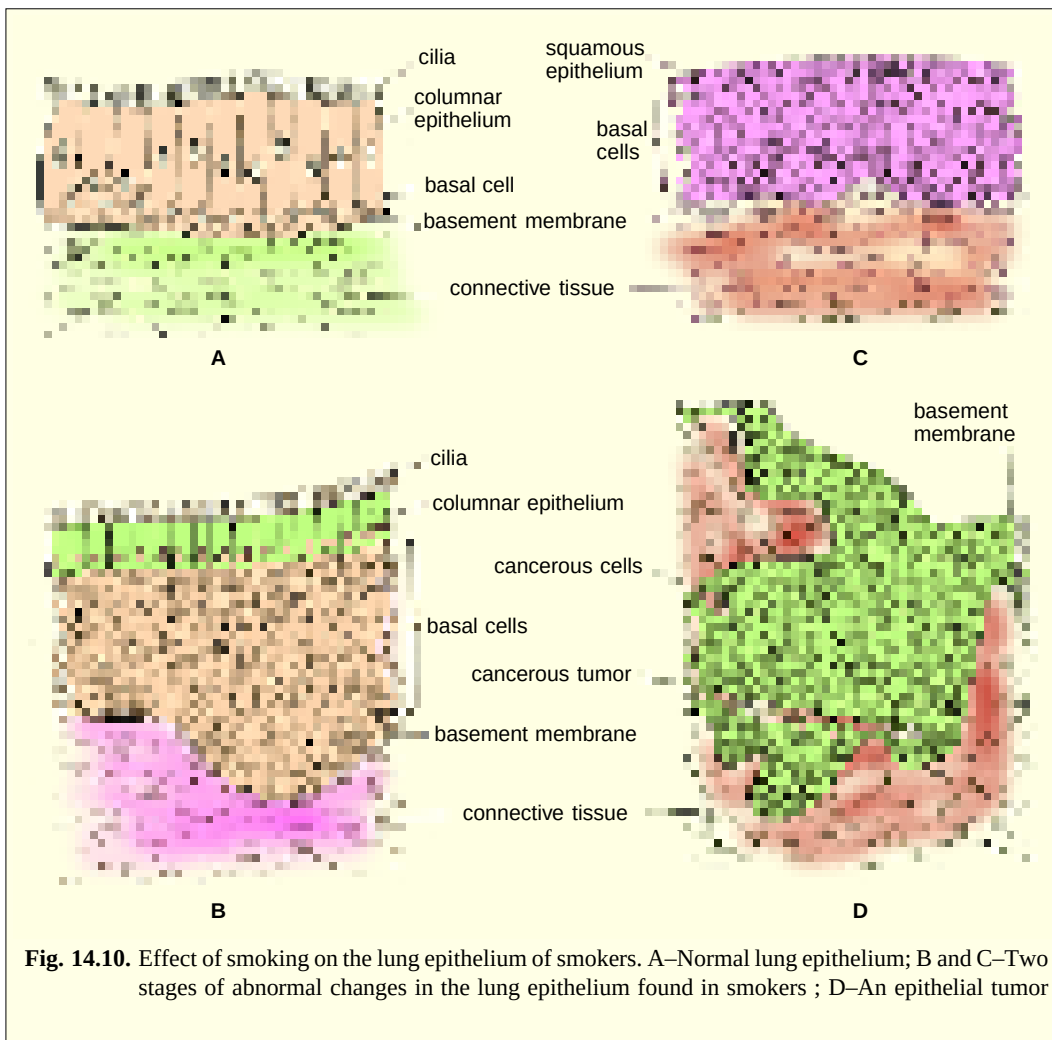


Fig. 14.9. A—The appearance of alveoli of a normal lung; B—The appearance of alveoli in the lung of the person suffering with emphysema.

9. Tobacco smoke. Tobacco smoke contains about 300 chemical compounds including nicotine and carcinogens such as tar (“aromatic hydrocarbon”). It is mainly produced by smoking cigarettes and bidis. It is gradually becoming a pollutant especially in closed atmospheres such as buses, trains, auditoria, discotheques and so on. **Nicotine** stimulates some types of synapse of nervous system, increase blood pressure and heart rate by the production of adrenaline, causes vasodilation in the muscles and vaso-constriction in the skin. When a person smokes, tiny particles in the smoke get caught on the lining of the windpipe and bronchial tubes. Extra mucus is produced and the cilia stop beating. The mucus collects in the bronchial tube and this gives rise to a “smoker’s cough”.

Repeated coughing may cause the delicate walls of the alveoli to break down into larger spaces. This cuts down the surface area over which gaseous exchange can take place, so the person gets very short of breath. Doctors call this condition **emphysema** (Fig. 14.9). Tobacco smoke also brings about thickening of bronchial epithelial layer, loss of ciliated cells and appearance of cells with bizarre nuclei, which are probably the precursors of cancerous cells (Fig. 14.10). Although smoking mainly affects the lungs, it can also cause cancer of other organs such as mouth, throat, oesophagus and bladder. It is also associated with heart disease and stomach ulcers and a woman who smokes while she is pregnant is more likely to have spontaneous abortion of still birth or to give birth to an under sized baby.



B. Particulate Pollutants

1. Fluorides. The particulate fluorides originate in the same way as the gaseous fluorides. They settle and accumulate on the grass and other vegetation. They are less toxic to these plants causing occasional leaf-tip burns. However, ingestion by cattle of various fluorine compounds falling on forage, causes **fluorosis**, a disease characterized by abnormal calcification of bones and teeth, eventually resulting in loss of teeth, body weight and in lameness.

Fluoride pollution in man and animals is mainly through water. In our country, fluorosis is a public health problem in states of Gujarat, Rajasthan, Punjab, Haryana, U.P., Andhra Pradesh, Tamil Nadu, Karnataka, and some areas of Delhi. Globally it is a problem of various other countries such as USA, Italy, Holland, France, Germany, Spain, Switzerland, China, Japan and some African and Latin American countries.

2. Lead. Lead, a heavy metal, is injected in the atmosphere mainly from automobile exhaust. Automobile gasoline contains tetraethyl lead that is used as an antiknock additive. Lead is emitted into the air with the exhaust as volatile **lead halides** (bromides and chlorides). About 75% of lead burnt in gasoline comes out as lead halides through tail pipe in exhaust gases. Of this about 40% settles immediately on the ground and the rest (60%) goes into air. That is why its concentration is higher in urban areas where automotive and industrial exhausts are more (Fig. 14.11).

Further, the use of lead-lined vessels for cooking and the storage of wine resulted in heavy lead burdens in the bodies of Roman citizens. Some have attributed the decline of the Roman empire to chronic lead poisoning. In fact, analysis of bones, of Roman citizens have revealed high lead concentrations (see **Kimball**, 1975). Lead pollution is also caused by occupational manipulation of putty and paints by painters; habitual or accidental nibbling of peels of old paints; smelting complexes, ceramics, pesticides and solder used for sealing. The lead level of air in air-quality guide of WHO is $2\mu\text{g}/\text{m}^3$. This level is already crossed in many Indian cities and in various countries of world. For example, in Kanpur and Ahmedabad lead level vary between 1.05 to $8.3\mu\text{g}/\text{m}^3$ and 0.59 to $11.38\mu\text{g}/\text{m}^3$, respectively. Roadside plants and meadow mice living along major highways are found to contain high concentrations of lead in tissues, and this has a sublethal effect on the health and longevity of the animal. Traffic policemen and others who are exposed for long periods to heavy traffic have higher than average lead in their blood and body tissues. High lead accumulation in the tissues of human body interferes with development and maturation of erythrocytes (RBC) and causes anaemia. Even 0.2 parts per million (0.2 ppm) concentration of lead in human body creates metabolic disturbances. Chronic exposure to lead leads to stippled red blood cells having impaired capacity for oxygen transportation. Being a cumulative poison, it disrupts the functioning of cells and organs of the muscular, circulatory and nervous systems binding with the cellular enzymes and coagulating proteins, thereby, results in nausea, weakness and dizziness. It also damages liver, kidney and gastro-intestine and induces abnormalities in fertility and pregnancy.

3. Mercury. It is a liquid volatile heavy metal which is found in rocks and soil. It is present in air due to human activities such as the use of mercury compounds in production of fungicides, paints, cosmetics, paper pulp, etc. Inhalation of $1\text{ mg}/\text{m}^3$ of mercury in air for three months may lead to human death. Nervous system, liver and eyes are damaged. Infant may be deformed. Other symptoms of mercury toxicity are headache, fatigue, anxiety, lethargy, loss of appetite, etc.

4. Zinc. Zinc in air occurs mostly as white zinc oxide fumes and is toxic to man. It exists in air around zinc smelters and scrap zinc refineries.

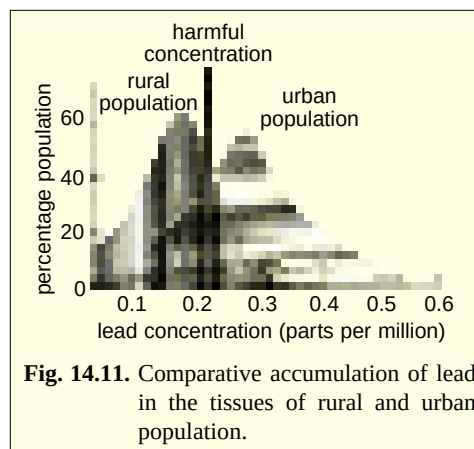


Fig. 14.11. Comparative accumulation of lead in the tissues of rural and urban population.

5. Cadmium. This metal is emitted to air by human activities and industries (e.g., electroplating and welding of cadmium containing materials; industries producing pesticides and phosphate fertilizers. Cadmium occurs in the air in the form of oxide, sulphate or chloride compounds. It is poisonous at very low levels and is known to accumulate in human liver and kidney, Cadmium causes hypertension, emphysema and kidney damage. It may also act as carcinogen in mammals.

6. Potassium salts. These particulate pollutants are derived mainly from potash mines and cause in plants abnormalities such as branch tip death, chlorosis and necrosis of leaves.

7. Sodium chloride. Certain de-icing salts such as sodium chloride, used to remove ice and snow from roads in winters, are found to cause multiple damage to the roadside trees such as leaf necrosis, defoliation, suppression of flowering, and dieback of terminal shoots in apple.

8. Agricultural chemicals. Several types of chemicals such as insecticides, herbicides, fungicides and pesticides, used widely in agriculture are found to result in foliar lesions, chlorosis and abscission of leaves and reduction in fruit set.

9. Particulate matter. The word particulate has been derived from particle and includes all solid or liquid substances primarily in the air. Particulate matter is usually divided into two categories : suspended particulates and dust fall.

(a) Suspended particulates. These are smaller than 10 microns. They are generated by various industrial processes, combustion processes and black soot. In polluted air, their concentration may reach up to 100,000 per cubic centimeter.

(b) Dust fall. Dust fall includes the particles larger than about ten microns. They are emitted into the air by physical processes such as grinding and abrasion, a soot and fly ash (or PFA = Pulverised fly ash) from fuel combustion. Dust particles settle at the ground quickly and cause nuisance to certain industries requiring aseptic conditions such as drug industries and food processing plants. The average level of dust and particulate matter in Dehi is about 600 mg/m³ in Bombay about 200 mg/m³, in Calcutta about 300mg/m³, whereas it is just 150 mg/m³ in other polluted western cities.

Dust may be inorganic and organic in nature: 1. **Inorganic dust** containing silica and trapped heavy metals is the main pollutant in mining, quarrying and stone cutting operations. 2. **Organic dust** raised in cotton textile mills, ginning plants, coir retting and processing, jute and hemp processing, saw mills and plywood industries are also potentially toxic with properties of sensitization of persons exposed to them.

Particulates emitted from cement manufacturing units cause in plants premature fall of needles, higher puberty of leaves, formation of more stomata and trichomes, reduction in number and size of cobs and weight of seeds and increase in number of infertile seeds. Dust from stone crushers, lime kilns and slate making units is also hazardous to plants and human beings. Particles of asbestos that are released from factories can wear off brakes and gears of cars. Coal dust and asbestos have been found to cause in plants, necrotic lesions, reduction in fruit sets and silicosis, asbestosis and lung cancer in human beings. The common dust particles sometime become health hazards as they may lead to diseases such as allergic asthma, bronchitis, emphysema and even fibrosis of lungs. Pulverised fly ash (PFA) is getting accumulated in such large quantities particularly in areas where thermoelectric generators are installed that it has proved to be one of the major sources of solid wastes. Since it contains boron, its deposition by wind in agricultural tracts often results in deleterious effects.

In fact, living organisms are rarely exposed to a single pollutant in nature. The influence of mixtures of pollutants is usually synergistic, i.e., greater than the additive effects of the different pollutants alone.

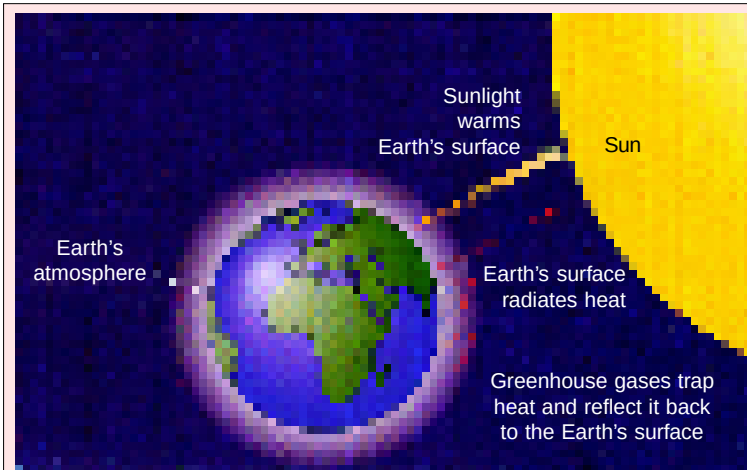
C. Effects of Air Pollution on Weather, Climate, and Atmospheric Processes

At gross level, air pollution causes two worldwide problems – contamination of the upper atmosphere and the alteration of weather and climate.

Air pollution also affects weather on a continental or global basis. Many gaseous pollutants and fine aerosols reach the upper atmosphere, where they have basic effects on the penetration and absorption of sunlight. **Brodine** (1973) and certain other modern environmental biologists feel that increasing particulate pollution may be reducing the amount of sunlight energy reaching the earth's surface, thereby, lowering solar radiation at the earth's surface and producing a cooling effect on world climates which could ultimately trigger another ice age. In fact, **Thompson** (1975) has reported a decrease in mean annual temperatures in the northern hemispheres and an increase in the north polar ice cap.

Green house effect.

Carbon dioxide is a natural constituent of the atmosphere, but, its concentration is increasing in the air with an alarming rate. A byproduct of the burning of fossil fuel, it is not necessarily a pollutant. It produces adverse physiological effects only at very high levels. It is estimated that approximately one-half of the CO_2 input stays in the atmosphere and other half of it is removed by the oceans and by plants. The increased amount of CO_2 in atmosphere is found to increase the temperature of earth.

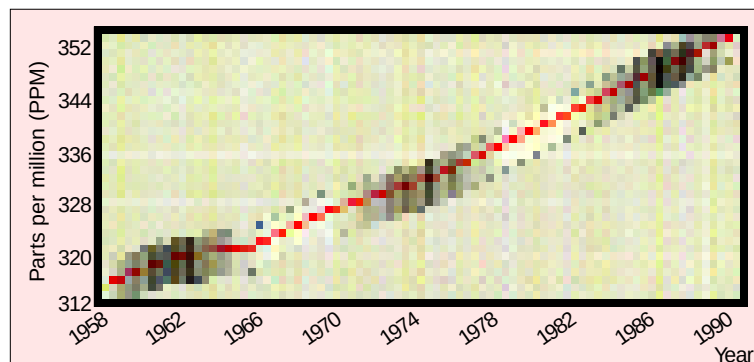


Green House Effect.

Energy from the Sun is either transmitted, reflected or trapped by the atmosphere. Energy radiated back into space is at longer wavelengths (in the infrared region) than the energy arriving from the Sun, and the consequence is that more heat enters than can escape.

Greenhouse gases (CO_2 , CH_4 , nitrous oxides, CFCs), together with water vapour in clouds, contribute to the 'blanketing' effect of the atmosphere.

The spectral properties of CO_2 in the atmosphere are such that it tends to prevent the long wave radiations (i.e., infra-red heat radiation) from earth from escaping into outer space and deflect it back to earth. The latter has an increased temperature at surface (**Turk et al.**, 1974). This phenomenon is called **atmospheric effect** (**Lee**, 1974) or **greenhouse effect** (see **Southwick**, 1976, **Smith**, 1977).



The rising level of carbon dioxide in the atmosphere. Data was collected at the Mauna Loa observatory in Hawaii. There is good evidence to suggest that changes in the average temperature at the surface of the Earth are closely linked to changes in the levels of greenhouse gases.

The simultaneous cooling and heating effects of air pollution on earth have increased variability in the world-wide weather patterns which may be a serious threat to global food production (**Thompson**, 1975). Recently, certain ecologists have tried to correlate air pollution with serious and prolonged droughts, heavier rains and floods, and more serious hurricanes and tornadoes (see **Southwick**, 1976).

Scientists fear that increased level of CO_2 will increase the greenhouse effect and thereby increase the temperature of the earth. Only a slight temperature rise would cause the polar ice caps to melt and to cause an enough rise in sea level submerging a number of major cities of the world, because many are along coasts.

Simultaneously, along with carbon dioxide man has been adding solid particles and droplets into the air which increase the **albedo** or shineness of the earth. This should act contrary to the greenhouse effect, reduce the sunlight reaching the earth and tend to lower the temperature of earth.

Peeling of ozone umbrella by CFMs. Certain fluorocarbon compounds which are called chlorofluoromethanes or CFMs or “freon” are used as propellants in pressurized aerosol cans. They are inert in normal chemical and physical reactions, but they get accumulated in greater amounts at high altitudes and there in the stratosphere these inert gaseous compounds (*i.e.*, CFMs) release chlorine atoms under the influence of intense short-wave ultraviolet radiation. Each atom of chlorine chain then reacts with more than 1,00,000 molecules of ozone, converting ozone to oxygen. The reduction in stratospheric ozone permits greater penetration of ultraviolet light, which intensifies UV radiation at the earth’s surface. Some scientists such as **Ahmed** (1975), **Brodeur** (1975), and **Russell** (1975), feel that this intensified radiation will cause a significant increase in skin cancer and eventually have a lethal effects on many organisms, including man. In plants such enhanced level of UV radiations are feared to cause stunted growth, short, thick stems, smaller leaves, plasmolysis of cells, destruction of anthocyanin, chlorophyll and nuclei. They are known to cause bronzing of leaves, injury to various fruits, somatic aberrations, discolouring of staminal hairs, inhibition of pollen germination and pollen tube growth.

The protective ozone layer of the stratosphere is also considered by many ecologists to be endangered by supersonic jets, the SSTs. The jet engines of supersonic aircraft flying at high altitudes release nitrogen oxides (NO_x) which catalytically destroy ozone molecules (see **Southwick**, 1976).

Control of Air Pollution

Most kinds of air pollutions can be controlled by modern technology, but the costs ultimately be borne by the public in the form of higher prices for manufactured goods, higher taxes, reduced profit margins in industry, and more restrictions on individual activities such as burning leaves and trash and use of automobiles. Following steps have to be taken to control pollution at source (prevention) as well as after the release of pollutants in the atmosphere.

1. Prevention and control of vehicular pollution. For controlling the air pollution because of vehicles, the following measures are adopted:

(a) Curbing the pollutant emission from vehicular exhaust. This type of control can be attained by (i) using new proportion of gasoline and air; (2) more exact timing of fuel feeding; (3) using gas additives to improve combustion; (4) by injecting air into the exhaust to convert exhaust compounds into less toxic substances, and by (5) correcting the engine design and/or fixing cessation device to improve combustion with the existing design. Complete elimination of three main pollutants, namely CO , NO_x and hydrocarbons can be attained by either updating the present design of engines or by making appropriate changes in devices for improving combustion. In recent years, **I.K.Bharati** of Mumbai has claimed to have devised a simple attachment (by the patent name **Thermoreactor**) to curb air pollution by motor vehicle. The reactor is fitted to the exhaust tail pipe and it converts carbon monoxide into oxygen. Various devices such as positive crankcase ventilation valve and catalytic convertor have been developed in USA to reduce exhaust emissions by automobiles.

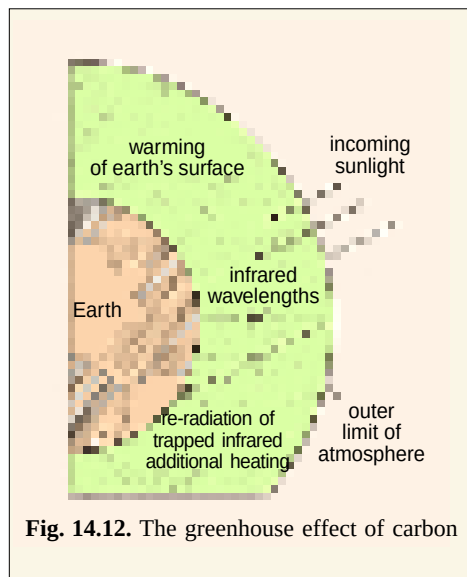


Fig. 14.12. The greenhouse effect of carbon

(b) Control of evaporation from fuel tank and carburettor. This can be performed by (1) collection of vapours with activated charcoal when the engine is turned off; (2) subjecting the gasoline in the tank to slight pressure to prevent the gas from evaporation; and (3) developing low volatile gasoline that does not evaporate easily.

(c) Use of filters. Filters can be used to capture and recycle the escaped gases (hydrocarbons) from the engine (i.e., the gas vapours which escape between walls and the piston and reach the crankcase).

2. Prevention and control of industrial pollution. To check the air pollution by industrial and power plant chimney wastes, measures are taken for the removal of particulate matter and gaseous pollutants. In different industries, air pollution can be checked at various steps (Fig.14.13).

(a) Removal of particulate matter.

This process involves collection of particulate matter under the influence of different forces, thereby, moving them continuously out of the gas stream. This step involves the use of following equipments :

(i) Cyclone collector. In this case the waste gas containing particles is subjected to **centrifugation**. The suspended particles move towards the wall of cyclone body and then to its bottom and finally are discarded out. The cyclone collectors can remove upto 70 per cent of the particles.

(ii) Electrostatic precipitators (ESPs). To remove the suspended particles from gas stream, the **electrical forces** are applied within the chamber in the precipitator. The particles become charged or ionized, and they are attracted to charged electrodes and removed. ESPs can remove 99 per cent of the particulate pollutants from the chimney exhaust. ESPs work very efficiently in power plants, papermills, cement mills, carbon black plants, etc. Sometimes, high resistivity dust may make separation in an ESP difficult. To overcome this difficulty, **fibric filters** or **bag filters** are employed. Further, tall chimneys are used for vertical dispersion of air pollutants.

(b) Removal of gaseous pollutants. These are removed from the chimney exhaust by the following three methods:

(i) Wet systems. These are used as **washing towers** in which alkali fluid circulate continuously. This liquid react with SO_2 to produce a precipitate. In **gas scrubber**, a fine spray of water can effectively separate many gases such as ammonia and sulphur dioxide.

(ii) Dry systems. In this case, the gas pollutants are allowed to react with an **absorbent** under a dry phase. Absorbents such as dolomite, lime (CaO) and limestone (CaOH) are placed in the way of the flowing gas (SO_2). The process is cheap and does not involve any spray of water.

(iii) Wet dry systems. In these cases water in the absorbent reacts with the acid components. The absorbent calcium hydroxide slurry is spread into the hot gas stream in the form of small droplets. Calcium reacts with SO_2 and the hot gases cause the water to evaporate simultaneously. The end product is a dry powder containing mostly fly ash and salts.

Control of air pollution through law. In our country there have been several legislative measures both at state and central Govt., levels to prevent and control different types of air pollutions:

1. Bengal smoke Nuisance Act, 1905 ;
2. The Motor Vehicle Act, 1938;
3. The Gujarat Smoke Nuisance Act, 1953 ;
4. The Air (prevention and Control of Pollution Act, 1981) (It was amended in 1987);

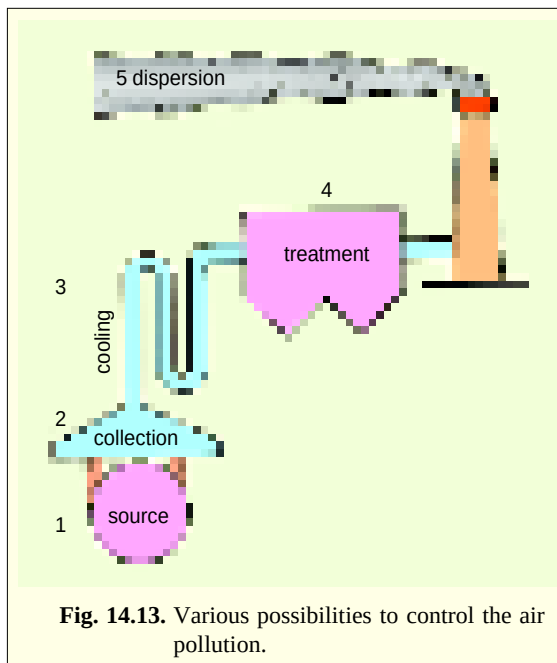


Fig. 14.13. Various possibilities to control the air pollution.

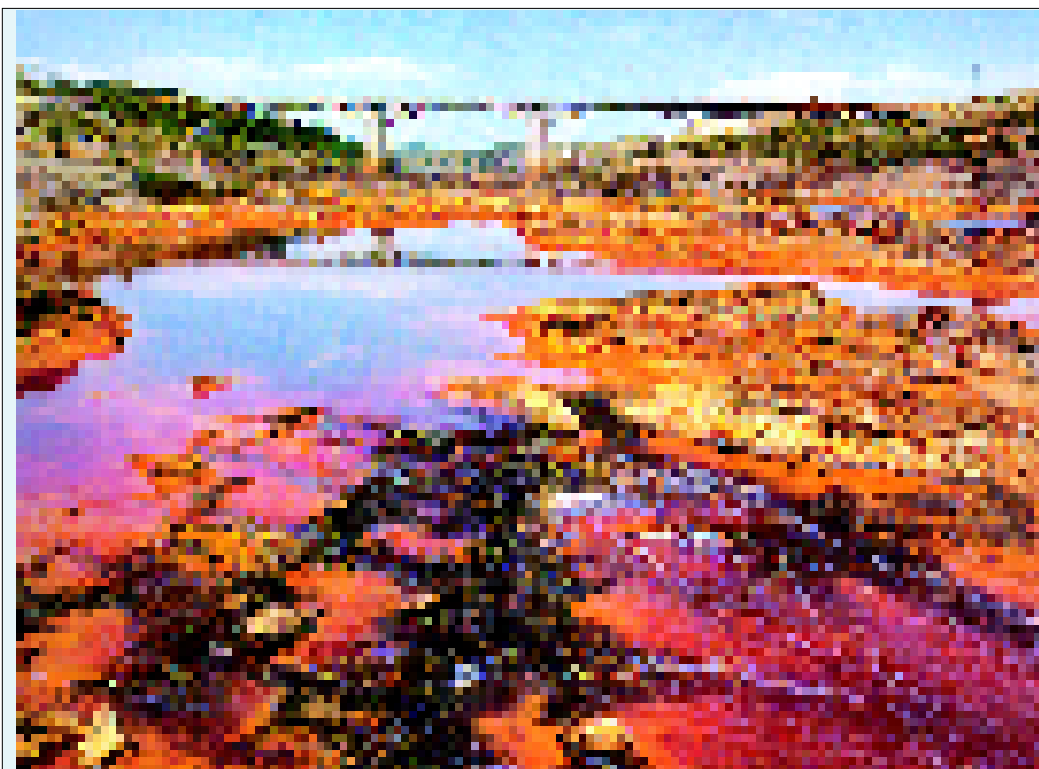
5. The Environment (protection) Act, 1986 ;
6. The Motor Vehicles Act, 1988 (This Act came into force from 1.7.1989).

2. WATER POLLUTION

The great solvent power of water makes the creation of absolutely pure water a theoretical rather than a practical goal. Even the highest-quality distilled water contains dissolved gases and to a slight degree, solids. The problem, therefore, is one of determining what quality of water is needed to meet a given purpose and then finding practical means of achieving that quality. The problem is further compounded because every use to which water is put—washing, irrigation, flushing away wastes, cooling, making paper, etc., adds something to the water. In fact for centuries rivers and lakes have been used as dumping grounds for human sewage and industrial wastes of every conceivable kind, many of them are highly toxic. Added to this are the materials leached and transported from land by water percolating through the soil and running off its surface to aquatic ecosystems.

The term **water pollution** is referred to any type of aquatic contamination between following two extremes: (1) a highly enriched, over productive biotic community, such as a river or lake with nutrients from sewage or fertilizer (**cultural eutrophication**), or (2) a body of water poisoned by toxic chemicals which eliminate living organisms or even exclude all forms of life (see **Southwick**, 1976).

Normally water contains two types of impurities—dissolved and suspended. Dissolved impurities are gases (H_2S , CO_2 , NH_3 , etc.) and minerals (Ca, Mg, Na, salts). Suspended matter includes clay, silt and sand and even microbes. Polluted waters are turbid, unpleasant, foul smelling, unfit for drinking, bathing and washing or other purposes. They are harmful and means of many diseases as cholera, dysentery, typhoid, hepatitis, etc.



The Rio Tinto in Spain is so called because of the color of its waters (*tinto* means “dark red” or “dyed”). The color comes from what were once Europe’s largest copper mines, close to the river. The mining produced a lot of spoil, which piled around the mine. Water running off the site picked up the copper, which colored it. Although the mine itself is now closed, much of the soil at the huge copper deposits in the river persist. Like other metals, copper is present in many organisms in tiny amounts, but a large dose is extremely toxic.

Types of Water Pollution

Types of water pollution may be classified by the *medium* in which they occur, such as **surface water pollution, ground water pollution, soil water pollution**, etc.; the *habitat* in which they occur, such as **river pollution, lake pollution, estuarine pollution, coastal water pollution, open ocean pollution**, etc.; and *source* or *type* of contamination, such as **nutrient pollution, bacterial pollution, viral pollution, metallic pollution, petrochemical pollution, pesticide pollution, thermal pollution, radioactive pollution**, etc.

Kinds and Sources of Water Pollutants

Pollutants entering water sources are classified broadly into following categories: domestic sewage and oxygen-demanding wastes; infectious agents; plant nutrients; chemicals such as insecticides, herbicides, and detergents; other minerals and chemicals; sediment from land erosions; radioactive substances; and heat from power and industrial plants.

These aquatic pollutants come from many sources. Excessive nutrients, such as nitrates and phosphates, commonly originate in domestic sewage, run-off from agricultural fertilizer, waste materials from animal feed lots, packing plants, etc. Toxic chemicals as agents of water pollution originate in industrial operations, acid mine drainage, surface erosion from strip mines, washing of herbicides and insecticides, radioactive fall out from atomic explosion, and commercial accidents such as oil spills or the rupture of chemical tanks. Besides the pollutants which come from point sources such as sewage, factory or industry, there are many pollutants which come from watershed run-off. Urban and sub-urban run-off, for example, contains many pollutants such as oil, pesticides, radioactive dust, salt, fertilizers, miscellaneous chemicals, and nematodes, pathogenic protozoans, bacteria and viruses. However, **Kimbal** (1975) recognized only three major sources of pollution: *domestic, industrial and agricultural*.

Ecology of Water Pollution

Each type of water pollution affects the abiotic and biotic factors of different aquatic systems in different degrees and its ultimate effect on man remains quite drastic in medical, aesthetic, and economical sense. Some of the well known ecological effects of water pollution are following:

1. Sewage Pollution

Contamination of freshwaters and shallow offshore seas by sewage is a common occurrence. Domestic sewage and waste-water is about 99.9 per cent water and 0.02-0.04 per cent solids of which proteins and carbohydrates each comprise 40–50 per cent and fats 5–10 per cent (**Simmons**, 1974). In other words, sewage includes mostly biodegradable pollutants such as human faecal matter, animal wastes, and certain dissolved organic compounds (*e.g.*, carbohydrates, urea, etc.) and inorganic salts such as nitrates and phosphates of detergents and sodium, potassium, calcium and chloride ions. Under natural processes most of the biodegradable pollutants of sewage are rapidly decomposed, but, when they accumulate in large quantities, they create problem, *i. e.*, when their input into environment exceeds the decomposition or dispersal capacity of the latter. Most cities of well developed countries such as USA, Britain, etc., and some cities of developing countries such as India have evolved certain engineering systems, such as, **septic tanks, oxidation ponds, filter beds, waste water treatment plants and municipal sewage treatment plants** for the removal of many harmful bacteria and other microbes, organic wastes and other pollutants from the sewage, before it is tipped into river or sea.

Sewage treatment is usually performed in following three stages: (1) **Primary treatment**, which removes large objects and suspended undissolved solids of raw sewage and converts them into a biologically inactive and aesthetically inoffensive state, the **sludge**, a valuable fertilizer. (2) **Secondary treatment**, which supplies aeration and bacteriological action to decompose organic compounds into harmless substances such as CO₂, sulphate and water. During later stages of secondary treatment,

whole waste water is chlorinated (*i.e.*, treated with chlorine) to reduce its content of bacteria. (3) **Tertiary treatment**, which removes nitrates and phosphates and releases pure water. These three stages of sewage treatment have become increasingly expensive and only in most advanced countries all the three treatments of sewage are done. However, most Indian cities either lack any sewage or waste-water treatment plant or have inadequate sewage treatment facilities. Consequently, normally and especially during heavy downpour and floods, raw sewage or incompletely treated sewage is dumped into rivers which cause severe water pollution problems in following ways:

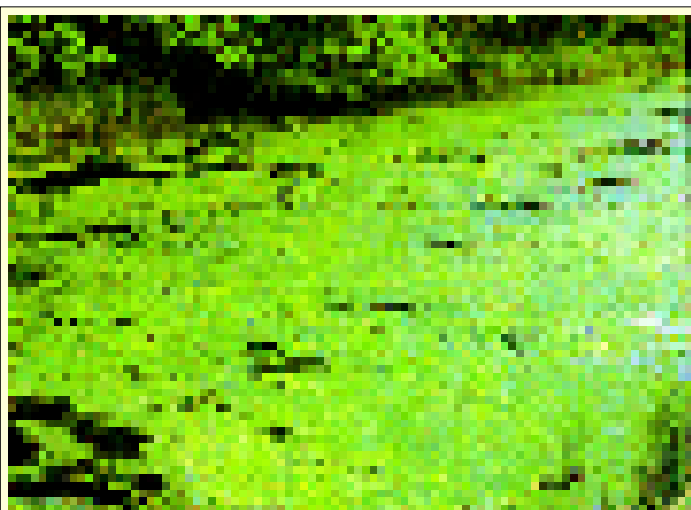
(i) **Bacterial and viral contamination.** Sewage wastes may contain pathogenic bacteria and viruses which are a threat to human health. Waterborne diseases such as typhoid, bacillary dysentery, amoebic dysentery, botulism, poliomyelitis, and hepatitis all represent potential health hazards in sewage-contaminated waters. Due to such kinds of sewage pollution waters of many ponds, lakes, rivers, sea beaches in India and abroad have been prohibited for human use, whether for drinking, bathing, swimming or other sort of water recreation.

BOD test. BOD or biological oxygen demand is the amount of oxygen required for biological oxidation by microbes in any unit volume of water. The test is done at 20°C for at least five days. BOD value generally approximates the amount of oxidisable organic matter (such as sewage and other organic wastes, animals and human excreta, all of which are called **oxygen demanding wastes**) and is, therefore, used as a measure of degree of water pollution and waste level. Thus, due to addition of sewage and waste, oxygen levels are depleted which are reflected in terms of BOD values of water. The number of microbes as *Escherichia coli* (bacterium) also increase tremendously and these also consume most of the oxygen. The number of bacteria as *E.coli* in unit volume of water is also taken (called **E.coli index**) as a parameter of water pollution. The quantity of oxygen in water (called **dissolved oxygen or DO**) along with BOD is indicated by the kind of organism present in water. For example, fish become rare at DO value of 4 to 5 ppm of water. Further decrease in DO value may lead to increase in anaerobic bacteria. Typical BOD value for raw sewage run from 200 to 400 mg of oxygen per liter of water (therefore, 200—400 ppm). Water for drinking should have a BOD less than 1.

(ii) **Eutrophication.** According to **Hutchinson** (1969), the eutrophication is a natural process which literally means “well nourished or enriched.” It is a natural state in many lakes and ponds which

have a rich supply of nutrients, and it also occurs as part of the aging process in lakes, as nutrients accumulate through natural succession. Eutrophication becomes excessive, however, when abnormally high amounts of nutrients from sewage, fertilizer, animal wastes and detergents, enter streams and lakes, causing excessive growth or ‘bloom’ of microorganisms and aquatic vegetation.

Most secondary sewage treatment plants, though, precipitate solids and inactivate most bacteria in domestic sewage, yet they do not remove the basic nutrients such as ammonia, nitrogen, nitrates, nitrites and phosphates. These nutrients



Nitrates stimulate algae growing in water as well as plants growing in soil. If runoff from fertilizer gets into a body of water, algae grow so profusely that they form a blanket over the surface. This usually happens in summer, when the light levels and warm temperatures favor growth.

stimulate algal growth and lead to **plankton blooms**. Some plankton blooms, particularly those of blue-green algae produce obnoxious odours and tastes in waters. Others, such as the dinoflagellate blooms or “**the red tide**” of southern coastal regions, produce toxic metabolic products which can result in major fish kills.

Plankton blooms of green algae do not always produce undesirable odours or toxic products, but still create problems of oxygen supply in the water. While these blooms exist under abundant sunlight, they contribute oxygen to the water through photosynthesis, but under conditions of prolonged cloudiness, they begin to decay and consume more oxygen and with heavy load the oxygen contents of the water may diminish below the point where most fish can not survive. As the conditions in the water become anaerobic due to increased oxygen depletion by bacterial decomposition or planktonic blooms, the breakdown products become reduced rather than oxidized molecules, many of which (*e.g.*, hydrogen sulphide) produce -offensive odours and tastes.

Excessive nutrient levels in aquatic systems can also cause two other kinds of ecological problems. Primarily, they may lead to extensive growth of aquatic weeds such as Eurasian milfoil, water hyacinth, water chestnut, etc. Excessive growth of these weeds can impair fishing, bathing, fish spawning, shell fish production, and even navigation (**Sculthorpe**, 1967). Secondly, nitrates can be converted in the human digestive tract by certain bacteria to nitrites, and the same transition may occur in opened cans of food even if they are subsequently refrigerated. Nitrites react with haemoglobin, forming methemoglobin which will not take up oxygen. Laboured breathing and occasional suffocation result most severely in human infants. Nitrites may also react with creatinine (present in the vertebrate muscles to form nitrosarcosine which can be carcinogenic (see **Simmons**, 1974; **Southwick**, 1976).

Effects of organic pollution on aquatic animal life. Organic pollution tends to bring about changes in faunal composi-

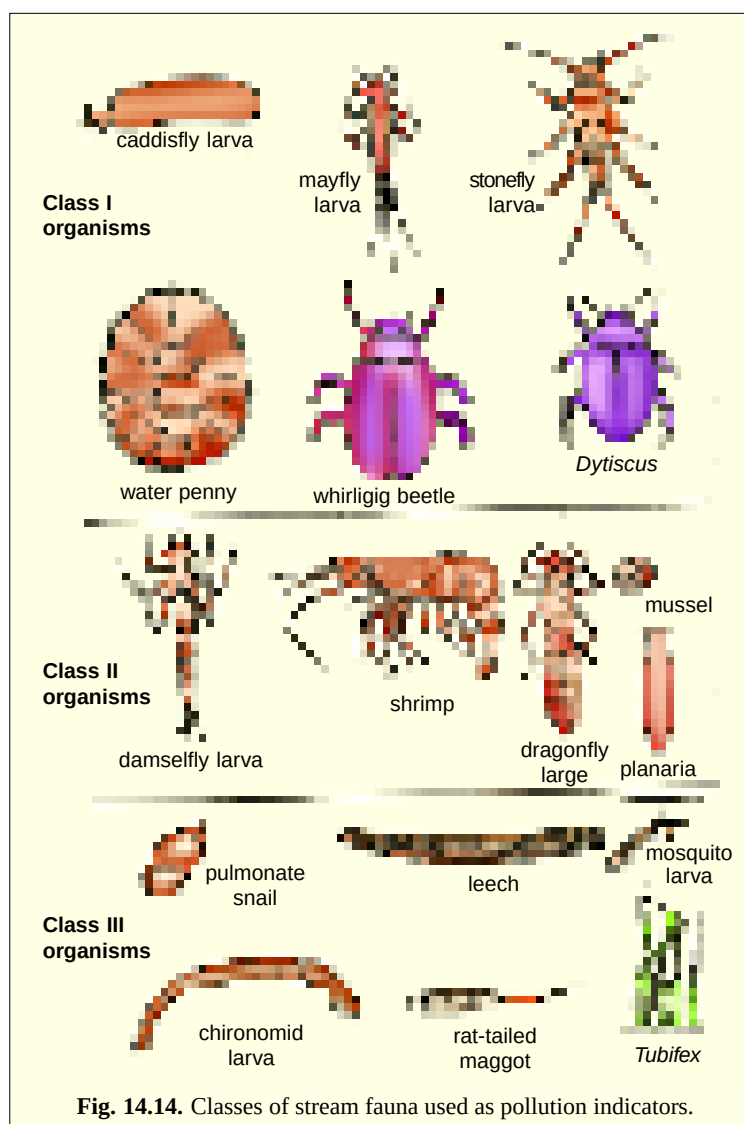


Fig. 14.14. Classes of stream fauna used as pollution indicators.

tion in a freshwater ecosystem. Nymphs of stone flies and may flies are the first to disappear from water which has high organic pollution. As pollution increases, caddishfly larvae and many fish which require high levels of environmental oxygen move into less polluted area of the stream. Shrimps, water fleas, leeches; snails and most of the fish vanish as the pollution becomes severe. At such levels of pollution there is very little of dissolved oxygen and the animals present are chironomid larvae (blood worms) and the oligochaete worms *Tubifex* (Fig. 14.14). Some decomposing plants are known to produce toxins as **strychnine** which kills animals including cattle.

2. Industrial Pollution

Most of the Indian rivers and freshwater streams are seriously polluted by industrial waste of effluents (see Table 14-3). **Effluents** are waste products in a liquid form resulting from industrial processes and domestic activities. They are released by different industries such as petro-chemical complexes; fertilizer factories; oil refineries; pulp, paper, textile, sugar and steel mills, tanneries, distilleries, coal washeries, synthetic material plants for drugs, fibres, rubber, plastics, etc. The

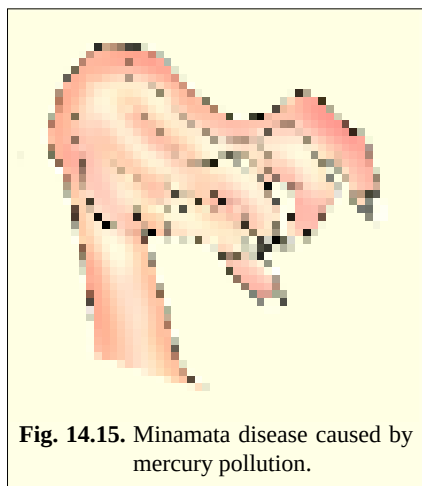


Fig. 14.15. Minamata disease caused by mercury pollution.

industrial wastes of these industries and mills include metals (copper, zinc, lead, mercury, etc.), detergents, petroleum, acids, alkalis, phenols, carbamates, alcohols, cyanide, arsenic, chlorine and many other inorganic and organic toxicants. All of these chemicals of industrial waste are toxic to animals and many cause death or sublethal pathology of the liver, kidneys, reproductive systems, respiratory systems, or nervous systems in both invertebrate and vertebrate aquatic animals (**Wilbur**, 1969). Chlorine which is added to water to control growth of algae and bacteria in the cooling system of power station, may persist in streams to cause mortality of plankton and fish. Heavy fish mortality in river **Sone** near Dehri-on-sone in Bihar is reported to cause by free chlorine content of the chemical wastes discharged by factories near Mirzapur in U.P.

Mercury like other heavy metals such as lead and cadmium has cropped up as a toxic agent of serious nature. Mercury, a byproduct of the production of vinyl-chloride, is used in many chemical industries and it is also a byproduct of some incinerators, power plants, laboratories and even hospitals, (**Aaronson**, 1971). In Japan, illness and even death occurred in the 1950s among fishermen who ingested fish, crabs, and shell-fish contaminated with methyl mercury from Japanese coastal industries. This mercury poisoning produced a crippling and often fatal disease called **Minamata disease** (Fig.14.15). Initial symptoms of minimata disease included numbness of the limbs, lips, and tongue, impairment of motor control, deafness, and blurring of vision. Cellular degeneration occurred in the cerebellum, midbrain, and cerebral cortex and this led to spasticity, rigidity, stupor and coma. In Japan in 1953, due to Minamata disease 17 persons died and 23 became permanently disabled (see **Southwick**, 1976).

Quit recently, **Dash** and coworkers (1988) studied mercury pollution due to effluents of a paper mill in the river Ib of the Mahanadi water system in Orissa. The paper mill uses mercury as a cathode in chloroalkali electrostatic cells and mercury leakage can be as high as 200 mg/tonne of chlorine produced (**Fergusson**, 1982). An analysis of the mercury in some fish showed 500 times biomagnification (or bioamplification) over the downstream surface water. Generally benthic fish and those higher in the food chain accumulated more mercury. People of the villages of that region, who used river water for drinking, suffered from diseases of eye, skin rashes and anemia.



Thermal pollution.

3. Thermal Pollution

Various industrial processes may utilize water for cooling, and resultant warmed water has often been discharged into streams or lakes. Coal-or oil-fired generators and atomic energy plants cause into large amount of waste heat which is carried away as hot water and cause **thermal pollution or calefaction** (warming). Thermal pollution produces distinct changes in aquatic biota. A body of water at 30-35°C is essentially a biological desert and many game fish require temperatures of $< 10^{\circ}\text{C}$ for successful reproduction, although they will survive above that temperature. A temperature rise of 10°C will double the rate of many chemical reactions and so the decay of the organic matter, the rusting of iron and the solution rate of salts are also accelerated by calefaction. Since the rate of exchange of salts in organisms increase, any toxin is liable to exert greater effects and temperature fluctuations are likely to affect organisms. Some plants and animals are killed outright by the very hot water. Other adverse effects of aquatic

pollution on aquatic life include (i) early hatching of fish eggs, (ii) failure of trout eggs to hatch; (iii) failure of salmon to spawn, (iv) increase in BOD, *i.e.*, solubility of oxygen is reduced causing deoxygenation; (v) change in diurnal and seasonal behaviour and metabolic responses of organisms; (vi) significant shift in algal forms and other organisms towards more heat tolerant forms (this leads to decrease in species diversity); (vii) affect changes in macrophytes and (viii) migration of some aquatic forms.

4. Silt Pollution

A result of intensive agriculture, earth moving for construction projects, poor conservation practices and downpour with resultant floods, is the increased production of silt in streams and lakes. This load of particulate matter cuts down primary productivity by decreasing the depth of light penetration. Silt may also interrupt or prevent the reproduction of fish, by smothering eggs laid on the bottom.

5. Water Pollution by Agrochemicals

Water that flows on the surface of crop fields, where agrochemicals such as fertilizers, pesticides and herbicides are used, contributes to heavy water and soil pollution. Pesticides and weedicides are used by human beings to control crop diseases by the pests or to kill the weeds and, hence, to increase the productivity. The use of these toxic chemicals has created health hazards not only for livestock and wild life but also for fish, other, aquatic organisms, birds and mammals including man.

Apart from killing the living organisms present on the surface of the soil, they reach even the deeper layers through tilling and irrigation of the land, killing still more living forms which might be involved in soil formation or humus formation, (*e.g.*, earthworms, centipede, millipede, etc.). With their continuous use the soil microorganisms lose their ability of nitrogen fixation. Moreover, when these chemicals find their way into water supplies, they contaminate and disrupt the aquatic ecosystem as well.

Any substance or mixture of substances which prevents, repels, destroys or mitigates any pest (*e.g.*, bacteria, viruses, fungus, insects, nematodes, rodents, etc.) is called a **pesticide**. Pesticides have proved tremendously beneficial to human populations, in reducing or eliminating the target organisms such as insects, snails, rats, and other animals which transmit disease, destroy agricultural crops, damage homes and stored products, and directly or indirectly affect human health and welfare. Hence,

the significance of chemical pesticides controlling mosquitoes, termites, houseflies, cockroaches, house crickets, weevils, locusts and grasshoppers, borers, snails, rats, rabbits, and a multitudes of other animals has been great, and it is difficult to imagine modern disease control and agricultural programmes without some forms of chemical control. Like pesticides, **herbicides** are specifically designed chemicals for the control of weed pests and unwanted plant growth.

Table 14-3.**Some Indian rivers and their major sources of pollution.**

Name of the river	Sources of pollution
1. Kali at Meerut (U.P.)	Sugar mills; distilleries; paint, soap, rayon, silk, yarn, tin and glycerine industries.
2. Jamuna near Delhi	D.D.T. factory, sewage, Indraprastha Power Station, Delhi.
3. Ganga at Kanpur	Jute chemical, metal, and surgical industries; tanneries, textile mills and great bulk of domestic sewage of highly organic nature.
4. Gomati near Lucknow (U.P.)	Paper and pulp mills; sewage.
5. Dajora in Bareilly (U.P.)	Synthetic rubber factories.
6. Damodar between Bokaro and Panchet	Fertilizers, fly ash from steel mills, suspended coal particles from washeries, and thermal power station.
7. Hooghly near Calcutta	Power Stations; paper pulp, jute, textile, chemical mills, paint, varnishes, metal, steel, hydrogenated vegetable oils, rayon, and soap, match, shellack, and polythene industries and sewage.
8. Sone at Dalmianagar (Bihar)	Cement, pulp and paper mills.
9. Bhadra (Karnataka)	Pulp, paper and steel industries.
10. Cooum, Adyar and Buckingham canal (Madras)	Domestic sewage, automobile workshops.
11. Cauvery (Tamil Nadu)	Sewage, tanneries, distilleries, paper and rayon mills.
12. Godavari	Paper mills.
13. Siwan (Bihar)	Paper, sulphur, cement, sugar mills.
14. Kulu (between Bombay and Kalyan)	Chemical factories, rayon mills and tanneries.
15. Suwao (in Balrampur)	Sugar industries.

Ecologically pesticides and herbicides have created two major serious problems which were not previously anticipated. In the first place many of them have persisted and accumulated in the environment and have harmed or contaminated numerous animals or plants not intended to be targets. Secondly, many of them have directly or indirectly affected human health.

Biomagnification or bioamplification. Many of pesticides, such as DDT, aldrin and dieldrin, have a long life time in the environment. They are fat-soluble and generally not biodegradable. They get incorporated into the food chain and ultimately deposited in the fatty tissues of animals and man. In the food chain, because of their build up, they get magnified in the higher trophic levels (called **biological magnification** or **biological amplification**). The pesticides have been in use during last 50 years. Their targets are insect pests, fungi, nematodes and rodents which damage crops. But these pesticides have created great problems for non-target organisms consisting largely beneficial species such as earthworms, honeybees, fish, amphibia, some reptiles, birds, mammals and man.

The phenomenon of biological magnification is also reported for certain other pollutants such as heavy metals such as lead, mercury and copper and radioactive substances (or radionuclids) as strontium-90.

Toxicity of pesticides. The toxicity of organo-chlorine pesticides (*i.e.*, DDT or Dichlorodiphenyl trichloroethane, hexachlorocyclohexane, chlordane, aldrin, dieldrin, etc.) lies in their inhibiting Na^+ ,

K⁺ and Mg⁺ adenosine triphosphatase activity in the nerve endings of animals particularly insects. It affects the sensory, motor nerve fibers and the motor cortex (Matsuma and Patil, 1961). In the giant axons of cockroach, DDT is known to influence the efflux of potassium ions from the axon.

DDT and other organochlorine pesticides are absorbed from the intestinal tract, from the alveoli of lungs and also through the skin, if the pesticides are in solution. A high concentration of DDT causes brain damage, centrilobular necrosis of the liver, and liver enlargement in small mammals. Concentration as low as 5—10 ppm in diet cause liver damage. In some birds, DDT concentration of 1—3 ppm destroys the female sex hormone and the egg shell becomes so thin that the eggs break when the parents sit on them for hatching.

Organophosphorus insecticides are absorbed by the gastrointestinal and respiratory tracts and the skin. These insecticides inhibit the **acetylcholinesterase (ACh E)** enzyme. Thus, an abnormal accumulation of endogenous acetylcholine occurs. This causes excessive activity of the para-sympathetic system, in the form of sweating, abdominal cramp, chest discomfort, vomiting, overactivity of smooth muscles, and so on. It also causes headache, nervousness, and over activity of voluntary muscles. The nervous system is also adversely affected.

The carbamate insecticides are absorbed by all portals, including the skin. These pesticides are reversible inhibitors of ACh E.

6. Marine Pollution

The marine ecosystem (*i.e.*, seas, oceans and estuaries) is so grand and vast but interacting in various ways with human life. How mighty and majestic ocean may appear to us, it was once challenged and tamed by Lord Rama during his march towards Ravana's Lanka. The oceans are used for navigation, fisheries, aquaculture, mining, acquisition of water, naval and military exercises as well as for the discharge and dumping of a variety of wastes. The very vastness of the ocean has led to the assumption that all wastes dumped into them can be harmlessly absorbed. Such an assumption now belongs to the past. The ever increasing range and volume of polluting activities and tend to seriously affect marine production. Marine pollution is most evident in coastal waters and estuarine areas. The oceans have in fact, become the final settling basin for millions of tons of waste products from human activities. For example, in the late 1960s West Germany was dumping 375 tons of sulphuric acid, 750 tons of iron sulphate, 20 tons of chlorinated hydrocarbons and 16,000 tons of gypsum wastes into the North sea and North Atlantic Ocean every day. Table 14-4 shows a few of the industrial and agricultural pollutants reaching the ocean annually in the mid- 1970s.

Thus, petroleum and its distillates, halogenated hydrocarbons including pesticides, organic compounds including detergents, domestic raw sewage, agricultural run-offs, inorganic chemicals including metals such as mercury, lead and copper, dumped containers, radioactive wastes, cooling waters from industries and hydel projects and coastal constructions are the major pollutants entering the sea. Often the noxious pollutants are not dispersed and diluted but tend to remain concentrated and localised, or when diluted may be reconcentrated by living organisms.

This tremendous burden of pollutants is evidently affecting the health and integrity of the world's oceans. Due to oceanic pollution the marine biota has been seriously affected. Cousteau (1974) reported a 30 to 40 per cent decline in the overall productivity of pelagic organisms from shrimps to whales in the past 25 years. He also observed a serious shrinkage of coral reefs in many tropical areas of world, and a displacement of these rich and varied communities with turbid and relatively barren waters. There has occurred a decline in populations of many fishes due to oceanic pollution. Oil spills have killed water birds, mammals, fish and vegetation (Hunt, 1965).

Oil is the most apparent pollutant of the ocean. It is discharged into the sea either deliberately when oil tankers are washed out prior to reloading or in accidental spills when tankers are wrecked (during war or otherwise). In coastal areas when oil spreads on the water surface, it clogs the feathers of diving birds making their flight impossible. White preening themselves in an attempt to clear the

plumage, they swallow enough oil to poison themselves. Apart from these, oil interferes with the insulation provided by the feathers and the birds die of cold or become susceptible to pneumonia. When oil covers the rocks and sea weeds, molluscs and crustaceans growing on them die. Oil-spills on the coasts of sea-side resorts, drive away holiday makers, affecting the economy of the place.

Table 14-4. Examples of industrial and agricultural pollutants discharged annually into the world's oceans (Source: southwick, 1976).

Pollutant	Estimated annual discharge, 1970-75 (in metric tons)	Source
1. Petroleum and industrial hydrocarbons	3,405,000	Offshore wells, oil tankers, industrial wastes.
2. Hydrocarbons (airborne)	15,000,000	Vehicles, industries, power plants.
3. Airborne lead	350,000	Vehicles.
4. Mercury	100,000	Industrial operations.
5. Aldrin-Toxaphene (converted to dieldrin)	25,000	Agricultural and public health operations.
6. Benzene hexachloride (BHC)	50,000	Agricultural and public health operations.
7. DDT	25,000	Agricultural and public health operations.
8. Polychlorinated biphenols (PCBs)	25,000	Plastic industries.

Control of Water Pollution

Most cities of world have evolved certain engineering systems such as **septic tanks, oxidation ponds, filter beds, waste water treatment plants** and **municipal sewage treatment plants** for the removal of various pollutants from the sewage before it is tripped into river or sea. It is essential to have

modern sewage treatment plants for every town and city of India so that the biodegradable as well as non-biodegradable pollutants can be removed from it and pure water obtained for recirculation.

The urban sewerage condition of India is quite distressing. At present only 15 class-I cities out of 142 are fully sewerred, whereas, only 7 of the 190 class-II towns claim to have this facility. Still the Indian cities continue to pour their waste water (286 thousand million cubic metre) into natural water sources.

Currently, water hyacinth (*Eichhornia crassipes*), an otherwise pernicious aquatic weed, has come into prominence for purifying domestic and industrial waste water. The plant regenerates rapidly and has a tremendous capacity to accumulate heavy and even radioactive metals. It is efficient in absorbing nitrogen, phosphorus and similar chemical pollutants. The polluted water fed into reservoirs or lagoons with water hyacinth, becomes markedly clean and free from 75-90 per



Water hyacinth plays an important role in controlling water pollution.

cent of its pollutants. This plant has also been used as a new source of food, fertilizer and biogas (energy).

In India, the enactment of 'Prevention and Control of Water Pollution Act' in 1974, has helped to a certain extent to prevent water pollution. However, adequate legislative measures have to be adopted by every state to ensure: (a) proper disposal of sewage and industrial wastes; (b) prevention of abuse of water resources; (c) recycling of waste waters through proper methods of purifications; and (d) punishment of erring industries which do not install effluent devices in their factories. A large number of international and inter-governmental agencies are now busy in controlling marine pollution. A number of highly hazardous pollutants are **black listed** and their discharge in the sea is totally prohibited. These are halogenated organic compounds, tin, mercury and cadmium compounds, radioactive isotopes and carcinogenic material. Somewhat less toxic substances are **gray listed** such as compounds of arsenic, zinc, and antimony, cyanides, organosilicon compounds, crude oils, foam forming detergents and surfactants (see **Ambasht**, 1990).

Realizing the devastation caused by the population of cities, towns and villages of 8 states and the union territory located along the coasts of river Ganga, through its run of 2525 km from Gangotri (Himalayas) to Ganga Sagar (Bay of Bengal), the Government of India has launched Ganga Action Plan to restore its water quality. The first phase of this gigantic cleaning operation had been initiated in year 1985 and may take several years in its completion.

Further, since xenobiotics, i.e., synthetic herbicides and pesticides and fertilizers pose a challenge to the natural degradative process. Therefore, we should

encourage a judicious, efficient and optimum use of organic manures. Recently, certain microbes have been used for the breakdown of these synthetic compounds. For instance, specific microorganisms are being evolved by molecular breeding process to convert the chief ingredients of herbicide agents, 2, 4, 5-T, into carbon dioxide and chloride.

The waste water polluted with cyanides and heavy metals can be purified by certain bacteria which cannot only withstand but are also able to grow at high level of cyanides. Millions of these bacteria are introduced in each of the several rotating discs of the plant's main processing units (Fig. 14.16). The sticky body surfaces of these bacteria pick up zinc, iron and other metals from the industrial waste water as it passes over the plates. These bacteria also absorb cyanide so toxic to the fish and other aquatic life. Subsequently this water is passed through clarifier, and filtered and then released into the streams.

In the past, attempts have been made to get rid of oil slicks by spraying them with **detergents** which break up the oil into drops. The trouble is that detergents are even more deadly than the oil and they kill many marine organisms that might have escaped the oil. Nowadays less destructive methods are used. Recently, a product "**bregoil**", resembling sawdust from paper industry waste has been employed. It rapidly absorbs oil. The resulting residue can be conveniently handled and even burnt as a fuel. Bacteria which thrive on oil eventually break it down and this brings about a regeneration of fauna and flora even within a year. Recently a strain of the bacterium *Pseudomonas putida* has been created by genetic engineering which breaks down octane, xylene and camphor (see **Green et al.**, 1990).

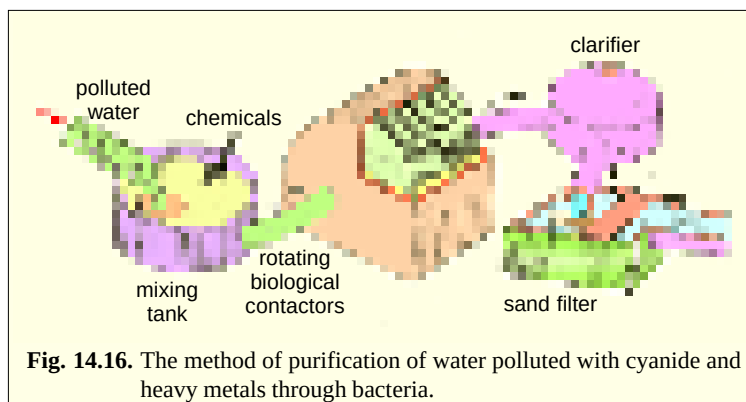


Fig. 14.16. The method of purification of water polluted with cyanide and heavy metals through bacteria.

3. LAND POLLUTANTS AND LAND POLLUTION

The land pollution is caused by solid wastes and chemicals. There are many examples of land that has been stripped of vegetation by industrial development and disposal of waste. Some common soil contaminants, their sources and wide range impacts on the biota have been summarized as follows:

The slag heaps from mines bear witness to the destructive effects which this can have on our environment. Areas around smelting and mining complexes are usually soiled by metals such as cadmium, zinc, lead, copper, arsenic and nickel. These are not only phytotoxic even in small amounts but also render plants unsafe for human and animal consumption. Zinc, often with cadmium, is released into the environment during the use or breakdown of lubricating oils, vehicle tyres, galvanized metals and fertilizers.

The major sources of land pollution are the industries such as pulp and paper mills, sugar mills, oil refineries, power and heating plants, chemicals and fertilizer manufacturing units, iron and steel plants, plastic and rubber producing complexes and so on. Huge amounts of solid wastes are either dumped, burnt, or emptied into rivers and seas. Most industrial furnaces produce a grey, powdery residue of unburnt material known as fly ash. The fly ash, cinders, solid wastes and litter all are thrown away by industries and form huge mounds which spoil the landscape.

Our households too contribute a large amount of solid rubbish in the form of domestic wastes. Some common examples are groceries, food scraps, vegetable remains, packing materials, cans, cardboard cartons, rags, paper, cinders, ash, broken gadgets, wood, worn-out furniture, metals, bones of dead animals, plastics, polythene bags, ceramics, glass, aluminium, rubber, leather, construction rubbish, brick, sand and other junk. Some man-made materials can be used again such as paper, scrap metal, glass, polythene, plastic, etc. But the majority of these cannot be reused and must be got rid of somehow. All these go to constitute heaps of municipal refuse. If not properly disposed, this rubbish can prove perilous, filthy and unhygienic. Such places often become a home for rats, flies, mosquitoes, bacteria and many other vectors, which may spread numerous human diseases.

Minimizing Land Pollution

It should be necessary for the industries to install collectors to remove the particulate wastes (fly ash) from the chimneys. Appropriate methods should be developed to dispose off or utilize the other types of pollutants. The garbage, instead of burning in the open, can be used not only to produce energy but also as filler for cement, bricks, asphalt and pavings. Some of these wastes, if properly sifted and separated, can even be recycled as raw material for other industries.

Another simplest method is crude **tipping** or **open dumping**, a common method used in most Indian cities. More satisfactory is **controlled tipping** or the **sanitary landfill**, which is recently adopted in Delhi for solid waste disposal. The **sanitary landfill** is a better remedy for larger objects because it brings about inexpensive biodegradation of such trash without causing much pollution, disease and ugliness. In sanitary landfill, a layer of about 2 meters of refuse is covered by at least 23 cm of earth, ash or other inert material, up to the level of the hole chosen. The chosen holes may be low lying watery areas and ditches. The land, thus, reclaimed can be used for making gardens, parks, playgrounds or apartment complexes. Before such filling, the wastes can be pulverized by machines to a uniform size— by this method the volume of refuse is reduced and some of the refuse is more quickly biodegraded. There are certain persons like **Nek Chand** who has found an aesthetic use of household refuse in the construction of world famous monuments such as “Rock Garden” of Chandigarh.

4. RADIOACTIVE POLLUTION

Radioactive isotopes, or radionuclides, are forms of elements with unstable atomic nuclei; that is, they decompose with ionizing radiation in the form of alpha or beta particles, or gamma rays. Many radioisotopes, such as radium-226, uranium-235 or 238, thorium-232, potassium-40, or carbon-14, occur naturally in rocks and soil. Other radioisotopes such as those of cesium, cobalt, iodine, krypton,

plutonium, and strontium, result primarily as fission products from atomic bomb fallout, nuclear reactors or other radiation sources. Of more than 450 radioactive isotopes which can occur as fission products, only a few are of major environmental concern. These are primarily argon-41, cobalt-60, cesium-137, iodine-131, krypton-85, strontium-90, tritium and plutonium-239 (Bebbington, 1973).

Within biotic communities and ecosystems, these radioactive elements may become dispersed or accumulated, depending upon the biological activity of the element and period of radioactivity of the isotope. **Strontium-90**, for example, normally occurs in radioactive fallout, has a half-life of 28 years and behaves like calcium in biogeochemical cycles. Thus it is absorbed by plants, ingested by animals and deposited in bone tissues close to blood forming tissue. Strontium-90 can also concentrate in natural biological systems in following method: water → bottom sediments → aquatic plants → freshwater clams → minnows and small fish → musk rats. It is demonstrated that due to this food chain musk rats concentrate strontium-90, 3500 times above the levels of the water in which they live. Grazing animals concentrate strontium-90 by ingesting it through grass and forage, and it can then be passed on to humans through milk.

Radioactive phosphate, cesium and iodine-132 also can readily accumulate in plants and animals through natural food chains. However, in food chains involving arthropods radioactive isotopes of potassium, sodium, and phosphorus accumulate, but isotopes of strontium and cobalt do not (Reichle *et.al.*, 1970).

Although isotopes may accumulate in human tissues as well as those of plants and animals, it is not established at the present time whether current levels of isotopes in human tissues represent serious health hazards to man (Southwick, 1976; Smith, 1977). Some medical scientists such as Gofman and Tamplin (1970) and Sternglass (1972), however, feel that man's radiation exposure from artificial sources is already sufficient to produce serious disease problems (leukemia and bone tumors), genetic damage, and infant mortality.

5. NOISE POLLUTION

Noise is primarily a feature of cities and is defined as 'sound without value' or 'any noise that is undesired by the recipient'. Noise levels in many urban-industrialized situations are known to be deleterious to human health and efficiency, with effects on the sense organs, cardiovascular, glandular and nervous systems.

High intensity sound or noise pollution is caused by many machines man has invented during his technological advancement. Thus, there exists a long list of sources of noise pollution including different machines of numerous factories, industries and mills, different kind of auto and motor vehicles such as scooters, motorbikes, cars, tempos, buses, trucks, tractors, aircrafts, motorboats, ships, loudspeakers, social gatherings, loud pop-music, supersonic aircrafts, etc. Noise can be measured by a sound metre and is expressed in a unit called the **decibel (dB)**. The quietest sound that the human ear can detect (zero decibels) is called the **threshold of hearing**. Calcutta, Bombay and Delhi are regarded to be noisiest cities in the world, where the average noise level was 90 decibels in 1975.

Noise pollution has certain well evident ecological and pathological effects on biota and human beings. For instance, the sonic boom path associated with SST projects such as Concorde produces noise of a very different order, in the form of sudden but repeated shock waves, which is suspected to cause disturbance to wild birds as well as domestic stock and buildings (see Simmons, 1974). Noise is not only annoying but if continued for a long time can also result in emotional and behavioural alterations in man.

Health Hazards of Noise Pollution

Noise causes disturbances in the atmosphere which in turn interferes with the systems of communication. It affects our peace of mind, health and behaviour. Sudden loud note can cause acute damage to the ear drum and the tiny hair cells in the internal ear, whereas prolonged noise results in

temporary loss of hearing or even permanent impairment. It causes headache irritability and impairs focussing. Noise is known to flush the skin, constrict stomach muscles and produce ulcers, heart disease, high blood pressure, nervousness and other defects in sensory and nervous systems.

Reducing Noise Pollution

Though it is almost difficult to completely get rid of the malady of noise pollution of current electronic age. However, there are certain methods by which deleterious influences of noise pollution can be minimized and intensity of noise level can be curbed. The means of noise control are : (a) to manipulate the source so as to reduce the noise at its origin; (b) to interrupt the path of transmission and (c) to protect the recipient.

Legislation and public awareness are essential. Nobody should be permitted to create noise in silent zones or during night. Noise producing traffic vehicles should be prevented from plying on the roads and their use of pressure horns should be entirely checked. Standards for noise control measures should be set up for industry and community, and a comprehensive safety programme enforced.

The path of the sound can be interrupted by using various materials which absorb the sound energy. Horticulturists should suggest adequate varieties of vegetation which can be planted around factories, hospitals, educational institutions, public libraries and houses which may reduce sound pollution and also may minimize dust pollution. Acoustic materials and mufflers can also be used to protect oneself. Under noisy situations, one may hold his hands over his ears or may run away from the source, or may simply stuff a bit of cotton or ear plugs in the ears to reduce much hazards of noise pollution.

REVISION QUESTIONS

1. Define the term 'pollution,' Write an essay on the 'problem' of ecological pollution and its control methods.
2. What is 'air pollution' ? How it affects the biota including man? Describe certain recent methods to control the air pollution.
3. What is 'water pollution' ? Describe different kinds of water pollutions and their ecological effects on the aquatic life and also on man.
4. Write an essay on 'the eutrophication of aquatic habitats'.
5. Write short notes on the following :
 - (i) Minamata disease; (ii) Greenhouse effect; (iii) Photochemical smog; (iv) Thermal pollution;
 - (v) Pesticide and herbicide contamination; (vi) Noise pollution; (vii) Radioactive contamination.

CHAPTER

15

Ecology and Human Welfare

(Natural Resource Ecology :
Natural Resources,
Conservation and Management)

Mother nature is quite generous. For thousands of years, it has provided us with all of our basic needs—air, water, sunlight, food, clothing and shelter. Up to the beginning of 20th century our thinking was that our natural treasures were eternal and virtually inexhaustible. However, by observing the present trend of ever increasing human population and mindless exploitation of these repositories, we can speculate for future that mankind seems to be heading for a grim time which might be full of shortages, scarcities and chaos. The fortune which the mankind had inherited from the nature, has been and is being exhausted so carelessly that it is not likely to last for very long, if not harnessed properly.

CLASSIFICATION OF NATURAL RESOURCES

We may classify resources into two major types according to how they are affected by their consumers :

1. Non-renewable resources. These resources once gone, have very little chance of recovery or resynthesis. Non-



These wind pumps use the power of the wind as their source of energy.

renewable resources are not altered by use. For example, space once occupied becomes unavailable; it is “replenished” only when the consumer (user of a commodity) leaves. Mineral deposits are formed slowly over million of years and once used cannot be regenerated, *e.g.*, fossil fuels, such as petrol and coal. Since the formation of soil takes thousands of years and is not renewable in the life span of many generations, it is thought of as a non-renewable resource.

2. Renewable resources. These resources are constantly regenerated or renewed. Births in a population of prey continually supply food items for predators. Plants (crops, forests, etc.) and animals (milk and meat production) form another good example of renewable resources; since their continued harvest is possible upon proper planning and management. Other examples of renewable resources are wild life, aquatic life, pastures and forests.

CONSERVATION OF NATURAL RESOURCES

Conservation has been defined as “the management for the benefit of life including human kind of the biosphere so that it may yield sustainable benefit to the present generation while maintaining its potential to meet the needs and aspirations of the future generations.” Thus, conservation is one of the most significant application of ecology. It avoids unplanned development which breaks ecological as well as human laws. A conservationist has two folds basic aims (i) to insure the preservation of a quality environment that considers esthetic and recreational as well as product needs and (ii) to insure a continuous yield of useful plants, animals, and materials by establishing a balanced cycle of harvest and renewal (**Odum**, 1971). Thus, conservation process remains chiefly concerned with the use, preservation and proper management of the **natural resources** of the earth and their protection from the destructive influences, misuse, decay, fire, or waste.

In fact, the preservation of rare species is certainly not all that conservation means. Conservation is never a “hoarding” or even “rationed use” of a limited source but a constant effort to increase the resources and rapid resynthesis of the materials. Thus, conservation of living resources has following three objectives : (1) to maintain essential ecological processes and life support system; (2) to preserve biological diversity; and (3) to ensure that any utilization of species and ecosystems is sustainable. Conservation, therefore, makes significant contributions to social and economic development.

Different kinds of natural resources and their conservation and management practices can be studied under following heads :

1. MINERALS AND THEIR CONSERVATION

Minerals are often called “**stock**” or “**non-renewable**” resources, because, their ‘new’ materials can only be extracted from the earth’s crust once. But even in the transformed state in which they are used, they are not lost to the planet and so are ideally available for reuse.

(a) Terrestrial Mineral Resources

For his industrial, technological and cultural growth man has required and still needs a great variety of inorganic materials, all of which come from the earth’s crust. Chief among these are the ores which are used on a large scale to yield metals such as **iron**, **aluminium**, **copper**, **silver**, **gold**, **platinum**, etc. To them must be added elements which may not be needed in large quantities but which are indispensable in many modern industrial processes, as for example catalysts and hardeners such as **vanadium**, **tungsten**, and **molybdenum**. Finally there are non-metallic materials which are vital to industrialized nations such as sand and gravel, cement, fluxes, clay, salt, sulphur, phosphorus, diamonds, and the chemical byproducts of petroleum refining.

(b) Marine Mineral Resources

The mineral resources of seas can be divided into three categories : those which are dissolved in water itself; sediments present on the sea-bed at various depths; and those present at some depth

below the sea-floor, beyond the sediments of relatively recent origin.

At present the utility of the dissolved elements is in the direct proportion to their abundance and to the relative cost from terrestrial sources. **Cloud** (1969) has enlisted commonest elements (*e.g.*, strontium, boron, silicon, fluorine, argon, nitrogen, lithium, rubidium, phosphorus, etc.) in sea water which could be extracted. Common salt is one of the resources that has been utilised since prehistoric times for its value in flavouring and meat preservation. At present only **salt, magnesium** and **bromine** are being extracted in commercial quantities and sea seems to remain inexhaustible for these elements. Except a few other elements, economically feasible recovery processes of most other elements from sea is very low.

Sediments and sedimentary rocks on the continental shelves are sources of certain materials. **Placer deposits** (= materials that has been concentrated in a particular place by mechanical action) contain workable quantities of **gold, tin**, and **diamonds**, and the other sediments which may be liable to exploitation include **sand, gravel** and **shells**. However, land-use problems by their extraction from the land would largely be prevented by the use of the sea as a source, provided that the ecosystems of the oceans were not too greatly damaged by the recovery processes, which create great quantities of silt and also eventuate imbalance in the sedimentary systems of the sea-floor. **Phosphates** are found as nodules and in crust. A further resource of the continental shelves is **freshwater**; large quantities of artesian water may be found in certain aquifers. Finally there are petroleum and natural gas.

The deep sea basins is found to contain enormous quantities of minerals such as **manganese, nickel, cobalt** and **copper**, but their extraction is not economically feasible (**Cloud**, 1969).

(c) Conservation of Terrestrial Mineral Resources

Until recently little attention was paid to conservation of terrestrial mineral resources because it was assumed that there were plenty for centuries to come and that nothing could be done to save them any way. But recently **Cloud** (1969) made it evident that both assumption are dead wrong. During his assessment of the situation, he has formulated following two situations about the exploitation of terrestrial mineral resources—(1) The first is the **demographic quotient** or **Q** :

$$Q = \frac{\text{total resources available}}{\text{population density} \times \text{per capita consumption}}$$

As this quotient goes down, so does the quality of modern life; it is going down at a frightening rate because available supplies can only go down as consumption goes up.

(2) The other concept introduced by **Cloud** is the graphic model of **depletion curve**, as shown in Figure 15.1. With the present procedure of “*mine, use and throw away*”, a huge boom and bust is projected, as shown in curve A. The time scale is uncertain because lack of data, but the “bust” could begin within this century since certain key metals and fuels such as zinc, tin, lead, copper, uranium-235, natural gas could be mined or extracted out within 20 years in so far as the readily exploitable reserves are concerned. If a programme of mineral conservation, substitutions (using less scarce minerals wherever possible), and partial recycle were to be started now, the depletion curve could be flattened as shown in curve B. Efficient recycling combined with stringent conservation and a reduction in per capita use could prolong depletion for a long time, as shown in curve C. It should be noted that even with perfect recycle, depletion would still occur, because small amounts of most metals lose by friction, rust, etc.

Consequently, terrestrial mineral conservationists have suggested following measures for terrestrial mineral conservation : 1. Technological advancement for extraction of minerals from sea which is unlimited supply depot. 2. Extraction of metals from lean ores with the use of vast quantities of cheap atomic energy. **Granite** is suggested as a source for the extraction of many metals. 3. Recovery

of mineral elements from scrap and waste (recycling).

Ecological Aspects of Mining

Most kinds of mining processes of man have side-effects of an ecological nature. Underground mining may include whole new towns among its surface installations (*e.g.*, the Khateri Copper Project, JhunJhunu, Rajasthan, India). Timber is cut in forested areas, often leading to soil erosion and the tailings and mine-

waste have to be discarded. Large-size solid wastes can be used as backfill or sold for aggregates, but tailings usually yield silt particles to wind and water and are often chemically unstable; the only suitable treatment appears to be to “fix” them with vegetation. Mine waters are often heavily contaminated and have to be treated chemically and physically, or injected into “safe” rock strata.

Further, in most countries open-pit mining is more widespread than extraction by shaft. In such mining waste disposal becomes a major problem to which back fill is the obvious solution; if the topsoil is saved then restoration of agriculture or recreational use is often possible. The processes of concentration, beneficiation and refining may all create biological changes if various products are released into nearby ecosystems, *i.e.*, they may create environmental pollution which affect adversely the ecosystems.

2. ENERGY AND ITS CONSERVATION

Energy utilization has become the modern key to human progress. At the present time, modern nations differ markedly in their energy consumption, with industrialized nations such as European nations like Britain, Germany, Poland, etc., United States, Japan, Hong Kong and Singapore, typically using 10 to 50 times as much energy per capita as agricultural and developing nations such as India, Bangladesh, Indonesia and Ethiopia (Southwick, 1976). Higher energy consumption is often associated with higher gross national product and personal income per capita, but it is also associated with higher levels of industrial pollution.

Fossil fuels such as oil, gas and coal form the non-renewable sources of energy and account for 90% of the world's production of commercial energy. Hydroelectric and nuclear power accounting for only 10%. The figures are :

Oil	39.5% energy
Natural gas	19.6% energy
Coal	30.3% energy
Hydro-electric	6.7% energy
Nuclear	3.9% energy

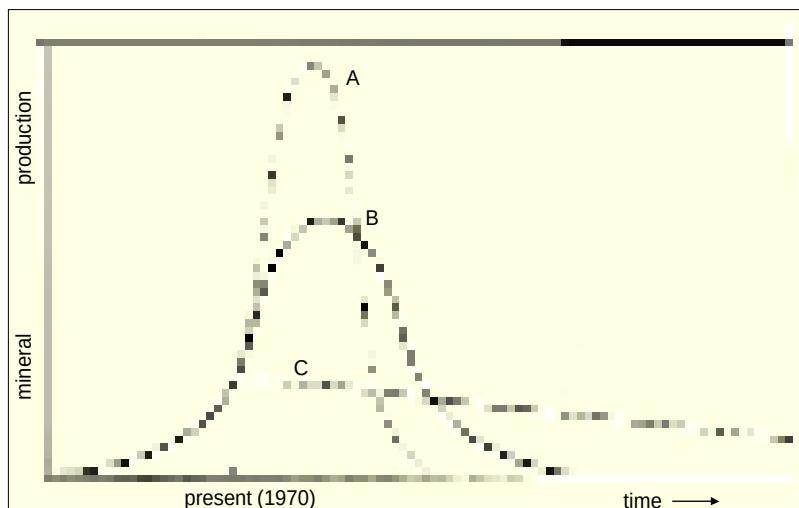


Fig. 15.1. Alternate depletion patterns for mineral resources. A—Pattern of rapid extraction and depletion of minerals that will occur under the present custom of unrestricted mining, use, and throw away ; B—Depletion time can be extended by partial recycle and less wasteful use ; C—Efficient recycle, combined with stringent conservation and substitutions can extend mineral depletion curves indefinitely (Cloud, 1969).

According to Advisory Board on Energy (ABE), the various forms of energy are likely to make the following contribution in India by year 2004–2005 : coal 450–540 Mt; oil 90.11 Mt; electricity 501–592 billion kW and non-commercial energy : 500 Mt.

A. Commercial Sources of Energy

I. Fuels. Fuel minerals include mainly the non-renewable fossil energy resources such as coal, petroleum (oil) and natural gas, all of which exist in large quantities and lie buried under the ground. Due to their great importance in generation of energy, these natural resources are receiving utmost attention. It is estimated that with ever increasing use of energy in present industrial—urban complexes, up to year 2000, global energy requirement would rise by 58 per cent. If the energy production fails to keep pace with the growth rate of our consumption, a catastrophic situation may arise. Fast depletion chances of these fuels and the impacts of their extraction and utilization on the environment are certain other serious problems.

1. Coal. Generally coal is regarded as our most plentiful and useful fossil fuel. This is largely carbon, mixed with incombustible mineral matter of the earth's crust. Coal is formed by the transformation of layers and layers of decomposed vegetable and animal matter under the combined effect of heat and pressure of earth. Coal deposits are probably being formed even today as in the Ganga river delta of our country. Coal deposits occur in abundance and are fairly evenly distributed over the various continents of world. Soviet Union, China and United States contain about 80 per cent of the world's total reserves. The rate of coal consumption in the world as calculated in 1976, was 3.7 billion short tons per year. On this basis, our total reserves would last for 212 years and recoverable resources for approximately 1700 years. Coal is the only long run fossil fuel which is the long-run source of gas and many coal-tar and petroleum products that can be derived from it.

2. Petroleum, oil and natural gas. The organic deposits of these fossil fuels were probably formed from tiny microorganisms rather than from the debris of large plants as was the case with coal. When these microscopic creatures settled on ocean floor and were later covered with mineral sediments, tiny droplets of body oil were squeezed out of them. This oil was trapped into large deposits during formation and was altered chemically by heat and pressure.

According to an estimate, the recoverable amount of petroleum of the world is approximately 2000 billion barrels. Of these, 339 billion barrels have already been produced, 646 billion barrels are reserves, and slightly over 1000 billion barrels are to be discovered as yet. About 60 per cent of the total world's reserves are concentrated in Persian Gulf and South West Asian countries. The industrialized nations, excluding the USSR, with the highest rate of petroleum consumption have the smallest reserves and resources. In comparison to other countries of the world, India's reserves of oil and production of crude petroleum are quite meagre. India produced only 21.7 billion barrels of oil in 1976. It is estimated that if the present oil production rate remained constant, global supply would last about 30 years and the entire resources no longer than 77 years.

However, before we reach that stage, we will undoubtedly have ravaged the great oil-laden shale deposits. A wide variety of laminated solidified mixtures of inorganic sediments and organic matter, called **oil-shales**, have the property of yielding oil on destructive distillation. Certain countries such as Brazil, China, USSR, Zaire and Scotland exploit this resource to get oil. It is speculated that 15,200 billion barrels of high grade oil (25 to 100 gallons per ton shale) and 3,18,000 billion barrels of low grade oil (10 to 25 gallons per ton material) constitute the total resources. Likewise, **tar sands** (also known as oil sands, bituminous sand and bituminous rocks) contain heavy asphaltic, viscous substance called **bitumen** (petroleum). These occur in Canada and have 731 billion barrels of inplace oil.

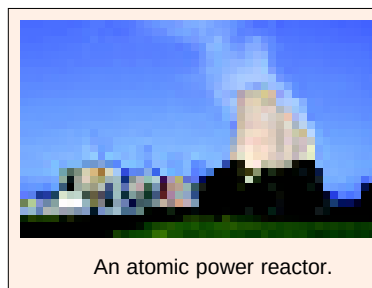
Natural gas is another energy resource. USA is its largest producer as well as consumer. About 50 trillion cubic feet natural gas was obtained in 1976 and if this pace continues constantly in future, the world's reserve of natural gas would last about 45 years.

3. Nuclear fuels. The nuclear energy of the atoms has been harnessed by three types of reactors: fission reactor, fast breeder reactor and nuclear fusion reactor. While fission and fast breeder reactors are based on the splitting of uranium and plutonium, the nuclear fusion reactor is latest and most powerful and involves fusion of hydrogen nuclei. Nuclear reactors produce enormous energy for driving ships and generating electric power.

Our planet is estimated to have 4900 thousand short tons of uranium, out of which 3510 are known from USA. These can fuel 800 gigawatts (800 billion watts) of light water reactor plants. In comparison to oil and gas, uranium represents a resource of limited potential under present state of technologies. However, new enrichment technologies may prolong these supplies. In fact, utmost care need to be exercised to produce energy from the nuclear fuel, radioactive contamination is a significant biological hazard. Any accidental fall-out or escape from nuclear wastes can produce long lasting disasters often lethal effects on man and other biotic components of the biosphere.

II. Electric energy production. The various sources of electric energy are (i) atomic (nuclear) power reactors; (ii) lignite; (iii) lean gas; (iv) hydropower plants; (v) coal plants, and (vi) small (micro-hydel plants).

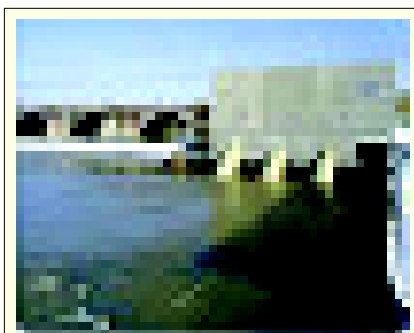
1. Atomic power reactors. Atomic power has become a principal source of energy in recent years, since the fossil fuel reserves are depleting very fast. A small quantity of radioactive material can produce an enormous amount of energy. For example, one tonne of uranium ^{235}U would provide as much energy as by three million tonnes of coal or 12 million barrels of oil. Besides getting the electricity, atomic power is also used as fuel for marine vessels, heat generation for chemical and food processing plants and for spacecrafts.



An atomic power reactor.

For the generation of atomic energy one requires a **nuclear reactor**. The decay of fissionable matter produces great amount of heat. This heat is used to make steam which is channeled through a turbine connected to **electric generator**.

2. Hydroelectric energy. Moving water has been harnessed as a source of energy for many years. Waterfalls turned paddle wheels to grind corn or wheat in early Europe, America and India. Later,



Hydroelectric power is an important renewable energy source.

moving water was backed up behind great dams and then released to turn turbines of huge generators for **hydroelectric energy**. Hydroelectric power is the most important renewable energy source. It is a cheapest source of power generation, since land is free and environmental costs are seldom calculated. It is regenerative, thus, helps in the conservation of fossil fuels. Potential of such hydroelectric energy is being developed in India through large-scale projects in the Himalayan region in North and North-East regions. Recently, however, these megaprojects have been criticized and vehemently opposed by the environmentalists such as **Sunder Lal Bahuguna** and **Medha Patekar**, for these have resulted (during development of their infrastructure) into deforestation, wildlife destruction, increased soil

erosion and other socio-political and economic problems.

3. Micro-hydels. In India, there is great scope for the development of micro-and minihydel power schemes. Micro-hydels are proper means of generating electricity in the remote areas. Such installations are usually developed on the run-off-the river schemes and do not require much infrastructure. They are more safe environmentally and are very useful for the local people. In fact, they do not involve approach roads, no pondage, no loss of vegetative cover, and are free from pollution hazards.

B. Non-commercial Sources of Energy

All materials which originate from photosynthesis form the **biomass**. It includes all new plant growth, residues and wastes (wood, short rotational trees); herbaceous plants, freshwater and marine algae, aquatic plants; agricultural and forest residues (straw husks, bagasse, corn cobs, bark, sawdust, wood shavings, roots, animal droppings); wastes (garbage, dung, night soil or faeces), industrial refuse, etc. Biodegradable organic effluents from industries such as canneries, sugar mills, slaughter house, meat packing plants, breweries, distilleries, etc., are also included in this category. Biomass-based energy can be obtained by the following sources :

1. Fire wood. According to Advisory Board on Energy or ABE the demand for firewood is going to go up in order of 300 – 330 Mt in the year 2004/ 2005 against the present level of 120 – 130 Mt.

Recently, efforts are going on to improve the firewood utilisation to get more and more energy. Some new techniques are the following :

(i) Briquetting. The Indian Institute of Technology, New Delhi has developed the technology of briquetting saw dust into smokeless fuel.

(ii) Gasification. Gasification of biomass is an important means of harvest energy through thermo-chemical conversion. It yields **biogas**, **producer gas** and **pyrogas**.

(iii) Improved chullahs. Present day chullahs (which are about 112 million in number) have a very low, 2–10% efficiency. This results into wastage of wood, deforestation, air pollution and health problems. Department of Non-conventional Energy Sources (DNES) have designed certain improved stoves which contain thermal efficiency up to 15 to 20%.

2. Petroplants. There are attempts to identify potential plant species as sources of liquid hydrocarbons, a substitute for liquid fuels. In this respect, 15 species hold promise and they belong to families Euphorbiaceae, Asclepiadaceae, Apocynaceae, Urticaceae, Convolvulaceae and Sapotaceae. There is a need to increase the biomass of these plants, and conversion of their hydrocarbons into petroleum fractions. The Indian Institute of Petroleum, Dehra Dun done a remarkable job in this area, particularly on hydrocracking of the crude products. The products, thus, obtained are gases, naphtha, kerosene, gas oil, coke, etc.

3. Biogas. Biogas provides a significant solution to present energy crisis, especially in rural areas. This is an environmentally clean technology. The generation of biogas involves decomposition of animal dung (even human faecal matter), leafy and agricultural residues, under anaerobic conditions by bacterial action, in a suitable capacity digester. This results in the production of methane gas which is used for cooking and even for lighting purpose. Biogas is composed of methane, CO_2 , H_2 and N_2 . At 40% methane content, calorific value of biogas is 3,214 kcal/m³, at 50% is 4,429 kcal/m³ and at 55% is 4,713 kcal/m³.

Dung of domestic herbivore is used in biogas production. According to **Dr. Archana Sharma** (1987), India annually produces about 1000 million tonnes of dung, from which about 22,425 million m³ gas can be produced. Besides the cooking gas, slurry (*i.e.*, residual organic matter obtained from the digester) can produce 206 million tonnes of organic manure (or bio-fertilizer) every year that can replace 1.4 million tones of N_2 , 1.3 million tonnes of P_2O_5 (phosphate) and 0.9 million tonnes of K_2O (potash). However, this source of bioenergy is not being fully tapped in India, though China has fully exploited this biotechnology.

Biogas can also be generated from sludge obtained from primary treatment of raw sewage and one such plant is in operation at Okhla. Water hyacinth, *Hydrilla*, duck weeds and algae can act as supplements for production of biogas.

At present, in India, there are about 6.1 lakh biogas plants. It is estimated that 1.5 lakh biogas plants will save 6 lakh tonnes of wood equivalent every year.

C. Non-conventional Renewable Sources of Energy

These sources can boost production in specific areas, in a decentralised manner and are of the following types :

1. Dendrothermal energy. In Philippines, energy is experimentally generated from energy plantation. There is a claim of 3.75 kw of electric energy production from wood growing on 7 hectares of land.

2. Solar Energy. Solar energy of sunlight is a highly potential renewable resource like the wind energy. It does not cause any pollution and can be harnessed to produce enough power by developing economical technologies. India receives abundant sun-shine about 1648–2108 kWh/m²/ year with nearly 250–300 days of useful sun-shine in a year. The daily solar energy incidence is between 5 to 7 kWh/m² at different parts of the country. This immense resource of solar energy can be converted into other form of energy through thermal or photovoltaic conversion routes. 1. The **solar thermal** uses solar radiation in the form of heat that in turn may be converted to mechanical, electric or chemical energy. Solar thermal devices such as solar cookers, solar water/air heaters, solar dryers, solar wood seasoning ovens or kilns and silicon systems have been developed. 2. The **photovoltaic conversion systems** convert solar radiation directly into electricity through **silicon solar cells**. These cells may be of various types such as single crystal silicon cell, polycrystalline cells, amorphous solar cells, etc. The systems have been exploited for community lighting, radio and TV sets, light houses, offshore platforms and installation in remote areas. A significant application of photovoltaic conversion systems is pumping of water for micro-irrigation and drinking purposes. These systems being decentralised on-the-spot electricity generation systems help to replace old, noisy and polluting diesel-utilising systems. They may be installed in remote areas such as forests, deserts and hills.

3. Wind energy. Wind has provided cheap and pollution free energy throughout much of western Europe, but windmills are presently often treated as quaint relics. Part of the problem is that wind is a rather local phenomenon and is extremely unpredictable in most places. In India, there are areas which are quite windy. Average annual wind density of 3 kW/m² /day are common at various places in peninsular India and also along coast line of Gujarat, Western Ghats and parts of central India. Wind densities exceeding 4 kW/m²/day are available for 5–7 months in a year. During winter, wind densities become even more than 10 kW/m²/day.

Wind energy can be converted into mechanical and electrical energy. National Aeronautical Limited, Bangalore is engaged in research and development of power generation through wind mills. Wind energy is useful in remote areas, helps saving fossil fuels, delivers on-the-spot small quantity of energy which is free from pollution and environmental degradation.

4. Ocean (tidal) energy. Tidal power is a form of hydroelectricity produced by harnessing the ebb and flow of the tides. Barriers containing reversible **turbines** are built across an estuary or gulf where the tidal range (*i.e.*, rise and fall of sea level) is great.

5. Geothermal energy. Geothermal energy provides the most promising sources of power. Geothermal energy utilisation is possible in volcanic regions or where hot springs and geysers occur. A cold storage unit and 5 mW power plant using geothermal energy have been set up at Manikaran (H.P.). About 350 geothermal springs have been located in the country (in 46 of them the water temperature exceeds 150°C; **Dash**, 1993). Data of the investigations on geothermal energy potential at Tapovan, Badrinath (U.P.) by Wadia Institute of Himalayan Geology and other agencies only are available (see **Dubey**, 1995).

3. FOOD, AGRICULTURE AND AQUACULTURE

Humans need food as a source of energy and for tissue replacement, like any other animal. The source of much of the food consumed by man is **terrestrial agriculture**. This represents the most manipulated of all the non-urban ecosystems, in which the energy and matter pathways are directed almost entirely to man and where he maintains a high level of input of matter and energy to keep the

system stable in order to yield his preferred crop. Not only is the ecosystem man-made, but the plant and animal components of it have usually been genetically altered by man in the course of their domestication. There are two main types of agriculture; **crop agriculture**, in which plant production is harvested for use by man either directly or after processing and **animal agriculture**, where a crop from highly manipulated ecosystem is fed to domesticated animals. Ecologically, terrestrial agriculture presents man either as a herbivore or as a third trophic level carnivore.

Terrestrial agriculture systems today exhibit a major division into shifting and sedentary types:

(i) Shifting Cultivation

In shifting cultivation, total manipulation of the natural system is practised over a limited area but for only a short (1–5 years) period of time. Thus, the agricultural path is spatially and temporarily enclosed by wild vegetation. Today, shifting cultivation is largely confined to tropical forests, savannas, and grasslands. In this type of cultivation crops are planted in a mosaic of different heights and times of fruition so that the plant cover of the soil remains as complete as possible throughout the year in order to reduce the leaching effect of heavy rain fall. ‘Slash and burn’ techniques provide mineral nutrients for their uptake by the crops. The natural diversity of the forests is imitated by the variety of crops which are grown by some shifting cultivators. For example, the tropic crops of Hanunoo of the Philippines include rice, beans, root crops, shrub, legumes, tree crops, yam, taro, sweet potato, vines, bananas and sugarcane, together with European-contact crops such as maize, ground nuts, tomatoes, melons and pumpkins.

Ecologists have found shifting cultivation as a well-adapted system in forested lands where the trees regenerate easily when the plots are deserted, and where an equilibrium population has established.

(ii) Sedentary Cultivation

Sedentary cultivation represents the permanent manipulation of an ecosystem; the natural biota are removed and replaced with domesticated plants and animals. Competition by the remnants of the original biota or by man-introduced organisms may still remain and considerable efforts may be needed to keep these weeds and pests at an acceptable level. In **dry land agriculture** the soil assumes an importance for it now becomes the long-term reservoir of all nutrients and is constantly depleted as crops are harvested and removed. The nutrients must be replenished either by the addition of organic excreta or chemical fertilizers. There is also the **paddy-culture** of rice, where the soil is very largely a mechanical rooting medium for the plants and the water supplies the essential mineral nutrients; it often contains blue-green algae which fix nitrogen, for example. The variety of crops grown under the various forms of sedentary cultivation is very high and changing patterns of agriculture together with shifting trade flows and altered rates of consumption make a world kaleidoscope of infinite variety.

Ecological training has been widely used in agriculture. To select particular climate and edaphic conditions for a particular crop is primarily an ecologist’s job. Ecology plays a vital role in weed control. The nitrogen and phosphorus requirements of soil, crop rotation practice, etc., have been learned from ecology. Due to application of ecologically sound principles agriculture has resulted into **Green Revolution**, i.e., increased agriculture production. In fact, the Green Revolution of the late 1960s, was based on new genetic strain of rice and wheat, improved irrigation, and better application of fertilizers and it has achieved a doubling and tripling of crop yields in many tropical countries. But recently, Green Revolution has exhibited certain ecological problems—the breakdown of soil structure in European countries under a regime of continuous cereal cropping with the use of physically heavy machinery and its increase dependence upon petroleum and fertilizer. Further, new strains of agricultural crops which are high yielding, protein-rich, and weather and pest resistant, require more careful management, more water, more fertilizer, more insecticides, and more intensive care.

New Sources of Food

Most modern ecologists hold the view that even if the rapid development of conventional agriculture is sustained, protein deficiencies will continue to exist (Simmons, 1974). A search for supplementary sources of plant and animal proteins is, therefore, in progress. Animal proteins are more preferred ones because their amino-acid make-up is closest to man's requirements. Animal flesh has the further advantage that it is usually the more easily assimilable, since the plant proteins are locked away behind a cell wall of cellulose not easily broken down by the action of the human stomach. Animals, thus, have been a means of harvesting the plant protein in a digestible form, and so have considerable dietary advantages. If animals are to be avoided, a food source which will yield plant protein in a digestible form or from which the majority of cellulose has been removed is clearly attractive. Fungi appear to be easily assimilated and contain a good deal of protein by comparison with some other sources such as beef, fish, etc. Other advantages of mushrooms are that they do not absorb much human or fossil energy in production, can be grown independently of environmental factors in places such as caves and abandoned railway tunnels, can readily be stored in dried form and require little sophisticated knowledge or technology (Pyke, 1970).

Animals retain many desirable characteristics as cellulose-converters and as saliva-inducers. As very small number of wild animals have been domesticated by man for milk, wool, skin, food, transportation, etc., recently, certain other wild animals are increasingly domesticated. Birds such as young colonial seabirds, reptiles, amphibians, insects, crustaceans, molluscs, rodents, and many other animals can be utilized as food. Freshwater and brackish-water fish are other sources of animal proteins. Marine and freshwater fisheries provide a best sources of animal proteins, many nations such as Peru, Japan, USSR, China, Norway, U.S.A., UK, Burma, Thailand and India depend on them for food requirements. Due to overfishing of certain favourite fish species, recently following trends are adopted for continuous abundant supply of fishes :

(i) Extension of fisheries. The fisheries have been extended from three sources : the utilization of untapped species, the cropping of hitherto unattractive areas, and the development of more novel methods of culture and harvesting. During recent years many new fisheries have started to flourish, such as Peruvian anchoveta, Alaska pollock, Bering sea flatfishes and herring, and several more. In future, fishing method may be extended in the cool temperate parts of southern hemisphere.

(ii) Krill. Uneaten food of whales is also found suitable for human consumption as food. Approximately 80 per cent of the prey of blue and fin whales is **krill**, the shrimp *Euphausia superba*, each of which contains 7% fat and 16% protein.

(iii) Aquaculture. To increase the yield of certain edible species of freshwater and marine fishes, molluscs and crustaceans and other aquatic organisms, **fish farming** or **aquaculture** that includes their culturing and herding, has been employed and at a commercial scale in Hongkong, Philippines and Japan. Framework are lowered into shallow offshore waters and allowed to colonize with sedentary molluscs like oysters and mussels with sense species, the individuals grow on ropes that hang clear of the action so that they are out of the reach of predators such as starfish. Although productive, such systems are very vulnerable to contamination, and since the organisms filter large quantities of water their ability to concentrate substances toxic either to themselves or to consumers is very high.

Livestock as renewable resource. The branch of agriculture which is mainly concerned with the breeding, feeding and caring of domestic animals is called **animal husbandry**. When it incorporates the study of proper utilization of economically important domestic animals, it is called **livestock management**. Livestock (viz., cattle, buffaloes, sheep, goats, pigs, camels, horses, donkeys, mules (ponies), etc.,) form an important renewable resource, as they provide milk, eggs, meat, skin (leather), horn, dung and other products. The total livestock in India amount to a huge population consisting of 185 million cattle, 97 million goats, 61 million buffaloes, 45 million sheep, 1 million horses and ponies, 1 million camels and about 1 million other livestock. There are also 156 million poultry (fowls) and other domestic birds (such as ducks, turkeys, etc.). All these provide 1 million tonnes of meat, 40 million tonnes of milk, 39 million kg of wool and 13 million eggs. The amount of dung produced is enormous and is used for biogas production and as manure. Livestock is also used for rural transport

and agriculture. Therefore to get a sustained yield from them, their proper breeding, health care and management and diet are essential.

4. WASTE MANAGEMENT

(Recycling of Resources and Vermitechnology)

Some types of solid wastes can now be recycled by the help of certain new technologies. For example, the fermentation of organic waste such as dung (of cattle, buffalo, etc.), animal excreta (including faeces or nightsoil and urine), garbage, aquatic weeds, etc., can be utilised to produce **biogas** which can be used for cooking, streetlighting, industry and so on. The **slurry** produced by this method is used as manure. Likewise, waste paper can be converted into toilet paper. Some solid wastes can be utilised in vermitechnology for the production of earthworm cast, earthworm tissue protein and so on. The worm casts can be utilised for the production of mushrooms (**Dash** and **Das**, 1989) and vegetable crops (**Senapati**, 1989). The earthworm tissue is used as feed material for the poultry (**Das** and **Dash**, 1990), fish, amphibia and pigs.



A domestic 'biogas' digester in an Indian village, using animal manure and producing methane for cooking and heating water.

Further, after extraction of metal from the ore, a waste product, called **slag** remains. The slag can be crushed and added to cement for construction work. **Flyash** is also used as a cementing material. **Scrap** metal produced in mills and factories, used metals of discarded vehicles (trucks, car, scooters, cycles, etc.), machines, ships and aeroplanes can be melted and recycled for various purposes. Thus, metals such as copper, zinc, lead, aluminium and platinum can be recovered from the used materials by recycling.

Ecologists have also suggested the following means of resource-recycling from the industrial effluents : 1. Waste heat (*i.e.*, heated water from power plants may maintain constant water temperatures in coastal temperate zones and there certain tropical fishes of high productivity can be raised besides native fishes. 2. Sewage and eutrophicated waste products might be used as the basis for algal production which forms the food of oysters which then filter the water as well. 3. Carbon dioxide from a power station chimney is used to enhance production of algae which are fed to clams.

Vermitechnology (Organic Waste Management through Earthworms or Verms)

India tends to produce about 2,500 million tonnes of organic waste annually, which can be utilised for the recovery of essential resources such as fertilizer, fodder, fuel and food. Earthworms (*e.g.*, *Dichogaster bolau*, *Drawida willsi*, *Lampito mauritii* and *Perionyx excavatus*) are now known to be a good biological elements for the recovery of vermifertiliser and vermiprotein (called **vermitin**) for use in agroecosystems, aquaculture and poultry. The earthworm is physically an aerator, crusher and mixer, chemically a degrader and biologically a stimulator in the decomposer subsystem (**Mitchell**, 1978). The degradation of organic waste by earthworm consumption is known as **vermicomposting** (**Dash** and **Senapati**, 1986).

5. FOREST RESOURCES

India has a rich flora and fauna (*i.e.*, rich biological diversity), much of which is present in forest areas. Forests occupy a special place in the life and thought of the people. They form an important

renewable natural resource. Forest ecosystem is dominated by trees, their species-content varying in different parts of the world. Forests contribute essentially to the economic development of our country by providing goods and services to the people and industry. They are intimately linked with our culture and civilization. The chief products that forests supply is **wood** which is used as fuel; raw materials for various industries as pulp, paper, news-print, board, plywood; timber for furniture items, toys, sports goods, musical instruments, wooden crates, boxes (for packing articles for fruit, tea, etc.), boats, truck bodies, carts, ploughs, railway sleepers, bridges, buildings; **fodder** for cattles, sheep, goats and camels and **bamboos**. Bamboos, called poor man's timber, are used in rafters, scaffolding, roofing, walling, flooring, basketry, cartwood and cordage. Industrially bamboos are used as a raw material in paper and rayon industry. Indian forests also supply **minor forest products** such as canes (rattans), gums, resins, rubber, dyes, tannins, fibres, flocs, medicines, katha, insecticides, camphor, essential oils (*e.g.*, rosha grasses, khas and sandal woods), cooking oils and spices. Tendu (kendu) leaves are used as wrappers for *bidis*, soap-substitutes such as *ritha* and *shikakai*, sola pith and ornamental seeds *rudraksha* are important commercial products of the forests. Lac, honey, wax, tusser or moga silk are obtained from forest insects. Feathers, horns, hides and ivory are also other significant forest product (obtained from forest wild life). For tribal people forests also provide food (tuber, roots, leaves, fruits, meat from birds and other animals).

Besides above discussed uses, forests are also a major factor of environmental concern, providing protection to wild life, help in gaseous (*i.e.*, CO₂, O₂) cycles of atmosphere, tend to enhance local rainfall and water holding capacity of soil, maintain the soil fertility, regulate the earth's temperature regimes and water cycle, check soil erosion, landslides, shifting of sand and silting and reduce the flood havoc. Forests play an important role in reducing atmospheric pollution by collecting the suspended particulate matter and by absorbing carbon dioxide. Lastly, forests have aesthetic and touristic values and serve as gene reserve of important species.

Forest Cover

According to **Brewbaker** (1984) the total forest area of the world in 1900 was nearly 7000 M ha (=million hectares). By 1975 it was reduced to 2890 M ha and if this trend continues by 2000 A.D., it would be merely 2370 M ha. It is also estimated that major reduction will be in tropics and subtropics (40.2%), being only 0.6% in temperate areas. In the tropics and subtropics, too, maximum reduction would be 50% in Asia and the Pacific. This occurs mainly due to population explosion in these areas.

According to **CFC** (=Central Forestry Commission), the forest cover of India in 1980 was around 74.8 Mha. (22.7% of the total land mass). This data have been contradicted by remote sensing surveys which have shown that it is hardly 14 per cent. Among 16 different forest types of India, the most common is the **tropical dry deciduous** (38.7%), followed by **tropical moist deciduous** (30.9%) both forming the major bulk (69.6%) of the forest area in India. Other forest types contain following percentage of forest cover : Tropical thorn (6.9%), Tropical dry evergreen (0.1%), Pure coniferous (or high mountain areas 6.3%), Sal forests (16%), Teak forests (13%), Broad leaved forests (excluding sal and teak, 55.8%), Bamboos (including in plantations, 8.8%). Further, according to the study made by the National Remote Sensing Agency (NRSA) in 1984, there has been a decrease in the forest area in India within a seven-year period, *i.e.*, between 1972–75 and 1980–82. In general, there has been a decrease in forest area in all states except in Sikkim and Arunachal Pradesh. Maximum decrease (11.3%) has been in Tripura, and least (less than 1%) is in Haryana, Delhi, Arunachal Pradesh and Nagaland. The overall forest area has decreased from 55.20 Mha to 45.70 Mha, *i.e.*, 2.89% during this period.

Deforestation (Destruction of Forests)

Deforestation is an alarming threat to the economy, quality of life and future of the environment. The main causes of the deforestation in India are : explosion of human and livestock population, increased require-ment of timber and fuel wood, expansion of croplands and enhanced grazing. Other

prominent causes of forest destruction are construction of roads along the mountain (hill-roads), dam building (valley projects) evermigrating graziers, shifting (*jhum*) cultivation, agriculturalisation, urbanisation and industrialisation. Stress on revenue earning and resources regeneration for pulp and match industries seems to have accelerated soil erosion. Resin tapping from chir-pine trees has choked the trees to death.

India consumes nearly 170 million tonnes of firewood annually and 10–15 million hectares

of forest cover is being stripped every year to meet fuel requirements. During a period of 20 years (1951 to 1971) forests have been cut for : agriculture (24.32 lakh hectares), river valley projects (4.01 lakh hectares), industrial uses (1.24 lakh hectares), road-construction (0.55 lakh hectares) and miscellaneous uses (3.88 lakh hectares). Thus, a total of 3.4 million hectares of forests were lost during this period. Due to deforestation, nearly 1 per cent of the land surface of India is turning barren and unproductive every year.

The devastating effects of deforestation in India include soil, water and wind erosions. Deforestation has a major impact on the productivity of our croplands. This happens in two ways : (i) soil erosion increases manifold and the soil actually gets washed, leading to an intensified cycle of floods and drought. (ii) the shortage of firewood has an obvious impact on the productivity of our croplands. Thus, when firewood becomes scarce, people begin to use cattle dung and crop wastes as fuel, mainly for cooking. In a way, every part of the plant gets used up gradually and goes back to the soil. Over a period of time, such a nutrient drain affects crop productivity due to loss in soil fertility. Further, the local cattles, goats, sheep, etc., not only destroy the vegetation but also pull out the roots of the plants.

Destruction of biotic potential of land also leads to **desertification**. Desertification is the process leading to desert formation. Removal of vegetal cover brings about marked changes in the local climates of the area. Thus, deforestation, overgrazing, etc., bring about changes in rainfall, temperature, wind velocity, etc., and also leads to soil erosion. Destruction often starts as patchy destruction of productive land. Increased dust particles in atmosphere lead to desertification and drought in margins of the zones that are not humid. For example, after denudation of Himalayas, the process of deforestation started in the Shivalik range (extending parallel to the Himalayas). Shivalik sal forests were exploited for industrial use (railway sleepers etc.). Foot hills of Shivaliks once covered with dense forests, are facing an acute water scarcity and semidesert conditions. When forests die, ecological balance maintained by nature breaks away; and floods and droughts are the dangerous consequences. In the Himalayan range, the rainfall has declined 3 to 4 per cent due to deforestation.

Afforestation

For the purpose of afforestation, following two types of methods of forestry are practiced :

A. Conservation or Protective Forestry

This includes the following three strategies :

(a) **Conservation or reserve forests**. These forests are the areas where our water regimes are



Several trees are cut every year for firewood which is a major cause of deforestation.

located (e.g., Himalayas and Western and Eastern Ghats together with catchment areas). They also include National Parks, Sanctuaries, Sacred Groves, Biosphere Reserves and all ecologically fragile areas. In these areas, no commercial exploitation can be allowed. These areas, in fact, need protection from fuel-starved villagers and fodder-starved cattle. The real goal of eco-development (conservation of reserve forests) cannot be achieved without getting support from public such as Chipko Movement and Honey Bee network of Prof. A.K. Gupta, AIIM, Ahmedabad.

Chipko movement. Chipko is a Hindi word which means to embrace or to hug. Hugging the trees is an age-old practice in the Himalayan region. In 1970, Gopeswar and some 20 villages of the Tehri-Garhwal district of Uttar Pradesh were devastated by flash flood in the Alakananda river. This flood occurred due to deforestation and was an eye-opener for the villagers. The people of these villages under the leadership of **Chandi Prasad Bhatt** pledged that they would not permit any more felling of trees. The people (mostly women) started hugging trees whenever forest contractors tried to cut them down. This movement became very successful and was popularised all over the world by **Sunderlal Bahuguna**. **Bahuguna** presented the plan of this chipko movement for the protection of soil and water through ban on tree felling in the Himalayas, at the meeting of UNEP (= United Nations Environment Programme) held in London in June 1982. According to him, every standing green tree in the forest is a sentry to protect us from avalanches and landslides to save our soils and conserve our water. Chipko movement advocates the slogan of planting five Fs—*food, fodder, fuel, fibre and fertiliser* trees—to make communities self-sufficient in all their basic needs.



Chipko movement.

(b) Limited production forests. These are less fertile areas at more than 1000 metres altitude with hilly topography. A part of the annual growth may be harvested in a very careful and controlled manner so as to avoid soil and tree damage.

(c) Production forests. Such types of forests lie on the flat land and are managed for high degree of production. Their working on scientific lines with proper logging methods does not pose environmental problems.

B. Commercial or Exploitative Forestry

The basic aim of commercial forestry is to supply goods and services and meet the needs of local people for firewood, fodder, food, fertilizer, fibre, timber, medicines, etc. These forests also supply material for industrial purposes as timber of all types, plywood, matchwood, fibre board, paper and pulp, rayon, silvichemicals, etc. This can be achieved by (i) intensive plantation and (ii) captive plantations.

(i) Intensive plantations. This type of forestry includes planting all the available land from villagers fields to commercial land, to road/rail sides and every available space. For removing pressure on the natural forests, indigenous and/or exotic species can be used for plantation. Intensive plantations include two types of programmes : social forestry and agro-forestry.

(a) Social forestry. This type of forestry started with NCA (National Commission on Agriculture) in 1976. **Social forestry** envisages use of community lands, individual holdings and other public lands, denuded/degraded lands for producing what the dependent rural and tribal communities need and for environmental purposes. There are two main objectives in social forestry : 1. use of public and common land to produce in a decentralised way fire wood, fodder and timber for the local poor men and also to manage soil and water conservation, and 2. to relieve pressure on conservation forests. This programme has been mainly aimed for the poor villagers and children who become involved in the intensification of nursery operation at village level for multipurpose species for firewood, fodder, fibre, pole, fruit, etc. In our country, social forestry programme become quite successful particularly in Gujarat. There are followed two systems internationally : Chinese system and South Korean system. The area under social forestry increased from 15 million hectare in First plan to 1524 million hectare in Sixth plan.

(b) Captive or Production plantations (Agro-forestry). This is absolutely commercial forestry which is developed to fulfil the needs of the various forest-based industries requiring large quantities of raw materials. The captive plantation is done on the fallow land which is not being used for agriculture, mostly on free grazing lands. A part of this plantation is used to produce fodder for the cattle. Further, in this type of plantation, short rotation of indigenous or exotic species are preferred over long duration sal or teak. In USA productivity of captive forests has been increased by adopting the following measures : 1. Proper manipulation of **silviculture** (The theory and practice of controlling the establishment, composition, and growth of stands of trees for any of the goods and benefits that they may be called upon to produce) and nutritional requirements such as use of fertilizers, irrigation, bacterial and mycorrhizal inoculations. 2. Management of diseases and pests. 3. Weed control 4. Adoption of advanced techniques in forest tree breeding for superior genetic strains. 5. Skilful use of tissue culture and cloning methods.

Few more measures for the conservation forestry. Following strategy can be adopted to prevent further depletion of tree cover in the conservation forests :

1. Extraction of timber should not interfere with watershed protection. Tree-felling should be matched with tree-planting programmes.
2. The use of firewood should be discouraged to reduce pressure on more valuable natural forests. Other sources of energy such as biogas, solar energy, etc., have to be provided to supplement practices.
3. A better understanding should be developed between the persons who manage and those utilising trees. Thus, there is a growing need for (i) creating a new cadre of forest managers and bare-foot foresters committed to conservation. (ii) involvement of rural/tribal people in forestry programmes and (iii) encouraging industries using forest products to obtain raw material at market price and help them to raise plantations much in advance to meet their demands.
4. Forest dwellers should have access to subsidised sources of fuel, fodder, building material, etc., so that they do not cut forest trees.
5. A ban or freeze (*i.e.*, moratorium) of 15 to 20 years should be imposed on commercial tree felling in fragile areas of Himalayas and other hilly areas.
6. During moratorium, an extensive afforestation programme with people's participation should be followed.
7. Environmentally sound action plans based on scientific research should be adopted.
8. By the help of people's cooperatives, community forests should be developed around villages.
9. Protection of standing forests should be done.
10. Creation of new stock should be made.
11. Masses and voluntary agencies should be involved in the task of tree planting.
12. Building of information base have to be made.

Forest conservation through law. The National Forest policy 1952 stated that one third of the geographical area of the country should be under forest. Forest (Conservation) Act, 1980 was enacted with a view to check indiscriminate dereservation and diversion of forest land to non-forest purposes. This act was amended in 1988 to incorporate stricter provisions against violators.

6. RANGE MANAGEMENT (GRASSLAND MANAGEMENT)

Grassland biomes are important to maintain the crops of many domesticated and wild herbivorous mammals such as horses, mules, asses, cattle, pigs, sheeps, goats, buffaloes, camels, deers, zebras, etc., all of which provide food, milk, wool, hide, or transportation, etc., to man. The range management involves an important application of ecological principles in maintaining the grassland biomes. The objective here is to preserve grasslands for maximum forage, *i.e.*, food for cattle. In India, most of the grasslands represent the seral stages in succession and if they are not maintained by grazing and fire they would develop into forest communities. A team of scientists at Indian Grasslands and Fodder Research Institute, Jhansi and Central Arid Zone Institute, Jodhpur, is mainly entrusted with such responsibility.

During range management a suitable degree of grazing is to be maintained. Some species are more palatable to the stock and are very sensitive for grazing and, thus, overgrazing usually results into their disappearance from the area where now some unpalatable, annuals, weeds, shrubs start to grow and turn the area into man-made desert. The palatable plant species, which are sensitive to grazing, have been called the **decreasers** whose disappearance from the area is an indicator of grazing stress and warning signal for range managers.

Overgrazing has certain other ecological effects—reduction of the mulch cover of the soil occur, microclimate becomes more dry and severe and is readily invaded by xerophytic plants. Due to absence of humus cover, mineral soil surface is heavily trampled when wet and produces puddling of the surface layers, which in turn reduces the infiltration of water into the soil and accelerates its runoff, producing drought. These changes all contribute to the reduction of the rate of energy flow, and the disruption of the stratification and periodicity of the primary producers results in a breakdown of the biogeochemical cycles of water, carbon and nitrogen. Water and wind erosion completely breakdown a very dry grassland-microclimate. Further, intensive grazing which results in increased areas of bare soil, creates a new habitat for burrowing animals such as mice, jackrabbits, gophers, prairie dogs, locusts, etc., which render sterile much areas of forage lands.

In range management, fire, plays an important role. Under moist conditions fire favours grass over trees and under dry conditions fire is often necessary to maintain grasslands against the invasion of desert shrubs. Burning of *Cynodon dactylon* increases forage yields.

7. WILD-LIFE MANAGEMENT

Wild-life management involves the following three main steps : 1. the restoration of the habitat; 2. enforcement of the law to stop poaching completely ; and 3. prevention of competition from domestic livestock regarding the grazing activity; the latter step may result in avoidance of transmission of diseases. Management programmes also involve the establishment of more national parks, wild-life sanctuaries and biosphere reserves (For details see Chapter 16).

8. WATER RESOURCE AND ITS MANAGEMENT

Water occurs naturally in gaseous, solid and liquid phases; man's use of it is nearly all concerned with the last state and is also dominated by his demand for water relatively low in dissolved salts, *i.e.*, freshwater. Freshwater is a precious resource and is obtained from the following three natural sources:

A. Rain water. India receives about 3 trillion m^3 of water from rainfall, which amounts to about 105 to 117 cm annually. This is a huge resource and largest in the world. But almost 90% of this precipitation falls between mid-June and October. Of the total annual precipitation, India utilises only 10% which may increase about 26% by 2025 A.D. There is a need of good management of rainwater resource.

B. Surface flow. In India surface flow takes place through 14 major river systems namely 1. Brahmani, 2. Brahmaputra, 3. Cauvery, 4. Ganga, 5. Godavari, 6. Indus, 7. Krishna, 8. Mahanadi, 9. Mahi, 10. Narmada, 11. Perriar, 12. Sabarmati, 13. Subarnarekha and 14. Tapi. Between them they share 83% of the drainage basin, account for 85% of surface flow. Apart from them, there are 44 medium and 55 minor river systems—these are fast flowing, monsoon fed and originate in the coastal mountains. Of the major rivers, Brahmaputra, Ganga and Indus basins along with Godavari cover more than 50% of the country. Only 4 rivers namely Brahmaputra, Ganga, Mahanadi and Brahmani are **perennial rivers** with a minimum discharge of $0.47 \text{ M m}^3/\text{km}^2/\text{year}$.

C. Ground water. Underground reservoirs of freshwater are called **aquifers**; they are continuously recharged through infiltration, seepage and evapo-transpiration. The total volume of ground water found in the aquifers is estimated to be $42.3 \times 10^{10} \text{ m}^3$. At present about 25% of the ground water is being used by man. Agriculture uses the maximum amount of water in the world. This amount is about 73% and leads to a lot of pressure on ground water. Excessive use of ground water depletes aquifers, lowers the water table and may lead to salinisation, waterlogging and alkalisation of the soils. As the human population goes on increasing, so do agriculture and industrial activities. As a result, there is a considerable demand for water resources.

Water Resource Management

Water resource management should ensure that (i) there is no wastage or misuse of water ; (ii) pure water is made available to man for various purposes and (iii) water storage and distribution is done in a scientific way. Usually water is wasted by leaking taps and excessive irrigation. Public awareness should be made against using more water than is necessary.

There are central and corresponding state organisations which are concerned with specific aspects of water resource management. Some of them have been tabulated in Table 15.2.

A. Quantitative management of water resource. Water is an integral part of land/soil

productivity base and its misuse can cause soil degradation and soil erosion. So water management becomes necessary for crop yields and other activities. Primary channel flow originates in upper catchments and these watersheds are very important for future..

1. Watershed managements. Availability of water in a given soil environment is a critical factor and is related to erosion, siltation, loss of plant cover and productivity. In India, **floods** bring considerable damage resulting in loss of life and property each year. Due to floods, the plains have become silted with mud and sand and affect the cultivable lands. The National Commission on Floods has calculated that the land areas prone to floods has doubled from 20 million hectares in 1971 to 40 million hectares in 1980. The worst flood-suffering states are Assam, Bihar, Orissa, U.P., and West Bengal. The management of rainfall and resultant runoff is very important and found to depend on **water sheds**. To recall, a watershed is an area bounded by the divide line of water. Thus, it may be a drainage basin or stream. The Himalayas are one of the most vital watersheds in the world. These



Water resource management.

watersheds are threatened by deforestation and other ecological malpractices and have resulted in the depletion of water resources. Thus, for watershed managements the following measures are adopted: 1. soil and land use survey; 2. soil conservation in catchments of River Valley Projects and flood prone areas; 3. afforestation, social forestry programmes and Drought prone area Development Programmes, 4. desert development and 5. control of shifting cultivation.

Table 15-2.**Some Indian organisations involved in water resource management.**

Name of agency	Nature of management
1. Central Water Commission	Surface water
2. Central Ground Water Board	Ground water
3. Indian Meteorological Department	Precipitation
4. Central Pollution Control Board	Water quality
5. Ministry of Agriculture and ICAR	Water use for agriculture
6. Department of Environment, Forests and Wildlife (Ministry of Environment and Forests).	Environmental impact assessment
7. Central Public Health and Environmental Engineering (Ministry of Urban development)	Water supplies, sanitation and sewage disposal
8. Department of Power	Hydro-electric power
9. Department of Forests	Watershed management

2. River valley projects. The environmental side effects of river valley and hydel power projects can be classified in following three categories : 1. Impacts within and around the area covered by dam and reservoir, *e.g.*, changes in microclimate, loss of vegetal cover, soil erosion, variation in water table and enhanced seismic activities due to pressure of water ; 2. Down stream effects caused by alteration in hydraulic regime ; and 3. Regional effects in terms of overall aspects including resource use and socio-economic aspects.

Other impacts caused by construction of dams and reservoirs are the following : 1. Blasting operations in hilly tracts for road construction can cause considerable damage to the environment through loosening of hill sides and resultant landslides, sedimentation of reservoirs, drying up of springs and flash floods. The creation of new settlements for the workman and rehabilitation of project oustees (= expelled population) in the watershed areas may increase the ecological hazards. In recent years, environmentalists like **Sunder Lal Bahuguna**, **Medha Patekar** and others have opposed the construction of the following river projects : (1) Tehri Dam Project, U.P. ; (2) Narmada Sagar Project, Madhya pradesh; (3) Sardar Sarovar Project, Gujarat; (4) Bodhghat Project, M.P.

3. Irrigation (waterlogging and salinity). Rain water is conserved through wells, reservoirs, budding of streams and canal systems. Canal irrigation increased by 6.7 million ha., from 1950-51 to 1978-79. It is found harmful to soils which degrade due to water logging and salinity. In India, more than 20% of irrigated land is damaged due to water logging and salinity.

B. Qualitative management of water resource. Availability of potable or drinking water is becoming increasingly difficult. There is a lot of industrial liquid waste, there are very poor urban sewerage and sewerage treatment facilities and there is heavy pollution load on water. Measures of qualitative management of water fall in two categories : (1) **Short term measures.** These include the following measures : (i) pollution control of urban settlement sources, (ii) sewerage regulation (discharge of effluents only in municipal sewers, etc.), (iii) proper pattern of sewage collection systems, (iv) pollution control of industrial sources, (v) environmental planning guides for industrial estate, (vi) protection of drinking water resources, and (vii) coastal management. (2) **Long term measures.** These are planned for each river basin and are of following nature : (i) preparation of water use maps (these are to classify and zone river waters based on best uses); (ii) evaluation of pollution potential

in the river basin, and (iii) preparation of water quality map on the basis of continuous water quality monitoring.

9. LAND USE PLANNING AND MANAGEMENT

Land is a precious resource, since it is put to diverse use by man. India with a land area of 32,88,000 km² which is about 2.4% of the world supports 15% of the world's population. There were about 238 million people in India in the year 1901 and are now about 850 million. The per capita land resource available now in India is less than 0.4 hectares, in comparison to more than 0.9 hectare in China and about 8.4 hectares in the USSR. About 44% of our land is used for agriculture, 23% is covered with forests, 4% is used for pastures and grazing fields, 8% for housing, agroforestry, industrial areas, roads and so on. The 14% land is barren and about 8% is used for miscellaneous purposes. The rapid increase of urbanisation and migration of population from rural areas to towns and cities has created many problems. All this has led to the utilization of agricultural land for housing, construction of office buildings, industries, and so forth.

The rational use of land resource is possible by adopting an integrated land-use policy which involves prevention of land misuse and reclamation of degraded and under-utilised land, wastelands, fallows, etc. Reclamation of abandoned mines and brick kilns may yield some much required land. Fertile agricultural land should not be sacrificed for non-agricultural purposes, such as roadbuilding, development of industries or construction of water reservoirs. Urban areas should not be developed on agricultural lands.

10. SOIL EROSION AND SOIL CONSERVATION

It is a common experience that the soil is liable of removal from one place to another whenever there is physical force such as storm or flood (or running water). The dust blows along with wind or flows in runoff water, usually after rains. This is called **erosion**. The term 'erosion' is derived from the Latin word *Erodere* which means to 'gnaw away' or 'tear away'. **Rama Rao** (1962) called soil erosion as **creeping death of the soil**. Soil erosion refers to physically detaching soil particles from their original place and transporting them to some other place. Thus, though it takes a very long time to build the soil, its erosion by the forces of rain and runoff water, wind action, etc., is a rapid process. Volume and intensity of precipitation, slope conditions, vegetation cover and wind speed are some important factors of soil erosion.



Soil erosion.

Type of Soil Erosion

Based on various **agents** that bring about soil erosion and the **form** in which the soil is lost during erosion, there are following types of soil erosion :

1. Water erosion. Water erodes or cuts and removes soil chiefly in four ways : (i) **Sheet erosion**. The soil is removed in small but uniform amounts from all over and therefore, does not leave a mark behind. (ii) **Rill erosion**. The run off water moves rapidly and cuts small stream-like structures. (iii)

Gully erosion. Several rills converge towards the steep slopes and join to form broad channels of water called **gullies**. (iv) **Riparian erosion** or **stream bank erosion**. The rivers during floods splash their water against the banks and thus, cuts through them.

2. Wind erosion. Erosion due to wind is very common and includes saltation, suspension and surface creep. During dust storms, a huge quantity of dust is raised high and transported to great distances. The wind lifts finer particles high up (called **suspension**) whereas the coarser and heavier particles roll along the surface (called **surface creep**). The rolling particles rub the ground and due to abrasive action help in loosening the soil. The process continues and more dust particles gather as the storm advances.

In the arid regions, where rainfall is low, drainage is poor and high temperatures prevail, water evaporates quickly leaving behind the salts. Salt accumulation occurs mainly in lowlands around the oceans. The salts are normally chloride, sulphates, carbonates and nitrates of potassium, magnesium and sodium. The salts also include chlorides and nitrates of calcium. The major portion of such salty soil is carried by wind in the form of small leaps.

3. Landslide erosion. The hydraulic pressure caused by heavy rains increases the weight of the rocks at cliffs which come under the gravitation force and finally slip or fall off.

4. Erosion due to overhunting, overgrazing and deforestation. Human beings are involved directly or indirectly in accelerating the rate of erosion. In the normal course of development of communities the edaphic conditions and other environmental conditions attain some sort of dynamic equilibrium with the vegetation and other biota. As a result of this the soil is held in position by vegetation cover. Killing of lions, tigers and other carnivores by man results in increased population of herbivore. The natural balance gets upset; herbivores in greater numbers damage the natural vegetation (by grazing) in greater quantities. Natural vegetation is also damaged by scraping and deforestation. All these lead to the removal of protective plant cover from the soil surface. The beating of rain drops on the bare soil raises soil particles selectively; finer particles of clay and humus being lighter are raised to greater heights than sand particles. Thus, the finer components which are accidentally more fertile brought upon the surface and these seal and clog the pores. The accumulated water fails to percolate down and therefore, runs down the slope. The fertile top soil also gets washed away.

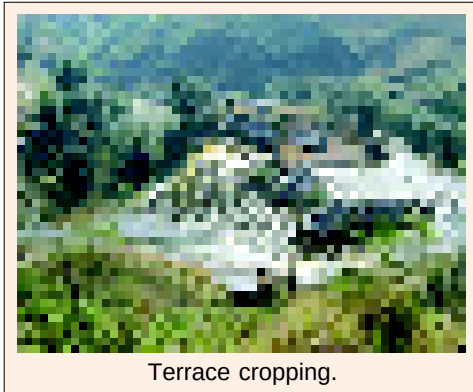
Soil Conservation

Soil conservation means protection, improvement and sustained renewal of the soil at any place.

1. Principles of soil conservation. The soil conservation depends on the following principles: (i) protection of soil from impact of rain drops, (ii) to prevent water from concentrating and moving down the slope, (iii) to slow down the water movement when it flows along the slope, (iv) to encourage more water to enter the soil, (v) to increase the size of soil particle, (vi) reduction in the wind velocity near the ground by growing vegetation cover, ridging the land, and (vii) to grow the strips of stubble or other vegetation cover which help to catch and hold the moving particles of soil.

2. Methods of soil conservation. For the conservation of soil (*i.e.*, to check the loss of soil during erosion) ecologists have devised the following methods :

A. Biological methods. In these methods, conservation is achieved by the use of plant vegetation cover : (i) **Agronomic practices.** Natural protection is provided to the soil by growing vegetation in a manner that reduces soil loss. These are the following : (a) **Contour farming.** This oldest method is found useful in those areas where the rainfall is low. It involves preparation of the field with alternate furrows and ridges (or contours). The water is caught and held in furrows and stored; this tends to reduce run off and erosion. Contour farming is also useful for lands with gentle slopes. In this method ploughing is done at right angles to the direction of the slope. (b) **Terrace cropping.** On hilly terrains where there is some good soil profile, terrace cropping is in practice since time immemorial to reduce soil erosion and obtain a reasonable productivity. In preparing **terrace**, one block of flat land abruptly ends and a new terrace begins at a lower level. The soil is cut and laid to make steps. On the margins

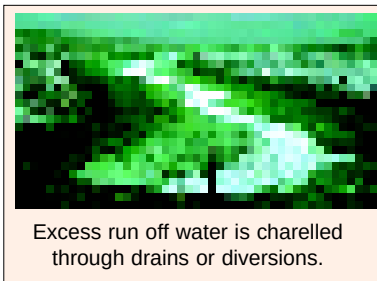


Terrace cropping.

decreases soil loss and preserves the productivity of land. The typical crop rotation involves one year of fallow, followed by winter heat. (e) **Strip cropping**. This involves the planting the crop in rows or strips to check flow of water. On more slopy lands for strip cropping, strips of land are ploughed at right angles of the direction of the slope and different crops are raised in adjacent strips. In different years the one crop of the strip is rotated with another crop.

(ii) **Dry farming**. This method is found useful for the croplands existing in the areas of low or moderate rainfall ; in these areas ordinary farming is at risk. Thus, for checking the soil erosion, there are only three alternatives : crop production, animal husbandary and growing grazing fields. Dry farming methods differ in different areas and involves the fallowing the land, strip cropping, crop rotation, contour farming, etc.

(iii) **Agrostological methods**. In these methods, grasses such as *Cynodon dactylon* are used as erosion-resisting plants. They are grown in strips between the crops. They act as stabilisers when grown in gully and sodding (*i.e.*, to cover with sod). These methods involve lay farming, retiring lands to grass, etc.



Excess run off water is charelled through drains or diversions.

between two steps the slope is vertical. This zone is planted with some perennial soil binding grass whereas the rest of the space is used for cultivation. The runoff of rainwater is greatly reduced and the soil becomes stable. (c) **Mulching**. In this method certain plants such as maize stalks, cotton stalks, potato tops, etc., are used as a mulch. Mulch is a protective layer formed by stubble (*i.e.*, the basal parts of herbaceous plants), especially cereals attached to the soil after harvest. Mulches (13 cm to 20 cm) bind the soil particles, reduce evaporation of soil moisture and increase the amount of soil moisture by adding organic matter to soil. (d) **Crop rotation**. This method



Strip cropping

B. Mechanical methods. They are used as supplements to biological methods and are of following types : (a) **Basin listing**. It involves the construction of small basin along the contours to retain water which also reduces its velocity. (b) **Contour terracing**. This involves digging a channel along the slope to intercept and divert the run off water.

C. Other methods. These are of following types : (1) **Gully control**. To check the widening of gullies by constructing bunds, dams, drains or diversions through which excess run off water is channelled. (2) **Stream bank protection**. For prevent-

ing the cutting and caving of river banks, vegetation is grown alongside the river bank, drains are constructed and concrete or pitching is done. (3) **Afforestation**. Trees as windbreaks are planted in deserts which check the velocity of wind. Plantation of trees in short blocks are called **wind breaks**. Extensive plantation of trees are called **shelter belts**. Windbreaks are planted across the area at 90° to the prevailing wind. They check the spread of sand dunes or desert conditions or blowing away of the fertile top soil. Windbreaks are planted in several rows. In Indian deserts, following species have been planted under the afforestation—*Lowsonia alba*, *Agava americana*, *Thevetia neriifolia*, *Calotropis gigantea*, *Ricinus communis*, *Zizyphus jujuba*, *Acacia catechu*, *Cassia*, *Acacia nelotica*, *Dalbergia*

sissoo, *Mangifera indica* and *Tamarindus indica*. Both types of plantations (wind breaks and shelter belts) have been grown in the region of Uttar Pradesh where desert is encroaching. The plantation is usually being done in two or three belts. Small sized plants planted on the wind-ward side in the region are *Saccharum munja*, *Calligonum polygonoides*, *Laptadenia spartium*, *Cenchrus ciliaris*, *Balanites roxburghii*, *Kochia indica*, *Panicum antidotale*, etc. On the lee ward are being planted the following – *Acacia leucophloea*, *Acacia senegal*, *Ricis communis*, *Tecoma undulata*, *Prosopis juliflora*, *Parkinsonia*. The roots of the plants (which are used in windbreaks or shelter belts) bind the soil and protect it from erosion. The shoot canopy acts as a physical barrier for wind and dust and greatly reduces the wind speed.

REVISION QUESTIONS

1. Write an essay on ecology and human welfare.
2. What are the natural resources ? Give an account of different natural resources and the means of their conservation.
3. Write short notes on the following :
 - (i) Aquaculture ;
 - (ii) Wildlife management ;
 - (iii) Soil conservation ;
 - (iv) Silviculture ;
 - (v) Waste-management parks;
 - (vi) Vermitechnology ;
 - (vii) Social forestry ;
 - (viii) Land use management.

CHAPTER

16

Wild Life Management

Wild life management which is an important branch of conservation, is concerned with assuring the maximum possible populations of game animals consistent with other land uses in the same area and with the number that the given habitat will support. This is attained by manipulating the balance of nature in such a way that the desired game species are favoured. Many sportsmen have the idea that the way to increase game animals is to kill off all of their predators, Wild life managers, of course, are well aware that predator elimination may well be the very thing that should not be done. Successful game management depends upon a number of factors including an understanding of ecological principles, an appreciation of the conflicts between game species and agricultural uses of land, and educating both the landowner and the hunter concerning the objectives, techniques and limitations of game management.

WILD LIFE OF INDIA

Indian subcontinent is unique in having great natural beauty in its different biomes and also in possessing a rich and diverse wild life fauna. It includes about 123 families of terrestrial vertebrates. According to an estimate, there are 400 species of mammals, 1200 species of birds, 350 species of reptiles and more than 29,70,000 species of insects (**Khajuria**, 1957). The animals such as black buck, Nilgiri tahr, pigmy hog, golden langur, lion-tailed macaque, etc., are unique to India. The other typical wild



Management of wild life assures the maximum possible populations of game animals consistent.

animals of India are elephant; rhinoceros; deer such as musk deer (kastura), barking deer (kotra), spotted deer (cheetal), hog deer (hoghira), mouse deer (Indian chevrotain), swamp deer (Barasingha), dancing deer (sambhar), Thamin, and Kashmir stag; antelopes such as black buck, four-horned antelope (Chausingha), Indian gazelle (chinkara), and blue bull (nilgai); bison; wild buffalo; Himalayan ibex or wild goat (*Capra siberica*); wild boar; wild ass; Nilgiri tahr; carnivores such as big cats such as lion, tiger and leopard, striped hyaena, jackal (*Canis aureus*) wolf (*Canis lupus*), desert cat (*Felis lybica*), desert fox (*Vulpes v. pusilla*); bears such as black Himalayan bear and sloth bear, monkeys and apes such as Nilgiri langur (*Ceropithecus johni*), lion tailed monkey (*Macaca silenus*), rhesus monkey (*Macaca mulatta*), and Hanuman monkey (*Semnopithecus entellus*); Indian giant squirrel (*Ratufa indica*); desert hare (*Lepus nigricollis*); porcupine; pangolin; birds such as peafowl (*Pavo cristatus*), jungle fowl, partridge, quail, great Indian bustard (*Choriotes nigriceps*), duck, pigeon, sandgrouse, storks and egrets, pelican, eagle, crane, owl, hornbill, etc., and reptiles such as crocodiles, gharials, lizards (*Uromastix*) and enormous varieties (about 216 species) of snakes. Wild life management requires autecological information about game animals and some basic facts of game species are the following:



Blue bull (nilgai).

1. Deer

The deer belong to the family 'Cervidae' of the mammalian order 'Artiodactyla' and they are herbivores and form significant component of grazing food-chains of the Indian forests. They control the overgrowth of forest vegetation and themselves become food of carnivores. Due to limited shelter and food supply, deforestation, urbanisation, mass hunting, etc., there has been a great reduction in the number of deer population in India. Sangai of Manipur or brow-antlered deer (*Cervus eldi eldi*) is at the verge of extinction. Mouse deer, *Tragulus meminna* stands just 30 cm off the forest floor. It has no antlers but male has downward pointed canines that look like tusks. Its small size helps them to evade predators by hiding beneath low bushes. Timid and gentle, these dwarfs lend enchantment to Indian forests and are subject of Indian folk tales.

The musk deer (*Moschus moschiferus*) is found at 2500 to 4000 metres altitudes in Himalayas, Himachal Pradesh, Northern Uttar Pradesh, Nepal and Sikkim. It is about 50 cm high, has long ears, short tail and gall-bladder, but no antlers and no pit-glands (suborbital glands) below the eyes. Male or buck has downwardly projecting canines of the upper jaw. It is a shy animal and lives a solitary life. This deer has been almost hunted out mainly for musk which is a valuable product used as perfume fixative and as important ingredient of several Ayurvedic and Homeopathic drugs and is obtained from the musk pouch existing below the skin of abdomen of buck. For the conservation of musk deer a National Park of 1,000 sq. km has been established at Manali in Kulu Valley by Himachal Pradesh Government.

The barking deer or rib-faced deer (*Cervulus muntjac*) is found at the altitudes of 2000 to 3000 metres and has four subspecies in India. It is 61 cm in height and has suborbital glands below eyes. Buck has antlers and non-protruding canines. It lives in pairs or small families in thickets and being diurnal, grazes in the morning and evening hours. The dancing deer or sambhar (*Cervus unicolor*) is large-sized (150–160 cm high), browsing and gregarious deer, occurring in the herds of 4 to 12 animals and is found in forests all over the country. It has very long (up to 100 cm) antlers. The spotted deer or cheetal (*Axis axis*) is 76 to 91 cm high, grazing and gregarious deer, living in the herds of 20 to 30. It is spotted with white on rich brown coat. The buck has long slender antlers, each of which has three branches or tines and is shed annually. It is mostly found in the forests of Indian plains and in hill upto 1333 metres

but are not found in Assam, Punjab and Rajasthan. It prefers open type forests with good grazing sites and running streams. The hog deer (*Axis porcinus*) is smaller and stouter in built and has hog-like appearance. It lives either solitary or in pair and prefers to live in grass patches bordering on forests. It is diurnal and comes out to feed only during morning and evening.

2. Antelopes and Other Herbivores

(1) Antelopes. In India there are four species of antelopes : the Indian gazelle, nilgai, black buck and chowsingha.

The Indian gazelle or chinkara (*Gazella dorcas bennettii*) is mostly found throughout desert region of southern part of Uttar Pradesh and Rajasthan. It lives in herds of 3 to 5 animals and has adaptations to live in extremely arid conditions, kilometres away from drinking water sources (**Prakash**, 1977). It prefers to live in open grasslands with thick bush cover. They feed on a variety of grasses and browse on shrubs.

The nilgai or blue bull (*Boselaphus tragocamelus*) occurs in plains and Rajasthan deserts but not in Bengal and Assam, living in small herds of 4 to 10. It prefers to stay near cultivated lands but is also found in desert grassland. It is 131–137 cm high, females being smaller. Young bulls and cows are tawny (= light yellowish-brown), but old males have iron gray colour. Both sexes have a short dark mane on the neck. Male has small cone-shaped curved horns. In Thar desert it is regarded sacred like 'cow', so is spared from hunting. However, it is killed under crop-protection act, because herds of nilgai inflict severe damage to crops.

The black buck has its four subspecies in India : *Antelope cervicapra rajputanae*, *Antelope cervicapra cervicapra*, *Antelope cervicapra rupicapra* and *Antelope cervicapra centralis*. Black buck is a graceful and fast-running animal which occurs in grassy plains of India and avoids forests. It inhabits regions having scrub vegetation in the vicinity of tanks and lakes where they pass the day hiding under bushes and become active during night. The male antelope is brown or black in colour while doe and fawns are yellowish brown in colour. Male has spirally ringed, unbranched, long (46 cm) and permanent horns. There occur large preorbital glands below the eyes of both sexes. Black buck is a polygamous and gregarious animal and its herd often include single male, several (50 or more) females and a few fawns. Almost all subspecies of black bucks have been hunted out from most part of India and now they occur in appreciable numbers only in certain pockets of Rajasthan, especially around the villages of 'Visnoi caste' of Hindus, who protect them from hunting.

The four-horned antelope or chausingha (*Tetracerus quadricornis*) has two pairs of horns, the front pair being shorter. It occurs in the foot hills of the Himalayas and in Bundel Khand region. It mostly leads a solitary life.

(2) Wild-buffalo. The wild buffalo (*Bubalus bubalis*) is a large-sized, robust animal having streamlined body. It prefers tall grassy forests close to marshy areas in the vicinity of rivers and lakes. It was once distributed over the grass jungles and river in forests of the Gangetic plain, the terai, Assam and eastern peninsular India down to the river Godavari. Great hunting of wild buffalo has restricted them now only to Assam (Manas and Kaziranga sanctuaries), the Nepal Terai and Bastar district in Madhya Pradesh. According to **Sheshadari** (1969), there are only 1425 wild buffaloes in India.

(3) Bison. Mithun, bull gaur or bison (*Bos gaurus*) is a relative of ox and is found in Andhra Pradesh, Orissa, Madhya Pradesh and Bihar. Due to certain epidemic diseases such as rinderpest and enteric viral disease numerous individuals were wiped out some years ago and now a few bisons are mainly confined to Bandipur in Karnataka.

(4) Wild goat and sheep. In India, wild goat and sheep are restricted to north. Ibex is found in the Himalayas. The Nilgiri tahr (*Hemitragus hylocrius*) is the only one to be in south, in Nilgiris, Tamil Nadu. It moves about in herds and grazes on grasses. This species is greatly endangered due to habitat disturbance, predation by black panther and wild dogs (dhole) and hunting by poachers and only 450 tahr are present today (**Davidar**, 1975).

(5) Elephant. The Indian elephant, *Elephas maximus* is a largest terrestrial mammal and is confined to the Terai and the foot-hills because of its dependence on succulent grass, bamboo and plenty of water. It is gregarious and its herds move constantly in the search of new feeding grounds. Its population is hunted out mainly for tusk and now Periyar sanctuary (Madras) has some elephants.

(6) Rhinoceros. The Indian rhinoceros (*Rhinoceros unicornis*) is found in the grasslands and jungle area of the foot hills of the Himalayas (central Nepal) and certain pockets of plains of West Bengal and Assam. It prefers swamps and open savannah, covered with the tall elephant grass. A full grown rhinoceros is 1.8 metres in height and 3–9 metres in length. It has a single horn on the nose. The horn is made of keratinized skin and matted hair, it is said to have aphrodisiac and other medicinal values, so that large sums are paid for it, this results in indiscriminate shooting and its near extinction in India. It is a slow, solitary animal and is strictly territorial.

(7) Wild ass. The wild ass (*Equus hemionus khur*) had a fairly wide distribution in the dry regions of North-West India (Jaisalmer and Bikaner), but their distribution now become restricted to the Little Rann in the southern part of the Thar desert, where **Gee** (1963) estimated their number to be 870. In past, the population of wild ass severely reduced due to hunting, catching for breeding mules, surra disease and south African Horse sickness disease. However, people around the Little Rann are orthodox and vegetarian and do not molest the wild ass inspite of the fact they inflict serious damages to crops. Because it is the only true 'wild ass' of the world, so it needs protection and conservation (**Prakash**, 1977).

(8) Wild boar. The wild boar, *Sus scrofa cristatus*, has a long mobile snout with terminal nostrils. The canines of both jaws grow continuously and form long triangular and up curved tusks for defence and digging roots, they are better developed in males. The wild boar or pig is omnivorous, it feeds on carrion, snakes, insects, roots, tubers and cultivated crops. It is gregarious living in families in grass or bush of marshy places and is now rare and is mostly found in small numbers along the Aravalli ranges in the eastern most part of Thar desert where they inhabit rocky slopes.



Wild boar.

3. Big Cats and Other Carnivores

India has a variety of world-known large-sized cats and other carnivores which are as follows:

The Asiatic lion, *Panthera leo persica* was found in the Thar desert, in the arid plains of Sind, and Rajasthan and Punjab. It is on record that the last lions were shot at Anadra and at Jaswantpura of Rajasthan desert in about 1876. In India lions are now preserved only in the Gir sanctuary, in Gujarat, on the South-West of the Thar desert. Its extermination is well known.

The cheetah, *Acinonyx jubatus venaticus* was found in Central India, Deccan region, and in Thar desert upto Jaipur in Rajasthan. **Sheshadri** (1969) stated that it disappeared from Rajasthan about the turn of this century, from Central India in 1920's and in Deccan it was last reported in 1952. Now the cheetah, one of the most beautiful, agile and fastest animal, is regarded to be extinct from India.

The tiger *Panthera tigris tigris* is a gracious carnivore which is distributed in Uttar Pradesh from the Himalayas to the Vindhya forests in the South. In Himalayas it is found up to an elevation of about 2,000 metres. It lives in a variety of habitats, from thorn forests to the dense terai forests. According to an estimate, in 1948, the tiger population was 20,000 to 25,000, but in 1958, about 4000 and in 1970 less than 3000 (merely 1540; **Daniel**, 1970). To save the Tiger from extinction in India, "**Project tiger**" has been launched in 1972; this project planned to create Tiger reserves in selected areas of India. Due to this effort, considerable improvement was observed in tiger population in 1973 (Table 16-1).

Table 16-1.

Increase in Tiger population due to 'Project Tiger' (News Letter, 1977, Vol. I, No.1).

Tiger Reserve	Population in 1972	Population in 1973
1. Manas	31	41
2. Palamau	22	30
3. Simlipal	17	50
4. Corbett National Park	44	55
5. Ranthambore	14	20
6. Kanha National Park	43	48
7. Melghat	27	32
8. Bandipur	10	19

The leopard, *Panthera pardus* is the smaller spotted cousin of tiger which can live in all types of forests. Once it was found all over India but now its number has been drastically reduced. Leopard is given full protection in the Mount Abu sanctuary, however, can be hunted under licence from the Indian Wild Life Act. The snow leopard (*Panthera unica*) is found in the high Himalayas from Kashmir to Sikkim near snow line. It has a creamy grey coat with large black rings. It is greatly hunted for its fur and now is endangered species. Other important cats are clouded leopard (*Neofelis nebulosa*), leopard cat (*Felis bengalensis*), pallas's cat (*Felis manul*), caracal (*Felis caracal*), etc. Caracal once was common in Rajasthan deserts, now it has become extremely rare and is suspected to occur only in the Sirohi-Jalore and Bikaner regions (Prakash, 1977).

The sloth bear, *Melursu ursinus*, is a stout carnivore which is found in the hilly forests like Mount Abu and Erinpura ranges. It lives mostly on fruits, honey and insects, it sucks up white ants from their hills which are broken by its claws with great speed. It is rapidly vanishing due to ruthless hunting and it needs conservation. The black Himalayan bear, has V-shaped white mark on the chest and is carnivorous.

Dolphin. In the Indus river, in the North-West and in the rivers Chenab and Sutlaj in the north of the Thar desert, the freshwater gangetic dolphin, *Platanista gangetica* (Cetacean mammal) was once found in good numbers (Murray, 1884), but is now totally extinct (Prakash, 1977).

4. Birds

The important wild bird-fauna of India include the following species : pea fowl (*Pavo cristatus*), grey jungle fowl (*Gallus sonnerati*), red spur fowl (*Gallus sonnerati*), ducks (*Anser indicus*, *Sarkidiornis melanotos*, *Anas poecilorhyncha*), pigeons (*Columba livia*), sand grouses (*Pterocles exustus*, *P. orientalis*, *Streptopelia decaocta*), storks (*Leptotilos dubuis*), egrets (*Egretta garetta*), grey partridge (*Francolinus pondieirianus*), black partridge (*Francolinus francolinus*), golden eagle (*Aquila chrysaetos*), vultures (*Gyps bengalensis*, *Neophron percnopterus*), spotted billed pelican (*Pelacanus philippensis*), sarus crane (*Grus antigone*), quails (*Coturnix coturnix*, *C. coromandelica*), and great Indian bustard (*Choriotes nigriceps*). Certain birds such as osprey, great Indian bustard, cranes, mountain quail, horned owl, red start avocet, pinkheaded duck, etc., are on the verge of extinction, if not saved. Great Indian Hornbill with its peculiar nesting behaviour is the pride of India and needs to be saved.

The great Indian bustard is a large Indian game bird having a heavy body, long neck and long bare running legs. It is essentially an inhabitant of wide, open dry scrubby plains and waste, broken undulating lands of the Rajasthan, West Punjab and Gujarat. It is ruthlessly hunted out for its delicious flesh and is almost at the verge of extinction. In 1980, an International Conference was held at Jaipur to assess its present position and to suggest conservation steps to save it from extinction.



Indian bustard is almost at the verge of extinction.

5. Crocodiles and Other Reptiles

The following three Indian aquatic reptiles have been found endangered : the estuarine crocodile (*Crocodilus porosus*) ; subnosed crocodile, freshwater or marsh crocodile or mugger (*Crocodilus palustris*) and gharial (*Gavialis gangeticus*). The estuarine crocodile is found in Bhitarkanika Island and adjacent mangrove areas of Orissa and in the Sunderbans of West Bengal. The marsh crocodile is widely distributed in India and has been greatly reduced in number in the past due to indiscriminate shooting for its valuable hide. *Gavialis* was once common in most large Indian rivers such as Indus, Ganga, Brahmaputra and Mahanadi, but now has very sparse distribution in Ganga and larger feeders in the Nepal-Terai, Bihar and Bengal. According to **Bustard** (1974), Indian gharials are on the verge of extinction and there are only 143 living gharials (**Subba Rao**, 1977). Mugger is also fast depleting and may be endangered, if not saved. In 1974, under the guidance of **Dr. H.R. Bustard** the project called 'Save the Crocodiles' was started for the conservation and management of crocodiles, through the incubation of eggs, rearing and release in preserves. Projects for preservation of mugger, estuarine crocodile and gharial are now being undertaken in Andhra Pradesh, Bihar, Gujarat, Kerala, Orissa, Rajasthan, Tamil Nadu and Uttar Pradesh.

Sea turtles migrate to "**Gahirmatha**" in Orissa to carry on nesting popularly known as "**Arribada**". The deep sea fishing is a continuous threat to this reptilian species. When turtles float on surface during copulation, certain tribals which feed on them trap them and sell in the market. Their nesting grounds are also disturbed by man who collect turtle's eggs for consumption.

A rare limbless lizard is the *Barkudia insularis* which is mistook as the snake and killed indiscriminately. It is subterranean in habit and found at Barkudia inland of Chilka lake and needs protection.



Gharial.

In India, snakes are killed by some and worshipped by others. Generally they are feared. They are believed to be farmer's best friends as a single rat snake can eat hundreds of rats and mice per year. Indiscriminate killing of snakes for skin results in a loss of ten million rat snakes, cobras and pythons every year.

6. Frog

The "State of India's Environment" 1984-85 reports that every year Indians kill an estimated 100 million frogs to satisfy the palates of well fed westerners. In 1981, 4368 tonnes of frogs legs were exported from India; these have been finding their way to Canada, Saudi-Arabia, Japan and UAE. Two common species of frogs viz., *Rana tigrina* and *Rana hexadactyla* have been affected by this trade and deserve protection.

CONCEPT OF THREATENED SPECIES

The International Union of Conservation of Nature and Natural Resources (IUCN) has categorized the rare species of plants and animals for conservation purpose. This classification is based on (i) the present and past distribution of the species or taxa; (ii) abundance and quality of natural habitat; (iii) the decline in the density of population in course of time; and (iv) the biology and ecological value of species. According to this system the following categories have been identified:

1. Endangered (E) species. These are species or taxa, which are in danger of extinction and which may not survive if the adverse factors continue to operate. The species whose numbers have been reduced to a critical level or their habitats have been drastically reduced in such a way that they are in the instant danger of extinction.

2. Vulnerable (V) species. The species or taxa likely to move into the endangered category in the near future if the causal adverse factors continue to operate. These are species whose populations have been seriously decreased and whose ultimate security is not assured and also those whose populations are still abundant but are under threat throughout their range.

3. Rare (R) species. These are species with small populations in the world. At present, these species are not endangered and vulnerable but are at risk. These species are usually localized within restricted geographical areas or habitats or are thinly scattered over a more extensive range.

4. Threatened (T) species. The term 'threatened' is used in the context of conservation of the species which are in any one of the above three categories, viz., endangered, vulnerable or rare.

The most threatened reptiles, birds and mammals of India have been listed in Table 16-2.

Table 16-2.

Table 16-2. Threatened reptiles, birds and mammals of India.

Amphibians and Reptiles	Birds	Mammals
1. <i>Tylotriton varrnncosus</i> (Himalayan Newt or Salamander) 2. <i>Varanus salvator</i> 3. <i>Crocodilus palustris</i> (Marsh crocodile) 4. <i>Crocodilus porosus</i> (Estuarine crocodile) 5. <i>Gavialis gangeticus</i> (Gharial)	1. <i>Choriotis nigriceps</i> (The great Indian bustard) 2. <i>Cairina cutalata</i> (The white winged wood duck)	1. <i>Presbytis pileatus</i> (Capped langur) 2. <i>Macaca silenus</i> (Lion tailed macaque) 3. <i>Hylobates hoolock</i> (White browed gibbon) 4. <i>Panthera tigris tigris</i> (Tiger) 5. <i>Panthera leo persica</i> (Lion) 6. <i>Panthera pardus</i> (Leopard) 7. <i>Panthera uncia</i> (Snow leopard) 8. Clouded leopard 9. <i>Felis marmorata</i> 10. <i>Felis temmincki</i> (Golden cat) 11. <i>Felis bengalensis</i> (Leopard cat) 12. <i>F. silvestris ornata</i> (Desert cat) 13. <i>F. viverrina</i> (Fishing cat) 14. <i>F. caracal</i> (Caracal) 15. <i>F. Lynx</i> (Lynx) 16. <i>Arctictis binturong</i> (Binturong) 17. <i>Prionodon pardicolor</i> (Spotted linsang) 18. <i>Canis lupus pallipes</i> (Indian wolf) 19. <i>Melursus ursinus</i> (Sloth bear)

REASONS FOR DEPLETION OF WILD LIFE

Many wild animals became extinct due to various human and natural activities : 1. Absence of cover or shelter to wild animals. 2. Due to deforestation for cultivation, road-building, railway routes, dam construction or for urbanisation, occurs reduction in the area for free movement of wild animals which retard reproductive capacity of certain wild animals such as deer, bison, rhino, tiger, etc. 3. Destruction of wild plants of forests for timber, charcoal and fire wood often deprive wild animals their most palatable food and affects their survival. 4. Noise pollution by different transporting media (trucks, buses, rails, aeroplanes, etc.) and polluting river water have adversely affected wild animals. 5. Various natural calamities such as floods, droughts, fires, epidemics, etc., have also caused great

destruction of wild life. 6. Hunting methods of all kinds and for any purpose (*i.e.*, for food, recreation, hide, fur, plumage, musk, tusk, horn, etc.) have caused destruction of wild life.

NECESSITY FOR WILD LIFE CONSERVATION

The conservation of wild life is required for the following benefits; 1. The wild life helps us in maintaining the 'balance of nature'. Once this equilibrium is disturbed it leads to many problems. The destruction of carnivores or insectivores often leads to the increase in herbivores which in turn affects the forest vegetation or crops. 2. The wild life can be used commercially to earn more and more money. It can increase our earning of foreign exchange, if linked with tourism. 3. The preservation of wild life helps many naturalists and behaviour biologists to study morphology, anatomy, physiology, ecology, behaviour biology of the wild animals under their natural surroundings. 4. The wild life provides best means of sports and recreation. 5. The wild life of India is our cultural asset and has deep-rooted effect on Indian art, sculpture, literature and religion.

MODES OF WILD LIFE CONSERVATION

In India, the following measures have been under taken for the wild life management :

1. Protection by Law

India was probably the first country to enact a Wild Life Protection Act. The Wild Birds and Animals Protection Act was passed in 1887 and repealed in 1912. For game protection in the states, in 1927, the Forest Act XVI was enacted. Indian Board for Wild Life was established in 1952 and this was followed by setting up of Wild Life Boards in different states. In 1972, new Wild Life Protection Act was passed. Under this Act, possession, trapping, shooting of wild animals alive or dead; serving their meat in eating houses; their transport and export are all controlled and watched by special staff (Chief Wild Life Warden and authorized officers). This Act prohibits hunting of females and young ones. Under this Act threatened species are absolutely protected and the rest afforded graded protection according to their state of population size.

2. Protected Species of Indian Wild Life

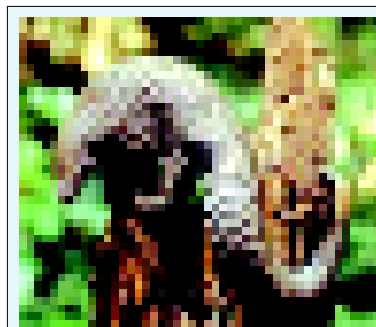
The following wild animals have been enlisted as threatened and protected species ; White eyed buck, Indian antelope or black buck, Four-horned antelope, Bharal, Swamp deer,



Water lizard.

Eastern pangolin, Elephant, Indian gazelle or Chinkara, Musk deer, Serow, Tahr, Golden cat, Snow leopard, Great Indian bustard, Pink

headed duck, Peafowl, Monal pheasant, Koklas pheasant, Chir pheasant, Horned pheasant, Monitor lizard, Water lizard, Gharial, Marsh crocodile, Python, and Leathery turtle. Albino and melanistic individuals of all species of wild animals are also protected.



Pangolin.

3. Establishment of Sanctuaries and National Parks

To protect, preserve and propagate rich natural bounty, the Government of India passed the Wild life (protection) Act, in 1972 under which national parks and sanctuaries could be created. Creation of biosphere reserves has also been put into practice since 1986. In India, nearly 434 national parks and sanctuaries have been established, out of which 17 have been selected as "Project Tiger" areas (on March 1, 1989, there were 445 national parks and sanctuaries, see **Rana**, 1994).

National parks are set up for preserving flora, fauna, landscapes and historic objects of an area. A national park is an area which is strictly reserved for the betterment of the wild life and where activities such as forestry, grazing or cultivation are not permitted. No private ownership right is allowed. There are 66 national parks in the country which are spread over an area of 33,988.14 square kilometres or nearly 1 per cent of the country's geographical area. **Sanctuaries** are places where the killing or capturing of any animal is prohibited except under orders of the authorities concerned. They provide protection and optimum living conditions to wild animals. Thus, in a sanctuary, protection is given only to the fauna and operations such as harvesting of timber, collection of minor forest products and private ownership rights are permitted so long as they do not interfere with the well-being of animals. There are 368 sanctuaries in India covering over 1,07,310,13 square kilometres, amounting to 3.2 per cent of India's total geographical area.

During the past three decades the concept of **Biosphere Reserves** has been evolved by the **Man and Biosphere (MAB) programme** of the UNESCO. India has identified 13 areas to be declared as biosphere reserves. Of these the Nilgiri Biosphere Reserve, including parts of Karnataka, Kerala and Tamil Nadu was declared in 1986 and the Nanda Devi Biosphere Reserve in 1988. Two more biosphere reserves, one at Uttarakhand (including the Valley of Flowers in North-Western Himalaya) and another at Nokrek (North-Himalaya) have also been recently included in the list of biosphere reserves. In a biosphere reserve, multiple land use is permitted by designating various zones. These zones are : the **core zone** (where no human activity is permitted), the **buffer zone** (where limited human activity is allowed) and the **manipulated zone** (where a large number of human activities would go on). In a biosphere reserve, wild populations as well as traditional life styles of tribals and varied domesticated plant and animals genetic resources are protected. Some of the well known wild life sanctuaries and national parks of India are as follows :

1. Kaziranga wild life sanctuary. It was established in 1926 in district Sibsagar, subdivision Jorhat (Assam) on the south bank of the Brahmaputra river. It consists of 430 sq km of forest grasslands and swamps and supports a fauna of 700 rhinoceros, besides elephant, wild buffalo, bison, tiger, leopard, sloth bear, sambhar, swamp deer, hog deer, barking deer, wild boar, gibbon and birds such as pelican, stork and ring-tailed fishing eagles.

2. Manas wild life sanctuary and Tiger reserve. It is located in district Kamrup in Assam, has an area of 540 sq km and is situated at an altitude of 80 metres. River Manas passes through it. It contains the following wild animals: tiger, panther, wild dog, wild boar, rhinoceros, gaur, wild buffalo, sambhar, swamp deer and golden langur.

3. Jaldapara wild life sanctuary. It is situated in Jalpaiguri district of West Bengal and is a 65 sq km stretch of grassland. Its wild life fauna includes animals such as rhinoceros, gaur, elephant, tiger, leopard, deer, and a variety of birds and reptiles.

4. Palamau national park. It is situated in Daltongunj district in Bihar and has an area of 345 sq km. Its flora is thick tropical forests. The fauna of this park includes tiger, panther, sloth bear, elephant, chital, gaur, nilgai, chinkara, chowsingha and mouse-deer.

5. Hazaribagh national park. This national park was established in 1954 in Bihar. It has an area of 184 sq km of thick tropical forests. The typical fauna of this park includes wild boar, sambhar, nilgai, tiger, leopard, sloth bear, hyena and gaur.

6. Simlipal national park. It is situated in district Mayuri Bhanj in Orissa and has an area of 2750 sq. km. It is covered over by dense sal forests and is chosen for project tiger. Its typical fauna includes tiger, elephant, deer, pea fowl, talking mynas, chital, sambhar, panther, gaur, hyena and sloth bear.

7. Chilka lake. This largest inland lake has an area of 1000 sq km and is about 100 km from Bhubaneswar (Orissa). Its typical fauna includes water fowls, ducks, cranes, ospreys, golden plovers, sandpipers, stone curlews, flamingoes, etc.

8. Kolameru bird sanctuary. This is a small bird sanctuary near Tadepallegudam in Andhra State. It is a breeding place for pelicans and many marine birds visit this place.

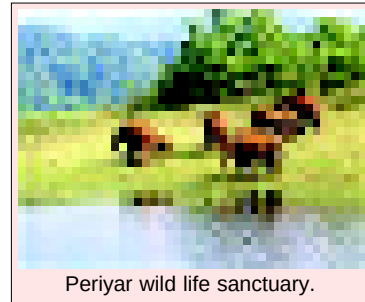
9. Vedanthangal bird sanctuary. This small-sized but very old sanctuary extend over a lake of about 0.30 sq km, 85 km south of Madras. Many migratory birds regularly visit this temporary lake. The regular visitors of this lake are birds such as spoon bill, open billed stork, egrets, ibis, cormorant, darter, grey heron, pelican, snipes and dab chick.

10. Guindy deer park. It is situated near Madras city and has mainly chitals and black bucks. A few albinos of black buck are also found here.

11. Point calimera wild life sanctuary. This is situated at the southern tip of the Thanjavur district of Tamil Nadu abutting the Palk strait. Its back water and lagoon is visited by flamingos and pelicans. Its nearby vedaranyam forests has a fauna of numerous black bucks, chitals and wild boars.

12. Mundanthurai sanctuary. It was established in 1962 in Tirunelveli district of Tamil Nadu and has an area of 520 sq km. It has evergreen forests and Tamaraparani river flows through this sanctuary. Its typical fauna includes panther, tiger, sambhar and chital.

13. Periyar wild life sanctuary. This is situated in Kerala State and has an area of 777 sq km. It was established in 1940 around the artificial lake which arose behind the dam built across the Periyar river in 1900. This sanctuary of great scenic beauty supports a fauna of wild elephants, gaurs, leopards, sloth bear, sambhars, barking deer, wild dogs, wild boars, black Nilgiri langur and water birds such as grey hornbills, egrets.



Periyar wild life sanctuary.

14. Mudumalai wild life sanctuary. This wild life sanctuary was established in 1940 in North Western part of Nilgiris in Tamil Nadu. It is known for its rich forests and diversity of fauna that includes wild elephant, gaur, sambhar, chital, barking deer, mouse deer, four-horned antelope, tiger, panther, bonnet monkey, langur, giant squirrel, flying squirrel, wild dog, jackal, wild cat, sloth bear, porcupine, pangolin, flying-lizard, monitor-lizard, rat snake, python and various birds.

15. Ranganthittoo bird sanctuary. It covers an area of 166 sq km and includes a series of islands in the Cauvery river 15 km off the Bangalore-Mysore Road, near Shrirangapatnam. Its avial fauna includes open bill stork, white ibis, egret, spoon bill, wild duck, peafowl, night heron birds.

16. Bandipur wild life sanctuary. This sanctuary was established in 1941 by the then ruler of Mysore (Karnataka State). It is situated 80 km south of Mysore city enroute to Ootacamund. It has an area of 874 sq km and is at an altitude of 1454.4 metres. Its forest is very thick and has plenty of rain fall. Its wild life fauna includes plenty of gaurs and animals such as elephant, leopard, sloth bear, wild dog, chital, panther, barking deer, porcupine and langur.

17. Cotigao wild life sanctuary. It is located in South Goa and has an area of 105 sq km. It has wet evergreen forest of Elve and Berla trees and supports rich fauna of the following animals : gaur, sambhar, chital, hog-deer, barking deer, wild boar, hyaena, panther, leopard, jackal, otter, porcupine and birds such as parakeet, lorikeet, woodpecker, kingfisher, bulbul, jungle fowl, egret, etc.

18. Bhagwan Mahadev wild life sanctuary. It is located in North Goa and has an area of 240 sq km and supports fauna almost similar to Cotigao wild life sanctuary.

19. Sesar Gir. This famous wild life sanctuary for the Asiatic lion is situated in Gujarat state, 468 km from Ahmedabad and 43 km from Veraval. It has an area of 1295 sq km of semiarid country with patches of thorn scrub and deciduous trees. Its fauna includes Asiatic lion, spotted deer, blue bull (nilgai), four-horned antelope, chinkara, striped hyena, wild boar, porcupine, langur, python, crocodiles and birds such as green pigeon, partridge, rock-grouse, etc.

20. Kanha national park. This national park was established in 1955 in former Banjar Valley Reserve (Madhya Pradesh). This park has an area of 939.94 sq km and includes hilly terrains and

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streams. It is 175 km away from Jabalpur and has forests of sal trees. Its typical fauna includes animals such as tiger, chital, panther, sambhar, black buck and bara singha.

21. Tandoba national park. It is located in Chandrapur district (Maharashtra) and has an area of 116 sq km. Its typical fauna includes tiger, sambhar, sloth bear, bison, chital, chinkara, barking deer, blue bull, four horned deer, langur, peafowl and few crocodiles.

22. Sariska. This is one of the most beautiful wild life sanctuary of Rajasthan near Alwar. It has an area of 800 sq km and has dense Dohokra and Solar forests. Its typical fauna includes tiger, leopard, spotted deer, jungle cat, four-horned antelope, langur, porcupine, hedgehog, peafowl, etc.

23. Bharatpur bird sanctuary (Keoldeo Ghana). It is located at Bharatpur (Rajasthan), has an area of 29 sq km and harbour all kinds of indigenous nesting water birds, water side birds and migratory birds. More than 328 varieties of birds including cormorants, spoonbills, white ibis, Indian darters, egrets, painted storks, open billed storks, great black necked storks, etc., Many migratory birds such as ducks, geese, siberian cranes, etc., regularly visit this sanctuary. Drier parts of this marshy sanctuary have animals like spotted deer, black buck, sambhar, blue bull, wild boar and python.



Bharatpur bird sanctuary.

24. Sultanpur lake bird sanctuary. This small sized (2 sq km area) bird sanctuary is located in Gurgaon district (Haryana) about 30 km from Delhi. Its typical avian fauna includes crane, sarus, spot-bill, ruddy shelduck and drake.

25. Shikari Devi wild life sanctuary. It is located in Mandi district in Himachal Pradesh. It has an area of 213 sq km and has the following animals : snow leopard, flying fox, black bear, barking deer, musk deer, chakor, partridge, etc.

26. Bir Motibagh sanctuary. It is located near Patiala in Punjab. The typical fauna of this wild life sanctuary includes black buck, blue bull, hog deer, hare, jackal and birds such as peafowl, partridge, sparrow, babbler, myna, parakeet, pigeon, dove, etc.

27. Dachigam wild life sanctuary. It was established in 1951 in Kashmir 26 km away from Srinagar. It has an area of 89 sq km and has two levels : upper Dachigam at 3692.3 metres altitudes and Lower Dachigam at 1846.2 metres altitudes. It mainly preserves hangul or Kashmir stag, but also has animals such as musk deer, leopard, black bear, brown bear and baboon.

28. Corbett national park. It is one of India's most famous wild life sanctuary and was constituted in 1935 as the first national park of India. It is situated between Nainital and Garhwal districts in Uttar Pradesh. It has an area of 525 sq km and is located within west-to-south bend of river Ramganga. It supports a rich and diverse fauna of the following animals : tiger, panther, sloth bear, hyaena, elephant, blue bull, swamp deer, barking deer, Indian antelope, porcupine, birds such as bulbul, woodpecker, barbet, babbler and bee eater, and reptiles such as crocodile, python, etc.

29. Shivpuri sanctuary. It is located in Madhya Pradesh and is an asylum for tigers.

30. Anamalai sanctuary. This sanctuary was established in 1972 in the southern part of Coimbatore District in Tamil Nadu. It has a vast area of 958 sq km and supports rich fauna of animals such as elephant, gaur, sambhar, spotted deer, barking deer, Nilgiri tahr, lion-tailed macaque, tiger, panther, sloth bear, langur, porcupine and pangolin.

Other Conservation Measures

Other important steps need to preserve wild life include the following measures : (i) For the preservation of species wild life management staff should have a correct idea about the exact habitat

which the species under consideration needs. (ii) Natural habitat of wild animals should be carefully protected. (iii) Habitats of wild life should be improved by constructing water holes, saltlicks and by raising plantation of better and nourishing fodder grasses and trees. (iv) Effective means of census operation should be adopted to measure population sizes of various wild animals. (v) The enforcement of Wild Life Protection Acts should be observed strictly. (vi) Shooting and hunting of endangered species should be totally banned. (vii) To take care of wild life in case of epidemics, veterinary efforts should be made. (viii) To understand the biology and behaviour of wild animals research on wild life should be encouraged. (ix) Public should be educated about the advantages and disadvantages of wild life.

REVISION QUESTIONS

1. Write an essay on the wild life of India.
2. Describe, why should wild life of India be protected ?
3. Describe various measures of wild life management.
4. Define the terms wild life sanctuary and national park and describe some important wild life sanctuaries and national parks of Indian Subcontinent.

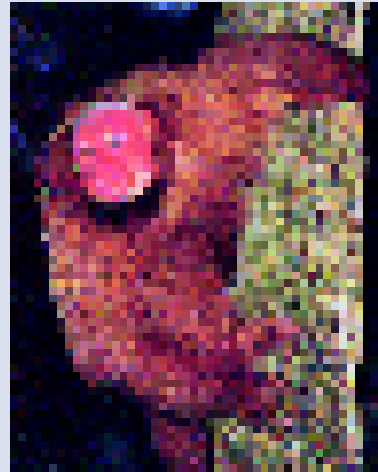
CHAPTER

17

Biogeography

(Distribution of Animals and Plants)

No organism occurs in uniform numbers throughout the world. Rather, specific organisms are restricted to specific communities or groups of communities. Three aspects of the distribution of an organism are generally recognized: **geographic range**, or the specific extent of land or water area where the organism normally occurs; the **geologic range**, or the distribution in time, past and present; and the **ecological distribution**, or the major biotic communities (*i.e.*, marine biome, freshwater biome and terrestrial biome) of which the organism is a member. Certain biologists have also made distinction between **geographic distribution** (horizontal or superficial distribution) and **bathymetric distribution** (vertical or altitudinal distribution). Bathymetric distribution includes the following three realms: (i) **Halobiotic**, or vertical distribution of organisms in marine (sea) habitat, (ii) **Limnobiologic**, or vertical distribution of organisms in freshwater habitat; and (iii) **Geobiologic** or altitudinal distribution of organisms on land. All the living organisms in a given region are termed the **biota** of that region. The animals of a given region are collectively termed the **fauna**, and the plants of a given region,



There is geographical distribution of species of all organisms on this earth.

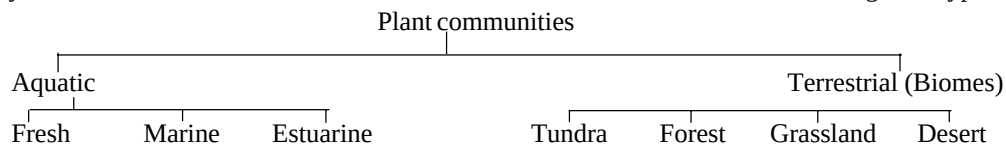
the **flora** (i.e., fauna+flora=biota). Here, a distinction can be made between the following two terms—flora and vegetation. **Flora** mainly refers to the botanical composition of a place, i.e., the names of different plant species, while **vegetation** means the totality of forms in which the emphasis is not on names of different plants but their life forms, number and coverage. The studies of the distribution of biota are collectively called **biogeography**; of animals only, **zoogeography**; and of plants only, **plant geography** or **phytogeography**. There are two major approaches to the study of biogeography or **geographical ecology** (Kendeigh, 1974): (i) **descriptive** or **static biogeography** which deals with the description of biota of different botanical and zoological areas of earth; and (ii) **interpretative** or **dynamic biogeography** which describes the forces which have brought about plant and animal distribution.

DESCRIPTIVE PHYTOGEOGRAPHY

With a descriptive approach of phytogeography, will be discussed the major plant communities (biomes) of the world, and different vegetational belts of the earth constituted by these biomes together with characteristic climatic conditions of the area.

1. Major Plant Communities (Biomes) of the World

As already discussed in sufficient details major plant communities of the world are classified chiefly on the basis of the kinds of habitat and environmental conditions into the following chief types:



2. Phytogeographical Regions of World (or Chief Vegetational Belts of the Earth)

On the basis of climatic and geographical conditions, the earth is usually divided into the following four broad vegetational belts:

A. Arctic zone. This zone is divided into the following two types:

(1) Arctic proper. This zone occurs around the north pole and remains covered with ice throughout the year. This zone in fact is Tundra biome and its chief components of vegetation include some algae, annual flowering plants, mosses and lichens.

(2) Subarctic. This is a less defined zone which extends from southern arctic to the northern limits of temperate zone. This region is very cold, contains abundant bogs and have vegetation including small height trees, shrubs and herbs in the months of June and July. Ground vegetation often includes some pteridophytes, orchids, insectivorous plants, mosses and lichens.

B. North temperate zone. This zone extends between 30° N latitude and 55° N latitude and includes following two major zones:

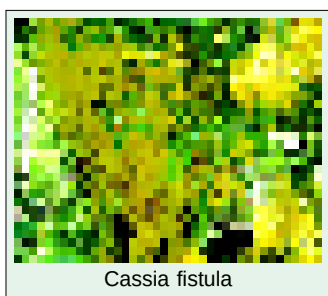
1. North temperate of the eastern zone. This zone is further subdivided into the following four zones: (i) **Western and central Europe.** This zone is demarcated in north by the subarctics and in south by Alps and British Islands. Its forests are dominated by several gymnospermous tall trees such as *Pinus*, *Picea* and *Abies*, and angiospermous trees such as oaks, maple and chestnuts. Ground vegetation comprises orchids, wild roses, buttercups, *Viola*, *Salvia*, *Dianthus*, etc. At high altitudes of this zone trees are replaced by grassy vegetation with some herbaceous flowering plants. (ii) **Mediterranean.** This zone extends between 30°N and 40°N latitudes, south of mountain ranges in Europe and in Asia around Mediterranean sea and is characterized by warm temperate type climate. Vegetation is chiefly composed of fruit trees, olives, nut trees, oranges, and also some foreign palms, cacti, acacias, etc. In the Asian region of Mediterranean as in Arab countries, rainfall is low, deserts are common and sparse vegetation include following species *Atriplex*, *Alhagi*, *Polygonum* and *Phoenix dactylifera*. (iii) **Northern Africa.** This zone includes the northern parts of Morocco, Algeria, Libya

and Egypt. It is characterized by scanty rainfall and sparse vegetation. In cooler areas (*i.e.*, mountains) some conifers and broad-leaved oaks are common. In deserts (including some portion of Sahara desert) some herbs, shrubs, woody acacias and succulent xerophytes occur. (iv) **Himalayas, eastern Asia and Japan.** Tibet, China and Japan have different type of vegetation. In China and Japan conifers such as *Cryptomeria*, *Sciadopitys*, *Cephalotaxus*, *Ginkgo biloba* and *Cycas* and angiosperms such as *Rhododendrons*, *Cinnamomum camphora* and *Begonia* are common. The vegetation of Himalayas will be described later in this chapter.

(2) North temperate of the western hemisphere. It includes the parts of United States and Canada lying mostly between north latitudes 30° to 55° . The eastern coastal regions of these countries in the temperate belt have some very characteristic plant species such as tropical fern (*Schizaea pusilla*). The forest communities are composed of conifers and deciduous trees. On lower altitudes some wild cherries, plums, roses and orchids are abundant. In the New England region trees of *Ulmus americana* and *Castania dentata* are abundant. Forests of conifers are common in southern parts and on western parts of Rocky mountains of USA. In north California grows *Sequoia sempervirens*, the tallest tree of world. Ground vegetation is composed of *Salicornia herbacea*, *Rumex maritima*, *Monotropa uniflora*, *Saxifragea*, *Primula*, etc. In the Colorado desert of Arizona and south-eastern California, there are several types of xerophytic plants.

C. Tropical zone. This zone is divided into two subzones :

(A) Palaetropic. It comprises old world or eastern tropics and has the following two botanical



Cassia fistula

regions : **(1) Tropical Africa.** This is a large-sized landmass of varied topography. It includes high altitudes and Sahara deserts which have little or no rainfall. In Africa, most remarkable plant is *Welwitschia mirabilis*, Eastern part of Central Africa has India-like vegetation of *Borassus flabelliformis*, *Tamarindus indica*, *Ficus*, *Asparagus*, *Clematis*, *Phaseolus*, *Cassia fistula*, *Erythrina*, *Acacia*, *Albizia*, *Zizyphus*, *Bauhinia*, etc. **(2) Tropical Asia.** It includes Arabia, Pakistan, India, Burma, Ceylon, Thailand, Indonesia, Philippines and Islands of Indian sea. In Arabia, the rainfall is low and most of the desert species are found, *Coffea arabica* is a native plant of Arabia. Ceylon is rich in

species diversity and ferns are the chief components of its sparse natural vegetation. Malaya, Java and Sumatra are characterized by heavy rainfall and their vegetation comprises varied types of palms, some types of ferns, tall trees, lianas and insectivorous plants. Some important plants of Java are *Albizia*, *Pterocarpus*, *Tamarindus*, *Bombax*, *Cassia*, *Dendrocalamus*, etc. The common trees of Burma and Thailand are jack fruit, orange, mango, betelnut, etc.

(B) Neotropics. It comprises Mexico and major part of South America. The low-rainfall areas of Mexico are rich in xerophytes. At higher cooler altitudes there is a forest of conifers such as *Pinus*, *Spruce*, *Quercus* and *Populus*. On mountain peaks grasses are most common. In wet areas of Mexico, there are mosses, palms, bamboos, orchids, etc. In South America, there are extensive forests of flood-resistant trees such as *Bertholletia excelsa*, *Maximiliana regia* and also mangrove vegetation. There are also found many epiphytes.

D. South temperate zone. This zone includes extreme southern region of Africa, Australia and New Zealand. In African area, the vegetation is chiefly made up of ferns and gymnosperms. On the hills conifers are present. Its lower wet regions contain *Salix* and *Phragmites* and dry regions have grasses such as *Andropogon* and trees such as *Acacia*. The vegetation of northern part of Australia is similar to that of south east Asia and includes trees of palms, nuts, *Eucalyptus*, *Acacia* and *Casuarina*. Some petridophytes are also found in the ground vegetation. In the South Australia *Araucarias* are common.

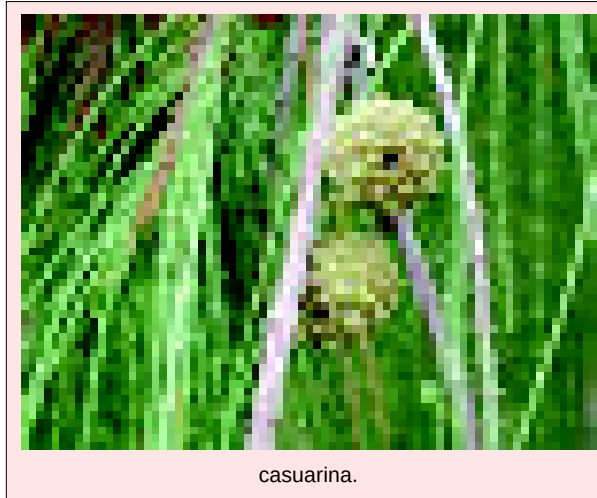
New Zealand forests are mostly made up of conifers together with ferns, palm such as *Rhopalostylis*, many species for *Metrosideros*. New Zealand has richest bryophytic flora.

3. Phytogeography of India

Indian subcontinent lying between 8° and 37° N, and 68° and 97° E has its own peculiar physiographic, climatic and biotic features. It is surrounded on its south, east and west by oceans and in the north by mountainous chains highest in the world. The subcontinent stretches out between tropical and subtropical belts. Its climate is chiefly modified by oceans and mountains. In the south and in the far east the climate is typically tropical, while it is temperate in the north, and highly arid in the northwest. The general climate of India is of monsoon type.

Vegetation of India. The most important factors used in the classification of Indian vegetation are rainfall, temperature, edaphic factors, biotic influences and life forms. There are two most common types of plant formation in India—forest and grassland.

(a) Forest vegetation. Champion and Seth (1968) have recognised the following 16 major types of forests in India.



casuarina.

I. MOIST TROPICAL FORESTS

(1) Tropical wet evergreen forests. These forests occur on the Western Ghats, Assam, Cachar, parts of Bengal, parts of Mysore and Andamans. These forests are climatic climax forests with very dense growth of tall trees (more than 45 metres high). The shrubs, lianas, climbers and epiphytes are abundant, however, grasses and herbs are rare on the forest floor. The rainfall exceeds 250 cm in these regions. The characteristic flora of these forests include *Dipterocarpus indica*, *Hopea*, *Artocarpus*, *Mangifera*, *Embllica*, *Michelia*, *Ervatamia*, *Lagerstroemia*, *Ixora*, etc.

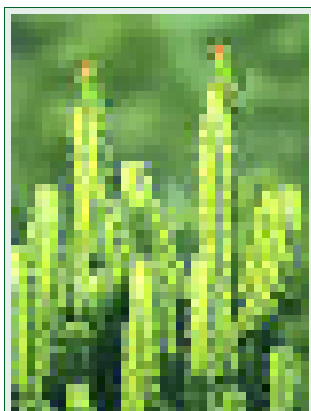
(2) Tropical moist semi-evergreen forests. These forests occur along Western Ghats, in parts of Upper Assam and Orissa. The rainfall of these regions is usually high, above 200 cm per year. These forests consist of dense and tall (25 to 35 metres) trees along with shrubs. In these forests, because deciduous species such as *Terminalia*, *Tetrameles* and *Shorea* grow intermixed with the evergreen species such as *Artocarpus*, *Michelia* and *Eugenia*, so they are called semi-evergreen forests.

(3) Tropical moist deciduous forests. These forests are distributed in a narrow belt along the foot of the Himalayas, on the eastern side of Western Ghats, Chhota Nagpur and Khasi Hills. These forests have high rainfall (150–200 cm) throughout the year except few very short dry periods. The trees are deciduous remaining leafless for one or two months. The most common species is teak (*Tectona grandis*) in the south and sal (*Shorea robusta*) in the north. Other common species of these forests are *Dalbergia*, *Cedrela*, *Salmalia*, *Albizzia*, *Terminalia*, *Cordia*, *Bombusa*, *Melia*, *Dillenia*, *Eugenia*, *Dendrocalamus*, etc.

(4) Littoral and swamp forests. These forests comprise mostly evergreen species and occur along sea coasts in wet and marshy areas and also in deltas of larger rivers on eastern coast. In saline swamps, mangroves are chiefly composed of *Rhizophora*, *Bruguiera*, *Ceriops*, *Nipa*, etc. In less saline areas, *Phoenix*, *Ipomoea*, *Phragmites*, *Casuarina*, etc., are found.

II. DRY TROPICAL FORESTS

(5) Tropical dry deciduous forest. These forests are distributed almost throughout the country except Kashmir, beyond Bengal, Rajasthan and Western Ghats. These occur in areas with annual rainfall ranging from 75 cm to 125 cm and with a dry season of about 6 months. The trees are small (10–15 metres) but canopy is open. Shrubs are abundant. These forests include following common plant species *Shorea robusta* (in north), *Tectona grandis* (in south), *Anogeissus*, *Terminalia*, *Semecarpus*, *Carissa*, *Emblia*, *Acacia*, *Zizyphus*, *Dalbergia*, *Dendrocalamus*, etc.



Euphorbia is a common plant of tropical thorn forests.

(6) Tropical dry deciduous forests of South-East Daccan Region. In some parts of Tamil Nadu and the Karnatic coasts evergreen forests of short (10–15 metres) but dense trees are found though the annual rainfall is relatively small. There are no bamboos but grasses are abundant. The common flora of this region includes species of *Memecylon*, *Maba*, *Pavetta*, *Feronia*, *Ixora*, *Sterculia*, etc.

(7) Tropical thorn forests. These forests are distributed in areas with very low rainfall as in Western Rajasthan, parts of Southern Punjab, Bundelkhand, Gujarat, Maharashtra, Madhya Pradesh and Tamil Nadu. The trees are short (8–10 meters), sparsely distributed, mostly thorny; shrubs are more common than trees. Plants remain leafless throughout the year. During rains, grasses and herbs become abundant. The common plants of these forests are *Acacia nilotica*, *Prosopis spicigera*, *Capparis*, *Albizia*, *Zizyphus*, *Euphorbia*, *Calotropis*, *Madhuca*, *Tephrosia*, *Crotalaria*, etc.

III. MONTANE (MOUNTAINOUS) SUBTROPICAL FORESTS

(8) Subtropical broad-leaved hill forests. These forests are dense, have predominantly evergreen species, and occur in relatively moist areas as lower slopes of eastern Himalayas, Bengal, Assam and on hill ranges as Khasi, Nilgiri, Mahabaleshwar, etc. In the north occur *Quercus*, *Castanopsis*, *Schima* and some temperate species. In eastern Himalayas due to higher humidity, bamboos, many epiphytes including orchids and ferns become abundant. In the south common trees are *Eugenia*, *Actinodaphne*, *Randia*, *Ficus*, *Populus*, *Canthium*, *Mangifera* and climbers are *Piper*, *Gnetum*, *Smilax*, etc.

(9) Subtropical pine forests. In Western Himalayas and Khasi Jayantia hills of Assam, up to an altitude of 1875 metres, open forests composed mainly of pine species such as *Pinus roxburghi* and *P. khasya*.

(10) Subtropical dry evergreen forests. Lower elevations of Himalayas have xerophytic forests containing mainly thorny and small-leaved evergreen species. These areas are characterized by low rainfall and low temperature. The common species are *Acacia modesta*, *Dodonea viscosa*, *Olea cuspidata*, etc.

IV. MONTANE TEMPERATE FORESTS

These forests are found at 1800 to 3800 metres altitude in the Himalayas.

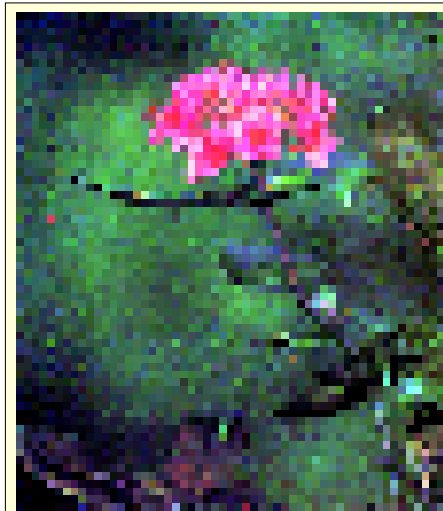
(11) Montane wet temperate forests. These forests are found in eastern Himalayas with high rainfall and also in Nilgiris in the south. The forests in the south are evergreen and are known as **Sholas**. The trees are 15 to 20 metres high with a dense growth forming closed canopy. Epiphytes are abundant. The common plant species of these forests are *Rhododendron nilagircum*, *Hopea*, *Artocarpus hirsuta*, *Salmalia*, etc.

In the north, the west temperate forests extend from eastern Nepal to Assam at an altitude of 1800 to 2900 metres. These forests are evergreen, dense and high (up to 25 metres). The common plant

species include *Quercus*, *Acer*, *Prunus*, *Ulmus*, *Begonia*, *Loranthus*, etc. In the western Himalayas *Quercus* species dominate all over. Conifers as *Abies*, *Cedrus*, *Picea*, etc., and broad-leaved species as *Betula*, *Alnus*, *Ulmus*, *Acer*, *Populus*, etc., are also widely distributed.

(12) Himalayan moist temperate forests. Central and Western Himalayas, have the forests of conifers or/and oaks at an altitude of 1500 to 3000 metres. The trees are tall (up to 45 metres), but undergrowth is thin and deciduous. These forests have plants such as *Cotoneaster*, *Berberis*, *Spiraea*, *Quercus*, *Cedrus*, *Pinus*, *Picea*, *Abies*, *Tsuga*, *Cupressus*, etc.

(13) Himalayan dry temperate forests. These forests occur in a narrow belt in the western Himalayas extending from parts of U.P. through Himachal Pradesh and Punjab to Kashmir. They are open, evergreen forests with scrub undergrowth. Oaks and conifers are dominant. In western drier areas *Pinus gerardiana* and *Quercus ilex* are common, while in the eastern moist zone *Abies*, *Picea*, *Larix*, *Juniperus* are found. Other associated plant species are *Cedrus*, *Picea*, *Daphne*, *Artemisia*, *Fraxinus*, *Alnus*, etc.



Rhododendron is widely found in alpine forests which occur throughout Himalayas.

V. ALPINE FORESTS

(14) Sub-alpine forests. These evergreen forests are open and occur throughout the Himalayas at high altitudes (above 3000 metres up to tree line). They comprise mostly conifers such as *Abies spectabilis*, broad-leaved species such as *Betula* and *Rhododendron* and shrubs as *Cotoneaster*, *Rosa*, *Lonicera*, *Smilax*, etc.

(15) Moist-alpine scrub. Throughout the Himalayas, above the timber line up to the height of 5500 metres, dwarf evergreen shrubby growth of conifers (*Juniperus*) and broad-leaved species (*Rhododendron*) are found.

(16) Dry-alpine scrub. In areas with low rainfall (below 35 cm) open xerophytic scrubs such as *Juniperus*, *Caragana*, *Eurotia*, *Salix*, etc., are found upto 5500 metres.

(b) Grassland vegetation. In India, natural grasslands (as climax formation) are not present but occur only during succession. The following three types of grasslands occur in India : (1) **Xerophilous** that occur in dry regions of north-west India under semidesert conditions : (2) **Mesophilous** which are extensive grass flats or savannas and occur in moist deciduous forests of Uttar Pradesh ; and (3) **Hygrophilous** which are called wet **savannas**. These three basic types of Indian grasslands have been further subdivided into the following eight subtypes, each of which is named on its dominant species: *Sehima–Dichanthium*, *Dichanthium–Cenchrus*, *Phragmites–Saccharum*, *Bothriochloa*, *Cymbopogon*, *Arundinella*, *Deyeuxia–Arundinella*, and *Deschampsia–Deyeuxia*.

4. Floristic (Botanical) Regions (Provinces) of India

The Indian sub-continent is characterised with a variety of climate types and the flora of the country is also correspondingly of different types in its different parts. The country has been divided into the following nine **floristic regions** for the study of flora :

A. Western Himalayas. It extends from Kumaon to Kashmir and has annual rainfall up to 200 cm. Altitudinally there are following three zones of vegetation corresponding to three climatic belts:

(i) Submontane zone. This extends up to 1500 metres altitude and comprises mostly of Siwalik ranges. The forests are tropical and subtropical having trees such as *Shorea robusta*, *Dalbergia sissoo*,

Cedrela toona, *Ficus glomerata*, *Eugenia jambolano*, *Acacia catechu*, *Butea monosperma* (dhak), *Zizyphus* and thorny succulent euphorbias on slopes.

(ii) Temperate zone. Above submontane zone extend montane temperate forests up to 3500 metres altitude. They are dominated by plant species such as *Quercus*, *Acer*, *Ulmus*, *Rhododendron*, *Betula* (birch), *Salix* (cane), *Populus*, *Cornus*, *Prunus*, *Fraxinus*, *Pinus*, *Cedrus*, *Picea* and *Taxus*.

(iii) Alpine zone. This zone extends from 3500 to 4500 metres altitudes (snow line) and is characterised with alpine forest vegetation with scrub and meadows. Most common tree species are *Abies*, *Betula*, *Juniperus* and *Rhododendrons*. The herbs which occur near the snow line include species such as *Primula*, *Potentilla*, *Polygonum*, *Geranium*, *Saxifraga*, *Aster*, etc.

B. Eastern Himalayas. It includes regions of Sikkim and NEFA and is characterised by more rainfall, less snow and higher temperature. This is also divided into the following three zones altitudinally :

(a) Tropical zone. Up to 1800 metres altitudes, this zone has tropical semi-evergreen or moist deciduous forests. These forests comprise the plants such as *Shorea robusta*, *Acacia catechu*, *Delbergia sissoo*, *Terminalia*, *Albizia*, *Cedrela*, *Dendrocalamus* (bamboo), etc.

(b) Temperate zone. This zone extends between 1800 metres to 3800 metres altitudes and has typical montane temperate forests which are dominated by oaks such as *Michelia*, *Quercus*, *Pyrus*, *Symplocos*, *Eugenia*, etc., at lower levels and by conifers as *Juniperus*, *Cryptomeria*, *Abies*, *Pinus*, *Larix* and *Tsuga* and also *Salix*, *Rhododendron* and *Arundinaria* (bamboo) at higher cooler levels.

(c) Alpine zone. Beyond the temperate zone extends alpine zone up to 5000 metres altitudes. It has alpine vegetation including *Juniperus* and *Rhododendron* with its other typical flora.

C. Indus plains. This zone includes the arid and semiarid regions of Punjab, Rajasthan, Kutch, part of Gujarat and Delhi. The rainfall is less than 70 cm. The vegetation is tropical thorn forest in semi-

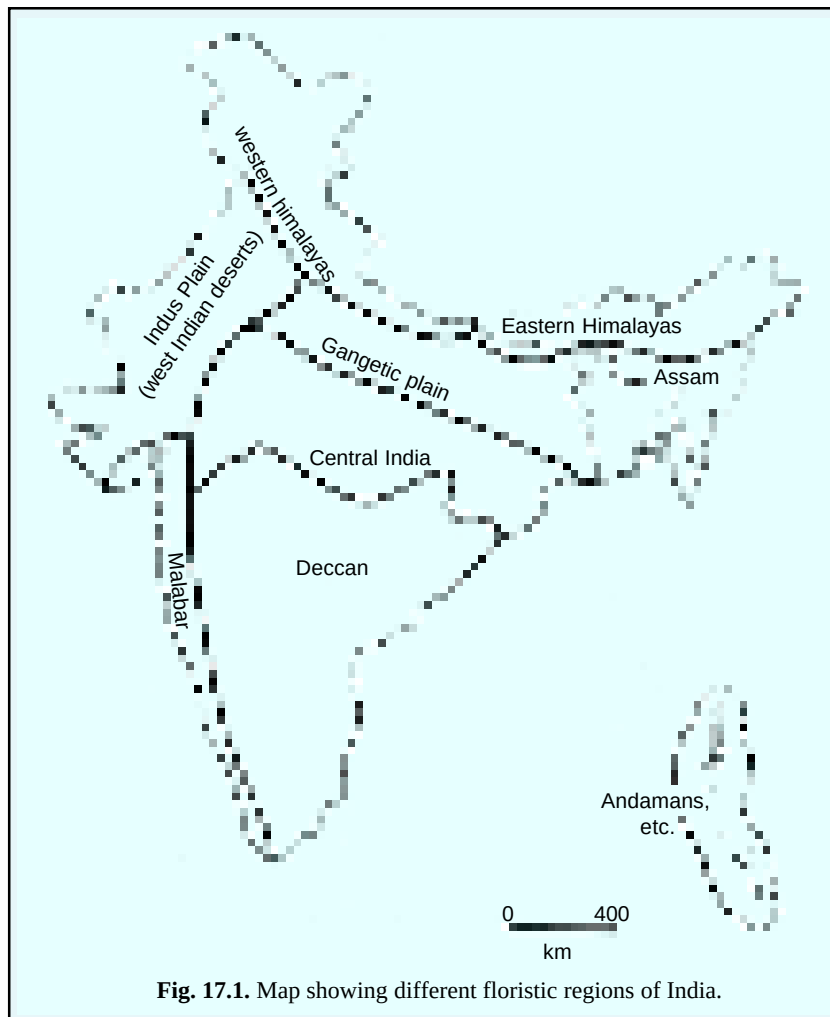


Fig. 17.1. Map showing different floristic regions of India.

arid region and is typical desert in the arid region. The plants of this zone are primarily xerophytic. The common plant species of this zone are *Acacia nelotica*, *Prosopis* sp., *Salvadora*, *Tecomella*, *Capparis*, *Tamarix*, *Zizyphus*, *Calotropis*, *Panicum*, *Saccharum*, *Cenchrus*, *Euphorbia*, etc.

D. Gangetic plains. This region extends over Uttar Pradesh, Bihar, Bengal and part of Orissa and is characterised by moderate amount of rainfall and most fertile (*i.e.*, alluvial) soils. Vegetation of this zone is chiefly of tropical moist and deciduous and dry deciduous forest type. The common plants of this zone are *Dalbergia sissoo* (shesham), *Acacia nelotica* (babul), *Saccharum munja*, *Butea monosperma*, *Madhuca indica* (mahua), *Terminalia arjuna* (arjuna), *Buchanania lanzan* (chiraunji), *Diospyros melanoxylon* (tendu), *Cordi myxa* (lisora), *Acacia catechu* (khair), *Azadirachta indica* (neem), *Ficus bengalensis* (bergad), *Ficus religiosa* (pipal), *Mangifera indica* (mango), and weeds and grasses such as *Xanthium*, *Cassia*, *Argemone*, *Amaranthus*, etc. In Gangetic delta (South Bengal) mangrove vegetation is common.

E. Central India. It comprises Madhya Pradesh, parts of Orissa and Gujarat. The rainfall is 150–200 cm and its vegetation is thorny, mixed deciduous and teak type. The chief plants of this region are *Tectona grandis*, *Madhuca*, *Diospyros*, *Butea*, *Dalbergia*, *Terminalia*, *Carissa*, *Zizyphus*, *Acacia*, *Mangifera*, etc.

F. Malabar (west coast). This region includes western coast of India from Gujarat to Cape Comorin and has heavy rainfall. The forests are tropical evergreen in west, semi-evergreen towards interior, subtropical or montane temperate evergreen forests in Nilgiris and mangroves near Bombay and Kerala coast.

G. Deccan Plateau. This region extends all over peninsular India (*i.e.*, Andhra Pradesh, Tamil Nadu and Karnataka) and has rainfall up to 100 cm. Its central hilly plateau has tropical dry deciduous forests of *Bowsellia serrata*, *Tectona grandis* and *Hardwickia pinnata*, while, the low eastern dry Coromandal Coast has tropical dry evergreen forests of *Santalum album* (chandan), *Cedrela toona* and plants such as *Acacia*, *Prosopis*, *Euphorbia*, *Capparis*, *Phyllanthus*, etc.

H. Assam. This region is characterised by heavy rainfall (200 to 1000 cm). The vegetation is either dense evergreen forest or sub-tropical. The evergreen forests include trees such as *Dipterocarpus macrocarpu*, *Mesua ferrea*, *Shorea robusta*, *Ficus elastica*, etc. bamboos as *Bambusa pallida*, *Dendrocalomus hamiltonii*, etc., grasses such as *Imperata cylindrica*, *Saccharum* sp., *Themeda* sp., insectivorous plants as *Nepenthes* sp., and also epiphytes (ferns and orchids). In the northern cooler regions, wet hill forests include plants such as *Alnus*, *Betula*, *Rhododendron*, *Magnolia*, etc., The hilly tracts also have pine forests of *Pinus khasiya* and *P. insularis*.

I. Andmans. This region possesses a varied type of vegetation : mangroves and beech forest at its coasts and evergreen forests of tall trees in the interior. Important plant species of this island are *Rhizophora*, *Mimusops*, *Calophyllum*, *Lagerstroemia*, etc.

PATTERNS OF DISTRIBUTION OF BIOTA

1. Distribution

The plants and animals exhibit specific patterns of distribution over the globe :

(i) Continuous distribution. A taxon distributed throughout the world in all climatic zones or in all the continents in at least one climatic zone is said to have continuous distribution. It is of the following types :

A. Cosmopolitan distribution. Such a taxon or a species occurs in all climatic zones. The common plants species of nearly cosmopolitan distribution are *Chenopodium album*, *Taraxcum officinale*, *Phragmites communis*, *Urtica dioica*, *Poa annua*, etc. Among the animals the following species have cosmopolitan distribution : *Tyto alba*, *Ardea cineria*, *Falco peregrinus*, *Vanessa cardui*, *Nonnophila noctuella*, and *Macrobiotus hufelandi*.

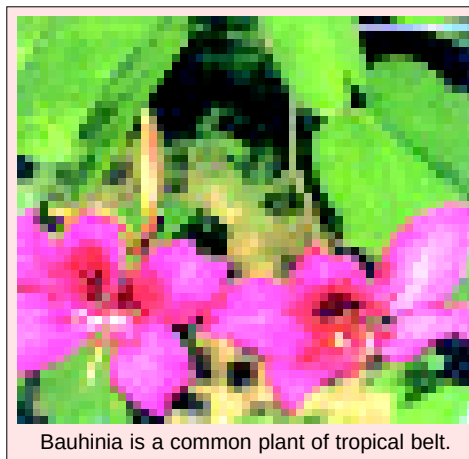
B. Circumpolar distribution. Certain species such as *Saxiphraga oppositaefolia*, *Carex lapponica*, *Ranunculus nivalis*, etc., are distributed in a belt around the north-pole.

C. Circumboreal and circumastral distribution. The species which are distributed in a near

continuous belt in the temperate region of northern or southern hemisphere are said to have circumboreal and circumastral distribution respectively. The genera *Alnus*, *Draba*, *Acer*, etc., are circumboreal in distribution and *Danthonia* is circumastral in distribution.

D. Pantropical distribution. Certain plant species such as *Bauhinia*, *Dalbergia*, *Dioscorea*, *Corchorus*, *Ocimum*, *Cassia*, *Eugenia*, *Phyllanthus*, etc., are distributed throughout the tropical belt.

(ii) Discontinuous distribution. A taxon distributed in two or more widely separated geographical areas, is said to have discontinuous distribution. Such taxa may occur in several small areas in the same continent or in two different continents of the same or different hemispheres. For example, two plant species—*Saxifraga* and *Silene* are distributed in Arctic region and high altitudes. Likewise, three genera of Dipnoi fish, *Epiceratodus*, *Protopterus* and *Lepidosiren* are distributed in Australia, Africa and South America. There are various other examples in plants and animals which show discontinuous distribution.



Bauhinia is a common plant of tropical belt.

2. Endemism

Species which are spread over a wide area or in different ecological conditions are said to be **cosmopolitan**. Some of them, called **endemics** are restricted in their distribution to a small region. This phenomenon of restricted distribution associated with some geographical or ecological factors is called **endemism**. Numerous plant families, genera and species are found to be highly endemic. In the Himalayas, some 28% (3,169 dicot species) of all the dicots are endemic (**Chatterjee**, 1939). For example, a great proportion of the taxa of the flora of oceanic islands is endemic to those islands only. Some important endemic plant genera of Indian deserts are *Omania*, *Xerotia*, *Leptadenia*, *Daemia*, etc., and of India are *Amphicome*, *Dittoceras*, *Dodecania*, *Ulteria*, *Cruddasia*, *Heylandia*, *Lagenandra*, *Zeylandium*, *Hitchenia*, *Blepharistemma*, etc. Certain important examples of endemic Indian plant species include *Ficus religiosa*, *Ficus bengalensis*, *Aegle marmelos*, *Artocarpus nobilis*, *Crotalaria juncea*, *Datura metal*, *Indigofera tinctoria*, *Elettaria*, *Eleusine coracana*, *Piper nigrum*, *Piper longum*, *Sesamum indicus*, *Hibiscus abelmoschus*, *Butea monosperma*, *Beaumontia grandiflora*, *Memecylon umbellatum*, *Holmskioldia sanguinea*, *Feronia elephantum*, *Saraca indica*, *Shorea robusta* and *Caryota urens*.

Among endemics, some species exhibit very localised distribution and are called **local endemics**. Sometimes mutants appear and vanish without being able to compete with parental species, and are called **pseudo-endemics**. Some species may show a restricted distribution but cover large areas in course of time. This is called **expanding** or **progressive endemics**. Some old species may be restricted to a small region because of a severe decline in their population, a phenomenon called **contracting** or **retrogressive endemics**. Lastly, there are following two more types of endemics : **1. Palaeoendemic or epibiotic or relics**. These endemics are supposed to have been the remnants of a once widely distributed taxon in the past, e.g., *Ginkgo biloba*, *Sequoia semipervirens*, *Trapa natans*. **2. Neoendemics or microendemics**. These endemic taxa are supposed to have evolved only during the recent times and did not have sufficient time to extend their ranges of distribution.



Ficus.

3. Centre of Origin

It is maintained that for every species, there is one region from where it came into existence. Thus, some areas might have given rise to many species. These regions are, therefore, called the centres of origin. Plants evolved much earlier than man. Paleobotanical evidence helps us to arrive at conclusions on the centre of origin. For example, evidently cincona, tomato (*Lycopersicon esculentum*) and tobacco (*Nicotiana tabacum*, *N. rustica*) originated in South America. Cotton (*Gossipium* sp.) and chilly (*Capsicum* sp.) originated in Central America. Lichi (*Litchi chinensis*), tea (*Camellia* or *Thea sinensis*) and brinjal (*Solanum melongena*) originated in China. India is the centre of origin of many crop plants such as rice (*Oryza sativa*), sugarcane (*Saccharum officinarum*) and gram (*Cicer arietinum*).

DESCRIPTIVE ZOOGEOGRAPHY

A few groups of animals are cosmopolitan in having world-wide distribution and most animals are restricted in distribution because of some kind of barrier or because of past history of origin and dispersal. An understanding of their present-day distribution takes us into zoogeography. Major units of distribution are the **zoological region** or **zoogeographical regions**, areas defined largely by the past and present relations of the continent to each other. Each region is further subdivided into **faunal** or **ecological units**, depending on the criteria used.

Zoogeographical Regions

There are six zoogeographical regions, each more or less embracing a major continental land mass and all have long been separated from one another by water (oceans and fresh-water bodies), mountain ranges or desert, so that, each region has evolved its distinctive and characteristic orders and families of animals. These major distributional units were first recognised by **Clater** (1858), modified by **Huxley** (1868), extended by **Wallace** (1876) and best described in a modern way by **Darlington** (1957) and others. The six zoogeographical regions are the **Palaearctic**, the **Nearctic**, the **Neotropical**, the **Ethiopian**, the **Oriental** and the **Australian**. Because some zoogeographers consider the Neotropical and the Australian regions to be so different from the rest of the world, these two are often considered as regions or realms equal to the other four combined. They are classified as **Neogea** (the Neotropical), **Notogea** (the Australian) and **Metagea** (the Palaearctic, Nearctic, Ethiopian and Oriental). Moreover, two zoogeographic regions, namely the Palaearctic and the Nearctic are quite closely related, so the two often considered as one, the **Holarctic**. In fact, these two regions are similar in climate, vegetation, and in their faunal composition having animals such as wolf, hare, moose (called elk in North America), caribou, wolverine and bison.

A. Palaearctic region. This largest region comprises the whole of Europe, Soviet Russia, Northern China, Japan, Northern Arabia (Persia) and narrow strip of coastal North Africa. It is subdivided into **European**, **Mediterranean**, **Siberian** and **Manchurian** subregions. Its eastern parts have characteristic fauna of the following animal species : *Rhacophorus* (flying frog), *Bombinator* (fire-bellied frog), *Agkistrodon halys* (pit viper), *Strix uralensis*, *Grus japonensis* (crane), *Aix galericulata*, *Erinaceus* (hedgehog), *Podces panderi*, *Capricornis sumatraensis*, *Camelus ferus* (camel), *Bos mutus*, *Ailuropus* (great panda), *Panthera tigris altaica* (tiger), *Uncia uncia*, *Phoca sibirica* (fresh-water seal), *Pteromys volans*, *Equus hemionus* (donkey) and *Macaca fuscata* (rhesus monkey). The common fauna of Western Palaearctic include animal species such as *Clupea harengus* (herring), *Sardina pilchardus* (fish), *Gadus morrhua* (codfish), *Vipera ammodytes* (viper snake), *Chamaeleon chamaeleon* (arboreal lizard), *Pelecanus onocrotalus* (pelecan), *Fratercula arctica*, *Phoenicopter ruber*, *Rangifer tarandus* (caribou), *Ovis ammon*, *Dama dama*, *Capra aegagrus*, *Odobenus rosmarus* (walrus), *Ursus arctos*, *Lynx lynx* (wild cat), *Castor fiber* (beaver), *Hystrix cristata* (porcupine), and *Panthera pardus* (leopard).

B. Nearctic region. This region comprises the North American continent south to the Tropic of Cancer (i.e., Greenland and North America). It is subdivided into **Californian**, **Rocky mountain**,

Allegany and **Canadian** subregions. The Nearctic is the home of many reptiles and has more endemic families of vertebrates. Characteristic fauna of this region include animal species as *Ambystoma tigrinum*, *Crotalus adamanteus* (rattle snake), *Branta canadensis* (Canadian goose), *Larus schistisagus* (gull), *Bucephala albeola*, *Meleagris gallopavo* (turkey), *Didelphis marsupialis* (opossum), *Ovibos moschatus* (musk-ox), *Rangifer tarandus arcticus* (caribou) *Bison bison* (North American bison), *Odobenus rosmarus divergens*, *Cystophora cristata*, *Martes pennanti* (fisher), *Lutra canadensis* (otter), *Dasybus novemcinctus* (armadillo), *Canis latrans*, *Procyon lotor* (raccoon), *Lagenorhynchus acutus*, *Delphinapterus leucas* and *Enhydra lutric*.

C. Neotropical region. It includes Central America, South America, part of Mexico and the West Indies. This region lacks well-developed ungulate fauna of the plains, but have rich, distinctive, varied endemic fauna of vertebrates. Characteristic animals of this region are *Lepidosiren* (lung-fish), *Lepidosteus* (Garpike), *Histrion histrio*, tortoises (*Dermatemys*, *Stourotypus*, *Peltocephalus*), *Heloderma horridum*, *Iguana iguana*, *Amazilia yucatanensis*, *Phoenicopterus ruber*, *Pelicanus occidentalis* (brown pelean), *Phalacrocorax harrisi*, *Sarcoramphus papa*, *Amazona amazonica*, *Spheniscus magellanicus*, *Chironectes minimus*, *Desmodus rotundus*, *Myrmecophaga tridactyla* (giant anteater), *Tapirus terrestris* (tapir), *Lama vicugna*, *Lama guanicoe*, *Panthera onca*, *Arctocephalus australis*, *Cavia aperea*, *Bradypus tridactylus* (three-toed sloth), *Vultur gryphus*, *Pterocnemia pennata* and *Ara macao*.



The jaguar (*Panthera onca*) maintains the populations of many prey species at low levels, establishing a "balance of nature" in the Neotropical forest.

D. Ethiopian region. The old world counterpart of the Neotropical is the Ethiopian, which includes the continent of Africa, south of the Atlas mountain and Sahara Desert, Southern Arabia, Madagascar and Mauritius. It embraces tropical forests in central Africa and in the mountains of East Africa, Savanna, grasslands and deserts. It includes a varied vertebrate fauna and several endemic families. The characteristic animals of this region are *Protopterus* (lung fish), *Crocodilus niloticus*, *Sagittarius serpentarius*, *Struthio camelus* (ostrich), *Oryx gazella* (African antelope), *Giraffa camelopardalis* (giraffe), *Loxodonta africana* (Africana elephant), *Equus quagga*, *Hippopotamus amphibius*, *Daubentonia madagascariensis* (ayeaye), *Lemur catta*, *Macaca sylvana*, *Pan troglodytes* (chimpanzee), *Gorilla gorilla*, *Acinonyx jubatus*, *Panthera leo* (lion), *Hyaena hyaena*, and *Delphinus delphis*. It lacks deer and bear among mammals and salamanders and tree-frogs among amphibians and contain several endemic families of birds.

E. Oriental region. It includes India, Indochina, South China, Malaya, and the western islands of the Malay Archipelago. It is bounded on the north by the Himalayas and on the other sides by the Indian and Pacific oceans. On the southeast corner, where the islands of the Malay Archipelago stretch out toward Australia, there is no definite boundary, although **Wallace's line** is often used to separate the Oriental from the Australian regions. This line runs between the Philippines and the Moluccas in the north, then bends southwest between Borneo and the Celebes, then south between the islands of Bali and Lombok. A second line, **Weber's line**, has been drawn to the east of Wallace's line ; it separates the islands with a majority of Oriental animals from those with a majority of Australian ones. Since the islands between these two lines form a transition between the Oriental and the Australian regions, some zoogeographers call the area **Wallacea** (Smith, 1977).

The Oriental region is divided into the following four subregions : **Indian sub-region**, **Ceylonese sub-region**, **Indo-Chinese subregion** and **Indo-Malayan sub-region**. Its characteristic fauna includes the following animal species : *Rhacophorus pardalis*, *Gavialis niloticus* (gharial), *Calotes versicolor* (garden lizard), *Draco volans* (flying lizard), *Draco dussumieri* (flying lizard), *Python reticulatus*, *Naja naja* (Indian cobra), *Ophiophagus hannah*, *Gallus gallus* (jungle fowl), *Pavo cristatus* (peacock), *Milvus migrans* (pariah kite), *Eudynamis* (Koel), *Psittacula krameri* (rose-ringed parakeet), *Bubo bubo* (great horned owl), *Coracias benghalensis* (blue jay), *Dinopium benghalense* (golden-backed woodpecker), *Corvus splendens* (crow), *Argusianus argus*, *Elephas maximus* (elephant), *Rhinoceros unicornis*, *Antelope cervicapra* (black buck), *Axis axis* (spotted deer), *Boselaphus tragocamelus* (blue bull), *Selenarctos tibetans* (black Himalayan bear), *Melursus ursinus* (sloth bear), *Sus cristatus* (wild boar), *Hystrix leucura* (Indian porcupine), *Herpestes* (mongoose), *Manis* (pangolin), *Soriculus* (Indian shrew), *Hyaena striata*, *Panthera tigris tigris*, *Panthera pardus*, *Babryrousa babryrousa*, *Macaca mulatta*, *Presbytis entellus* (langoor), *Pongo pygmaeus* (orangutan), *Ailurus fulgens*, and *Hylobates lar* (gibbon).

F. Australian region. This region includes Australia, Tasmania, New Guinea, and a few smaller islands of the Malay Archipelago. New Zealand and the Pacific Islands are excluded, for these are regarded as oceanic islands separate from the major faunal regions. Partly tropical and partly south temperate, the Australian region is noted for its lack of a land connection with other regions; the poverty of freshwater fish, amphibians and reptiles; the absence of placental mammals and dominance of marsupials. Included are the egg-laying mammals (monotremes) and the spiny ant-eaters. The characteristic fauna of this region includes following animal species : *Neoceratodus* (fish), *Chelmon rostratus*, *Phyllopteryx eques*, *Ornithorhynchus anatinus* (platypus or duck mole), *Zaglossus bruijnii* (proechidna), *Techyglossus aculeatus* (echidna, spiny ant-eater), *Macropus giganteus* (kangaroo), *Notoryctes typhlops*, *Macrotis lagotis*, *Dromaius novaehollandiae*, *Petaurus australis*, *Paradisaea rubra*, *Pteridophora alberti* and *Phaseolarchos cinereus* (koala).

New Zealand has *Varanus komodensis*, *Sphenodon punctatus*, *Abduryx owenii haasti* (kiwi), etc. Sometime, south polar region called the **Antarctic** or **Archinotic** is also included in these major distributional units. Antarctic region has the following characteristic animal species : *Balaenoptera musculus* (blue whale), *Orcinus orcam*, *Megadyptes antipodes*, *Aptenodytes patagonica* (penguin), *A. forsteri* (emperor penguin), *Pygoscelis adeliae*, *Diomedea exulans* (albatross), *Chionis alba*, and *Lobodon carcinophagus*.

REVISION QUESTIONS

1. Describe different patterns of distribution of biota.
2. What is phytogeography ? Describe chief vegetational belts of Earth.
3. Write an essay on the phytogeography of India.
4. What is zoogeography ? Describe different zoogeographical regions of Earth.
5. Describe the process of dispersal of organisms.
6. Write short notes on the following : Endemic species ; Oriental region ; range expansion ; broadcasting; barriers; continental drift theory; land bridges; migration; and ecesis.

CHAPTER

18

Adaptations

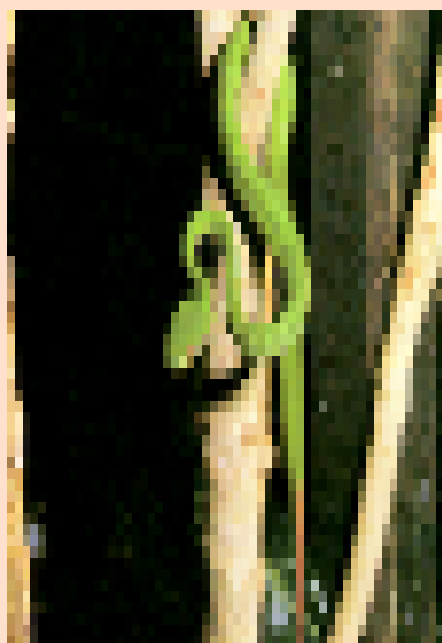
(Aquatic Adaptations, Volant Adaptations And Desert Adaptations)

The idea of adaptation maintains that organisms (animals and plants) are structurally and functionally designed for meeting the needs of life in the habitats in which they live. Thus, adaptations include adjustments by which an organism accommodates itself to its environment. These may occur by natural selection. In this chapter only following three types of adaptations of animals have been described : 1. Aquatic adaptations ; 2. Volant adaptations and 3. Desert adaptations. (For other sort of adaptations of plants and animals see Chapter 3).

AQUATIC ADAPTATIONS

Aquatic adaptations occur in those animals which live in water habitat, viz., fresh, brackish or sea water. They are called **aquatic animals** or **hydrocoles**. Based upon the phylogenetic history of the aquatic animals, following two types of hydrocoles have been recognised :

1. Primary aquatic animals. The primarily aquatic animals are those in which the phylogenetic history is restricted to water as habitat. Therefore, all their adaptations are originally designed to meet the necessities of aquatic life. Generally, by



Pit Viper has heat sensitive pits between each eye and nostril. This enables them to pick up changes in temperature around them.

primarily aquatic forms is meant the fishes, which have never had a terrestrial ancestry, but have evolved from more primitive aquatic progenitors. As a consequence their adaptation to a dense watery medium is perfect and they do not suffer as those secondarily adapted do through their inability to breathe water. They are, therefore, the primitive gill breathing vertebrates. Primary aquatic animals include protozoans, sponges, coelenterates, some annelids, molluscs and arthropods ; echinoderms ; and chordates such as cephalochordates (*Amphioxus*), urochordates (*Herdmania*, *Doliolum*, *Salpa*, etc.), cyclostomates (*Petromyzon*, *Myxine*), fishes, etc.

2. Secondary aquatic animals. These are those hydrocoles which have a record of terrestrial life in their phylogeny (i.e., they are descended from ancestors which led a life on land). Secondary aquatic animals are lung-breathers, mainly amphibious vertebrates, which through stress of circumstances such as inhospitable lands, where food was scarce or severe competitions were forced to return once more to the water. Consequently these animals show in their body structure, the evidences which speak of their ancestry from land-living animals, e.g., *Pila*, frog and other amphibians, swamp river turtles (*Notosaurs*, *Phytosaurs*), crocodiles, birds such as *Ichthyornis*, penguins, albatrosses, petrels, ducks, geese, etc., and mammals such as *Hippopotamus*, *Otter*, whales, porpoises, etc.

A. Primary Aquatic Adaptations

1. Body contour. The form of body depends upon the habits of life. The majority of fixed and partly sedentary forms have radially symmetrical body forms, e.g., sponges (*Sycon*, *Euplectella*, *Hyalonema*, etc.), *Hydra*, *Obelia*, *Aurelia*, *Metridium* (sea anemone), echinoderms such as *Holothuria*, *Echinus*, *Astropecten* or star fish. The active locomotor type have fusiform, spindle-shaped elongated and worm-like bodies. The spindle form is the characteristic of fishes and wavy, worm-like form is found in the annelids (*Nereis*).

The piscine body is designed for fast locomotion in water. There occurs a side to side compression of head, body and tail into a beautifully curved streamlined fish form. Head is sub-conical. There is no protuberance over the body, which would retard the swift passage of the animal through water. Further, since the relative weight is more on the upper half of the fish body (on account of the myotomes) a dead fish will float ventral side turned upwards.

Regarding the body form of planktonic organisms (which float passively at or about the surface), some are globular such as *Noctiluca*, some have umbrella shape (medusa of *Obelia*, *Aurelia*) and some have barrel-like shape (tunicates such as *Doliolum* and *Salpa*).

2. Swimming organs. Some aquatic animals float passively and do not possess much powers of movement, e.g., dinoflagellates (Protozoa). The medusae and most siphonophores move by alternative contraction and expansion of their sub-umbrellar side.

Organs for active swimming exist in arthropods (e.g., prawn or *Palaemon*), annelids

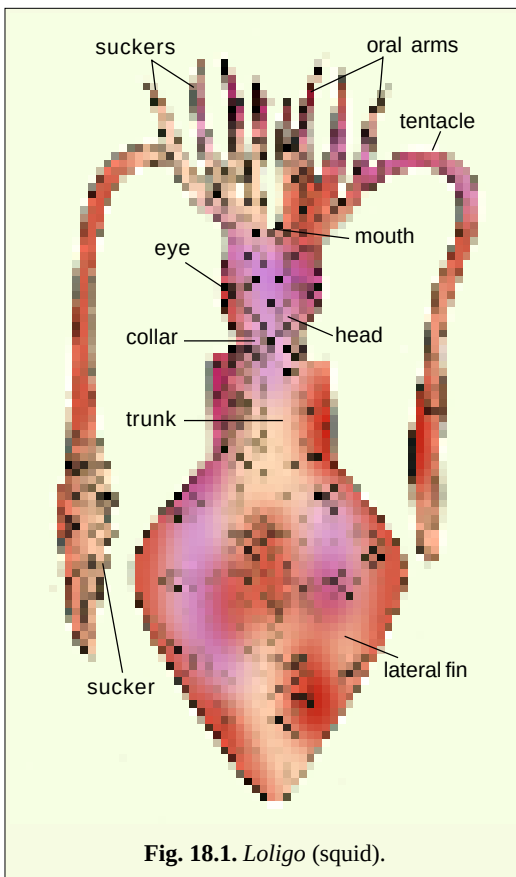


Fig. 18.1. *Loligo* (squid).

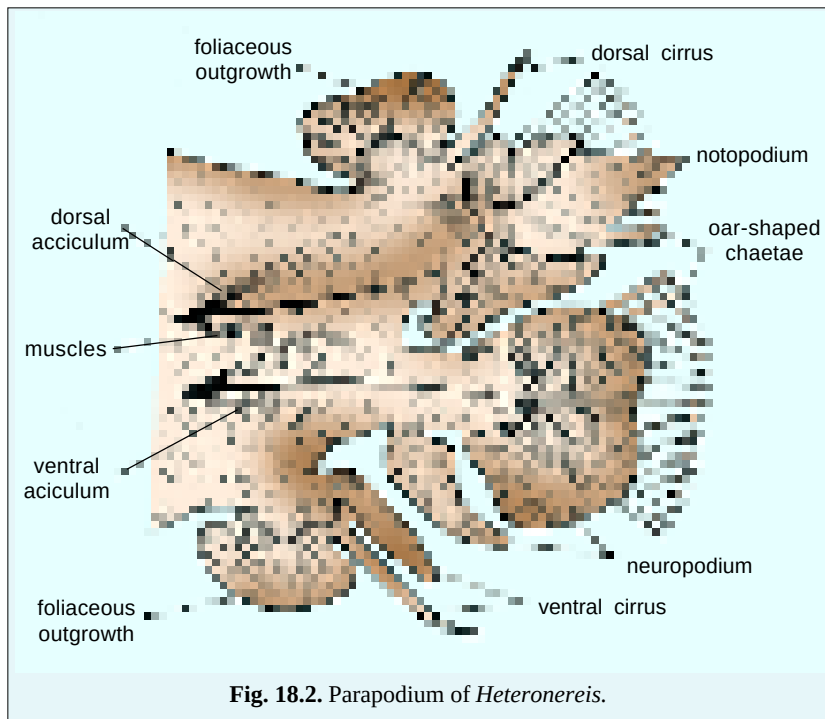


Fig. 18.2. Parapodium of *Heteronereis*.

(e.g., *Heteronereis*), cephalopods (e.g., *Sepia*, *Octopus* and *Loligo*) and vertebrates. For example, the **parapodia** of *Heteronereis* (Fig. 18.2) are well adapted for active swimming. Each parapodium has a flattened structure divided into the noto-podium and neuropodium bearing cirri and setae. The prawn has specially adapted abdominal appendages, called **pleopods** and **uropods** for swimming.

In vertebrates, the primary aquatic animals are the fishes. The fish move (swim) by the help of fins and also by lateral undulations of the flexible body. The fins of fish are of two types, the median fins and paired fins. The median fins include dorsal fins, caudal fins and ventral fins. The paired fins are the pectorals and pelvics (Fig. 18.3). The caudal fin is the chief propeller, the dorsal and ventral fins help to keep body vertical and the pectoral and pelvic fins help in propulsion and in making changes in direction.

3. Respiration. The primary aquatic animals are able to respire inside the water, without the need to come up to the surface. The exchange of respiratory gases takes place between the blood of these animals and the water outside. There are two methods of aquatic breathing : 1. through diffusion through general body surface, e.g., protozoans, coelenterates and planktonic larvae, and 2. with the help of special organs called branchia or **gills**, e.g., prawns, and other crustaceans; *Unio*, *Pila* and other molluscs ; and many vertebrates such as fishes, tadpole of frog and salamanders. Indeed, gills of fishes are most remarkable aquatic breathing organs utilizing dissolved oxygen of water.

4. Air bladder. Advanced

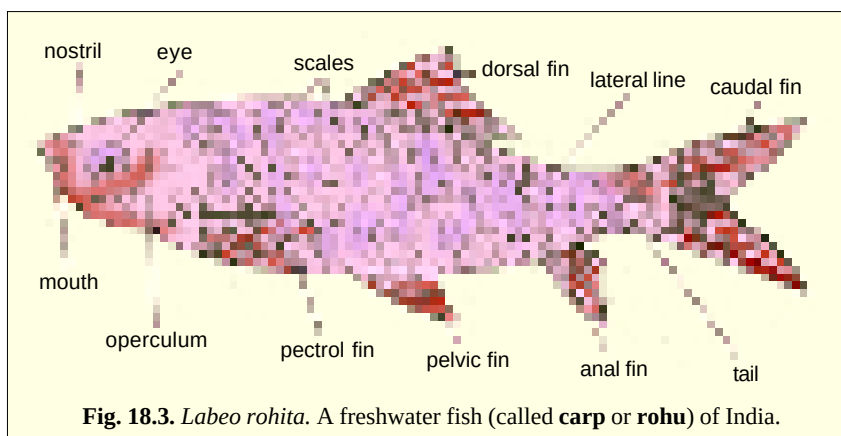


Fig. 18.3. *Labeo rohita*. A freshwater fish (called **carp** or **rohu**) of India.

bony fishes contain air bladder (or swim bladder) which serve as an accessory respiratory organ and hydrostatic organ. Air bladder is a hollow outgrowth of the alimentary canal and is filled with gas or air.

5. Lateral line sense organs. Fishes have lateral line systems extending all over the body. It contains neuromast organs which act as rheoreceptors (*i.e.*, detect pressure changes in surrounding water).

6. Skin. Skin of fishes is rich in mucous glands and/or is protected with scales.

B. Secondary Aquatic Adaptations

1. Stream-lined body. The body shape is stream-lined like primarily adapted forms : neck constriction disappears and tail enlarges, *e.g.*, Ichthyosauria (extinct fish-lizards), Cetacea (whales, dolphins, and porpoises), Sirenia (manatees and dugongs), Pinnipedia (walruses and seals). Frog also contains stream-lined body.

2. Enlargement of size. Aquatic vertebrates tend to be larger in size because in these creatures energy, which in terrestrial form is exhausted in gravitational forces, is turned into growth. For example, largest sulphur-bottom whale (*Balaenoptera musculus*) is several times bigger than the largest elephant. Other example include giant sharks and squids.

3. Submergence. All secondary aquatic animals need to develop capacity of submergence since swimming below water surface demands such an adaptation. For example, in whales the ribs are strongly arched, the lungs are massive, the external nostrils communicate with the median "blow hole" which is closable.

Certain adult aquatic insects too are able to increase their period of submergence by storing air inside the subelytral space, *e.g.*, *Nepa*.

4. Shortening of neck. There occurs reduction of length and mobility of neck. In whales cervical vertebrae (which are seven in number like other mammals) are fused to form a solid and compressed mass of bone.

5. Disappearance of excrescences. The external ears (pinnae) which hinder water locomotion tend to disappear, since they collect sound waves in air medium and are useless in aquatic forms. Thus, ears are reduced in amphibious mammals and are lost in whales, true seals and walruses. The nostrils (nares) move towards the apex of head as in whales, ichthyosaurs, phytosaurs, etc. Nares are often capable of being closed (*e.g.*, otter). Likewise, eyes become water-adapted by shifting higher on the face as in hippopotamus.

6. Occurrence of locomotory paddles (fins). There occur fleshy and fin-like expansions of the body wall in whales and ichthyosaurs which help in propulsion. These fins may be dorsal or caudal. Dorsal fin is present in killer whale, while absent in *Delphinopterus* and *Balaena*. Caudal fin (also called **caudal** or **tail fluke**) of marine mammals in horizontal (vertical in reptiles) and the bone divides the tail into two equal parts rather than running into one lobe. In turtles **oar propulsion** occurs by fin-like limbs (Fig. 18.4); but in whales, sirenians, etc., **tail propulsion** takes place as their hindlimbs become disappeared. Pectoral paddles of whales and sirenians exhibit the following

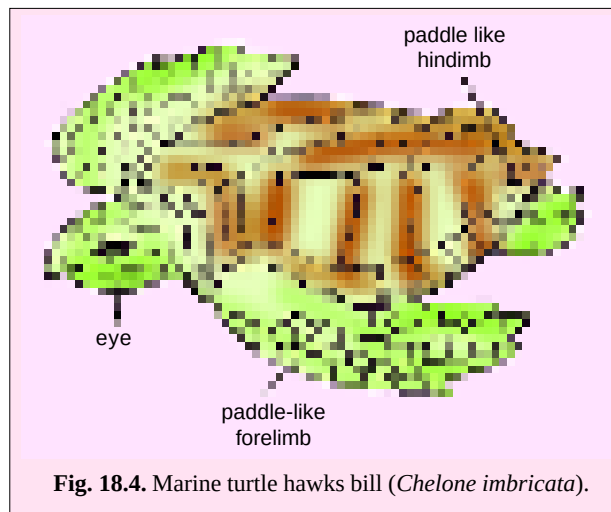


Fig. 18.4. Marine turtle hawks bill (*Chelone imbricata*).

adaptations : (1) the restriction of movements corresponding to the elbow and wrist joints ; (2) the fusion between digits ; (3) increase in the number of phalanges, called hyperphalangy, and (4) increase in the number of digits for increase of expanse of paddling surface, called hyperdactyly.

In *Nepa* (Insecta) legs are flattened and oar-like.

7. Disappearance of hairs, skin glands, etc. In whales and sirenians, the skin becomes naked due to loss of hairs. The hair loss is compensated by the formation of a fatty layer below the skin (**blubber**) for the retention of the bodily heat. The blubber also has a hydrostatic advantage (*e.g.*, it helps in floatation or to keep positions in the water and act in combination with the buoyancy of the aquatic medium).

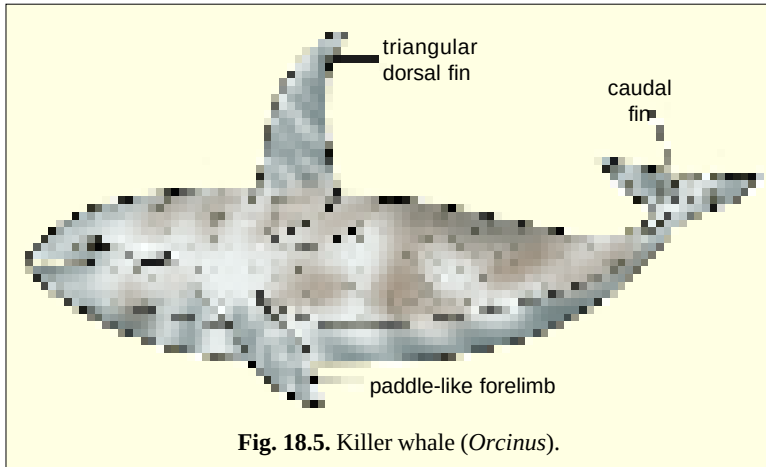


Fig. 18.5. Killer whale (*Orcinus*).

Sweat or oil glands disappear as they have nothing to do with the aquatic mode of life. Muscles and nerves also atrophy from the integument due to its thickening and immobility.

8. Mouth armament. Since jaws are not used for mastication in whales they lost the power of movement. Teeth become simplified (homodont in dolphins)

and greater in number. In sperm whale, teeth are present only on one jaw or entirely absent from both the jaws (*e.g.*, baleen whale).

9. Skull modification. In certain aquatic mammals (*e.g.*, Dolphins, porpoises) the cranium is shortened and front part of the skull becomes elongated to acquire the shape of a **rostrum**. In the skull of the cetacea, the zygomatic arch is reduced to a vestige.

10. Simplification of vertebrae. In secondarily aquatic forms (vertebrates) the vertebrae tend to be simple. In Ichthyosaurs, vertebrae are simple with biconcave centra like the fishes. Various secondary articulations or zygapophysis become reduced, as body weight is supported by water. The chest too become cylindrical. The rib articulations are modified and are central, *i.e.*, they are articulated to the centrum and are not articulated to the transverse processes.

Sacrum in cetaceans and sirenians is more or less reduced, since it does not withstand and transmit the supporting impact of the hindlimbs, as does in terrestrial forms.

11. Lightness of bones. The bones in aquatic forms are light and spongy. In whales, their interstices are filled with oil.

VOLANT ADAPTATIONS

The volant adaptations are concerned with the flight. The flight is a form of locomotion in the air under which the body has to be firstly prevented from falling down and secondly moved forwards, the speedier the better. Thus, volant adaptations must include modifications in the animals body for reducing the weight of the body and also for the formation of organs capable of executing the flight.

The flight may be of following two types :

1. Passive or gliding type flight. This type of movement involves no propulsion other than the initial force of jumping. Gliding is characterised by leaping or jumping from a high point and held up by some sustaining organs, then to glide to lower level. Thus, there is no locomotive force other than

gravity. Here the 'wings' are made of **patagia** which do not flapped (*i.e.*, do not move up and down) by the muscular action.

The gliding flights are performed by various lizards (*e.g.*, *Draco volans* or flying dragon, Fig. 18.6), fishes (*e.g.*, *Exocoetus*, Fig. 18.7), birds (*e.g.*, ostriches, etc.), mammals such as flying phalangers (*Petaurus sciurens*, a marsupial, Fig. 18.8), flying squirrel (Fig. 18.8) etc., and amphibians (*e.g.*, *Rhacophorus*, Fig. 18.8). All these forms are found to possess the following adaptations for the gliding :

(i) Development of patagia. The sustaining surface ("wing") for the gliding is a fold or series of folds of skin, called **patagium**. The patagium lies between forelimbs and hindlimbs and can be folded like a fan against the body when is not in use. In *Draco*, the patagium is supported by ribs (Fig. 18.6). *Ptychozoon* (Fig. 18.8) is another gliding lizard which is commonly known as "the flying or fringed gecko" and in which lateral expansion of skin (patagium) extends along the side of neck, body, tail and limbs and between toes. Flying snakes (*e.g.*, *Chrysopelea*) leap by the concave ventral side of body. The extinct reptiles pterodactyls (Fig. 18.8) of Mesozoic era were volant creatures akin to birds, containing true flight patagia. Their patagia were extensions between limbs supported by ribs.

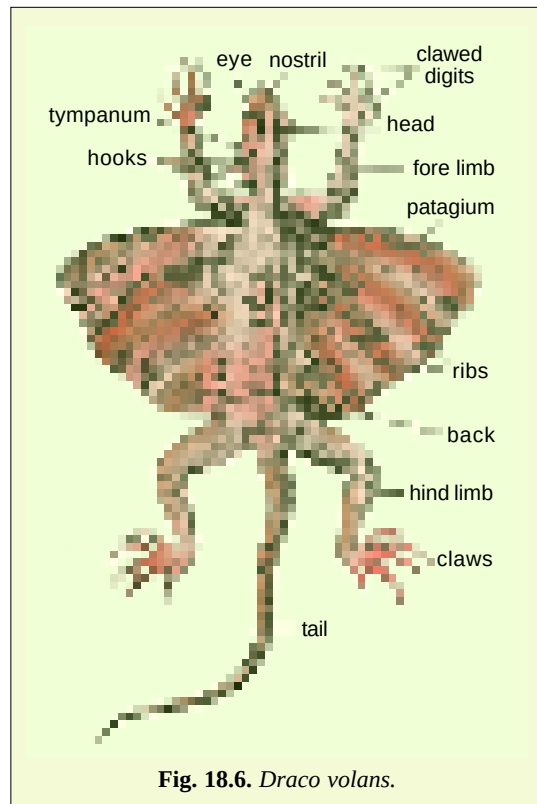


Fig. 18.6. *Draco volans*.

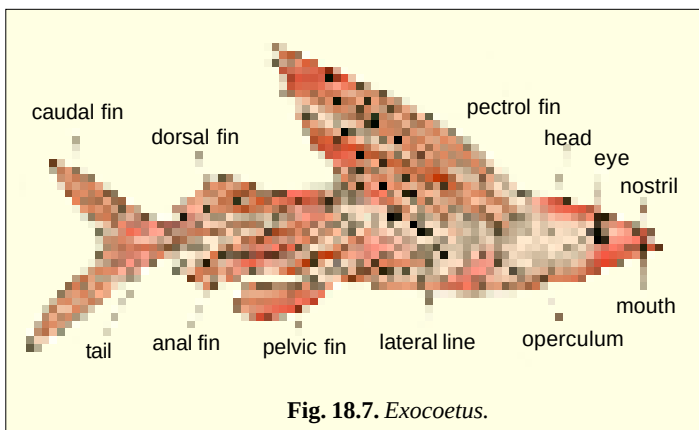


Fig. 18.7. *Exocoetus*.

Among mammals, flying squirrel, *Sciuropterus* (Fig. 18.9) has highly developed patagium. Its wide hairy tail is further supplemented by the hairy fringe on the patagium and along the rear of the thighs. In flying lemur (*Galeopithecus volans*; Fig. 18.9) the patagium extends from side of the neck to the tip of the tail even including digits, which are also webbed. In the bats, the patagium is supported mainly by the elongated forelimbs and the

second, third and fifth digits. The first digit remains free.

Traces of patagia are also found in front and behind the arms (forelimbs) in birds which have adequate supporting function. Flying frog too contains rudiments of patagia in front and behind the limbs.

(ii) Enlargement and high insertion of pectoral fins. In flying fishes (*Exocoetus*) are trim-built creatures with large parachute-like pectoral fins which are highly inserted on the body. The pelvic fins are much smaller in size. The lower lobe of tail is also invariably longer, helping in leaping. The pectoral

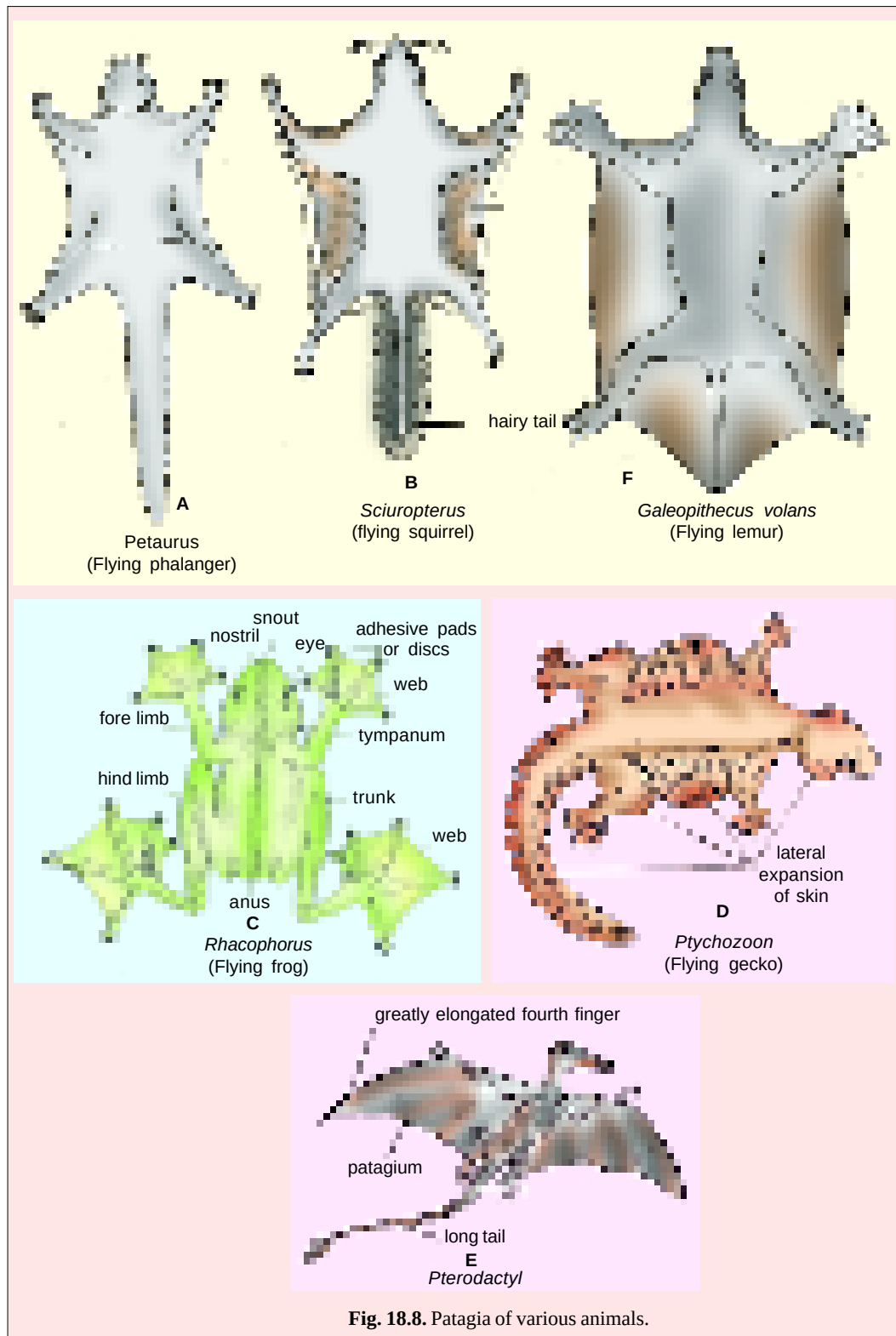


Fig. 18.8. Patagia of various animals.

fins do vibrate. *Exocoetus* can fly up to 200–300 metres to escape from the large fishes such as tunny and albacore). Other genera of flying fishes are *Dactylopterus*, *Pantodon* and *Pegasus*. *Pegasus* is a little marine fish found along the coasts of Japan, China, India and Australia. They skim along the surface of the water for 40 feet or more.

(iii) Webbing of feet. In flying frog (*Rhacophorus paradailis*, Fig. 18.10), the feet are webbed which sustain the prolonged leaps. Flying frog's digits terminate in adhesive pads which help in adhesion to trees.

2. Active or true flight. It is the aerial flight caused by the action of wings. True flight is found in insects, pterodactyls, birds and bats. In all of them the nature of development and structure of wings are quite different and their analogy suggest that the flight has evolved independently in different groups. In true flights the power is implied and the movement in air is sustained.

Varieties of Wings

1. Insect wings. The flying insects develop wings at their last moult. The mode of origin of the wing of the insects is different in the Exopterygota and Endopterygota. The wings are made up of cuticle and are strengthened by thickenings, called **veins**. The insect flight is affected by the flapping movements of the wings. Typically, there are two pairs of wings developed on the dorsolateral sides of the meso- and meta-thoracic segments (e.g., Orthoptera, Isoptera, Coleoptera, Ephemeroptera, etc). In Diptera only the mesothoracic wings are developed. The wing muscles are highly striated and are rich in mitochondria.

In fact, insect's body is ideally suited for the evolution of aerostatic adaptations because of the system of tracheae which contain air and penetrate to the different parts of body. The highest degree of adaptation is met with in the dragon flies, may flies and damsel flies. In may flies, the adults are short lived and has its alimentary canal filled with air. In the honey-bees, there are air sacs associated with the tracheal system, these function in reducing the weight of the body.

2. Bat wing. In a bat wing, humerus bone is well developed, radius is long and curved and ulna is vestigial. The pollex (thumb) is free and clawed for crawling and climbing. In smaller insectivorous bats (Microptera; Fig. 18.9), the second finger is not

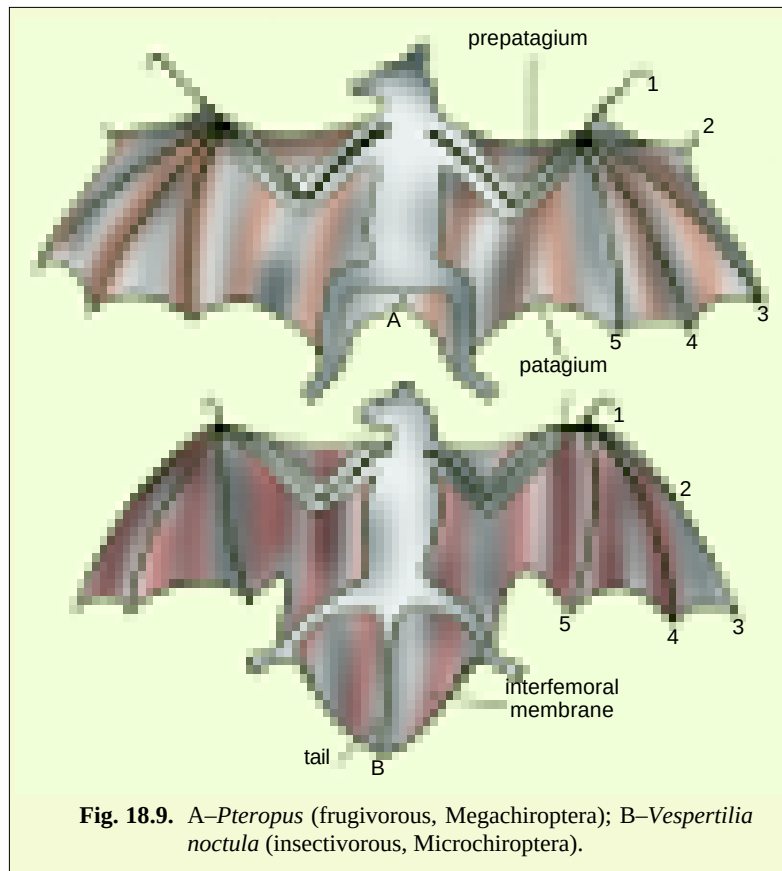


Fig. 18.9. A–*Pteropus* (frugivorous, Megachiroptera); B–*Vespertilia noctula* (insectivorous, Microchiroptera).

free from third, but is attached to it distally. These two support the anterior margin of the wing. The fourth and fifth digits are well developed. In frugivorous bats (Megachiroptera, Fig. 18.9), the second digit is clawed and free from the third.

3. Pterodactyl wing. In pterodactyl's wings, the radius and ulna bones are nearly equal. The next segment consists of a heavy fourth metacarpal bearing great wing finger and three small metacarpals supporting the first, second and third clawed digits. There is also present a **pteroid** bone which is directed towards the shoulder and supposed to support the anterior margin of a prepatagium which lies in front of arm from the wrist to the neck. The single wing finger (*i.e.*, fourth) is huge and forms the entire anterior support of the patagium beyond the wrist.

4. Bird wing. In birds forelimbs are modified into flight structures, the wings. The wings of birds are more specialized of all modern wings. In a wing of bird, digits are reduced to three and these are fused together to help in flight. The metacarpals are three which are unequally developed and co-ossified. The digits are represented by one or two phalanges which support the so-called **bastard quills** (*i.e.*, three, four or five feathers on the first digit (homologous of the thumb) of a bird's wings, (Fig. 18.10).

Volant adaptations of birds. Bird flight is characterized by the flapping of the wings. In flying, the bird lifts its body and drives forward by beating its wings in a characteristic way. An object moved swiftly in the air is affected by two forces : 1. upward pushing (**lifting**) and 2. downward pull (**drag**) due to the action of gravity. Hence, for successful flight, enough force must be applied to neutralize the drag and also to move forward. Birds wings are slightly concave, so are able to produce the air-current for producing the lifting force.

Birds have the following adaptations for true flight :

1. Body contour. The streamlined body is spindle-shaped or boat shaped, encountering least aerial resistance and can easily be passed through the air. Thus, the beak is pointed; head is compact; neck is long and mobile and wings are attached high upon the thorax.

2. Development of feathers. The entire body of birds is invested with a close covering of feathers, constituting the **plumage**. The feathers form the exoskeleton of birds. These nature's "master pieces" are light, elastic, waterproof and most important in flight. Bird's feathers are classified into quill feathers, contour feathers, down feathers and filoplumes. Quills are flight feathers. Flight feathers of wings are called **remiges** and those of tail are called **rectrices**. Structurally, flight feather consists of a basal **quill** or **calamus** and distal **rachis** or **shaft**. The rachis bears a leaf-like **vane** (or vexillum) consisting of series of lateral **barbs**. Barbs consist of double row of **barbules** connected with each other by **barbicles** or **hooklets** (through interlocking arrangement). The barbs, barbules and barbicles form a sort of net which help in flight.

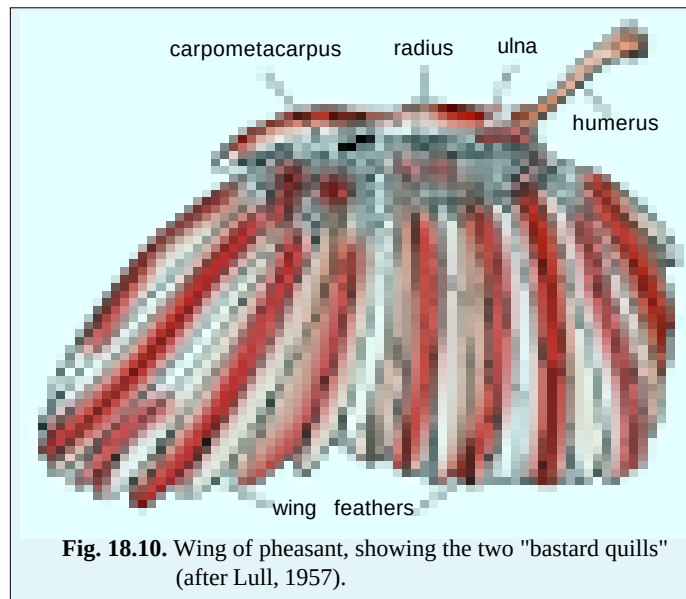


Fig. 18.10. Wing of pheasant, showing the two "bastard quills" (after Lull, 1957).

3. Presence of wings. In birds the **fore-limbs** are modified into **wings** helping in flying. The hind-limbs or legs are large and variously adapted for walking, running, scratching, perching, food-capturing, swimming.

4. Pneumatization of bones. The bones of birds are hollow and air filled. They also contain many

air cavities. These add buoyancy during flight.

5. Occurrence of flight muscles and keeled sternum. In birds, specific flight muscles are developed which connect the wings with limb bones. Each wing is depressed or lowered by an enormous muscle called **pectoralis major**. It is elevated or raised by **pectoralis minor** (the tendon of which is inserted on the head of humerus). The **sternum** or breast bone is well developed and bears a median keel or carina for the attachment of pectoralis muscles.

6. Development of air sacs. They act as air reservoir during respiration and serve as balloons to provide lightness and buoyancy to the body. Air sacs also help in internal perspiration, thus, helping in the regulation of body temperature.

7. Brain and sense organ's specificity. Cerebrum is well developed (for controlling manoeuvrability) and optic lobes become enlarged (for controlling the great development of sight) and olfactory lobes are reduced (*i.e.*, power of smell is reduced). Bird's eyes are large and bear characteristic sclerotic plates to resist variable air pressure. Eyes also contain pectens which are comb-like, vascular and pigmented structures to regulate fluid pressure within the eye (*i.e.*, accommodation).

8. Beak. The conversion of forelimbs into wings is compensated by the presence of a **bill** or **beak**. The beak is horny and lacks teeth.

9. Mobile neck. The neck of birds is very long and flexible.

10. Single ovary. Presence of a single functional ovary of the left side in the female bird also leads to reduction of weight which is very essential for flight.

11. Absence of urinary bladder. Birds do not have a urinary bladder which is present to store the urine temporarily in other animals. Further, birds excrete a semisolid excreta which chiefly contains the insoluble uric acid and urates (**ureotelic excretion**). These features help in reducing the weight of body.

DESERT ADAPTATIONS

Desert animals or xerocoels have adaptations for following three sorts :
1. Moisture getting; 2. Moisture conservation and 3. Self-defence against physical and organic environment.

1. Moisture getting. As already described in Chapter 3, the securing of adequate quantity of moisture is the prime need of desert forms and its scarcity is the characteristic of the desert. At certain places of desert is found deep ground water, even then the surface sand is dry and burning. Hence, for getting the deep water, rain water from the superficial layers of sand, absorption of dew water, the xerophytic plants (*e.g.*, date palm) and xerocoels (*e.g.*, *Moloch*) have varied adaptations. The date palm has long deep exploring and horizontally spreading roots. The sand lizard or *Moloch* has hygroscopic skin to absorb water like the

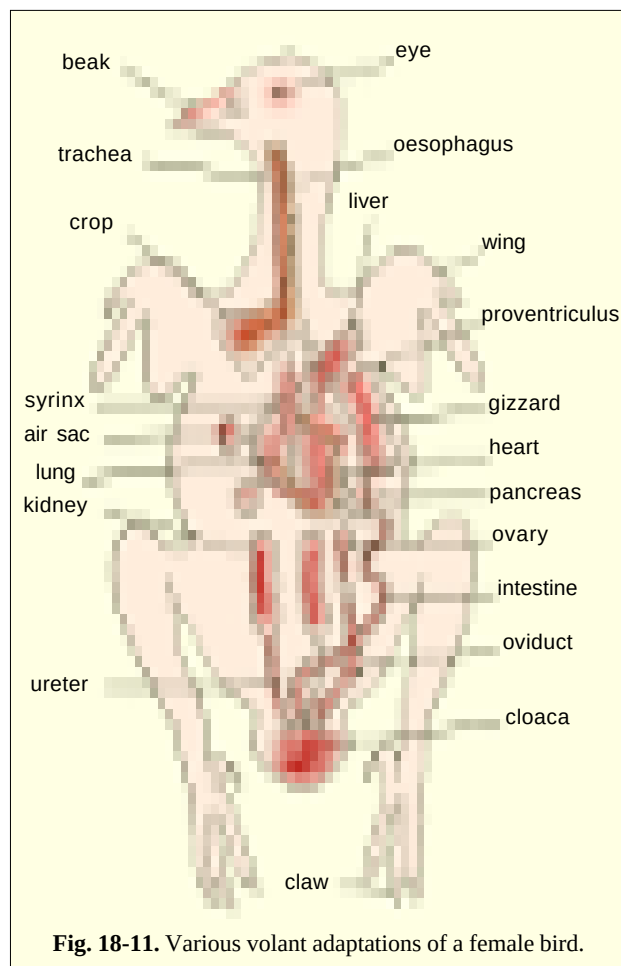


Fig. 18-11. Various volant adaptations of a female bird.

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blotting paper. Its surface is covered with thorn-like scales (Fig. 18.12). A few animals absorb dew drops along with vegetation food. Barrel cactus affords drinkable water to thirsty desert people.

Desert rabbit, tortoises and wood rat (*Neotoma*) eat succulent plant for their water need. Jerboas, kangaroo rat (*Dipodomys*), pocket mouse (*Perognathus*) and certain other rodents eat dry seeds and vegetation and quench their thirst. Carnivorous animals feed on their prey for food and water. Grasshopper mice of western North America and long eared hedgehog of Sahara desert eat insects and small rodents for getting water.

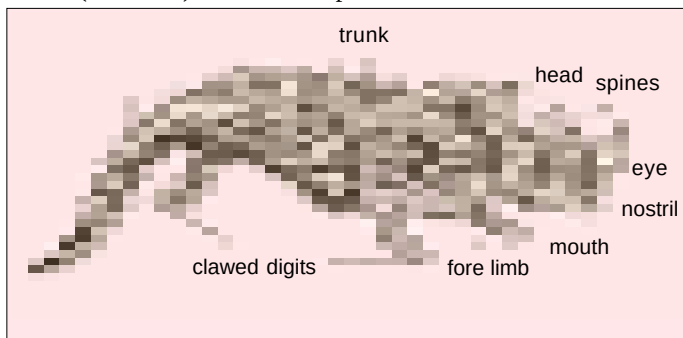
2. Moisture conservation.

After getting moisture it becomes essential for the xerocoles to conserve it. As a result, desert animals contain the following types of adaptations for the water conservation.

- (i) Camel exhibits many adaptations for water conservation (For details see Chapter 3).
- (ii) To avoid water loss through skin, the horned toad (*Phrynosoma*,) has hard and rough skin and *Moloch* has scales and spines all over the body surface (i.e., skin).
- (iii) *Uromastix* (desert lizard) store water in large intestine.
- (iv) Desert mammals has thick skin to avoid water loss by perspiration. In them the number of sweat glands in the skin are reduced or totally absent.
- (v) Desert insects are wax proof.
- (vi) To conserve water, desert animals remain in burrows during the day time and come outside during night when the percentage of moisture in their burrow and outside is equal. Certain animals plug the mouth of their burrows during day time.

3. Self defence against scorching sun. The desert animals protect themselves from the extreme heat of the sand and arid climate by the following type of adaptations:

- (i) In burrow-digging species of reptiles, the nostrils are directed upward instead of forward. In most snakes nostrils are either reduced to pin holes or protected by complicated valves.
- (ii) The eyes of *Typhlops* (a burrowing desert snake) are overhung by the sheath. In ostrich and camel, the eyes are protected by reflecting heat of the sand by possessing the long neck.
- (iii) In lizard *Mabuya*, the lower lip becomes enlarged with a transparent window in it.
- (iv) The ear opening of desert animals is either small or protected by fringes or scales or they may be abolished.
- (v) Possession of venom is another desert adaptation (self-defence). Almost all desert snakes are poisonous. All poisonous spiders are also found in desert. Scorpions too are common in deserts.
- (vi) *Gazelle* (antelope) resembles the general colouring of landscape (white on sand, dark grey on volcanic rocks). It provides a good example of self-defence.



Horned toad (*Phrynosoma*) has hard and rough skin to avoid water loss.

REVISION QUESTIONS

1. Write an account of primary aquatic adaptation ?
2. What is adaptation ? Give an account of volant adaptation.
3. Give an account of desert adaptations.

286 ventral
 aciculum

In vertebrates, the primary aquatic animals are the fishes. The fish move (swim) by the help of fins and also by lateral undulations of the flexible body. The fins of fish are of two types, the median fins and paired fins. The median fins include dorsal fins, caudal fins and ventral fins. The paired fins are the pectorals and pelvics (Fig. 18.3). The caudal fin is the chief propeller, the dorsal and ventral fins help to keep body vertical and the pectoral and pelvic fins help in propulsion and in making changes in direction.

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adaptations : (1) the restriction of movements corresponding to the elbow and wrist joints ; (2) the fusion between digits ; (3) increase in the number of phalanges, called hyperphalangy, and (4) increase in the number of digits for increase of expanse of paddling surface, called hyperdactyly.